

# Transaction Security Services Processing Guide

## BASE24<sup>®</sup>



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# Preface

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The Transaction Security Services process provides all hardware security module functions for BASE24 processes. The Transaction Security Services process is built using ACI's BASE24-es object-oriented architecture. Files used by the process are maintained through a separate user interface. The ***BASE24 Transaction Security Services Processing Guide*** describes how the Transaction Security Services process is integrated into a BASE24 system. It describes how BASE24 hardware security data is converted and maintained in data files, how BASE24 processes are configured to use the Transaction Security Services process, and the implementation requirements for using the process. It also describes the operations facilities used to maintain the Transaction Security Services process and provides several transaction message flows.

## Audience

This processing guide is intended for the system administrators and operational staff who determine how the BASE24 system is to handle transaction security and who set up the BASE24 database.

## Prerequisites

This processing guide is intended to be used in conjunction with the ***BASE24 Transaction Security Manual*** and the context-sensitive ACI Online Help provided with the ACI desktop user interface. The ***BASE24 Transaction Security Manual*** provides detailed information on BASE24 processing for transaction security functions including configuration, processing steps, and internal message content. It also provides detailed message authentication transaction flows. You can access context-sensitive online help directly from within the ACI desktop user interface. The ACI Online Help for the ACI desktop user interface describes the screens, fields, and procedures used to maintain hardware security data. Users of this processing guide should become familiar with the topics discussed in the ***BASE24 Transaction Security Manual*** and the relevant topics discussed in ACI Online Help for the ACI desktop user interface.

Also important is an understanding of the basic functionality of the hardware security modules to be integrated with BASE24. Hardware security modules are all integrated into the BASE24 system in a similar fashion; however, the individual functions of each module and the ways in which these functions are accessed vary by vendor and module. Thus, users of this processing guide should have a working knowledge of the security modules they plan to use, especially in the area of key generation and management.

## Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarized and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers who do not need the additional detail and at the same time provide the source for readers who require the additional information. This processing guide contains references to the following BASE24 publications:

- The ***BASE24-atm Diebold 10XX/478X Device Support Manual*** provides detailed information on exchanging certificates and downloading the ATM master key for the BASE24-atm Automated Key Distribution System add-on product.
- The ***BASE24-atm NCR 5XXX Device Support Manual*** provides detailed information on authenticating and exchanging public keys and downloading the ATM master key for the BASE24-atm Automated Key Distribution System add-on product.
- The ***BASE24-pos Files Maintenance Manual*** and ***BASE24-pos Standard POS Device Configuration Manual*** describe BASE24-pos Terminal Data files (PTD) and ACI Standard Device Configuration File (ACNF) screen fields, respectively, used to configure data field encryption.
- The ***BASE24 Transaction Security Manual*** provides detailed information on configuration, processing steps, and internal message content as well as on using triple DES in a *translate mode* only in certain Interchange Interface processes without using the Transaction Security Services process.
- The ***BASE24 Release 6.0 Implementation Guide*** and ***BASE24-es and TSS HP NonStop Install Guide*** provide detailed procedures on running the BASE24 database conversion programs and installing the Transaction Security Services process and its environment.

This processing guide also references the following ACI publications:



- The ***ACI Desktop User Interface Manual*** provides information on installing and configuring the ACI desktop user interface, setting up user security, and auditing read, add, update, and delete operations on Transaction Security Services files.
- The ***ACI Java Server Framework Reference Guide*** describes db.audited.db-service-name properties, the report.userauditlog.loc property, and other configuration properties supported for the TSS environment. This guide also describes the files and ACI desktop user interface windows specific to applications based on the ACI Java Server Framework. The Java User Interface servlets (ESWEB) process is based on the ACI Java Server Framework.
- The ***BASE24-es and TSS HP NonStop Install Guide*** provides step-by-step procedures for initial startup of the various TSS environment components.
- The ***NET24-XPNET Network Control Command Reference Manual*** describes the process control commands that you can use to control the Transaction Security Services process.
- The ***NET24-XPNET TCP/IP Protocol Implementation Guide*** describes the Common Transport System (CTS), which is used to connect the user interface and User Auditing (UAUD) device, lines, and stations to the TCP/IP subsystem.
- The ***Remote Application Manager Implementation and Installation Guide*** provides information on configuring and using the Remote Application Manager process (referred to as the Racal Access Module process when used with Thales e-Security devices) with Transmission Control Protocol/Internet Protocol (TCP/IP). This informal document is a TGAL document provided in the BA10RAM.RAMUPDT file on the HP NonStop system and is also available from ACI upon request.
- The ***RAM/UDP Update Guide*** describes the use of the URAM process with User Datagram Protocol/Internet Protocol (UDP/IP) for Thales e-Security devices. This informal document is a Word document provided in the BA10RAM.URAMUDOC file on the HP NonStop system and is also available from ACI upon request.

This processing guide references the Hewlett-Packard (HP) ***NonStop TM/MP Operations and Recovery Guide*** for detailed information on Transaction Management Facility (TMF) operations.

This processing guide references the following Insession Technologies' Intersystem Communications Environment (ICE) publications:

- The ***ICE Messages and Codes Reference*** documents the event messages that can be generated by the WebGate HTTP Server.
- The ***NOF Command Reference and Resource Definition Guide*** provides detailed information on the commands used to start the WebGate HTTP Server and its resources.
- The ***WebGate Administrators Guide*** provides detailed information on configuring the WebGate HTTP Server.

This processing guide references the following security device vendor publications:

- The Atalla ***A8000 Installation, Operations, and Command Reference Guide*** and the ***Banking Command Reference Manual*** provide information on enabling banking commands for A8000 devices.
- The Atalla ***Secure Configuration Assistant User Guide*** provides detailed procedures on using the Secure Configuration Assistant (SCA)
- The Bull ***BNT EMV Cryptographic Principles*** document provides a detailed description of key identifiers and key usage control elements used for Bull BNTng key headers as well as descriptions of encryption methods and the types of keys encrypted under each method.
- The Eracom ***ProtectHost White Communications Guide*** provides detailed information on protocol configuration requirements.
- The Eracom ***ProtectHost White Mark II Console User Guide*** and ***ProtectHost White AMB Console User Guide*** provide procedures for configuring the ProtectHost White Eracom Mark II and AMB host security modules (HSMs), respectively, for each of the supported protocols.

## Software

This processing guide documents standard processing as of its publication date. Software that is not current and custom software modifications (CSMs) may result in processing that differs from the material presented in this processing guide. The customer is responsible for identifying and noting these changes.

## Manual Summary

The following is a summary of the contents of this processing guide.

**Section 1, “Introduction,”** introduces the reader to the Transaction Security Services process and why it is being used in the BASE24 environment. It includes overviews of security data maintenance after conversion, configuration, implementation requirements, integration into a BASE24 system, and operations.

**Section 2, “Database Conversion and Security Data Maintenance,”** describes the relationships between the existing BASE24 database files that are converted when the Transaction Security Services process is implemented and the files used by the Transaction Security Services process. It also describes how security data is maintained after the files are converted.

**Section 3, “Configuration,”** describes the configuration changes required to implement the Transaction Security Services process.

**Section 4, “Security Module Requirements,”** describes the security module requirements needed to support the functions available with the Transaction Security Services process.

**Section 5, “Transaction Processing Message Flows,”** presents message flows showing how the Transaction Security Services process is incorporated into online BASE24 transaction processing.

**Section 6, “Data Field Encryption,”** describes the data encryption support provided by the Transaction Security Services process for message field data.

**Section 7, “Environment Management,”** discusses the configuration of environment attributes and creating the System Interface Message Delivery Service Destination File (MDSDEST) for the TSS environment.

**Section 8, “Event Management,”** discusses the facilities and procedures used to monitor event messages generated by components in the TSS environment.

**Section 9, “Process Control,”** describes the process control facilities provided in the TSS environment, including the commands that can be entered for TSS components.

**Section 10, “Process Tracing,”** describes the process tracing facilities provided in the TSS environment, including the Trace Configuration and Trace Filter Profile Configuration windows in the ACI desktop user interface.

**Appendix A, “Database Conversion,”** provides an overview of the database conversion programs used to convert an existing BASE24 database for use with the Transaction Security Services process.

**Appendix B, “Automated Key Distribution System,”** describes security-related configuration and implementation requirements for the BASE24-atm Automated Key Distribution System add-on product.

**Appendix C, “Security Device Utilities,”** describes the AKB Conversion utility and the Master File Key Conversion utility. These utilities are provided in the TSS environment for security device management.

**Appendix D, “Europay, MasterCard, and Visa (EMV),”** contains miscellaneous topics related to EMV processing when using the Transaction Security Services process.

**Appendix E, “Diebold PIN Verification,”** describes Transaction Security Services support for the Diebold PIN verification method.

## Publication Identification

Three entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (for example, BA-AE000-14 for the ***BASE24 Transaction Security Services Processing Guide***) appears on every page to assist readers in identifying the manual from which a page of information was printed. The publication date (for example, Dec-2005 for December, 2005) indicates the issue of the manual. The software release information (for example, R6.0v6, TSS 05.4 for release 6.0, version 6, TSS release 05.4) specifies the software that the manual describes. This information matches the document information on the copyright page of the manual.

# 1: Introduction

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This section introduces the Transaction Security Services process. The Transaction Security Services process, built using ACI's Enterprise Services architecture, provides hardware security module services to BASE24 processes.

This section describes the Transaction Security Services process, explains why it was created to serve as an interface between the security utilities bound into BASE24 processes and hardware security modules, and introduces the following topics related to implementing the Transaction Security Services process into a BASE24 system:

- Security data maintenance
- Configuration
- Integration in a BASE24 system
- Advanced ATM key management
- Dutch security extensions
- Data field encryption and decryption
- Implementation requirements
- Transaction Security Services process
- Transaction Security Services environment
- Operations considerations
- Transaction processing

# Transaction Security Services Process

The Transaction Security Services process runs as an integrated server process in the Transaction Security Services environment. Integrated server processes are constructed from object-oriented components bound together into an executable process. Transaction Security Services processes use the TSSISLO executable object. The component within the Transaction Security Services process that performs most transaction security functions is called the Transaction Security component. The Key Conversion component performs key maintenance functions in hardware security modules such as changing the master file key. Other components within the Transaction Security Services process perform essential administrative or processing tasks. For example, the Process Control component enables you to enter process control commands for integrated server processes and the Environment component enables you to set attributes for processing.

**Note:** The AKDS process is a special Transaction Security Services process that contains all of the components of the Transaction Security Services process plus the components required to support the BASE24-atm Automated Key Distribution System add-on product. The AKDS process uses the AKDSISLO executable object.

The Transaction Security Services environment is comprised not only of Transaction Security Services processes, but also a graphical user interface called the ACI desktop user interface, User Interface processes, a Timer process, a database, and user security and auditing functions for the ACI desktop user interface. The functions of the User Interface processes are divided between a C++ User Interface (XML Server) process running under the Guardian operating system and Java User Interface servlets running within a Servlet Engine process under Open System Services (OSS). Together, the C++ User Interface (XML Server) process and Java User Interface servlets are collectively referred to as the User Interface process throughout this processing guide, unless explicitly stated otherwise. Each component of the Transaction Security Services environment is described in more detail later in this section.

# Overview

The Transaction Security Services process serves as an interface between the security utilities bound into BASE24 processes and hardware security modules.

The Transaction Security Services process was created to ease the migration to triple DES. It also supports the Atalla Key Block (AKB) format and serves as the foundation for all future transaction security enhancements such as automated key distribution.

Beginning with version AKB 1.2 software, Atalla requires that all keys be stored in the database in the following format:

|                  |                          |                |
|------------------|--------------------------|----------------|
| Header (8 bytes) | Encrypted Key (48 bytes) | MAC (16 bytes) |
|------------------|--------------------------|----------------|

The AKB format protects stored keys against known attacks. The Header field contains eight one-byte values describing the attributes of the key. Before a key in AKB format is used in an Atalla Network Security Processor, the content of the header block is validated. The Encrypted Key field contains the encrypted key block. Single-length and double-length keys are padded before encryption to hide their length. The MAC field contains a message authentication code (MAC) of the header and encrypted key. The MAC value represents a MAC of the 8 byte header and the 48 byte key and is generated by an Atalla hardware security module (HSM) when the key is generated. This MAC value cryptographically binds the header and the encrypted key together.

The header ensures the proper use of the key, the padding of the encrypted key block disguises the length of the key, and the MAC value ensures that the key block has not been tampered with.

Because Atalla does not support both the old key block format and the new key block format in a single HSM, all keys must be converted to the new key block format at the same time.

A set of database files used by the Transaction Security Services process supports extended length keys required for triple DES as well as the AKB key block format. These files are accessed by the Transaction Security Services process and are maintained using the ACI desktop user interface. If you are migrating to this release from a BASE24 release earlier than release 6.0, version 2, refer to [appendix A](#) or the **BASE24 Release 6.0 Implementation Guide** for database conversion information.

The Transaction Security Services process supports all of the following transaction security functions currently supported in BASE24 systems:

- PIN encryption using single DES ensures that PIN information is maintained in a confidential manner.
- PIN verification ensures that the true cardholder is performing the transaction. The Transaction Security Services process supports the following methods of PIN verification:
  - ❑ Diebold (this method is not supported for Eracom host security modules)
  - ❑ IBM DES (3624-format)
  - ❑ Identikey (this method is supported only with Atalla security devices)
  - ❑ NCR (this method is supported only with Thales e-Security security devices and requires a custom HSM command)
  - ❑ Visa PVV (this method is the only PIN verification method supported for Bull BNTng HSMs)
- Message authentication ensures that data remains tamper-free while passing between an EFT device and its authorization destination.

**Note:** If using triple DES for message authentication, the encrypt operations performed by the sender and receiver of the message are replaced by encrypt/decrypt/encrypt operations using double or triple length keys.
- Card verification ensures that a valid card is being used to initiate the transaction.
- Chip authentication ensures that a credit or debit card with an embedded microprocessor chip is valid.

Besides the existing security-related functions described above, the Transaction Security Services process provides the following security enhancements to a BASE24 system:

- Triple DES PIN encryption for stronger PIN protection and reduced potential fraud losses
- Longer keys for protection against key exhaustion attacks
- Stronger key storage techniques to protect against key compromise
- Data field encryption and decryption
- RVT/RVS key management (this key management scheme is supported only with Thales e-Security security devices and requires custom HSM commands)



- Public key cryptography for the BASE24-atm Automated Key Distribution System add-on product

The Transaction Security Services process runs as an application process under the XPNET process (i.e., it runs under control of the XPNET process for the purpose of process restart and recovery only; the XPNET process does not send transaction processing messages to the Transaction Security Services process, except for the BASE24-atm Automated Key Distribution System add-on product). The same XPNET process control mechanisms available to other BASE24 processes can be used on the Transaction Security Services process. The process can be replicated as needed and can be started, stopped, etc., from a network control facility. Security utilities bound into BASE24 application processes interface with the Transaction Security Services process using a direct write/read interface.

## Database Relationships

When using the Transaction Security Services process, hardware-encrypted key information is maintained in the Transaction Security Services database. The hardware-encrypted key fields in the BASE24 database are set to a nonencrypted value known as a *key locator*. Key locator values are set when a new record is added to a BASE24 file (e.g., when adding a new terminal to the BASE24-atm Terminal Data files) or when existing records in a BASE24 file are converted. For more information on converting BASE24 files, refer to [appendix A](#).

Key locator values are either set to the primary key value for the record (e.g., the terminal ID for hardware key fields in the BASE24-atm Terminal Data files and BASE24-pos Terminal Data files) or an index value created by the conversion program for the file (i.e., for the Key File). The Transaction Security Services process uses the key locator value as the primary key value to retrieve the actual hardware-encrypted key information from its database.

**Note:** If you are using the LN-IND param to partition the Transaction Security Services database by logical network, the two-character value of this param is added to the beginning of the key locator value. For more information on the LN-IND param, refer to [section 3](#).

The following table shows the relationship of existing BASE24 database files with key locator values to the corresponding Transaction Security Services database files. The first column lists the existing BASE24 files and the second column lists the files accessed by the Transaction Security Services process.

| BASE24 File                          | TSS File                        |
|--------------------------------------|---------------------------------|
| BASE24-atm Terminal Data files (ATD) | Channel Security File (CSECD)   |
| BASE24-pos Terminal Data files (PTD) |                                 |
| Derivation Key File (KEYD)           | Derivation Key File (DRVKD)     |
| Key File (KEYF)                      | Interface Security File (ISECD) |

When using the Transaction Security Services process, two BASE24 files are replaced by Transaction Security Services files and are no longer accessed in processing for verification in hardware security modules. These BASE24 files are used only if you need to support verification in software. The following table lists

the Transaction Security Services files that replace these BASE24 files for verification processing in a hardware security module. These Transaction Security Services files use the same primary keys as the BASE24 files they replace.

| BASE24 File                         | TSS File                                |
|-------------------------------------|-----------------------------------------|
| Key Authorization File (KEYA)       | Card Verification File (CRDVD)          |
|                                     | Diebold Number Table File (DNTD)        |
|                                     | IBM DES Verification File (IDESD)       |
|                                     | Identikey PIN Verification File (IDNTD) |
|                                     | Visa PVV Verification File (VPVVD)      |
| Dutch Key Authorization File (KEYV) | NCR PIN Verification File (NCRD)        |
| ICC Key File (KEYI)                 | EMV Authorization File (EMVSD)          |

The Transaction Security Services database files are created during preparation for installation using the Metadata Manager (Metaman) utility. For the Transaction Security Services process, the Metaman utility creates both FUP command files used to create the Transaction Security Services files and Comma Separated Values (CSV) files that list and define all elements, tables, assigns, and params. All Metaman utility output files are created by ACI prior to installation at a customer site.

The Metadata Configuration File (CONFCSV) table on the data subvolume defines the assigns for Transaction Security Services files. The assigns specify the file locations of the Enscribe disk files, the swap files for external memory tables, and the maximum size allowed for each external memory table. The CONFCSV also contains parameters that control the System Interface Services (SIS) layer of the TSS environment. These parameters are described in [section 7](#) of this guide. This file must be modified during installation for site-specific assign and parameter information.

The Meta Database Comma Separated Values (MDBCSV) file on the data subvolume defines the attributes and data elements for each TSS file and is the equivalent of a DDL file in the BASE24 environment.

Separate CONFCSV and MDBCSV files must exist on a different subvolume for the USEC and UAUD applications. This subvolume, referred to as the USEC/UAUD data subvolume, contains the user security and user auditing database files.

The Environment File (ENVMTD) and Event File (EVENTD) assigns in this CONFCSV must point to the ENVMTD and EVENTD residing on the TSS data subvolume. The USEC and UAUD applications use the same ENVMTD and EVENTD used by the Transaction Security Services process.

**Note:** With the advent of TSS release 04.1, the UAUD auditing database is no longer used. In TSS release 04.1, the UAUD application forwarded all auditing messages from the C++ User Interface (XMLS) process to the Java User Interface servlets (ESWEB) process for logging in the User Audit Log File (UALOGD). With the advent of TSS release 05.4, the C++ User Interface (XMLS) process logs all auditing messages directly to the UALOGD.

During installation, it is important to make sure that entries in the CONFCSV and MDBCSV are not truncated. Truncation can occur when transferring the files from the installation CD-ROM to the HP NonStop system using the HP NonStop Information Xchange Facility (IXF) file transfer program without increasing the default record size (132 characters) to a value large enough to accommodate the longest CONFCSV and MDBCSV entries. Truncation can also occur when using the HP NonStop TEDIT program to edit CONFCSV and MDBCSV entries. TEDIT also truncates lines to 132 characters. Therefore, use the HP NonStop EDIT program rather than TEDIT to modify CONFCSV and MDBCSV entries on the HP NonStop system.

**Note:** The CONFCSV and MDBCSV are specific to the TSS database for which they are used. For example, if you have different TSS databases for different XPNET systems (e.g., Production, Test, QA, Certification), you must configure a separate CONFCSV and MDBCSV on a separate subvolume for each system.

The key and key locator relationships between BASE24 files and the Transaction Security Services database files are described in detail in [section 2](#).

# Security Data Maintenance

Hardware encrypted key information for the Transaction Security Service process is maintained using the ACI desktop user interface. The following table shows you the relationships between ACI desktop user interface windows and the files used by the Transaction Security Services process.

| User Interface Windows                     | TSS File                                     |
|--------------------------------------------|----------------------------------------------|
| Authentication Profile Configuration       | Card Verification File (CRDVD)               |
|                                            | Diebold Number Table File (DNTD)             |
|                                            | EMV Authorization File (EMVSD)               |
|                                            | IBM DES Verification File (IDESD)            |
|                                            | Identikey PIN Verification File (IDNTD)      |
|                                            | NCR PIN Verification File (NCRD)             |
|                                            | Visa PVV Verification File (VPVVD)           |
| BASE24 KEYF Configuration <sup>*</sup>     | Key File (KEYF)                              |
| Channel Security Configuration             | Channel Security File (CSECD)                |
| DUKPT Derivation Key Profile Configuration | Derivation Key File (DRVKD)                  |
| Interface Security Configuration           | Interface Security File (ISECD)              |
| Security Device Configuration <sup>†</sup> | Security Device Configuration File (SECD CD) |

<sup>\*</sup> The BASE24 KEYF Configuration window is used to maintain data in the BASE24 Key File (KEYF) rather than a Transaction Security Services file.

<sup>†</sup> This window maintains security device configuration parameters similar to the LCONF assigns and params used to configure hardware security modules (HSMs) in an existing BASE24 system.

Context-sensitive ACI Online Help is provided for each of the above windows in the ACI desktop. All database maintenance activity performed from the ACI desktop user interface can be audited. Java server properties, in the form of db.audited.db-service-name, for the Java User Interface servlets process, and the UI\_AUDIT\_LEVEL environment attribute for the C++ User Interface process, allow delete, insert, update, and view actions on TSS files to be logged. For the db.audited.db-service-name properties, db-service-name is the name of a Java database service that maintains the data in a particular TSS file. For more information on db.audited.db-service-name properties and a list of the properties supported for the TSS environment, refer to the ***ACI Java Server Framework Reference Guide***. For more information on auditing, refer to the ***ACI Desktop User Interface Manual***.

Security data maintenance through the ACI desktop user interface is discussed in more detail in [section 2](#).

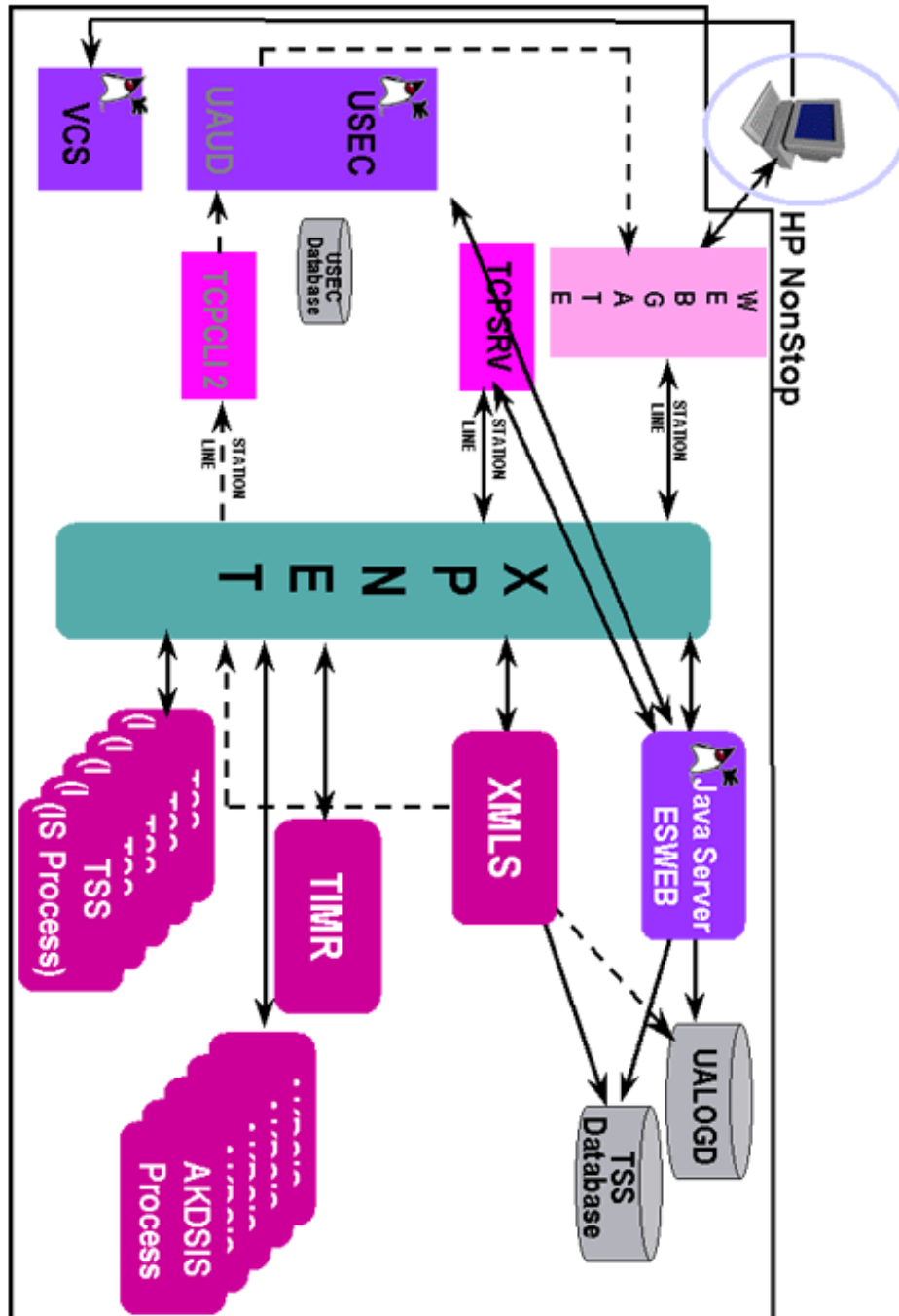
# Configuration

Each Transaction Security Services process and AKDS process must be configured as an application process to the XPNET system. In addition, Timer processes and User Interface Server processes must also be configured. The Timer processes set timers for keys stored in the database accessed by the Transaction Security Services process. The User Interface Server processes interface with the ACI desktop user interface, the User Security (USEC) application, and the TSS database.

ACI desktop user interface clients communicate with the HP NonStop system using the WebGate function of Insession Technologies' ICE product, a WebGate interface to the XPNET process, and a servlet container.

WebGate provides Web server capability to the HP NonStop system and connects to ACI desktop user interface clients using Hypertext Transfer Protocol (HTTP) and Transmission Control Protocol/Internet Protocol TCP/IP). The NET24-XPNET product manages Servlet Engine processes, which are XPNET satellite applications provided by ACI that control communication between the XPNET process and a Java Virtual Machine (JVM) provided by the HP NonStop Open System Services (OSS). Java User Interface servlets run within a servlet container under the Servlet Engine (ESWEB) processes. Because the Servlet Engine processes run under the XPNET process, the Java User Interface servlets running within a servlet container under the Servlet Engine processes are persistent. If the Java User Interface servlets receive a request that needs to be processed by the C++ User Interface (XML Server) process, the request is *passed through* XPNET to the C++ User Interface (XML Server) process and the response is sent back through XPNET to the Servlet Engine (ESWEB) process. The USEC and UAUD applications run as general Java processes under OSS. XPNET stations use the Web Services Application Programming Interface (WSAPI) protocol to interface with WebGate. The servlet container executes in a JVM and is responsible for invoking and managing resources. The relationships among all of the ACI desktop user interface components in the TSS environment are depicted in the illustration on the following page.

**Note:** The ESWEB process, along with the Version Checker and USEC applications, can reside on or off the HP NonStop platform. If they reside off the HP NonStop platform, they must run in a Windows environment with Tomcat version 5.0.28 or greater. A TDAL process is used to access the Enscribe database on the HP NonStop platform. The following illustration assumes these applications are installed on the HP NonStop platform.





The detailed configuration for all of these entities is performed during installation and is beyond the scope of this guide.

Use of the Transaction Security Services process by BASE24 processes is controlled through LCONF assigns and params. BASE24 processes send messages directly to the Transaction Security Services processes identified in the SECURE-DEV1 and SECURE-DEV2 assigns, or in the SECURE-DEV-*xx* assigns, using a write/read interface when the SECURE-DEV-TYP param is set to a value of TSS in the LCONF. The SECURE-DEV-ACCESS-TYPE param indicates whether a BASE24 application process communicates with a primary and secondary Transaction Security Services process identified by the SECURE-DEV1 and SECURE-DEV-2 assigns or with up to 20 Transaction Security Services processes identified by SECURE-DEV-*xx* assigns, where *xx* is any two characters. The SECURE-DEV1 assign specifies the primary Transaction Security Services process, while the SECURE-DEV2 assign specifies the secondary Transaction Security Services process. The SECURE-DEV-*xx* assigns specify up to 20 Transaction Security Services processes accessed on a *round-robin* (i.e., rotating) basis.

Some LCONF assigns and params are not used when implementing the Transaction Security Services process. For example, the SECURE-DEV-CONF-CMD100 and SECURE-DEV-CONF-TXT params are not used because they are replaced by similar parameters entered from the Security Device Configuration window.

A detailed discussion of how each LCONF [assign](#) and [param](#) related to HSMs need to be configured when using the Transaction Security Services process is provided in section 3.

## Integration in a BASE24 System

When a BASE24 process requires the use of an HSM, the security utilities call the Transaction Security Services process specified in the SECURE-DEV1, SECURE-DEV2, or SECURE-DEV-xx assign with the key locator value and the requested function to be performed by the HSM. The Transaction Security Services process uses the key locator value to retrieve the actual hardware key value from the Transaction Security Services database.

The process uses the value of the transaction origin (based on the value in the ORIGINATOR field in the STM, PSTM, or TSTM) to determine whether it retrieves the PIN encryption key from the CSECD or the ISECD. For example, if the ORIGINATOR field is set to a value of 1 or 2, the process retrieves the PIN encryption key from the CSECD. If the ORIGINATOR field is set to a value of 7, the process would retrieve the PIN encryption key from the ISECD.

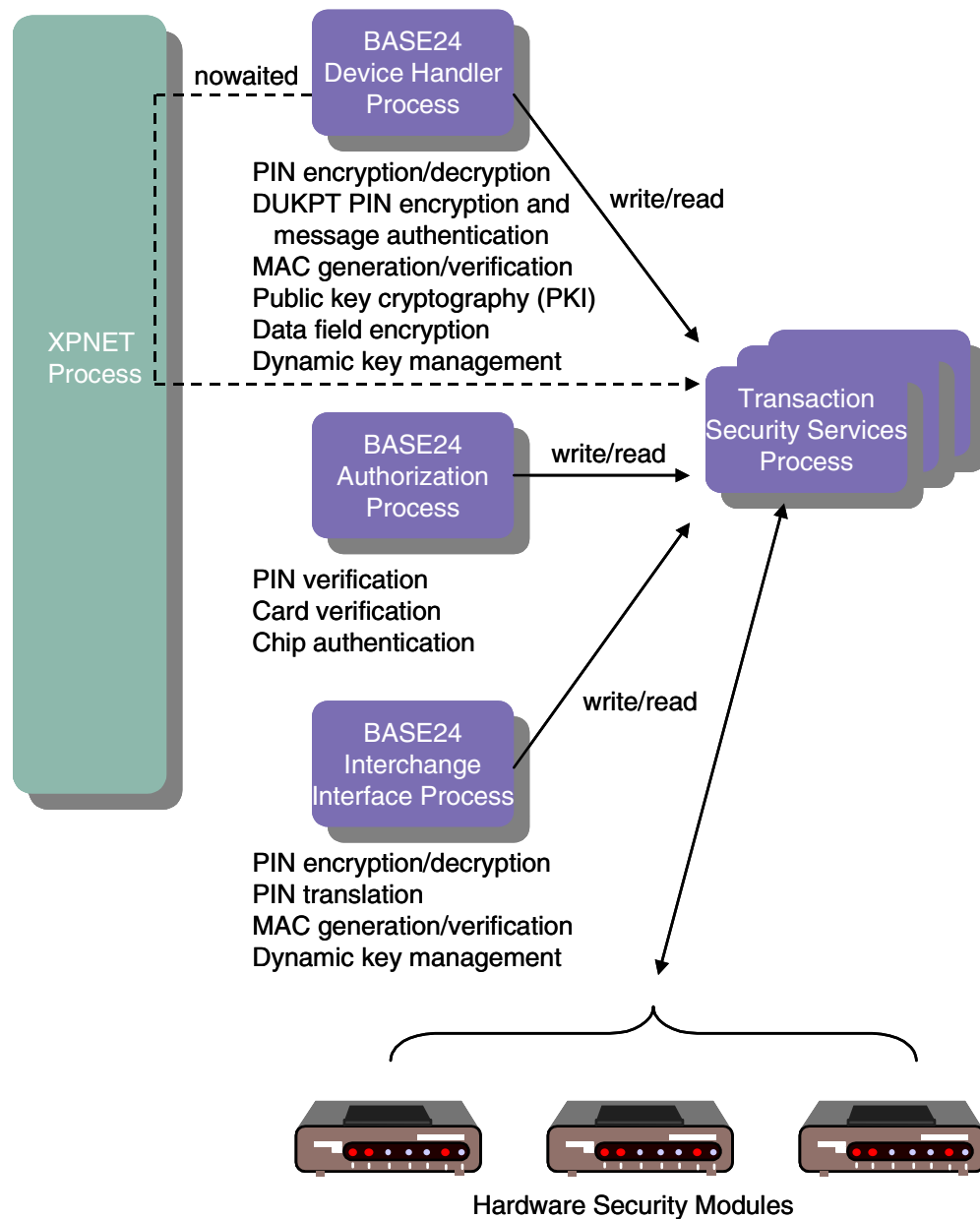
The process uses the HSM command information passed by the calling process to retrieve other key information needed to process the request. For example, if the HSM command requested specifies IBM DES (3624-format) PIN verification, the process retrieves the PIN verification key information from the IDESD.

After all the required key information is retrieved, the process formats the request to the HSM with the correct key values and returns the response to the requesting BASE24 process.

The illustration on the following page depicts how the Transaction Security Services process integrates into BASE24 transaction security processing.

The security utilities bound into BASE24 application processes communicate with a Transaction Security Services process using a direct write/read interface when a security function requiring the use of a hardware security module is needed, except for some public key cryptography (PKI) functions used with the BASE24-atm Automated Key Distribution System add-on product. These functions require nowaited I/O communication with the Transaction Security Services process.

Each BASE24 process can communicate with either two Transaction Security Services processes (one primary and one secondary) or with up to 20 Transaction Security Services processes on a round-robin (i.e. rotating) basis. The Transaction Security Services process can be replicated as needed for load balancing and transaction throughput.



The XPNET process is used for nowaited PKI communication.

## Implementation Requirements

Use of the Transaction Security Services process is required in the following situations:

- When implementing the new Atalla key block format (i.e., migrating to AKB version software in Atalla HSMs)
- When implementing triple DES in ATM and POS devices
- When implementing triple DES across the BASE24 system
- When implementing the BASE24-atm Automated Key Distribution System add-on product

The Transaction Security Services process supports both single DES and triple DES operations. It also supports Atalla keys stored in either variant or Atalla Key Block (AKB) format; however, it does not support a mix of keys in variant and AKB formats. After the decision is made to convert to the new AKB format, all keys in a BASE24 system must be converted at the same time.

## Software Support

The Transaction Security Services process requires a minimum of BASE24 release 6.0, version 2 software. ACI recommends that the BASE24 user be on a current Hewlett-Packard (HP) NSK-supported operating system to obtain maximum performance and support. The Transaction Security Services process is currently compatible with the HP NSK Guardian D46 (or higher) operating system on a K-series server. On an S-series server, it is compatible with the HP NSK Guardian Gxx (or higher) operating system. In addition, BASE24 release 6.0 requires the Transaction Management Facility (TMF). BASE24 release 6.0 software also requires that the BASE24 user be on the XPNET 3.0 or later system.

The TSS environment uses the XPNET process primarily for process startup and management. Except for certain public key commands required for the BASE24-atm Automated Key Distribution System add-on product, the TSS environment does not use the XPNET process for messaging. Messaging is performed directly using interprocess I/O.

**Note:** The ACI desktop user interface requires additional software. These software requirements are discussed under the “Transaction Security Services Environment Support” topic provided later in this section.

For BASE24 release 6.0, version 6, four releases of Transaction Security Services (TSS) are supported: 04.1, 04.4, 05.2, and 05.4. Some functions of the Transaction Security Services process are supported only with TSS release 05.4, while others were introduced in earlier releases. The functions introduced in a TSS release are supported in all subsequent TSS releases. For example, the functions introduced in release 04.1 are supported in all three subsequent TSS releases, while the functions or capabilities introduced in release 05.2 are supported only in release 05.2 and 05.4. The following paragraphs describe the functions or capabilities introduced in each release and list the releases in which the functions or capabilities are supported. Because the functions introduced in releases prior to 04.1 are available in all four TSS releases, they are not listed here.

## **Critical Fixes**

The following Transaction Security Services process functions or capabilities were introduced in a critical fix and are supported in all available TSS releases:

- The PIN\_VRFY\_FAIL\_RETRY\_IND Environment attribute specifies whether the Transaction Security Services process retries a PIN verification HSM operation using the previous key from the Channel Security Configuration File (CSECD) when the HSM returns a valid fail error (i.e., the cardholder PIN is considered to be invalid).

## **Release 05.4**

The following Transaction Security Services process functions or capabilities are supported only when TSS release 05.4 is installed.

- The C++ User Interface (XML Server) process logs auditing messages directly to the User Audit Log File (UALOGD). The UAUD application, Audit Store and Forward File (ASAFD), and Audit Store and Forward Configuration File (ASAFCD) are no longer used.
- Support for Bull BNTng hardware security modules.
- BASE24-atm Automated Key Distribution System add-on product support for shared Wincor Nixdorf signature devices (i.e., Wincor Nixdorf devices running in 912 emulation or NCR NDC emulation mode) and the AKDS Capacity Report.

## Release 05.2

The following Transaction Security Services process functions or capabilities are supported when TSS release 05.4 or 05.2 is installed.

- All user security validation origination from the Java User Interface servlets (ESWEB process). This eliminates the configuration formerly required for the C++ User Interface (XML Server) process to communicate with the USEC application.
- Access of Channel Security data source (CSECD) records from memory or from disk (described in [section 2](#)).
- Additional security device configuration parameters, including the ability to automatically re-establish communication with a hardware security device after the device has reached its maximum consecutive errors limit (described in [section 3](#)).
- Enhanced process tracing facilities. Enables you to configure, filter, and trace C++ processes in the TSS environment using new ACI desktop user interface windows instead of entering process control commands to execute traces (described in [section 10](#)).

## Release 04.4

The following Transaction Security Services process functions or capabilities are supported only when TSS release 05.4, 05.2, or 04.4 is installed.

- American Express card security code (CSC) verification
- Automatic restart by the calling BASE24 process when communications with a Transaction Security Services process fails
- Derived unique key per transaction (DUKPT) PIN encryption for the BASE24-pos Hypercom Device Handler (HPDH) (described in [section 2](#))
- EMV 2000 authorization (described in [section 2](#) and [appendix D](#))
- Encrypted IBM DES 3624-format decimalization table support
- Eracom ProtectHost White Australian Major Banks (AMB) host security module (HSM) support (described in [section 4](#))
- AS2805 6.2 and AS2805 6.4 PIN blocks (described in [section 4](#))
- Master File Conversion utility support for the BASE24-atm Automated Key Distribution System add-on product (described in [appendix C](#))

- *Round-robin* (rotating) access to up to twenty Transaction Security Services processes
- Triple DES support for Triton Device Handler processes

## Release 04.1

The following Transaction Security Services process functions or capabilities are supported only when TSS release 05.4, 05.2, 04.4 or 04.1 is installed.

- Derived unique key per transaction (DUKPT) PIN encryption and message authentication for the BASE24-pos Standard POS Device Handler (SPDH) (described in [section 2](#))
- EMV derivation key index support (described in [section 2](#))
- Interac key management (described in [section 2](#))
- ISO format 1 PIN blocks (described in [section 4](#))
- Check Digit Validation and Generation utility (described in [appendix C](#))
- File partitioning (described later in this section)

**Note:** The Node Controller application and the auditing database for the UAUD application are no longer used or required by the ACI desktop user interface in this release. The UAUD application forwards all auditing messages it receives from the C++ User Interface (XML Server) process to the Java User Interface servlets (ESWEB) process to be logged to the User Audit Log File (UALOGD).

## Security Module Support

A number of vendors offer Tamper Resistant Security Modules (TRSMs) that are designed to perform PIN protection and verification, as well as other security-related functions within a physically secure environment. BASE24 currently supports the following security module models from four vendors—Atalla, Bull, Eracom, and Thales e-Security:

- Atalla Network Security Processor (NSP) models A8000, A8100, A9000, A9100, A10000E, and A10100.

All Axx100 models can run using the older variant key format or the Atalla Key Block (AKB) format. These devices can be configured to support the variant format or the AKB format simply by installing the appropriate CD-ROM in the device and rebooting. All Axx100 models come standard with an

AKB software CD-ROM. Appending the letter V to the Axx100 model number when ordered (e.g., A10100-V, A9100-V, and A8100-V) ensures that the module is shipped from the factory with the variant software CD-ROM as well. If the letter V is not appended to the model number when ordering, the Axx100 model is shipped from the factory only with the standard AKB software CD-ROM.

All other Atalla models can also be ordered with variant or AKB version software preloaded. While on-site upgrades from variant to AKB software are possible on the other Atalla models, this process is not as convenient as on the Axx100 series.

- Bull BNTng family of hardware security modules (HSMs)
- Eracom ProtectHost White Mark II and ProtectHost White Australian Major Banks (AMB) host security modules (HSMs).
- Thales e-Security (Racal) Host Security Module models RG7100 and RG7110, RG7300 and RG7310, RG7400, and HSM 8000.

Thales e-Security Host Security Modules were formerly named Racal-Guardata Host Security Modules.

If you are implementing full triple DES or the Atalla Key Block (AKB) format, Atalla TSRMs require AKB version 2.30 or 1.3 software. The 2.30 version software is used for A10100 AKB models while the 1.3 version software is used for A10000 AKB models. Note that Atalla recommends version 2.9 software as the minimum version when using the older variant key block format. This version is available from Atalla as a free upgrade for a limited time. It is the only version that allows an upgrade to the AKB versions of software, and it allows customers to make future upgrades in-house, without an HP CE visit. Note that the 2.90 equivalent software on the Axx100-V models is version 1.10.

Bull BNTng HSMs require software with a minimum version of 7.

Eracom HSMs require software with a minimum release date of 01/01/2000 or later. All software releases dated 01/01/2000 or later support triple DES.

Thales e-Security RG7000-series HSMs require a minimum of version 5.05/1.05 firmware to support full triple DES. The HSM 8000 requires a minimum software version of 2.0 to support full triple DES. (Triple DES support for DUKPT PIN encryption and message authentication is only provided in version 2.0 or greater software.)

If you are not implementing full triple DES, Atalla NSPs can use version 2.90 software, and Thales e-Security HSMs can use version 5.05/1.05 firmware. Atalla models A9000 and A10000E with version 2.90 software, Atalla models A8100-V,



A9100-V, and A10100-V with version 1.10 software, and all Thales e-Security HSMs with 5.05/1.05 firmware, allow triple DES encrypted PIN blocks to be translated to single DES encrypted PIN blocks, and vice versa. The Atalla A8000 model does not support this functionality.

If you are implementing the BASE24-atm Automated Key Distribution System add-on product, you must use an Atalla NSP model with a minimum of AKB version 2.40 or 1.4 software or a Thales e-Security HSM with a minimum of version 6.02 firmware and an optional DSP plug-in (for RSA functions). For Atalla NSPs, the public key cryptography commands needed to support the BASE24-atm Automated Key Distribution System add-on product are available on Axx100 models (i.e., A8100, A9100, and A10100) only.

The hardware security module requirements for the Transaction Security Services process are provided in [section 4](#).

## ATM Device Support

The following BASE24-atm Device Handler processes currently support the use of the Transaction Security Services process, although the Transaction Security Services process is not required when using single DES for devices supported by these processes:

- Diebold 10XX/478X (includes Wincor Nixdorf devices running in 912 emulation mode)
- Cash Source Plus
- IBM 3624, Version 8
- NCR 5XXX (includes 56XX and *personaS* devices as well as Wincor Nixdorf devices running in NDC emulation mode; triple DES is not supported on 50XX devices)
- Tidel
- Triton

The Transaction Security Services process is required when ATM devices supported by these Device Handler processes are configured for triple DES. Contact your ACI account manager for information on the availability of other ATM Device Handler processes for use with the Transaction Security Service process.

The Diebold 10XX/478X and NCR 5XXX Device Handler processes support the BASE24-atm Automated Key Distribution System add-on product in both leased line and dial-up modes.

## POS Device Support

BASE24-pos Device Handler modules will be enhanced as needed to support the use of the Transaction Security Services process for triple DES, although the Transaction Security Services process is not required when using single DES. The Transaction Security Services process is required when POS devices are configured for triple DES.

The BASE24-pos Standard POS Device Handler (SPDH) module supports the use of the Transaction Security Services process for both triple DES or single DES, DUKPT, and dynamic key management.

The BASE24-pos Hypercom Device Handler (HPDH) module supports the use of the Transaction Security Services process for both triple DES or single DES PIN encryption and DUKPT. PIN keys can be single- or double-length. Hypercom does not support triple-length keys. MAC encryption keys are not supported because Hypercom utilizes a proprietary message authentication algorithm that supports only single-length keys.

The following BASE24-pos Device Handler modules support the use of the Transaction Security Services process for single DES only:

- BASE24-pos VISA II Device Handler (VVDH)
- BASE24-pos APACS 30/40 Device Handler without message authentication

## Interface Support

The following Interchange Interface processes support the use of the Transaction Security Services process:

- BASE24 ISO (BIC ISO) Interchange Interface
- Banknet ISO Interchange Interface
- EPS-Net Interchange Interface
- Interpay Interchange Interface
- LINK ISO Interchange Interface

- MDS/Cirrus ISO Interchange Interface
- MDS Maestro Interchange Interface
- VisaNet ISO Interchange Interface

Other Interchange Interface processes also support the use of the Transaction Security Services process. Refer to the ***BASE24 Release 6.0 Implementation Guide*** for a complete list of all Interchange Interface processes that support the Transaction Security Services process.

The Transaction Security Services process is not supported in releases prior to release 6.0; however, triple DES can be supported in certain release 6.0, version 3 Interchange Interface processes without the Transaction Security Services process in a *translate mode* only. For more information on this latter option, refer to the ***BASE24 Transaction Security Manual***.

## New Key Activation

When an acquirer Interface process generates a new key using the Transaction Security Services process, the new key is not activated (i.e., the current index for the key is changed to point to the new key) until after the Interface process receives a response from the secondary key processor or co-network indicating that the new key was successfully processed. In TSS releases prior to 02.4, the new key is activated immediately after it is generated. New key activation was implemented in a critical fix for TSS release 02.4.

## PIN Blocks and Pad Characters

The Transaction Security Services process supports the following PIN block formats:

- AS2805 6.2 and 6.4
- PIN/PAN (ANSI or ISO Format 0)
- PIN/PAD
- DUKPT

DUKPT PIN blocks are supported between a BASE24 Standard POS Device Handler (SPDH)-compliant device and the SPDH module and between a Hypercom device and the BASE24-pos Hypercom Device Handler (HPDH) module. Both the SPDH and HPDH modules translate DUKPT PIN blocks into PIN/PAN PIN blocks.

- ISO Format 1 (for Bull BNTng and Thales e-Security HSMs only)

The Transaction Security Services process also supports the destination PIN block formats required for the BASE24-atm EMV add-on product. These PIN block formats are described in [appendix D](#) of this guide. The ISO Format 1 PIN block format is described in [section 4](#) of this guide. The AS2805 6.2 and AS2805 6.4 PIN block formats are also described in [section 4](#) of this guide. The PIN/PAN, PIN/PAD, and DUKPT PIN block formats are described in the ***BASE24 Transaction Security Manual***.

A pad character is used for both PIN/PAD and ISO format 1 PIN blocks. The TSS database allows for the pad character to be set to any valid hexadecimal character (i.e., 0–9 and A–F) for PIN/PAD and ISO format 1 PIN blocks, except when using Thales e-Security security modules. The Thales e-Security security module restricts the pad character to hexadecimal F. Therefore, you must set the PIN Pad Character field to a value of F when using Thales e-Security HSMs for both PIN/PAD PIN blocks and ISO format 1 PIN blocks.

For PIN/PAN PIN blocks, the hexadecimal character F is always used for padding.

# BASE24-atm Automated Key Distribution System

With the industry-wide move to improve key management practices and recent mandates requiring unique keys per terminal, security administrators have been searching for a cost-effective way to generate, manage, and distribute increasing numbers of ATM keys. To address this need, ACI has worked closely with leading ATM vendors, security device vendors, and major card associations to develop the BASE24-atm Automated Key Distribution System add-on product. This add-on product automates the otherwise error-prone and labor-intensive process of generating, protecting, and hand-entering ATM master keys at each ATM in your system.

The BASE24-atm Automated Key Distribution System add-on product uses public-key cryptography, often referred to as PKI, to remotely and securely establish the initial master key (i.e., the A-key) in the encrypting PIN pad (EPP) of an ATM. The BASE24-atm Automated Key Distribution System add-on product is a PKI implementation based on the Rivest-Shamir-Adleman (RSA) scheme, which is an algorithm based on prime number mathematics.

A public key scheme is one in which the sender and receiver each have their own unique pair of keys. One key in the pair is the *secret* (private) key. As its name implies, only its owner can know the secret key—it cannot be shared with any other party. The other key in the pair is the *public* key. The public key can be shared between the sender and receiver without threat to the integrity of the key pair. A mathematical relationship exists between the secret key and the public key, but it is virtually impossible to derive one from the other. When data (e.g., an ATM master key) is encrypted by one of the keys in the pair, only the other key can decrypt it. Thus, public-key cryptography differs from traditional DES cryptography in that the key used to encrypt data is different than the one used to decrypt the data. For this reason, PKI is referred to as an *asymmetrical* security scheme while DES is referred to as a *symmetrical* security scheme. More importantly, the key used to encrypt the data cannot be reused to decrypt the data. This characteristic means that the encrypting key (i.e., the public key) has no value to an adversary if exposed or intercepted while being transmitted.

The BASE24-atm Automated Key Distribution System add-on product is available for NCR 5XXX and Diebold 10XX/478X Device Handler processes for ATMs connected over leased lines or running in dial-up mode. These Device Handler processes must use a special Transaction Security Services process called the AKDS process to communicate with security modules for BASE24-atm Automated Key Distribution System add-on product cryptographic functions. The AKDS process contains all of the components of the Transaction Security Services process plus the components required to support the BASE24-atm Automated Key Distribution System add-on product. AKDS functions are supported in hardware

only; they are not supported in software. Atalla NSPs and Thales e-Security HSMs are supported for the BASE24-atm Automated Key Distribution System add-on product; Bull BNTng and Eracom HSMs are not supported for the BASE24-atm Automated Key Distribution System add-on product.

The configuration requirements and procedures for implementing the BASE24-atm Automated Key Distribution System add-on product are discussed in appendix B of this processing guide. This appendix also lists the security module commands required to support the BASE24-atm Automated Key Distribution System add-on product.

The procedures for downloading a public key or master key from BASE24 to an ATM are provided in the ***BASE24-atm NCR 5XXX Device Support Manual*** for NCR devices and Wincor Nixdorf devices (running in NDC emulation mode), and in the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for Diebold devices and Wincor Nixdorf devices (running in 912 emulation mode).

# Dutch Security Extensions

The Transaction Security Services process provides cryptographic support in Thales e-Security HSMs for the following security functions used by interfaces to Interpay Nederland B.V.—a national switch in the Netherlands:

- Large message MAC generation and verification
- NCR PIN verification method \*
- RVT/RVS key management \*

The cryptographic functions used to support the above functions are available only on Thales e-Security HSMs. The functions identified with an asterisk (\*) require custom HSM commands that are not supported in the standard Thales e-Security product. All of these cryptographic functions are described in section 4.

## Large Message MAC Generation and Verification

Interpay supports message authentication codes (MACs) for large messages using the ANSI X9.19 method and double-length MAC encryption keys. The data encrypted can be binary or expanded hexadecimal format. This type of message authentication is provided by the Thales e-Security MS (Generate MAC using ANSI X9.19 Method for a Large Message) host command. The MS command is used primarily for support of the ANSI X9.19 method.

**Note:** The Transaction Security Services process does not support sending multiple blocks of data to the HSM with the MS command. Therefore, the maximum length of data for which a MAC can be generated is 1975 bytes.

## NCR PIN Verification

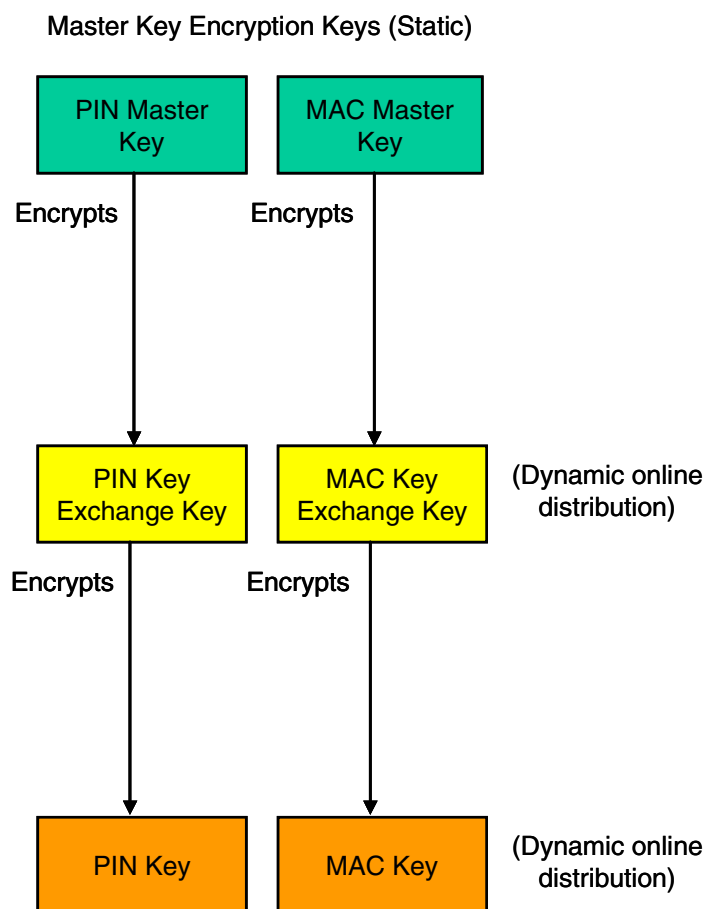
Interpay uses the NCR PIN verification method. The NCR PIN block can be encrypted under a terminal PIN key or under an interface encryption key. The Transaction Security Services process uses keys stored in either the CSECD or ISECD to decrypt the PIN block and the PIN verification key stored in the NCRD to verify the PIN. The NCR PIN verification key is entered on the NCR tab of the Authentication Profile Configuration window. This tab is described in section 2.

The cryptographic function used to support NCR PIN verification is available only on customized Thales e-Security HSMs. This cryptographic function is described in section 4.

## RVT/RVS Key Management

Interpay uses a three-tier hierarchy of keys for key management instead of the two-tier key hierarchy used throughout the rest of BASE24. This three-tier key management scheme uses static master keys and random values to encrypt and decrypt transport (key exchange) and session (working) keys for dynamic online distribution.

A two-tier hierarchy of keys is used for most interfaces where working keys (session keys) are encrypted under key exchange keys (transport keys) for online distribution. For an Interpay interface, a third tier of keys called master key encryption keys are used to encrypt key exchange keys for online distribution. This three-tier hierarchy of keys is depicted in the following illustration.



Master key encryption keys comprise the top tier in the three-tier key hierarchy. Key exchange keys comprise the second tier, while working keys make up the third tier. Keys in the second and third tiers are dynamically generated using a 64-bit or



128-bit random value. The random value used to generate key exchange keys (transport keys) is referred to as the RVT. The random value used to generate working keys (session keys) is referred to as the RVS. Both the RVT and the RVS can be generated by the Interpay interface or can be received from Interpay.

The static offline master keys used to encrypt key exchange keys are entered on the Master Keys tab of the Interface Security Configuration window. This tab is described in [section 2](#).

The cryptographic functions used to support RVT/RVS key management are available only on customized Thales e-Security HSMs. These cryptographic functions are described in [section 4](#).

# Transaction Security Services Environment Support

A number of additional facilities are required in the Transaction Security Services environment to support the Transaction Security Services process. These facilities are described below.

## User Interface

C++ User Interface Server processes (XML Server processes) and Java User Interface servlets are required to support requests from ACI desktop user interface clients running on user workstations. The WebGate HTTP Server and XPNET Servlet Engine processes serve as an interface between multiple ACI desktop user interface clients and the server-side User Interface Server processes. If the Java User Interface servlets running in a Servlet Engine process receive a request that needs to be processed by the C++ User Interface (XML Server) process, the request is routed through the XPNET process to the C++ process. You can configure multiple URLPATHs to the WebGate HTTP Server to access different TSS networks (e.g., test, quality assurance, production, etc.). In this case, desktop users can select the URLPATH to be accessed after they log on to the ACI desktop. For detailed information on configuring URLPATHs, refer to the *WebGate Administrators Guide*.

All Java-based ACI desktop user interface components, including clients, servlets, Version Checker server, and User Security (USEC) application require the Java Virtual Machine (JVM) of 1.2 or higher runtime environment. In addition, the JVM must be 32-bit rather than 64-bit for compatibility with the TSS database.

All ACI desktop user interface components, except for clients, run on the HP NonStop system. The minimum NonStop Java release on an HP NonStop system that supports JVM release 1.2 is NonStop Java 2 (NSJ2.1). On ACI desktop user interface client workstations, the URLPATHs a user can connect to are configured in a Remote.cfg file while the ACI desktop user interface client is configured in a Local.cfg file.

On the HP NonStop system, the Java User Interface servlets must run under the Open System Services (OSS) environment, which coexists with the Guardian environment and runs under the NonStop Kernel operating system. The OSS environment has a structure and feel similar to a UNIX system.

The C++ User Interface (XML Server) processes run under the Guardian environment.

When an ACI desktop user interface client is started, the Version Checker client communicates with the Version Checker server, also running in the JVM runtime environment, to compare the current version of Java archive files (JARs) used by ACI desktop client applications to the current published versions of these files in the JAR Repository. If the current published versions are newer than the client versions or are missing from the client set, the Version Checker server returns the required JARs to the Version Checker client. The client then notifies the user that the JAR files need to be updated. The user can then select the JARs to be updated. The Version Checker server communicates with Desktop clients over a raw socket TCP/IP connection.

The C++ User Interface Server processes and Java User Interface servlets (running within a servlet container provided by Servlet Engine processes running under the OSS environment) perform the following functions:

- Validate ACI desktop user interface user requests with the USEC application
- Read, add, update, and delete items in TSS files
- Load TSS files into memory tables and warmboot the processes that read the memory tables
- Initialize, enable, disable, or status hardware security modules
- Create optional audit records for logging in the User Audit Log File (UALOGD)

## Timer Process

A Timer process is required to set the expiration times for old (i.e., “previous”) keys in the TSS database. If needed, you can add Timer processes for performance and throughout purposes or for redundancy. When a Transaction Security Services process needs to set a new timer, it accesses the available Timer processes in a round-robin manner.

**Note:** The Timer process does not set the timers used for dynamic key management thresholds. These timers are still maintained within the respective Interface or Device Handler processes or modules.

## Java Server Framework

The Java User Interface servlets for the ACI desktop user interface are based on ACI's Java Server Framework, which includes the following files:

- Application Configuration File (APPLD)
- Event Definition File (EVTDD)
- External Connection File (ETCND)
- Listener File (LSTND)
- Process File (PROCD)

All of the above files are populated during the installation process. After installation, you can view the data contained in the Java Server Framework files from the Java Server Framework windows (accessed from **System Operations > Server Management**) and the Integrated Server Event Configuration and Management window (accessed from **System Operations > Event > Event Management**). If you have any custom software modifications (CSMs), you may need to change some of the configuration settings on the Java Server Framework windows. The following table shows you the ACI desktop user interface windows that enable you to view or change data in the Java Server Framework database files. A user interface window does not exist for the ETCMD.

| ACI Desktop Window                 | Java Server Framework Database File    |
|------------------------------------|----------------------------------------|
| Process Configuration              | Process File (PROCD)                   |
| Event Configuration and Management | Event Definition File (EVTDD)          |
| Application Configuration          | Application Configuration File (APPLD) |
| Listener Configuration             | Listener File (LSTND)                  |
| Connection Configuration           | External Connection File (ETCND)       |

You can also monitor Java Server Framework services using the server administration windows accessed from **System Operations > Server Management > Server Administration**. From these windows, you can view various statistics on how Java Server Framework services are being used, such as how many database connections are open, what external connections are established, etc.

Refer to the *ACI Java Server Framework Reference Guide* for detailed information on the Java Server Framework configuration requirements for the Java User Interface servlets used for the ACI desktop user interface.

## User Security (USEC) Application

The User Security (USEC) application provides operator security for ACI desktop user interface functions.

Each time the Java User Interface servlets receive a message from an ACI desktop user interface client, they send a waited request to the USEC application server to verify that the message is authentic and that the operator has the appropriate security to perform the requested task or function. For failover purposes with a multiple node configuration, all USEC applications share a common security database. In the event of a failure, the Java User Interface servlets can switch over to another USEC application, without requiring the user to log in again. In addition, the security database is backed up. If the USEC application is unable to access the primary security database, it can switch over to the backup database, without interruption.

Additional software is required for the USEC application to run on the HP NonStop system. The application must run under the Open System Services (OSS) environment, which coexists with the Guardian environment and runs under the NonStop Kernel operating system on an HP NonStop system.

The OSS environment has a structure and feel similar to a UNIX system. In addition, the USEC application requires one of the following, depending on whether you are implementing the USEC database using SQL or Enscribe:

- If you are implementing an SQL database, runtime NonStop SQL/MP with Java bound in and either the NonStop SQL/MP database driver or the NonStop SQL/MX database driver is required. ACI recommends the NonStop SQL/MX database driver because of its better error reporting capabilities. This option is not available for K-series servers.
- If you are implementing an Enscribe database, ACI's Java Database Connectivity (JDBC) driver called the Java DAL Database Connectivity (JDAL) driver is required for Java server applications to interface with the Enscribe database.

The USEC application must be configured as a general process in the OSS configuration file.

## Default User IDs

Four preconfigured user IDs are provided for the Transaction Security Services environment at installation. These user IDs and their passwords are listed in the following table and described below.

| User ID (Name) | Password    |
|----------------|-------------|
| TSSSysAdmin    | TSYSAdmin01 |
| SecAdmin       | SecAdmin01  |
| SysAdmin       | SysAdmin01  |
| TSSAdmin       | TSAdmin01   |

The TSSSysAdmin and TSSAdmin preconfigured user IDs are specific to the Transaction Security Services environment while the SecAdmin and SysAdmin preconfigured user IDs are generic to all BASE24-es applications. The preconfigured TSSSysAdmin user ID has access to all TSS windows and functions available on the ACI desktop, including the User Security and Java Server Framework windows and functions. The preconfigured TSSAdmin user ID has access to all TSS and Java Server Framework windows and functions on the ACI desktop, except for the User Security windows and functions. The SecAdmin user ID has access to the User Security windows and functions only. The SysAdmin user ID has access to almost all windows and functions in a BASE24-es system, including all TSS and User Security, and Java Server Framework windows and functions. Therefore, you can use the TSSSysAdmin, SecAdmin, or SysAdmin user IDs to set up initial security access for other users, however, the SysAdmin user ID has access to several windows and functions that are not applicable to the TSS environment. You can retain or delete default user IDs as needed for your security environment.

**Note:** For security reasons, the system administrator or security administrator should change the passwords for these preconfigured user IDs immediately after installation.

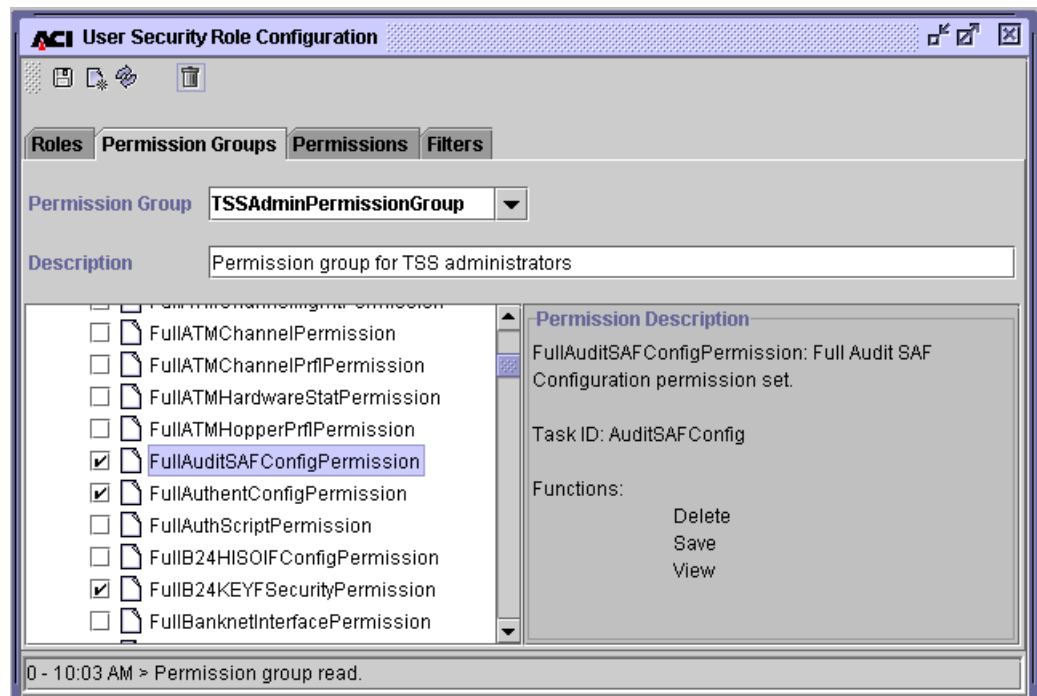
The following table shows you the default roles configured for each of the default user IDs provided at installation.

| User ID  | Role         |
|----------|--------------|
| SecAdmin | SecAdminRole |

| User ID     | Role            |
|-------------|-----------------|
| TSSAdmin    | TSSAdminRole    |
| TSSSysAdmin | TSSSysAdminRole |
| SysAdmin    | SysAdminRole    |

You can view the individual permissions configured for each of these default user IDs on the Permission Groups tab of the User Security Role Configuration window. The TSSSysAdminPermissionGroup is assigned to the TSSSysAdmin user ID, the TSSAdminPermissionGroup is assigned to the TSSAdmin user ID, the SysAdminPermissionGroup is assigned to the SysAdmin user ID, and the SecAdminPermissionGroup is assigned to the SecAdmin user ID.

The individual permissions configured to these permission groups are displayed in a list in the lower left-hand portion of the window. Check marks are displayed next to permissions configured to the group. You can right click on a selected permission to view a description of the permission, the task IDs associated with the permission, and the functions available, as shown in the following illustration.



**Note:** The permissions list displays all permissions available in a BASE24-es system. The permissions checked for the TSSSysAdminPermissionGroup include all of the full permissions available for the TSS environment. If a full permission is not checked for this permission group, you can ignore it.

The following table summarizes the permissions available for each permission group configured for default user IDs. All permission IDs for the TSS environment are listed in an appendix to the *ACI Desktop User Interface Manual*.

| User ID     | Permission Group           | Permissions                                                                                                                                                     |
|-------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TSSSysAdmin | TSSSysAdminPermissionGroup | All permission IDs listed for TSS.                                                                                                                              |
| TSSAdmin    | TSSAdminPermissionGroup    | All permissions IDs listed for TSS except for the following:<br><br>FullUSCtlPermission<br>FullUSGCfgPermission<br>FullUSPPPermission<br>FullUSRCfgPermission   |
| SecAdmin    | SecAdminPermissionGroup    | FullUSCtlPermission<br>FullUSGCfgPermission<br>FullUSPPPermission<br>FullUSRCfgPermission                                                                       |
| SysAdmin    | SysAdminPermissionGroup    | All permission IDs listed for TSS as well as additional permission IDs listed for BASE24-es, that are not relevant to the TSS environment for BASE24 customers. |

A specific task ID is associated with each window on the ACI desktop user interface. Task IDs are combined with permissions to provide delete, save, and/or view functions to users for each window. For windows that retrieve a list of values for a field from another window, you must at least give a user view access to the window from which the list of values are retrieved. In addition, some windows that are based on a channel or interface framework, also require access to the server component for the framework. A specific task ID (i.e., ChannelConfiguration and InterfaceConfiguration) is associated with each of these server components.



Refer to the *ACI Desktop User Interface Manual* and ACI Online Help for the ACI desktop for a complete list of the task IDs needed to access each window available for TSS users.

## User Auditing

Optional user auditing enables you to audit all online operator activity performed through the ACI desktop user interface. The database auditing mechanism enables you to track all adds, updates (before and after images), and deletes to the application database by user ID and time of change. If you update or delete a profile, all data records for the profile are updated or deleted and are included in the audit trail. User logouts, password verifications, and password changes are also recorded. When a TSS file is audited, records are written to a User Audit Log File (UALOGD). You can configure the files audited when your ACI system is installed and control the extent to which files are audited. For full auditing, header information as well as detail information about the before and after images of the affected record are logged to the UALOGD. For medium auditing, only the header information is logged to the UALOGD.

The Java User Interface servlets (ESWEB) process and the C++ User Interface (XML Server) process write auditing records directly to the UALOGD. The option for the XML Server process to write nowaited messages to an Audit Store and Forward File (ASAFD) for transmission to the UAUD application is no longer available beginning in TSS release 05.4. Therefore, in TSS release 05.4 and beyond, the USE\_LOCAL\_USER\_AUDIT environment attribute must always be set to a value of Y, which indicates the process writes auditing records directly to the UALOGD.

You can extract user auditing records from the UALOGD to a specified extract file using the Audit Extract window on the ACI desktop user interface (accessed from **Tools > Audit Extract**). An application configuration property named report.userauditlog.loc specifies the directory where the extract file is created (e.g.: /user/B24es/ESWeb/reports/). Refer to the *ACI Desktop User Interface Manual* for detailed procedures on using the Audit Extract window. The *ACI Java Server Framework Reference Guide* provides a detailed description of the report.userauditlog.loc property.

The UAUD application must be configured as a general process in the OSS configuration file.

Additional software is required for the UAUD application to run on the HP NonStop system. The application must run under the Open System Services (OSS) environment, which coexists with the Guardian environment runs under the NonStop Kernel operating system on an HP NonStop system.

Most of the user interface auditing in the TSS environment is controlled by db.audited.db-service-name properties configured for the Java User Interface servlets on the Application Configuration window of the ACI desktop user interface, where db-service-name is the name of a particular Java database service. For each Java database service, you can configure the servlets to perform no auditing, headers only auditing, or full auditing. For files maintained by the C++ User Interface process, the UI\_AUDIT\_LEVEL attribute for the Transaction Security Services environment indicates the level of user interface auditing performed by the C++ User Interface process. Using this attribute, you can configure the C++ User Interface process to perform no auditing, headers only auditing, or full auditing. Each of these options is explained in detail in [section 3](#).

Refer to the *ACI Java Server Framework Reference Guide* for a list of all possible db.audited.db-service-name properties that apply to the TSS environment. If a TSS file and associated ACI desktop window is not represented in the list in the *ACI Java Server Framework Reference Guide*, auditing for the file is controlled by the UI\_AUDIT\_LEVEL attribute.

The *ACI Desktop User Interface Manual* describes UAUD processing in further detail.

## Secure Sockets Layer (SSL) Protocol

Messaging endpoints of the ACI desktop user interface (i.e., Desktop clients, WebGate HTTP Server, USEC, and UAUD applications) provide support for the Secure Sockets Layer (SSL) protocol. You can optionally configure any or all of the above ACI desktop user interface components to use the SSL protocol when the ACI desktop is installed. The ACI desktop uses Sun Microsystems' Java Secure Sockets Extension (JSSE) version 1.0.3\_01. This version of JSSE supports SSL version 3.0.

**Note:** If you want to provide SSL support for the WebGate HTTP Server, you must purchase it directly from Insession Technologies.

The SSL protocol helps ensure that user interface message data is not read by an unauthorized party (through the use of certificates) or tampered with during transmission over the user interface network (using data encryption). Hashing and encryption algorithms can be used to encrypt data transferred between network

endpoints. Using SSL with Remote Method Invocation (RMI) or basic socket connections provides encryption security to the transmitted data. When an SSL-enabled server authenticates itself to an SSL-enabled client, the client receives a certificate to accept or reject. If the client accepts the certificate, both the client and server establish an encrypted connection.

If desired, you can use the trace facility provided with the ICE product to trace SSL negotiation sequences for WebGate. The ICE Probe utility provided on the WebGate installation CD-ROM is a Windows-based tool that provides a user friendly interface for reading trace files.

SSL configuration requirements are discussed in the *ACI Desktop User Interface Manual* and the *ACI Java Server Framework Reference Guide*.

## Files

The Transaction Security Services environment requires the following additional files other than those security-related TSS files listed earlier. The assigns and params for these files, as well as the security-related TSS files listed earlier, are configured in the CONFCSV.

- The Environment File (ENVMTD) contains attributes for the Transaction Security Services environment similar to params defined in the LCONF. The Environment Management window in the ACI desktop user interface enables you to configure these attributes. The Transaction Security Services attributes required for support of the Transaction Security Services process are described in [section 3](#).
- The Event File (EVENTD) contains event messages that can be generated by processes in the Transaction Security Services environment. The Event Document File (EVD OCD) contains event message and token documentation. The Event Configuration window in the ACI desktop user interface enables you to view the cause and remedy of event messages, modify the text and documentation for event messages, and suppress the generation of certain event messages.
- The Exception Log File (EXLOGD) contains Extensible Markup Language (XML) messages that the C++ User Interface process could not interpret. This file can be used as a debugging tool.

- The System Interface Message Delivery Service Destination File (MDSDEST) maps security device assign names to PPD names (e.g., ATALLA1 to \$BOX1). It also maps Transaction Security Services processes to PPD names to enable the use of INFO and STATUS process control commands for these processes.
- The Trace File (TRCD) contains the output of process traces generated for traces configured in the Trace Configuration File (TRCCFD) and the Trace Filter Profile Configuration File (TRCFLD).

The OSS environment and Servlet Engine JVM environment also require several additional files other than those security-related TSS files listed earlier.

Several files are required to support the Java Server Framework. See the “Java Server Framework” topic provided earlier in this section for a list of these files.

Several files are required to support file partitioning. See the “File Partitioning” topic provided later in this section for a list of these files.

For detailed information on user security and auditing files, refer to the ***ACI Desktop User Interface Manual***.

# Operations

As stated previously, the Transaction Security Services process can be controlled through commands entered from an XPNET network control facility. Additional operational controls are available for the Transaction Security Services environment. Each of these operations functions is described below.

## Startup

Transaction Security Services environment components should be started up as described below. These procedures are for routine startup and assume the Transaction Security Services environment has already been started up initially according to the startup procedures in the *BASE24-es and TSS HP NonStop Install Guide*.

**Note:** The subvolume and symbolic processes names referenced in these procedures may differ from your site-specific subvolume and symbolic process names. Please adjust your startup procedures accordingly.

1. Start the TCP/IP Server process by executing the GOTCPSRV file.  

```
> VOLUME ESTCNFG  
> RUN GOTCPSRV ppd-name
```
2. Start the TCP/IP Client process by executing the GOTCPCLI file.  

```
> VOLUME ESTCNFG  
> RUN GOTCPCLI ppd-name
```
3. Start the WebGate HTTP Server application.
4. Start the Version Checker server and User Security (USEC) application from a TACL prompt or from the OSS environment.

To start the Version Checker server and USEC application from a TACL prompt, perform the following steps:

- a. Start the Version Checker server. Press the Enter key again after entering these commands to return to the TACL prompt.  

```
> VOLUME ESTCNFG  
> OBEY RUNVCS
```
- b. Start the USEC application.  

```
> VOLUME ESTCNFG  
> OBEY RUNUSEC
```

To start the Version Checker server and USEC application from the OSS environment, perform the following steps:

- a. Start the Version Checker server.

```
> OSH
$ cd ..
$ cd VersionChecker
$ RunVerChkSrv
```

- b. Start the USEC application.

```
> OSH
$ cd ..
$ cd ES/USEC
$ RunUSEC
```

5. Start the XPNET node, including the Transaction Security Services processes (ISLO and AKDSLO), Timer process (TIM); C++ User Interface (XML Server) process, device, lines, and stations; and ESWEB Server process, device, lines, and stations.

**Note:** If your TSS environment is configured in a separate node from your BASE24 processes, start the TSS node before starting the BASE24 node. Also, replace the sample symbolic names provided in these sample commands with the actual symbolic names configured for your system and account for multiple copies of a process.

```
> VOLUME network-control-subvolume
> O GONCP
  NODE 1> START NODE P1A^NODE
  NODE 2> START PROCESS P1A^TCP
  NODE 3> START PROCESS P1A^XML
  NODE 4> START PROCESS P1A^ISLO
  NODE 5> START PROCESS P1A^TIM
  NODE 6> START PROCESS P1A^AKDSLO01 (optional, only if
        using AKDS)
  NODE 7> START PROCESS P1A^ESWEB01
```

**Note:** The EWEB Server process should take several minutes to initialize. You need to check the following log files in OSS to determine whether any errors occurred during initialization: STDERR, STDOUT, ESWeb\_XXX.LOG, SERVLET\_XXX.LOG, ESWebCriticalEvents, etc. For more information on troubleshooting, refer to section 5.4, “Troubleshooting and Debugging Strategies,” in the *NET24-XPNET Web Application Services Implementation Guide*.

Due to a timing issue, the P1A^ESWEB01 process might not start the first time. If the process status does not change into the *started* state after 10 minutes, kill the PPD from the TACL prompt and restart the process in NCPCOM.

```
NODE 8> START LINE L1A^ESWEB
NODE 9> START LINE I1A^GUI
```

**Note:** This last line remains in a STARTING state until it processes a message.

6. Check EMSperus to make sure all processes, lines, and stations are successfully started.
7. Start the ACI desktop user interface client. On a workstation where the ACI desktop is installed, perform one of the following to start the ACI desktop:
  - Access and click the Desktop application from the Windows Start menu.

```
Start > Programs > ACI_GUI > Desktop
```

- Access Internet Explorer and enter the following URL address:

```
http://<system IP address or DNS name>:<port-number>/esweb
```

For example:

```
http://172.16.23.61:8004/esweb
```

**Note:** The Transaction Management Facility (TMF) should be started before performing any functions from the ACI desktop user interface or sending a message to a Transaction Security Services process.

The XPNET node and the processes, lines, stations, and devices configured to the node are started from an XPNET network control facility. The TCP/IP Server process on the HP NonStop system is started by executing the GOTCPSRV obey file on the object subvolume. The TCP/IP Client processes are started by executing the GOTCPCLI obey file on the object subvolume. The line associated with a TCP/IP client process must be started before the client process is started.

On a client workstation, the ACI desktop user interface client is started by executing a batch file (i.e., RunDesktop.bat) on the client workstations. The USEC, UAUD, and Version Checker applications are started with commands entered within OSS.

You start the WebGate HTTP Server and the URLPATHs used to connect to User Interface processes using the Node Operator Facility (NOF) of the Intersystem Communications Environment (ICE) product. Refer to the ***NOF Command Reference and Resource Definition Guide*** in the ICE manual library for detailed information on the commands used to start these resources.

After the Java User Interface servlets and C++ User Interface process and stations are started, ACI desktop user interface stations remain in a STARTING state until a connection is made from the WebGate HTTP Server (i.e., when a START

URLPATH command is entered from NOF). When the WebGate HTTP Server connects to the Java User Interface Server process, the user interface stations move to a STARTED state. When the WebGate HTTP Server is stopped, the user interface stations revert to the STARTING state.

The configuration requirements for the GOTCPSRV and GOTCPCLI obey files are provided in the ACI desktop user interface and Transaction Security Services Installation section of the ***BASE24 Classic 6.0 Installation Guide***. Sample network control facility commands to configure the Transaction Security Services process, Timer process, User Interface Server process, and ACI desktop user interface device, lines, and stations, are also provided in this processing guide.

## Process Control

You can control all TSS-related processes, lines, and stations configured to the XPNET system from a network control facility. Refer to the ***NET24-XPNET Network Control Command Reference Manual*** for a discussion of process, line, and station control commands. Typically, these commands are used to configure and start the Transaction Security Services processes, Timer process, C++ User Interface processes, Java User Interface servlets (ESWEB) process, and the UAUD components.

You can control the WebGate HTTP Server and associated resources using the Node Operator Facility (NOF) of the Intersystem Communications Environment (ICE) product. Refer to the ***NOF Command Reference and Resource Definition Guide*** for detailed information on the commands used to control these resources.

Additional operational control mechanisms are available through the Process Control facility of the Transaction Security Services environment. Through this facility, security personnel operators can enter commands to perform the following functions:

- Rebuild external memory tables
- Warmboot the Transaction Security Services process to begin using a rebuilt external memory table or start rebuilding an internal memory table
- Initialize, enable, and disable all hardware security modules (HSMs) or a single HSM
- Obtain the current status of one or all HSMs
- Download the Diebold number table (DNT) into the memory of an HSM

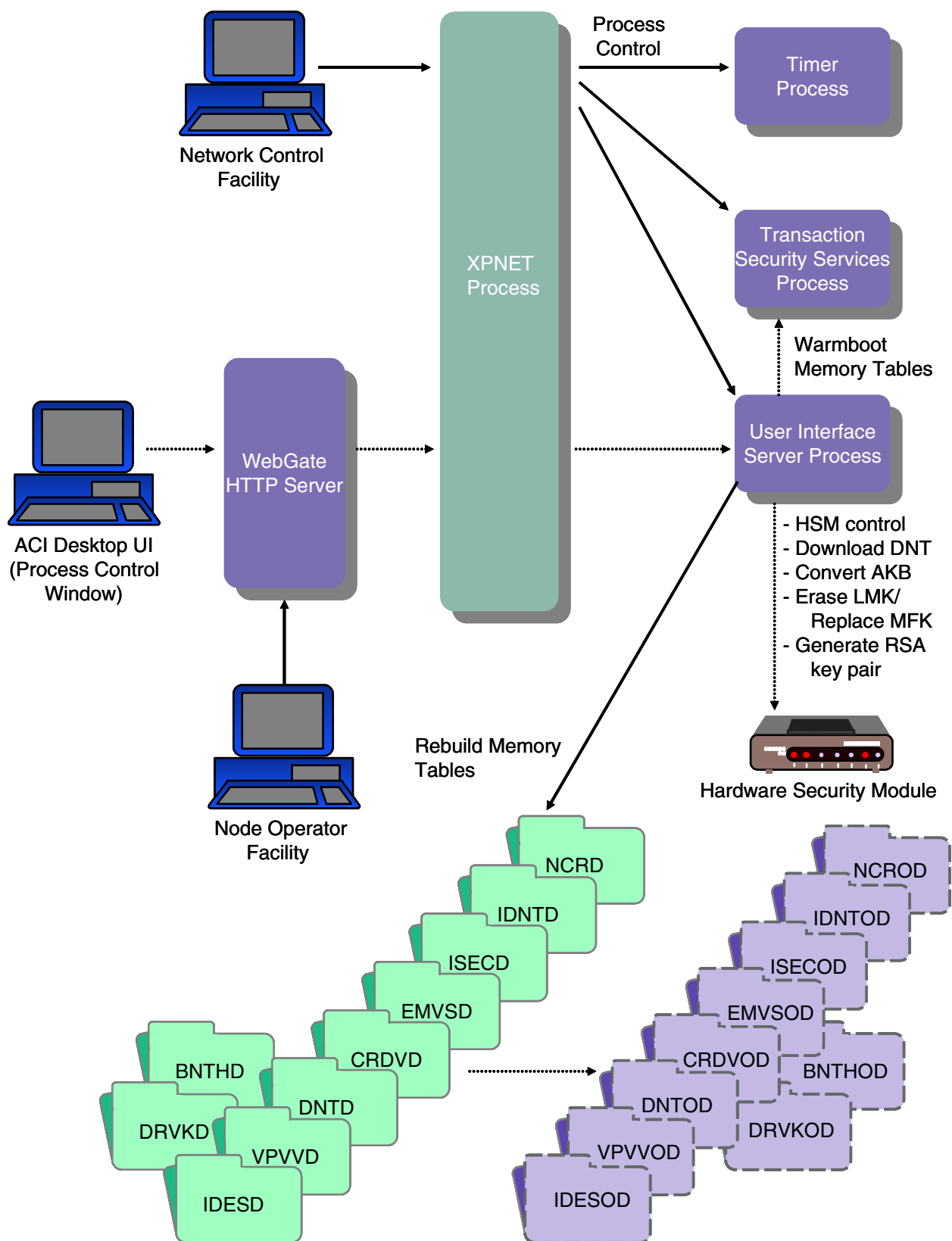


- Convert keys in the TSS database from non-AKB format (i.e., variant format) to AKB format
- Erase the old LMK key pair for a Thales e-Security HSM or replace the current MFK with the pending MFK for an Atalla NSP after converting keys using the Master File Key Conversion utility.
- Generate a new RSA key pair for the host BASE24 system

This process control function is only available if you have purchased the BASE24-atm Automated Key Distribution System add-on product.

All of these commands are entered from the Process Control window in the ACI desktop user interface. For detailed information on using the Process Control window to rebuild external memory tables, warmboot the Transaction Security Services process to start using rebuilt external memory tables, and warmboot the Transaction Security Services process to start rebuilding internal memory tables, refer to [section 2](#). For detailed information on entering HSM commands, refer to [section 9](#) and [appendix C](#) in this guide. For detailed information on generating a host RSA key pair, refer to [appendix B](#) of this guide. For more information on the Process Control window, refer to [section 9](#) and the ACI Online Help provided with the ACI desktop user interface.

The process control mechanisms available for the Transaction Security Services process are depicted in the following illustration.



## Process Tracing

Through the ACI desktop user interface, you can configure, execute, and filter traces of C++ processes (e.g., Transaction Security Services processes) in the TSS environment. The Trace Configuration window (accessed from **System Operations > Trace > Trace**) enables you to configure and execute a process trace. The Trace Filter Profile Configuration window (accessed from **System Operations > Trace > Trace Filter Profile**) allows you filter the messages for which trace output is provided. For detailed information on using these trace facilities, refer to [section 10](#).

## Master File Key Translation

ACI provides a Master File Key Conversion tool that enables you to translate hardware-encrypted keys in the TSS database from encryption under one master key file (i.e., MFK for Atalla or LMK pair for Thales e-Security) to encryption under another master key file. This conversion tool is executed from the Master File Key Conversion window, accessed as **Configure > Transaction Security > Master File Key Conversion** on the ACI desktop. Using this tool, you can translate all the keys in the TSS database or just the keys of a selected type. For detailed information on using this tool, refer to the [Master Key File Conversion](#) topic provided in appendix C.

The Master File Key Conversion tool is not currently supported for Eracom or Bull BNTng HSMs.

## Event Management

Event messages generated by processes running in the Transaction Security Services environment on the HP NonStop system are sent to an EMS collector process and are displayed on the EMS log just like all BASE24 modules. You should also monitor the EMS log for Transaction Management Facility (TMF) events that require attention. You can use the Event Configuration window in the ACI desktop user interface to view event message documentation for the Transaction Security Services environment. For detailed information on managing events for the TSS environment, refer to [section 8](#).

**Note:** Event messages generated by the WebGate HTTP Server are documented in the *ICE Messages and Codes Reference* in the ICE manual library.

Security device errors are mapped to error messages provided by the security utilities bound into BASE24 processes.

## Transaction Management Facility (TMF)

All of the Enscribe files maintained by the Transaction Security Services process are audited by TMF. The Transaction Security Services process uses TMF to back out updates on a per-transaction basis. More sophisticated features of TMF, such as rollback and roll forward, are not required or used. For more information on TMF operations, refer to HP's *NonStop TM/MP Operations and Recovery Guide*.

## File Partitioning

File partitioning enables you to partition a single TSS database file across multiple data sets. On an HP NonStop system, file partitioning allows you to expand the Enscribe limit of 16 partitions for a single file. File partitioning also spreads I/O across multiple data sets, thereby reducing the bottleneck associated with a single data set. Any database file in the TSS environment that is not accessed from an external memory table can be partitioned. You use the File Partition Configuration window (accessed from **System Operations > Database Management > File Partition**) to configure file partitions. Refer to the *BASE24-es File Partitioning User Guide* for detailed information on using the File Partition Configuration window and procedures for partitioning files. Data entered on the File Partition Configuration window is stored in the following files:

- File Partition Catalog File (FPCTGD)
- File Partition Fields File (FPFLDD)
- File Partition List File (FPLSTD)

Data in the FPFLDD and FPLSTD is accessed from external memory tables during processing.

# Disaster Recovery or Dual-Site Configuration

**Note:** The following is an example of TSS files to replicate in a disaster recovery or dual-site configuration. Your files may differ based on your specific disaster recovery or dual-site processing requirements.

Typically, you would replicate all TSS files that are audited by the Transaction Monitoring Facility (TMF) for disaster recovery purposes or dual-site use. In a typical TSS installation that also includes the BASE24-atm Automated Key Distribution System add-on product, this would include the following files:

- BNTng Header File (BNTHD)
- BNTng Initialization File (BNTID)
- Card Verification File (CRDVD)
- Channel Public Key Security File (CHNPKD)
- Channel Security File (CSECD)
- Derivation Key File (DRVKD)
- Diebold Number Table File (DNTD)
- EMV Authorization File (EMVSD)
- Environment File (ENVMTD)
- EPP Serial Number File (EPNUMD)
- Event Document File (EVDOCD)
- Host Public Key Security Primary File (HSPKPD)
- Host Public Key Security Secondary File (HSPKSD)
- IBM DES Verification File (IDESD)
- Identikey PIN Verification File (IDNTD)
- Interface Security File (ISECD)
- NCR PIN Verification File (NCRD)
- Security Device Configuration File (SECDCD)
- Trace Filter Profile Configuration File (TRCFLD)
- Visa PVV Verification File (VPVVD)

Typically, you would not replicate the following files, including all external memory table swap files:

- Audit Store and Forward Configuration File (ASAFCD)\*
- Audit Store and Forward File (ASAFD)\*
- Audit Store and Forward Outstanding File (ASAFOD)\*
- BNTng Header external memory table swap file (BNTHOD)
- Card Verification external memory table swap file (CRDVOD)
- Derivation Key external memory table swap file (DRVKOD)
- Diebold Number Table external memory table swap file (DNTOD)
- EMV Authorization external memory table swap file (EMVSOD)
- Event File (EVENTD)
- Exception Log File (EXLOGD)
- IBM DES Verification external memory table swap file (IDESOD)
- Identikey PIN Verification external memory table swap file (IDNTOD)
- Interface Security external memory table swap file (ISECOD)
- NCR PIN Verification external memory table swap file (NCROD)
- Visa PVV Verification external memory table swap file (VPVVOD)

\* These files are no longer used in TSS release 05.4.

# Transaction Processing Examples

The Transaction Security Services process is used in the processing of every transaction that requires a hardware security module function. For transaction security functions performed in software, the Transaction Security Services process is not used.

Example transaction processing flows using the Transaction Security Services process are provided in [section 5](#).

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## **2: Database Relationships and Security Data Maintenance**

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This section describes the relationships between BASE24 database files and the Transaction Security Services database files. It also describes how security data is maintained.

## BASE24-atm Terminal Data Files

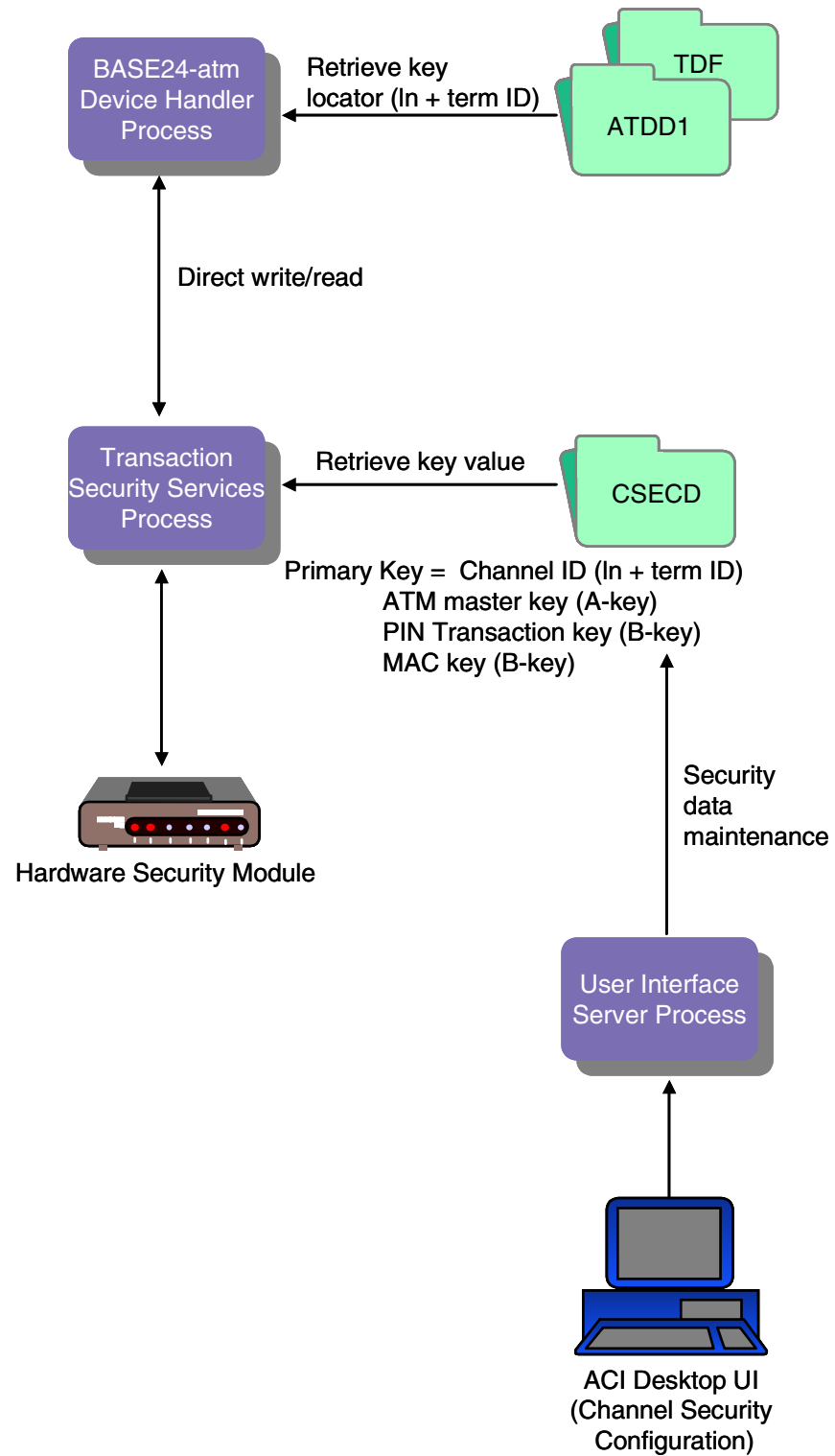
When the BASE24-atm Terminal Data files are converted, the MASTER KEY field on the device-specific ATD screen 9 for records that use hardware-encrypted keys when the Transaction Security Services process is implemented (i.e., the SECURE-DEV-TYP param is set to a value of TSS) are set to key locator values. In addition, these fields are not checked for all hexadecimal characters. If you change the key locator value in the MASTER KEY field of a converted record to a value other than the terminal ID, you must add a corresponding record for this key locator value on the Channel Security Configuration window.

Regardless of whether a record is converted, all encryption type fields (i.e., PIN ENCRYPT TYPE, PIN CHANGE ENCRYPT TYPE, and MAC SUPPORT TYPE fields on the device-specific ATD screen 9) for a particular terminal's record must contain compatible values. You cannot mix software encryption and hardware encryption types in the same record. For example, if the PIN encryption type value specifies to use a security device, the MAC support type value cannot specify that MAC support is in software, and vice versa.

The following diagram illustrates the relationship between BASE24-atm Terminal Data files and the Channel Security File (CSECD) after file conversion. The hardware encrypted key fields in the BASE24-atm Terminal Data files are set to the value of the two-character logical network identifier (specified by the LN-IND LCONF param if you are partitioning the CSECD) added to the beginning of the terminal ID, and the actual key values are stored in the CSECD. Data in the CSECD is maintained from the Channel Security Configuration window. The terminal ID value entered in the TERM ID field on BASE24-atm Terminal Data files (ATD) screen 1 must match the value entered in the Channel ID field on the Channel Security Configuration window. If you add additional terminal records to the BASE24-atm Terminal Data files, you enter the encryption keys into corresponding CSECD records from the Channel Security Configuration window on the ACI desktop user interface. Refer to [section 3](#) for more information on the LN-IND param.

**Note:** If an ATM is configured in the BASE24-atm Terminal Data files to send PINs in the clear to any interchanges on the BASE24 system, a CSECD record must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an Interchange Interface process supports the conversion of incoming encrypted PINs to clear PINs internally, an ISECD record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.



## Channel Security Configuration Window

The Channel Security Configuration window enables you to configure keys and parameters required by ATM devices for PIN encryption and message authentication. The Data Encryption tab on this window is not currently used for any ATM devices. Samples of the PIN Information tab and MAC Information tab of Channel Security Configuration window for a Diebold 10XX/478X device are shown below. For ATM devices, the ATM master key is used as the exchange key for both PIN and MAC encryption keys. Therefore, the same ATM master key must be used in the Exchange Key field on both the PIN Information and MAC Information tabs. Refer to the ACI Online Help provided with the ACI desktop user interface for detailed descriptions of this window and its fields.

**Note:** You must configure a security device using the Security Device Configuration window before you can configure channel security information using the Channel Security Configuration window. The Data Encryption tab is disabled for Eracom host security modules.

## Old Key Timer Limit

The Old Key Timer Limit field located at the top of the Channel Security Configuration window specifies the maximum length of time, in seconds, that a PIN, MAC, or data encryption key can be used after a successful dynamic key exchange before it is automatically cleared.

This timer limits the amount of time that an old key can be used for PIN verification, message authentication, or data encryption after a successful key exchange has taken place.

Currently, this limit is only used for Triton ATMs.

## Eracom ProtectHost White AMB HSMs

When you use Eracom ProtectHost White AMB host security modules (HSMs) in your system, the fields displayed on the Channel Security Configuration window differ significantly from those shown and described in this section. For detailed information on configuring channel keys for Eracom ProtectHost White AMB HSMs, refer to [section 4](#).

## PIN Information Tab

An example of the PIN Information tab on the Channel Security Configuration window is shown below for an NCR 5XXX device.

The screenshot displays the 'ACI Channel Security Configuration - NCR001' window. The 'PIN Information' tab is selected, showing the following configuration:

- Channel ID: NCR001
- External Key Block Format: Standard Atalla
- Old Key Timer Limit: 0
- PIN Block Format: ANSI PIN (ISO Format 0)
- PIN Pad Character: F
- Exchange Key: Key Variant 5 - ATM Key Variant, Key Length Single
- Inbound Key: Key Length Single

The Exchange Key table shows the following data:

|  | Header   | Key Part 1          | Key Part 2          | Key Part 3          | MAC                 | Check Di |
|--|----------|---------------------|---------------------|---------------------|---------------------|----------|
|  | 1KDNE000 | 7C4E 5AF9 CBD0 4AB2 | FC8B AEC9 75DB 7782 | 30A6 0E9A B833 5446 | 32C1 78A5 23B3 62A1 | 4353     |

The Inbound Key table shows the following data:

|  | Header   | Key Part 1          | Key Part 2          | Key Part 3          | MAC                 | Check Di |
|--|----------|---------------------|---------------------|---------------------|---------------------|----------|
|  | 1PUNE000 | 2D34 EEF7 D353 EB22 | 3C94 0234 EA35 33BE | 9001 BDBD 57FE 95B3 | 8FA1 85E1 2B5B 92A4 |          |

The status bar at the bottom indicates: 0 - 04:56 PM > Channel Security Information Retrieved

## MAC Information Tab

An example of the MAC Information tab on the Channel Security Configuration window is shown below for an NCR 5XXX device.

The screenshot shows the 'MAC Information' tab of the 'ACI Channel Security Configuration - NCR001' window. The 'Channel ID' is 'NCR001'. The 'External Key Block Format' is 'Standard Atalla'. The 'Old Key Timer Limit' is '0'. The 'MAC Option' is 'Single length DES Cipher Block Chai...' and the 'MAC Length' is '32 bits'. The 'Exchange Key' is '0 - Key E...' and the 'Key Variant' is '0 - Key E...'. The 'Key Length' is 'Single'. The 'Export Variant' is '0 - Key E...'. The 'Key Part 1' is '7C4E 5AF9 CBD0 4AB2', 'Key Part 2' is 'FC8BAEC975DB7782', 'Key Part 3' is '30A60E9AB8335446', and 'Check Digits' is '4353'. The 'Inbound Key' is '3336 C804 293E 46EE'. The 'Key Length' is 'Single'. The status bar at the bottom shows '0 - 04:41 PM > Channel Security Information Retrieved'.

| Key Part 1          | Key Part 2       | Key Part 3       | Check Digits |
|---------------------|------------------|------------------|--------------|
| 7C4E 5AF9 CBD0 4AB2 | FC8BAEC975DB7782 | 30A60E9AB8335446 | 4353         |

| Key Part 1          | Key Part 2 | Key Part 3 | Check Digits |
|---------------------|------------|------------|--------------|
| 3336 C804 293E 46EE |            |            |              |

The optional Export Variant field on this tab specifies the variant under which a new device MAC key is generated when you are using Atalla security devices without AKB software. Otherwise, this field is not used. If the MAC key is to be downloaded under variant 0 of the KEK, the Transaction Security Services process uses Atalla command 11D to generate the new device MAC key. If the MAC key is to be downloaded under variant 5 of the KEK, the Transaction Security Services process uses Atalla command 10 (Generate Working Key, Any Type) and the specified export variant to generate the new device MAC key. If you set this field

to a value of 0, the Key Variant field must be set to a value of 0 or 5. If you set this field to a value of 3, the Key Variant field must be set to a value of 0. Valid values are as follows:

- 0 = Key Exchange Key variant (default)
- 3 = MAC Encryption Key variant

## BASE24-pos Terminal Data Files

When the BASE24-pos Terminal Data files are converted, the PIN MASTER KEY and MAC MASTER KEY fields on PTD screen 7 for records that use hardware-encrypted keys when the Transaction Security Services process is implemented (i.e., the SECURE-DEV-TYP param is set to a value of TSS) are set to key locator values. In addition, these fields are not checked for all hexadecimal characters. If you change the key locator value in the PIN MASTER KEY or MAC MASTER KEY field of a converted record to a value other than the terminal ID, you must add a corresponding record for this key locator value on the Channel Security Configuration window.

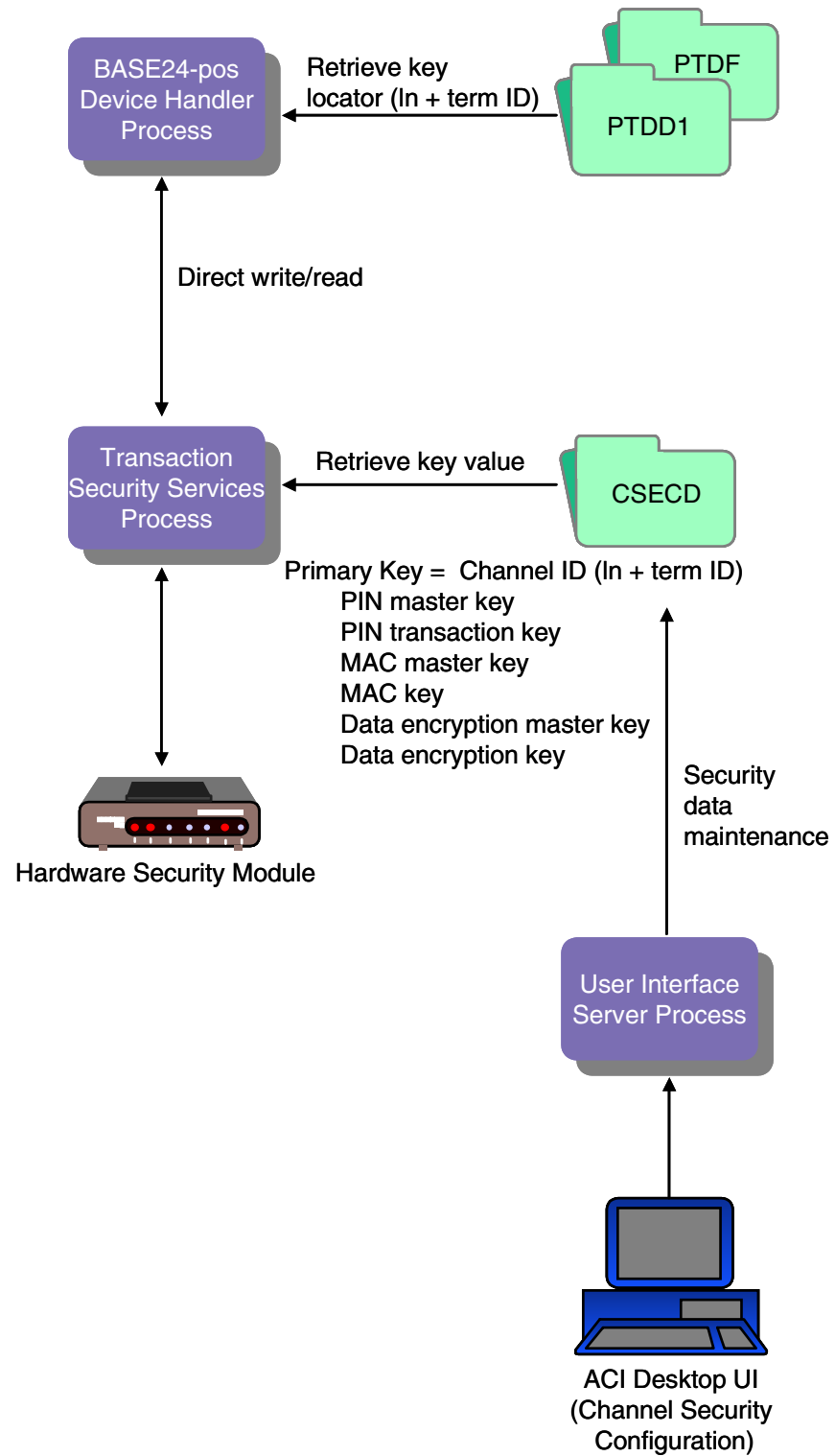
Regardless of whether a record is converted, the encryption type fields (i.e., PIN ENCRYPTION TYPE and MAC TYPE fields on PTD screen 7) for a particular terminal's record must contain compatible values. You cannot mix software encryption and hardware encryption types in the same record. For example, if the PIN encryption type value specifies to use a security device, the MAC type value cannot specify that MAC support is in software, and vice versa.

The following diagram illustrates the relationship between BASE24-pos Terminal Data files and the Channel Security File (CSECD) after file conversion. The hardware encrypted key fields in the BASE24-pos Terminal Data files are set to the value of the two-character logical network identifier (specified by the LN-IND LCONF param if you are partitioning the CSECD) added to the beginning of the terminal ID, and the actual key values are stored in the CSECD. Data in the CSECD is maintained from the Channel Security Configuration window. The terminal ID value entered in the TERMINAL ID field on BASE24-pos Terminal Data files (PTD) screen 1 must match the value entered in the Channel ID field on the Channel Security Configuration window. If you add additional terminal records to the BASE24-pos Terminal Data files, you enter the encryption keys into corresponding CSECD records from the Channel Security Configuration window on the ACI desktop user interface. Refer to [section 3](#) for more information on the LN-IND param.

**Note:** If a POS device is configured in the BASE24-pos Terminal Data files to send PINs in the clear to any interchanges on the BASE24 system, a CSECD record must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an Interchange Interface process supports the conversion of incoming encrypted PINs to clear PINs internally, an ISECD record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.





## Channel Security Configuration Window

The Channel Security Configuration window enables you to configure keys and parameters required by POS devices for PIN encryption, message authentication, and data encryption. The Data Encryption tab is currently used for SPDH-compliant devices only. The PIN, MAC, and data encryption master keys are entered in the Exchange Key field on the appropriate tab. For POS devices, these keys can all be different. Samples of the PIN Information, MAC Information, and Data Encryption tabs of the Channel Security Configuration window for an SPDH-compliant device are shown below. Refer to the ACI Online Help provided with the ACI desktop for detailed descriptions of this window and its fields.

**Note:** You must configure a security device using the Security Device Configuration window before you can configure channel security information using the Channel Security Configuration window.

## Old Key Timer Limit

The Old Key Timer Limit field located at the top of the Channel Security Configuration window specifies the maximum length of time, in seconds, that a PIN, MAC, or data encryption key can be used after a successful dynamic key exchange before it is automatically cleared.

This timer limits the amount of time that an old key can be used for PIN verification, message authentication, or data encryption after a successful key exchange has taken place.

## Eracom ProtectHost White AMB HSMs

When using Eracom ProtectHost White AMB host security modules (HSMs) in your system, the fields displayed on the Channel Security Configuration window differ significantly from those shown and described in this section. For detailed information on configuring channel keys for Eracom ProtectHost White AMB HSMs, refer to [section 4](#).

## PIN Information Tab

An example of the PIN Information tab on the Channel Security Configuration window is shown below for an SPDH-compliant device.

**Channel Security Configuration - SPDH001**

Channel ID: SPDH001

External Key Block Format: Standard Atalla

Old Key Timer Limit: 30

**PIN Information** | MAC Information | Data Encryption

PIN Block Format: ANSI PIN (ISO Format O) | PIN Pad Character: F

Exchange Key | Key Variant: 0 - Key Exchange Key Variant | Key Length: Triple

| Header   | Key Part 1          | Key Part 2          | Key Part 3          |
|----------|---------------------|---------------------|---------------------|
| 1KDNE000 | 9C25 C8E2 B1FD C7E8 | 2C39 395B D218 BF4A | 1447 C0B0 FCDD 7ACF |

Inbound Key | Key Length: Triple

| Header   | Key Part 1          | Key Part 2          | Key Part 3          |
|----------|---------------------|---------------------|---------------------|
| 1PUNE000 | 2D34 EEF7 D353 EB22 | 3C94 0234 EA35 33BE | 9001 BDBD 57FE 95B3 |

**Note:** If you are configuring PIN encryption for an SPDH-compliant device or Hypercom device using the derived unique key per transaction (DUKPT) method of encryption, you must set the PIN Block Format field to a value of DUKPT, and the PIN encryption key entered in the Inbound Key table must be encrypted under variant 1 for an Atalla NSP or under key pair 06/07 for a Thales e-Security HSM.

## MAC Information Tab

An example of the MAC Information tab on the Channel Security Configuration window is shown below for an SPDH-compliant device.

The screenshot shows the 'MAC Information' tab for device 'NCR001'. The 'Channel ID' is 'SPDH-01'. The 'External Key Block Format' is 'Standard Atalla'. The 'Old Key Timer Limit' is '0'. The 'MAC Option' is 'Single length DES Cipher Block Chai...'. The 'MAC Length' is '32 bits'. The 'Exchange Key Export Variant' is '0 - Key E...'. The 'Key Variant' is '0 - Key E...'. The 'Key Length' is 'Single'. The 'Inbound Key' section shows a table with four columns: Key Part 1, Key Part 2, Key Part 3, and Check Digits. The values are: 7C4E 5AF9 CBD0 4AB2, FC8BAEC975DB7782, 30A60E9AB8335446, and 4353. The 'Inbound Key' section also has a 'Key Length' dropdown set to 'Single'. The status bar at the bottom indicates '0 - 04:41 PM > Channel Security Information Retrieved'.

| Key Part 1          | Key Part 2       | Key Part 3       | Check Digits |
|---------------------|------------------|------------------|--------------|
| 7C4E 5AF9 CBD0 4AB2 | FC8BAEC975DB7782 | 30A60E9AB8335446 | 4353         |

| Key Part 1          | Key Part 2 | Key Part 3 | Check Digits |
|---------------------|------------|------------|--------------|
| 3336 C804 293E 46EE |            |            |              |

**Note:** If you are configuring message authentication for an SPDH-compliant device using the derived unique key per transaction (DUKPT) method of message authentication, you must set the MAC Option field to a value of “Old Style Visa DUKPT.”

The optional Export Variant field on this tab specifies the variant under which a new device MAC key is generated when you are using Atalla security devices without AKB software. Otherwise, this field is not used. If the MAC key is to be downloaded under variant 0 of the KEK, the Transaction Security Services process uses Atalla command 11D to generate the new device MAC key. If the MAC key is to be downloaded under variant 5 of the KEK, the Transaction Security Services process uses Atalla command 10 (Generate Working Key, Any Type) and the specified export variant to generate the new device MAC key. If you set this field

to a value of 0, the Key Variant field must be set to a value of 0 or 5. If you set this field to a value of 3, the Key Variant field must be set to a value of 0. Valid values are as follows:

- 0 = Key Exchange Key variant (default)
- 3 = MAC Encryption Key variant

## Data Encryption Tab

An example of the Data Encryption tab on the Channel Security Configuration window is shown below for an SPDH-compliant device.

The screenshot shows the 'Channel Security Configuration - SPDH001' window with the 'Data Encryption' tab selected. The window includes fields for 'Channel ID' (SPDH001), 'External Key Block Format' (Standard Atalla), and 'Old Key Timer Limit' (30). Below these are tabs for 'PIN Information', 'MAC Information', and 'Data Encryption'. The 'Data Encryption' section contains 'DES Mode' (ECB) and 'Data Type' (Unpacked ASCII hexadeci...). The 'Exchange Key' section shows 'Key Variant' (0 - Key Exchange Key Variant) and 'Key Length' (Triple), with a table of key parts. The 'Outbound Keys' section shows 'Encryption Layers' (1) and 'Key Length' (Triple), with a table of key parts. The 'Initialization Vector' section shows a table of key parts.

Channel ID: SPDH001

External Key Block Format: Standard Atalla

Old Key Timer Limit: 30

PIN Information | MAC Information | **Data Encryption**

DES Mode: ECB | Data Type: Unpacked ASCII hexadeci...

Exchange Key | Key Variant: 0 - Key Exchange Key Variant | Key Length: Triple

|  | Header   | Key Part 1          | Key Part 2          | Key Part 3             |
|--|----------|---------------------|---------------------|------------------------|
|  | 1KDNE000 | 9C25 C8E2 B1FD C7E8 | 2C39 395B D218 BF4A | 1447 C0B0 FCDD 7ACF 9E |

Outbound Keys | Encryption Layers: 1 | Key Length: Triple

|   | Header   | Key Part 1          | Key Part 2          | Key Part 3             |
|---|----------|---------------------|---------------------|------------------------|
| 1 | 1DDEE000 | B8BA DA3D 2FE2 4CC9 | 69ED 3710 713B E566 | 73A4 8F96 FAE5 EB8E 3D |
| 2 |          |                     |                     |                        |

Initialization Vector

|  | Header | Key Part 1 | Key Part 2 | Key Part 3 |
|--|--------|------------|------------|------------|
|  |        |            |            |            |

# Maintaining Channel Security Data

The following paragraphs describe procedures for adding, modifying, and deleting channel ID records in the Channel Security File (CSECD).

## Adding a New Channel ID Record

Use the following procedure when adding a new ATM terminal or POS device to your BASE24 system.

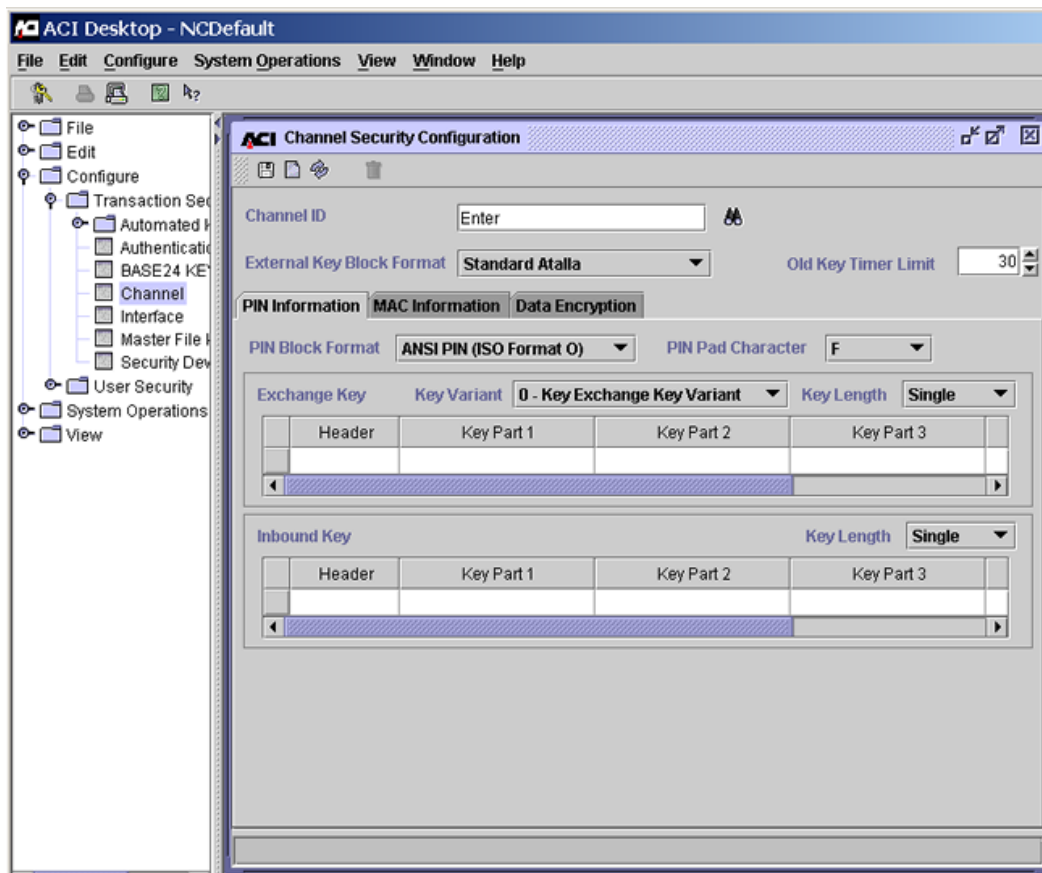
1. Log on to the BASE24 user access system and add the new terminal record on the BASE24-atm Terminal Data file (ATD) or BASE24-pos Terminal Data file (PTD) screens. Set the PIN encryption type and the MAC type for the record to perform both PIN encryption and message authentication in a security device (i.e., in hardware rather than software).

When you add an ATD record, the MASTER KEY field on device-specific ATD screen 9 is automatically set to the terminal ID, which is used as the key locator value (channel ID) for CSECD records.

When you add a PTD record, the PIN MASTER KEY and MAC MASTER KEY fields on PTD screen 7 are automatically set to the terminal ID, which is used as the key locator value (channel ID) for CSECD records.

2. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.

3. Display the Channel Security Configuration window by accessing **Configure > Transaction Security > Channel**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



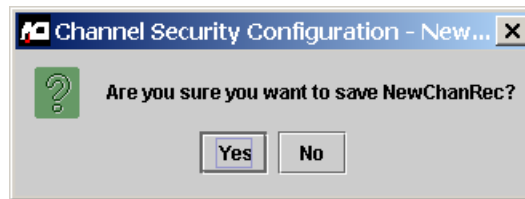
4. Type the terminal ID from step 1 in the Channel ID field. If you press the **Enter** key at this point, a message similar to the following is displayed at the bottom of the window. This message indicates that the record does not exist.

0 - 09:13 AM > Unable to retrieve NewChanRec. Channel not found. Enter data and Save to insert.

5. Enter or change data in the appropriate fields for your security requirements. Refer to the “Configure Security Information for a Channel” procedure provided in the ACI Online Help for the Channel Security Configuration window for details.



6. Click Save (  ) and answer Yes to the “Are you sure you want to save” dialog box.



7. The following message is displayed at the bottom of the window to indicate that the channel ID record was successfully saved.



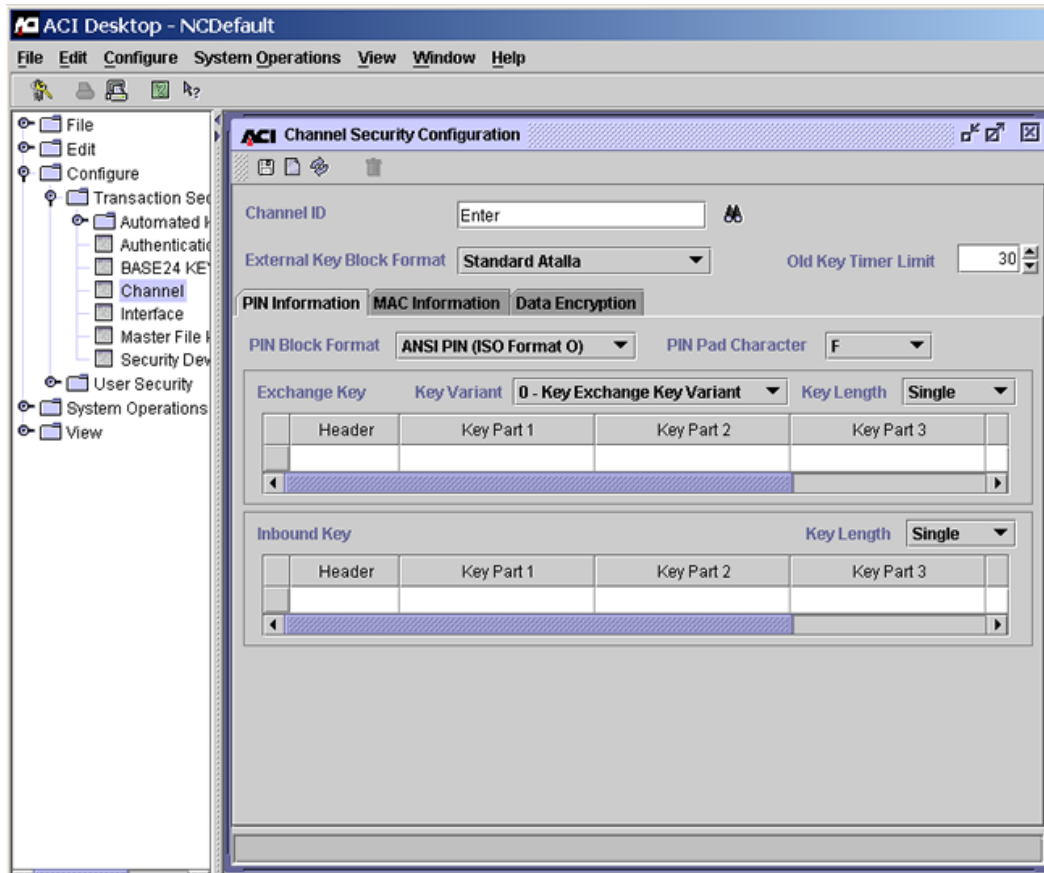
**Note:** Because CSECD records are read dynamically from disk into an internal memory table as they are needed in online transaction processing, you do not need to warmboot the Transaction Security Services process when adding a new CSECD record.

## Modifying an Existing Channel ID Record

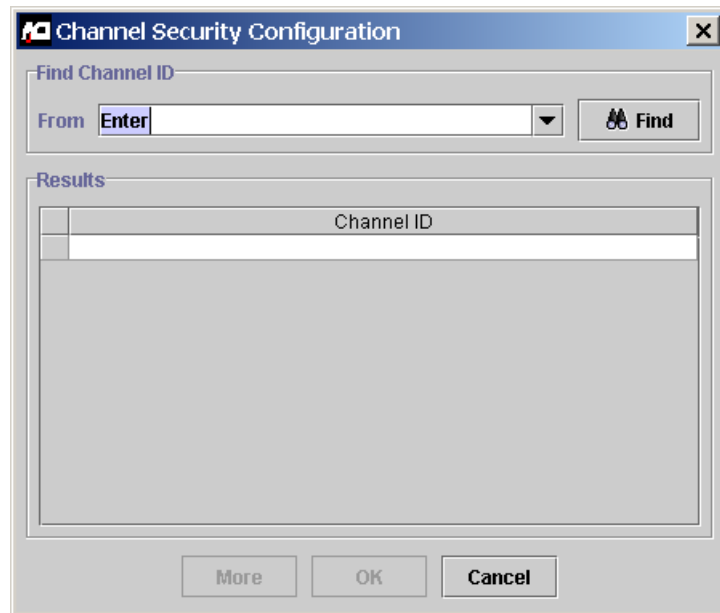
Use the following procedure when modifying an existing channel ID record for an ATM terminal or POS device in your BASE24 system.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.

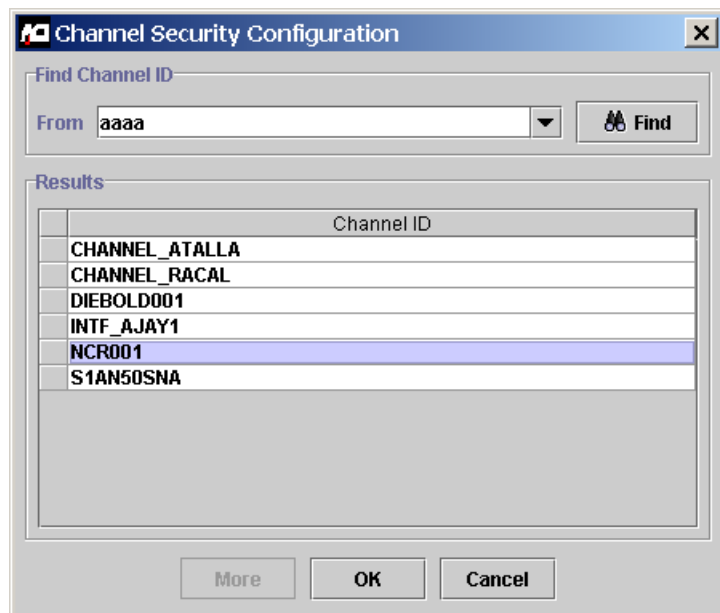
2. Display the Channel Security Configuration window by accessing **Configure > Transaction Security > Channel**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



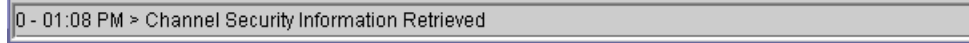
3. Type the channel ID in the Channel ID field and press the **Enter** key. If you do not know the channel ID value, click Find (  ). If you click Find, the following dialog box is displayed.



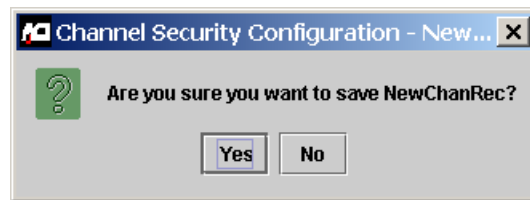
To find the channel ID, enter a partial value less than what you think the channel ID value might be in the From field and click the Find button (  ). All configured channel IDs equal to or greater than the partial value you entered are displayed in the Results area. For searching, characters are ordered from 0–9 and A–Z, where 0 is the lowest and Z is the highest.



Highlight the record you want to retrieve and click OK. The record is retrieved and displayed on the Channel Security Configuration window. A message similar to the following is displayed at the bottom of the window. This message indicates that the record was retrieved.



4. Change data as needed in the appropriate fields for your security requirements. Refer to the “Configure Security Information for a Channel” procedure provided in ACI Online Help for the Channel Security Configuration window for details.
5. Click Save (  ) and answer Yes to the “Are you sure you want to save” dialog box.



6. The following message is displayed at the bottom of the window to indicate that the channel ID record was successfully saved.



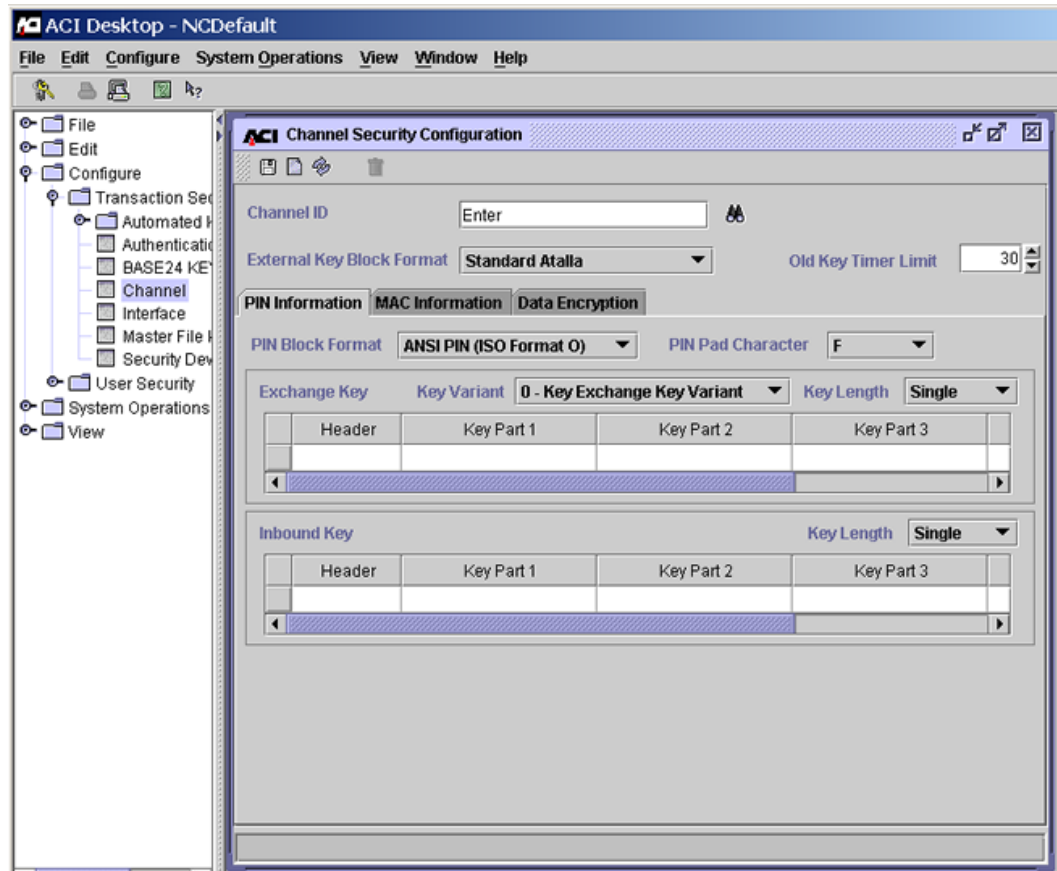
7. Warmboot the Transaction Security Services process to begin using the changed data in processing. Refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section.
8. Using the Device Control Terminal (DCT) or an XPNET network control facility, download the updated key information to the ATM terminal or POS device.

## Deleting a Channel ID Record

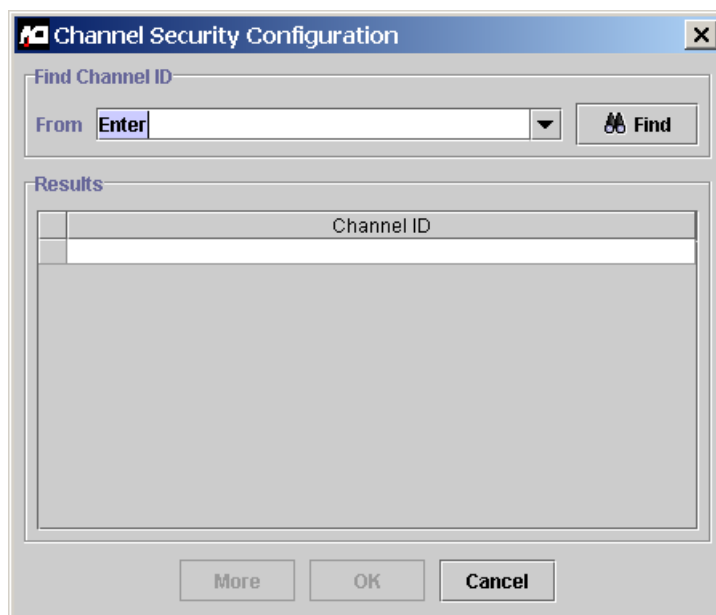
Use the following procedure when deleting a channel ID record for an ATM terminal or POS device in your BASE24 system.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.

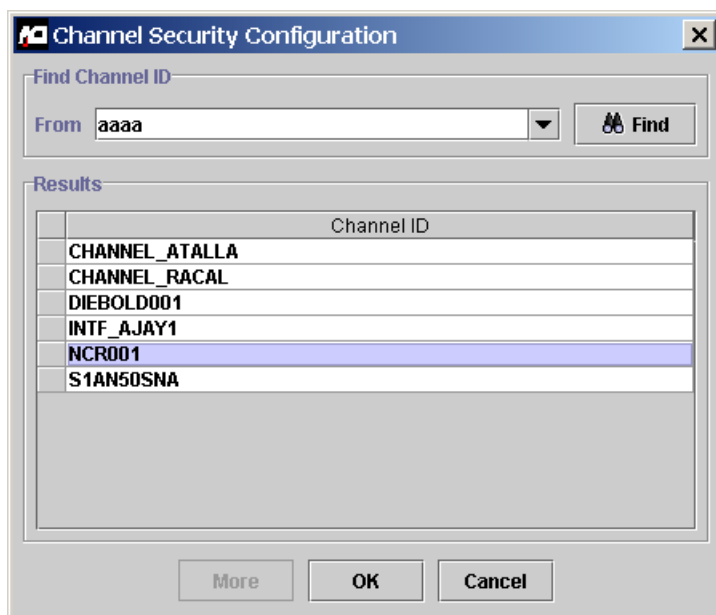
2. Display the Channel Security Configuration window by accessing **Configure > Transaction Security > Channel**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



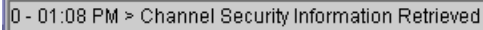
3. Type the channel ID in the Channel ID field and press the **Enter** key. If you do not know the channel ID value, click Find (  ). If you click Find, the following dialog box is displayed.



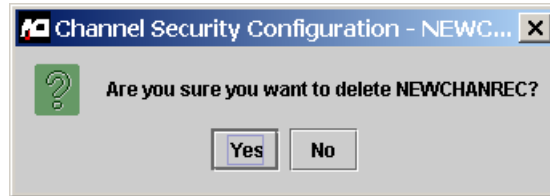
To find the channel ID, enter a partial value less than what you think the channel ID value might be in the From field and click the Find button (  ). All configured channel IDs equal to or greater than the partial value you entered are displayed in the Results area. For searching, characters are ordered from 0–9 and A–Z, where 0 is the lowest and Z is the highest.



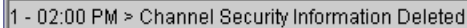
Highlight the record you want to delete and click OK. The record is retrieved and displayed on the Channel Security Configuration window. A message similar to the following is displayed at the bottom of the window. This message indicates that the record was retrieved.

A horizontal status bar with a light gray background and a thin border. It contains the text "0 - 01:08 PM > Channel Security Information Retrieved" in a small, black, sans-serif font.

4. Click Delete (  ) and answer Yes to the “Are you sure you want to delete” dialog box.



5. The following message is displayed at the bottom of the window to indicate that the channel ID record was successfully deleted.

A horizontal status bar with a light gray background and a thin border. It contains the text "1 - 02:00 PM > Channel Security Information Deleted" in a small, black, sans-serif font.

6. If you are deleting the terminal data completely from the BASE24 system, log on to the BASE24 user access system and delete the ATD or PTD record for the terminal associated with the channel ID record.

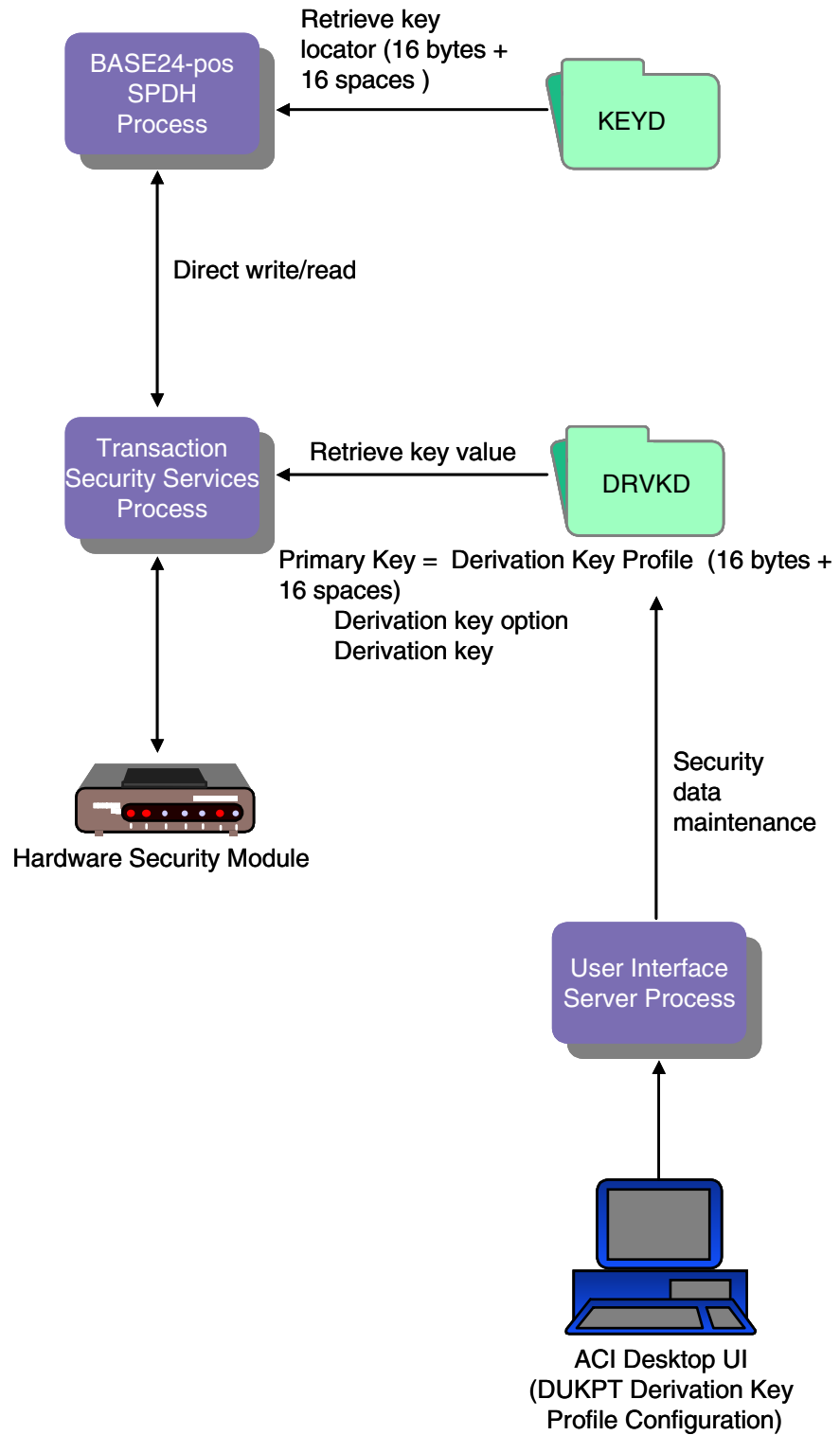
## Derivation Key File (KEYD)

When the Derivation Key File (KEYD) is converted, the DERIVATION KEY field on KEYD screen 1 is set to a key locator value. This key locator value consists of the first 16 bytes of the original derivation key value appended with 16 bytes of spaces. This key locator value is used as the derivation key profile, which is the primary key to the Derivation Key data source (DRVKD) used by the Transaction Security Services process.

If you change the key locator value in the DERIVATION KEY field of a converted record to a value other than the first 16 bytes of the original derivation key value appended with 16 bytes of spaces, you must add a corresponding record for this key locator value on the DUKPT Derivation Key Profile Configuration window.

The following diagram illustrates the relationship between the Derivation Key File (KEYD) and the Derivation Key File (DRVKD) after file conversion. Data in the DRVKD is maintained from the DUKPT Derivation Key Profile Configuration window. The key locator value entered in the DERIVATION KEY field on KEYD screen 1 must match the value entered in the Derivation Key Profile field on the DUKPT Derivation Key Profile Configuration window. If you add additional records to the KEYD after it is converted, you enter the derivation keys into corresponding DRVKD records from the DUKPT Derivation Key Profile Configuration window on the ACI desktop user interface.





## DUKPT Derivation Key Profile Configuration window

The DUKPT Derivation Key Profile Configuration window (access **Configure > Transaction Security > DUKPT Derivation Key Profile**) enables you to configure derivation keys and parameters used by the BASE24-pos Standard POS Device Handler (SPDH) module and the BASE24-pos Hypercom Device Handler (HPDH) module for derived unique key per transaction (DUKPT) PIN block translation and DUKPT message authentication.

The DUKPT Derivation Key Profile Configuration window maintains data in the Derivation Key data source (DRVKD). The Derivation Key Profile specifies the primary key to DRVKD records and must match a key locator value in the KEYD. The Derivation Key Option field specifies whether the security device is to use the single key derivation algorithm or the double key derivation algorithm with this derivation key. This option does not specify the length of the derivation key. The length of the derivation key is specified in the Key Length field. For the single key derivation algorithm, the derivation key can be a single or double length key. For the double key derivation algorithm, the derivation key must be a double length key. The Derivation Key table contains the encrypted derivation key. Refer to the ACI Online Help provided with the ACI desktop for detailed descriptions of this window and its fields.

**Note:** You must configure a security device using the Security Device Configuration window before you can configure derivation key information using the DUKPT Derivation Key Profile Configuration window.

An example of the DUKPT Derivation Key Profile Configuration window is shown below.

ACI DUKPT Derivation Key Profile Configuration - DLSTEST1

Derivation Key Profile: DLSTEST1

Derivation Key Option: Single Key Length: Double

Derivation Key

| Key Part 1          | Key Part 2          | Key Part 3 | Check Digits |
|---------------------|---------------------|------------|--------------|
| 3D46 2F8A E268 B0EB | 892A 753F CFC3 7670 |            |              |

5 - 09:00 AM > Successful view of DLSTEST1

## Key File (KEYF)

The following diagram illustrates the relationship between the BASE24 Key File (KEYF) and the Interface Security File (ISECD) after file conversion. The hardware encrypted key fields in the KEYF are set to the value of the two-character logical network identifier (specified by the LN-IND LCONF param if you are partitioning the ISECD) added to the beginning of an index value generated by the conversion program, and the actual key values are stored in the ISECD. The only exception is the hardware encrypted intermediate key (refer to the note below). Refer to [section 3](#) for more information on the LN-IND param.

Data in the ISECD is maintained from the Interface Security Configuration window. Also after file conversion, the remaining data in the KEYF, such as the PIN and MAC encrypt types, PIN block format, all dynamic key information, and IMNI interface-specific information, is maintained from the BASE24 KEYF Configuration window on the ACI desktop user interface. This allows all security data to be maintained from a single user interface. The Key Locator field on the BASE24 KEYF Configuration window is used to access records in the ISECD on the Interface Security Configuration window. The value in the Interface Name field on the Interface Security Configuration window must match the value in the Key Locator field on the BASE24 KEYF Configuration window. You can have multiple records in the KEYF with the same key locator value as long as the external interface or interchange treats all the links to the BASE24 system as a single logical connection and shares the same keys among all the links.

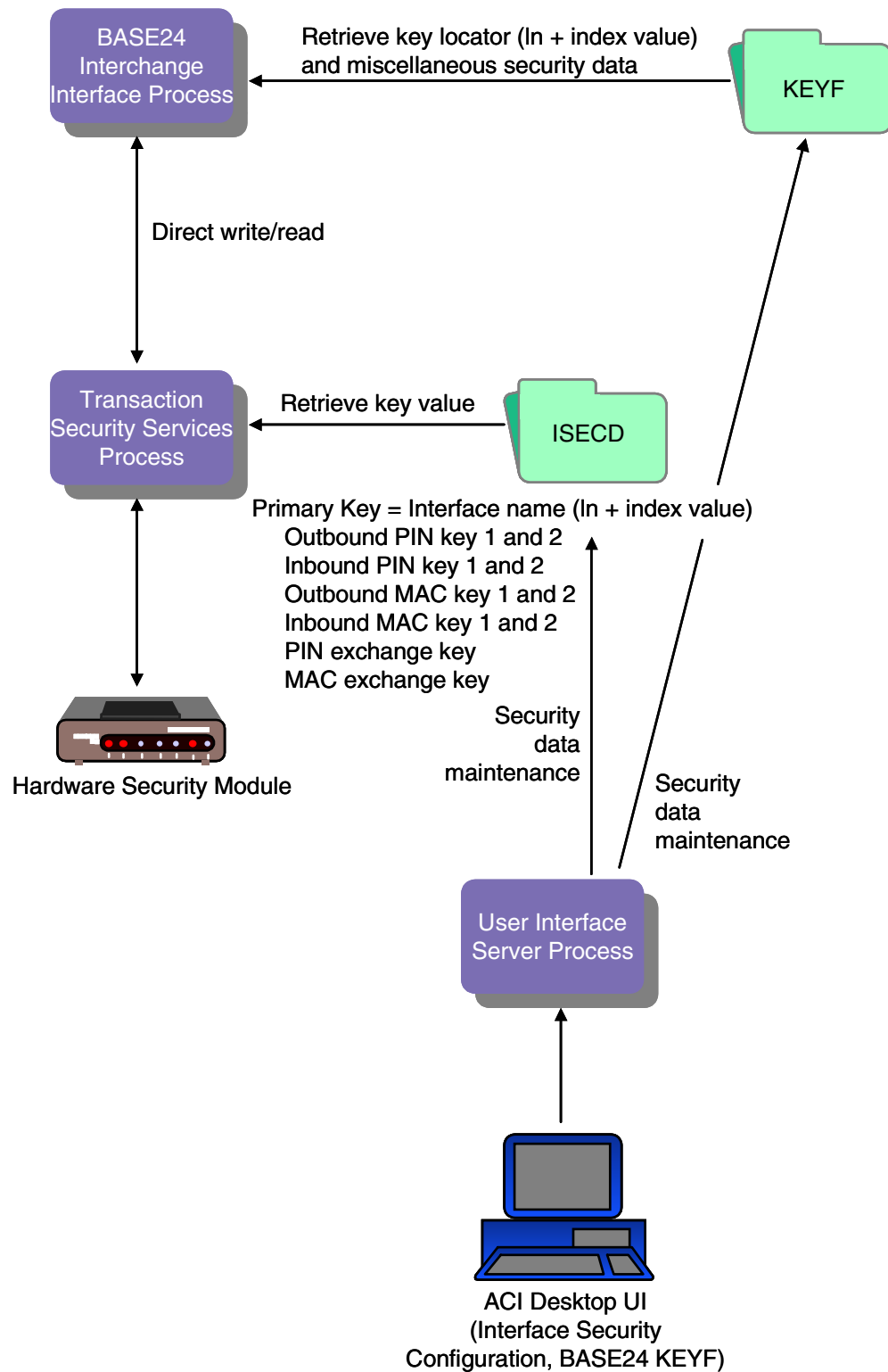
If you need to add KEYF and ISECD records after conversion, you must add a new KEYF record from the BASE24 KEYF Configuration window before you can add the new ISECD record from the Interface Security Configuration window.

**Note:** If an interchange encrypts PINs in hardware and the BASE24 system processes PINs internally in the clear, or if an interchange uses clear PINs while the BASE24 system uses PINs encrypted in software, an intermediate key must be configured in the KEYF using the Pathway-based Key File (KEYF) screens. The ACI desktop user interface does not grant access to intermediate keys.

If an acquiring host or Interchange Interface process is configured to send PINs in the clear to any interchange in the BASE24 system that supports encrypted PINs, an ISECD record must be manually added for the encrypted intermediate key with a PIN block type of PIN/PAD.

If an acquiring host or Interchange Interface process encrypts PINs and an interchange interface supports clear PINs, an ISECD record must be manually added for the encrypted intermediate key with a PIN block type of ANSI.

If you are using the MULT-LN-PIN-KEY LCONF param, an ISECD record must be manually added for the common PIN encryption key. Refer to [section 3](#) for more information on the MULT-LN-PIN-KEY param.



## BASE24 KEYF Configuration Window

The BASE24 KEYF Configuration window on the ACI desktop user interface enables you to maintain PIN and MAC encrypt types, PIN block format, and all dynamic key information stored in the BASE24 Key File (KEYF). Fields are displayed on five tabs on this window: PIN Information, MAC Information, Dynamic Key Limits, Dynamic Key Management, and Interac Key Management. Fields displayed on the Interac Key Management tab are specific to IMNI interfaces. Samples for each one of these tabs on the BASE24 KEYF Configuration window are shown below. Refer to the ACI Online Help provided with the ACI desktop for detailed descriptions of this window and its fields.

If your TSS environment is implemented into a BASE24 system with multiple logical networks, you can select the logical network for the KEYF you want to access by entering the logical network identifier in the Logical Net field. A BASE24\_KEYF assign must be defined in the Metadata Configuration File (CONFCSV) for each logical network in your BASE24 system. The format of these assigns is as follows:

**"BASE24\_KEYF $\textit{logical-net}$ "**

where:

*logical-net* is 1 to 4 alphanumeric characters identifying a logical network. For consistency, ACI recommends that this logical network identifier match the value of the LOGICAL-NET param in the Logical Network Configuration File (LCONF) for each logical network.

In addition, the CONFCSV must contain one BASE24\_KEYF assign without the logical network identifier appended to it. This assign defines the default KEYF accessed if you do not specify a logical network in the Logical Net field.

Sample entries in the CONFCSV for the KEYF are shown below. In these entries, the default KEYF accessed is for the PRO1 logical network. Users could access the KEYF for the PRO1, PRO2, and PRO3 logical networks in the Logical Net field.

---

```
"BASE24_KEYF", "BASE24_KEYF", "1.0", "$DATA.PRO1DATA.KEYF", "Enscribe", "", ""
"BASE24_KEYFPRO1", "BASE24_KEYF", "1.0", "$DATA.PRO1DATA.KEYF", "Enscribe", "", ""
"BASE24_KEYFPRO2", "BASE24_KEYF", "1.0", "$DATA.PRO2DATA.KEYF", "Enscribe", "", ""
"BASE24_KEYFPRO3", "BASE24_KEYF", "1.0", "$DATA1.PRO3DATA.KEYF", "Enscribe", "", ""
```

---

If you attempt to access a KEYF that is not properly defined in the CONFCSV, a data not found error is displayed.

**Note:** Any time you change the CONFCSV, you must stop and restart the process that uses the changed data. To ensure that no transactions currently in progress are lost, you should stop the process in an orderly fashion. The change does not take effect until after the process is stopped and restarted.

## PIN Information Tab

An example of the PIN Information tab on the BASE24 KEYF Configuration window is shown below.

The screenshot displays the 'BASE24 KEYF Configuration' window. The 'Keys Section' at the top includes fields for 'Logical Net', 'Interface Process Name', 'Key Locator', and 'DPC/IIID'. Below this, the 'PIN Information' tab is selected, showing various configuration options for PINs, including 'Encrypt Type', 'BASE24 Encrypt Type', 'PIN Block Format', 'ANSI PAN Format', 'Keys Supported', 'Notarization', 'Key Length', and 'PIN Pad Character'.

| Keys Section           |                      |          |              |
|------------------------|----------------------|----------|--------------|
| Logical Net            | <input type="text"/> | DPC/IIID | Select/Enter |
| Interface Process Name | Select/Enter         |          |              |
| Key Locator            |                      |          |              |

| PIN Information     |                          |                   |               |
|---------------------|--------------------------|-------------------|---------------|
| Encrypt Type        | Security Module          | Keys Supported    | Combined Keys |
| BASE24 Encrypt Type | Security Module          | Notarization      | Not Supported |
| PIN Block Format    | ANSI PIN (ISO Format 0)  | Key Length        | Single        |
| ANSI PAN Format     | 12 Right, no Check Digit | PIN Pad Character | F             |

## MAC Information Tab

An example of the MAC Information tab on the BASE24 KEYF Configuration window is shown below.

The screenshot shows the 'BASE24 KEYF Configuration' window. The 'Keys Section' is visible at the top, containing fields for 'Logical Net', 'Interface Process Name', and 'Key Locator'. Below this, the 'MAC Information' tab is selected, showing options for 'MAC Encrypt Type' (set to 'No MAC Processing'), 'Fields included in MAC' (set to 'Selected Fields'), and 'MAC Data Character Set' (set to 'ASCII'). Other tabs visible include 'PIN Information', 'Dynamic Key Limits', 'Dynamic Key Management', and 'Interac Key Management'.

**BASE24 KEYF Configuration**

**Keys Section**

Logical Net:  DPC/FID:

Interface Process Name:

Key Locator:

**PIN Information** **MAC Information** **Dynamic Key Limits** **Dynamic Key Management** **Interac Key Management**

MAC Encrypt Type:

Fields included in MAC:

MAC Data Character Set:



## Dynamic Key Limits Tab

An example of the Dynamic Key Limits tab on the BASE24 KEYF Configuration window is shown below.

The screenshot shows the 'BASE24 KEYF Configuration' window with the 'Dynamic Key Limits' tab selected. The window has a title bar with the ACI logo and standard window controls. Below the title bar is a toolbar with icons for file operations. The main content area is divided into several sections. At the top is the 'Keys Section' with fields for 'Logical Net', 'Interface Process Name', 'Key Locator', and 'DPC/FID'. Below this is a tabbed interface with five tabs: 'PIN Information', 'MAC Information', 'Dynamic Key Limits' (selected), 'Dynamic Key Management', and 'Interac Key Management'. The 'Dynamic Key Limits' tab contains two sub-sections: 'PIN Key' and 'MAC Key'. Each sub-section has five fields with spinners and a 'Timer Interval' dropdown set to 'Minutes'. The fields are: 'Transaction Limit', 'Translation Error Limit', 'Consecutive Error Limit', 'Timer Limit', and 'KMAC Synchronization Error Limit'. All limit fields are currently set to 0.

| Section                          | Field                            | Value   |
|----------------------------------|----------------------------------|---------|
| PIN Key                          | Transaction Limit                | 0       |
|                                  | Translation Error Limit          | 0       |
|                                  | Consecutive Error Limit          | 0       |
|                                  | Timer Limit                      | 0       |
|                                  | Timer Interval                   | Minutes |
| MAC Key                          | Transaction Limit                | 0       |
|                                  | Transaction Error Limit          | 0       |
|                                  | Consecutive Error Limit          | 0       |
|                                  | Timer Limit                      | 0       |
|                                  | Timer Interval                   | Minutes |
| KMAC Synchronization Error Limit | KMAC Synchronization Error Limit | 0       |

## Dynamic Key Management Tab

An example of the Dynamic Key Management tab on the BASE24 KEYF Configuration window is shown below.

The screenshot shows the 'BASE24 KEYF Configuration' window. The title bar includes the ACI logo and standard window controls. Below the title bar is a toolbar with icons for file operations. The main content area is divided into two sections. The top section, titled 'Keys Section', contains fields for 'Logical Net', 'Interface Process Name', 'Key Locator', and 'DPC/IIID'. The bottom section contains tabs for 'PIN Information', 'MAC Information', 'Dynamic Key Limits', 'Dynamic Key Management' (which is selected), and 'Interac Key Management'. The 'Dynamic Key Management' tab displays fields for 'Exchange Receiver', 'Exchange Originator', and 'Management Type'.

| Keys Section           |                      |
|------------------------|----------------------|
| Logical Net            | <input type="text"/> |
| Interface Process Name | Select/Enter ▼       |
| Key Locator            | <input type="text"/> |
| DPC/IIID               | Select/Enter ▼       |

| Dynamic Key Management |                      |
|------------------------|----------------------|
| Exchange Receiver      | <input type="text"/> |
| Exchange Originator    | <input type="text"/> |
| Management Type        | None ▼               |

## Interac Key Management Tab

The Interac Key Management tab displays exchange key sequence number fields for inbound and outbound keys, fields indicating the index (i.e., 1 or 2) of the inbound and outbound PIN, MAC, and data encryption keys to use for the interface, and a field for entering a new key exchange key. All of the fields on this tab are specific to the Inter Member Network Interchange (IMNI) Interface. Financial institution members of the Interac association in Canada use this interface to route transactions to and from other Interac association members. This tab is enabled only if the Security Device Type field on the Security Device Configuration window is set to Atalla and only applies to Interac association members.

### Exchange Key Sequence Numbers

Two exchange key sequence number counters are maintained in the KEYF—one for the inbound (import) key exchange keys and one for the outbound (export) key exchange keys. These counters are displayed in the Inbound KEK Counter and Outbound KEK Counter fields, respectively, on the IMNI tab. When the IMNI interface exchanges an inbound or outbound working key (i.e., PIN, MAC, or data) the associated counter value is increased by one. When an inbound working key is exchanged, the inbound key exchange key sequence number counter is increased by one. When an outbound working key is exchanged, the outbound key exchange key sequence number counter is increased by one. The valid values for each counter are 0–999999999999999. You can manually reset the values in either field by entering a new value in the field and pressing the Save icon (📁).

### Key Indexes

This tab displays an index field for each inbound and outbound working key stored in the KEYF. These fields indicate which of the two keys (1 or 2) stored for each inbound or outbound PIN, MAC, or data encryption key is currently in use for the interface. You can manually change the index value for a key by selecting one of the selectable list values in the associated key index field and pressing the Save icon (📁).

Transaction acquirers pass the index of a PIN, MAC, or data encryption key to be used in a transaction request instead of repeating an attempt with more than one key. BASE24 stores the specified indexes to be used for a transaction in the TSS Index token (BC) and passes this information in calls to the Transaction Security Services process.

## Key Exchange Key

From this tab you can also change the current key exchange keys in the Interface Security File (ISECD) by entering a new key exchange key value in the New Key Exchange Key field and pressing the Save icon ( ). If the new key exchange key is formatted correctly, all of the inbound and outbound key exchange keys for the interface in the ISECD (i.e., PIN key exchange keys, MAC key exchange keys, and data encryption key exchange keys) are changed to the specified key value. When this occurs, both the inbound and outbound key counters in the KEYF are reset to 0.

An example of the Interac Key Management tab on the BASE24 KEYF Configuration window is shown below.

The screenshot shows the 'BASE24 KEYF Configuration' window with the 'Interac Key Management' tab selected. The window has a title bar with the ACI logo and standard window controls. Below the title bar is a 'Keys Section' with fields for 'Logical Net', 'Interface Process Name', and 'Key Locator', along with a 'DPC/FID' dropdown. The main area contains five tabs: 'PIN Information', 'MAC Information', 'Dynamic Key Limits', 'Dynamic Key Management', and 'Interac Key Management'. The 'Interac Key Management' tab is active and displays a note: 'Note: This tab is to be used only for Interac Security setup.' Below the note are six input fields arranged in two columns: 'Inbound KEK Counter' and 'Outbound KEK Counter' (both set to 0), 'Inbound PIN Index' and 'Outbound PIN Index' (both set to 1), 'Inbound MAC Index' and 'Outbound MAC Index' (both set to 1), and 'Inbound KME Index' and 'Outbound KME Index' (both set to 1). Below these fields is a 'New Key Exchange Key' section with a 'Key Length' dropdown set to 'Single'. At the bottom is a table with four columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The table has one row with empty input fields for each column.

| Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|------------|------------|------------|--------------|
|            |            |            |              |

## Interface Security Configuration Window

The Interface Security Configuration window enables you to configure keys and parameters required by interfaces for PIN encryption, message authentication, and data encryption. Key exchange keys for PIN and MAC keys are entered on the Exchange Information tab. Key exchange keys for data encryption keys are entered on the Data Encr Exch Intf tab. You can use the same key exchange key (import key) for both decrypting keys received from the interface and sending keys to the interface, or you can use a separate key (export key) for sending keys to the interface. PIN, MAC, and data encryption working keys are entered on the PIN Information, MAC Information, and Data Encr Key Intf tabs, respectively. Static master PIN and MAC keys used to encrypt and decrypt transport (key exchange) and session (working) keys for dynamic online distribution are entered on the Master Keys tab. The Data Encr Exch Intf and Data Encr Key Intf tabs are currently used for the IMNI interface only. The Master Keys tab is currently used for the Interpay interface only.

Samples of the Exchange Information, PIN Information, MAC Information, Data Encr Exch Intf, Data Encr Key Intf, and Master Keys tabs of the Interface Security Configuration window are shown on the following pages. Refer to the ACI Online Help provided with the ACI desktop for detailed descriptions of this window and its fields.

**Note:** You must configure a security device using the Security Device Configuration window before you can configure interface security information using the Interface Security Configuration window.

## Old Key Timer Limit

The Old Key Timer Limit field located at the top of the Interface Security Configuration window specifies the maximum length of time, in seconds, that a PIN, MAC, or data encryption key can be used after a successful dynamic key exchange before it is automatically cleared.

This timer limits the amount of time that an old key can be used for PIN verification, message authentication, or data encryption after a successful key exchange has taken place.

If you set this field to a value of 0, the previous key is not deleted from the Interface Security File (ISECD). When a key exchange occurs, the current key becomes the previous key and the new key becomes the current key.

## Eracom ProtectHost White AMB HSMs

When using Eracom ProtectHost White AMB host security modules (HSMs) in your system, the fields displayed on the Interface Security Configuration window differ significantly from those shown and described in this section. For detailed information on configuring interface keys for Eracom ProtectHost White AMB HSMs, refer to [section 4](#).

## Memory Usage

Interface security key data is accessed from different sources during transaction processing, depending on how each interface is configured on the Interface Security Configuration window. The security key data can be obtained from an internal process memory collection and the ISECD, which is an Enscribe disk file, or from the Interface external memory table, which is a hash table designed to improve performance for interfaces that do not exchange keys.

The Transaction Security Services process always first attempts to access an interface record from the Interface external memory table. If the Option field on the Exchange Information tab of the Interface Security Configuration window for the record is set to a value of “No keys exchanged”, the security key data for the interface is obtained from the external memory table during processing. If the Option field is set to any other value, the interface exchanges keys. In this case, the Transaction Security Services process attempts to access the interface record from an internal process memory collection to obtain the security key data. If the key data is not present in the internal memory collection, the process reads the record from the ISECD disk file and loads it into the internal memory collection.

When you change values on the Interface Security Configuration window for an interface that does not exchange keys, you must rebuild the Interface external memory table and warmboot, or stop and restart, the Transaction Security Services process to reread the Interface external memory table. The new values will be used in processing for the next transaction for the interface that requires security key data.

Refer to the “[External Memory Tables](#)” and “[Internal Memory Tables](#)” topics provided later in this section for procedures on rebuilding external memory tables and warmbooting Transaction Security Services processes to reread external or internal memory tables.

## Exchange Information Tab

The Exchange Information tab enables you to configure key exchange key information for an interface.

The Keys Generated Online field specifies whether key exchange keys and working keys for the interface are generated online (dynamically). If the interface uses a three-tier key management hierarchy, such as for the Dutch Security Extensions, you must enter a check mark in this field and enter the appropriate static offline PIN and MAC master keys on the Master Keys tab. If you have not licensed the Dutch Security Extensions, this field is disabled.

When the Keys Generated Online field is checked, you can enter information in the Option, Key Variant, and Key Length fields for both PIN and MAC key exchange keys, but not in the Import or Export key fields, since the keys in these fields are generated dynamically. When the Keys Generated Online field is not checked or disabled, you can enter data in all fields on this tab.

To support the Dutch Security Extensions, set the Option fields for both PIN keys and MAC keys to “Use import key for both functions”, and the Key Length fields for both PIN keys and MAC keys to either Single (for single DES) or Double (for triple DES).

**Note:** When implementing the Dutch Security Extensions, the Key Length fields on this tab, the PIN Information tab, and the MAC Information tab must all be set to the same value (i.e., either Single or Double).

An example of the Exchange Information tab on the Interface Security Configuration window is shown below.

The screenshot shows the 'Interface Security Configuration - 955614B0EBFE4358' window. The 'Exchange Information' tab is selected. The 'Interface Name' is '955614B0EBFE4358'. The 'External Key Block Format' is 'Standard Atalla'. The 'Old Key Timer Limit' is '30'. The 'Keys Generated Online' checkbox is unchecked. The 'PIN Keys' section has 'Option' set to 'Use import k...', 'Key Variant' set to '0 - Key Excha...', and 'Key Length' set to 'Single'. Below this is a table with columns 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The 'Import' row is highlighted. The 'MAC Keys' section has the same settings as the 'PIN Keys' section. Below this is a similar table with columns 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The 'Import' row is highlighted. At the bottom, a status bar reads: '0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button'.

Interface Name: 955614B0EBFE4358

External Key Block Format: Standard Atalla

Old Key Timer Limit: 30

Exchange Information | PIN Information | MAC Information | Data Encr Exch Inf | Data Encr Key Inf | Master Keys

Keys Generated Online: ☐

PIN Keys

Option: Use import k... Key Variant: 0 - Key Excha... Key Length: Single

|        | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|--------|------------|------------|------------|--------------|
| Import |            |            |            |              |
| Export |            |            |            |              |

MAC Keys

Option: Use import k... Key Variant: 0 - Key Excha... Key Length: Single

|        | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|--------|------------|------------|------------|--------------|
| Import |            |            |            |              |
| Export |            |            |            |              |

0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button



## PIN Information Tab

The PIN Information tab enables you to enter outbound and inbound PIN encryption keys for an interface.

**Note:** When the Keys Generated Online field on the Exchange Information tab is checked, you can enter information in the Key Length and Current Index fields for both outbound and inbound PIN keys, but not in the Key Part or Check Digits fields since these keys are generated dynamically. To support the Dutch Security Extensions for an interface, set the Key Length fields for both outbound and inbound PIN keys to either Single (for single DES) or Double (for triple DES) and the Current Index fields for both outbound and inbound PIN keys to a value of 1 or 2. The Key Length fields on this tab, the Exchange Information tab, and the MAC Information tab must all be set to the same value (i.e., either Single or Double).

You can enter data in the PIN Block Format and PIN PAD Character fields, regardless of how the Keys Generated Online field is set.

The Current Index fields indicate which of two outbound PIN encryption keys or two inbound PIN encryption keys are currently used for an interface. Some interfaces, such as the IMNI Interface, can specify which key to use for a transaction, regardless of which key is identified as currently being used.

An example of the PIN Information tab on the Interface Security Configuration window is shown below.

The screenshot shows the 'ACI Interface Security Configuration - 955614B0EBFE4358' window. The 'PIN Information' tab is selected. The 'Interface Name' is '955614B0EBFE4358'. The 'External Key Block Format' is 'Standard Atalla'. The 'Old Key Timer Limit' is '30'. The 'PIN Block Format' is 'ANSI PIN (ISO Format...)'. The 'PIN PAD Character' is 'F'. The 'Outbound Keys' section shows 'Key Length' as 'Single', 'Current Index' as '1', and 'Key Counter' as '0'. Below this is a table with 4 columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The table has 2 rows. The first row is empty. The second row has a '2' in the first column. The 'Inbound Keys' section shows 'Key Length' as 'Single', 'Current Index' as '1', and 'Key Counter' as '0'. Below this is a table with 4 columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The table has 2 rows. The first row is empty. The second row has a '2' in the first column. At the bottom, a status bar reads: '0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button'.

ACI Interface Security Configuration - 955614B0EBFE4358

Interface Name: 955614B0EBFE4358

External Key Block Format: Standard Atalla

Old Key Timer Limit: 30

Exchange Information | **PIN Information** | MAC Information | Data Encr Exch Inf | Data Encr Key Inf | Master Keys

PIN Block Format: ANSI PIN (ISO Format...) | PIN PAD Character: F

Outbound Keys | Key Length: Single | Current Index: 1 | Key Counter: 0

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |

Inbound Keys | Key Length: Single | Current Index: 1 | Key Counter: 0

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |

0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button

## MAC Information Tab

The MAC Information tab enables you to enter outbound and inbound MAC encryption keys for an interface.

**Note:** When the Keys Generated Online field on the Exchange Information tab is checked, you can enter information in the Key Length and Current Index fields for both outbound and inbound MAC keys, but not in the Key Part or Check Digits fields since these keys are generated dynamically. To support the Dutch Security Extensions for an interface, set the Key Length fields for both outbound and inbound MAC keys to either Single (for single DES) or Double (for triple DES) and the Current Index fields for both outbound and inbound MAC keys to a value of 1 or 2. The Key Length fields on this tab, the Exchange Information tab, and the PIN Information tab must all be set to the same value (i.e., either Single or Double).

You can enter data in the MAC Option and MAC Length fields, regardless of how the Keys Generated Online field is set.

When viewing MAC keys on this window that were generated dynamically, the Key Length field must be set to the same length as the key that was dynamically generated. Otherwise, an “incorrect key size” error message is displayed.

The Current Index fields indicate which of two outbound MAC encryption keys or two inbound MAC encryptions keys are currently used for an interface. Some interfaces, such as the IMNI Interface, can specify which key to use for a transaction, regardless of which key is identified as currently being used.

An example of the MAC Information tab on the Interface Security Configuration window is shown below.

The screenshot shows the 'MAC Information' tab of the 'Interface Security Configuration' window for interface '955614B0EBFE4358'. The window includes tabs for 'Exchange Information', 'PIN Information', 'MAC Information', 'Data Encr Exch Inf', 'Data Encr Key Inf', and 'Master Keys'. The 'MAC Option' is set to 'Single length DES Cipher Block Chaining' and the 'MAC Length' is '32 bits'. The 'External Key Block Format' is 'Standard Atalla' and the 'Old Key Timer Limit' is '30'. The 'Outbound Keys' section shows a table with columns 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The 'Inbound Keys' section also shows a similar table. Both tables have a 'Current Index' of 1 and a 'Key Counter' of 0. The status bar at the bottom indicates the time is 08:46 AM and prompts the user to enter new security data and push the Save button.

ACI Interface Security Configuration - 955614B0EBFE4358

Interface Name: 955614B0EBFE4358

External Key Block Format: Standard Atalla Old Key Timer Limit: 30

Exchange Information PIN Information **MAC Information** Data Encr Exch Inf Data Encr Key Inf Master Keys

MAC Option: Single length DES Cipher Block Chaining MAC Length: 32 bits

Outbound Keys Key Length: Single Current Index: 1 Key Counter: 0

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |

Inbound Keys Key Length: Single Current Index: 1 Key Counter: 0

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |

0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button

## Data Encr Exch Intf Tab

An example of the Data Encr Exch Intf tab on the Interface Security Configuration window is shown below. This tab is only valid for Atalla security devices.

The screenshot shows the 'Data Encr Exch Intf' tab in the 'ACI Interface Security Configuration - 955614B0EBFE4358' window. The 'Interface Name' is '955614B0EBFE4358'. The 'External Key Block Format' is 'Standard Atalla' and the 'Old Key Timer Limit' is '30'. The 'Exchange Information' tab is selected, showing 'Exchange Keys' with 'Option' set to 'Use import...', 'Key Variant' set to '0 - Key Exc...', and 'Key Length' set to 'Single'. Below this is a table for key parts and check digits, with 'Import' and 'Export' buttons. The 'Initialization Vector' section contains a table with three columns: 'Key Part 1', 'Key Part 2', and 'Key Part 3'. A status bar at the bottom reads: '0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button'.

|        | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|--------|------------|------------|------------|--------------|
| Import |            |            |            |              |
| Export |            |            |            |              |

|  | Key Part 1 | Key Part 2 | Key Part 3 |
|--|------------|------------|------------|
|  |            |            |            |
|  |            |            |            |
|  |            |            |            |

0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button

## Data Encr Key Intf Tab

The Data Encr Key Intf tab enables you to enter outbound and inbound data encryption keys for an interface.

The Current Index fields indicate which of two outbound data encryption keys or two inbound data encryptions keys are currently used for an interface. Some interfaces, such as the IMNI Interface, can specify which key to use for a transaction, regardless of which key is identified as currently being used.

An example of the Data Encr Key Intf tab on the Interface Security Configuration window is shown below. This tab is only valid for Atalla security devices.

The screenshot shows the 'Data Encr Key Intf' tab in the 'Interface Security Configuration' window for interface '955614B0EBFE4358'. The window title is 'ACI Interface Security Configuration - 955614B0EBFE4358'. The 'Interface Name' is '955614B0EBFE4358'. The 'External Key Block Format' is 'Standard Atalla'. The 'Old Key Timer Limit' is '30'. The 'Data Encr Key Intf' tab is selected, along with 'Exchange Information', 'PIN Information', 'MAC Information', 'Data Encr Exch Intf', and 'Master Keys'. The 'DES Mode' is 'ECB', 'Data Type' is 'Binary', and 'Encryption Layers' is '1'. The 'Inbound Keys' section has 'Key Length' set to 'Single' and 'Current Index' set to '1'. It contains a table with 4 columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The first row is highlighted with a blue cursor. The 'Outbound Keys' section also has 'Key Length' set to 'Single' and 'Current Index' set to '1'. It contains a similar table with 4 columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. The first row is highlighted with a blue cursor. At the bottom, a status bar reads: '0 - 08:46 AM > Please enter new security data for Interface Security 955614B0EBFE4358 and push the Save button'.

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |
|   |            |            |            |              |

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |
|   |            |            |            |              |

## Master Keys Tab

The Master Keys tab is used to configure offline static PIN and MAC master keys used with random values to encrypt and decrypt transport (key exchange) and session (working) keys for dynamic online distribution. This tab is only available when you license the Dutch Security Extensions and use Thales e-Security HSMs. Otherwise, the tab is greyed out and cannot be accessed.

**Note:** The ACI desktop does not validate check digits entered on this window.

An example of the Master Keys tab on the Interface Security Configuration window is shown below.

The screenshot shows the 'ACI Interface Security Configuration' window with the 'Master Keys' tab selected. The window has a title bar with standard icons. Below the title bar, there is a section for 'Interface Name' with a dropdown menu set to 'Select'. To the right of this are icons for saving, loading, and deleting. Below this is a section for 'External Key Block Format' with a dropdown menu set to 'Variant method' and an 'Old Key Timer Limit' spinner set to 30. A row of tabs is visible: 'Exchange Information', 'PIN Information', 'MAC Information', 'Data Encr Exch Inf', 'Data Encr Key Inf', and 'Master Keys' (which is active). The 'Master Keys' tab contains two sections: 'PIN Key' and 'MAC Key'. Each section has a table with four columns: 'Key Part 1', 'Key Part 2', 'Key Part 3', and 'Check Digits'. Below each table is a large text area for additional information.

| Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|------------|------------|------------|--------------|
|            |            |            |              |

| Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|------------|------------|------------|--------------|
|            |            |            |              |

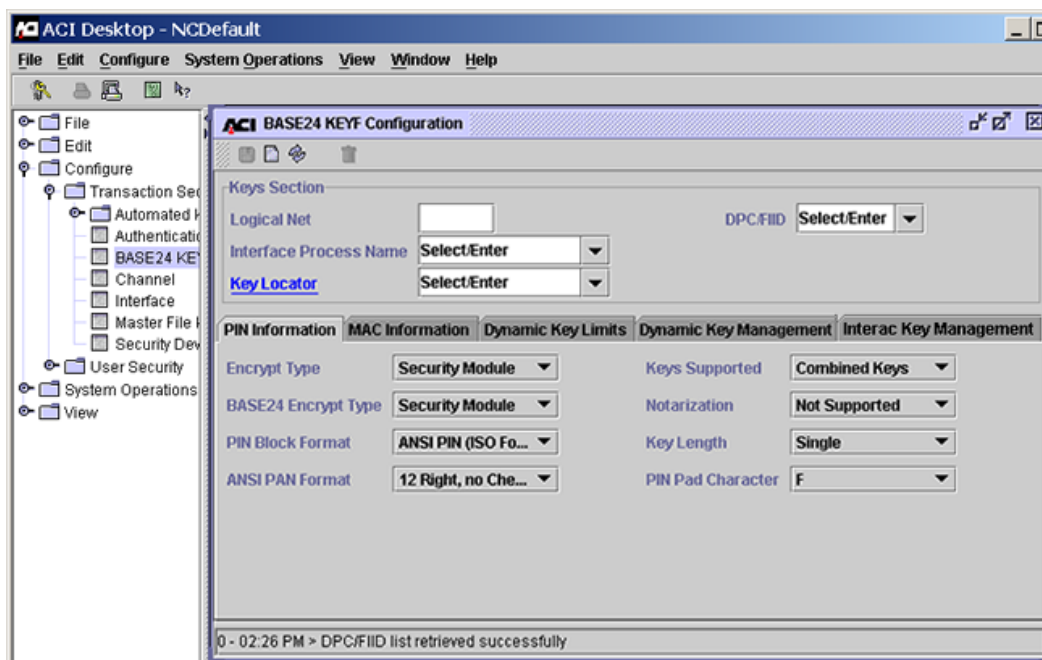
## Maintaining KEYF and Interface Security Data

The following paragraphs describe procedures for adding, modifying, and deleting interface records in the Key File (KEYF) and in the Interface Security File (ISECD).

### Adding New KEYF and ISECD Records

Use the following procedure to add a new KEYF and ISECD record when adding an interface to your BASE24 system.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the BASE24 KEYF Configuration window by accessing **Configure > Transaction Security > BASE24 KEYF Configuration**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Optionally enter the logical network of the KEYF you want to access in the Logical Net field.
  - If you do not specify a logical network value, the application uses the first entry that is defined for the BASE24\_KEYF assign in the Metadata Configuration File (CONFCSV).



- If you do specify a logical network value, it must match one of the BASE24\_KEYF assigns in the CONFCSV (the logical network value is appended to the BASE24\_KEYF assign). If you specify a logical network value that does not exist in a BASE24\_KEYF assign, the application will notify you that the data could not be found.
4. Place the cursor in the DPC/FIID field, type a new DPC or FIID value, and press the **Enter** key. The Interface Process Name field becomes active.

5. Click once in the Interface Process Name field, type the symbolic name of the new Interface process, and press the **Enter** key.

ACI BASE24 KEYF Configuration - P1A^HISO01

Keys Section

Logical Net  DPC/IIID 0001

Interface Process Name P1A^HISO01

Key Locator

PIN Information MAC Information Dynamic Key Limits Dynamic Key Management Interac Key Management

Encrypt Type No Encryption Keys Supported Separate Keys

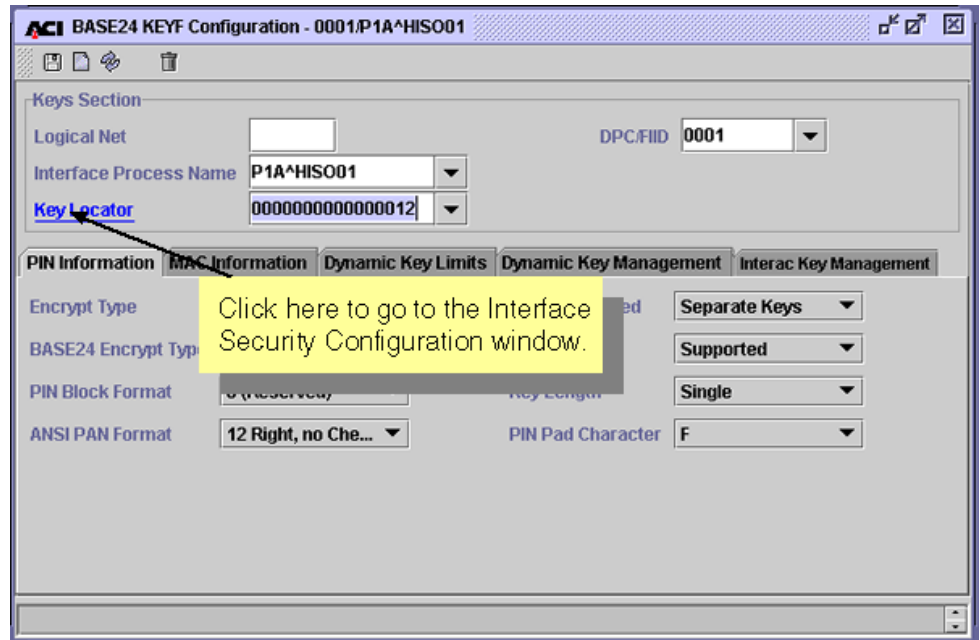
BASE24 Encrypt Type No Encryption Notarization Supported

PIN Block Format 0 (Reserved) Key Length Single

ANSI PAN Format 12 Right, no Check Digit PIN Pad Character F

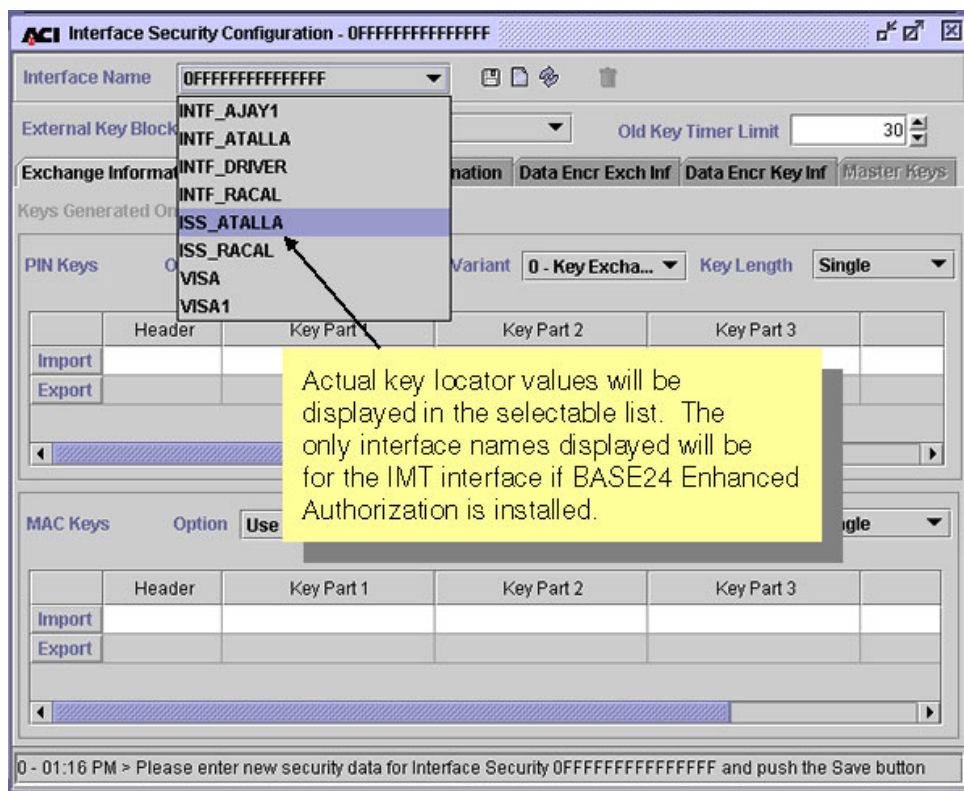
3 - 02:30 PM > Successful view of 0001 P1A^HISO01

6. Enter any hexadecimal value in the Key Locator field and click on the Key Locator field label hypertext link to display the Interface Security Configuration window. If the value you enter is less than 16 characters, the system pads the value with the hexadecimal F character.



7. Determine the next available key locator value. Click on the selectable list for the Interface Name field to find the last key locator value used for the logical network you are logged on to. If you have multiple logical networks in your system, the first two characters of the key locator value identifies the logical network. For example, if you are adding an interface to logical network 01 (i.e., the LN-IND param is set to 01) and the last key locator value used is 0100000000000009, the next key locator value would be 0100000000000010.

**Note:** If BASE24 Enhanced Authorization with the IMT Interface is installed on your system, the interface names associated with the IMT Interface will also be listed in the selectable list for the Interface Name field.



8. Exit the Interface Security Configuration window without saving changes and return to the BASE24 KEYF Configuration window.

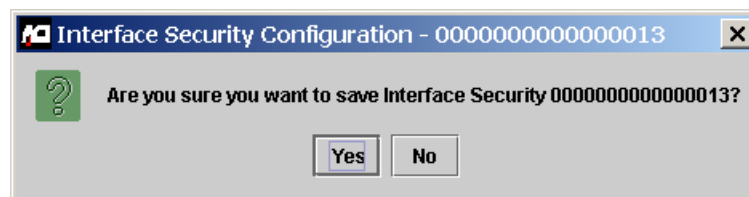
9. Click in the Key Locator field, enter the new key locator value determined from step 7, and press the **Enter** key.

The screenshot shows the 'ACI BASE24 KEYF Configuration - 0001/P1A^HIS001' window. The 'Keys Section' is active, displaying fields for 'Logical Net', 'Interface Process Name' (P1A^HIS001), 'Key Locator' (0000000000000013), and 'DPC/FID' (0001). Below this, the 'PIN Information' tab is selected, showing settings for 'Encrypt Type' (No Encryption), 'BASE24 Encrypt Type' (No Encryption), 'PIN Block Format' (0 (Reserved)), 'ANSI PAN Format' (12 Right, no Che...), 'Keys Supported' (Separate Keys), 'Notarization' (Supported), 'Key Length' (Single), and 'PIN Pad Character' (F). The 'Dynamic Key Limits', 'Dynamic Key Management', and 'Interac Key Management' tabs are also visible.

10. Click on the Key Locator field label again. The new key locator value is displayed in the Interface Name field on the Interface Security Configuration window.

11. Enter all required and optional data as needed for the ISECD record and click Save ( ). Answer Yes to the “Are you sure you want to save” dialog box.

Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields for each tab on this window.



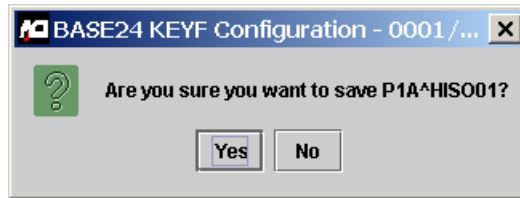
If the record is successfully added, a message similar to the following is displayed at the bottom of the window.

13 - 02:43 PM > Successful insert of Interface Security 0100000000000010 .

12. Exit the Interface Security Configuration window and return to the BASE24 KEYF Configuration window.

13. Enter all required and optional data as needed for the KEYF record and click Save ( ). Answer Yes to the “Are you sure you want to save” dialog box.

Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields for each tab on this window.



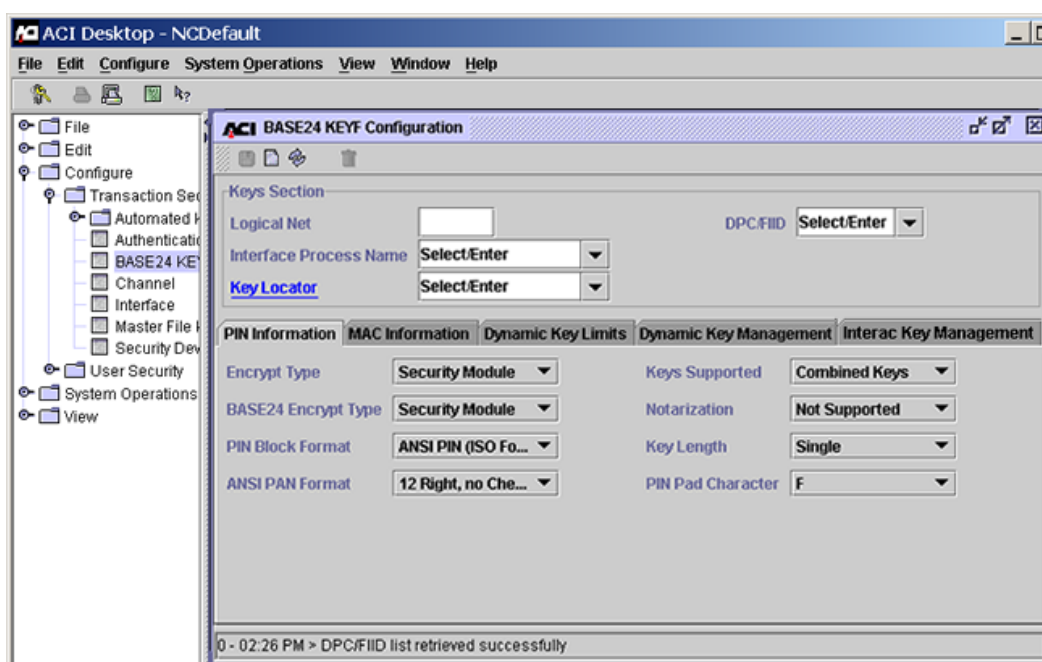
If the record is successfully added, a message indicating that “BASE24 KEYF Information Inserted” is displayed at the bottom of the window.

14. If the interface for which you added an ISECD record does not support dynamic key management (indicated by a value of “No keys exchanged” in the Options field on the Exchange Information tab of the Interface Security Configuration window), the ISECD record is accessed in an external memory table, which must be rebuilt. Otherwise, the ISECD record is accessed as an internal memory table, which does not need to be rebuilt. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for instructions on rebuilding external memory tables.
15. Warmboot the Transaction Security Services process to begin using the new data in processing. This step is required only if the ISECD is accessed as an external memory table. Refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for external memory tables.

## Modifying KEYF and ISECD Records

Use the following procedure to modify KEYF and ISECD records for an interface to your BASE24 system.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the BASE24 KEYF Configuration window by accessing **Configure > Transaction Security > BASE24 KEYF Configuration**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter the logical network ID (e.g., PRO1, PRO2, etc.) in the Logical Net field and click Refresh (🔄). The system determines whether the logical network is defined. If not, an error message is displayed.
4. Select the DPC of the host or FIID of the interchange from the selectable list for the DPC/FIID field.
5. Select the interface process symbolic name from the selectable list for the Interface Process Name field. Click once in the Interface Process Name field to refresh the window. The KEYF record to be modified should be displayed.
6. Update any fields as needed for the KEYF record.



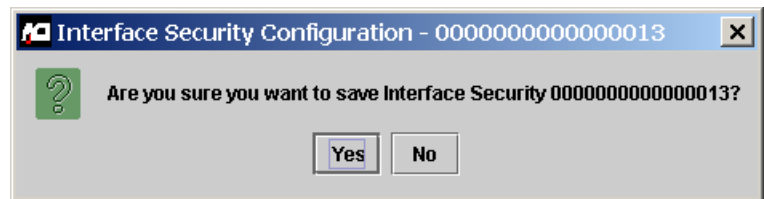
Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields and their values for each tab on this window.

7. If you need to update any key information for this KEYF record, click on the Key Locator field label. The Interface Security Configuration window is displayed with the related ISECD record.

If you need to update any key values, double-click in the appropriate field (e.g., Key Part 1). The key value is unformatted so that you can edit it. When you are finished editing the key value, press the **Enter** key to reformat the value.

Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields and their values for each tab on this window.

When you are finished updating the ISECD record, click Save (💾). Answer Yes to the “Are you sure you want to save” dialog box.



If the record is successfully updated, a message indicating that update was successful is displayed at the bottom of the window.

Exit the Interface Security Configuration window.

If you made any changes on the BASE24 KEYF Configuration window, click Save (💾). Answer Yes to the “Are you sure you want to save” dialog box.

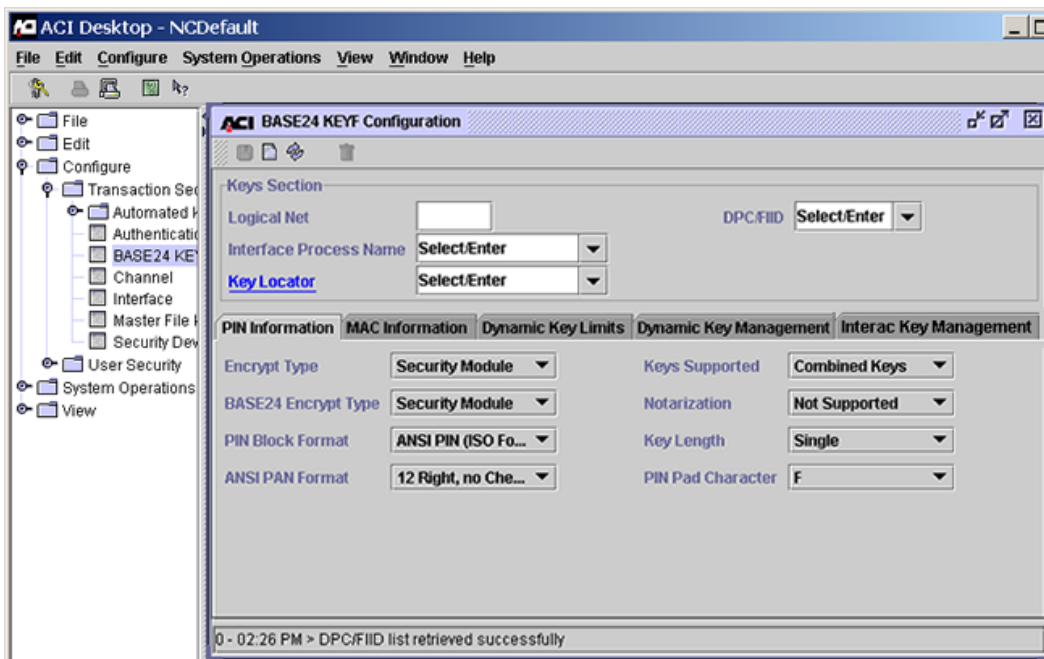
8. If you updated an ISECD record and the interface does not support dynamic key management (indicated by a value of “No keys exchanged” in the Options field on the Exchange Information tab of the Interface Security Configuration window), the ISECD record is accessed in an external memory table, which must be rebuilt. Otherwise, the ISECD record is accessed as an internal memory table, which does not need to be rebuilt. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for instructions on rebuilding external memory tables.
9. Warmboot the Transaction Security Services process to begin using the new data in processing. This is required whether the ISECD is accessed as an external memory table or an internal memory table.

- If the ISECD is accessed as an external memory table, refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for external memory tables.
- If the ISECD is accessed as an internal memory table, refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for internal memory tables.

## Deleting KEYF and ISECD Records

Use the following procedure to delete KEYF and ISECD records for an interface from your BASE24 system.

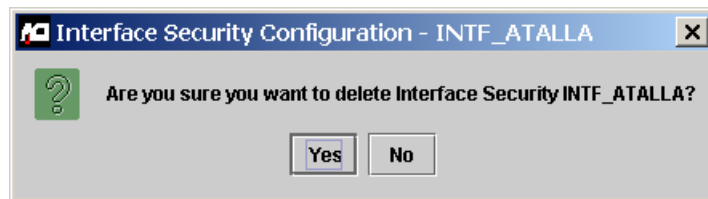
1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the BASE24 KEYF Configuration window by accessing **Configure > Transaction Security > BASE24 KEYF Configuration**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter the logical network ID (e.g., PRO1, PRO2, etc.) in the Logical Net field and click Refresh (🔄). The system determines whether the logical network is defined. If not, an error message is displayed.

4. Select the DPC of the host or FIID of the interchange from the selectable list for the DPC/FIID field.
5. Select the interface process symbolic name from the selectable list for the Interface Process Name field. Click once in the Interface Process Name field to refresh the window. The KEYF record to be deleted should be displayed.
6. If you need to delete any key information associated with this KEYF record, click on the Key Locator field label. The Interface Security Configuration window is displayed with the related ISECD record.

Click Delete (  ). Answer Yes to the “Are you sure you want to delete” dialog box.



If the record is successfully deleted, a message indicating that the delete was successful is displayed at the bottom of the window.

Exit the Interface Security Configuration window.

7. On the BASE24 KEYF Configuration window, click Delete (  ). Answer Yes to the “Are you sure you want to delete” dialog box.

If the record is successfully deleted, a message indicating that the delete was successful is displayed at the bottom of the window.

8. If you deleted an ISECD record and the interface does not support dynamic key management (indicated by a value of “No keys exchanged” in the Options field on the Exchange Information tab of the Interface Security Configuration window), the ISECD record is accessed in an external memory table, which must be rebuilt. Otherwise, the ISECD record is accessed as an internal memory table, which does not need to be rebuilt. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for instructions on rebuilding external memory tables.
9. Warmboot the Transaction Security Services process to begin using the new data in processing. This is required whether the ISECD is accessed as an external memory table or an internal memory table.
  - If the ISECD is accessed as an external memory table, refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for external memory tables.

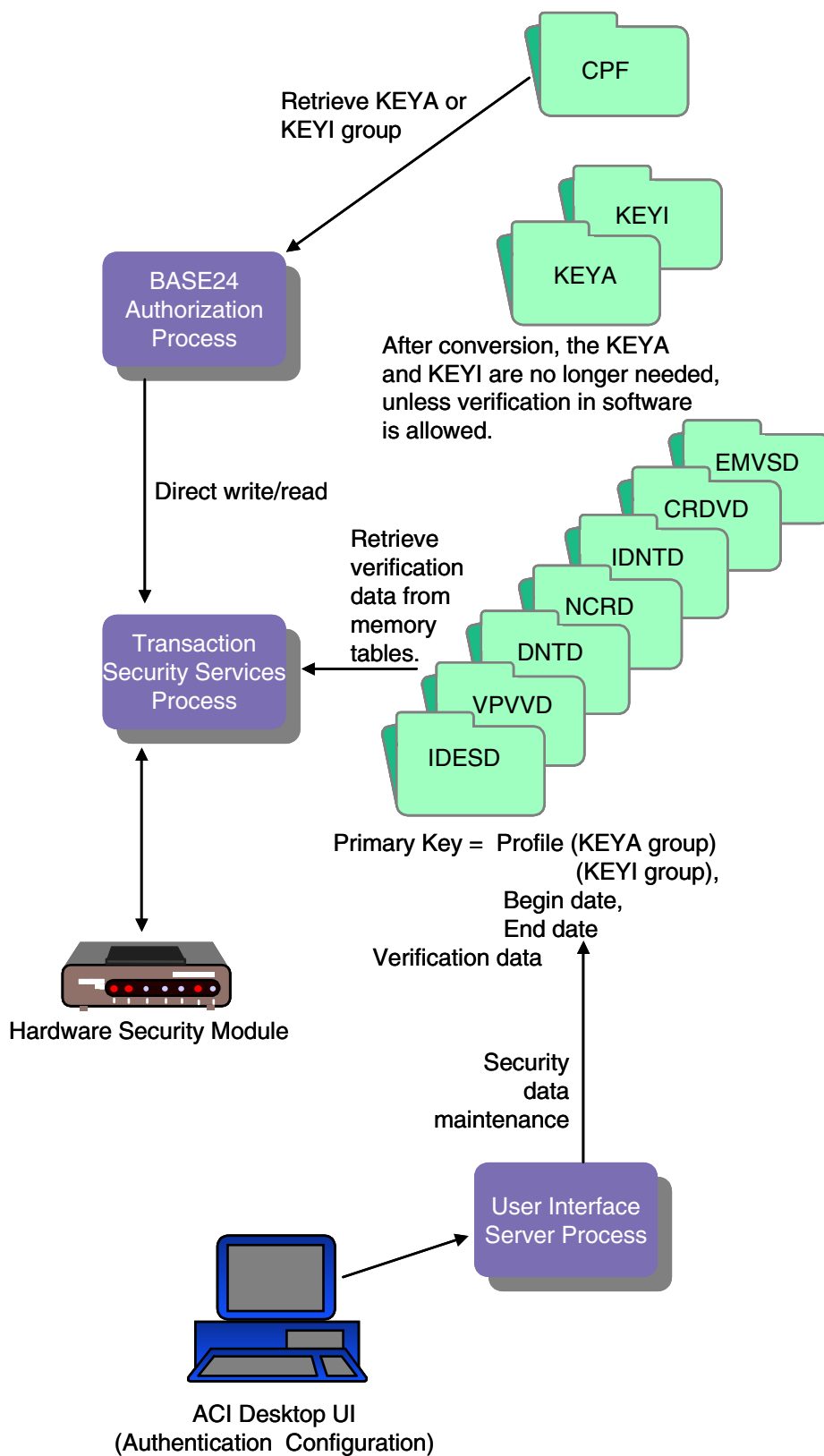
- If the ISECD is accessed as an internal memory table, refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for internal memory tables.

## Key Authorization Files

The following diagram illustrates the relationships between the BASE24 key authorization files—Key Authorization File (KEYA) and ICC Key File (KEYI) and the key authorization files maintained by the Transaction Security Services process after file conversion. The hardware encrypted key fields in the KEYA and KEYI are set to binary zeros and the actual key values are stored in the IBM DES Verification File (IDESD), Diebold Number Table File (DNTD), Identikay PIN Verification File (IDNTD), NCR PIN Verification File (NCRD), Visa PVV Verification File (VPVVD), EMV Authorization File (EMVSD), or Card Verification File (CRDVD). The same primary key values used in the converted KEYA and KEYI records are copied to the IDESD, DNTD, IDNTD, VPVVD, CRDVD, or EMVSD. Data in the IDESD, DNTD, IDNTD, VPVVD, CRDVD, and EMVSD is maintained from the Authentication Profile Configuration window. After the KEYA and KEYI have been converted, they are no longer accessed during transaction processing, unless PIN verification, card verification, or PIN generation in software is required and it is allowed. If PIN verification, card verification, or PIN generation in software is required and allowed, the Authorization process still accesses the KEYA and KEYI for key information.

The following table shows you the fields in the Card Prefix File (CPF) that are used to access the appropriate records in the TSS database. The value in these fields must match a value entered in the Profile field on the Authentication Profile Configuration window.

| CPF Field                                   | TSS File                        |
|---------------------------------------------|---------------------------------|
| PIN VERIFICATION KEYA GROUP (CPF screen 2)  | IDESD, DNTD, IDNTD, NCRD, VPVVD |
| CARD VERIFICATION KEYA GROUP (CPF screen 2) | CRDVD                           |
| MANUAL CARD VERIF KEYA GROUP (CPF screen 2) |                                 |
| SIV KEYA GROUP (CPF screen 6)               |                                 |
| KEYI GROUP (CPF screen 11)                  | EMVSD                           |



## Authentication Profile Configuration Window

The Authentication Profile Configuration window on the ACI desktop user interface enables you to maintain PIN, card, and EMV verification keys and parameters. This window supports the entry of keys and parameters for the following methods of PIN verification: IBM DES (3624-format), Visa PVV, Diebold, NCR, and Identikey. You can also enter card verification key pairs as well as keys and parameters required for processing EMV transactions. Fields are displayed on seven tabs on this window: DES (IBM 3624) PIN Verification, Visa PVV Keys, Diebold, NCR, Identikey, Card, and EMV. Samples for each one of these tabs on the Authentication Profile Configuration window are shown below. Refer to the ACI Online Help provided with the ACI desktop for detailed descriptions of this window and its fields.

**Note:** You must configure a security device using the Security Device Configuration window before you can configure authentication security information using the Authentication Profile Configuration window.

## Eracom ProtectHost White AMB HSMs

When you use Eracom ProtectHost White AMB host security modules (HSMs) in your system, the fields displayed on the Authentication Profile Configuration window differ significantly from those shown and described in this section. For detailed information on configuring keys for Eracom ProtectHost White AMB HSMs, refer to [section 4](#).

## DES (IBM 3624) PIN Verification Tab

The DES (IBM 3624) PIN Verification tab of the Authentication Profile Configuration window contains the input data used in IBM DES (3624-format) PIN verification. Data entered on this tab is stored in the IBM DES Verification File (IDESD).

**Note:** There has been a well publicized attack on customer PINs that can be facilitated by repeatedly altering the decimalization table and sending PIN verification commands to the HSM. To protect your cardholder's PINs from such an attack, you can store the decimalization table in the memory of the security device or encrypt the decimalization table value in the TSS database. The options available depend on the type of security devices installed in your system as described below.

- For Atalla NSPs, the decimalization table value sent in commands must be in the clear. You can store the clear decimalization table in the memory of the NSP using the SCT or SCA. In this case, you enter the index of this memory location instead of the clear decimalization table value in the Decimalization Table field.
- For Eracom host security modules (HSMs), the actual value of the decimalization table is stored in the memory of the Eracom HSM. Therefore, the Decimalization Table field is not used.
- For Thales e-Security HSMs, decimalization table value sent in commands can be encrypted or in the clear. When an HSM configure security parameter is set to support encrypted decimalization tables, you can encrypt the clear decimalization table value using a console command and enter the encrypted value in the Decimalization Table field. Both the clear decimalization table and the encrypted decimalization table consist of 16 hexadecimal characters. Note that you cannot mix clear and encrypted decimalization tables. Either all decimalization tables must be in the clear or must be encrypted. For more information on this configure security parameter, refer to [section 4](#).



An example of the DES (IBM 3624) PIN Verification tab on the Authentication Profile Configuration window is shown below.

The screenshot displays the 'Authentication Profile Configuration - ATALLA\_PROFILE' window. The 'Profile' dropdown is set to 'ATALLA\_PROFILE'. Below this, a table shows the 'Begin Date' as 'January 1971' and 'End Date' as 'December 2025'. The 'PIN Verification' section includes checkboxes for 'IBM DES' (checked), 'Visa PVV' (checked), 'Diebold' (unchecked), 'NCR' (unchecked), and 'Identikey' (unchecked). The 'Card Verification' section has checkboxes for 'Card' (unchecked) and 'EMV' (unchecked). The 'DES (IBM 3624) PIN Verification' tab is active, showing a 'Key Length' dropdown set to 'Single'. Below this, a table for key parts is shown:

| Key Part 1          | Key Part 2 | Key Part 3 | Check Digits |
|---------------------|------------|------------|--------------|
| 1111 1111 1111 1111 |            |            |              |

Below the key parts table, there are fields for 'Decimalization Table' (1234567890123456), 'PAN Verify Length' (5), 'PAN Pad Character' (F), and 'PAN Verify Offset' (1). The status bar at the bottom indicates: '3 - 10:30 AM > Successful view of Profile ATALLA\_PROFILE January 1971 - December 2025 .'

## Visa PVV Keys Tab

The Visa PVV Keys tab of the Authentication Profile Configuration window contains the PIN verification key pairs input data used in Visa PVV PIN verification. Data entered on this tab is stored in the Visa PVV Verification File (VPVVD).

An example of the Visa PVV Keys tab on the Authentication Profile Configuration window is shown below.

ACI Authentication Profile Configuration - ATALLA\_PROFILE

Profile: ATALLA\_PROFILE

|  | Begin Date   | End Date      | PIN Verification                    |                                     |                          |                          |                          | Card Verification        |                          |
|--|--------------|---------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|  |              |               | IBM DES                             | Visa PVV                            | Diebold                  | NCR                      | Identikey                | Card                     | EMV                      |
|  | January 1971 | December 2025 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

DES (IBM 3624) PIN Verification | **Visa PVV Keys** | Diebold | NCR | Identikey | Card | EMV

| Key Pair | Key Part 1          | Key Part 2 | Key Part 3 | Check Digits |
|----------|---------------------|------------|------------|--------------|
| 1        | 6FAF F58C A1F5 1E39 |            |            | 000000       |
|          | 6FAF F58C A1F5 1E39 |            |            | 000000       |
| 2        | 00FF FFFF FFFF FFFF |            |            | 000000       |
|          | 00FF FFFF FFFF FFFF |            |            | 000000       |
| 3        | 6FAF F58C A1F5 1E39 |            |            | 000000       |
|          | 6FAF F58C A1F5 1E39 |            |            | 000000       |
| 4        | 00AF F58C A1F5 1E39 |            |            | 000000       |
|          | 00AF F58C A1F5 1E39 |            |            | 000000       |
| 5        | 00AF F58C A1F5 1E39 |            |            | 000000       |
|          | 00FF FFFF FFFF FFFF |            |            | 000000       |
| 6        | 00FF FFFF FFFF FFFF |            |            | 000000       |
|          | 00FF FFFF FFFF FFFF |            |            | 000000       |

3 - 10:30 AM > Successful view of Profile ATALLA\_PROFILE January 1971 - December 2025 .

## Diebold Tab

The Diebold tab of the Authentication Profile Configuration window contains the Diebold number table (DNT) information used in Diebold PIN verification. Data entered on this tab is stored in the Diebold Number Table File (DNTD).

An example of the Diebold tab on the Authentication Profile Configuration window is shown below.

**ACI Authentication Profile Configuration - ATALLA\_PROFILE**

Profile: **ATALLA\_PROFILE**

| Begin Date   | End Date      | PIN Verification                    |                                     |                                     |                          |                          | Card Verification        |                          |
|--------------|---------------|-------------------------------------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|              |               | IBM DES                             | Visa PVV                            | Diebold                             | NCR                      | Identikey                | Card                     | EMV                      |
| January 1971 | December 2025 | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

DES (IBM 3624) PIN Verification | Visa PVV Keys | **Diebold** | NCR | Identikey | Card | EMV

Relative Location:  Algorithm Number:

**Diebold Number Table**

|   | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|---|------------|------------|------------|--------------|
| 1 |            |            |            |              |
| 2 |            |            |            |              |
| 3 |            |            |            |              |
| 4 |            |            |            |              |
| 5 |            |            |            |              |
| 6 |            |            |            |              |
| 7 |            |            |            |              |
| 8 |            |            |            |              |
| 9 |            |            |            |              |

27 - 10:37 AM > Successful view of Profile ATALLA\_PROFILE January 1971 - December 2025 .

## NCR Tab

The NCR tab of the Authentication Profile Configuration window contains the encryption keys used for NCR PIN verification. Data entered on this tab is stored in the NCR PIN Verification File (NCRD). This tab is only available when you license the Dutch Security Extensions and use Thales e-Security HSMs. Otherwise, the tab is greyed out and cannot be accessed.

**Note:** On this tab, the Begin Date and End Date fields are not used. The Version field is used instead, in combination with the Profile field, to access records in the NCRD. The version is a unique number from 000-999 that is assigned by the vendor to the key that you are using to configure PIN verification. After the version number is entered and saved, it cannot be changed.

The Key Scheme field in the Encryption Keys table specifies whether the Thales e-Security HSM is configured to use the old (ANSI X9.17 method) non-variant (32H) key encryption scheme or the variant (1A+32H) key encryption scheme. For more information on these encryption schemes, refer to the Thales ***Host Security Module, RG7000 Programmer's Manual***.

An example of the NCR tab on the Authentication Profile Configuration window is shown below.

The screenshot displays the 'ACI Authentication Profile Configuration - RACAL\_PROFILE' window. The 'Profile' dropdown is set to 'RACAL\_PROFILE'. The 'Begin Date' is 'January 1971' and the 'End Date' is 'December 2025'. The 'PIN Verification' section includes checkboxes for 'IBM DES' (checked), 'Visa PW' (unchecked), 'Diebold' (unchecked), 'NCR' (checked), and 'Identikey' (unchecked). The 'Card Verification' section includes checkboxes for 'Card' (unchecked) and 'EMV' (unchecked). The 'NCR' tab is selected, showing the 'Encryption Keys' section with a table for key management.

|  | Version | Key Scheme | Key Part 1 | Key Part 2 | K |
|--|---------|------------|------------|------------|---|
|  |         |            |            |            |   |

4 - 04:12 PM > Successful view of Profile RACAL\_PROFILE January 1971 - December 2025 .

## Identikey Tab

An example of the Identikey tab on the Authentication Profile Configuration window is shown below.

The screenshot shows the 'ACI Authentication Profile Configuration - BK30' window. The 'Profile' dropdown is set to 'BK30'. The 'Begin Date' is 'January 1971' and the 'End Date' is 'December 2025'. The 'PIN Verification' section includes checkboxes for 'IBM DES' (checked), 'Visa PVV' (unchecked), 'Diebold' (unchecked), 'NCR' (unchecked), and 'Identikey' (checked). The 'Card Verification' section includes checkboxes for 'Card' (unchecked) and 'EMV' (unchecked). The 'Identikey' tab is selected, showing fields for 'Bank ID' (empty), 'PAN Verify Offset' (0), 'PAN Verify Length' (4), and 'Compare Indicator' (Not Used). The status bar at the bottom reads: '31 - 11:05 AM > Successful view of Profile BK30 January 1971 - December 2025'.

| Begin Date   | End Date      | PIN Verification                    |                          |                          |                          |                                     | Card Verification        |                          |
|--------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|--------------------------|
|              |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikey                           | Card                     | EMV                      |
| January 1971 | December 2025 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

**Identikey Tab Fields:**

- Bank ID:
- PAN Verify Offset:
- PAN Verify Length:
- Compare Indicator: **Not Used**

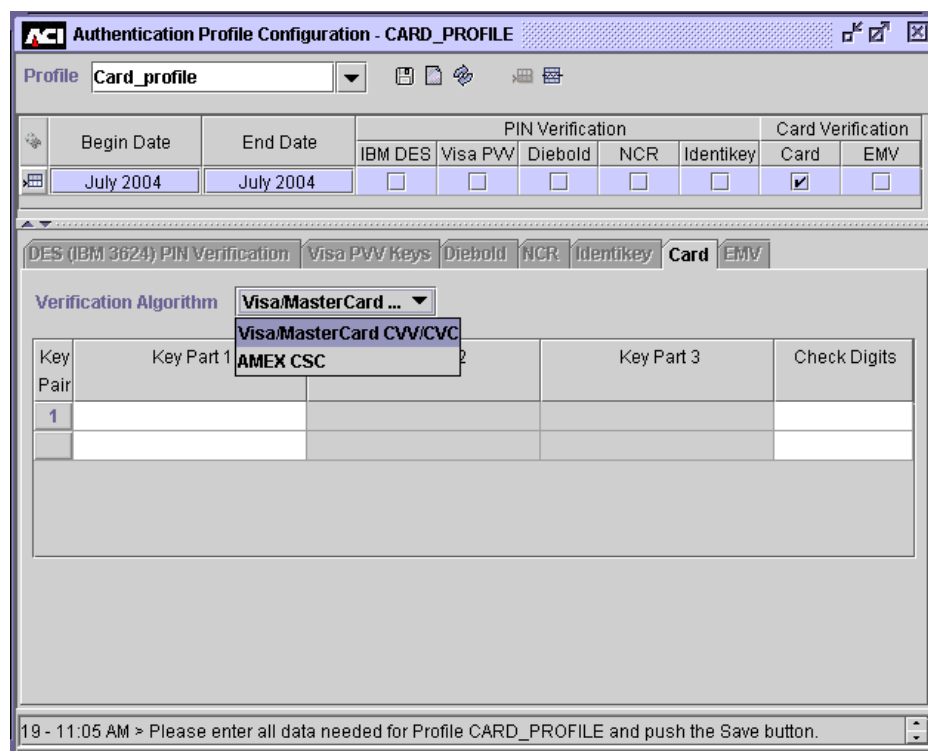
31 - 11:05 AM > Successful view of Profile BK30 January 1971 - December 2025

## Card Tab

The Card tab enables you to maintain the card verification key or key pairs used for Card Verification Value (CVV), Cardholder Authentication Verification Value (CAVV), and Card Security Code (CSC) verification.

**Note:** CSC verification is supported only when you are using an Atalla security device with AKB or variant (i.e., non-AKB) software, a Bull BNTng HSM, or a Thales e-Security HSM. It is not currently supported for Eracom host security modules.

An example of the Card tab on the Authentication Profile Configuration window is shown below.



Authentication Profile Configuration - CARD\_PROFILE

Profile: Card\_profile

| Begin Date | End Date  | PIN Verification         |                          |                          |                          |                          | Card Verification                   |                          |
|------------|-----------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|
|            |           | IBM DES                  | Visa PWV                 | Diebold                  | NCR                      | Identikey                | Card                                | EMV                      |
| July 2004  | July 2004 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

DES (IBM 3624) PIN Verification | Visa PWV Keys | Diebold | NCR | Identikey | **Card** | EMV

Verification Algorithm: Visa/MasterCard ...

| Key Pair | Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|----------|------------|------------|------------|--------------|
| 1        |            |            |            |              |

19 - 11:05 AM > Please enter all data needed for Profile CARD\_PROFILE and push the Save button.

The Verification Algorithm field specifies whether the card verification key or key pairs are used for verification of cards issued by Visa or MasterCard, or cards issued by American Express.

**Note:** The Verification Algorithm field is only displayed when you are using an Atalla NSP with variant software or a Thales e-Security HSM. This field is not displayed when you are using an Atalla NSP with AKB software, a Bull BNTng HSM, or an Eracom host security module.

Atalla NSPs (with AKB or variant software), Bull BNTng HSMs, and Thales e-Security HSMs use a different algorithm for verifying the American Express card security code (CSC) from the algorithm used to verify either the Visa card verification value (CVV) or MasterCard card verification code (CVC). Two single-length keys (key pair) are used to verify a CVV or CVC. A single, double-length key is used to verify a CSC. The key entry fields displayed vary according to the value you select in the Verification Algorithm field. When the Verification Algorithm field is set to a value of AMEX CSC, you must enter a single double-length card verification key. When the Verification Algorithm field is set to a value of Visa/MasterCard CVV/CVC, you must enter two single-length card verification keys (key pair).

Although Atalla NSPs with AKB software and Bull BNTng HSMs also use a different algorithm for verifying the American Express card security code (CSC) from the algorithm used to verify either the Visa card verification value (CVV) or MasterCard card verification code (CVC), the key fields displayed do not change for the different algorithms. Therefore, the Verification Algorithm field is not displayed when using Atalla NSPs with AKB software or Bull BNTng HSMs.



## EMV Tab

An example of the EMV tab on the Authentication Profile Configuration window is shown below

**ACI Authentication Profile Configuration - ATALLA\_MCHIP**

Profile: **ATALLA\_MCHIP**

| Begin Date   | End Date      | PIN Verification         |                          |                          |                          |                          | Card Verification        |                                     |
|--------------|---------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|
|              |               | IBM DES                  | Visa PWV                 | Diebold                  | NCR                      | Identikey                | Card                     | EMV                                 |
| January 2000 | December 2025 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> |

**DES (IBM 3624) PIN Verification** | **Visa PWV Keys** | **Diebold** | **NCR** | **Identikey** | **Card** | **EMV**

**EMV Scheme** ☐ Visa ☒ MasterCard/Europay **Branch/Height Parameters** Branch factor 2; Tree Height 16  
Branch factor 4; Tree Height 8

**Secure Message Key for Integrity**

| Key Part 1          | Key Part 2          | Key Part 3 |
|---------------------|---------------------|------------|
| 72DD DBE3 F5B4 1E13 | 61A3 E01C 37C3 0EEC |            |

**Secure Message Key for Confidentiality** **Destination PIN Block Type** **Standard EMV PIN BL...**

| Key Part 1 | Key Part 2 | Key Part 3 | Check Digits |
|------------|------------|------------|--------------|
|            |            |            |              |

**Cryptographic Keys**

| Key Index | Key Part 1          | Key Part 2          | Key Part 3 | Check Dig |
|-----------|---------------------|---------------------|------------|-----------|
| 1         | 0111 88F2 DFEO 25E8 | 155F 09F5 440D F8A4 |            |           |

5 - 09:51 AM > Successful view of Profile ATALLA\_MCHIP January 2000 - December 2025.

## Branch/Height Parameters

The Branch/Height Parameters field sets the branch factor and tree height parameters used for EMV 2000 processing. Refer to the “[EMV 2000](#)” topic in appendix D of this guide for detailed information on how these parameters are used in processing.

## Cryptographic Keys

**Note:** You can enter up to 24 cryptographic keys on this window, which are stored in the EMVSD. When the Transaction Security Services process receives a request that requires a cryptographic key, it uses the derivation key index value of 0–255 passed in the command request to retrieve the cryptographic key used for

the function from the EMVSD. Therefore, you must specify a unique key index value of 0–255 in the Key Index column of the Cryptographic Keys table for each cryptographic key entered on this tab.

If a cryptographic key is not entered in one of the 24 rows of the Cryptographic Keys table, the Key Index column for the table entry will be blank. For these entries, a default key index value of -1 is stored in the EMVSD.

## Maintaining Authentication Profile Security Data

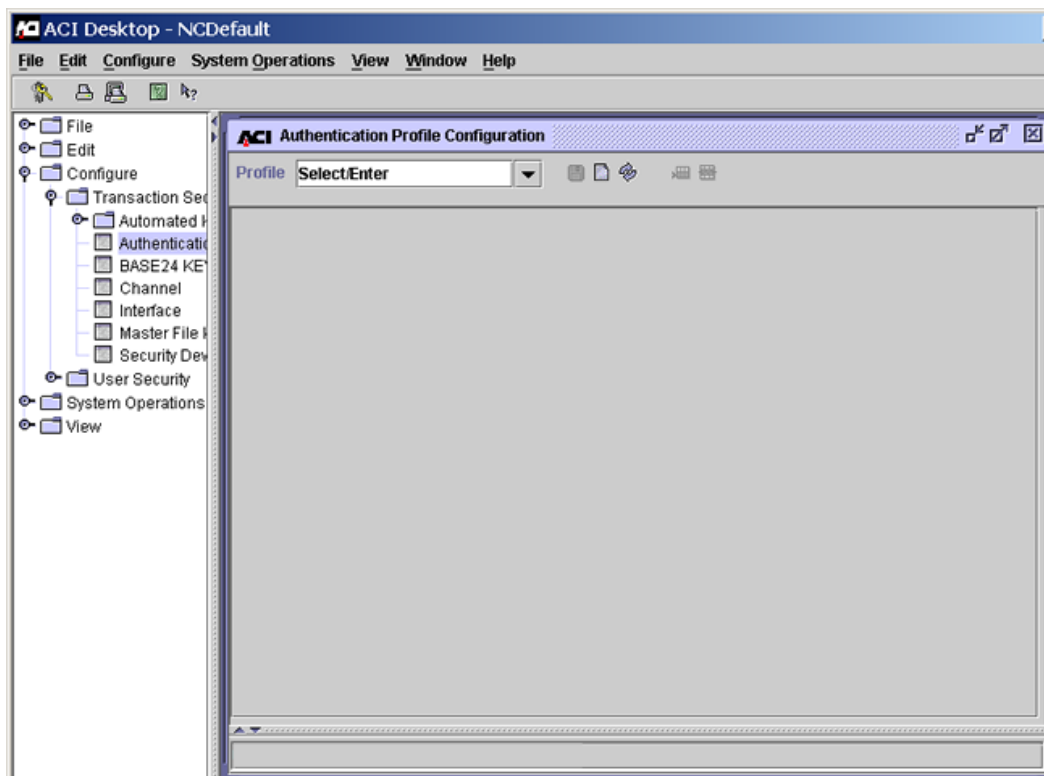
The following paragraphs describe procedures for displaying, adding, modifying, and deleting authentication profile security data from the Authentication Profile Configuration window. Data entered on the Authentication Profile Configuration window is stored in different files, depending on the tab from which the data is entered. The following table shows you the security data files that correspond to each tab.

| <b>Tab</b>                      | <b>File</b>                             |
|---------------------------------|-----------------------------------------|
| DES (IBM 3624) PIN Verification | IBM DES Verification File (IDESD)       |
| Visa PVV Keys                   | Visa PVV Verification File (VPVVD)      |
| Diebold                         | Diebold Number Table File (DNSTD)       |
| NCR                             | NCR PIN Verification File (NCRD)        |
| Identikey                       | Identikey PIN Verification File (IDNTD) |
| Card                            | Card Verification File (CRDVD)          |
| EMV                             | EMV Authorization File (EMVSD)          |

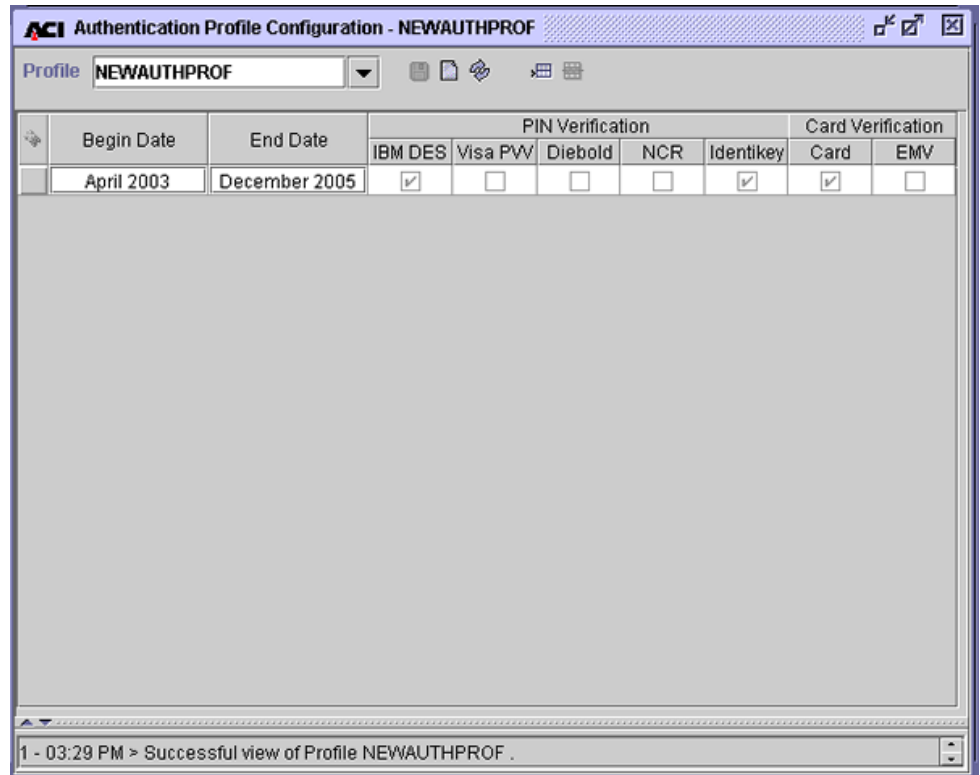
## Displaying an Authentication Profile

To display an authentication profile, perform the following steps:

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the Authentication Profile Configuration window by accessing **Configure > Transaction Security > Authentication Profile**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter an authentication profile name in the Profile field and press the **Enter** key or select a name from the selectable list. The table below the field displays the various entries that make up the profile. The verification types that have been defined for the profile are indicated by a check mark in a box that is labeled with a verification type name.



ACI Authentication Profile Configuration - NEWAUTHPROF

Profile: **NEWAUTHPROF**

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                                     | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikey                           | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

1 - 03:29 PM > Successful view of Profile NEWAUTHPROF .

4. To display the configuration details for the verification types in an entry, select a row by clicking anywhere in the row. Only the tabs below the table for the verification types that have been configured for the row are active; the inactive tabs appear dimmed so that you cannot access them.

**ACI Authentication Profile Configuration - NEWAUTHPROF**

Profile: **NEWAUTHPROF**

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          | Card Verification                   |                                     |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|
|  |            |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikay                           | Card                                |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

**DES (IBM 3624) PIN Verification** | Visa PVV Keys | Diebold | NCR | Identikay | Card | EMV

Key Length: **Triple**

| Key Part 1          | Key Part 2          | Key Part 3          | Check Digits |
|---------------------|---------------------|---------------------|--------------|
| 1111 1111 1111 1111 | 1111 1111 1111 1111 | 1111 1111 1111 1111 |              |

Decimalization Table:  PAN Verify Length:

PAN Pad Character:  PAN Verify Offset:

4 - 03:33 PM > Successful view of Profile NEWAUTHPROF April 2003 - December 2005 .

5. Click the active tabs to display the configurations of the various verification types.

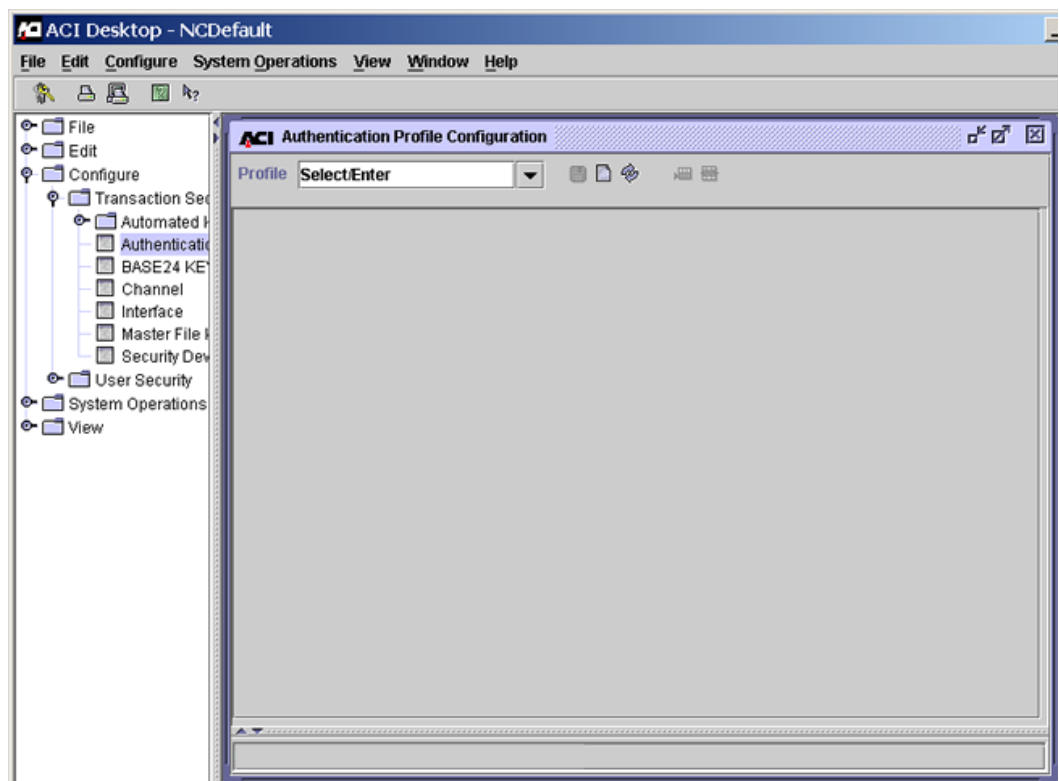
## Adding a New Authentication Profile

You can add a new authentication profile in two ways. You can create a new authentication profile from the very beginning or you can use an existing authentication profile as a *template* for creating a new profile. Procedures for both means of adding a new authentication profile are provided below.

## Creating a New Authentication Profile from the Beginning

Use the following procedure to add a new authentication profile from the very beginning.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the Authentication Profile Configuration window by accessing **Configure > Transaction Security > Authentication Profile**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



- Enter the name of the authentication profile to be added in the Profile field and press the **Enter** key. The main body of the window is displayed and you are prompted to enter all the data needed for the new profile as shown below.

- Select the beginning date and ending date for the profile in the Begin Date and End Date fields. You can configure multiple date range entries for the authentication profile by inserting additional rows in the table near the top of the window. This step and all remaining steps in this procedure pertain to each row in this table.

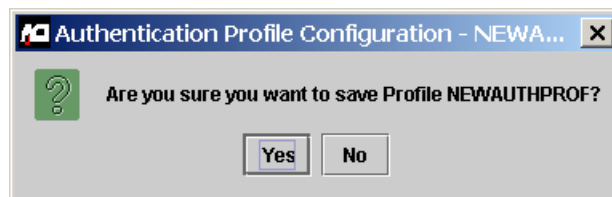
These fields specify the range of card expiration dates to which this authentication profile applies. The expiration date in the transaction data is compared to the range of card expiration dates to determine which entry of the authentication profile to use, and therefore, which PIN verification, card verification, and EMV parameters to use. For example, if there are two authentication profile entries and one has a range of April 2002–2003, and another has a range of April 2003–2004, and a transaction contains an expiration date of September 2003, the application would use the second entry to authenticate the transaction.

Valid values are January 1000 through December 9999. You can change the month and year in this field by selecting either value and using the up and down arrows to change them.



5. Select the PIN verification and Card Verification boxes for the verification types to be configured in this profile by clicking in the appropriate boxes. You must select at least one type of verification for the profile. The tabs for the verification types selected are activated.

6. Enter the data required for each type of verification selected for this authentication profile on the active tabs and click Save ( ). Answer Yes to the “Are you sure you want to save” dialog box.



Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields for each type of verification.

7. If the authentication profile information is successfully saved, a message similar to the following is displayed at the bottom of the window.

9 - 09:41 AM > Successful insert of Profile NEWAUTHPROF April 2003 - December 2005 .

8. Rebuild the external memory tables for the verification types included in the new authentication profile. A separate external memory table is maintained for each tab on the Authentication Profile Configuration window. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for procedures.
9. Warmboot the Transaction Security Services process to begin using the new data in processing. Refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for procedures.
10. Log on to the BASE24 user access system and access the Card Prefix File (CPF) screens. Add or update the appropriate fields as shown in the following table with the new authentication profile value. These fields contain the key locator values used by the Transaction Security Services process to access authentication profile information.

The following table shows you the fields in the Card Prefix File (CPF) that are used to access the appropriate records in the TSS database. The value in these fields must match a value entered in the Profile field on the Authentication Profile Configuration window.

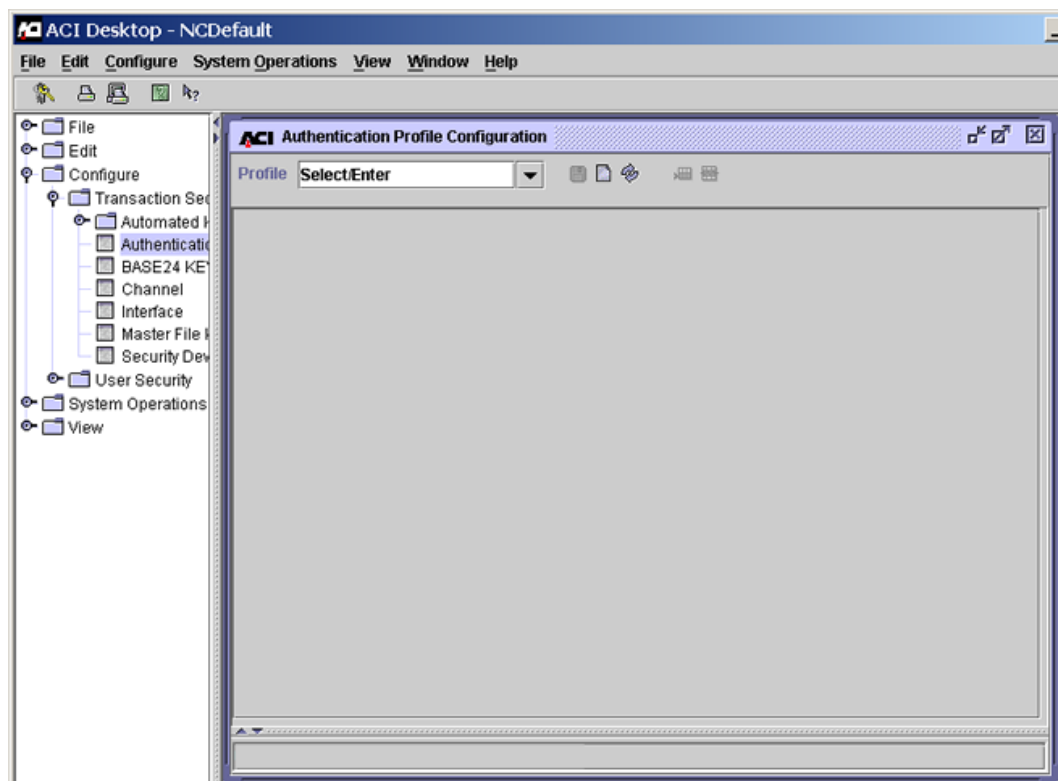
| CPF Field                                   | TSS File                        |
|---------------------------------------------|---------------------------------|
| PIN VERIFICATION KEYA GROUP (CPF screen 2)  | IDESD, DNTD, IDNTD, NCRD, VPVVD |
| CARD VERIFICATION KEYA GROUP (CPF screen 2) | CRDVD                           |
| MANUAL CARD VERIF KEYA GROUP (CPF screen 2) |                                 |
| SIV KEYA GROUP (CPF screen 6)               |                                 |
| KEYI GROUP (CPF screen 11)                  | EMVSD                           |

The PIN VERIFICATION KEYA GROUP field specifies the authentication profile to use to retrieve PIN verification method information. The CARD VERIFICATION KEYA GROUP, MANUAL CARD VERIF KEYA GROUP, and SIV KEYA GROUP fields specify the authentication profile to use to retrieve card verification keys. Finally, the KEYI GROUP field specifies the authentication profile from which EMV processing information is retrieved.

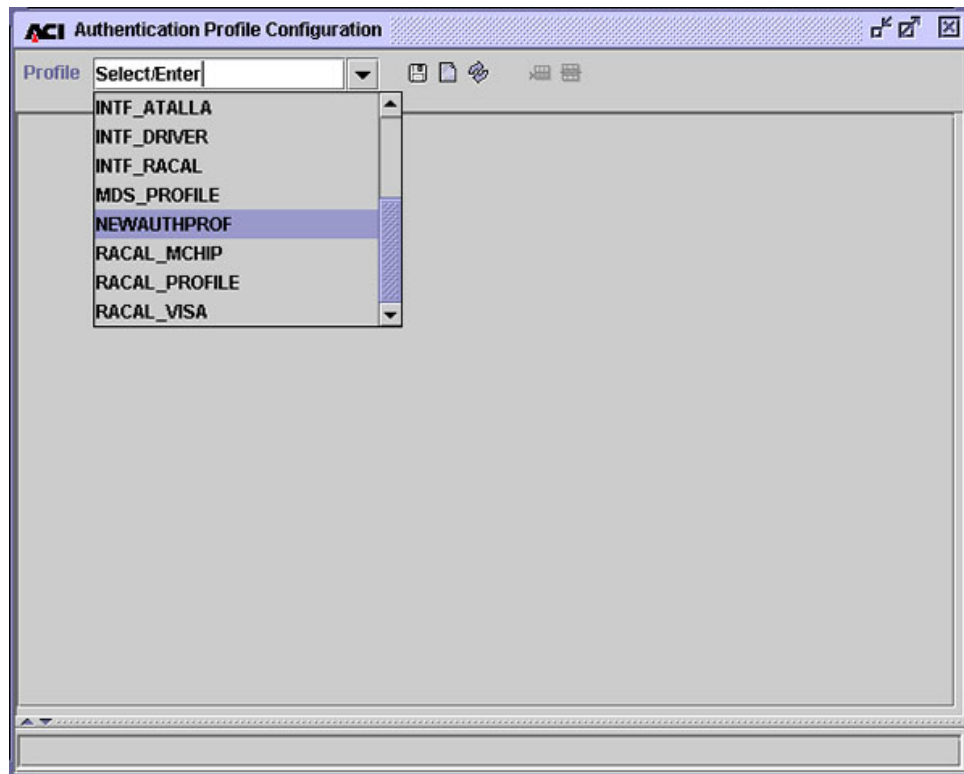
## Adding a New Authentication Profile from a Template

Use the following procedure to create a new authentication profile by using an existing authentication profile as a *template*.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the Authentication Profile Configuration window by accessing **Configure > Transaction Security > Authentication Profile**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter the name of the authentication profile you want to use as a template in the Profile field and press the **Enter** key or select the profile from the selectable list



4. All the date range entries for the profile are displayed. In this example, only one data range has been specified for the NEWAUTHPROF authentication profile. This profile includes IBM DES PIN verification, Identikey PIN verification, and card verification key information.

ACI Authentication Profile Configuration - NEWAUTHPROF

Profile: NEWAUTHPROF

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                                     | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PW                  | Diebold                  | NCR                      | Identikey                           | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

1 - 03:29 PM > Successful view of Profile NEWAUTHPROF .

- Click anywhere in a date range row to retrieve the configuration data for that data range.

ACI Authentication Profile Configuration - NEWAUTHPROF

Profile: NEWAUTHPROF

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                                     | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikey                           | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

DES (IBM 3624) PIN Verification

Key Length: Triple

| Key Part 1          | Key Part 2          | Key Part 3          | Check Digits |
|---------------------|---------------------|---------------------|--------------|
| 1111 1111 1111 1111 | 1111 1111 1111 1111 | 1111 1111 1111 1111 |              |

Decimalization Table: 1111111111111111

PAN Pad Character: F

PAN Verify Length: 5

PAN Verify Offset: 0

4 - 03:33 PM > Successful view of Profile NEWAUTHPROF April 2003 - December 2005 .

6. Change the name in the Profile field to the name of the new profile. Enter the beginning and ending dates for the date range of this new profile and re-select the verification types to be included in the profile. Verification data on the tabs that were active for the profile being used as a template is still displayed.

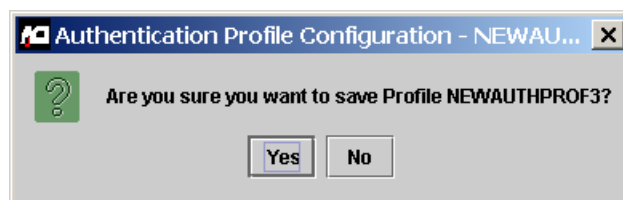
| Begin Date |            | End Date |                | PIN Verification                    |                          |                          |                          |                                     | Card Verification                   |                          |
|------------|------------|----------|----------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
|            |            |          |                | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikey                           | Card                                | EMV                      |
|            | April 2003 |          | September 2006 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

| Key Pair | Key Part 1          | Key Part 2 | Key Part 3 | Check Digits |
|----------|---------------------|------------|------------|--------------|
| 1        | 1111 2222 3333 4444 |            |            |              |
|          | 4444 3333 2222 1111 |            |            |              |

12 - 04:33 PM > Please enter all data needed for Profile NEWAUTHPROF3 and push the Save button.

7. Click Save (💾), and answer Yes to the “Are you sure you want to save” dialog box.



If the new profile is successfully saved, a message similar to the following is displayed at the bottom of the window.

14 - 04:45 PM > Successful insert of Profile NEWAUTHPROF3 April 2003 - September 2006 .

8. Rebuild the external memory tables for the verification types included in the new authentication profile. A separate external memory table is maintained for each tab on the Authentication Profile Configuration window. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for procedures.
9. Warmboot the Transaction Security Services process to begin using the new data in processing. Refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for procedures.
10. Log on to the BASE24 user access system and access the Card Prefix File (CPF) screens. Add or update the appropriate fields as shown in the following table with the new authentication profile value. These fields contain the key locator values used by the Transaction Security Services process to access authentication profile information.

The following table shows you the fields in the Card Prefix File (CPF) that are used to access the appropriate records in the TSS database. The value in these fields must match a value entered in the Profile field on the Authentication Profile Configuration window.

| CPF Field                                   | TSS File                        |
|---------------------------------------------|---------------------------------|
| PIN VERIFICATION KEYA GROUP (CPF screen 2)  | IDESD, DNTD, IDNTD, NCRD, VPVVD |
| CARD VERIFICATION KEYA GROUP (CPF screen 2) | CRDVD                           |
| MANUAL CARD VERIF KEYA GROUP (CPF screen 2) |                                 |
| SIV KEYA GROUP (CPF screen 6)               |                                 |
| KEYI GROUP (CPF screen 11)                  | EMVSD                           |

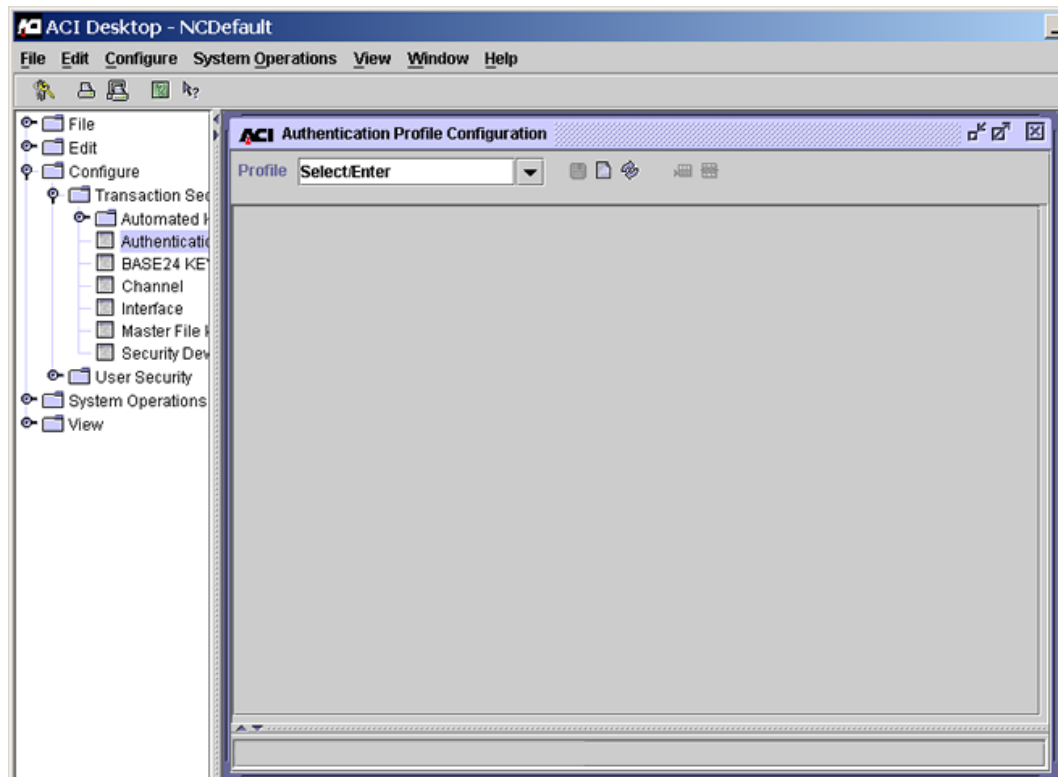
The PIN VERIFICATION KEYA GROUP field specifies the authentication profile to use to retrieve PIN verification method information. The CARD VERIFICATION KEYA GROUP, MANUAL CARD VERIF KEYA GROUP, and SIV KEYA GROUP fields specify the authentication profile to use to retrieve card verification keys. Finally, the KEYI GROUP field specifies the authentication profile from which EMV processing information is retrieved.



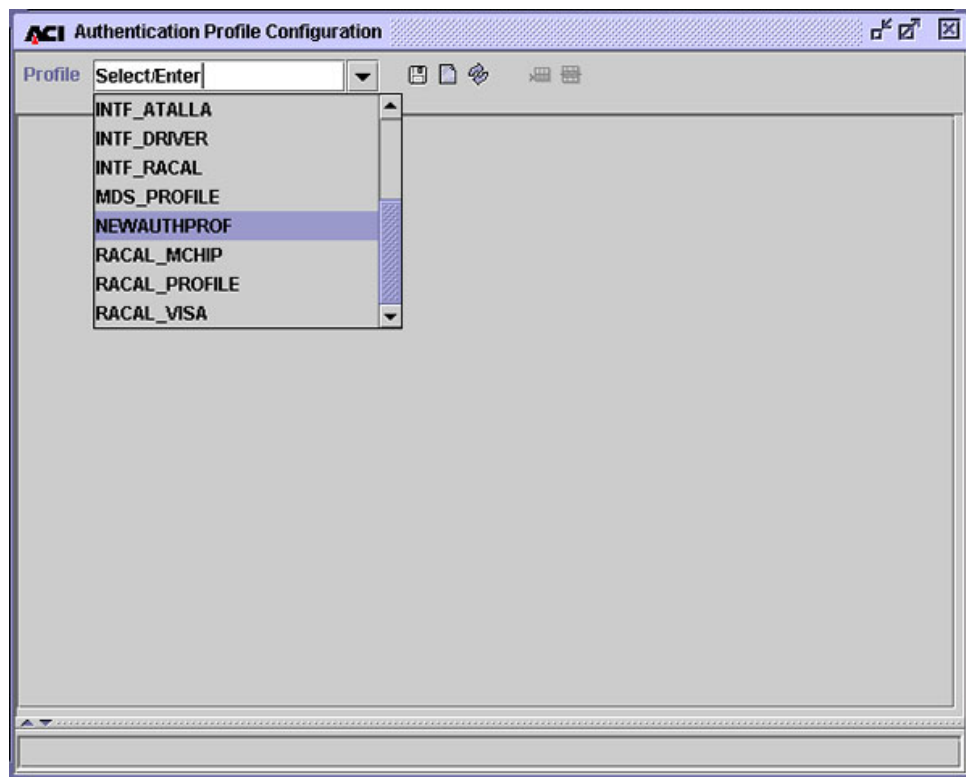
## Modifying an Existing Authentication Profile

Use the following procedure to make changes to an existing authentication profile.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the Authentication Profile Configuration window by accessing **Configure > Transaction Security > Authentication Profile**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter the authentication profile you want to change in the Profile field and press the **Enter** key or select the profile from the selectable list



4. All the date range entries for the profile are displayed. In this example, only one data range has been specified for the NEWAUTHPROF authentication profile. This profile includes both IBM DES PIN verification and card verification key information.

ACI Authentication Profile Configuration - NEWAUTHPROF

Profile: NEWAUTHPROF

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                          | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PW                  | Diebold                  | NCR                      | Identikey                | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

1 - 01:02 PM > Successful view of Profile NEWAUTHPROF .

- Click anywhere in a date range row to retrieve the configuration data for that data range.

**ACI Authentication Profile Configuration - NEWAUTHPROF**

Profile: **NEWAUTHPROF**

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                          | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikey                | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

**DES (IBM 3624) PIN Verification** | Visa PVV Keys | Diebold | NCR | Identikey | Card | EMV

Key Length: **Triple**

| Key Part 1          | Key Part 2          | Key Part 3          | Check Digits |
|---------------------|---------------------|---------------------|--------------|
| 1111 1111 1111 1111 | 1111 1111 1111 1111 | 1111 1111 1111 1111 |              |

Decimalization Table:  PAN Verify Length:

PAN Pad Character:  PAN Verify Offset:

3 - 01:10 PM > Successful view of Profile NEWAUTHPROF April 2003 - December 2005 .

- Update the existing data for the profile on activated tabs as needed. If you need to add an additional verification type to the profile, select the verification type in the date range row and enter the necessary data on the activated tab.

Refer to the procedures and field descriptions in ACI Online Help for detailed information on the required and optional fields for each type of verification.

In this example, Identikay PIN verification data is being added to the NEWAUTHPROF authentication profile.

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          | Card Verification                   |                                     |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|
|  |            |               | IBM DES                             | Visa PVV                 | Diebold                  | NCR                      | Identikay                           | Card                                |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> |

DES (IBM 3624) PIN Verification | Visa PVV Keys | Diebold | NCR | Identikay | Card | EMV

Bank ID: 123456

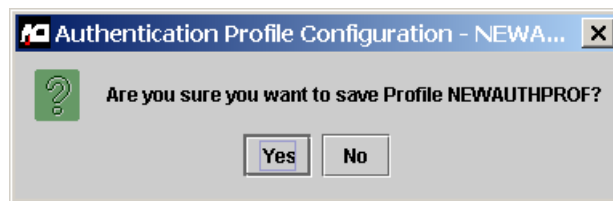
PAN Verify Offset: 2

PAN Verify Length: 4

Compare Indicator: Right

4 - 03:33 PM > Successful view of Profile NEWAUTHPROF April 2003 - December 2005 .

7. Click Save (  ), and answer Yes to the “Are you sure you want to save” dialog box.



8. If the authentication profile information is successfully updated, a message similar to the following is displayed at the bottom of the window.

6 - 01:32 PM > Successful update of Profile NEWAUTHPROF April 2003 - December 2005 .

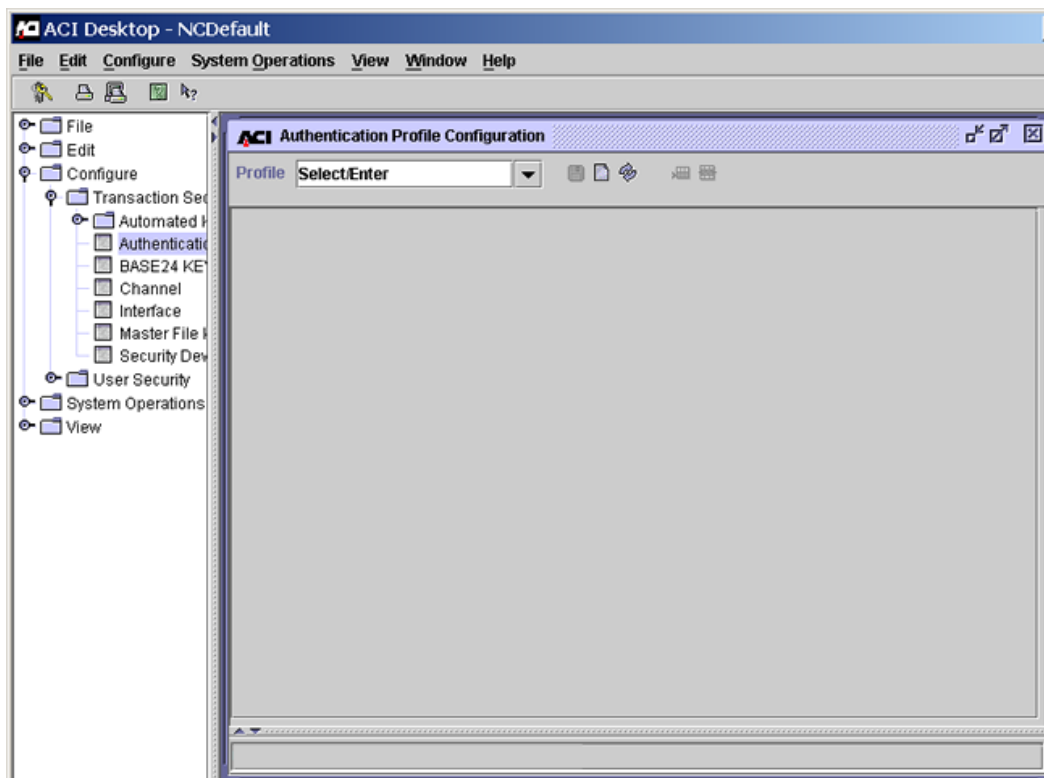
9. Rebuild the external memory tables for the verification types in the updated authentication profile. A separate external memory table is maintained for each tab on the Authentication Profile Configuration window. Refer to the “Rebuilding External Memory Tables” topic provided later in this section for procedures.

10. Warmboot the Transaction Security Services process to begin using the new data in processing. Refer to the “Warmbooting the Transaction Security Services Process” topic provided later in this section for procedures.

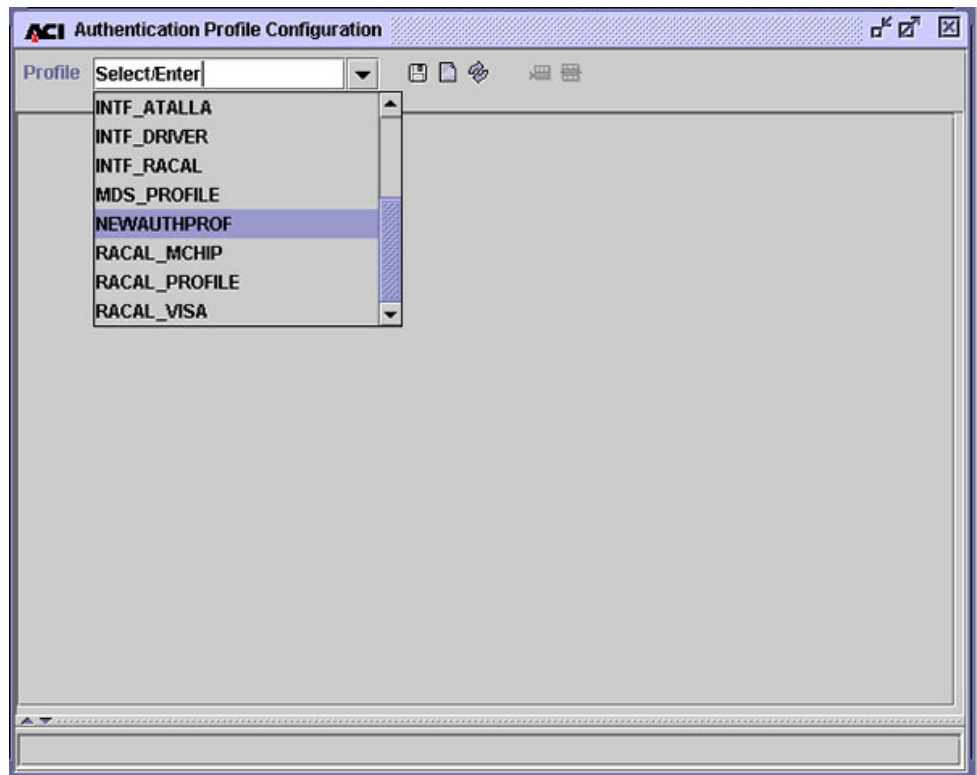
## Deleting an Authentication Profile

Use the following procedure to delete an authentication profile.

1. Log on to the ACI desktop user interface. Refer to the *ACI Desktop User Interface Manual* for instructions on logging on.
2. Display the Authentication Profile Configuration window by accessing **Configure > Transaction Security > Authentication Profile**. You can use the Configure drop-down menu at the top of the user interface or the navigation pane on the left to access this window.



3. Enter the name of the authentication profile you want to delete in the Profile field and press the **Enter** key or select the profile from the selectable list.



4. All the date range entries for the profile are displayed. In this example, only one data range has been specified for the NEWAUTHPROF authentication profile. This profile includes IBM DES PIN verification, Identikey PIN verification, and card verification key information.

The screenshot shows the 'ACI Authentication Profile Configuration - NEWAUTHPROF' window. The 'Profile' dropdown is set to 'NEWAUTHPROF'. Below the dropdown is a table with the following data:

|  | Begin Date | End Date      | PIN Verification                    |                          |                          |                          |                                     | Card Verification                   |                          |
|--|------------|---------------|-------------------------------------|--------------------------|--------------------------|--------------------------|-------------------------------------|-------------------------------------|--------------------------|
|  |            |               | IBM DES                             | Visa PWV                 | Diebold                  | NCR                      | Identikey                           | Card                                | EMV                      |
|  | April 2003 | December 2005 | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input checked="" type="checkbox"/> | <input checked="" type="checkbox"/> | <input type="checkbox"/> |

At the bottom of the window, a status bar displays: '1 - 03:29 PM > Successful view of Profile NEWAUTHPROF .'



5. Click anywhere in a date range row to select the configuration data for that date range.

6. Click Delete Row (🗑️). The date range row is now marked for deletion.

7. Click Save (💾), and answer Yes to the “Are you sure you want to delete” dialog box.

If the profile is successfully deleted, a message similar to the following is displayed at the bottom of the window.

20 - 05:11 PM > Successful delete of Profile NEWAUTHPROF .

## Key Variant Fields

The Key Variant fields on various tabs of the Channel Security Configuration window and the Interface Security Configuration window specify the key variant to be applied to the PIN, MAC, or data encryption key exchange key (KEK) when translating PIN, MAC, or data encryption keys.

For example, if the working key provided by an external processor has been encrypted with the KEK using a non-Atalla security device and the BASE24 system uses Atalla security devices, a variant must be used. The variants available for each type of working key are described below.

### PIN Working Key Variants

The following variants are available in the Key Variant field for PIN key exchange keys.

| Key Variant                    | Description                                                                                                                                                                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 - Key Exchange Key Variant   | Translate the PIN key directly from the PIN KEK parts.                                                                                                                                                                                                           |
| 1 - PIN Encryption Key Variant | Include the variant 1 constant with the data encryption KEK parts when translating the data encryption key. The Atalla variant 1 constant is hexadecimal 0800 0000 0000 0000. This value is displayed only when Atalla security devices are used in your system. |
| 5 - ATM Key Variant*           | Include the variant 5 constant with the PIN KEK parts when translating the PIN key. The Atalla 5 constant is hexadecimal 2800 0000 0000 0000.                                                                                                                    |

\* This variant is only available on the Channel Security Configuration window.

## MAC Working Key Variants

The following variants are available in the Key Variant field for MAC key exchange keys.

| Key Variant                                 | Description                                                                                                                                                                                                                      |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 - Key Exchange Key Variant                | Translate the MAC key directly from the MAC KEK parts.                                                                                                                                                                           |
| 3 - MAC Encryption Key Variant <sup>*</sup> | Include the variant 3 constant with the MAC KEK parts when translating the MAC key. The Atalla 3 constant is hexadecimal 1800 0000 0000 0000. This value is displayed only when Atalla security devices are used in your system. |
| 5 - ATM Key Variant <sup>†</sup>            | Include the variant 5 constant with the MAC KEK parts when translating the MAC key. The Atalla 5 constant is hexadecimal 2800 0000 0000 0000.                                                                                    |

<sup>\*</sup> This variant is only available on the Interface Security Configuration window.

<sup>†</sup> This variant is only available on the Channel Security Configuration window.

**Note:** If the variant specified in the [Export Variant](#) field on the MAC Information tab of the Channel Security Configuration window is set to a value of 0, then you must set this field to a value of 0 or 5. If the variant specified in the Export Variant field on the MAC Information tab of the Channel Security Configuration window is set to a value of 3, then you must set this field to a value of 0.

## Data Encryption Working Key Variants

The following variants are available in the Key Variant field for data encryption key exchange keys.

| Key Variant                  | Description                                                                    |
|------------------------------|--------------------------------------------------------------------------------|
| 0 - Key Exchange Key Variant | Translate the data encryption key directly from the data encryption KEK parts. |

| <b>Key Variant</b>              | <b>Description</b>                                                                                                                                                                                                               |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 - Data Encryption Key Variant | Include the variant 2 constant with the PIN KEK parts when translating the PIN key. The Atalla 2 constant is hexadecimal 1000 0000 0000 0000. This value is displayed only when Atalla security devices are used in your system. |

# External Memory Tables

Several of the files maintained by the Transaction Security Services process are accessed as external memory tables during online transaction processing. These external memory tables are similar to extended memory tables in the BASE24 processing environment. As with extended memory tables, external memory tables must be built or rebuilt when the underlying file is changed (through an add, update, or delete) and the application process that accesses the external memory table (i.e., the Transaction Security Services process) must be warmbooted before it starts using the new external memory table. The following files for the Transaction Security Services process are accessed as external memory tables:

- BNTng Header File (BNTHD)
- Card Verification File (CRDVD)
- Derivation Key File (DRVKD)
- Diebold Number Table File (DNTD)
- EMV Authorization File (EMVSD)
- IBM DES Verification File (IDESD)
- Identikey PIN Verification File (IDNTD)
- Interface Security File (ISECD)\*
- NCR PIN Verification File (NCRD)
- Visa PVV Verification File (VPVVD)

\* Data for an interface in the ISECD is accessed from an external memory table (i.e., ISECOD) only when the interface does not support dynamic key management. This is indicated by a value of “No keys exchanged” in the Options field on the Exchange Information tab of the Interface Security Configuration window. Otherwise, data for an interface is accessed from an internal memory table (or the ISECD disk file).

When you change values on the Interface Security Configuration window for an interface that does not exchange keys, you must rebuild the Interface external memory table and warmboot, or stop and restart, the Transaction Security Services process to reread the Interface external memory table. The new values will be used in processing for the next transaction for the interface that requires security key data.

When you change key values on the Interface Security Configuration window for an interface that does exchange keys, you do not have to warmboot the Transaction Security Services process to reread the ISECD, or stop and restart the Transaction Security Services process. The Transaction Security Services process reads the new key values from the ISECD disk data source for the next transaction for the interface that requires security key data if it encounters a key synchronization error using the key in the internal memory table or if the key is an outbound key. If, however, you add or delete a record for an interface that does exchange keys, or if you change the value of the Options field on the Exchange Information tab for an interface to start or stop exchanging keys, you must rebuild the Interface extended memory table before warmbooting the Transaction Security Services process to reread the ISECD, or stopping and restarting the Transaction Security Services process. As an alternative to changing key exchange key values from the ACI desktop user interface, you can change them by entering a KEY CHANGE text command from a network control facility. In this case, the changed key data is automatically reread from the ISECD disk file and there is no need to warmboot the Transaction Security Services process to reread the ISECD, or to stop and restart the Transaction Security Services process.

Refer to the [“Internal Memory Tables”](#) topic for more information on warmbooting the Transaction Security Services process for the ISECD internal memory table.

A separate data source assign is required for each file that is accessed as an external memory table. These data source assigns specify the location of *swap* files used when loading the file data into external memory. By convention, these assigns always have \_OLTP appended to the assign name and cannot specify the same file location as the data source assigns for the associated disk files from which the external memory tables are built. Also as a convention, ACI inserts the letter O into the swap file names specified by these \_OLTP assigns. For example, the external memory table swap file for the ISECD file is ISECOD.

All of these files are maintained from the Authentication Profile Configuration window, except for the ISECD, which is maintained from the Interface Security Configuration window. You can rebuild external memory tables using the Process Control window from the ACI desktop user interface or from an XPNET network control facility as described below.

## Rebuilding External Memory Tables

You can rebuild external memory tables in the TSS environment in two ways. You can rebuild external memory tables by entering a DELIVER PROCESS command from an XPNET network control facility or by entering a command from the Process Control window on the ACI desktop user interface. Procedures for both means of rebuilding external memory tables are provided below. Commands to rebuild external memory tables should be sent to the User Interface (XML Server) process instead of a Transaction Security Services process to avoid any impact to online transaction processing performance.

### XPNET Network Control Facility

To rebuild an external memory table from an XPNET network control facility, use the following command.

#### **Syntax**

DELIVER PROCESS *object-name*, "DELIVER DSLDR *oltp-assign-name*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*oltp-assign-name* — The assign name specified for the external memory table in the CONFCSV on the data subvolume. The following table lists the assign names for each of the files accessed as external memory tables.

| Assign Name               | TSS File                                 |
|---------------------------|------------------------------------------|
| BNTNG_HEADER_OLTP         | BNTng Header File (BNTHOD)               |
| CARD_VERIFICATION_OLTP    | Card Verification File (CRDVOD)          |
| DERIVATION_KEY_OLTP       | Derivation Key File (DRVKD)              |
| DIEBOLD_NUMBER_TABLE_OLTP | Diebold Number Table File (DNTOD)        |
| EMV_SECURITY_OLTP         | EMV Authorization File (EMVSOD)          |
| IBM_DES_OLTP              | IBM DES Verification File (IDESOD)       |
| IDENTIKEY_OLTP            | Identikey PIN Verification File (IDNTOD) |

| Assign Name               | TSS File                            |
|---------------------------|-------------------------------------|
| INTERCHANGE_SECURITY_OLTP | Interface Security File (ISECOD)    |
| NCR_PIN_VERIFICATION_OLTP | NCR PIN Verification File (NCROD)   |
| VISA_PVV_OLTP             | Visa PVV Verification File (VPVVOD) |

**Example**

The following is an example of a DELIVER PROCESS command sent to the User Interface process to rebuild the external memory table for the Visa PVV Verification File.

```
deliver process p1a^xmll01,"deliver dsldr VISA_PVV_OLTP"
```

**Process Control Window**

To rebuild an external memory table from the Process Control window on the ACI desktop user interface, perform the following steps:

1. After logging on, display the Process Control window by selecting System Operations and then Process Control.
2. In the Destination field, select XMLS from the list of values, or enter the symbolic name of the User Interface process.
3. In the Command field, select Deliver from the list of values. The XPNET process sends Deliver commands to a single process.
4. In the Component field, select Data Source Loader from the list of values.
5. In the Command Text field, enter the name of the file to be loaded into memory. This is the assign name specified for the external memory table in the CONFCSV on the data subvolume. The following table lists the assign names for each of the files accessed as external memory tables.

| Assign Name            | TSS File                        |
|------------------------|---------------------------------|
| BNTNG_HEADER_OLTP      | BNTng Header File (BNTHOD)      |
| CARD_VERIFICATION_OLTP | Card Verification File (CRDVOD) |

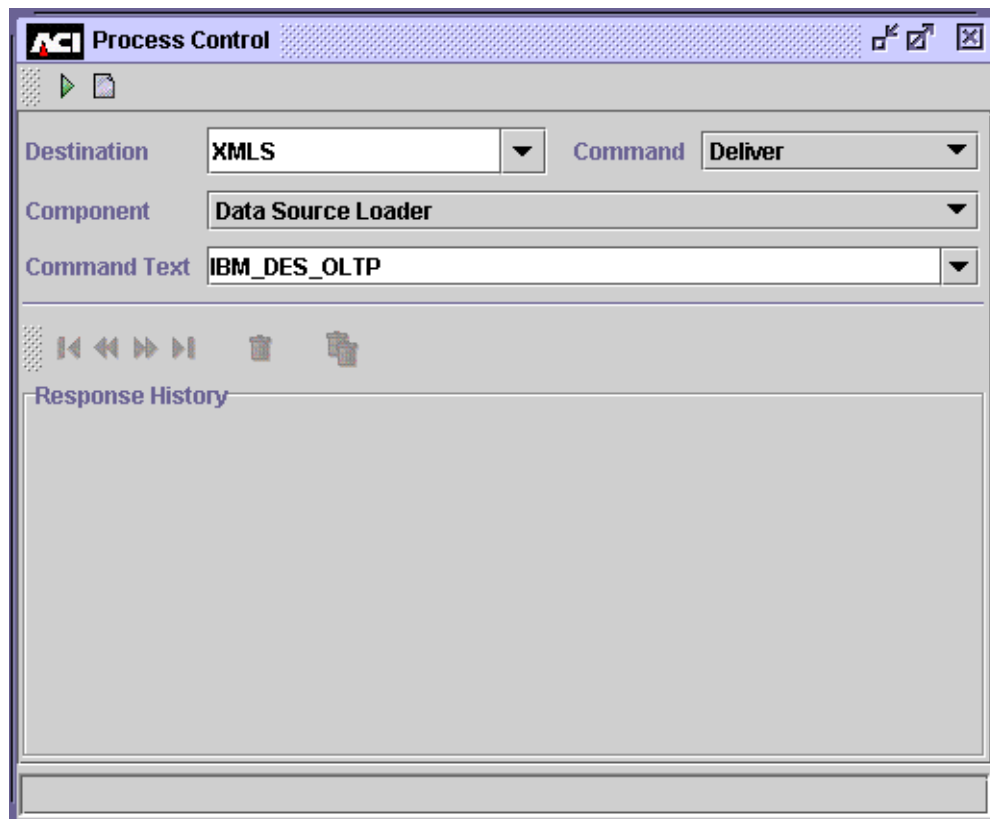


| Assign Name               | TSS File                                 |
|---------------------------|------------------------------------------|
| DERIVATION_KEY_OLTP       | Derivation Key File (DRVKD)              |
| DIEBOLD_NUMBER_TABLE_OLTP | Diebold Number Table File (DNTOD)        |
| EMV_SECURITY_OLTP         | EMV Authorization File (EMVSOD)          |
| IBM_DES_OLTP              | IBM DES Verification File (IDESOD)       |
| IDENTIKEY_OLTP            | Identikey PIN Verification File (IDNTOD) |
| INTERCHANGE_SECURITY_OLTP | Interface Security File (ISECOD)         |
| NCR_PIN_VERIFICATION_OLTP | NCR PIN Verification File (NCROD)        |
| VISA_PVV_OLTP             | Visa PVV Verification File (VPVVOD)      |

6. Click Submit (▶) at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window. If the table was successfully rebuilt, an event message similar to the following will be displayed on the EMS log.

```
MM-DD-YY;HH:MM:SS.HHH \SYS.$B4XM ACI.1033.1000 60
Mgr PPD:\SYS.$BB4AN Generator: P1A^XML01
DSLDR: File IBM_DES_OLTP successfully loaded.
```

An example of the Process Control window is shown below.



## External Memory Table Size

If the build of an external memory table fails, it's possible that the memory size allowed for the table isn't big enough. If so, you can increase the size allotted for the external memory table in its assign configuration (e.g., "`-MT_SIZE=500000`") by editing the Metadata Configuration File (CONFCSV) on the data subvolume. See the "Calculating Maximum Hash Table Memory Requirements" topic below for detailed information on calculating the size for hash tables accessed in external memory.

**Note:** Any time you change the CONFCSV, you must stop and restart the process that uses the changed data. The change does not take effect until after the process is stopped and restarted. In addition, the process should be shut down in an orderly fashion so that any transactions currently in progress are not lost.

## Calculating Maximum Hash Table Memory Requirements

Currently, all external memory tables used by the Transaction Security Services process are [hash tables](#). When determining the amount of memory required for a hash table, you must consider the following three data areas required to build a hash table:

- The hash table header, which is always a fixed length of 4184 bytes
- The data records and hash data record headers. The space required for data records depends on the [aligned length of each record](#) in the data source and the total number of records. The hash data record header is a fixed length of 22 bytes.
- The bucket space, which depends on the total number of records in the data source and the [hash algorithm](#) selected to build the hash table. The size of each [hash bucket](#) is a fixed length of 22 bytes.

Calculating the size required for the data records and data record headers is simply a matter of multiplying the total number of records in the data source by the length of each record, including the hash data record header.

Calculating the bucket space required is more problematic because the Data Source Loader component attempts 108 different algorithms to determine the most efficient hash algorithm. The most efficient algorithm is the one eventually used but the component requires space to attempt all 108 algorithms, one at a time. Some algorithms require more space than others and the most efficient algorithm does not necessarily conserve the most space.

The most important function of the hash algorithm is to eliminate key [collisions](#) when calculating hash keys to optimize the speed of data retrieval. Keys are distributed across buckets. Therefore the best algorithm results in the least number of keys in a bucket to eliminate the additional overhead required for searching a chain of keys that hash to the same bucket (when searching for an exact match to a record).

## Modulo Divide Hash Algorithm

The modulo divide hash algorithms requires the number of buckets to be a prime number greater than or equal to the number of hash records times the bucket factor. The maximum bucket factor is always 6. Therefore, this algorithm has the potential to allocate a number of buckets equal to the next prime number greater than or equal to the total number of data records in the data source multiplied by a factor of 6.

## Multiplicative Hash Algorithm

For the multiplicative hash algorithm, the number of buckets need only be greater than or equal to the number of hash records times the maximum bucket factor of 6. For example, if there are 10,000 data records in a file, the multiplicative hash algorithm requires 10,000 or more buckets times 6, or 60,000 or more buckets.

## Mid-Square Hash Algorithm

For the mid-square hash algorithm, the number of buckets must be a power of 2 (i.e., 2, 4, 8, 16, 32, 64, 128, 256, etc.), greater than or equal to the number of hash records times the maximum bucket factor of 6. For example, if there are 50 data records in a file, the mid-square hash algorithm requires a power of two greater than or equal to 50 times 6 buckets, or 300 buckets. The next power of 2 greater than or equal to 300 is 512.

## Data Alignment

Although the total number of data records is multiplied by 6, the data itself is only loaded once. However, records need to be aligned on an 8-byte boundary. Therefore, you must add enough bytes to the record length so that each record starts on the proper boundary. This length is referred to as the aligned record length. Structures are adjusted to be on 4-byte boundaries.

## Example Calculation

The following is an example for calculating the maximum space required for a hash table using the modulo divide hash algorithm.

|                          |                                                                                        |
|--------------------------|----------------------------------------------------------------------------------------|
| Hash table header:       | 4184 bytes                                                                             |
| Hash data record header: | 24 bytes (the actual definition is 22 bytes, but the next 8-byte boundary is 24 bytes) |
| Data records:            | $nn$ bytes (total number of data records times the aligned record length)              |
| Hash bucket size:        | 24 bytes (the actual definition is 22 bytes, but the next 8-byte boundary is 24 bytes) |

For the sake of this example, assume the following for the source data source to be hashed:

Record length: 121 bytes

Number of records: 2031

In this case, the aligned record length is 128 bytes because the record length of 121 bytes does not fall on an 8-byte boundary. You must add 7 bytes to reach the next 8-byte boundary, which is 128 bytes.

The amount of space required for data records and hash data record headers is calculated as follows:

Data record space = (hash data record header + aligned record length)  $\times$  number of data records

$$= (24 + 128) * 2031$$

$$= 308,712$$

The maximum number of buckets is the next prime number equal to or greater than the number of records in the data source(2031) times the maximum bucket factor of 6.

Next prime  $\geq$  (maximum bucket factor  $\times$  number of records):

$$= \text{next prime } \geq (6 \times 2031)$$

$$= \text{next prime } \geq 12186$$

$$= 12197$$

Therefore the maximum bucket space required is 12,197 buckets times the size of each bucket (24 bytes).  $12,197 \times 24 = 292,728$ .

After you have determined the size of each of the three data areas, you can add them together to obtain the maximum size required for the hash table in memory.

Table size = Hash table header + Data records and hash data headers + Bucket space

$$= 4184 + 308,712 + 292,728$$

= 605,624

Therefore, the `-MT_SIZE` param for the external memory table in this example must be set to a value equal to or greater than 605624. If it is set to a lesser value, such as 500000, which is the default value set for most external memory tables in the CONFCSV at installation, the build of the external memory table will fail. Note that when the Data Source Loader component has finished determining the most efficient hash algorithm to use, it resizes the memory segment to the actual number of bytes used to build the table. This memory table is then flushed to disk (i.e., in the disk location specified by the `_OLTP` assign) as an unstructured data source where it only occupies the space needed.

## CONFCSV Parameters

By setting certain optional parameters for an external memory table assign (e.g., `_OLTP`) in the CONFCSV, the Data Source Loader component generates a hash table processing summary report when you build an external memory table. By default, the report is not suppressed. When the `-SUPPRESS_RPT` param is not present or is set to a value of N, the summary report is generated and written to a spooler or write-only data source location specified for stream data in the assign name specified in the `-STATS_DSRC_ASSIGN` parameter. For example, you could set the `-STATS_DSRC_ASSIGN` to “MTRCRC,” which is write-only data source (WOF) for stream data provided in the default installed CONFCSV configuration. Samples of these assigns and params in the CONFCSV are shown below.

### Assign Entries:

```
"MTRCRC", "Stream", "1.0", "$bard.EDUCdata.mtrcrc", "WOF", "", ""  
  
"INTERCHANGE_SECURITY_OLTP", "Interchange_Security_OLTP", "1.  
0", "$bard.EDUCdata.isecod", "HASHDS", "-MT_SIZE=500000", "-SUPPRESS_  
RPT=N", "-STATS_DSRC_ASSIGN=MTRCRC", "ICHG_SECURITY"
```

### Param Entries:

```
"isecod_MT_SIZE", "1.0", "500000", "bt_stringf", 1, 1, False, ""  
"SUPPRESS_RPT", "1.0", "N", "bt_stringf", 1, 1, False, ""  
"STATS_DSRC_ASSIGN", "1.0", "MTRCRC", "bt_stringf", 1, 32, False, ""
```

Note that the hash table processing summary report is only written if it is not suppressed and a valid write-only location is specified. This summary processing report lists the bucket factor used by the optimum algorithm. You can replace the

maximum bucket factor used in the example calculations above with the bucket factor listed on this report to obtain a close approximation of the required size of the final table built.

Two other optional parameters for external memory table assigns in the CONFCSV control how the Data Source Loader component builds external memory tables. The -KEY\_TYPE parameter indicates whether the key for the hash table is alphanumeric (ASCII or A), binary (BINARY or B), or both alphanumeric and binary (ANY or \*). This parameter defaults to a value of NONE.

The -BLD\_DSRC\_ASSIGN parameter is an optional parameter that is used by those applications that need to connect to a read-only external memory table and that memory table has not already been built. Typically the application could check the error from the data source to determine that the memory table did not exist. If it does not exist, the application could make use of the PreLoad From attribute of the assign and recreate the data source with the read-only flag set to false. This would cause the table to be built using the PreLoad from file as input. After the table was built the application would then disconnect so that the memory table is flushed to disk, and then reconnect with the read-only flag set to true.

## **Hash Table Processing Summary Report**

When the processing summary report is not suppressed and a valid write-only location is specified for an external memory table, the Data Source Load component generates a hash table processing summary report when you rebuild an external memory table. The hash table summary report details the hashing

algorithms and bucket factors used to build the optimum hash table for a file. A sample hash table summary report is shown below, followed by descriptions of the fields that appear on the report.

```
1: load oltp <oltp-assign-name>;
   Table Size:                    5288 bytes

   Total Data Records:           9
   Maximum Record Length:       46 bytes

   Hash Statistics:
   Key Type:                     NONE
   Key Length:                   16 bytes
   Key Algorithm:                DOUBLE BYTE XOR
   Hash Algorithm:               MODULO DIVIDE

   Bucket Factor:                2
   Total Buckets:                19
   Total Buckets Used:           9
   Percent Buckets Used:         47%
   Total Barrels:                1

   Total Collisions:             0
   Percent Collisions:           0%
   Total Collision Chains:        0
   Longest Collision Chain:       0
   Average Collision Chain:       0

   Build Attempts:               20
Load of <oltp-assign-name> successful.
```

**Table Size** — The total size, in bytes, required for the external memory table using the optimum hash algorithm. Note that this may not be the maximum space used because the Data Source Load component tests 108 different algorithms one at a time when building the hash table. Some algorithms may require more space than the optimum algorithm.

**Total Data Records** — The total number of data records included in the external memory table.

**Maximum Record Length** — The maximum length, in bytes, of each record included in the external memory table.



**Key Type** — The type of record key specified for the build of this external memory table. Valid types are: ALL ASCII (alphanumeric), ALL BINARY (the binary data cannot exceed four bytes), ASCII AND BINARY (combination of alphanumeric and binary data), and NONE.

**Key Length** — The total length, in bytes, of the primary key to this external memory table.

**Key Algorithm** — The optimum key algorithm used to hash the key from this data source in building the optimum hash table. The Data Source Loader component tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid key algorithms include the following: SINGLE BYTE XOR, DOUBLE BYTE XOR, TRIPLE BYTE XOR, QAUDRUPLE BYTE XOR, SINGLE BYTE SUMMATION, DOUBLE BYTE SUMMATION, and SINGLE BYTE POLYNOMIAL.

**Hash Algorithm** — The optimum algorithm used to hash key values from this data source in building the optimum hash table. The Data Source Loader component tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. Valid hash algorithms include the following: MULTIPLICATIVE, MID-SQUARE, and MODULO DIVIDE.

**Bucket Factor** — The bucket factor used in building the hash table. The bucket factor is the ratio of the number of buckets that will exist to the actual number of data records in the hash table. The Data Source Loader component tries all combinations of allowed hash algorithms, key algorithms, and bucket factors until a table with the fewest collisions is built. For example, a bucket factor of 3 indicates that there would be three times as many buckets as there are hash records determined by the hash algorithm.

**Total Buckets** — The total number of buckets used for this combination of hash algorithm, key algorithm, and bucket factor.

**Total Buckets Used** — The total number of buckets that actually contain data records for this combination of hash algorithm, key algorithm, and bucket factor.

**Percent Buckets Used** — The percentage of buckets containing data items in the optimum hash table.

**Total Barrels** — The total number of barrels used in the optimum hash table. A barrel is equal to a group of buckets.

**Total Collisions** — The total number of collisions that occurred on the optimum hash table. A collision occurs when two or more data records hash to the same bucket.

**Percent Collisions** — The percentage of collisions that occurred on the optimum hash table. A collision occurs when two or more data records hash to the same bucket.

**Total Collision Chains** — The total number of collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records hash to the same bucket.

**Longest Collision Chain** — The longest collision chain that occurred on the optimum hash table. A collision chain occurs when two or more data records hash to the same bucket.

**Average Collision Chain** — The average length of all collision chains that occurred on the optimum hash table. A collision chain occurs when two or more data records hash to the same bucket.

**Build Attempts** — The total number of combinations attempted to obtain the optimum hash table.

## Terminology

The following terms are essential to understanding hash tables and how they are used in processing.

**Hash table** — A hash table uses a hash function to provide rapid access to data items which are distinguished by some key. Each data item to be stored is associated with a key (e.g, the name of a person). A hash function is applied to the item's key and the resulting hash value is used as an index to select one of a number of hash buckets in a hash table. Records in a hash table are retrieved by exact key value. A hash bucket contains a list of records that were hashed to that specific bucket. The list of records is search sequentially until an entry matches the key value being used.

**Hash bucket** — A notional receptacle, a set of which might be used to apportion data items for sorting or lookup purposes. For example, when you look up a name in the phone book, you typically hash it by extracting its first letter; the hash buckets are the alphabetically ordered letter sections. This term is used as techspeak with respect to code that uses actual hash functions; in jargon, it is used

for human associative memory as well. Thus, two things *in the same hash bucket* are more difficult to discriminate, and may be confused. For example, if you hash English words only by length, you get too many common grammar words in the first couple of hash buckets. Compare to hash collision.

**Hash collision** — A hash collision occurs when two or more different keys hash to the same value, that is, to the same location in a hash table.

**Hash function** — A hash coding function which assigns a data item distinguished by some key into one of a number of possible hash buckets in a hash table. The hash function is usually combined with another more precise function.

For example a program might take a string of letters and put it in one of twenty six lists depending on its first letter. Ideally, a hash function should distribute items evenly between the buckets to reduce the number of hash collisions. If, for example, the strings were names beginning with “Mr.”, “Miss” or “Mrs.” then taking the first letter would be a very poor hash function because all names would hash the same.

## Warmbooting the Transaction Security Services Process

After you have successfully rebuilt an external memory table, you must either warmboot the Transaction Security Services process, or stop and restart the process, before it will begin using the new external memory table. You can stop and restart a Transaction Security Services process from an XPNET network control facility using the STOP PROCESS and START PROCESS commands. You can warmboot Transaction Security Services processes by entering a DELIVER PROCESS command from an XPNET network control facility or by entering a command from the Process Control window on the ACI desktop user interface. Procedures for both means of warmbooting Transaction Security Services processes are provided below. For detailed information on entering STOP PROCESS and START PROCESS commands, refer to the *NET24-XPNET Network Control Command Reference Manual*.

### XPNET Network Control Facility

To warmboot a Transaction Security Services process from an XPNET network control facility for an external memory table, use the following command.

**Syntax**

---

DELIVER PROCESS *object-name*, "ALTER DASRC *oltp-assign-name*"

*object-name* — The symbolic name assigned to a Transaction Security Services process in the XPNET node.

*oltp-assign-name* — The assign name specified for the external memory table in the CONFCSV on the data subvolume. For example, the assign name for the IDESOD external memory table is IBM\_DES\_OLTP.

**Example**

---

The following is an example of a DELIVER PROCESS command sent to a Transaction Security Services process to reread the external memory table for the IBM DES Verification File.

```
deliver process p1a^tss01,"alter dasrc IBM_DES_OLTP"
```

## Process Control Window

To warmboot Transaction Security Services processes from the Process Control window on the ACI desktop user interface for an external memory table, perform the following steps:

1. Log on to the ACI desktop user interface and display the Process Control window by selecting System Operations and then Process Control.
2. In the Destination field, select IS or PCTL from the list of values. For the IS destination, the ALTER command is broadcast to all Transaction Security Services processes configured in the XPNET node for the "IS" service. The destination of PCTL sends commands to all components registered to use the command, regardless of the type of process in which they are found. The PCTL destination causes processes of all types to close and reopen (re-read) the memory table, thereby accessing the new data in processing. When using the PCTL destination, you may see informational event messages logged for any processes that attempt to close and reopen the memory table if they do not actually use the table.
3. In the Command field, select Alter from the list of values.
4. In the Component field, select Data Source from the list of values.

5. In the Command Text field, enter the assign name of the file to be warmbooted. This is the assign name specified for the external memory table in the CONFCSV on the data subvolume. For example, the assign name for the IDESOD external memory table is IBM\_DES\_OLTP.
6. Click Submit (▶) at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window.

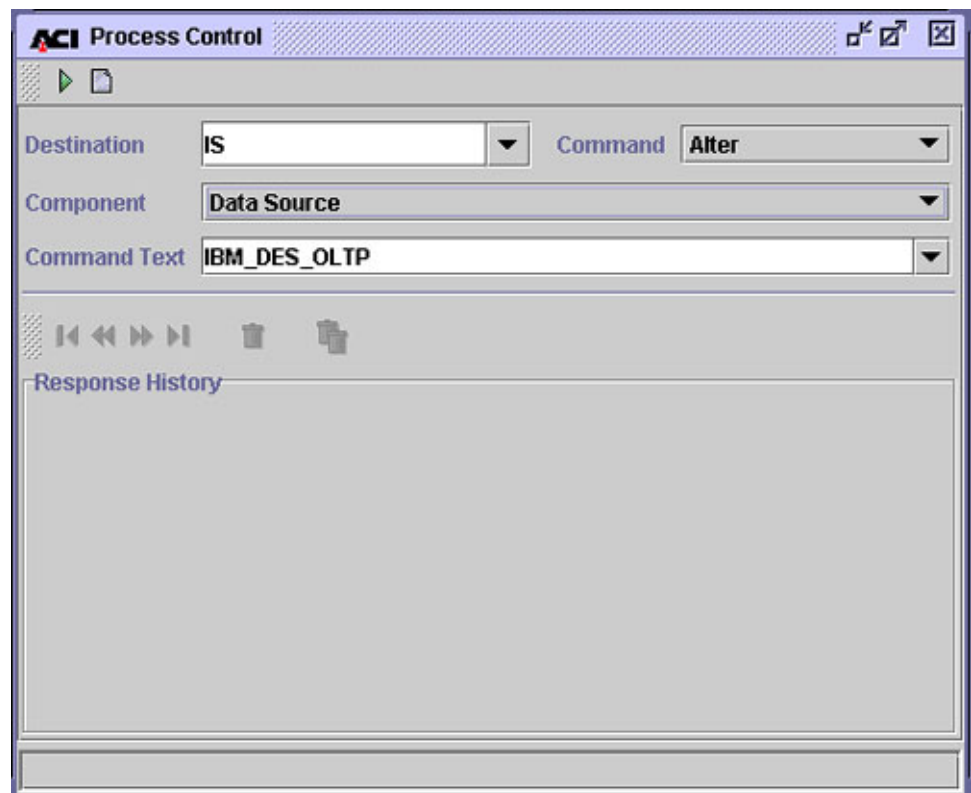
If the table was successfully warmbooted for a process, an event message similar to the following will be displayed on the EMS log.

```
MM-DD-YY;HH:MM:SS.HHH \SYS.$B4XM ACI.1033.1000 60
Mgr PPD:\SYS.$BB4AN Generator: P1A^IAS01
DASRC: Successful warmboot of data source IBM_DES_OLTP
```

If a process does not have the specified external memory table open, the following message is displayed:

```
Warmboot not needed
```

An example of the Process Control window is shown below.



## Internal Memory Tables

Some of the files maintained by the Transaction Security Services process are accessed as internal memory tables during online transaction processing. The Transaction Security Services processes that access these files must be warmbooted when you make a change to the file from which the internal memory tables are built. The following files for the Transaction Security Services process are accessed internally as memory tables:

- Channel Security File (CSECD)\*
- Interface Security File (ISECD)†
- Security Device Configuration File (SECDCD)

\* Data for a channel in the CSECD can be accessed from an internal memory table or from disk, depending on how the CSEC\_MEMORY\_USE Environment attribute is set. When this attribute is set to Y, a limited number of CSECD records are maintained in internal memory. In this case, the MAX\_CSEC\_IN\_MEM Environment attribute specifies the maximum number of CSECD records that can be maintained in memory. When the CSEC\_MEMORY\_USE Environment attribute is set to N, all CSECD records are accessed from disk.

Due to hardware memory and swap space limitations, it may not be possible for a single Transaction Security Services process to maintain all CSECD records in memory if the channel (terminal) population exceeds a maximum number of devices. If you have more CSECD records than can be handled in memory for your hardware configuration, you must either always access CSECD records from disk by setting the CSEC\_MEMORY\_USE attribute to N or limit the number of CSECD records that can be maintained in memory. The MAX\_CSEC\_IN\_MEM Environment attribute limits the number of CSECD records that can be maintained in internal memory when the CSEC\_MEMORY\_USE attribute is set to Y. When the number of CSECD records added to memory reaches this limit and a CSECD record is read from disk, the first entry is deleted from the memory table and the record just read from disk is inserted into the memory table. In essence, records are swapped in and out of memory when this limit is reached.

For CSECD records maintained in memory, recursive function calls are made when either a PIN verify or PIN translate function fails. This is done because the PIN key in memory may not be current. In this case, the CSECD record is re-read from disk and the security device call is made again. For CSECD records accessed from disk, recursive function calls are not made when either a

PIN verify or PIN translate function fails. For MAC verify, PVV generate, MAC generate, and data encrypt/decrypt/translate function calls, the first call is always forced to disk so no recursive calls are ever made.

- † Data for an interface in the ISECD is accessed from an internal memory table only, except for outbound keys, when the interface supports dynamic key management. When an interface supports dynamic key management, outbound keys are always retrieved from the ISECD disk file. Dynamic key management is indicated by a value of “Use import key for both options” or “Use separate keys” in the Options field on the Exchange Information tab of the Interface Security Configuration window. Otherwise, data for an interface is accessed from the external memory table.

When you change key values on the Interface Security Configuration window for an interface that does exchange keys, you do not have to warmboot the Transaction Security Services process to reread the ISECD, or stop and restart the Transaction Security Services process. The Transaction Security Services process reads the new key values from the ISECD disk data source for the next transaction for the interface that requires security key data if it encounters a key synchronization error using the key in the internal memory table or if the key is an outbound key. If, however, you add or delete a record for an interface that does exchange keys, or if you change the value of the Options field on the Exchange Information tab for an interface to start or stop exchanging keys, you must rebuild the Interface extended memory table before warmbooting the Transaction Security Services process to reread the ISECD, or stopping and restarting the Transaction Security Services process.

When you change values on the Interface Security Configuration window for an interface that does not exchange keys, you must always rebuild the Interface extended memory table and warmboot, or stop and restart, the Transaction Security Services process to reread the Interface extended memory table. The new values will be used in processing for the next transaction for the interface that requires security key data.

Refer to the “External Memory Tables” topic for more information on rebuilding external memory tables and warmbooting the Transaction Security Services process for the ISECD external memory table.

The CSECD is maintained from the Channel Security Configuration window, the ISECD is maintained from the Interface Security Configuration window, and the SECD is maintained from the Security Device Configuration window.

When you warmboot a Transaction Security Services process for a data source accessed as an internal memory table using the Alter Data Source process control command, the process deletes all records from internal memory and reloads the memory table as described below. For the CSECD, you can also warmboot a Transaction Security Services process to reread a single record from disk using the CSEC Refresh process control command. In this case, all of the other records in internal memory are retained.

For the CSECD (when the CSEC\_MEMORY\_USE Environment attribute is set to Y) and ISECD, the process adds entries to the internal memory table as each transaction is processed. When a CSECD record is needed, the process first searches for the record in the internal memory table. If the record is not found in memory, the process reads the record from the disk data source and adds it to the internal memory table. If adding the record to memory exceeds the limit imposed by the MAX\_CSEC\_IN\_MEM Environment attribute, the first entry is deleted from the memory table before the record just read from disk is inserted into the memory table. When an ISECD record is needed for an interface that supports dynamic key management, except for outbound keys, the process first searches for the record in the internal memory table. If the record is not found in internal memory, the process reads the record from the disk data source and adds it to the internal memory table. For an outbound key, the record is always read from the disk data source. Thus, the internal memory tables for CSECD and ISECD records are rebuilt dynamically as transactions are processed.

For the SECDCD, the process immediately reloads all records from the disk data source to the internal memory table.

## **Warmbooting the Transaction Security Services Process**

After you have changed data in the CSECD, ISECD, or SECDCD, you must warmboot the Transaction Security Services process before it will begin using the new data in processing. You can warmboot Transaction Security Services processes by entering a DELIVER PROCESS command from an XPNET network control facility or by entering a command from the Process Control window on the ACI desktop user interface. Procedures for both means of warmbooting Transaction Security Services processes are provided below.

### **XPNET Network Control Facility**

To warmboot a Transaction Security Services process from an XPNET network control facility for an internal memory table, use the following command.



**Syntax**

DELIVER PROCESS *object-name*, "ALTER DASRC *data-source-name*"

*object-name* — The symbolic name assigned to a Transaction Security Services process in the XPNET node.

*data-source-name* — The assign name specified for the disk file in the CONFCSV on the data subvolume. The internal memory table is rebuilt from the disk file. The following table lists the assign names for each of the disk files accessed as internal memory tables.

| Assign Name      | TSS File                                     |
|------------------|----------------------------------------------|
| CHANNEL_SECURITY | Channel Security File (CSECD)                |
| ICHG_SECURITY    | Interface Security File (ISECD)              |
| SEC_DEV_CONFIG   | Security Device Configuration File (SECD CD) |

**Example**

The following is an example of a DELIVER PROCESS command sent to a Transaction Security Services process to reload the internal memory table for the Channel Security File.

```
deliver process p1a^tss01,"alter dasrc CHANNEL_SECURITY"
```

**Process Control Window**

To warmboot Transaction Security Services processes from the Process Control window on the ACI desktop user interface for an internal memory table, perform the following steps:

1. Log on to the ACI desktop user interface and display the Process Control window by selecting System Operations and then Process Control.
2. In the Destination field, select IS or PCTL from the list of values. For the IS destination, the ALTER command is broadcast to all Transaction Security Services processes configured in the XPNET node for the "IS" service. The destination of PCTL sends commands to all components registered to use the command, regardless of the type of process in which they are found. The PCTL destination causes processes of all types to close and reopen (re-read) the memory table, thereby accessing the new data in processing. When using

the PCTL destination, you may see informational event messages logged for any processes that attempt to close and reopen the memory table if they do not actually use the table.

3. In the Command field, select Alter from the list of values.
4. In the Component field, select Data Source from the list of values.
5. In the Command Text field, enter the assign name of the file to be warmbooted. This is the assign name specified for the disk file in the CONFCSV on the data subvolume from which the internal memory table is built. The following table lists the assign names for each of the disk files accessed as internal memory tables.

| Assign Name      | TSS File                                    |
|------------------|---------------------------------------------|
| CHANNEL_SECURITY | Channel Security File (CSECD)               |
| ICHG_SECURITY    | Interface Security File (ISECD)             |
| SEC_DEV_CONFIG   | Security Device Configuration File (SECDCD) |

6. Click Submit (▶) at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window.

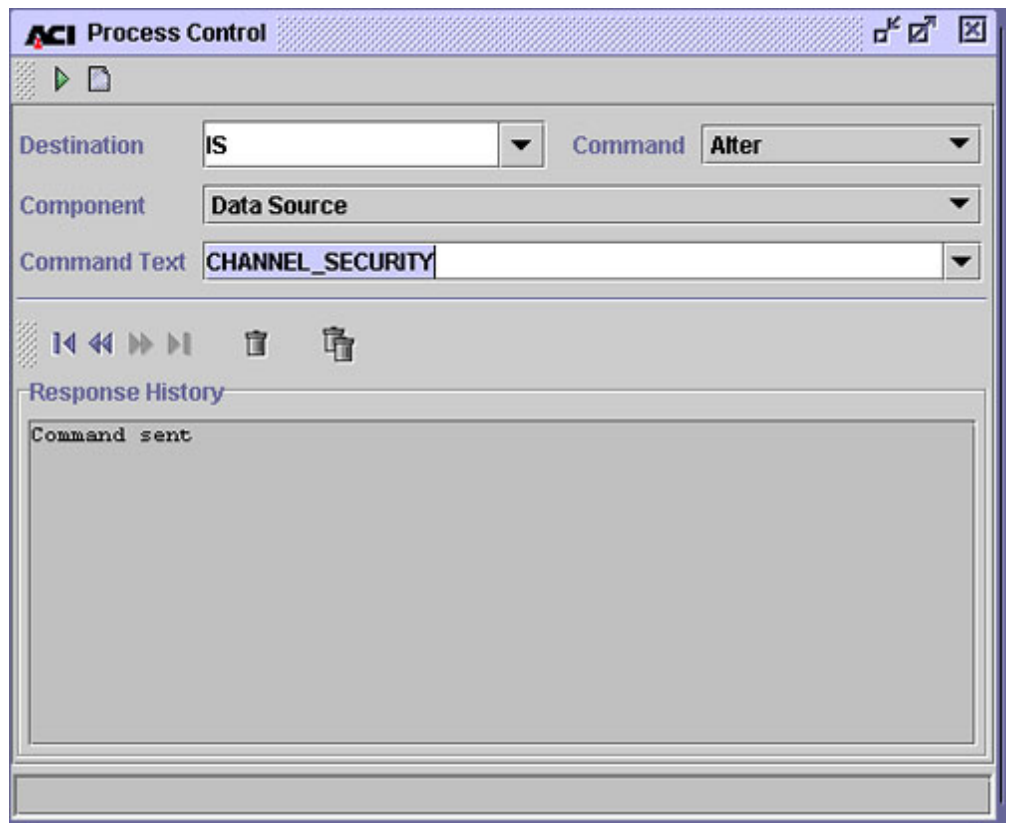
If the table was successfully warmbooted for a process, an event message similar to the following will be displayed on the EMS log.

```
MM-DD-YY;HH:MM:SS.HHH \SYS.$B4XM ACI.1033.1000 60
Mgr PPD:\SYS.$BB4AN Generator: P1A^IAS01
DASRC: Successful warmboot of data source CHANNEL_SECURITY
```

If a process does not have the specified internal memory table open, the following message is displayed:

```
Warmboot not needed for CHANNEL_SECURITY
```

An example of the Process Control window is shown below.



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## 3: Configuration

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This section describes the application configuration requirements for implementing the Transaction Security Services process. Additional configuration requirements for XPNET, TCP/IP, HTTP, and OSS should be determined during installation and are beyond the scope of this guide.

## LCONF Assigns

When implementing the Transaction Security Services process, all SECURE-DEV1, SECURE-DEV2, or SECURE-DEV-xx assigns must be set to the process pair directory (PPD) name of a Transaction Security Services process. BASE24 application processes send messages directly to the Transaction Security Services process or AKDS process identified in the SECURE-DEV1, SECURE-DEV2, or SECURE-DEV-xx assigns using a write/read interface when the SECURE-DEV-TYP param is set to a value of TSS in the LCONF. The SECURE-DEV-ACCESS-TYPE param indicates whether a BASE24 application process communicates with a primary and secondary Transaction Security Services process identified by the SECURE-DEV1 and SECURE-DEV-2 assigns or with up to 20 Transaction Security Services processes identified by SECURE-DEV-xx assigns, where xx is any two characters. When using the SECURE-DEV-xx assigns, the BASE24 application process accesses Transaction Security Services processes on a *round-robin* (i.e., rotating) basis.

**Note:** The AKDS process is a special Transaction Security Services process that contains all of the components of the Transaction Security Services process plus the components required to support the BASE24-atm Automated Key Distribution System add-on product. NCR 5XXX or Diebold 10XX/478X Device Handler processes must use the AKDS process instead of the regular Transaction Security Services process when using the BASE24-atm Automated Key Distribution System add-on product. Configuration requirements for the AKDS process are the same as that of the regular Transaction Security Services process.

## Primary and Secondary TSS Process Access Method

When the SECURE-DEV-ACCESS-TYPE param in the LCONF is set to a value of 0, the BASE24 application process uses the SECURE-DEV1 and SECURE-DEV2 assigns to access Transaction Security Services processes.

The SECURE-DEV1 assign specifies the PPD name of the primary Transaction Security Services process for a BASE24 application process, while the SECURE-DEV2 assign specifies the PPD name of the secondary Transaction Security Services process for a BASE24 application process. You can assign different Transaction Security Services processes to different BASE24 application processes by configuring multiple SECURE-DEV1 and SECURE-DEV2 assigns in the LCONF and assigning each to a specific symbolic process name in the READ BY field on the LNCF Assign screen. In addition, the USER FIELD for the SECURE-DEV1 and SECURE-DEV2 assigns on the LNCF Assign screen must be left blank. The security utilities bound into BASE24 processes use this

value to define the device subtype. Since there are no valid device subtypes associated with calls to the Transaction Security Services process, the utilities generate an error event message when a value is present.

When a transaction is received by a BASE24 process that requires the use of a security module, the BASE24 process first attempts to communicate with its primary Transaction Security Services process. If the request to the primary Transaction Security Services process times out or the primary Transaction Security Services process is considered inoperative, the BASE24 process attempts to communicate with its secondary Transaction Security Services process to complete the transaction.

The security utilities for a BASE24 process can report a 9100 error message with hardware error 201 (the current path to the device is down) when the Transaction Security Services processes specified by the SECURE-DEV1 and SECURE-DEV2 assigns have been stopped and restarted after the BASE24 process is initialized. In this case, you must warmboot, or stop and restart, the BASE24 process reporting the error.

**Note:** When the security device associated with the primary Transaction Security Services has reached the [maximum consecutive error count](#) and the security device associated with the secondary Transaction Security Services is in use, you must warmboot the BASE24 process to start using the security device associated with the primary Transaction Security Services process again.

You can also use the SECURE-DEV-MONITOR-THRESHOLD param to check the secondary Transaction Security Services process at specified intervals or balance requests evenly between the primary and secondary Transaction Security Services processes.

## Checking the Secondary Transaction Security Services Process

To minimize the possibility of both Transaction Security Services processes becoming inoperative, the BASE24 system can be configured to occasionally perform one request using the secondary Transaction Security Services process instead of the primary Transaction Security Services process. The request is sent to the secondary Transaction Security Services process at intervals based on the number of requests performed by the primary Transaction Security Services process.

Each BASE24 process using the Transaction Security Services process maintains its own request counter. The value specified in the SECURE-DEV-MONITOR-THRESHOLD param is the number of requests processed by the primary Transaction Security Services process before the BASE24 process sends a single request to the secondary Transaction Security Services process. For example, if this param were set to a value of 1000, the BASE24 process would send 999 requests to the primary Transaction Security Services process before it would send one request to the secondary Transaction Security Services process, thereby checking its availability. The request counter is reset each time the threshold is reached.

Occasionally using the secondary Transaction Security Services process permits the identification and resolution of any problems with the secondary Transaction Security Services process while the primary Transaction Security Services process is functioning properly.

When using the SECURE-DEV-MONITOR-THRESHOLD param to periodically check the availability of the secondary Transaction Security Services process, ACI recommends that the primary Transaction Security Services process assigned to a BASE24 application process run in the same CPU as the requesting BASE24 process for performance reasons. By sending the majority of requests to a Transaction Security Services process running in the same CPU, the additional overhead associated with communicating over the inter processor bus between CPUs is avoided.

## **Balancing Requests Between Transaction Security Services Processes**

Although you could use the SECURE-DEV-MONITOR-THRESHOLD param to balance the number of requests processed by the primary and secondary Transaction Security Services processes, it is not recommended because the Transaction Security Services process automatically balances requests across all security devices. The Transaction Security Services process alternates requests among all security devices (or interface module processes) configured in the Security Device Configuration File one request at a time.

When this param is set to a value of 1, each BASE24 process alternates requests between the primary and secondary Transaction Security Services processes. Each BASE24 process uses the primary Transaction Security Services process for one transaction, uses the secondary Transaction Security Services process for one transaction, and then repeats the cycle.



## Round-Robin TSS Process Access Method

When the SECURE-DEV-ACCESS-TYPE param in the LCONF is set to a value of 1, the BASE24 application process uses the SECURE-DEV-*xx* assigns to access Transaction Security Services processes. You can configure a BASE24 application process to access up to twenty different Transaction Security Services processes on a *round-robin* (i.e., rotating) basis by configuring up to twenty SECURE-DEV-*xx* assigns, where *xx* is any two characters, with each assign specifying a different Transaction Security Services process.

For this method of Transaction Security Services process access, the security utilities in the BASE24 application process maintains the status of each Transaction Security Services process in the Security Device Control Block (SDCB) as well as a time to attempt to reopen the Transaction Security Services process. In addition, the SDCB contains fields indicating the next Transaction Security Services process to use, the number of Transaction Security Services processes open, and the time to delay before attempting to reinstate a Transaction Security Services process. This reinstate delay time is set by the SECURE-DEV-REINSTATE-TIME param.

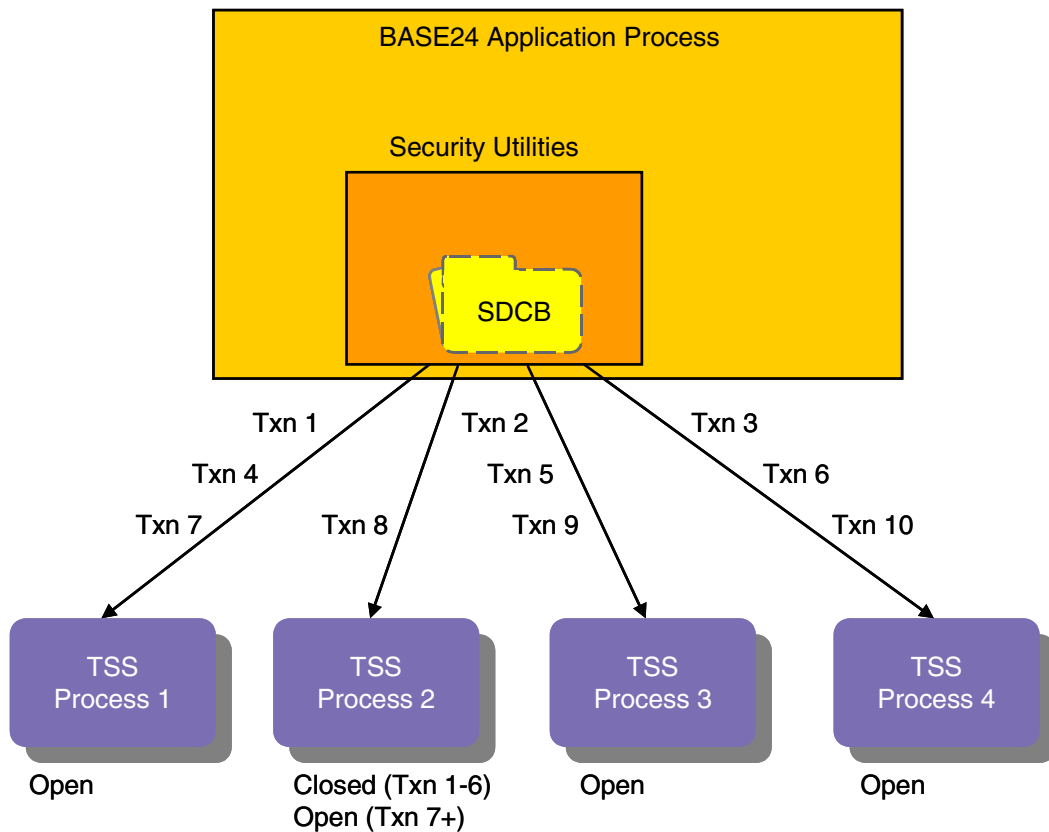
As each call to a Transaction Security Services process is made, the security utilities check the index in the SDCB to determine which Transaction Security Services process to use. The security utilities then increment the index to point to the next Transaction Security Services process to be used for the next transaction. If the SDCB flag indicates that a particular Transaction Security Services process is down, that Transaction Security Services process is skipped in the rotation.

If a Transaction Security Services process is marked down, the security utilities checks the reinstate time. If the reinstate time has passed, the security utilities attempt to restart the Transaction Security Services process. If the restart is successful, the SDCB is updated to indicate that the Transaction Security Services process is in an open state.

The security utilities generate an event message each time a Transaction Security Services process state is modified. The security utilities also log the status of the Transaction Security Services process and how many Transaction Security Services processes are up. If no Transaction Security Services processes are up, the utilities generates an event message indicating this fact.

The security utilities reads the LCONF for SECURE-DEV-*xx* assigns until no more assigns are found or until twenty assigns have been read. For each assign, the utilities initializes the SDCB with the Transaction Security Services process name, file number, subtype, fail count, state and reopen time.

Use of the round-robin Transaction Security Services process access method is depicted in the following illustration. In this example, there are four Transaction Security Services (TSS) processes. TSS process 2 is marked as closed in the SDCB when the BASE24 application process requires a hardware security device for transactions 1–6. Therefore, it is excluded from the round-robin distribution. By the time transaction 7 needs to be processed, TSS process 2 is now marked as open and is reinstated into the round-robin distribution and receives transaction 8.



## PPD Names

The PPD names entered for the SECURE-DEV1, SECURE-DEV2, or SECURE-DEV-xx assigns must match the PPD names assigned to the primary, secondary, or multiple Transaction Security Services processes in the XPNET configuration.

The Security Device Name field on the Security Device Configuration window of the ACI desktop contains an assign name for a particular security device. All Transaction Security Services processes can access all of the security devices defined in this window. This assign name is associated with the actual PPD name

of the security device (i.e., the SYSGEN utility, DSC, or SCF name of the device) or the interface module process (i.e., Boxcar, Racal Access Module, HSM Interface) in the System Interface Message Delivery Service Destination File (MDSDEST) on the data subvolume. This file must be created on the HP NonStop system when installing the ACI desktop user interface. For detailed information on creating the MDSDEST, refer to [section 7](#).

**Note:** All Atalla implementations using the TCP/IP protocol require the Boxcar process.

## LCONF Params

As stated previously, the SECURE-DEV-TYP param in the LCONF must be set to a value of TSS when implementing the Transaction Security Services process.

## Unused Params

When using the Transaction Security Services process, two params in the LCONF are not used because they are replaced by similar parameters specified in the Security Device Configuration window of the ACI desktop. The security utilities generate a warning event message when either of the following params are read from the LCONF and the SECURE-DEV-TYP param is set to TSS:

- SECURE-DEV-CONF-CMD100  
This param is replaced by the Initialization String field on the Security Device Configuration window of the ACI desktop.
- SECURE-DEV-CONF-TXT  
This param is replaced by the Configuration String field on the Security Device Configuration window of the ACI desktop.

The RSM-TERM-MASTER-KEY param is replaced by the RSM\_TERM\_MASTER\_KEY attribute set in the Transaction Security Services environment. Refer to the [“Transaction Security Services Attributes”](#) topic later in this section for a description of this attribute.

## Access Method

The SECURE-DEV-ACCESS-TYPE param indicates whether a BASE24 application process communicates with a primary and secondary Transaction Security Services process identified by the SECURE-DEV1 and SECURE-DEV-2 assigns or with up to 20 Transaction Security Services processes identified by SECURE-DEV-xx assigns. Valid values are as follows:

- 0 = Access primary and secondary Transaction Security Services processes using the SECURE-DEV1 and SECURE-DEV2 assigns. Default.
- 1 = Access multiple Transaction Security Services processes on a round-robin basis using the SECURE-DEV-xx assigns.

## Reinstate Time

The SECURE-DEV-REINSTATE-TIME param indicates the number of minutes from the time a Transaction Security Services process is marked closed to the time the BASE24 application process attempts to restart the Transaction Security Services process. A Transaction Security Services process is marked closed when three or more consecutive critical errors have occurred since initialization or warmboot. The valid range for this param is 0–32000. A value of 0 indicates that the BASE24 application process does not attempt to reopen a Transaction Security Services process based on the time interval. This param is only used when the SECURE-DEV-ACCESS-TYPE param is set to a value of 1.

The SECURE-DEV-REINSTATE-TIME param provides an additional means of determining when the BASE24 application process attempts to reopen a Transaction Security Services process. The reopen is attempted either when the given time interval specified by this param has been reached or when the number of times the device has come up in the SDCB round-robin list since the last attempt to open the process is equal to the value of the SECURE-DEV-FAIL-NOTIFY param. You can configure both the SECURE-DEV-REINSTATE-TIME and SECURE-DEV-FAIL-NOTIFY when using the SECURE-DEV-xx assigns. If both params are configured, then the Transaction Security Services process is reopened when either threshold is reached.

## Software PIN Verification

For BASE24-atm Authorization processes and BASE24-pos Device Handler/Router/Authorization processes, a param called SOFTWARE-PIN-VERIFICATION-ALWD, indicates whether PIN verification, card verification, or PIN offset generation is allowed to be performed in software or in the clear within any BASE24-atm Authorization processes and BASE24-pos Device Handler/Router/Authorization process in the BASE24 system when the SECURE-DEV-TYP param is set to a value of TSS. A value of Y in this param indicates that PIN verification, card verification, or PIN offset generation is allowed in software or in the clear when the Transaction Security Services process is used. A value of N in this param indicates that PIN verification, card verification, or PIN offset generation is not allowed in software or in the clear when the Transaction Security Services process is used. In the latter case, for any transactions in which PIN verification, card verification, or PIN offset generation is attempted in software or in the clear, the transaction is denied. This param defaults to a value of N. The Transaction Security Services process itself does not perform PIN verification, card verification, or PIN offset generation in software or in the clear. This processing must be performed within a BASE24 authorization process.

## Diebold PIN Verification

When using Diebold PIN verification, the ATM-4000-MASTER param for BASE24-atm Authorization processes and the POS-4000-MASTER param for BASE24-pos Device Handler/Router/Authorization processes are not used and do not need to be configured. The Transaction Security Services process controls downloading of the Diebold number table rather than the BASE24 authorization processes.

## Partial Triple DES

For Interchange Interface processes, the DES-TRIPLE-SINGLE param must be set to a value of N. This prevents an Interchange Interface process from attempting to read the Key 6 File (KEY6), which is not used in processing by the Transaction Security Services process.

## Changed Param Usage

Three existing LCONF params, SECURE-DEV-MONITOR-THRESHOLD, SECURE-DEV-FAIL-NOTIFY, and SECURE-DEV-TIM-LMT have different meanings and usage when the Transaction Security Services process is implemented.

The SECURE-DEV-MONITOR-THRESHOLD param specifies the number of requests a BASE24 process sends to its primary Transaction Security Services process before it sends a single request to its secondary Transaction Security Services process. You can use this param to check the secondary Transaction Security Services process at specified intervals or balance requests evenly between the primary and secondary Transaction Security Services processes. Refer to the “Checking the Secondary Transaction Security Services Process” and “Balancing Requests Between Transaction Security Services Processes” topics provided earlier for detailed information on how this param is used with Transaction Security Services processes.

The SECURE-DEV-FAIL-NOTIFY param specifies the intervals (i.e., the number of transactions processed) at which the security utilities attempt to automatically reopen a Transaction Security Services process when the process has been marked closed due to three or more consecutive critical errors since initialization or warmboot. For example, if this param is set to 20, the security utilities will attempt to reopen the Transaction Security Services process once after the third failure to

communicate with it, and then once every twenty transactions afterward. This param applies when using both the primary/secondary and round-robin methods to access Transaction Security Services processes.

The SECURE-DEV-TIM-LMT specifies the maximum time, in hundredths of a second, to wait for a response from the Transaction Security Services process before the request is considered to be timed out. This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in seconds, to wait for a response from an HSM before the Transaction Security Services process considers the request to be timed out. When a call to the HSM times out, the Transaction Security Services process increments the consecutive retries counter by 1. If the maximum retries (hardcoded to 3) has not been reached, the process attempts the HSM call again using the next available HSM from the pool of HSMs, which may or may not be the same HSM on which the timeout occurred.

The SECURE-DEV-TIM-LMT param value must be set to a value large enough to include the Timeout Limit field value plus the time required to send a request and receive the response from the Transaction Security Services process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window.

## Logical Network Identifier

In certain logical network configurations, it may be desirable for performance reasons to partition the CSECD and ISECD by logical network. Partitioning is also required if you have duplicate device names between logical networks. For example, when separate BASE24 logical networks reside on separate HP NonStop systems, each logical network would have its own CRDVD, DNTD, IDESD, EMVSD, VPVVD, SECD, and KEYF. The CSECD and ISECD could be partitioned by logical network using the LN-IND param as described below.

The LN-IND param contains a two-character hexadecimal code representing the logical network name specified in the LOGICAL-NET param for this LCONF. The value of this param is used to support the Transaction Security Services process for multiple BASE24 logical networks residing on separate HP NonStop systems. It enables the physical partitioning of the CSECD and ISECD accessed by the Transaction Security Services process. Valid values are 0–F.

When adding or updating a BASE24-atm or BASE24-pos Terminal Data files record, the ATD and PTD requester and server processes add the value of this param to the beginning of the terminal ID to create key locator values that are

placed in hardware-encrypted key fields (if the SECURE-DEV-TYP param is set to a value of TSS). If the terminal ID for the record being added or updated has more than 14 characters, the last one or two characters are truncated when the key locator value is created. If the LN-IND param does not exist, key locators are set to the full value of the terminal ID.

TSS database conversion programs also use a runtime parameter called LN-IND to create a key locator value when converting existing BASE24 files. When they are run on a BASE24-atm Terminal Data file or BASE24-pos Terminal Data file, the two-character LN-IND value is added to the beginning of the 1–16 character terminal ID. If the terminal ID for the terminal record being converted has more than 14 characters, the last one or two characters are truncated when the key locator value is created.

When converting Key File (KEYF) or Key 6 File (KEY6) records to the TSS database, the two-character LN-IND runtime param value is added to the beginning of a 14-character key locator index value generated by the conversion program.

If you need to partition the TSS database by logical network, the value of the LN-IND LCONF param and the LN-IND runtime parameter used by the TSS database conversion programs must be set to the same value.

If you do not need to partition the TSS database, the LN-IND LCONF param should not be configured and the LN-IND runtime parameter used by the TSS database conversion programs must be set to a value of 00.

The following BASE24 processes use the LN-IND LCONF param:

BASE24-atm Terminal Data files requester and server objects  
BASE24-pos Terminal Data files requester and server objects

## Multiple Logical Networks

If the Transaction Security Services process is implemented into a BASE24 system with multiple logical networks and a separate TSS database for each network, the MULT-LN-PIN-KEY LCONF param is required to ensure that PINs are properly translated when a transaction is SPROUTE-routed from one logical network to another using the CP file type.

In BASE24 logical networks using the Transaction Security Services process, the MULT-LN-PIN-KEY param contains a key locator value to a common single-length PIN encryption key in the ISECD. You must manually add this record to



the ISECD in each TSS database. In BASE24 logical networks that do not use the Transaction Security Services process, the param contains an encrypted single-length PIN encryption key. The MULT-LN-PIN-KEY param can contain up to 16 hexadecimal characters. PIN blocks translated using the MULT-LN-PIN-KEY param are always converted to ANSI PIN block format.

The following processes use this param, however, it must be configured with a READ BY value of all asterisks.

BASE24-atm Authorization  
BASE24-atm ISO Host Interface  
BASE24-pos Device Handler/Router/Authorization  
BASE24-pos ISO Host Interface

BASE24-atm Authorization processes and BASE24-pos Device Handler/Router/Authorization processes use this param when processing transactions that meet the following conditions:

- The prefix for the transaction was not found in the Card Prefix File (CPF) for the current logical network.
- The transaction is SPROUTE-routed to another logical network using the LGNT routing method and the CP file type.
- The transaction contains an encrypted PIN block.
- The MULT-LN-PIN-KEY param was successfully read during initialization or warmboot processing and contained valid hexadecimal characters.

If the above conditions are met, the process translates the PIN block in the internal message to encryption under the PIN encryption key specified by the MULT-LN-PIN-KEY and adds the Multiple Logical Network token to the transaction. If the transaction is a PIN Change transaction, the new PIN block is translated in the same manner.

When a BASE24-atm Authorization, BASE24-pos Device Handler/Router/Authorization, BASE24-atm ISO Host Interface, or BASE24-pos ISO Host Interface process receives a transaction containing the Multiple Logical Network token, it sets the ORIGINATOR field in the internal message to 1 (interface). This serves two purposes. If you are using a Thales e-Security HSM, the common PIN encryption key must be a Zone (interface) PIN key. If the Transaction Security Services process is used in the current logical network, this value allows the process to retrieve the common PIN encryption key from the ISECD.

## Security Device Configuration Window Parameters

Each hardware security device in your system must be configured on the Security Device Configuration window. This window enables you to set several parameters specific to the type of hardware security device used. A sample of the Security Device Configuration window is shown below, followed by descriptions of its fields.

**Note:** When you add a new security device or change the parameters for a security device, you must [warmboot the Transaction Security Services process to reread the internal table for the Security Device Configuration File \(SECD CD\)](#). When you add or upgrade security device, you must also [initialize the security device](#) before the firmware version and maximum buffer length are displayed on this window for the device.

**Security Device Configuration - ATALLA1**

Security Device Name: **ATALLA1**

Security Device Type: **Atalla** Hardware Model: **Atalla Ax000**

Protocol/Interface: **Async - NSK** Key Scheme: **AKB - Atalla**

Status: **Active** Reset Timer Limit: **0**

Maximum Consecutive Errors: **5** Error Notify Interval: **5**

Timeout Limit: **5** Enable PKI: ☒

Maximum Buffer Length: **0** Firmware Version:

Initialization String: **<100#2C533EF4938B7E3F625344049FEF#2A7B 2445#>**

Configuration String:

1 - 08:51 AM > Successful view of ATALLA1 .

## Security Device Name

The Security Device Name field contains an assign name for a particular security device. All Transaction Security Services processes can access all of the security devices defined in this window. This assign name is associated with the actual PPD name of the security device (i.e., the SYSGEN utility, DSC, or SCF name of the device) or the interface module process (i.e., Boxcar, Remote Application Manager, Racal Access Module, HSM Interface) in the System Interface Message Delivery Service Destination File (MDSDEST) on the data subvolume. This file must be created on the HP NonStop system when installing the ACI desktop user interface. For detailed information on creating the MDSDEST, refer to [section 7](#).

## Security Device Type

The Security Device Type field identifies the manufacturer of the hardware security device. The Transaction Security Services process currently supports devices manufactured by Atalla, Thales e-Security, Eracom, and Bull (BNTng models).

**Note:** You cannot configure devices from more than one vendor in the same logical network. For example, you cannot combine Thales and Atalla devices because they each only recognize their own scheme for storing cryptograms. The two HSM brands are not interoperable. For example, a Thales HSM cannot interpret the format of an Atalla-encrypted key block. You can, however, mix different devices in the same BASE24 system using separate logical networks if a single vendor type is used in each logical network and each logical network is configured with its own TSS database. To route between the logical networks, you would need to configure LGNT routing (CPF routing) over SPROUTE routing and configure the Authorization process or module to translate the PIN block to encryption under a common intermediate key configured on all logical networks.

## Hardware Model

The Hardware Model field indicates the specific model of the hardware security device for the vendor specified in the Security Device Type field. The Transaction Security Services process uses the settings in this field and the Protocol/Interface

and Key Scheme fields to determine the command functions and function parameters it can send to the security device. The possible hardware model values for each vendor are described in the following table.

| <b>Vendor</b> | <b>Hardware Model Value</b> | <b>Description</b>                                                    |
|---------------|-----------------------------|-----------------------------------------------------------------------|
| Atalla        | Atalla Ax000                | Specifies Atalla A8000, A9000, and A10000E NSPs.                      |
|               | Atalla Ax100                | Specifies Atalla high performance A8100, A9100, and A10100 NSPs.      |
| Thales        | Thales RG7400               | Specifies Thales e-Security RG7400 HSMs.                              |
|               | Thales RG71x0               | Specifies Thales e-Security RG7100 and RG7110 HSMs.                   |
|               | Thales RG73x0               | Specifies Thales e-Security RG7300 and RG7310 HSMs.                   |
|               | Thales HSM8000              | Specifies Thales e-Security HSM 8000 devices.                         |
| Eracom        | Eracom Mark II              | Specifies Eracom ProtectHost White Mark II host security modules.     |
|               | Eracom AMB                  | Specifies Eracom ProtectHost White AMB host security modules.         |
| Bull BNTng    | BNTng V7                    | Specifies Bull BNTng HSMs running with version 7 or greater software. |

## Protocol/Interface

The Protocol/Interface field indicates the protocol or interface process used to communicate with the hardware security device. Only values that end with “NSK” are used with the Transaction Security Services process. Values that end with

“UNIX” or “IBM” are only applicable to BASE24-es systems. The possible values for this field depend on the hardware vendor and model as shown in the following table.

| Vendor     | Hardware Model   | Protocol/Interface Value | Description                                                                                                                                        |
|------------|------------------|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Atalla     | N/A              | Async – NSK              | Specifies an asynchronous connection to the NSP.                                                                                                   |
|            |                  | TCP/IP – Boxcar NSK      | Specifies the Boxcar process to communicate with the NSP using TCP/IP.                                                                             |
| Thales     | Thales RG7400    | Async – NSK              | Specifies an asynchronous connection to an RG7400 HSM.                                                                                             |
|            | Thales RG73x0    | SDLC - HIP NSK           | Specifies the HSM Interface Process (HIP) to communicate with an RG73x0 HSM using a synchronous data-link control (SDLC) emulation mode connection |
|            | All other models | TCP/IP – RAM NSK         | Specifies the RAM process to communicate with the HSM using TCP/IP.                                                                                |
|            |                  | UDP/IP – URAM NSK        | Specifies the URAM process to communicate with the HSM using UDP/IP.                                                                               |
| Eracom     | N/A              | Async – NSK              | Specifies an asynchronous connection to the HSM.                                                                                                   |
|            |                  | Ethernet - Blocker NSK   | Specifies the Blocker process to communicate with the HSM using the Ethernet protocol.                                                             |
| Bull BNTng | BNTng V7         | TCP/IP – RAM NSK         | Specifies the RAM process to communicate with the HSM using TCP/IP.                                                                                |

**Note:** For Eracom and Thales e-Security devices, the value of the Protocol/Interface field also indicates whether the Transaction Security Services process communicates with the device with bracketed or unbracketed messages. Bracketed messages include start-of-text (STX) and end-of-text (ETX) control characters. Bracketed messages are used with the asynchronous protocol (i.e., Async – NSK) and are not used with the TCP/IP or UDP/IP protocols.

## Key Scheme

The Key Scheme field indicates the key block format used by Atalla NSPs or the key encryption scheme used by Thales e-Security HSMs. This field is not applicable for Eracom host security modules (HSMs) and Bull BNTng hardware security modules (HSMs). The possible values for this field are described in the following table.

| Key Scheme Value    | Description                                                                                                                |
|---------------------|----------------------------------------------------------------------------------------------------------------------------|
| N/A                 | Not applicable. This value is set for Eracom host security modules (HSMs) and Bull BNTng hardware security modules (HSMs). |
| AKB – Atalla        | Specifies the Atalla Key Block (AKB) format.                                                                               |
| Variant – Atalla    | Specifies the standard Atalla (variant) key block format.                                                                  |
| ANSI X9.17 – Thales | Specifies Thales' old (ANSI X9.17 method) non-variant (32H) key encryption scheme.                                         |
| Variant – Thales    | Specifies Thales' variant (1A+32H) key encryption scheme.                                                                  |

## Status

The Status field displays the current logical status of a device. This field is set to Active when the device is intentionally placed into service (e.g., by sending an ENABLE process control command to the device) or Inactive when the device is intentionally removed from service (e.g., by sending a DISABLE process control command to the device). Security devices are also marked as Inactive while RSA key generation in progress for the Automated Key Distribution System.

The value of the Status field is only read from disk when a security device is initialized or when you warmboot the Security Device Configuration File (SECD CD). Thus, you can preconfigure security devices on the Security Device Configuration window before they are put into service.

**Note:** The status of a device must be Active and the consecutive error count must be less than the maximum consecutive limit for a security device to be used by the Transaction Security component in transaction processing. If the consecutive error

count for a device is exceeded, a device may still display a logical status of Active even though the device will no longer be used in processing until the error counter is reset to 0.

## **Reset Timer Limit**

The Reset Timer Limit field specifies the number of seconds the Transaction Security component waits before automatically attempting to re-establish communications with a hardware security device after the device has reached its maximum consecutive errors limit. Valid values are 0–9999 seconds. A value of 0 in this field indicates that the Transaction Security component does not automatically attempt to reconnect with the device. If you set this field to a value greater than 0, the Transaction Security component sends an echo test, or HSM status request message to the device at intervals defined by this field until the device successfully responds to the request. For example, if this field is set to 30 seconds, the Transaction Security component waits 30 seconds after the device is initially marked inactive and then sends an echo test or HSM status message request to the device. If Transaction Security component does not receive a successful response to the message after another 30 seconds have elapsed, it sends another echo test or HSM status message to the device. This continues indefinitely until a successful response is received. At that time, the consecutive error count is reset to 0 and the Status field is set to Active. Note that a device may still display a logical status of Active after the maximum consecutive error limit is reached until communications are re-established with the device.

## **Maximum Consecutive Errors**

The Maximum Consecutive Errors field defines the maximum number of consecutive critical errors that can be returned from a security device before it is marked down and taken out of service. When a security device is taken out of service, it is no longer included in the rotation (distribution in a round-robin manner) of transactions sent from the Transaction Security Services process. The value of the Reset Timer Limit field indicates whether the process attempts to automatically re-establish communications with the device when the maximum consecutive error limit is reached.

## **Timeout Limit**

The Timeout Limit field value specifies the maximum time, in seconds, to wait for a response from a security device before the Transaction Security Services process considers the request to be timed out. When a call to the security device times out,

the Transaction Security Services process increments the consecutive retries counter by 1. If the maximum retries (hardcoded to 3) has not been reached, the process attempts the security device call again using the next available security device from the pool of security devices, which may or may not be the same security device on which the timeout occurred.

## Error Notify Interval

The Error Notify Interval field specifies the interval (i.e., the number of times the error occurs) at which critical event messages are generated indicating the error. Valid values are 1–99. The default is 10.

## Public Key Cryptography

A check mark in the Enable PKI field indicates that the security device is used for public key cryptography (PKI) functions required for the BASE24-atm Automated Key Distribution System add-on product. Refer to the “[Automated Key Distribution System](#)” appendix for detailed information on PKI command functions.

## Maximum Buffer Length

The Max Buffer Length field displays the maximum input buffer length, in bytes, supported by the device. This is a display-only field that is automatically set when a security device is initialized. No data is displayed in this field until the device is initialized.

For Atalla NSPs, this field displays the maximum MAC data buffer length (e.g., 4096 bytes).

For Eracom ProtectHost White Mark II host security modules (HSMs), the maximum buffer length depends on the protocol used to connect to the HSM as follows:

| Protocol | Maximum Buffer Length |
|----------|-----------------------|
| Ethernet | 1496 bytes            |
| TCP/IP   | 2048 bytes            |
| Async    | 4096 bytes            |



For Thales e-Security HSMs, the value of this field is set according to the I/O Buffer Size value returned from the HSM in response to the HSM status message sent to the HSM when it is initialized as shown below.

| <b>I/O Buffer Size</b> | <b>Maximum Buffer Length</b> |
|------------------------|------------------------------|
| 0                      | 2000 bytes                   |
| 1                      | 8000 bytes                   |
| 2                      | 16000 bytes                  |
| 3                      | 32000 bytes                  |

**Note:** Although the maximum buffer length for Thales e-Security HSMs allows from 2000–32000 bytes, the MAC Generate/Verify (MS) command is restricted to 1975 bytes of MAC data.

For Eracom ProtectHost White AMB host security modules, the maximum buffer length returned in the response to an Echo Test (0000) function is displayed in this field. If the Echo Test (0000) function fails, the default values listed above for Eracom ProtectHost White Mark II host security modules are used.

For Bull BNTng HSMs, the maximum buffer length for the TCP/IP protocol is 2048 bytes.

## **Firmware Version**

The Firmware Version field is a display-only field that displays the version of software for Atalla NSPs (e.g., 2.62), the firmware version (e.g., 0352.E001 or 1053.0801) for Thales e-Security HSMs, the specification, revision, and customization level for Eracom host security modules, and the version, release, build, and software type for Bull BNTng HSMs. This field is automatically set for a security device when that device is initialized. No data is displayed in this field until the device is initialized:

- For Atalla NSPs, the Transaction Security Services process sends the 1110 command to all configured NSPs during initialization to obtain the software version information.
- For Thales e-Security HSMs, the process sends the HSM Status (N0) command to all configured HSMs to obtain the firmware version.

- For Eracom ProtectHost White AMB host security modules (HSMs), the process sends the Echo Test (0000) function to all configured HSMs to obtain the specification, revision, and customization level. The Eracom ProtectHost White Mark II host security module does not support an equivalent function.
- For Bull BNTng HSMs, the component sends the self-test (F2) function to all configured Bull BNTng HSMs to obtain the major version number, release number, build number, and software type identifier (e.g., I for International). These values are returned in the DF 1E tag of the FF 12 self-test response and are displayed in *VVRR-EEL* format, where *VV* is the major version number, *RR* is the release number, *EE* is the build number, and *L* indicates the software type (e.g., I=International).

Because vendors sometimes support different functionality across software, specification and revision, or firmware releases for a given security device model, the Transaction Security Services process uses this information to determine which command functions are available for each configured device.

## Initialization String

The Initialization String field defines the text that the Transaction Security Services process sends to an Atalla Network Security Processor (NSP) model A10000E in a 100 command during initialization or warmboot. This field does not apply to Thales e-Security, Eracom, or Bull BNTng hardware security devices or Atalla NSP models other than the A10000E. This field replaces the LCONF param SECURE-DEV-CONF-CMD100 when using the Transaction Security Services process with an Atalla A10000E model. Refer to the description of the SECURE-DEV-CONF-CMD100 param in the ***BASE24 Transaction Security Manual*** for detailed information on the data to enter in this field.

## Configuration String

The Configuration String field defines the configuration text the Transaction Security Services process sends to an Atalla Network Security Processor (NSP) model A10000E in a 101 command during initialization or warmboot. This field does not apply to Thales e-Security, Eracom, or Bull BNTng hardware security devices or Atalla NSP models other than the A10000E. This field replaces the LCONF param SECURE-DEV-CONF-TXT when using the Transaction Security Services process with an Atalla A10000E model. Refer to the description of the SECURE-DEV-CONF-TXT param in the ***BASE24 Transaction Security Manual*** for detailed information on the data to enter in this field.

**Note:** When using the Boxcar process to interface with Atalla NSPs, you should configure all command options in the NSP's CONFIG.PRM file rather than using this field. For more information, refer to the "Option 023 and the Boxcar Interface Module" topic in the *BASE24 Transaction Security Manual*.

## Transaction Security Services Attributes

Configuration of the following attributes for the Transaction Security Services environment must be considered when implementing the Transaction Security Services process. These attributes are set on the Environment Management window of the ACI desktop user interface. If you change an environment attribute on the Environment Configuration window, the new value does not take effect until after you send an [Alter Attribute](#) command (using the Process Control window or an XPNET network control facility) to the processes that use the attribute, or stop and restart the processes that use the attribute. Each of these attributes is described below.

**Note:** The Atalla Key Block Conversion utility uses three Environment attributes to specify the new AKB-formatting pending MFK under which keys are encrypted when translating keys from non-AKB format to AKB format. This utility also uses the AKB\_KEY\_IMPORT\_VARIANT\_SOURCE attribute to convert keys to AKB format using the key variant or the key type and several optional Environment attributes to specify the AKB header value used for different key types when the keys are translated. If an attribute is not configured, a default header value is used instead. All of these attributes and their associated default values are described for the [“AKB Conversion Utility”](#) topic in appendix C.

**ATALLA\_CHK\_DGTS\_LGTH** — Specifies whether the Check Digit Validation and Generation utility, AKB Conversion utility, and Master File Key Conversion utility generate and validate four or six check digits when using Atalla NSPs. Valid values are as follows:

- 4 = The Check Digit Validation and Generation utility, AKB Conversion utility, and Master File Key Conversion utility generate and validate four check digits. Default.
- 6 = The Check Digit Validation and Generation utility, AKB Conversion utility, and Master File Key Conversion utility generate and validate six check digits.

If this attribute contains an invalid value or is not configured, the Check Digit Validation and Generation utility, AKB Conversion utility, and Master File Key Conversion utility generate and validate four check digits.

**Note:** Although users can enter four to six check digits on the ACI desktop user interface, only the first four digits are validated.

Default Value: 4  
Component: Key Conversion  
Process Usage: XML Server (XMLS)

**CSEC\_MEMORY\_USE** — A flag indicating whether the Transaction Security Services process maintains Channel Security File (CSECD) records in internal memory. Valid values are as follows:

Y = Yes, CSECD records are maintained in internal memory.  
N = No, CSECD records are not maintained in internal memory. They are always read from disk.

Default Value: Y  
Component: Transaction Security  
Process Usage: XML Server (XMLS) and Transaction Security Services (IS)

**DECIMAL\_TBL\_ENCRYPT** — An optional attribute indicating whether you are using encrypted decimalization tables in your system when using Thales e-Security HSMs. The Master File Key Conversion utility uses this attribute to properly translate decimalization tables from the old LMK pair to the new LMK pair. Valid values are as follows:

Y = Yes, decimalization tables are encrypted.  
N = No, decimalization tables are not encrypted.

If this attribute is not configured or contains an invalid value, it defaults to a value of N.

Default Value: N  
Component: Key Conversion  
Process Usage: XMLS

**EVT\_DEST** — Specifies the name of the EMS Collector process (e.g., \$0) to which event messages generated by processes in the Transaction Security Services environment are sent. You may want to set this attribute to the same value set for the XPNET node, so that all event messages are sent to the same location.

Default Value:     \$0  
Component:         Event  
Process Usage:     XML Server (XMLS) and Transaction Security Services (IS)

**MAX\_CSEC\_IN\_MEM** — An optional attribute specifying the maximum number of CSECD records that the Transaction Security Services process can maintain in memory when the CSEC\_MEMORY\_USE attribute is set to a value of Y. If the CSEC\_MEMORY\_USE attribute is set to a value of N, the Transaction Security Services process uses a value of 1 for this attribute. Valid values are 1–9999999999.

Default Value:     20000 if the CSEC\_MEMORY\_USE attribute is set to Y.  
                    1 if the CSEC\_MEMORY\_USE attribute is set to N.  
Component:         Transaction Security  
Process Usage:     XML Server (XMLS) and Transaction Security Services (IS)

**NON\_ATALLA\_KEY\_EXPORT** — A code indicating the variant to use for exporting Atalla keys to non-Atalla interchange endpoints. When this attribute is set to a value of F (False), keys are exported using the key type (i.e., 1 = PIN, 2 = Data Encryption, and 3 = MAC) as the variant. When this attribute is set to a value of T (True), keys are exported using the key exchange key (KEK) variant (set in the Key Variant field on the Exchange Information tab of the Interface Security Configuration window) if it is set to a value of 0 (Key exchange key). If it is not set to a value of 0, the key type is used for the variant. Valid values are as follows:

T = True. Use the KEK variant from the Interface Security data source (ISECD) if it is 0 (Key exchange key). If it is not set to 0, use the key type for the variant.  
F = False. Use the key type (i.e., 1 = PIN, 2 = Data Encryption, and 3 = MAC) as the variant.

Default Value:     F  
Component:         Transaction Security  
Process Usage:     XML Server (XMLS) and Transaction Security Services (IS)

**PIN\_LGTH\_ERR\_RETRY\_IND** — A code specifying whether the Transaction Security Services component retries an HSM operation using the previous key from the Channel Security Configuration File (CSECD) when the HSM returns a sanity check error or a PIN length error. Valid values are as follows:

- Y = Yes, retry both sanity check errors and PIN length errors using the previous key.  
N = No, do not retry PIN length errors using the previous key.

Default Value: Y  
Component: Transaction Security  
Process Usage: Transaction Security Services (IS)

**PIN\_VRFY\_FAIL\_RETRY\_IND** — A code specifying whether the Transaction Security Services process retries a PIN verification HSM operation using the previous key from the Channel Security Configuration File (CSECD) when the HSM returns a valid fail error (i.e., the cardholder PIN is considered to be invalid).

When using the PIN/PAD PIN block format, an HSM can sometimes return a valid fail error even though the correct PIN was entered. In this case, the HSM successfully decrypts the PIN block (using the wrong key) and then fails the PIN validation. To ensure that correctly entered PINs are processed successfully, you should always set this attribute to Y when using the PIN/PAD PIN block format. Because this situation rarely occurs with other PIN block formats, you can leave this attribute set to the default value of N when using any other PIN block format. Valid values are as follows:

- Y = Yes, retry valid fail errors using the previous key.  
N = No, do not retry valid fail errors using the previous key.

Default Value: N  
Component: Transaction Security  
Process Usage: Transaction Security Services (IS)

**PIN\_XLATE\_DISK\_READ** — A code specifying whether the Transaction Security Services process always reads keys from the Channel Security or Interface Security disk data sources rather than from the associated internal memory table when translating a PIN. Reading the key from disk ensures that the correct key is always used.

Some security devices, such as an Eracom host security module (HSM), can return a successful result for translating a PIN/PAD PIN block even if the wrong key is used after a key change. For PIN/PAD PIN blocks, an Eracom HSM checks only for a hexadecimal pad character to delimit the PIN. As long as the decrypted PIN block contains an A–F hexadecimal character, the Eracom HSM considers the PIN block to have been successfully translated, even if the wrong key is used. After a key change on a BASE24-es system with more than one Integrated Server process, all processes other than the one that initiated the change will continue to attempt to use the old key. Forcing the process to always read the key from disk ensures that the correct key is always used.

Because this situation rarely occurs with PIN block formats other than the PIN/PAD PIN block format, you can leave this attribute set to the default value of N when translating any other PIN block format. Valid values are as follows:

- Y = Yes, read the key from disk instead of the internal memory table for all PIN translation functions.  
N = No, do not read the key from disk instead of the internal memory table for all PIN translation functions.

Default Value: N  
Component: Transaction Security  
Process Usage: Transaction Security Services (IS)

**RSM\_EMV\_KEY\_SCHEME** — A code specifying the key scheme used by Thales e-Security security devices for EMV commands. Valid values are as follows:

- 0 = Use old key scheme (16 or 32 hexadecimal characters)  
1 = Use new key scheme (1A + 32 or 48 hexadecimal characters)

For the new key scheme, the following rules apply to the first character of a key value:

- If the first character of the key value is a hexadecimal character (i.e., 0–9, A–F), the key is the default length for its intended function (single length or double length as appropriate).
- If the first character of the key value is “U”, the key is a double length key and must be followed by 32 hexadecimal characters.



- If the first character of the key value is “T”, the key is a triple length key and must be followed by 48 hexadecimal characters.

Default Value: 1  
Component: Transaction Security  
Process Usage: XML Server (XMLS) and Transaction Security Services (IS)

**RSM\_KEY\_STORE\_LGTH** — A code indicating the length of keys stored in the memory of a Thales e-Security security device, including Diebold number table entries. Valid values are as follows:

- 1 = Single length keys
- 2 = Double length keys
- 3 = Triple length keys

Default Value: 1  
Component: Transaction Security  
Process Usage: XML Server (XMLS) and Transaction Security Services (IS)

**RSM\_TERM\_MASTER\_KEY** — The encrypted terminal master key used with Thales e-Security security devices. This key is required when the Transaction Security Services process uses Thales e-Security security devices to generate message authentication code (MAC) encryption keys on behalf of an Interchange Interface process. This attribute is not required if the Transaction Security Services process only generates MAC encryption keys on behalf of BASE24 Device Handler processes or when Thales e-Security security devices are not used to generate MAC encryption keys.

The BASE24 ISO Host Interface process and the BIC ISO Interface process use this key in an intermediate step that is performed when BASE24 is responsible for generating and distributing MAC encryption keys. The key in this attribute does not coincide with any other keys being used by the external processor. For ease of maintenance, ACI recommends using one RSM\_TERM\_MASTER\_KEY attribute for a BASE24 network.

When this attribute is required, the default value of zeros must be changed. The Thales e-Security console command K can be used to obtain the encrypted key for this attribute. Refer to the appropriate Thales e-Security manual for information regarding the console command.

Default Value: 0000000000000000  
Component: Transaction Security  
Process Usage: XML Server (XMLS) and Transaction Security Services (IS)

**UI\_AUDIT\_LEVEL** — A code indicating the level of user interface auditing performed by the C++ User Interface Server process. This attribute is optional. When performing full auditing, the C++ User Interface Server process includes all record fields and their values in auditing messages written directly to the User Audit Log File (UALOGD). When performing headers only auditing, the User Interface process includes the record's primary key fields only in auditing messages. When performing full auditing, the User Interface process includes all primary key and modified fields for deletes and inserts, all modified fields for updates, and primary key fields only for views (reads).

**Note:** Most of the user interface auditing in the TSS environment is controlled by `db.audited.db-service-name` properties configured for the Java User Interface servlets on the Application Configuration window of the ACI desktop user interface, where `db-service-name` is the name of a particular Java database service. The `UI_AUDIT_LEVEL` attribute controls auditing only for data sources accessed by C++ servers. Refer to the *ACI Java Server Framework Reference Guide* for a list of all possible `db.audited.db-service-name` properties that apply to the TSS environment. Thus, if a data source and associated ACI desktop window is not represented in the list in the *ACI Java Server Framework Reference Guide*, auditing for the data source is controlled by the `UI_AUDIT_LEVEL` attribute.

Note that if you have never used the ACI desktop user interface to view, add, update, or delete data and then change the value of the `UI_AUDIT_LEVEL` attribute, an event message indicating an "Unknown Attribute Name" is generated. This is normal and the updated value will take effect on the first request that requires auditing. Valid values are as follows:

- 0 = No user interface auditing
- 1 = Headers only auditing (primary key fields only)
- 2 = Full auditing (all primary key and modified fields)

Default Value: 1  
Component: User Interface Auditing Implementation  
Process Usage: XML Server (XMLS)

**UI\_AUDIT\_OUTPUT\_DATA\_TYP** — A flag indicating whether the C++ User Interface (XML Server) process converts the format (character set) of auditing detail when it writes auditing records directly to the User Audit Log File (UALOGD) and whether it converts the format to ASCII or EBCDIC. Valid values are as follows:

- A = Yes, convert the format of audit detail to ASCII.
- E = Yes, convert the format of audit detail to EBCDIC.
- N = No, do not convert the format of audit detail written to the UALOGD.  
Write the detail in the native format of the XML Server process.

Default Value: N  
Component: User Interface Auditing Implementation  
Process Usage: XML Server (XMLS)

**USE\_LOCAL\_USER\_AUDIT** — A flag indicating whether the C++ User Interface (XML Server) process writes auditing records directly to the User Audit Log File (UALOGD). If not, the process writes nowaited auditing messages to the Audit Store and Forward File (ASAFD) for transmission to the UAUD application. When the UAUD application receives an auditing message, it routes the message through WebGate to the ESWEB process to be logged to the UALOGD.

**Note:** With the advent of TSS release 05.4, you should always set this attribute to a value of Y. The configuration required to write auditing messages to the Audit Store and Forward File (ASAFD) for transmission to the UAUD application and subsequent transmission to the ESWEB process is not installed in this release.

Valid values are as follows:

- Y = Yes, the C++ User Interface (XML Server) process writes auditing records directly to the UALOGD.
- N = No, the C++ User Interface (XML Server) process does not write auditing records directly to the UALOGD.

Default Value: N  
Component: User Interface Auditing Implementation  
Process Usage: XML Server (XMLS)

## TSS Process WorkLoad Balancing

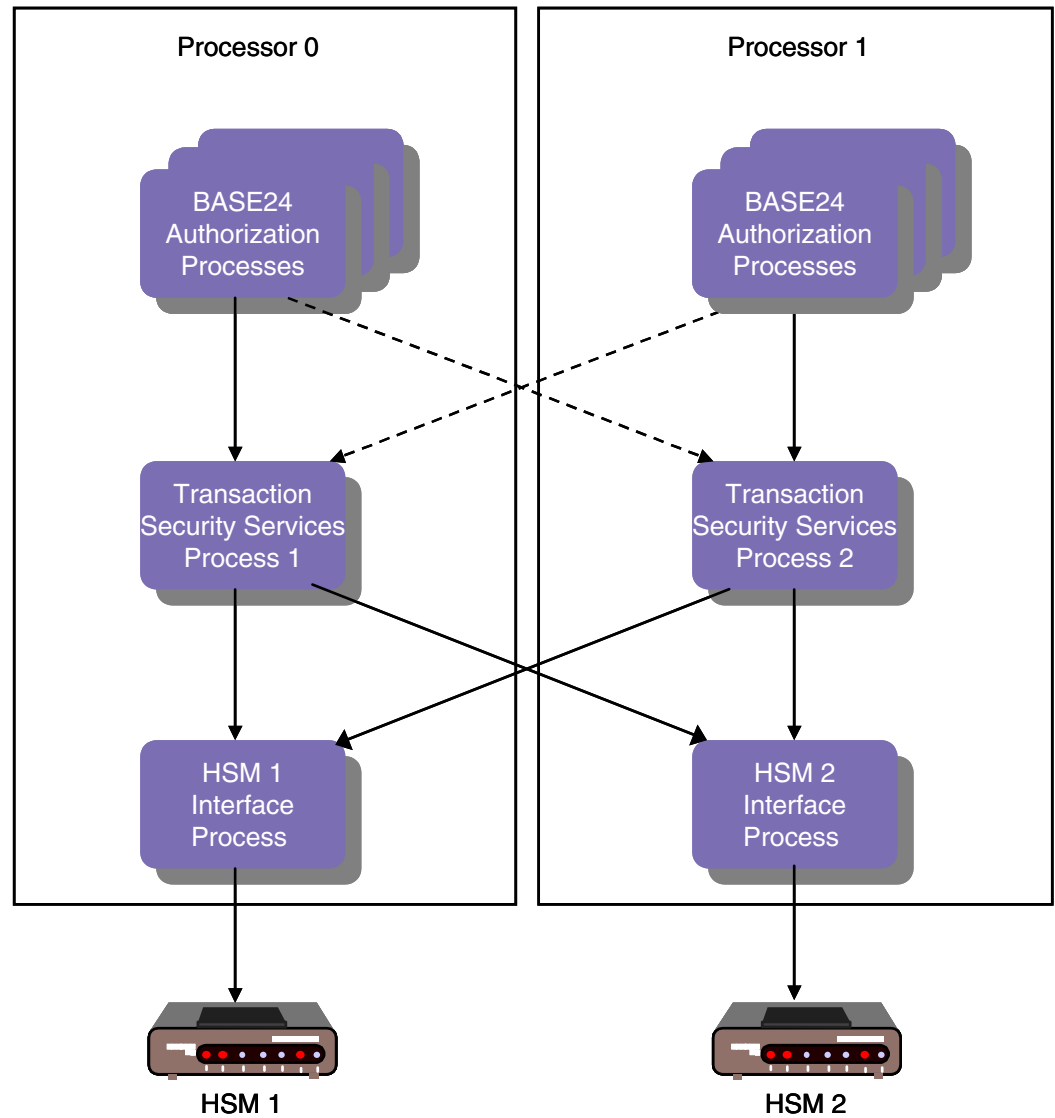
**Note:** Because hardware security device requests are automatically distributed evenly among all available Transaction Security Services processes when you are using the round-robin access method (i.e., SECURE-DEV-ACCESS-TYPE param set to 1), the following paragraphs only apply when using the primary and secondary Transaction Security Services process access method (i.e., SECURE-DEV-ACCESS-TYPE param set to 0).

When you implement the Transaction Security Services process using the primary and secondary Transaction Security Services process access method, two levels of workload balancing are used. You can change the number of requests sent to primary and secondary Transaction Security Services processes using the SECURE-DEV-MONITOR-THRESHOLD param. Each Transaction Security Services process can send requests to any security module (or HSM interface process) defined in the SECD CD. Each instance of the Transaction Security Services process maintains a list of security modules (or HSM interface process) in memory and alternates requests one at a time to each device. Because the Transaction Security Services process alternates requests to HSMs one at a time, ACI recommends that you do not use the SECURE-DEV-MONITOR-THRESHOLD param to alternate requests between primary and secondary Transaction Security Services processes running in separate CPUs. Otherwise, a single request may travel across the inter processor bus twice, resulting in a possible degradation of performance.

You can monitor the number of transaction messages processed by a Transaction Security Services process using the Start, Status, and Stop Statistics process control commands. For more information on these commands, refer to [section 9](#).

The following diagram depicts a highly simplified example distribution of Transaction Security Services processes and security modules in a BASE24 system with two CPUs and two HSMs. The primary path for requests is denoted with solid lines while the secondary path for request is denoted with dotted lines. In this example, the SECURE-DEV-MONITOR-THRESHOLD param would be configured to intermittently send a single request (e.g., once every 5000 transactions) to the secondary Transaction Security Services process to make sure the process is available.

**Note:** This example is for illustrative purposes only, the optimum configuration for any BASE24 system can only be obtained through ongoing resource allocation analysis and performance tuning.



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## 4: Security Module Requirements

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This section describes the security module requirements needed to support the functions available with the Transaction Security Services process.

## ISO Format 1 PIN Block Format

Bull BNTng and Thales e-Security HSMs support the ISO format 1 PIN block format in addition to the PIN/PAN, PIN/PAD, and DUKPT PIN block formats described in the *BASE24 Transaction Security Manual*. To use the ISO format 1 PIN block for a terminal device, set the PIN Block Format field on the PIN Information tab of the Channel Security Configuration window to a value of ISO Format 1. To use the ISO format 1 PIN block for an interface, set the PIN Block Format field on the PIN Information tab of the Interface Security Configuration window and BASE24 KEYF Configuration window to a value of ISO Format 1.

The ISO format 1 PIN block is represented by the following 16 hexadecimal values:

*1NppppppppppppppRR*

Where:

1 is a control field character that identifies the PIN block as an ISO format 1 PIN block.

N is length of the PIN in hexadecimal representation. Since PINs can be from 4 to 12 digits in length, the hexadecimal representation of the length is 4–C as shown in the following table:

| PIN Digits | Hexadecimal Representation |
|------------|----------------------------|
| 4          | 4                          |
| 5          | 5                          |
| 6          | 6                          |
| 7          | 7                          |
| 8          | 8                          |
| 9          | 9                          |
| 10         | A                          |
| 11         | B                          |
| 12         | C                          |



$p$  is the 4–12 digit PIN. If the PIN is less than 12 digits, characters to the right are randomly padded with any valid hexadecimal character (i.e., 0–9 and A–F).

$R$  is a random padding hexadecimal character (i.e., 0–9 and A–F).

**Note:** The random padding characters should not be transmitted and are not required in order to translate the PIN block to another format since the PIN length is known.

## Atalla Security Modules

Atalla security functions are available through separate utilities, the ACI desktop user interface, the Atalla NSPDIAG tool and the Secure Configuration Terminal (SCT), or the Atalla Secure Configuration Assistant (SCA). You can only use the SCA with Atalla x100 HSMs with software version 2.10 or later.

Check digits are automatically validated by the security device when the keys and check digits are entered from the ACI desktop user interface. This provides immediate validation of keys, rather than having to use a batch utility to validate keys. You can also use the Check Digit Validation and Generation utility to validate or generate check digits for all the keys in one or more TSS files. Refer to [appendix C](#) for more information on this utility.

## External Key Block Format

The external key block format is the format of the key block that an external entity (e.g., device or interface) supports—Standard Atalla (variant format) or Atalla Key Block (AKB) format. The format is set in the External Key Block Format field on the Channel Security Configuration window and the Interface Security Configuration window. Currently, no devices or interfaces in the BASE24 system support the AKB format. Therefore, this field should always be set to Standard Atalla.

## Preparing Data for the TSS Files

A secure key loading device such as the Secure Configuration Terminal (SCT) or Secure Configuration Assistant (SCA) is used to perform offline key management and the programming of the Master File Key (MFK) directly into the NSP.

A secure key loading device, such as the SCT or SCA, can be used to encrypt or translate the keys under the proper MFK variant for storage in the TSS files, as shown in the following table. The table identifies the TSS field name, the window on the ACI desktop user interface on which the field appears, MFK variant, key type, and AKB header value. The AKB header values are shown in parentheses in the Key Type column of the table.

**Note:** Although the AKB NSP supports both triple- and double-length MFKs, Atalla strongly encourages that you use a triple-length MFK. If you do use a double-length MFK, you cannot use any triple-length keys.

| Field Name on User Interface Window          | Window               | MFK Variant | Key Type (AKB Header)                                                             |
|----------------------------------------------|----------------------|-------------|-----------------------------------------------------------------------------------|
| PIN Information: Exchange Key                | Channel              | 5           | ATM A key (1ADNE000)                                                              |
| PIN Information: Inbound Key                 | Channel              | 1           | PIN encryption key (1PUNE000)*<br>AS2805-compliant PIN encryption key (1PUNN000)  |
| MAC Information: Exchange Key                | Channel              | 0           | Key exchange key (1KDNE000)                                                       |
| MAC Information: Inbound Key                 | Channel              | 3           | Message Authentication Code Key (1MDNE000)<br>AS2805-compliant MAC key (1MDGN000) |
| Data Encryption: Exchange Key                | Channel              | 0           | Key exchange key (1KDNE000)                                                       |
| Data Encryption: Outbound Keys               | Channel              | 2           | Data Encryption Key for both encryption and decryption (1DDNE000)                 |
| Derivation Key                               | DUKPT Derivation Key | 8           | Derivation key (1dDNE000)                                                         |
| Cryptographic Keys                           | Authentication       | 9           | EMV ACC master keys (1mENE000)                                                    |
| Secure Message Key For Integrity             | Authentication       | 13          | EMV MAC master key (1mENE00M)                                                     |
| Secure Message Key for Confidentiality       | Authentication       | 14          | PIN encryption key (1mENE00E)                                                     |
| DES (IBM 3624) PIN Verification <sup>†</sup> | Authentication       | 4           | PIN verification key (1V3NE000)                                                   |
| Diebold Number Table <sup>‡</sup>            | Authentication       | –           | Diebold Number Table Row (1ndNE000)                                               |

| Field Name on User Interface Window   | Window         | MFK Variant | Key Type (AKB Header)                                                                              |
|---------------------------------------|----------------|-------------|----------------------------------------------------------------------------------------------------|
| VISA PVV Keys                         | Authentication | 4           | PIN verification key (1VVNE000)                                                                    |
| Card Verification <sup>§</sup>        | Authentication | 3           | Card verification key A or B (1CDNE000)<br><br>American Express Card Security Code KCSC (1mXNE000) |
| Exchange Information: PIN Key: Import | Interface      | 0           | Key exchange key (1PUNN0I0)                                                                        |
| Exchange Information: PIN Key: Export | Interface      | 0           | Key exchange key (1KDEE000)                                                                        |
| Exchange Information: MAC Key: Import | Interface      | 0           | Key exchange key (1MDNN0I0)                                                                        |
| Exchange Information: MAC Key: Export | Interface      | 0           | Key exchange key (1KDEE000)                                                                        |
| PIN Information: Outbound Keys        | Interface      | 1           | PIN encryption key (1PUNE000)<br><br>AS2805-compliant PIN encryption key (1PUNN000)                |
| PIN Information: Inbound Keys         | Interface      | 1           | PIN encryption key (1PUNE000)<br><br>AS2805-compliant PIN encryption key (1PUNN000)                |
| MAC Information: Outbound Keys        | Interface      | 3           | Message authentication key (1MDNE000)<br><br>AS2805-compliant MAC key (1MDGN000)                   |

| Field Name on User Interface Window       | Window    | MFK Variant | Key Type (AKB Header)                                                        |
|-------------------------------------------|-----------|-------------|------------------------------------------------------------------------------|
| MAC Information: Inbound Keys             | Interface | 3           | Message authentication key (1MDNE000)<br>AS2805-compliant MAC key (1MDVN000) |
| Data Encr Exch Intf: Exchange Key: Import | Interface | 0           | Key exchange key (1DDNN0I0)                                                  |
| Data Encr Exch Intf: Exchange Key: Export | Interface | 0           | Key exchange key (1KDNE000)                                                  |
| Data Encr Key Intf: Inbound Keys          | Interface | 2           | Data Encryption Key for both encryption and decryption (1DDNE000)            |
| Data Encr Key Intf: Outbound Keys         | Interface | 2           | Data Encryption Key for both encryption and decryption (1DDNE000)            |

\* For ATM or POS devices configured to send PINs in the clear to any interchanges on the BASE24 system, the intermediate key value of 9999999999999999 (16 hexadecimal 9s) must be stored in a CSECD record.

† For security purposes, you can store the clear decimalization table value in the memory of the NSP using the SCT or SCA. In this case, you enter the index of this memory location instead of the clear decimalization table value in the Decimalization Table field on the ACI desktop.

‡ Anytime you change the DNT in the TSS database, you must rebuild the Diebold Number Table external memory table, warmboot the Transaction Security Services processes that access the external memory table, and execute the DNTLOAD process control command to reload the DNT in the memory of the NSP. Refer to [section 2](#) for information on rebuilding external memory tables and warmbooting processes that access them. Refer to [section 9](#) and [appendix E](#) for more information on entering the DNTLOAD command.

- § For verification of MasterCard or Visa cards, you must enter a key pair of two single-length keys encrypted under MFK variant 3 or with an AKB header of 1CDNE000. For verification of American Express cards, you must enter a single double-length key encrypted under MFK variant 3 or with an AKB header of 1mXNE000.

With AKB version software, Atalla has made several enhancements to the Secure Configuration Terminal (SCT), including a PC-based “helper application” to simplify key generation. In addition, Atalla has developed the Secure Configuration Assistant (SCA), which is a secure configuration device based on the HP iPAQ Personal Data Assistant. The SCA is used with Smartcards that perform cryptographic operations and store security-relevant data. The implementation of Smartcards used with the SCA provides identity-based authorization.

The security windows in the ACI desktop user interface are designed to allow you to cut and paste key cryptograms from these Atalla applications directly into the BASE24 database, thereby eliminating the manual entry of cryptograms. The ACI desktop automatically parses parts of the key cryptogram into the correct fields (i.e., Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits) on the user interface window. Copy key text and paste key text options are available from a pop-up menu when you position the cursor in a key field and right click. For more information, refer to ACI Online Help for the ACI desktop user interface.

## Generate Check Digits Command

The User Interface process uses the 7E (Generate Check Digits) command when you enter a new key from the ACI desktop user interface, or when you validate or generate check digits using the Check Digit Validation and Generation utility. Refer to [appendix C](#) for more information on this utility. The 7E command accepts single-length keys only if option 6C is enabled in your security policy using commands 108 and 109. This option applies only to Atalla NSPs using AKB software.

The Transaction Security Services process also uses the 7E command to generate check digits for a specified key on behalf of BASE24 AS2805-compliant Device Handler or Interface processes or modules.

The Transaction Security Services process uses the 7E command to generate check digits using the leftmost four digits or the leftmost six digits method as specified in the TSS request message. You must enable option 88 in your security policy to use the leftmost six digits method.

The User Interface process, Check Digit Validation and Generation utility, AKB Conversion utility, and Master File Key Conversion utility use the 7E command to generate check digits using the leftmost four digits or the leftmost six digits method. The ATALLA\_CHK\_DGTS\_LGTH Environment attribute specifies whether to generate and validate four or six check digits. Note that the ACI desktop user interface only validates the first four digits if six check digits are entered.

## Changing the Master File Key

ACI provides a Master File Key Conversion tool that enables you to translate hardware-encrypted AKB- and non-AKB-formatted keys in the TSS database from encryption under one Atalla Master File Key (MFK) to another MFK. This conversion tool is executed from the Master File Key Conversion window, accessed as **Configure > Transaction Security > Master File Key Conversion** on the ACI desktop. Using this tool, you can translate all the keys in the TSS database or just the keys of a selected type (i.e., card verification, PIN verification, EMV, channel, or interface). For detailed information on using this tool, refer to the [Master File Key Conversion](#) topic provided in appendix C.

Good security practice as well as several network and card associations require the MFK to be replaced periodically. This conversion tool enables you to convert keys from encryption under the old MFK to encryption under the new MFK and automatically replace them in the TSS database files. NSPs must contain the old MFK as the current MFK and contain the new MFK as the pending MFK before this utility is used. The Master File Key Conversion utility requires the 9E command (Translate Working Key From Under the Current MFK to the Pending MFK) to translate keys. After keys are converted, use the Erase LMK or MFK command on the Process Control window make the pending MFK the current MFK. This process control command executes the Atalla 9A command (Network Security Processor Status Key) and 9F command (Replace the Current MFK with the Pending MFK) on the specified NSPs.

## Importing and Exporting Keys

If you are using an Atalla NSP with AKB software, you must enable the key types of working keys that can be imported and exported as well as key exchange keys that can be used to both import and export working keys. Refer to the “Importing and Exporting Working Keys” topic in the *Atalla Key Block Banking Command Reference Manual* for detailed information on the options used to define the keys that can be imported or exported.

## Import Keys

You must enable any of the following key types to be imported to your BASE24 system in option E0 using the 108 and 109 security policy commands:

| Key Type | Description                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------|
| D        | Data Encryption (currently only used for Interac support in Canada)                                   |
| M        | Message Authentication (MAC)                                                                          |
| P        | PIN Encryption Key                                                                                    |
| V        | PIN Verification Key (although supported, this key type typically is not imported in a BASE24 system) |

## Export Keys

You must enable any of the following key types to be exported from your BASE24 system in options E1 using the 108 and 109 security policy commands:

| Key Type | Description                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------|
| D        | Data Encryption (currently only used for Interac support in Canada)                                   |
| M        | Message Authentication (MAC)                                                                          |
| P        | PIN Encryption Key                                                                                    |
| V        | PIN Verification Key (although supported, this key type typically is not exported in a BASE24 system) |

## Key Exchange Keys

For ATM master keys to be used as key exchange keys for both importing and exporting working keys, you must enable options E3 and E2 using the 108 and 109 security policy commands. In option E2, you must enable key type A (ATM master keys). Option E3 enables ATM master key headers.



## Triple DES Enabled Banking Commands

With AKB version software, the following banking commands used by BASE24 are triple DES enabled for all Atalla security device models: 10, 11, 13, 15, 31, 32, 33, 37, 5E, 7E, 98, 99, 9E, 335, 350, 351, 352, BD, and D0. Note that the 335 and BD commands are not available for the A8000 model. The A8100 and A8100-V models support the 335 and BD commands; however, you cannot use binary data in the BD command when connected to a security device over an asynchronous port.

The 11D command has been retired in AKB version software. This command is replaced by the 10 and 1A commands in AKB software. For Atalla NSPs running with non-AKB software, the [Export Variant](#) field on the MAC Information tab of the Channel Security Configuration window specifies whether the 11D or 10 command is used to generate a new MAC key for a device. If this field is set to a value other than 0, the 10 command is used. Otherwise, the 11D command is used. The 10 command is always used to generate a new data encryption key.

AKB versions of Atalla security device models also support the 1A and 11B commands. The 1A command is used to export a working key in AKB format to non-AKB format. The 11B command is used to import a working key in non-AKB format to AKB format. Neither of these commands are enabled in the default factory security policy. To use these commands you must add them to your security policy using commands 108 and 109.

Currently, all Atalla Network Security Processors (NSPs) running AKB software support double- and triple-length keys for triple DES operations in the default factory security policy. The NSP default factory security policy does not allow single length keys for triple DES operations. If you must support single length keys in triple DES operations, you must enable option 6C in your security policy using commands 108 and 109.

Triple DES encryption using a double length key is accomplished by replicating the first key as the third key. The data is encrypted using the first key, the result is decrypted using the second key, and this second result is encrypted again using the first key. When decrypting data, the process is reversed. The encrypted data is decrypted using the first key, the result is encrypted using the second key, and this second result is decrypted again using the first key.

For triple DES encryption using triple length keys, all three key parts must be unique. Atalla NSPs ensure this is true when requested to generate a triple length DES key.

When encrypting data using triple length keys, the data is first encrypted using the first key, the result is decrypted using the second key, and this second result is encrypted using the third key. When decrypting data, the process is reversed. The encrypted data is first decrypted using the third key, the result is encrypted using the second key, and this second result is decrypted using the first key.

## Triple DES Migration Software Requirements

The following table summarizes the software version requirements for Atalla devices when migrating a BASE24 system to triple DES and use of the Transaction Security Services process. The first column of the table lists the BASE24 software releases for which triple DES functionality and the Transaction Security Services (TSS) process are available. The second column lists the minimum software versions required to support single DES. The third column lists the minimum software versions required to support the translation of single DES to triple DES PIN blocks for interchanges using the Key 6 File (KEY6). This functionality enables you to meet triple DES mandates for individual interchanges before full triple DES encryption is required in acquiring devices. The fourth column lists the minimum software versions required to support triple DES across the entire BASE24 system.

| <b>BASE24 Release and TSS Usage</b> | <b>Single DES</b> | <b>Triple DES Translation*</b> | <b>Full Triple DES</b> |
|-------------------------------------|-------------------|--------------------------------|------------------------|
| 5.x without TSS                     | Variant           | Variant                        | Not available          |
| 6.0 without TSS                     | Variant           | Variant                        | Not available          |
| 6.0 with TSS                        | Variant or AKB    | Not available                  | AKB only               |

\* This function requires command 335, which is available in all Atalla models and minimum software versions described below (except the A8000).

| <b>Software</b> | <b>A8000, A9000, A10000</b> | <b>A8100, A9100, A10100</b> |
|-----------------|-----------------------------|-----------------------------|
| Variant         | 2.9                         | 2.3                         |
| AKB             | 1.4                         | 2.30 (2.31 for AKDS)        |

Note that Atalla devices running variant software cannot interoperate with Atalla devices running AKB software because of the difference in key block formats between the versions. Therefore, when you convert Atalla devices to AKB

software, all your devices must be converted at the same time. ACI provides a conversion program to convert keys to the new key block format. Refer to the “AKB Conversion” topic below for more information.

Triple DES translation of ANSI PIN blocks is supported for both the Atalla minimum variant and AKB software versions when using the Transaction Security Services process. When using Atalla variant software, triple DES PIN Pad PIN verification is not supported.

Only the AKB version of Atalla software can be used in a full triple DES environment. With few exceptions, older versions of Atalla software do not support triple DES PIN verification. One exception is triple DES Visa PIN verification using double length keys.

## AKB Conversion

ACI provides a conversion program to convert working keys stored in the TSS database in the variant key format (non-AKB format) to the Atalla Key Block (AKB) format. The program enables you to convert the keys in one or more files by entering the Convert AKB command on the Process Control window of the ACI desktop. For detailed information on the AKB Conversion utility, refer to the [AKB Conversion](#) topic provided in appendix C. You can use the Check Digit Validation and Generation utility to validate or generate check digits for all the keys in one or more TSS files prior to converting keys.

## ATM Master Key Exchange

The Transaction Security Services process supports the generation of a new ATM master key encrypted under the previous master key. At the request of a BASE24-atm Device Handler process, the Transaction Security Services process calls an NSP to generate a new ATM terminal master key encrypted under the previous ATM terminal master key for downloading to a terminal. The new master key is stored as both the PIN master key and the MAC master key (key exchange keys) in the Channel Security File (CSECD). This is done because ATMs only support one master key. Therefore, the PIN key exchange key and the MAC key exchange key must be the same in the CSECD. If the download fails, the Device Handler process can request that the new key be backed out, up to the expiration of the old key timer limit in the CSECD.

In an NSP with the AKB version of Atalla software, the new ATM master key is generated using the 10 command (Generate Working Key, Any Type). In an NSP with variant (non-AKB) version of Atalla software, the new ATM master key is generated using both the 10 command and the 14#1 command (Load ATM Master Key, Primary Node – Diebold).

**Note:** Support of this functionality requires a CSM for the BASE24-atm Device Handler process.

## PIN Translation

Atalla's support for PIN translation function varies based on the PIN block formats, key lengths, and version of Atalla software used as described below.

If the destination PIN Block is ANSI, the 31 command is used. When using AKB software, the 6C command must be enabled to allow the use of single-length keys. Variant software on xx100 NSPs accepts both single- and double-length keys as is. Variant software version 2.84 only accepts single-length keys.

If the destination PIN Block is PIN/PAD and the destination key length is single, the 33 command is used. When using AKB software, the 6C command must be enabled to allow the use of single-length keys. Variant software on xx100 NSPs accepts both single- and double-length keys as is. Variant software version 2.84 only accepts single-length keys.

If the destination PIN Block is PIN/PAD and the destination key length is double, or the source key length is double, the 335 command is used. If the source key length is single, the 6C option must be enabled. If the destination key length is single, the 6A option must be enabled because the destination key length in the 335 command must be double length and the 6A option allows a replicated double-length key. Note that the 335 command is not used when the Atalla Key Block (AKB) is used.

The following table shows you the scenarios for which options 6A or 6C must be enabled for the 335 command.

| Source PIN Block Type | Source Key Length | Destination PIN Block Type | Destination Key Length | Option 6A Required? | Option 6C Required? |
|-----------------------|-------------------|----------------------------|------------------------|---------------------|---------------------|
| ANSI                  | Single            | PIN/PAD                    | Double                 |                     | ✓                   |
| ANSI                  | Double            | PIN/PAD                    | Double                 |                     |                     |

| Source PIN Block Type | Source Key Length | Destination PIN Block Type | Destination Key Length | Option 6A Required? | Option 6C Required? |
|-----------------------|-------------------|----------------------------|------------------------|---------------------|---------------------|
| PIN/PAD               | Single            | PIN/PAD                    | Double                 |                     | ✓                   |
| PIN/PAD               | Double            | PIN/PAD                    | Single                 | ✓                   |                     |
| PIN/PAD               | Double            | PIN/PAD                    | Double                 |                     |                     |
| PIN/PAD               | Double            | ANSI                       | Single                 | ✓                   |                     |

The Transaction Security Services process uses command 335 in some cases where commands 31 or 33 could have been used. For example, if the NSP software is variant, but is not version 2.84 or 2.9, the Transaction Security Services process could have used the 31 and/or 33 commands. However, since the Transaction Security Services process does not differentiate between different versions of variant software, command 335 is used for consistency.

**Note:** The following PIN translation information applies only to Atalla NSPs running variant software. Triple-length keys are not supported by Atalla variant software.

In order to translate a PIN block from encryption under a double-length acquirer key (in any valid PIN block format, excluding DUKPT) to encryption under a single-length issuer key in PIN/PAD PIN block format, you must use a double-length key with both key parts set to the same value that is passed to the Atalla NSP in the 335 command. This requires that you enable the Atalla premium option 6C in the NSP. If this option is not enabled, these PIN translations are not allowed. The following table shows you which PIN translation scenarios are supported by the Transaction Security Services process. The first column indicates the PIN block format of the acquirer. The second column indicates the PIN block format of the issuer. The third column indicates the Atalla command used to perform the PIN translation or indicates “N/A” if the PIN translation scenario is not available.

| Acquirer             | Issuer               | Command |
|----------------------|----------------------|---------|
| ANSI (single-length) | ANSI (single-length) | 31      |
| ANSI (single-length) | ANSI (double-length) | 31      |
| ANSI (single-length) | ANSI (triple-length) | N/A     |

| <b>Acquirer</b>         | <b>Issuer</b>           | <b>Command</b> |
|-------------------------|-------------------------|----------------|
| ANSI (double-length)    | ANSI (single-length)    | 31             |
| ANSI (double-length)    | ANSI (double-length)    | 31             |
| ANSI (double-length)    | ANSI (triple-length)    | N/A            |
| ANSI (triple-length)    | ANSI (single-length)    | N/A            |
| ANSI (triple-length)    | ANSI (double-length)    | N/A            |
| ANSI (triple-length)    | ANSI (triple-length)    | N/A            |
| ANSI (single-length)    | PIN/PAD (single-length) | 33             |
| ANSI (single-length)    | PIN/PAD (double-length) | 335*           |
| ANSI (single-length)    | PIN/PAD (triple-length) | N/A            |
| ANSI (double-length)    | PIN/PAD (single-length) | 33             |
| ANSI (double-length)    | PIN/PAD (double-length) | 335            |
| ANSI (double-length)    | PIN/PAD (triple-length) | N/A            |
| ANSI (triple-length)    | PIN/PAD (single-length) | N/A            |
| ANSI (triple-length)    | PIN/PAD (double-length) | N/A            |
| ANSI (triple-length)    | PIN/PAD (triple-length) | N/A            |
| PIN/PAD (single-length) | ANSI (single-length)    | 335*           |
| PIN/PAD (single-length) | ANSI (double-length)    | 335            |
| PIN/PAD (single-length) | ANSI (triple-length)    | N/A            |
| PIN/PAD (double-length) | ANSI (single-length)    | 31             |
| PIN/PAD (double-length) | ANSI (double-length)    | 31             |
| PIN/PAD (double-length) | ANSI (triple-length)    | N/A            |
| PIN/PAD (triple-length) | ANSI (single-length)    | N/A            |
| PIN/PAD (triple-length) | ANSI (double-length)    | N/A            |

| <b>Acquirer</b>         | <b>Issuer</b>           | <b>Command</b> |
|-------------------------|-------------------------|----------------|
| PIN/PAD (triple-length) | ANSI (triple-length)    | N/A            |
| PIN/PAD (single-length) | PIN/PAD (single-length) | 33             |
| PIN/PAD (single-length) | PIN/PAD (double-length) | 335*           |
| PIN/PAD (single-length) | PIN/PAD (triple-length) | N/A            |
| PIN/PAD (double-length) | PIN/PAD (single-length) | 335*           |
| PIN/PAD (double-length) | PIN/PAD (double-length) | 335*           |
| PIN/PAD (double-length) | PIN/PAD (triple-length) | N/A            |
| PIN/PAD (triple-length) | PIN/PAD (single-length) | N/A            |
| PIN/PAD (triple-length) | PIN/PAD (double-length) | N/A            |
| PIN/PAD (triple-length) | PIN/PAD (triple-length) | N/A            |

\* Refer to the previous table for the options required by the 335 command.

## Device Options and Banking Commands

Some device options and banking commands required by BASE24 must be configured or enabled in Atalla devices before they can be used. These options and commands are discussed below.

### Device Options

For all Atalla devices, the option for omitting carriage return and line feed characters from the messages they return to BASE24 must be configured in a 101 (Configure Security Processor Options) command in the Configuration String field on the Security Device Configuration window of the ACI desktop user interface. For the A10000E device, the options for setting the minimum PIN length and specifying the response code the NSP returns for a PIN length error are also configured in a 101 command in the Configuration String field on the Security Device Configuration window. BASE24 products require a minimum PIN length of 4 digits. For all other Atalla devices, these options must be configured using the 108 (Define Security Policy) and 109 (Confirm Security Policy) commands in the NSPDIAG tool or using the NSP Option Value Configuration screen in the Atalla

Secure Configuration Assistant (SCA). These commands are entered using the NSPDIAG tool with the (A0) and (A1) options, respectively. For more information on the 101 command, defining a security policy, and the SCA, refer to the appropriate Atalla documentation.

If you need to convert ATM master keys to the AKB format, you must enable option E3 for commands 10 and 1A to support ATM master key headers. This option is not enabled in the default factory security policy. To use this option, you must add it to your security policy using commands 108 and 109.

If you need to use the Generate Check Digits (7E) command to generate six check digits instead of four check digits, you must enable option 88 for command 7E. This option is not enabled in the default factory security policy. To use this option, you must add it to your security policy using commands 108 and 109.

The following table lists premium options, which must be purchased from Atalla in a command 105 or command 100 (A10000E only) before they can be used in an Atalla NSP. The first column of the table lists each premium option, while the second column identifies the BASE24 function for which the premium option is required. You only need to purchase the options for the BASE24 functions implemented in your system.

| Premium Option                              | BASE24 Function                                                                                                                                                                                                                                                     |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 62 (Allow command 31 DUKPT)                 | Required for the derived unique key per transaction (DUKPT) encryption method. This option allows you to use the Translate PIN—Visa DUKPT (31#7) command to translate DUKPT-formatted PIN blocks to ANSI PIN blocks encrypted under an outbound PIN encryption key. |
| 66 (Do not validate old PIN for command 37) | Required for PIN Change transaction processing. BASE24 processes require the 37 command to be performed without the old PIN, regardless of the PIN verification method used.                                                                                        |



| Premium Option                                                      | BASE24 Function                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8C (Allow a replicated single length Visa PVV PIN verification key) | Premium option 8C must be enabled for the Verify PIN—Visa PVV Method (32#3) and PIN Change—Visa PVV (37#3) commands to allow single-length Visa PVV keys to be replicated in triple DES operations. Therefore, if you convert single-length Visa PVV keys to AKB format using the AKB Conversion utility, you must enable option 8C to use the 32#3 and 37#3 commands. You can obtain option 8C, free of charge, in a 105 command from Atalla. |

## Device Banking Commands

Some of the Atalla NSP banking commands used in BASE24 processing may need to be enabled before transaction processing begins. All Atalla NSPs, except for the A8000 and A10000E NSPs, support the 105 (Enable Premium Value Command and Options), 108 (Define Security Policy) and 109 (Confirm Security Policy) commands to enable Atalla NSP banking commands. For the A10000E NSP, premium commands are enabled using the 100 command (Configure Security Processor Commands). The A8000 NSP does not support the 100 or 105 commands because it does not have any premium value commands or options. For all other NSPs, premium commands are enabled using the 105 command.

Some of the banking commands that BASE24 supports are disabled when an Atalla device is delivered from the factory because of the security exposure involved with the command. These commands can be enabled using only the 108 and 109 security policy commands. For example, the D0 (Verify Clear PIN) command is disabled in Atalla devices due to security exposure. Other banking commands that BASE24 supports are identified as *premium* commands for Atalla NSPs. These commands are disabled when an Atalla device is delivered from the factory and can only be enabled in the security policy commands by purchasing a 105 command from Atalla. The 105 command is based on the device serial number and is unique to each device.

The following table lists premium commands, which must be purchased from Atalla in a command 105 or command 100 (A10000E only) before they can be used in an Atalla NSP. The first column of the table lists each premium command,

while the second column identifies the BASE24 function for which the premium command is required. You only need to purchase the commands for the BASE24 functions implemented in your system.

| Premium Command                                            | BASE24 Function                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11D (Generate ATM MAC Key)*                                | <p>This command is needed if message authentication is performed, the <a href="#">Export Variant</a> field on the MAC Information tab of the Channel Security Configuration window is set to a value of 0, and the NSP is used to generate a new MAC transaction key for a terminal.</p> <p>This command is not needed to generate a new MAC key for an interface process.</p> |
| 12B (Translate Public or Private Key AKB from MFK to PMFK) | Required to support the conversion of public and private keys by the Master File Key Conversion utility on behalf of the BASE24-atm Automated Key Distribution System add-on product.                                                                                                                                                                                          |
| 31#7 (Translate PIN—Visa DUKPT)                            | <p>Required for the derived unique key per transaction (DUKPT) encryption method. This command translates DUKPT-formatted PIN blocks to ANSI PIN blocks encrypted under an outbound PIN encryption key.</p> <p>You must purchase option 62 in the form of a command 105, and enable it in the NSP's security policy to use this command.</p>                                   |
| 37 (PIN Change)                                            | Required for PIN Change transaction processing. BASE24 processes require the 37 command to be performed without the old PIN, regardless of the PIN verification method used.                                                                                                                                                                                                   |

| Premium Command                                           | BASE24 Function                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 75 (Enter Key Part)                                       | This command is not recommended because of security risks, but is needed if you must enter Diebold number table values in the clear on the Authentication Profile Configuration window. For more information on this command, refer to <a href="#">appendix E</a> . |
| 114 (Import 2805 Working Key)                             | Required to import a PIN or MAC key from an AS2805-compliant interface using cipher block chaining (CBC) encryption. Electronic code book (ECB) encryption is not currently supported.                                                                              |
| 115 (Generate Non-AKB Working Key for AS2805 Export)      | Required to generate and download/export a non-AKB PIN or MAC key to an AS2805-compliant device or interface using CBC encryption. ECB encryption is not currently supported.                                                                                       |
| 120 (Generate RSA key pair)                               | Required for the BASE24-atm Automated Key Distribution System (AKDS) add-on product. Refer to <a href="#">appendix B</a> for more information on these commands.                                                                                                    |
| 122 (Generate AKB for root public key)                    |                                                                                                                                                                                                                                                                     |
| 123 (Verify public key and generate AKB of public key)    |                                                                                                                                                                                                                                                                     |
| 124 (Generate digital signature)                          |                                                                                                                                                                                                                                                                     |
| 125 (Verify digital signature)                            |                                                                                                                                                                                                                                                                     |
| 126 (Generate message digest)                             |                                                                                                                                                                                                                                                                     |
| 12F (Generate ATM master key and encrypt with public key) |                                                                                                                                                                                                                                                                     |
| 351 (EMV PIN Change)                                      | Required for validation of offline PIN changes to EMV cards.                                                                                                                                                                                                        |

| Premium Command       | BASE24 Function                                                                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| D0 (Verify Clear PIN) | This command is not recommended because of security risks, but is needed if you require clear PIN processing using the Identikey PIN verification method. |

- \* The 11D command is replaced by the 10 and 1A commands in AKB version software. The 10 and 1A commands are not premium commands. For Atalla NSPs running non-AKB software, the [Export Variant](#) field on the MAC Information tab of the Channel Security Configuration window specifies whether the 11D or 10 command is used to generate a new MAC key for a device. The 10 command is always used to generate a new data encryption key.

All of the NSP banking commands to be enabled in an Atalla device, except for the A10000E, must be specified in the security policy established using the 108 and 109 commands in the NSPDIAG tool or the NSP Configuration Management menu in the Atalla Secure Configuration Assistant (SCA).

The 105, 108 and 109 commands must be entered outside of BASE24. The 105 command to enable the 37 premium command must be obtained from Atalla and entered into the Atalla device using the NSPDIAG tool or the SCA. After this is done, you can establish the security policy using the 108 command in the NSPDIAG tool or the NSP Configuration Management menu in the SCA. When using the NSPDIAG tool, the 208 command response returned for the 108 command then must be entered at the NSP using the Secure Configuration Terminal (SCT). The NSP encrypts this result using variant 30. The result from this is then entered as a 109 command using the NSPDIAG tool. For detailed information on the 105, 108, and 109 commands, NSPDIAG tool, SCT, and SCA, refer to the appropriate Atalla documentation.

To enable banking commands for A8000 devices, use the 108 and 109 commands entered from the SCT. For more information, refer to the *A8000 Installation, Operations, and Command Reference Guide* and the *Banking Command Reference Manual*.

If you are translating PIN/PAD PIN blocks from single DES to triple DES, and vice versa, as an interim step in migrating interchanges to triple DES, the 335 (PIN Translate) command and options 6A (allow the outgoing double-length PIN encryption key to be a replicated single-length key) and 6C (allow one key-triple DES (single-length) working keys) are required. Option 6A is required if the destination uses single-length keys. Options 6C is required if the source uses single-length keys. This banking command is available in all Atalla models

(except the A8000) with AKB 2.1.3 or later software and in the following models for software version 2.9: A8100-V, A9000, A9100-V, A10000E, and A10100-V. Note that the 2.90 equivalent software on Axx100-V models is version 1.10.

If you need to convert keys from variant format to AKB format and any of the keys (KPE, KEK, MAC, etc.) to be converted to AKB format are single length keys or if any keys to be converted are comprised of key pairs (CVV, PVV) and have the same value for key 1 and key 2, option 6C must be enabled in your security policy for the 11C (Convert Atalla DES Key to 3DES AKB Format) command.

If you need options 6A or 6C for the 335 or 11C commands in your BASE24 system, you must enable them in the NSP's security policy using the 108 and 109 commands. These options are disabled in the NSP's default factory security policy. Note that with AKB version 2.51 or later software, you must enable the 11C command with a specific count of the number of times the command can be executed. For more information on the 335 and 11C commands and their options, refer to Atalla documentation.

**Note:** For NSPs running software prior to release 2.40 or 1.4, you must obtain option 6A in a 105 command, free of charge from Atalla, before you can enable it.

## NSP Commands

The following table lists the Atalla NSP commands required to support BASE24 security functions using the Transaction Security Services process. The first two columns of the table list the Atalla commands and associated message ID. The third column describes the BASE24 function for which the command is used.

| Atalla NSP Command                     | Message ID | BASE24 Function                                                                                                                                                                    |
|----------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Echo Test Message                      | 00         | Enabling a security device                                                                                                                                                         |
| Generate Working Key, Any Type         | 10         | PIN and MAC key management (device and interface)<br><br>ATM master key exchange (requires a CSM for the Device Handler process)<br><br>Also used to generate data encryption keys |
| Translate Working Key for Distribution | 11         | Key management (interface)                                                                                                                                                         |

| <b>Atalla NSP Command</b>                                  | <b>Message ID</b> | <b>BASE24 Function</b>                                                                                       |
|------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------|
| Import Non-AKB formatted Working Keys                      | 11B <sup>†</sup>  | Key management                                                                                               |
| Convert Atalla DES Key to 3DES AKB Format                  | 11C               | Key management (AKB conversion utility)                                                                      |
| Generate ATM MAC or Data Encryption Key                    | 11D <sup>*§</sup> | MAC key management (device)                                                                                  |
| Translate Working Key for Distribution to Non-Atalla Node  | 1A                | Key management (device and interface)                                                                        |
| Translate Public or Private Key AKB from MFK to PMFK       | 12B               | BASE24-atm Automated Key Distribution System add-on product<br>(Master File Key Conversion utility)          |
| Generate ATM Master Key and Encrypt with Public Key        | 12F               | BASE24-atm Automated Key Distribution add-on product<br><br>This command is supported only in Axx100 models. |
| Translate Working Key for Local Storage (switch-to-switch) | 13                | Key management (interface)                                                                                   |
| Load ATM Master Key, Primary Node – Diebold (non-AKB only) | 14#1#55           | ATM master key exchange (requires a CSM for the Device Handler process)                                      |
| Change Transaction Key                                     | 15#2              | PIN key management (device)                                                                                  |
| Translate PIN—ANSI format                                  | 31#1              | PIN translation (interface)                                                                                  |
| Translate PIN—PIN Pad format                               | 31#3              |                                                                                                              |
| Translate PIN—Visa DUKPT                                   | 31#7              | Derived unique key per transaction (DUKPT) PIN encryption                                                    |

| <b>Atalla NSP Command</b>                  | <b>Message ID</b> | <b>BASE24 Function</b>                                                                                                                                                                                                                                              |
|--------------------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Verify PIN—Identikey method                | 32#1              | PIN verification                                                                                                                                                                                                                                                    |
| Verify PIN—IBM 3624-format method          | 32#2              |                                                                                                                                                                                                                                                                     |
| Verify PIN—Visa PVV method                 | 32#3              |                                                                                                                                                                                                                                                                     |
| Verify PIN—Diebold method                  | 32#5              |                                                                                                                                                                                                                                                                     |
| Translate PIN—ANSI to PIN Pad              | 33#13             | PIN translation (interface)                                                                                                                                                                                                                                         |
| Translate PIN—PIN Pad to PIN Pad           | 33#33             |                                                                                                                                                                                                                                                                     |
| Verify AMEX CSC                            | 35A               | Card verification (American Express cards)                                                                                                                                                                                                                          |
| Encrypt Or Decrypt Data Or Translate       | 55                | Data field encryption                                                                                                                                                                                                                                               |
| Generate MAC and Encrypt or Translate Data | 59                |                                                                                                                                                                                                                                                                     |
| Load Diebold Number Table                  | 74                | Downloading the Diebold Number Table (DNT)                                                                                                                                                                                                                          |
| Enter Key Part                             | 75                | This command is not recommended because of security risks, but is needed if you must enter Diebold number table values in the clear on the Authentication Profile Configuration window. For more information on this command, refer to <a href="#">appendix E</a> . |
| Reformat Initialization Vector             | 95                | Data field encryption                                                                                                                                                                                                                                               |
| Encrypt Or Decrypt Data                    | 97                |                                                                                                                                                                                                                                                                     |
| Generate MAC                               | 98                | Message authentication (device and interface)                                                                                                                                                                                                                       |
| Verify MAC                                 | 99                |                                                                                                                                                                                                                                                                     |

| <b>Atalla NSP Command</b>                                           | <b>Message ID</b> | <b>BASE24 Function</b>                                                                                          |
|---------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Network Security Processor Status Key                               | 9A                | Master File Key Conversion utility                                                                              |
| Translate Working Key From Under the Current MFK to the Pending MFK | 9E                |                                                                                                                 |
| Replace the Current MFK with the Pending MFK                        | 9F                |                                                                                                                 |
| Import 2805 Working Key                                             | 114               | Key management (AS2805-compliant interface)                                                                     |
| Generate Non-AKB Working Key for AS2805 Export                      | 115               | Key management (AS2805-compliant device or interface)                                                           |
| Generate RSA Key Pair                                               | 120               | BASE24-atm Automated Key Distribution add-on product<br><br>These commands are supported only in Axx100 models. |
| Generate AKB for Root Public Key                                    | 122               |                                                                                                                 |
| Verify Public Key and Generate AKB of Public Key                    | 123               |                                                                                                                 |
| Generate Digital Signature                                          | 124               |                                                                                                                 |
| Verify Digital Signature                                            | 125               |                                                                                                                 |
| Generate Message Digest                                             | 126               |                                                                                                                 |
| PIN Translate—Double Length KPE                                     | 335 <sup>§</sup>  | Triple DES/DES translation (interface)                                                                          |
| PIN Translate—Visa DUKPT to 3DES KPE with DUKPT MAC Verify          | 346 <sup>**</sup> | DUKPT message authentication                                                                                    |
| DUKPT to DES PIN Translate with DES MAC Generate                    | 347 <sup>**</sup> |                                                                                                                 |
| DUKPT MAC Verify                                                    | 348 <sup>**</sup> |                                                                                                                 |
| Verify EMV ARQC                                                     | 350 <sup>§</sup>  | EMV processing                                                                                                  |
| EMV PIN Change                                                      | 351 <sup>§</sup>  |                                                                                                                 |
| Generate EMV MAC                                                    | 352 <sup>§</sup>  |                                                                                                                 |



| Atalla NSP Command                     | Message ID        | BASE24 Function                                                                                                                                           |
|----------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| PIN Change—Identikey                   | 37#1 <sup>‡</sup> | PIN change processing                                                                                                                                     |
| PIN Change—IBM 3624                    | 37#2 <sup>‡</sup> |                                                                                                                                                           |
| PIN Change—Visa PVV                    | 37#3 <sup>‡</sup> |                                                                                                                                                           |
| PIN Change—Diebold                     | 37#5 <sup>‡</sup> |                                                                                                                                                           |
| DUKPT MAC Generate                     | 386 <sup>**</sup> | DUKPT message authentication                                                                                                                              |
| Get System Configuration Information   | 1110              | NSP initialization (to obtain software version)                                                                                                           |
| Generate CVV/CVC                       | 5D#3              | Requires a CSM in the security utilities to use.                                                                                                          |
| Verify CVV, Track 2, Online            | 5E#3              | Card verification (MasterCard and Visa cards)                                                                                                             |
| Generate Check Digits                  | 7E                | ACI desktop key entry<br>Check Digit Validation and Generation utility<br>Key management (AS2805-compliant device or interface)                           |
| Translate PIN and Generate MAC—ANSI    | BD#1 <sup>§</sup> | PIN translation and message authentication (interface)                                                                                                    |
| Translate PIN and Generate MAC—PIN Pad | BD#3 <sup>§</sup> |                                                                                                                                                           |
| Verify Clear PIN                       | D0#1              | This command is not recommended because of security risks, but is needed if you require clear PIN processing using the Identikey PIN verification method. |

\* The 11D command has been retired in AKB version software. This command is replaced by the 10 and 1A commands in AKB software. For Atalla NSPs running non-AKB software, the [Export Variant](#) field on the MAC Information tab of the Channel Security Configuration window specifies whether the 11D or 10 command is used to generate a new MAC key for a device. The 10 command is always used to generate a new data encryption key.

- † The 11B command translates a working key in non-AKB format from encryption under a KEK, to encryption under the MFK in AKB format.
- ‡ The 37 message ID can be configured by Atalla to be performed with or without the old PIN, depending on the value used in command 100/105 or in security policy commands (108 and 109) when this command is activated. For the BASE24 system, it must be configured to be performed without the old PIN. This configuration is called option 66.
- § The BD, 11D, 335, 350, 351, and 352 functions are not supported by the A8000 model. The A8100-V model supports all of these commands; however, you cannot send binary data in the BD command when sending the command over an asynchronous port. The A8100 model does not support the 11D command when it is configured with AKB version software.

Additional parameters are passed in the 350, 351, and 352 functions for EMV 2000 processing. Refer to [appendix D](#) for more information.

- \*\* These functions are only supported on NSP's running with AKB release 2.5.1 or later software. The 347 and 386 commands are defaulted to off due to high security exposure. You must enable these commands in your NSP's security policy before you can use them in processing.

# Bull BNTng Hardware Security Modules (HSMs)

This subsection describes how Bull BNTng HSMs must be configured to work with the Transaction Security Services process. It does not describe the proprietary workings of the security modules, but rather the functions of each that are supported by the Transaction Security Services process and the information a user will require for configuring the modules for use by the Transaction Security Services process.

Bull BNTng HSM security functions are available through separate utilities, the ACI desktop user interface, or the Key Management Center (KMC), which consists of an application on a Linux personal computer and a BNTng HSM. Keys can be arranged by group and assigned to HSMs either individually or by group.

## Connectivity

The Transaction Security Services process connects to a Bull BNTng HSM using a single TCP/IP connection. You must use a Remote Application Manager (RAM) process between the Transaction Security Services processes and the HSM to facilitate multiple Transaction Security Services processes communicating with a single Bull BNTng HSM.

## Supported Functions

BNTng HSMs used with the Transaction Security Services process currently support PIN translation, PIN verification (for the Visa PVV PIN verification method only), PVV generation (for Visa PVV only), card verification, message authentication, and EMV functions. The IBM DES, Diebold, Identikey, and NCR methods of PIN verification, PIN encryption, and data encryption and decryption are not currently supported. In addition, the Automated Key Distribution System (AKDS) and the Master File Key Conversion utility are not currently supported by the Transaction Security Services process for BNTng HSMs. The RSA commands needed for AKDS are supported by the BNTng HSM, but the Transaction Security Services process does not yet support these functions for the BNTng HSM.

## Key Storage Method

Bull provides an application to create the keys for the BNTng HSM. After a key is created, you must store it in the TSS database using the key token storage method described below.

For the key token storage method, you must enter a *header locator* value in the Header field for the key that corresponds to a BNTng header record in the BNTHD, the key cryptogram in the Key Part 1 and Key Part 2 fields, and the cryptogram check digits in the Check Digits field. The Header field for a key cryptogram must contain 8 alphanumeric characters. You must also enter the header record on the BNTng Header Configuration window where the Header Key field contains the header locator value you entered in the Header field for the key. The Transaction Security Services process sends the *key token*, which contains the 32- or 42-byte binary header, the key cryptogram and the key check value, in request messages to the Bull BNTng HSM.

**Note:** Binary data is entered in hexadecimal character format in the TSS database.

## BNTng Key Header

When storing key cryptograms in the TSS database, all key cryptograms used by the BNTng HSM are stored with a 32- or 42-byte binary\* key header that specifies the following for each key stored in the TSS database:

- Key ID type
- Decryption key ID (i.e., the identifier of the storage or transport key used to encrypt this key)
- Exchanged key ID (i.e. the identifier of this key)
- Key usage
- Encryption method

\* Binary values are entered and stored as hexadecimal characters in the TSS database. Key headers are referred to as *labels* in the Bull BNTng nomenclature.

Because the size of the BNTng header is too large to fit in the existing TSS database with the key cryptograms, it is stored in a separate data source and is mapped from the key cryptogram record using a *header locator* value. You enter the header locator value in the Header field for a particular key cryptogram. This

value must match the value you enter in the Header Key field on the BNTng Key Header Configuration window. An example of a header locator value is shown below.

| Exchange Key           |   | Key Variant | 0 - Key Exchange Key Variant |      |      |      | Key Length |      |      |      | Double     |  |    |
|------------------------|---|-------------|------------------------------|------|------|------|------------|------|------|------|------------|--|----|
|                        |   | Header      | Key Part 1                   |      |      |      | Key Part 2 |      |      |      | Key Part 3 |  | Ch |
|                        | 1 | KTKatmmk    | 3333                         | 3333 | 3333 | 3333 | 4444       | 4444 | 4444 | 4444 |            |  |    |
| <div><div></div></div> |   |             |                              |      |      |      |            |      |      |      |            |  |    |

The data for BNTng headers is entered on the BNTng Key Header Configuration window (accessed from **Configure > Transaction Security > BNTng Header** on the ACI desktop user interface). The data entered on this window is stored in the BNTng Header File (BNTHD) and is accessed in online transactions processing from the BNTng Header external memory table (BNTHOD). An example of the BNTng Key Header Configuration window is shown below, followed by descriptions of its fields.

**ACI BNTng Key Header Configuration - KTKatmmk**

Header Key: KTKatmmk ▼

Key ID Type: 03 ▼

Decryption Key ID: FF0011223344556677889900AABBCCDDEEFF

Exchange Key ID: AA0011223344556677889900AABBCCDDEEFF

Key Usage: 1000

Encryption Method: DES ECB (Single or Double length keys) ▼

## Header Key

The Header Key field specifies a header locator value defined for the key in the Header field of the TSS database record to which this header record pertains. The value in this field serves as the primary key to BNTHD records. Because the Header field for a key cryptogram must contain 8 alphanumeric characters, the value you enter in the Header Key field must also contain 8 alphanumeric characters.

## Key ID Type

The Key ID Type field specifies a value of 01 or 03 that indicates the format of the key. The 01 value is used for interbanking identifiers of the type 02 in the issuing environment for French interbanking and is not used for RSA keys. The 03 value is recommended for all other cases. Bull strongly recommends that you only use the 03 value.

## Decryption Key ID

The Decryption Key ID field contains the BNTng key identifier of the key used to encrypt this key for local storage. Refer to the Bull ***BNT EMV Cryptographic Principles*** document for a detailed description of key identifiers.

The value in this field must consist of 28 hexadecimal characters if the Key ID Type field is set to 01, or 38 hexadecimal characters if the Key ID Type field is set to 03.

## Exchange Key ID

The Exchange Key ID field contains the BNTng key identifier of the key used to encrypt this key for transport and storage. Refer to the Bull ***BNT EMV Cryptographic Principles*** document for a detailed description of key identifiers.

The value in this field must be consist of 28 hexadecimal characters if the Key ID Type field is set to 01, or 38 hexadecimal characters if the Key ID Type field is set to 03.

## Key Usage

The Key Usage field contains two bytes of binary controls that limit the usage of the key to the functions available to its key type. These two bytes of binary data are represented by four hexadecimal characters in the BASE24-es database. Therefore, you must enter the hexadecimal representation of the binary data in this field. Refer to the Bull ***BNT EMV Cryptographic Principles*** document for a detailed description of these key usage control elements.

## Encryption Method

The Encryption Method field specifies the encryption method used for the key. The encryption method defines the computational algorithm of the key cryptogram on the basis of its clear value and possibly other key attributes from the key header. Valid values are as follows:

|                                                   |                                                                                                                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| DES ECB (Single or double length keys)            | = DES electronic code book (ECB) encryption of a single or double length key by a double length key with a control vector.            |
| DES CBC (Double length keys)                      | = DES cipher block chaining (CBC) encryption by a double length key with a control vector.                                            |
| DES ECB (Double length keys with additional data) | = DES electronic code book (ECB) encryption of a double length key with additional data by a double length key with a control vector. |

Refer to the Bull ***BNT EMV Cryptographic Principles*** document for a detailed description of these encryption methods and the types of keys encrypted under each method.

## Check Digits

Check digits are required for any keys entered into the TSS database for Bull BNTng HSMs because they are part of the key token used to represent keys. Check digits are automatically validated by the HSM when you enter new keys and check digits from the ACI desktop user interface. This provides immediate validation of keys, rather than having to use a batch utility to validate keys.

The User Interface (XML Server) process uses the ED (Secret key verification) command when you enter a new key from the ACI desktop user interface.

## BNTng Initialization Data

Data required to initialize a BNTng is stored in the BNTng Initialization File (BNTID). The BNTID contains a separate record for each BNTng HSM in your BASE24 system. The primary key to each record is the assign name of the HSM as defined in the Security Device Name field on the Security Device Configuration window and a user-defined initialization data identifier.

Each record contains the data to be loaded in BNTng key distribution file format (i.e., tag DF 14). Each HSM must be loaded with a particular KDK file containing the master key of the institution (the equivalent of the Atalla MFK) encrypted under the master key of the HSM. Note that it is valid to load KDKs for two distinct institutions into one HSM.

The data is held as raw binary in the BNTID. Because it is assumed that the BNTID will always be loaded programmatically, there is no configuration window on the ACI desktop user interface for this data source.

When you enter a process control INIT command for a Bull BNTng HSM, the initialization data for the HSM from the BNTID is loaded into the HSM.

## Preparing Data for the TSS Database

To aid in the entry of keys, the following table lists the relevant key entry fields in the TSS database with the key mnemonic (e.g., KTK, KT) associated with that key in the Bull BNTng documentation. Refer to the Bull ***BNT EMV Cryptographic***



*Principles* document for a detailed description of these key mnemonics, the header data associated with each type of key, and the KMC commands used to generate the key. All keys are double-length keys unless specified otherwise.

| Field or Tab Name on User Interface Window                       | Window                      | Bull Key Mnemonic              |
|------------------------------------------------------------------|-----------------------------|--------------------------------|
| PIN Information > Exchange Key                                   | Channel <sup>*</sup>        | KTK                            |
| PIN Information > Inbound Key                                    |                             | KT<br>KT1 (single length)      |
| MAC Information > Exchange Key                                   |                             | KTK                            |
| MAC Information > Inbound Key                                    |                             | KSC<br>KSC1 (single length)    |
| VISA PVV Keys                                                    | Authentication <sup>†</sup> | KPV                            |
| Card Verification > Card                                         |                             | KCVX, KCVC, or KCSC            |
| Card Verification > EMV > Secure Message Key for Integrity       |                             | IMK <sub>sm</sub> <sup>‡</sup> |
| Card Verification > EMV > Cryptographic Keys                     |                             | IMK <sub>ac</sub>              |
| Card Verification > EMV > Secure Message Key for Confidentiality |                             | IMK <sub>sm</sub> <sup>‡</sup> |
| Exchange Information > PIN Keys > Import                         | Interface <sup>*</sup>      | KTK                            |
| Exchange Information > PIN Keys > Export                         |                             | KTK                            |
| Exchange Information > MAC Keys > Import                         |                             | KTK                            |
| Exchange Information > MAC Keys > Export                         |                             | KTK                            |
| PIN Information > Outbound Keys                                  |                             | KT<br>KT1 (single length)      |
| PIN Information > Inbound Keys                                   |                             |                                |
| MAC Information > Outbound Keys                                  |                             | KSC<br>KSC1 (single length)    |
| MAC Information > Inbound Keys                                   |                             |                                |

- \* The Data Encryption tabs on the Channel Security Configuration window and the Interface Security Configuration window as well as the Master Keys tab on the Interface Security Configuration window are not supported when using a Bull BNTng HSM.
- † When using a Bull BNTng HSM, the DES (IBM 3624) PIN Verification, Diebold, NCR, and Identikey tabs on the Interface Configuration window are not supported when using a Bull BNTng HSM.
- ‡ For the key stored in the Secure Message Key for Integrity field, the U\_mac bit must be set in the Key Usage field for the key header on the BNTng Key Header Configuration window. For the key stored in the Secure Message Key for Confidentiality, the U\_conf bit must be set in the Key Usage field.

## Copy/Paste Key Text

When using a BNTng HSM, the Copy Key Text and Paste Key Text functions are disabled on Transaction Security windows.

## Supported BNTng Commands

The following table lists the Bull BNTng HSM request functions required to support Transaction Security Services functions. The first two columns of the table list the Bull BNTng HSM request functions and associated function codes. The third column describes the TSS function for which the command is used. If a Bull BNTng HSM function is not listed, it is not currently supported by the Transaction Security Services process.

| Bull BNTng HSM Function                                                                          | Function Code | TSS Function                                    |
|--------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------|
| MAC computation                                                                                  | E4            | EMV processing (Script MAC generation)          |
| Application cryptogram (AC) verification<br>Authorization Response Cryptogram (ARPC) computation | E2 and E3     | EMV processing (Card and issuer authentication) |
| Token verification                                                                               | ED            | Check digit generation                          |
| Self-test                                                                                        | F2            | HSM initialization                              |

| <b>Bull BNTng HSM Function</b>     | <b>Function Code</b> | <b>TSS Function</b>                      |
|------------------------------------|----------------------|------------------------------------------|
| Card verification                  | FD                   | Card verification (CVV and CVC)          |
| CVX2 verification                  | FE                   | Card verification (CVV2 and CVC2)        |
| CSC verification                   | FF 43                | American Express card verification (CSC) |
| CVX computation                    | FF 2A                | CVD generation (CVV and CVC)             |
| CVX2 computation                   | FF 2B                | CVD generation (CVV2 and CVC2)           |
| DES key generation                 | E9                   | Key generation                           |
| DES key import                     | EB                   | Key translation                          |
| Generic MAC computation            | F8                   | MAC generation                           |
| Generic MAC verification           | F9                   | MAC verification                         |
| Code verification                  | FC                   | PIN verification (Visa PVV method)       |
| PIN translate                      | FB                   | PIN translation                          |
| PIN verification value computation | FF 29                | PVV generation                           |

## Eracom Host Security Modules (HSMs)

Eracom HSM security functions are available through separate utilities, the ACI desktop user interface, or the ProtectHost White console.

### Connectivity

The Transaction Security Services process currently supports both the Eracom ProtectHost White Mark II host security module and the Eracom ProtectHost White AMB host security module.

#### ProtectHost White Mark II HSM

The ProtectHost White Mark II host security module (HSM) can be connected to the BASE24 host computer using either of the following two communications interfaces:

- Asynchronous Serial (V.24/RS232)
- Ethernet (IEEE 802.3)

Refer to Eracom's *ProtectHost White Communications Guide* and the *BASE24 ERACOM Security Module (ESM) Reference Manual* for detailed information on the configuration requirements for each of these protocols. Procedures for configuring the HSM for each of the above interfaces are provided in Eracom's *ProtectHost White Mark II Console User Guide*.

#### ProtectHost White AMB HSM

The ProtectHost White AMB host security module (HSM) can be connected to the BASE24 host computer using the Ethernet (IEEE 802.3) communications interface or a TCP/IP protocol interface.

Refer to Eracom's *ProtectHost White Communications Guide* for detailed information on the configuration requirements for these protocols. Procedures for configuring the HSM for the above interfaces are provided in Eracom's *ProtectHost White AMB Console User Guide*.

## Supported Functions

Eracom HSMs require software with a minimum release date of 01/01/2000 or later. All software releases dated 01/01/2000 or later support triple DES.

The functions supported by the Transaction Security Services process differs between the Eracom ProtectHost White Mark II host security module (HSM) and the ProtectHost White AMB HSM. Therefore, the functions supported are described separately for each type of HSM.

Although both the ProtectHost White Mark II HSM and ProtectHost White AMB HSM generate Key Verification Codes (KVCs) (i.e., check digits) when generating keys, they do not support a *generate check digits* command that generates Key Verification Codes (KVCs) for working or session keys. The ACI desktop user interface requires such a command to validate the check digits entered by a user on Transaction Security windows. Therefore, when a ProtectHost White Mark II or ProtectHost White AMB HSM is used, the Desktop has no means of validating the check digits value entered by a user. As such, the Check Digit fields on ACI desktop user interface windows are either disabled (i.e., greyed-out) or not displayed when the Type field on the Security Device Configuration window is set to Eracom. This prevents users from entering check digit values.

### ProtectHost White Mark II HSM

ProtectHost White Mark II host security modules (HSMs) used with the Transaction Security Services process support PIN encryption, PIN verification, PVV generation, card verification (for MasterCard and Visa cards), and message authentication. For PIN verification, the HSM supports the IBM DES (3624-format) and Visa PVV PIN verification methods. The Diebold, Identikey, and NCR methods of PIN verification, card verification (for American Express cards), EMV functions, data encryption and decryption, the BASE24-atm Automated Key Distribution System add-on product, and the Master File Key Conversion utility currently are not supported for Eracom ProtectHost White Mark II HSMs. Although the ProtectHost White Mark II HSM does not support the translation of keys from encryption under the local storage key (Domain Master Key) to encryption under a key exchange key for sending to an interchange, key generation is supported. The ProtectHost White Mark II HSM supports the generation of PIN encryption and MAC keys for terminals and interchanges. The generation of data encryption keys is not supported. Therefore, the ProtectHost White Mark II HSM supports key exchanges to a terminal or interchange as well as key change network control commands.

## ProtectHost White AMB HSM

ProtectHost White AMB host security modules (HSMs) used with the Transaction Security Services process support PIN encryption, PIN verification, PVV generation (for Visa PVV only), card verification (for MasterCard and Visa cards), message authentication, and proof-of-end-point processing. For PIN verification, the HSM supports the IBM DES (3624-format) and Visa PVV PIN verification methods. The Diebold, Identikey, and NCR methods of PIN verification, card verification (for American Express cards), EMV functions, data encryption and decryption, the BASE24-atm Automated Key Distribution System add-on product, and the Master File Key Conversion utility currently are not supported by the Transactions Security Services process for ProtectHost White AMB devices. Data encryption and decryption are supported by the ProtectHost White AMB HSM, but the Transaction Security Services process does not currently support these functions for the ProtectHost White AMB HSM.

The ProtectHost White AMB HSM supports key generation for sending keys to an interchange and the translation of keys from encryption under a key exchange key to encryption under the local storage key (Domain Master Key) for keys received from an interchange. The ProtectHost White AMB HSM supports the generation of PIN encryption and MAC keys for terminals and interchanges. The ProtectHost White AMB HSM supports key exchanges to a terminal or interchange as well as key change network control commands.

Although the ProtectHost White AMB HSM does not return check digits when working or session keys are generated; check digits can be generated using the KVC Request function. Interfaces use the KVC Request function when they need check digits (e.g., for a Key Verification message).

## External Key Block Format

Eracom security modules do not use different key block formats. Therefore, the External Key Block Format field on the Channel Security Configuration window and the Interface Security Configuration window should always be set to Standard Eracom when using either ProtectHost White Mark II or ProtectHost White AMB host security modules (HSMs).

## AS2805 6.2 and 6.4 PIN Block Formats

ProtectHost White AMB HSMs support the AS2805 6.2 and 6.4 PIN block formats defined by Standards Australia in the AS 2805.3-2000 standard, *Electronic funds transfer - Requirements for interfaces - PIN management and*

**security.** Refer to this standard for detailed information on these PIN block formats. Besides the ANSI PIN (ISO Format 0) PIN block format, you can select these PIN block formats in the PIN Block Format field on the PIN Information tab of the Channel Security Configuration window using one of the following values:

AS2805 6.2 = AS2805 6.2 PIN block format

AS2805 6.4 = AS2805 6.4 PIN block format

**Note:** When you select one of the above PIN block format values on the PIN Information tab of the Channel Security Configuration window, the only other field on this tab for which you can enter information is the Key Length field. Also, the MAC Information tab is disabled.

## Proof-of-Endpoint Processing

The ProtectHost White AMB HSM performs proof-of-endpoint processing as prescribed by Standards Australia in the AS 2805.3-2000 standard, *Electronic funds transfer - Requirements for interfaces - PIN management and security*. The Transaction Security Services process supports the following functions for proof-of-endpoint processing for AS2805 compliant POS devices and interfaces: TERMPROOF 6.4, NODEPROOF, NODERESP.

The TERMPROOF - 6.4 function is designed to perform the cryptographic part of the terminal proof-of-endpoint processing. The PIN Pad Acquirer Serial Number (PPASN) is enciphered using the \*KEK and the left-hand 32 bits are compared with the value received from the terminal.

The NODEPROOF function generates the single-length random key to be forwarded to the remote node as part of the inter-nodal proof-of-endpoint processing. The Random Key (KRs) is inverted to form KRr. KRs and KRr are returned to the host encrypted by the KEKs in the request.

The NODERESP function performs the response part of the inter-nodal proof-of-endpoint processing. The function decrypts a single-length key (KRs) using the KEKr in the request. KRr is formed by inverting KRs and is returned encrypted under KEKr.

## Preparing Data for the TSS Files

Keys for Eracom ProtectHost White Mark II and AMB HSMs are entered, stored, and displayed in different formats in the TSS database. The following paragraphs describe the different formats.

## ProtectHost White Mark II HSM

Offline HSM Domain Master Key loading and key management is accomplished using a user-supplied VGA-compatible video monitor and standard 104-type keyboard attached to the video output and keyboard input HSM ports. This terminal is known as the HSM console. For detailed procedures on entering keys, refer to the *ProtectHost White Mark II Console User Guide*.

The table below summarizes how keys must be stored in the TSS database for a ProtectHost White Mark II HSM. Key fields listed with “N/A” in the Domain Master Key Variant column of the table are stored as four-hexadecimal-character index values in the TSS database, with the exception of the card verification key, which is stored as a two-numeric-character (00–99) index value. When a command request is sent to the HSM, the HSM uses the index value for the key sent in the request to retrieve the actual single- or double-length key value from memory in the HSM. The number of index locations available for a key depends on the key type. The Domain Master Key variant is not applicable for keys that are stored as index values in the TSS database.

**Note:** Some of the ProtectHost White Mark II HSM key index values entered on the Channel Security Configuration, Interface Configuration, and Authentication Profile Configuration windows are entered and displayed as five decimal characters that are converted to hexadecimal characters for storage in the TSS database. This applies to the PIN and MAC exchange keys on the Channel Security Configuration window, the PIN and MAC import and export exchange keys on the Interface Security Configuration window, and the IBM DES and Visa PVV keys on the Authentication Profile Configuration window.

For key fields in which the keys are stored as index values, the index value is entered and displayed in the Key Part 1 field for the key field on the ACI desktop window. The Key Part 2 and Key Part 3 fields are disabled (i.e., greyed-out).

Key fields listed with a number in the Domain Master Key Variant column of the table below must be encrypted under the specified Domain Master Key variant for storage. For these key fields, the actual single- or double-length key value is entered on the ACI desktop window and stored in the TSS database.



Because the HSM does not support the generation of check digits, the Check Digits fields for all key fields are greyed-out when an Eracom HSM is configured on the Security Device Configuration window. This prevents you from entering or displaying check digit information in any of the key fields listed in this table.

| Field Name on User Interface Window          | Window                      | Domain Master Key Variant |
|----------------------------------------------|-----------------------------|---------------------------|
| PIN Information > Exchange Key               | Channel <sup>*</sup>        | N/A                       |
| PIN Information > Inbound Key                |                             | 1                         |
| MAC Information > Exchange Key               |                             | N/A                       |
| MAC Information > Inbound Key                |                             | 2                         |
| DES (IBM 3624) PIN Verification <sup>†</sup> | Authentication <sup>‡</sup> | N/A                       |
| VISA PVV Keys                                |                             |                           |
| Card Verification                            |                             | N/A <sup>§</sup>          |
| Exchange Information > PIN Keys > Import     | Interface <sup>*</sup>      | N/A                       |
| Exchange Information > PIN Keys > Export     |                             | N/A                       |
| Exchange Information > MAC Keys > Import     |                             | N/A                       |
| Exchange Information > MAC Keys > Export     |                             | N/A                       |
| PIN Information > Outbound Keys              |                             | 1                         |
| PIN Information > Inbound Keys               |                             | 1                         |
| MAC Information > Outbound Keys              |                             | 2                         |
| MAC Information > Inbound Keys               |                             | 2                         |

<sup>\*</sup> The Data Encryption tabs on the Channel Security Configuration window and the Interface Security Configuration window as well as the Master Keys tab on the Interface Security Configuration window are not supported when using an Eracom security module.

<sup>†</sup> The actual value of the decimalization table is stored in the memory of the Eracom security module.

- ‡ When using an Eracom ProtectHost White Mark II host security module, the Secure Message Key For Integrity, Secure Message Key for Confidentiality, and Cryptographic Keys fields under the EMV tab are not supported.
- § This field is stored as a two-numeric-character index field instead of a four-hexadecimal-character index field.

## ProtectHost White AMB HSM

Offline AMB HSM Domain Master Key loading and key management is accomplished using a user-supplied VGA-compatible video monitor and standard 104-type keyboard attached to the video output and keyboard input AMB HSM ports. This terminal is known as the AMB console. For detailed procedures on entering keys, refer to the *ProtectHost White AMB Console User Guide*.

The table below summarizes how keys must be stored in the TSS database for an AMB HSM when using ANSI PIN (ISO Format 0) blocks.

When keys that are stored in the TSS database and sent in a command to the AMB HSM, the key field must include the format code, KM index or key index (if applicable), and encrypted key (if applicable).

The Format Code column specifies the format codes you can enter in the Format Code field when entering and storing a key in the TSS database. Format codes specify attributes of the key such as the key length (i.e., single, double, or triple), mode of operation (i.e., encrypted, electronic code book (ECB) encrypted, or cipher block chain (CBC) encrypted) used to encrypt the key, algorithm used to encrypt the key, etc. Possible format codes are:

- 02 = Key stored in AMB HSM in BCD
- 03 = Key stored in AMB HSM in binary
- 20 = Encrypted key – single length – 64 bit
- 21 = CBC encrypted key – double length – 128 bit
- 22 = CBC encrypted key – triple length – 192 bit
- 23 = ECB encrypted key – double length – 128 bit
- 30 = Encrypted key – single length – 64 bit (no KM index; KEK encrypted)
- 31 = CBC encrypted key – double length – 128 bit (no KM index, KEK encrypted)
- 32 = CBC encrypted key – triple length – 192 bit (no KM index, KEK encrypted)
- 33 = ECB encrypted key – double length – 128 bit (no KM index, KEK encrypted)

Format codes 22 and 30–33 are not currently used and are reserved for future use.

**Note:** Because the key length is specified in the Format Code field value for AMB HSMs, the Key Length field is disabled.

The KM Index/Key Index column lists the valid key index values you can enter on ACI desktop windows for these fields. Key index values must be entered as either four numeric characters or two hexadecimal characters. They are stored in the TSS database as either binary coded decimal (BCD) or binary values.

When a command request is sent to the AMB HSM, the AMB HSM uses the index value for the key sent in the request to retrieve the actual single-length, double-length, or triple-length key value from memory in the AMB HSM. The number of index locations available for a key depends on the key type.

You enter the format code and the numeric or hexadecimal key index value in the Format Code and KM Index/Key Index fields, respectively, on ACI desktop transaction security windows.

For keys stored in the TSS database, you enter the encrypted key values in the Key Part 1, Key Part 2, and Key Part 3 fields as required for the key length. For key fields in which the keys are stored as index values in the TSS database, the Key Part 1, Key Part 2, and Key Part 3 fields are not displayed.

Because the AMB HSM does not support the generation of check digits for working or session keys, the Check Digits fields for all key fields are not displayed when an AMB HSM device is configured on the Security Device Configuration window. This prevents you from entering or displaying check digit information in any of the key fields listed in this table.

| Field Name on User Interface Window | Window   | Format Code | KM Index/Key Index |
|-------------------------------------|----------|-------------|--------------------|
| PIN Information > Exchange Key      | Channel* | 20, 21, 23  | 00–FF              |
| PIN Information > Inbound Key       |          | 20, 21      |                    |
| MAC Information > Exchange Key      |          | 20, 21, 23  |                    |
| MAC Information > Inbound Key       |          | 20, 21      |                    |

| Field Name on User Interface Window                   | Window                      | Format Code | KM Index/Key Index |
|-------------------------------------------------------|-----------------------------|-------------|--------------------|
| DES (IBM 3624) PIN Verification <sup>†</sup>          | Authentication <sup>‡</sup> | 02          | 0000–9999          |
| VISA PVV Keys                                         |                             | 03          | 0000–FFFF          |
| Card Verification                                     |                             | 20, 21, 23  | 00–FF              |
| Exchange Information > PIN Keys > Import              | Interface <sup>*</sup>      | 20, 21      | 00–FF              |
| Exchange Information > PIN Keys > Export              |                             |             |                    |
| Exchange Information > MAC Keys > Import <sup>§</sup> |                             |             |                    |
| Exchange Information > MAC Keys > Export <sup>§</sup> |                             |             |                    |
| PIN Information > Outbound Keys                       |                             |             |                    |
| PIN Information > Inbound Keys                        |                             |             |                    |
| MAC Information > Outbound Keys                       |                             |             |                    |
| MAC Information > Inbound Keys                        |                             |             |                    |

\* The Data Encryption tabs on the Channel Security Configuration window and Interface Security Configuration window, and the Master Keys tab on the Interface Security Configuration window, are not supported when using an Eracom security module.

If you select a value of AS2805 6.2 or AS2805 6.4 in the PIN Block Format field on the PIN Information tab, enter a value in the Key Length field only on this tab. None of the key table fields are displayed on this tab for these PIN block formats. In addition, the MAC Information tab is disabled.

<sup>†</sup> The actual value of the decimalization table is stored in the memory of the Eracom security module.

<sup>‡</sup> When using an Eracom ProtectHost White AMB host security module, the Secure Message Key For Integrity, Secure Message Key for Confidentiality, and Cryptographic Keys fields under the EMV tab are not supported.

<sup>§</sup> The exchange of MAC keys is reserved for future use. Therefore, these fields are currently disabled.

## Supported Eracom Online Functions

This section lists the Eracom ProtectHost White Mark II and AMB host security module (HSM) functions required to support Transaction Security Services functions.

### ProtectHost White Mark II HSM

The following table lists the ProtectHost White Mark II HSM functions required to support Transaction Security Services functions. The first two columns of the table list the ProtectHost White Mark II HSM functions and associated function codes. The third column describes the TSS function for which the command is used. If a ProtectHost White Mark II HSM function is not listed, it is not currently supported by the Transaction Security Services process.

| Eracom Function                         | Function Code | TSS Function                     |
|-----------------------------------------|---------------|----------------------------------|
| HSM Status                              | 01            | Enabling a security device       |
| Generate initial device PIN key         | EE0400        | PIN key management (device)      |
| Generate initial device MAC key         | EE0400        | MAC key management (device)      |
| Generate initial interface PIN key      | EE0402        | PIN key management (interface)   |
| Generate initial interface MAC key      | EE0402        | MAC key management (interface)   |
| Translate PIN key for storage           | EE0403        | Key management (interface)       |
| Translate MAC key for storage           | EE0403        | Key management (interface)       |
| Verify PIN—IBM 3624 (ANSI PIN block)    | EE0603        | PIN verification—IBM 3624 method |
| Verify PIN—IBM 3624 (PIN/Pad PIN block) | EE0603        | PIN verification—IBM 3624 method |
| Verify PIN—Visa PVV (ANSI PIN block)    | EE0605        | PIN verification—Visa PVV method |
| Verify PIN—Visa PVV (PIN/Pad PIN block) | EE0605        | PIN verification—Visa PVV method |

| <b>Eracom Function</b>                                          | <b>Function Code</b> | <b>TSS Function</b>                              |
|-----------------------------------------------------------------|----------------------|--------------------------------------------------|
| Translate PIN (both PIN block format and/or PIN encryption key) | EE0602               | PIN translation                                  |
| PIN offset generate—IBM 3624 (ANSI PIN block)                   | EE0604               | PIN change and PIN selection                     |
| PIN offset generate—IBM 3624 (PIN/Pad PIN block)                | EE0604               | PIN change and PIN selection                     |
| PIN offset generate—Visa PVV                                    | EE0607               | PIN change and PIN selection                     |
| Generate CVV/CVC                                                | 9B                   | Requires a CSM in the security utilities to use. |
| Verify CVV/CVC                                                  | 9C                   | Card verification                                |
| Generate MAC                                                    | EE0701               | Message authentication                           |
| Verify MAC                                                      | EE0702               | Message authentication                           |

## ProtectHost White AMB HSM

The following table lists the ProtectHost White AMB host security module (HSM) functions required to support Transaction Security Services functions. The first two columns of the table list the ProtectHost White AMB HSM functions and associated function codes. Function codes on the ProtectHost White AMB HSM that conforms to Australian Payments Clearing Association (APCA) standards are denoted by “(APCA)” in the Function Code column. The third column describes the TSS function for which the command is used. If a ProtectHost White AMB HSM function is not listed, it is not currently supported by the Transaction Security Services process.

| <b>Eracom Function</b>                                     | <b>Function Code</b> | <b>TSS Function</b>                                                       |
|------------------------------------------------------------|----------------------|---------------------------------------------------------------------------|
| Echo Test                                                  | 0000                 | Security module initialization                                            |
| TERMKEYGEN1 (Generate Terminal Session Keys and Roll KEK1) | 3000<br>3500 (APCA)  | Key management: active terminal master key (AS2805-compliant POS channel) |

| <b>Eracom Function</b>                                                                                      | <b>Function Code</b> | <b>TSS Function</b>                                                       |
|-------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------|
| TERMKEYGEN2 (Generate Terminal Session Keys using KEK2 and Roll KEK1/KEK2)                                  | 3010<br>3510 (APCA)  | Key management: backup terminal master key (AS2805-compliant POS channel) |
| NODEKEYGEN - 6.3 (Generate                                                                                  | 3A00 (APCA)          | Key management (interface)                                                |
| MKEYGEN (Generate ATM Master Key)                                                                           | 3B20 (APCA)          | Key management (ATM channel)                                              |
| RTMK (Key Translation - Receive)<br>(Translate received interchange session keys)                           | 4500 (APCA)          | Key translation (AS2805-compliant interface)                              |
| MACGEN - 6.3 and 6.4                                                                                        | 5500 (APCA)          | Message authentication (POS channel and interface)                        |
| MACVERIFY - 6.3 and 6.4                                                                                     | 5600 (APCA)          | Message authentication (POS channel and interface)                        |
| PINVERIFY 2000 (transaction from interchange; ANSI PIN Block format; DES (IBM3624) PIN Verify)              | 6500 (APCA)          | PIN verification (interface acquired)—IBM 3624 method                     |
| PINVERIFY 2000 – VISA 6.3<br>(transaction from interchange; ANSI PIN Block format; Visa PVV PIN Verify)     | 6501 (APCA)          | PIN verification (interface acquired)—Visa PVV method                     |
| PINVERIFY - 6.4 (transaction from POS AS2805 6.4 terminal; ANSI PIN Block format; DES (IBM3624) PIN Verify) | 6510 (APCA)          | PIN verification (POS AS2805 channel acquired)—IBM 3624 method            |
| PINVERIFY – VISA 6.4 (transaction from POS AS2805 6.4 terminal; ANSI PIN Block format; Visa PVV PIN Verify) | 6511 (APCA)          | PIN verification (POS AS2805 channel acquired)—Visa PVV method            |
| PVVGen - using given PIN                                                                                    | 65B4 (APCA)          | PIN change and PIN selection                                              |
| PINBLOCKTRANS - 6.3 to 6.3 (PIN Block translation – interchange to interchange)                             | 6600 (APCA)          | PIN key management (interface)                                            |
| PINBLOCKTRANS - 6.4 to 6.3 (PIN Block translation – POS terminal to interchange)                            | 6610 (APCA)          | PIN key management (interface)                                            |

| <b>Eracom Function</b>                                           | <b>Function Code</b> | <b>TSS Function</b>                                                                              |
|------------------------------------------------------------------|----------------------|--------------------------------------------------------------------------------------------------|
| KVC Request (Return Verification Code of a key)                  | 7510 (APCA)          | Check digit verification for Domain Master Key (KM) (interface and POS channel under AS2805 6.2) |
| CVVVERIFY                                                        | 8520 (APCA)          | Card verification                                                                                |
| TERMPROOF - 6.4 (Proof-of- endpoint processing – POS terminal)   | E500 (APCA)          | Proof-of-endpoint (AS2805-compliant POS channel) processing                                      |
| NODEPROOF (Proof-of-endpoint processing – request – interchange) | E520 (APCA)          | Proof-of-endpoint (interface) processing                                                         |
| NODERESP (Proof-of-endpoint processing – response - interchange) | E530 (APCA)          | Proof-of-endpoint (interface) processing                                                         |

## Considerations

Currently, the Transaction Security Services process requests one key at a time, even though the “EE” key generation commands for a ProtectHost White Mark II HSM allow multiple keys to be generated in a single request message.

The Transaction Security Services process always uses the 12 rightmost digits, excluding the check digit, for the PAN string used to build an ANSI PIN block. It does not support the 12 rightmost digits, including the check digit, or the 12 leftmost digits, for the PAN string used to build an ANSI PIN block.

For keys stored in the TSS database for a ProtectHost White Mark II HSM, the TSS application uses the Eracom key specifier format of 10 (encrypted key—single length) for single-length keys and the Eracom key specifier format of 11 (encrypted key—double-length—ECB) for double-length keys.

For keys stored in the ProtectHost White Mark II HSM, the TSS application assumes a key specifier format of 03 (index—long/binary).



# Thales e-Security Modules

Thales security functions are available through separate utilities, the ACI desktop user interface, or the HSM console.

Management of key check values (check digits) has been enhanced so that check digits are automatically validated by the security device when the keys and check digits are entered from the ACI desktop user interface. This provides immediate validation of keys, rather than having to use a batch utility to validate keys.

## External Key Block Format

The external key block format is the format of the key block that an external entity (e.g., device or interface) supports. The format is set in the External Key Block Format field on the Channel Security Configuration window and the Interface Security Configuration window. If the external entity uses a Thales e-Security HSM, you can set this field to ANSI X9.17 method or Variant method. Refer to the Thales e-Security vendor documentation for descriptions of these key encryption schemes and how they are used in processing.

## Double- and Triple-Length Key Encryption Scheme

The Transaction Security Services process requires all double- and triple-length keys for Thales e-Security HSMs stored in the TSS database to be encrypted using Thales' variant (1A+32H) key encryption scheme instead of the old (ANSI X9.17 method) non-variant (32H) key encryption scheme. For more information on these encryption schemes, refer to the Thales *Host Security Module, RG7000 Programmer's Manual*.

If a variant of the LMK is required for the key type (see the Master Key Pair column in the table for the "Preparing Data for the TSS Files" topic on the following page), this variant is always applied regardless of the key encryption scheme used (ANSI X9.17 or variant) and is applied to the left half of the LMK.

For double- and triple-length keys encrypted using the variant key encryption, another variant (two or three different ones depending on the length of the key being encrypted or decrypted) is applied to the right half of the LMK. Refer to the "Local Master Key Triple DES Variant Scheme" topic in the Thales *Host Security Module, RG7000 Programmer's Manual* for a description of how these variants are applied.

**Note:** Although Thales e-Security HSMs treat key pairs like double-length keys, the Transaction Security Services process views key pairs as two single-length keys. Because of this, the Transaction Security Services process does not support the variant (1A+32H) key encryption scheme for these keys. Therefore, Visa PVV PIN verification and card verification key pairs must be encrypted using the non-variant (32H) key encryption scheme in the TSS database.

## Decimalization Table Security

For Thales e-Security HSMs, the decimalization table value sent in commands can be encrypted or in the clear. If you are using the IBM DES (3624-format) PIN verification method, you can encrypt the clear decimalization table value using a console command, and enter the encrypted value in the Decimalization Table field on the DES (IBM 3624) PIN Verification tab of the Authentication Profile Configuration window. The Configure Security Parameter “Encrypted/Plaintext Decimalization Tables” indicates whether the HSM supports Encrypted or Plaintext decimalization tables. The valid values for this parameter are:

E = Encrypted decimalization tables are used. Default.  
P = Plaintext (in the clear) decimalization tables are used.

The “Encrypted/Plaintext Decimalization Tables” configure security parameter is available beginning in the following firmware/software version numbers for all HSM ranges.

| HSM Type | Version Number |
|----------|----------------|
| HSM 8000 | 2.0            |
| RG7x10   | 3.0            |
| RG7x00   | 7.0            |

If you choose to use encrypted decimalization tables, all decimalization tables must be encrypted. You cannot have some decimalization tables in the clear and some encrypted. In addition, you must set the optional environment attribute DECIMAL\_TBL\_ENCRYPT to a value of Y for the Master File Key Conversion utility to be able to properly translate encrypted decimalization tables from the old LMK pair to the new LMK pair.

## Preparing Data for the TSS Files

Offline HSM Master Key loading and key management are accomplished using a user-supplied asynchronous ASCII terminal connected to the HSM asynchronous port. This asynchronous ASCII terminal is known as the HSM console.

The table on the following pages summarizes the HSM console commands used to create, encrypt, and translate keys for storage in the TSS database. Note that the HSM Master Keys must have already been loaded and the HSM placed in the authorized state before the commands can be performed.

| Field Name on User Interface Window    | Window         | Master Key Pair    | HSM Console Command                                                          |
|----------------------------------------|----------------|--------------------|------------------------------------------------------------------------------|
| PIN Information: Exchange Key          | Channel        | 14–15              | K = Encrypt a terminal master key                                            |
| PIN Information: Inbound Key           | Channel        | 14–15              | K = Encrypt a terminal PIN key *                                             |
| MAC Information: Exchange Key          | Channel        | 14–15              | K = Encrypt a terminal master key                                            |
| MAC Information: Inbound Key           | Channel        | 16–17              | FK = Format key from components                                              |
| Derivation Key                         | Derivation Key | 28–29              | DE = Form a ZMK from clear components<br>DG = Generate a base derivation key |
| Cryptographic Keys                     | Authentication | Variant 1 of 28–29 | FK = Format key from components                                              |
| Secure Message Key For Integrity       | Authentication | Variant 2 of 28–29 | FK = Format key from components                                              |
| Secure Message Key for Confidentiality | Authentication | Variant 3 of 28–29 | FK = Format key from components                                              |
| DES (IBM 3624) PIN Verification        | Authentication | 14–15              | K = Encrypt a PIN verification key                                           |
| Decimalization Table                   | Authentication | 18–19              | ED=Encrypt decimalization table                                              |

| <b>Field Name on<br/>User Interface Window</b> | <b>Window</b>  | <b>Master<br/>Key<br/>Pair</b> | <b>HSM<br/>Console<br/>Command</b>                                                              |
|------------------------------------------------|----------------|--------------------------------|-------------------------------------------------------------------------------------------------|
| Diebold Number Table                           | Authentication | 10–11                          | R = Load and encrypt a clear Diebold table <sup>**</sup>                                        |
| VISA PVV Keys                                  | Authentication | 14–15                          | K = Encrypt a PIN verification key                                                              |
| Card Verification                              | Authentication | Variant of 14–15 <sup>††</sup> | KA = Generate a CVK pair                                                                        |
| NCR: Encryption Keys                           | Authentication | Variant of 20–21               | UD or UI= Translate PVK from KTR to LMK <sup>‡‡</sup>                                           |
| Exchange Information:<br>PIN Keys: Import      | Interface      | 04–05                          | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key <sup>†</sup> |
| Exchange Information:<br>PIN Keys: Export      | Interface      | 04–05                          | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key <sup>†</sup> |
| Exchange Information:<br>MAC Keys: Import      | Interface      | 04–05                          | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key <sup>†</sup> |
| Exchange Information:<br>MAC Keys: Export      | Interface      | 04–05                          | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key <sup>†</sup> |
| PIN Information:<br>Outbound Keys              | Interface      | 06–07                          | FK = Format key from components <sup>‡</sup>                                                    |
| PIN Information: Inbound Keys                  | Interface      | 06–07                          | FK = Format key from components <sup>‡</sup>                                                    |

| Field Name on User Interface Window | Window    | Master Key Pair | HSM Console Command                                                                |
|-------------------------------------|-----------|-----------------|------------------------------------------------------------------------------------|
| MAC Information: Outbound Keys      | Interface | 16–17           | FK = Format key from components <sup>§</sup>                                       |
| MAC Information: Inbound Keys       | Interface | 16–17           | FK = Format key from components <sup>§</sup>                                       |
| Master Keys: PIN Key                | Interface | 06–07           | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key |
| Master Keys: MAC Key                | Interface | 16–17           | EC = Encrypt clear component<br>FK = Format key from components<br>KI = Import key |

- \* The intermediate key value of 9999999999999999 (16 hexadecimal 9s) encrypted under the HSM Master Key pair 06–07 must be stored in a CSECD record for ATM or POS devices configured to send PINs in the clear to any interchanges on the BASE24 system.

For SPDH-compliant devices or Hypercom devices using the DUKPT method of PIN encryption, this field must contain an intermediate key encrypted under the HSM Master Key pair 06–07. Although the HSM Master Key pair 06–07 represents an interchange rather than a terminal key, it is required because Thales e-Security only supports a command to translate from DUKPT encryption to encryption under an interchange master/session key.

- † A key exchange key is the key under which working keys are encrypted for exchange between BASE24 and an external processor. It equates to the Thales e-Security Zone Master Key (ZMK).

An encrypted ZMK need not be placed in the Exchange Information fields on the Interface Security Configuration window unless automatic key exchange is supported. (BASE24 only supports automatic key exchange with DPCs and certain interchanges.)

To generate a ZMK, the console command EC must be used first to obtain three encrypted ZMK components. Console command FK is then used to format the key from the three encrypted ZMK components. Finally, the KI command is used to translate the key from encryption under the ZMK to encryption under the LMK for storage in the Exchange Information fields.

- ‡ If the BASE24 operations staff is responsible for generating the inbound and outbound keys for PIN encryption, console command B or FK can be used to generate and encrypt these keys under the HSM Master Key pair 06–07. These keys can then be placed in the PIN Information fields on the Interface Security Configuration window, and the ZMK-encrypted forms of these keys can be conveyed to the external processor.

If the inbound and outbound working keys are generated by an external processor, the keys can be distributed automatically or manually. If these keys are distributed automatically when the interface process is first brought up, inbound and outbound entries in the PIN Information fields must be left as zeros. The ZMK must be provided in the Exchange Information fields on the Interface Security Configuration window (refer to the first footnote for more information about the ZMK).

Otherwise, the BASE24 operations staff receives ZMK-encrypted versions or clear versions of these keys. The ZMK-encrypted versions of the keys must be used to obtain versions encrypted under the HSM Master Key pair 06–07 for storage in the TSS database. Since the inbound and outbound keys must be encrypted or decrypted using the ZMK, depending on whether the keys are being sent to or received from an external processor, the ZMK must be provided in the HSM console command B or FK.

If the operations staff receives clear versions of these keys, they must be encrypted using the HSM console command BK to generate the encrypted zone PIN keys (ZPKs). The key type must be set to a value of 2 for this command. If only one component is used, it should be entered three times. Otherwise, at least two different components should be entered. The resulting cryptogram can then be placed in the PIN Information fields, and the ZMK-encrypted forms of these keys can be conveyed to the external processor.

- § If zone authentication keys are not distributed automatically, they must be generated manually and encrypted using the HSM console FK command.

- \*\* Handling the Diebold Number Table — Initially, the clear Diebold number table (DNT) must be manually entered using the RSM console command R. This command causes the clear DNT to be encrypted and loaded into the RSM. It also returns an encrypted version of the DNT to the BASE24 system

administrator. The BASE24 system administrator can then copy the encrypted form of the DNT and enter it on the Authentication Profile Configuration window.

Anytime you change the DNT in the TSS database, you must rebuild the Diebold Number Table external memory table, warmboot the Transaction Security Services processes that access the external memory table, and execute the DNTLOAD process control command to reload the DNT in the memory of the HSM. Refer to [section 2](#) for information on rebuilding external memory tables and warmbooting processes that access them. Refer to [section 9](#) and [appendix E](#) for more information on using the DNTLOAD command.

†† The HSM encrypts card verification keys under a variant of key pair 14-15. This variant, unlike the variants used with Atalla security devices, is an unknown value. HSM card verification keys must be encrypted using HSM console command KA. This command incorporates the variant that is required to obtain the proper results.

‡‡ Use the UA–UD commands for single DES encryption and the UF–UI commands for two-key triple DES encryption.

## Generate Check Digits Command

The User Interface process uses the BU (Generate a Key Check Values) command when you enter a new key from the ACI desktop user interface.

## Changing Local Master Key Pairs

ACI provides an optional Master File Key Conversion tool that enables you to translate hardware-encrypted keys in the TSS database from encryption under one Local Master Key (LMK) pair value to another LMK pair value, automating what otherwise is a manual process. This conversion tool is executed from the Master File Key Conversion window, accessed as **Configure > Transaction Security > Master File Key Conversion** on the ACI desktop. Using this tool, you can translate all the keys in the TSS database, or just the keys encrypted under a particular LMK pair (i.e., 04–05, 06–07, 14–15, 16–17, 18–19, 28–29, 34–35, or 36–37). For detailed information on using this tool, refer to the [Master File Key Conversion](#) topic provided in appendix C.

Good security practice as well as several network and card associations require the LMK pairs to be replaced periodically. This conversion tool enables you to convert keys from encryption under the old LMK pair to encryption under the new

LMK pair and automatically replace them in the TSS database files. HSMs must contain the old LMK pair as the current LMK pair and contain the new LMK pair as the pending LMK pair before this utility is used. The Master File Key Conversion utility requires the BW command (Translate Keys From Old LMK to New LMK) to translate keys. After keys are converted, use the Erase LMK or MFK command on the Process Control window to erase the old LMK pair. This process control command executes the Thales e-Security BS command (Erase the Key Change Storage) on the specified HSMs.

## Triple DES Enabled Commands

All HSM commands required for BASE24 processing are triple DES-enabled in firmware version 5.05/1.05. This release also includes an option to disable single DES in HSM commands. A minimum of version 6.02 firmware and an optional DSP plug-in (for RSA functions) is required to support the BASE24-atm Automated Key Distribution System add-on product.

## Racal Access Module (RAM) Process

The RAM process provides direct IP support and supports the User Datagram Protocol/Internet Protocol (UDP/IP), a connectionless protocol, in addition to TCP/IP. These features eliminate the I/O to the XPNET process and the check-pointing overhead associated with the XPNET system. Thus, BASE24 environments using the RAM process should see performance improvement, even with the additional I/O to the Transaction Security Services process. For more information on the RAM process, refer to the ***BASE24 Transaction Security Manual*** and the following informal publications:

- The ***Remote Application Manager Implementation and Installation Guide*** describes the use of the RAM process with TCP/IP. This guide is a TGAL document located in the RAMUPDT file on the BA10RAM subvolume.
- The ***RAM/UDP Update Guide*** describes the use of the URAM process with UDP/IP. The RAM process is referred to as the URAM process when used with UDP/IP. This guide is a Microsoft Word document. The binary format of this document is located in the URAMUDOC file on the BA10RAM subvolume. To view this document, perform a binary transfer of the URAMUDOC file to your PC and add a .DOC extension to the file name. You can then open the document using the Word application.



## HSM Commands

The following table lists the Thales e-Security commands required to support BASE24 security functions using the Transaction Security Services process. The first two columns of the table list the Thales e-Security commands and associated transaction codes. The third column describes the BASE24 function for which the command is used.

| Thales e-Security Command                                                                  | Transaction Code | BASE24-es Function                                                                                         |
|--------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| Generate a key                                                                             | A0               | BASE24-atm Automated Key Distribution System add-on product                                                |
| Translate a TMK, TPK, or PVK from LMK to Another TMK, TPK, or PVK                          | AE               | PIN key translation (device)                                                                               |
| Erase the Key Change Storage                                                               | BS               | Master File Key Conversion utility                                                                         |
| Generate a Key Check Value                                                                 | BU               | ACI desktop key entry                                                                                      |
| Translate Keys From Old LMK to New LMK                                                     | BW               | Master File Key Conversion utility                                                                         |
| Translate PIN (encryption under LMK pair 14-15 to encryption under LMK pair 06-07)         | CA               | PIN translation—terminal originated (interface)                                                            |
| Translate PIN (encryption under LMK pair 06-07 to encryption under LMK pair 06-07)         | CC               | PIN translation—interface originated (interface)                                                           |
| Generate a Diebold PIN Offset                                                              | CE               | PIN change—Diebold PIN verification method (device and interface)                                          |
| Verify PIN from Terminal—Diebold method                                                    | CG               | Diebold PIN verification (device)                                                                          |
| Translate PIN (BDK encryption under LMK pair 28–29 to ZPK encryption under LMK pair 06–07) | CI               | Derived unique key per transaction (DUKPT) PIN block translation using the single key derivation algorithm |
| Generate CVV/CVC                                                                           | CW               | Requires a CSM in the security utilities to use.                                                           |

| Thales e-Security Command                                                                 | Transaction Code | BASE24-es Function                                                                                         |
|-------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| Verify a Visa CVV                                                                         | CY               | Card verification (verify CVV/CVC)<br>(MasterCard and Visa cards)                                          |
| Verify PIN from Terminal—IBM 3624-format                                                  | DA               | IBM 3624 PIN verification method<br>(device)                                                               |
| Verify PIN from Terminal—Visa PVV                                                         | DC               | Visa PVV PIN verification method<br>(device)                                                               |
| Generate IBM 3624 PIN Offset                                                              | DE               | PIN change—IBM 3624 PIN verification<br>method (device and interface)                                      |
| Generate Visa PVV                                                                         | DG               | PIN change—Visa PVV PIN verification<br>method (device and interface)                                      |
| Interchange PIN Verify—IBM 3624-format                                                    | EA               | IBM 3624 PIN verification method<br>(interface)                                                            |
| Interchange PIN Verify—Visa PVV                                                           | EC               | Visa PVV PIN verification method<br>(interface)                                                            |
| Interchange PIN Verify—Diebold<br>method                                                  | EG               | Diebold PIN verification method<br>(interface)                                                             |
| Generate an RSA Key Set                                                                   | EI               | BASE24-atm Automated Key<br>Distribution System add-on product                                             |
| Generate a MAC on a Public Key                                                            | EO               |                                                                                                            |
| Validate a Certificate and Generate<br>MAC on its Public Key                              | ES               |                                                                                                            |
| Generate a Signature                                                                      | EW               |                                                                                                            |
| Validate a Signature                                                                      | EY               |                                                                                                            |
| Translate a Secret Key from the Old<br>LMK to a New LMK                                   | EM               | BASE24-atm Automated Key<br>Distribution System add-on product<br><br>(Master File Key Conversion utility) |
| Translate a MAC on a Public Key                                                           | EU               |                                                                                                            |
| Interchange Key Translate<br>(encryption under ZMK to<br>encryption under LMK pair 06-07) | FA               | PIN key translation (interface)                                                                            |

| Thales e-Security Command                                                            | Transaction Code | BASE24-es Function                                                                                                                         |
|--------------------------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Translate DUKPT PIN to DES PIN – Triple DES                                          | G0               | Derived unique key per transaction (DUKPT) PIN block translation using the double key derivation algorithm                                 |
| Interchange Key Translate (encryption under LMK pair 06-07 to encryption under ZMK)* | GC               | PIN key translation (interface)                                                                                                            |
| Export a DES key                                                                     | GK               | BASE24-atm Automated Key Distribution System add-on product                                                                                |
| Hash a Block of Data                                                                 | GM               |                                                                                                                                            |
| MAC Verify/MAC Generate†                                                             | GW               | Derived unique key per transaction (DUKPT) message authentication                                                                          |
| Generate TAK under TMK, TPK, or LMK                                                  | HA               | MAC key management (device and interface)                                                                                                  |
| Generate a New Terminal Master Key or PIN Key                                        | HC               | PIN key management (device)<br>Changing ATM master key (device)                                                                            |
| Interchange Key Generate (encryption under ZMK and LMK pair 06-07)                   | IA               | PIN key management (interface)                                                                                                             |
| Translate PIN from TMK or TPK to LMK                                                 | JC               | PIN change—Diebold, IBM 3624, or Visa PVV PIN verification methods (device)                                                                |
| Translate PIN from ZWK to LMK                                                        | JE               | PIN change—Diebold, IBM 3624, or Visa PVV PIN verification methods (interface)                                                             |
| Generate a Key Check Value (Not Double Length ZMK)                                   | KA               | ACI desktop key entry                                                                                                                      |
| ARQC (or TC/ACC) Verification or ARPC Generation                                     | KQ               | EMV authorization                                                                                                                          |
| Generate Secure Message with Integrity and optional Confidentiality and PIN Change   | KU               | The enhanced KU command (available in Thales version 6 or higher software) is required for validation of offline PIN changes to EMV cards. |

| Thales e-Security Command                                                                     | Transaction Code | BASE24-es Function                                                    |
|-----------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------|
| Generate Secure Message with Integrity and optional Confidentiality and PIN Change (EMV 2000) | KY               | EMV 2000 authorization                                                |
| ARQC (or TC/ACC) Verification or ARPC Generation (EMV 2000)                                   | KW               |                                                                       |
| Verify Encrypted Counters - M/Chip 4                                                          | K0               |                                                                       |
| Load Data to User Storage                                                                     | LA               | Diebold PIN verification—loading the encrypted DNT to HSM memory      |
| Translate Decimalization Table from Old to New LMK                                            | LO               | Master File Key Conversion utility (Encrypted decimalization table)   |
| Translate TAK from LMK to ZMK                                                                 | MG               | MAC key management (interface)<br>MAC key translation (interface)     |
| Translate TAK from ZMK to LMK                                                                 | MI               | MAC key translation (interface)                                       |
| Generate MAC <sup>‡</sup>                                                                     | MK               | Message authentication                                                |
| Verify MAC <sup>‡</sup>                                                                       | MM               | Message authentication                                                |
| MAC Generate/Verify <sup>§</sup>                                                              | MS               | Dutch security extensions (message authentication for large messages) |
| HSM Status                                                                                    | NO               | Enabling a security device                                            |
| Verify Card Security Codes                                                                    | RY               | Card verification (American Express cards)                            |
| PIN Generate RVT <sup>**</sup>                                                                | SA               | Dutch security extensions (Interpay interface key management)         |
| PIN Receive RVT <sup>**</sup>                                                                 | SC               |                                                                       |
| PIN Generate RVS <sup>**</sup>                                                                | SE               |                                                                       |
| PIN Receive RVS <sup>**</sup>                                                                 | SG               |                                                                       |

| Thales e-Security Command           | Transaction Code | BASE24-es Function                                                       |
|-------------------------------------|------------------|--------------------------------------------------------------------------|
| Verify PIN—NCR method <sup>**</sup> | SI               | Dutch security extensions<br>NCR PIN verification (device and interface) |

\* In some HSM firmware versions, the GC transaction code may be available only when the HSM is in the authorized state. As a result, the Repeat Key command for dynamic key management may not work. Check with your Thales e-Security customer support representative.

† Only eight-byte MACs are supported for the GW command.

‡ The MK and MM transaction codes are not supported in the standard Thales e-Security product. They must be obtained from Thales e-Security.

§ This command uses the X.19 method of message authentication, also known as pseudo triple DES.

<sup>\*\*§</sup> These commands are not available in standard Thales e-Security products and must be specially ordered.

## Asynchronous Protocol and Thales HSM Commands

The following commands for Thales e-Security hardware security modules (HSMs) are not supported over an asynchronous protocol:

- ARQC (or TC/ACC) Verification or ARPC Generation (KQ)
- Generate Secure Message with Integrity and optional Confidentiality and PIN Change (KU)
- Generate Secure Message with Integrity and optional Confidentiality and PIN Change (EMV 2000) (KY)
- ARQC (or TC/ACC) Verification or ARPC Generation (EMV 2000) (KW)
- Verify Encrypted Counters - M/Chip 4 (K0)
- MAC Verify/MAC Generate (GW)
- Generate RSA Key Set (EI)
- Generate a MAC on a Public Key (EO)

- Validate a Certificate and Generate a MAC on its Public Key (ES)
- Generate a Signature (EW)
- Validate a Signature (EY)
- Export a DES Key (GK)
- Hash a Block of Data (GM)

## 5: TSS Processing Examples

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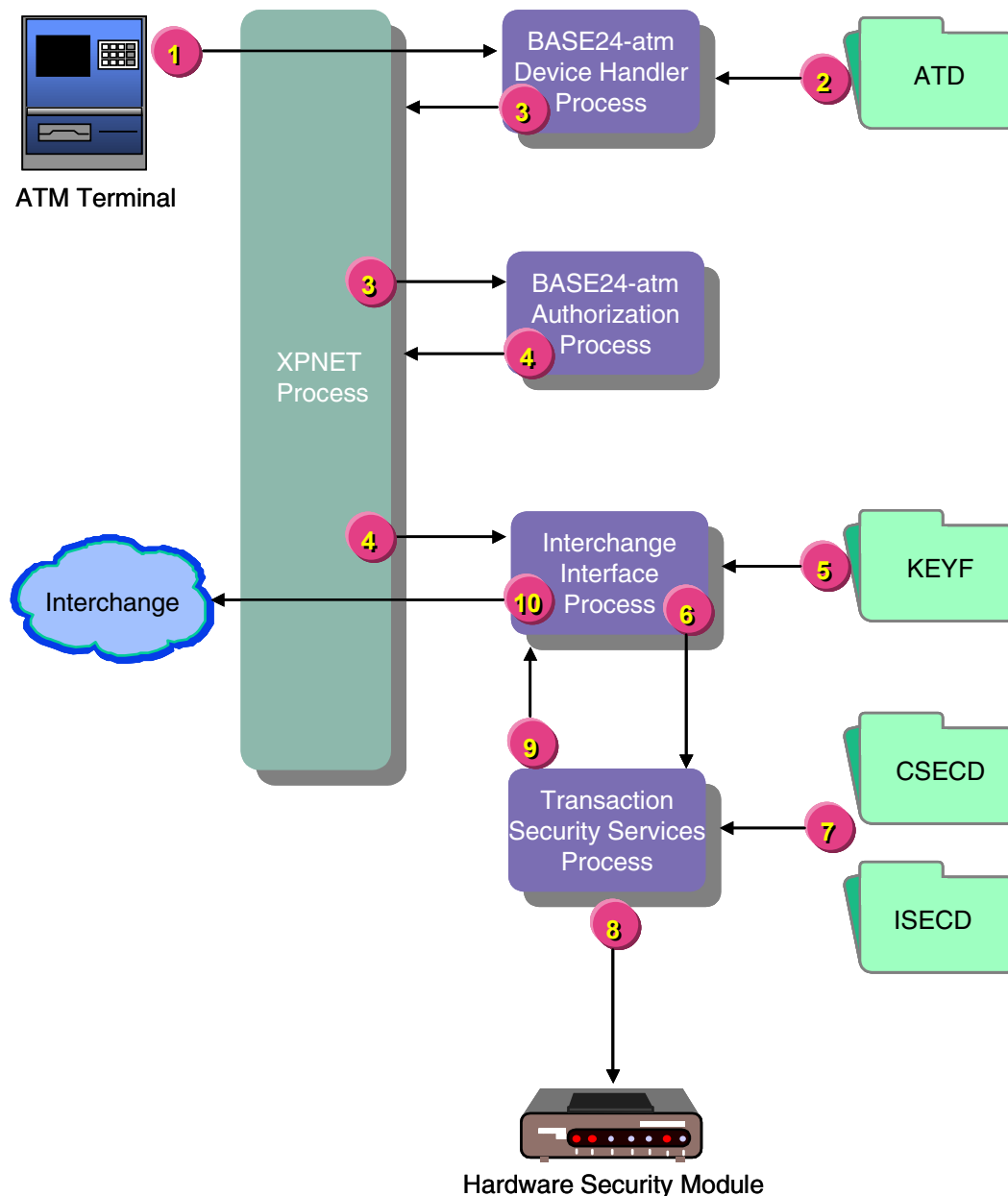
This section presents example message flows showing how the Transaction Security Services process is incorporated into online BASE24 transaction processing. All message flows assume hardware encryption for PINs and MACs, except where noted.

The message flows provided in this section are intended to provide an overall understanding of how the Transaction Security Services process fits in the flow of transaction security processing. This section is not comprehensive and is not intended to depict Transaction Security Services processing for every possible transaction security function.

**Note:** The flows in this section depict transaction processing steps integral to transaction security only.

## Terminal Acquired—Not-on-Us Card

The diagram below illustrates the flow of PIN block processing for a transaction that originates at an ATM and is authorized by an interchange. The steps depicted in the illustration are described following the diagram.



1. A transaction including a PIN originates at an ATM terminal. The XPNET process passes the request containing a single DES or triple DES encrypted PIN block to the BASE24-atm Device Handler process.

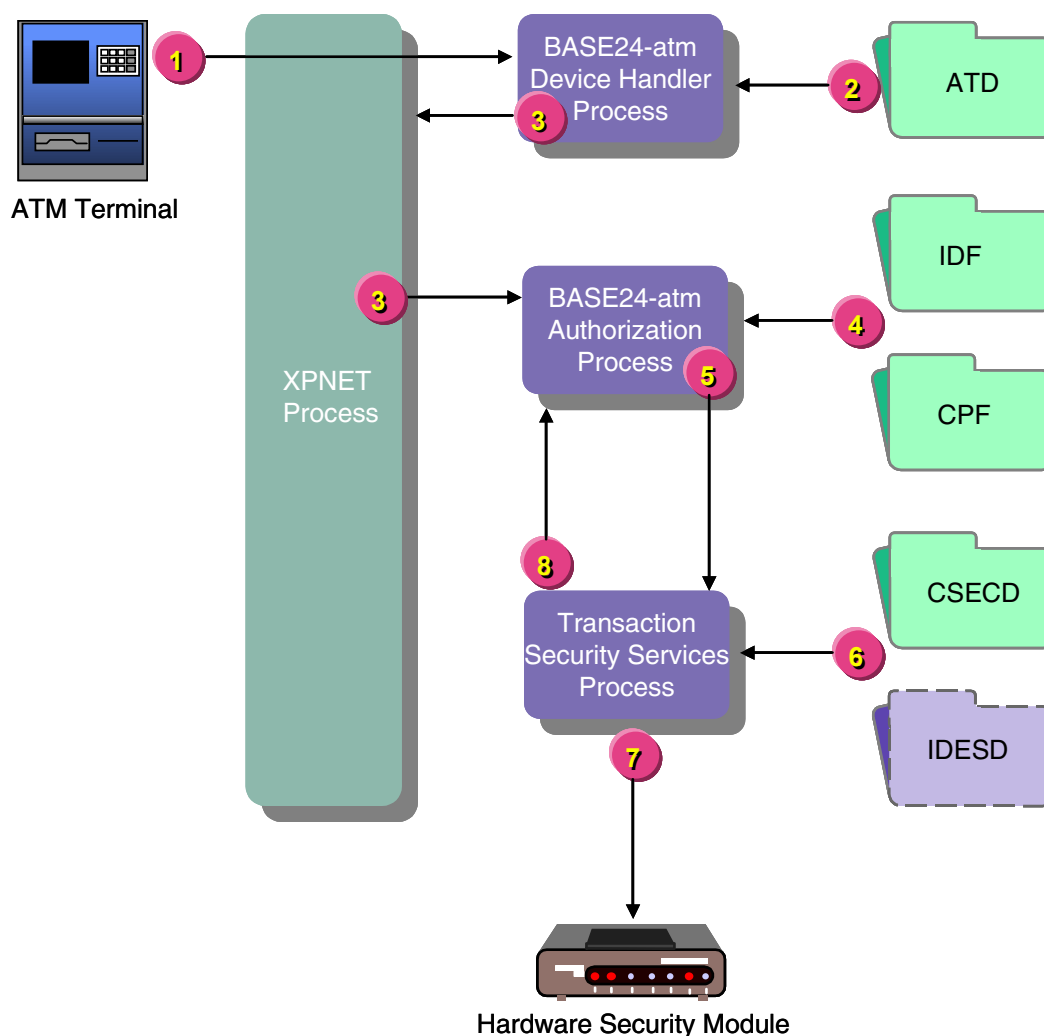


2. The Device Handler process retrieves the key locator value (i.e., the terminal ID) for the terminal's PIN key from the BASE24-atm Terminal Data files and places this value in the STM along with the encrypted PIN block.
3. The Device Handler process sends the transaction to the XPNET process for delivery to the Authorization process for authorization and routing.
4. The Authorization process determines that the transaction needs to be routed to an Interchange Interface process. The Authorization process passes the transaction to the XPNET process for delivery to the Interchange Interface process.
5. The Interchange Interface process retrieves the key locator value (i.e., an index value created by the KEYF or KEY6 conversion program) for the outbound PIN key from the Key File (KEYF).
6. The security utilities bound into the Interchange Interface process pass the ATM PIN key locator value, outbound interface PIN key locator value, the encrypted PIN block and other PIN-related information to the Transaction Security Services process. The request does not pass through the XPNET process.
7. The Transaction Security Services process uses the key locator values to retrieve the actual ATM PIN key from the CSECD and the outbound PIN key from the ISECD.
8. The Transaction Security Services process calls the HSM to translate the encrypted PIN block from encryption under the ATM PIN key to encryption under the interface's outbound PIN key. Depending on the lengths of the keys retrieved, the translation can be any of the following: single DES to triple DES, single DES to single DES, triple DES to single DES, or triple DES to triple DES.
9. The Transaction Security Services process passes the translated PIN block back to the Interchange Interface process.
10. The Interchange Interface process sends the transaction, with the translated PIN block, to the XPNET process for delivery to the interchange for authorization.

## Terminal Acquired—On-Us Card

The diagram below illustrates the flow of PIN verification processing for a transaction that originates at an ATM and is authorized on the BASE24 system. The steps depicted in the illustration are described following the diagram.

In this transaction flow, the issuing institution uses the IBM DES (3624-format) PIN verification method. If a different PIN verification method were used, the flow would be essentially the same except that the Transaction Security Services process would access a different external memory table to retrieve the PIN verification information.

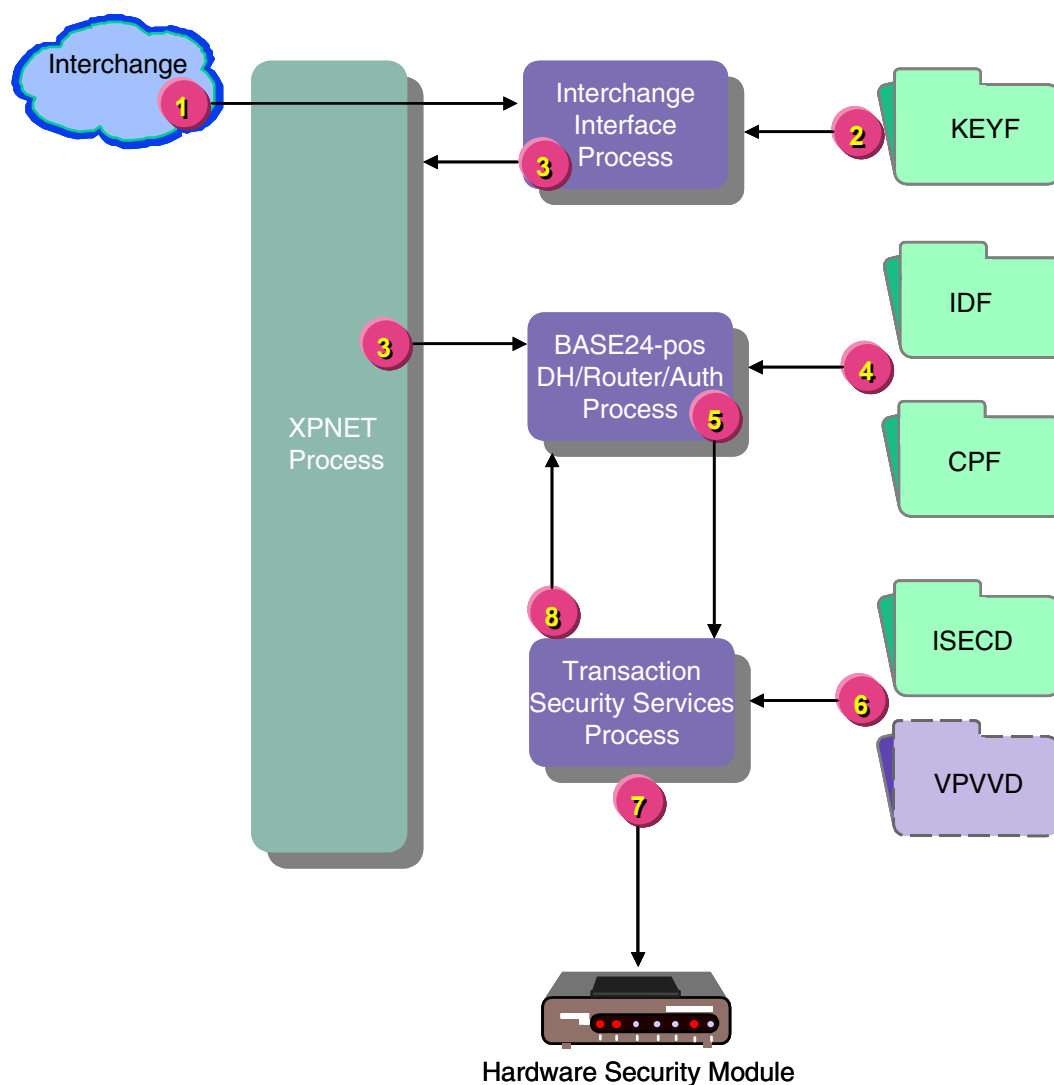


1. A transaction including a PIN originates at an ATM terminal. The XPNET process passes the request containing a single DES or triple DES encrypted PIN block to the BASE24-atm Device Handler process.
2. The Device Handler process retrieves the key locator value (i.e., the terminal ID) for the terminal's PIN key from the BASE24-atm Terminal Data files and places this value in the STM along with the encrypted PIN block.
3. The Device Handler process sends the transaction to the XPNET process for delivery to the Authorization process for authorization and routing.
4. The Authorization process determines that the transaction needs to be authorized on the BASE24 system and retrieves the PIN check type value (i.e., the PIN verification method) from the Institution Definition File (IDF) or the Card Prefix File (CPF) and the PIN verification KEYA group value for the transaction from the CPF. Note that the Authorization process does not access the KEYA at all.
5. The security utilities bound into the Authorization process pass the ATM PIN key locator value, the PIN verification type, KEYA group, card expiration date, the encrypted PIN block and other PIN-related information to the Transaction Security Services process. The request does not pass through the XPNET process.
6. The Transaction Security Services process uses the ATM PIN key locator value to retrieve the actual ATM PIN key from the CSECD and the KEYA group and card expiration date values to retrieve the IBM DES (3624-format) PIN verification information from the IDESD external memory table.
7. The Transaction Security Services process calls the HSM to decrypt the encrypted PIN block using the ATM PIN key and verify the PIN using the PIN verification key. Depending on the lengths of the keys retrieved, the HSM can perform single or triple DES decryption and verification.
8. The Transaction Security Services process passes the result of the PIN verification back to the Authorization process.

## Interface Acquired—On-Us Card

The diagram below illustrates the message flow of PIN verification processing when an interchange sends a transaction to the BASE24 system for authorization. The steps depicted in the illustration are described following the diagram.

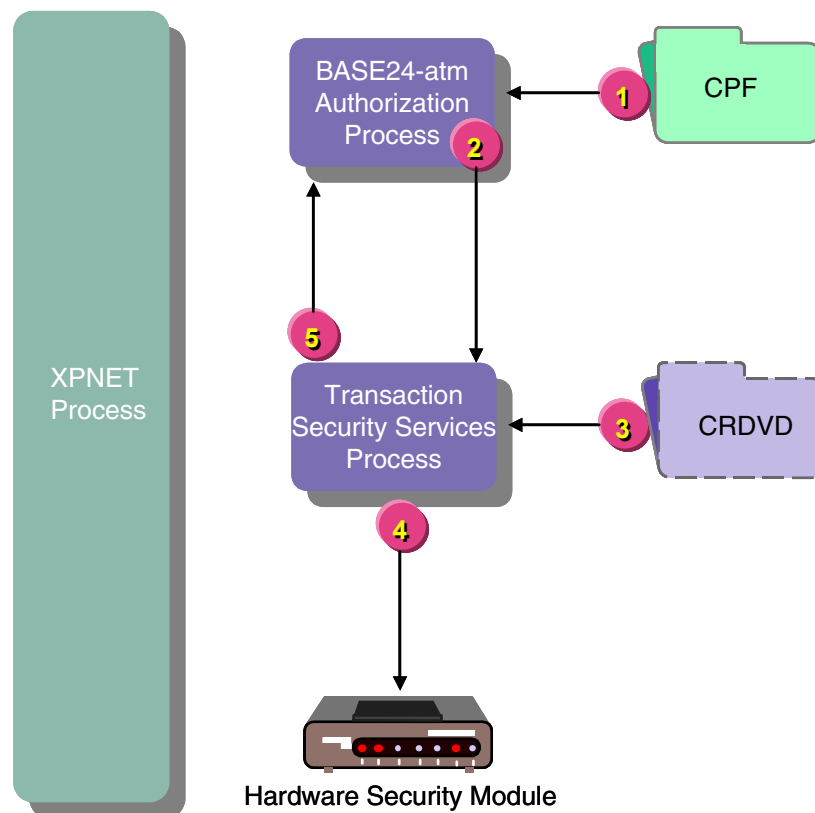
In this transaction flow, the issuing institution uses the Visa PVV PIN verification method. If a different PIN verification method were used, the flow would be essentially the same except that the Transaction Security Services process would access a different external memory table to retrieve the PIN verification information.



1. An interchange sends a transaction request to the BASE24 system with an encrypted PIN. The XPNET process delivers the request to the Interchange Interface process.
2. The Interchange Interface process checks the value in the ENCRYPT TYPE and BASE24 ENCRYPT TYPE fields on Key File (KEYF) screen 1. In this case, both fields are set to a value of 1 indicating that PIN encryption and PIN management are performed in a security module. The Interchange Interface process retrieves the key locator value (i.e., an index value created by the KEYF or KEY6 conversion program) for the inbound PIN key from the Key File (KEYF) and places it in the PSTM.
3. The Interchange Interface process passes the transaction to the XPNET process for delivery to the BASE24-pos Device Handler/Router/Authorization process for authorization.
4. The Authorization module in the BASE24-pos Device Handler/Router/Authorization process retrieves the PIN check type value (i.e., the PIN verification method) from the Institution Definition File (IDF) or the Card Prefix File (CPF) and the PIN verification KEYA group value for the transaction from the CPF. Note that the Authorization process does not access the KEYA at all.
5. The security utilities bound into the BASE24-pos Device Handler/Router/Authorization process pass the inbound PIN key locator value, the PIN verification type, KEYA group, card expiration date, the encrypted PIN block and other PIN-related information to the Transaction Security Services process. The request does not pass through the XPNET process.
6. The Transaction Security Services process uses the inbound PIN key locator value to retrieve the actual inbound PIN key from the ISECD and the KEYA group and card expiration date values to retrieve the Visa PVV PIN verification information from the VPVVD external memory table.
7. The Transaction Security Services process calls the HSM to decrypt the encrypted PIN block using the inbound PIN key and verify the PIN using the PIN verification key. Depending on the lengths of the keys retrieved, the HSM can perform single or triple DES decryption and verification.
8. The Transaction Security Services process passes the result of the PIN verification back to the Authorization module.

## Card Verification

The diagram below illustrates the message flow for card verification. This message flow assumes that PIN verification has already been performed. The steps depicted in the illustration are described following the diagram.

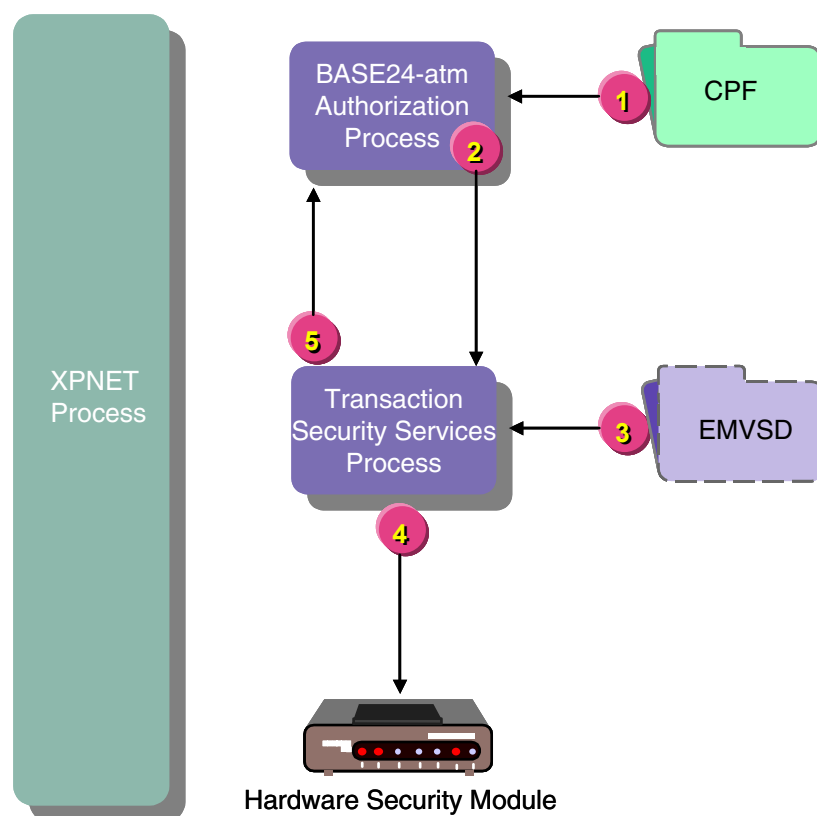


1. The Authorization process retrieves the card verification check type and the card verification KEYA group values for the transaction from the CPF. Assume that card verification is required for this transaction.
2. The security utilities bound into the Authorization process pass the card verification KEYA group, card expiration date, and the card verification data to the Transaction Security Services process. The request does not pass through the XPNET process.
3. The Transaction Security Services process uses the card verification KEYA group value and card expiration date to retrieve the security module-encrypted Card Verification Key pair (CVK pair) from the CRDVD external memory table.

4. The Transaction Security Services process calls the HSM to verify the card using the key pair retrieved in the previous step.
5. The Transaction Security Services process passes the result of the card verification back to the Authorization process.

## Chip Authentication

The diagram below illustrates the security processing message flow for chip authentication. This message flow assumes that PIN verification has already been performed. The steps depicted in the illustration are described following the diagram.



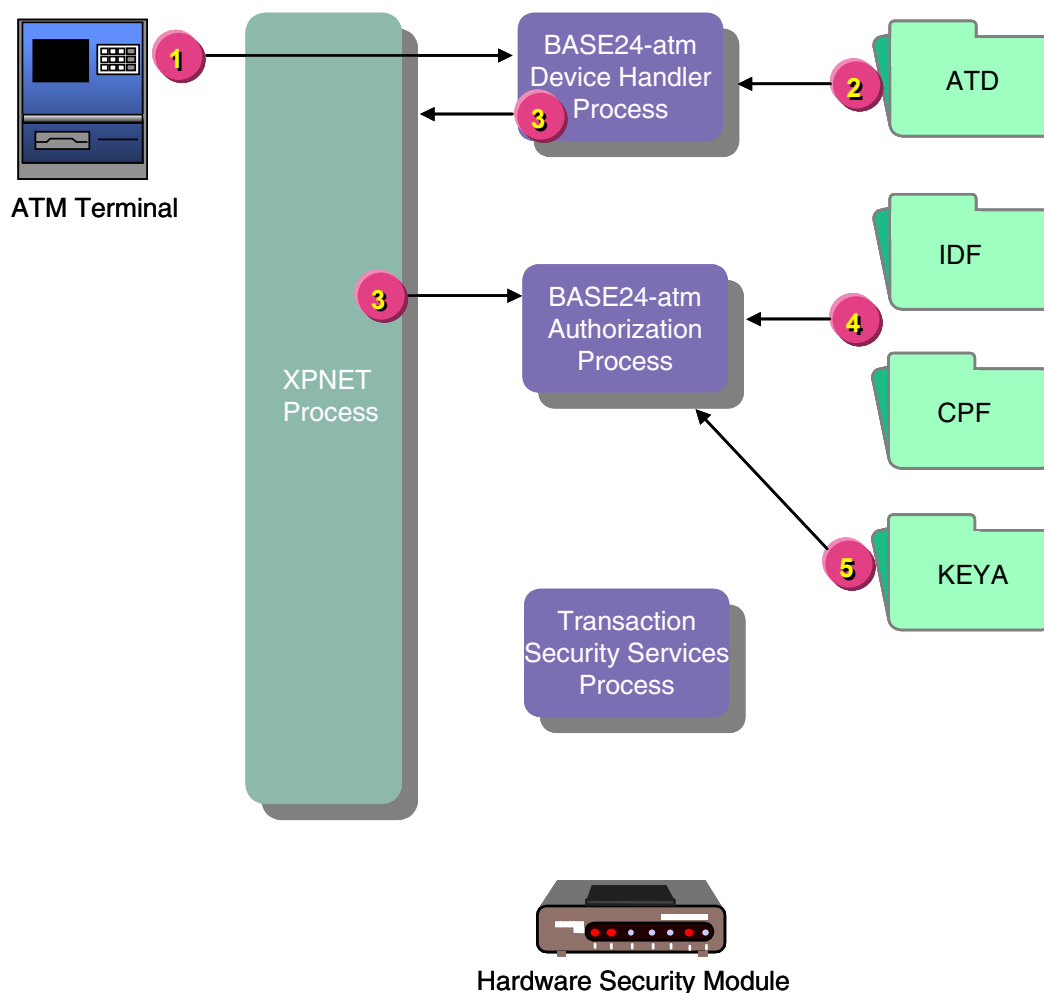
1. The Authorization process retrieves the KEYI group and EMV check values for the transaction from the CPF. Assume that EMV checking is required for this transaction.
2. The security utilities bound into the Authorization process pass the KEYI group, card expiration date, and the EMV authentication data to the Transaction Security Services process. The request does not pass through the XPNET process.
3. The Transaction Security Services process uses the KEYI group value and card expiration date to retrieve the appropriate issuer master key or secure messaging key from the EMVSD external memory table.



4. The Transaction Security Services process calls the HSM to verify the request cryptogram and generate the response cryptogram.
5. The Transaction Security Services process passes the result of the authentication processing back to the Authorization process.

## Terminal Acquired—On-Us Card—Software Verification Allowed

The diagram below illustrates the flow of PIN verification for a transaction that originates at an ATM and is authorized on the BASE24 system. In this case, the issuing institution verifies PINs in software and software verification is allowed (i.e., the SOFTWARE-PIN-VERIFICATION-ALWD param is set to a value of Y). The steps depicted in the illustration are described following the diagram.

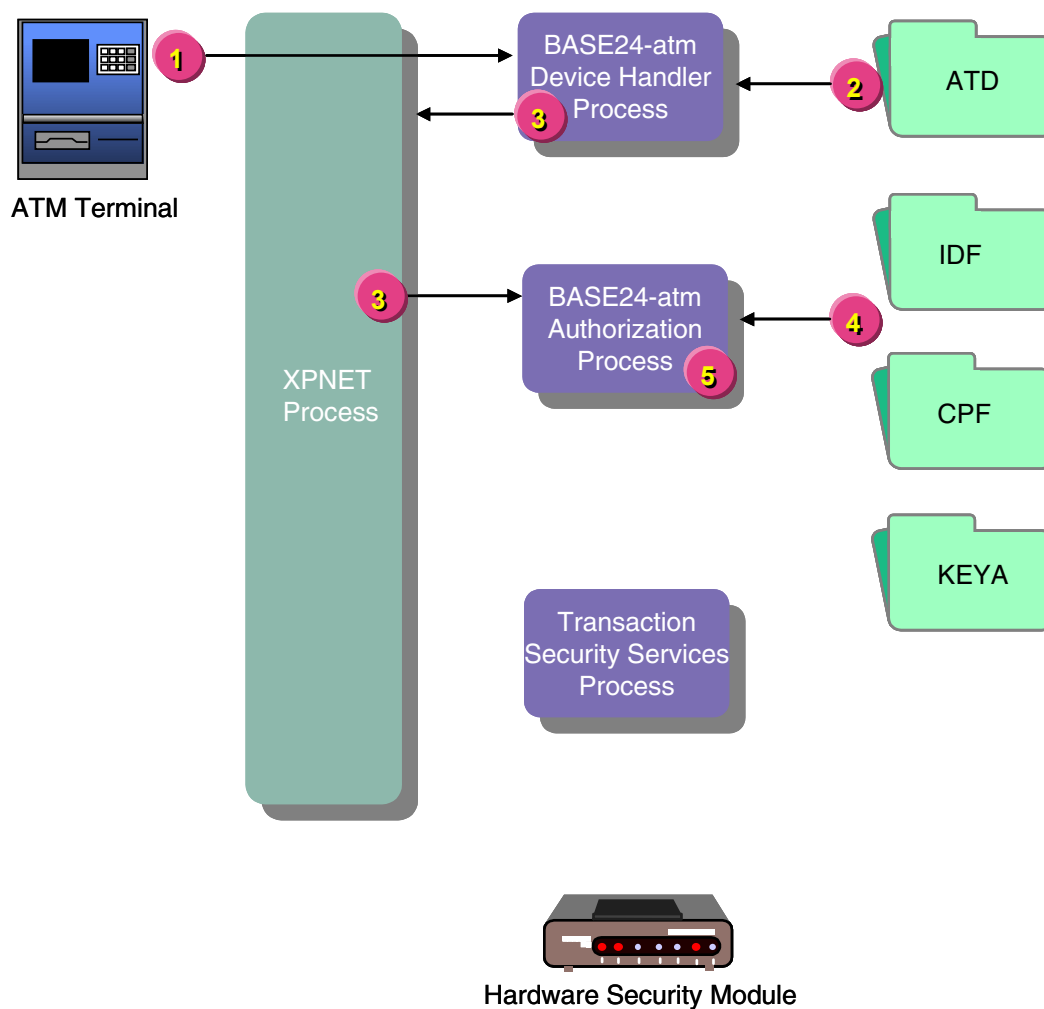


1. A transaction including a PIN originates at an ATM terminal. The XPNET process passes the request containing a clear PIN block to the BASE24-atm Device Handler process.

2. The Device Handler process checks the PIN encrypt type value for the terminal from the BASE24-atm Terminal Data files and determines no PIN encryption is used. The process places this value in the STM along with the clear PIN block.
3. The Device Handler process passes the transaction to XPNET process for delivery to the Authorization process for authorization and routing.
4. The Authorization process retrieves the PIN check type value (i.e., the PIN verification method) from the Institution Definition File (IDF) or the Card Prefix File (CPF) and the PIN verification KEYA group value for the transaction from the CPF.
5. The Authorization process retrieves the clear key for the PIN verification method specified in the STM from the KEYA and verifies the PIN in software using the specified verification method. The security utilities and the Transaction Security Services process are not involved in processing.

## Terminal Acquired—On-Us Card—Software Verification Not Allowed

The diagram below illustrates the flow of PIN verification for a transaction that originates at an ATM and is authorized on the BASE24 system. In this case, the issuing institution verifies PINs in software, but software verification is not allowed (i.e., the SOFTWARE-PIN-VERIFICATION-ALWD param is set to a value of N). The steps depicted in the illustration are described following the diagram.



1. A transaction including a PIN originates at an ATM terminal. The XPNET process passes the request containing a clear PIN block to the BASE24-atm Device Handler process.

2. The Device Handler process checks the PIN encrypt type value for the terminal from the BASE24-atm Terminal Data files and determines no PIN encryption is used. The process places this value in the STM along with the clear PIN block.
3. The Device Handler process passes the transaction to the XPNET process for delivery to the Authorization process for authorization and routing.
4. The Authorization process retrieves the PIN check type value (i.e., the PIN verification method) from the Institution Definition File (IDF) or the Card Prefix File (CPF) and the PIN verification KEYA group value for the transaction from the CPF.
5. The Authorization process determines that PIN verification in software is not allowed and denies the transaction.

*ACI Worldwide Inc.*

## 6: Data Field Encryption

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This section describes the data encryption support provided by the Transaction Security Services process for message field data. It describes message field encryption processing, the available options, and how to configure the BASE24 and TSS databases to enable it.

## Overview

Data encryption and decryption enables you to encrypt sensitive data, other than PINs, in messages transmitted between endpoints. An endpoint can be an ATM, POS device, interchange, or host, and the BASE24 system. Both the transmitting endpoint and the receiving endpoint must share the same working key called a data encryption key. The transmitting endpoint uses the data encryption key to encrypt a selected data field. The receiving endpoint uses the data encryption key to decrypt the data field. As with other working keys in the BASE24 system, you can change data encryption keys dynamically using dynamic key management thresholds.

When encrypting a data field for transmission, you can use two different modes of the DES algorithm to encrypt and decrypt the data—cipher block chaining (CBC) or electronic code book (ECB). You can also use either one or two layers of data encryption.

## Cipher Block Chaining (CBC)

Cipher block chaining (CBC) is a mode of operation that uses feedback in the encryption process. The result of one encryption operation is used to *feed* the encryption of the next block. The plaintext is combined with the previous ciphertext block using an exclusive-OR operation before it is encrypted. Thus, the encryption of each block depends on all the previous blocks. This also requires that all encrypted blocks be processed sequentially when decrypting the data.

CBC mode requires a random initialization vector, which is combined with the first data block using an exclusive-OR operation before it is encrypted. An initialization vector is a sequence of random bytes appended to the front of the plaintext block before encryption. The initialization vector does not have to be kept secret. The Transaction Security Services process uses a static initialization vector. The process calls the security module to generate the initialization vector the first time one is needed for a CBC operation. The initialization vector does not change thereafter.

BASE24 system calls the security module to automatically generate a new initialization vector each time one is needed for a CBC operation.

One of the key characteristics of the CBC mode is that it uses a chaining mechanism that causes the decryption of a block of ciphertext to depend on all the preceding ciphertext blocks. As a result, the entire validity of all preceding blocks



is contained in the immediately previous ciphertext block. A single bit error in a ciphertext block affects the decryption of all subsequent blocks. Rearrangement of the order of the ciphertext blocks also causes decryption to yield a corrupted value.

## Electronic Code Book (ECB)

Electronic Code Book (ECB) is a mode of operation for a block cipher, with the characteristic that each possible block of plaintext has a defined corresponding ciphertext value and vice versa. In other words, for a given data encryption key, the same plaintext value will always result in the same ciphertext value. You can use the ECB mode when a volume of plaintext is separated into several blocks of data, each of which is then encrypted independently of other blocks.

ECB mode is not considered to be effective with small block sizes (for example, smaller than 40 bits) and identical encryption modes. This is because some words and phrases can be reused often enough so that the same repetitive part-blocks of ciphertext can emerge, laying the groundwork for a codebook attack where the plaintext patterns are fairly obvious. However, security can be improved if random pad bits are added to each block. On the other hand, 64-bit or larger blocks should contain enough unique characteristics to make a codebook attack unlikely to succeed.

## Encryption Functions

The Transaction Security Services process supports data encryption, data decryption, and data translation. Using the Transaction Security Service process, you can encrypt a specified data field for transmission to a POS device or certain interfaces and decrypt data fields received from certain interfaces. In addition, you can translate encrypted data fields from encryption under one data encryption key to encryption under another data encryption key. The BASE24 system can carry the data field internally either in the clear or in encrypted format as required.

The following table lists the various combinations of encrypted or clear values supported for the BASE24 system as the sending endpoint, the receiving endpoint, and internally. It also lists the encrypt, decrypt, or translate actions required for each valid combination.

| Receiving Endpoint |         | Internal  |         | Sending Endpoint |           |
|--------------------|---------|-----------|---------|------------------|-----------|
| Data               | Action  | Data      | Action  | Data             | Action    |
| Clear              |         | Clear     |         | N/A              |           |
| Clear              |         | Clear     |         | Clear            |           |
| Clear              |         | Clear     |         | Encrypted        | Encrypt   |
| Clear              | Encrypt | Encrypted | Decrypt | N/A              |           |
| Clear              | Encrypt | Encrypted |         | Clear            | Decrypt   |
| Clear              | Encrypt | Encrypted |         | Encrypted        | Translate |
| Encrypted          | Decrypt | Clear     |         | N/A              |           |
| Encrypted          | Decrypt | Clear     |         | Clear            |           |
| Encrypted          | Decrypt | Clear     |         | Encrypted        | Encrypt   |
| Encrypted          |         | Encrypted | Decrypt | N/A              |           |
| Encrypted          |         | Encrypted |         | Clear            | Decrypt   |
| Encrypted          |         | Encrypted |         | Encrypted        | Translate |

## Data Encryption Keys

The same data encryption keys used to encrypt data by the sending endpoint must be used by the receiving endpoint to decrypt the encrypted data field. The BASE24 system supports single, double, or triple length keys and both single DES or triple DES encryption. For one layer of data encryption, only one data encryption key is needed. For two layers of data encryption, two data encryption keys are required.

## Data Encryption Layers

When encrypting data, you can use one or two layers of encryption. When using one layer, the data is encrypted and decrypted using a single data encryption key. When using two layers, the data encrypted or decrypted using a data encryption key is encrypted or decrypted again using a second data encryption key. The number of data encryption layers required by the endpoint typically dictates the number of layers to use.

## Encryption Data Types

You can encrypt data in two different data types, binary or unpacked ASCII hexadecimal, depending on the DES encryption mode used. For the CBC mode, only binary data is supported. For the ECB mode, only unpacked ASCII hexadecimal data is supported. The data type required by the endpoint usually dictates the type of encryption mode to use.

## Message Authentication Codes (MACs)

If an endpoint uses message authentication codes (MACs), the BASE24 system generates a MAC for encrypted data it transmits to an endpoint and verifies the MAC for encrypted data it receives from an endpoint.

## Tokens

The data encryption key locator and encryption mode (i.e., CBC or ECB) used to encrypt a data field is carried in the Data Encryption Key (BN) token. Currently, the only data field that the SPDH module can encrypt is the Available Balance field (FID J). The IMNI interface may encrypt this balance before it is sent to the SPDH module. If it is encrypted, the SPDH module either translates or decrypts the balance as needed.

The TSS Key Index token indicates the index (1 or 2) of the PIN, MAC, or data encryption keys to use for a transaction when specified by the sending or acquiring interface. If the key index in the TSS Key Index token specifies a value of 1 for a key, the Transaction Security Services process uses the first key defined in the database. If the key index value in the TSS Key Index token specifies a value of 2 for a key, the Transaction Security Services process uses the second key defined in

the database. If the key index value is not specified or the TSS Key Index token is not present for a transaction, the Transaction Security Services process uses the key currently defined as the current key in the database.

# Configuring Data Encryption for SPDH Devices

Currently, BASE24 only supports the encryption of the available balance field returned in an authorization response to a BASE24 Standard POS Device Handler (SPDH-compliant) POS device. To support encryption of the available balance field in responses, you must configure the terminal to return the available balance in a response, perform ASCII or binary encryption, generate and store the data encryption key, and indicate whether to use the CBC or ECB mode of encryption. These configuration options and values are set in the BASE24 database and TSS database as described below.

## BASE24 Database Settings

The following paragraphs discuss the data encryption configuration requirements for the BASE24 database.

The available balance must be included in a response for it to be encrypted. The BALANCE RETURNED field on BASE24-pos Terminal Data files (PTD) screen 2 indicates whether the BASE24 system includes the available balance in response messages sent to the specified POS terminal.

To enable data encryption for a POS device, set the DATA ENCRYPTION TYPE field on PTD screen 7 to a value of 01 (ASCII encryption) or 02 (binary encryption). To disable data encryption for a POS device, set this field to a value of 00 (no encryption).

To support dynamic key management of the data encryption key, you must set either the DATA KEY THRESHOLD or DATA KEY ERROR THRESHOLD field on the Key Threshold screen of the ACI Standard Device Configuration File (ACNF) to a nonzero value.

The PIN MASTER KEY field on PTD screen 7 must be set to a key locator value. This value is used to access the data encryption key configuration in the TSS database.

For detailed information on PTD screens, refer to the *BASE24-pos Files Maintenance Manual*. For detailed information on ACNF screens, refer to the *BASE24-pos Standard POS Device Configuration Manual*.

## TSS Database Settings

The following fields must be configured on the Data Encryption tab of the Channel Security Configuration window on the ACI desktop user interface to support data encryption and decryption:

- DES Encr Mode (specifies whether to use the ECB or CBC mode of DES)
- Data Type (specifies whether you are using unpacked ASCII hexadecimal or binary data encryption)
- Exchange Key (contains the transport key for downloading a new data encryption key to a device). This key must be encrypted under variant 0 (Key Exchange Key Variant) of the MFK.
- Inbound Keys (contains one or two data encryption keys, depending on whether you are using one or two layers of data encryption)
- Encr Layers (specifies whether to use one or two layers of data encryption)

The Initialization Vector field is a display only field. The BASE24 system calls the security module to generate the initialization vector the first time one is needed for a CBC operation. The value does not change thereafter.

All of this information is stored in the Channel Security File (CSECD). A sample of the Data Encryption tab on the Channel Security Configuration window is shown in [section 2](#). Refer to the ACI Online Help provided for the ACI desktop user interface for detailed information on this window and its fields.

# Configuring Data Encryption for Interfaces

Data encryption support is provided for the following interface processes in a BASE24 system:

- IMNI Interface
- Merchant Link Interface

Configuration support for these interfaces is provided on the Interface Security Configuration window and the BASE24 KEYF Configuration window.

## Interface Security Configuration

To support data encryption for the IMNI Interface process, you must configure the data encryption keys and data encryption key exchange keys on the Interface Security Configuration window of the ACI desktop user interface. This record must correspond to an existing record in the KEYF. If a record does not already exist in the KEYF for the interface, you must first add one on the BASE24 KEYF Configuration window on the ACI desktop user interface.

The Merchant Link Interface simply carries the encrypted data (i.e., available balance and data encryption key information in tokens between the transaction acquiring BASE24 system and the transaction accepting BASE24 system.

**Note:** Dynamic key management thresholds for the data encryption key are not supported in the KEYF. Therefore, the exchange of a new data encryption key can only be initiated by the interface.

Several fields must be configured on two tabs of the Interface Security Configuration window on the ACI desktop user interface to support data encryption and decryption.

Configure the Exchange Keys fields on the Data Encr Exch Inf tab for the key exchange keys to import or export a new data encryption key. The import key is used to decrypt a new data encryption key received from an interface. The export key is used to encrypt a new data encryption key transmitted to an interface. If desired, the import key can be used for both operations.

The following fields on the Data Encr Key Intf tab must be configured for the inbound and outbound data encryption keys. The inbound key is used to decrypt or translate encrypted data received from the interface. The outbound key is used to encrypt or translate data to be sent to the interface.

- Inbound Keys (contains one or two data encryption keys, depending on whether you are using one or two layers of data encryption). These keys are used to decrypt or translate encrypted data received from the interface.
- Outbound Keys (contains one or two data encryption keys, depending on whether you are using one or two layers of data encryption). These keys are used to encrypt or translate encrypted data transmitted to the interface.
- DES Encr Mode (specifies whether to use the ECB or CBC mode of DES)
- Data Type (specifies whether you are using unpacked ASCII hexadecimal or binary data encryption)
- Encr Layers (specifies whether to use one or two layers of data encryption)

All of this information entered on the Interface Security Configuration window is stored in the Interface Security File (ISECD). Samples of the Data Encr Exch Inf and Data Encr Key Intf tabs on the Interface Security Configuration window are shown in [section 2](#).

## BASE24 KEYF Configuration

On the Interac Key Management tab of the BASE24 KEYF Configuration window, you can view and update counters for inbound and outbound key exchange keys, change the index of the inbound or outbound data encryption key, and change the key exchange key for the IMNI interface. When the IMNI interface exchanges a new working key, the associated inbound or outbound key counter is incremented by one. These counters are reset to 0 when you enter and save a new key exchange key in the New Key Exchange Key field. Key exchange keys entered in this field are validated and stored in the Interface Security File (ISECD).



# Security Module Configuration

Currently, data encryption is available only with Atalla NSPs. Data encryption is supported for NSPs using AKB or non-AKB (variant) software. The following paragraphs discuss the configuration requirements for an Atalla NSP to support data encryption.

## Data Encryption Keys

For storage in the CSECD or ISECD, the data encryption keys must be encrypted under variant 2 of the MFK or be generated with an AKB header value of 1DDNE000, depending on whether the NSP is using variant or AKB software. The same data encryption key can be used for both encryption and decryption.

## Key Exchange Keys

As with other key exchange keys, the key exchange keys used for data encryption keys must be encrypted under variant 0 of the MFK or be generated with an AKB header value of 1KDNE000, depending on whether the NSP is using variant or AKB software.

## Commands

The mode of the DES algorithm used determines which commands are required in the Atalla NSP to support data encryption and decryption. To encrypt and decrypt data using the ECB mode of DES and generate the static initialization vector, you must enable the 55 command (Encrypt Or Decrypt Data Or Translate) and the 59 command (Generate MAC and Encrypt or Translate Data). To encrypt and decrypt data using the CBC mode of DES, you must enable the 55 command, the 97 command (Encrypt Or Decrypt Data), the 95 command (Reformat Initialization Vector), and the 59 command. The 10 command is always used to generate a new data encryption key. All of these commands are described below.

The 55 command (Encrypt Or Decrypt Data Or Translate) provides data encrypt or decrypt functions using the ECB mode. This command is also required to generate the static initialization vector for the CBC mode. This command is not enabled in the default factory security policy due to its high security exposure. You must purchase this command in the form of a 105 command, and enable it in

the NSP's security policy using the SCT and the 108 and 109 commands, or the NSP Configuration Management menu in the SCA, if you are using the ECB mode of DES.

The 97 command (Encrypt Or Decrypt Data) provides data encrypt or decrypt functions using the CBC mode. This command is not enabled in the default factory security policy and must be enabled in the NSP's security policy using the SCT and the 108 and 109 commands, or the NSP Configuration Management menu in the SCA, if you are using the CBC mode of DES.

The 95 command (Reformat Initialization Vector) is used to reformat an initialization vector. As with the 55 command, this command is also not enabled in the default factory security policy and must be enabled in the NSP's security policy using the SCT and the 108 and 109 commands, or the NSP Configuration Management menu in the SCA, if you are using the CBC mode of DES.

The 59 command (Generate MAC and Encrypt or Translate Data) generates a MAC and encrypts or translates data. This command supports both the ECB and CBC modes of DES. This command is not enabled in the default factory security policy and must be enabled in the NSP's security policy using the SCT and the 108 and 109 commands, or the NSP Configuration Management menu in the SCA.

The 10 command (Generate Working Key, Any Type) generates a new data encryption key when the exchange of a new data encryption key is initiated by the interface. Dynamic key management thresholds for the data encryption key are not supported in the KEYF.

# Dynamic Key Management

You can also configure dynamic key management thresholds for which a new data encryption key is automatically downloaded when one of the specified thresholds is exceeded. At a minimum, the BASE24 system forces the data encryption key to be changed at least once every daily cutover period.

For an SPDH-compliant device, you can configure two thresholds for dynamic key management of the data encryption key. These thresholds are configured on the Key Thresholds screen of the ACNF.

There are no data encryption key dynamic key management thresholds for an interface.

For detailed information on configuring dynamic key management of data encryption keys, refer to the ***BASE24 Transaction Security Manual***.

*ACI Worldwide Inc.*

## 7: TSS Environment Management

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This section discusses the configuration of environment attributes, creating the System Interface Message Delivery Service Destination File (MDSDEST), and modifying Metadata Configuration File (CONFCSV) parameters for the TSS environment. It also includes procedures for adding security devices, Transaction Security Services processes, and Timer processes to your BASE24 system.

## Environment Configuration Overview

The Environment Configuration facility enables configuration of environment-level control attributes. Environment-level control attributes affect general processing. For example, the environment control attribute EVT\_DEST is used to control the destination for all event messages generated by the application servers.

The environment attributes specific to the TSS environment are described in [section 3](#) and [appendix C](#) of this processing guide. The attributes described in section 3 are relevant to the entire TSS environment while the attributes described in appendix C are attributes used by the AKB Conversion utility only.

All application servers, regardless of whether they handle online transactions or requests from the ACI desktop user interface, access the Environment Configuration control attribute. In the event the appropriate control attribute is missing or is invalid, the application uses a default value.

# Configuring Environment Control Attributes

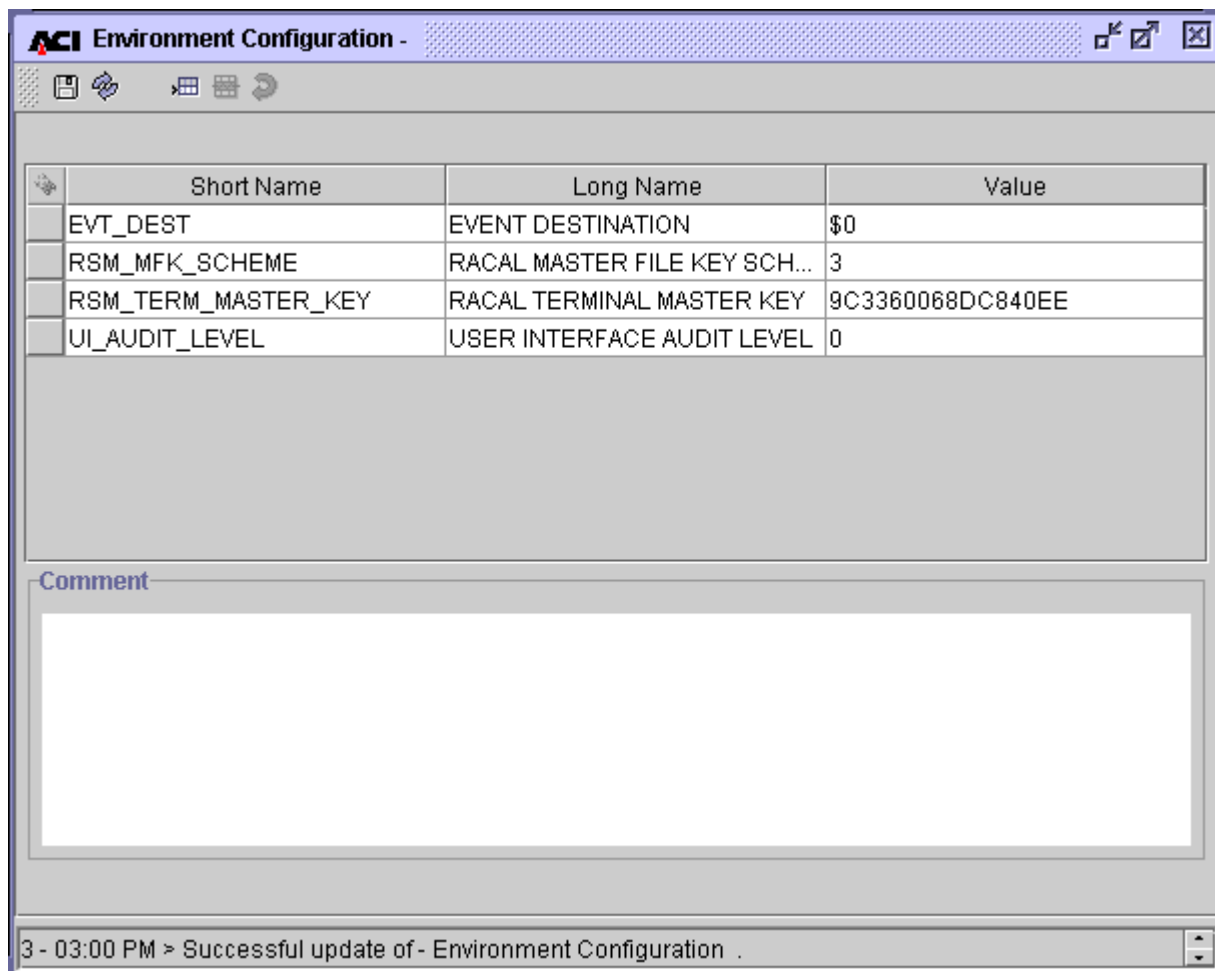
The Environment Configuration user interface window is the only window to use for the configuration of environment control attributes.

A complete description of this window and configurable attributes is available from the ACI desktop user interface.

Using the ACI desktop user interface, the Environment Configuration window facilitates the configuration of environment control attributes. Environment control attributes are referred to by their *short name*.

Upon entry into the window, all environment control attributes configured in the Environment File (ENVMTD) are displayed. The list below is not necessarily a complete list. Refer to the “[Transaction Security Services Attributes](#)” topic in section 3 for descriptions of all environment attributes relevant to the TSS environment. When you select a row, a description of the attribute is displayed in the Comment field. If an attribute is not configured, it’s default value is used.

**Note:** Although the RSM\_MFK\_SCHEME attribute is displayed in this example, it is no longer used in release 05.2.



The screenshot shows the 'ACI Environment Configuration' window. It contains a table with four columns: Short Name, Long Name, and Value. The table lists four attributes: EVT\_DEST, RSM\_MFK\_SCHEME, RSM\_TERM\_MASTER\_KEY, and UI\_AUDIT\_LEVEL. Below the table is a 'Comment' field, which is currently empty. At the bottom of the window, a status bar displays the message: '3 - 03:00 PM > Successful update of - Environment Configuration .'

| Short Name          | Long Name                    | Value            |
|---------------------|------------------------------|------------------|
| EVT_DEST            | EVENT DESTINATION            | \$0              |
| RSM_MFK_SCHEME      | RACAL MASTER FILE KEY SCH... | 3                |
| RSM_TERM_MASTER_KEY | RACAL TERMINAL MASTER KEY    | 9C3360068DC840EE |
| UI_AUDIT_LEVEL      | USER INTERFACE AUDIT LEVEL   | 0                |

Comment

3 - 03:00 PM > Successful update of - Environment Configuration .



# System Interface Message Delivery Service Destination File (MDSDEST)

The System Interface Message Delivery Service Destination File (MDSDEST) is accessed by the System Interface Services (SIS) layer of the integrated server (IS). The MDSDEST is used to map remote destinations to HP NonStop NSK Process Pair Directory (PPD) names. These include the following remote destinations:

- Hardware security device
- Timer process
- IS processes (when an XML Server process needs to send a message to an IS process for STATUS and INFO commands only)

## Contents

The MDSDEST contains one entry for each remote destination. Each entry is made up of three fields; assign, PPD name, and a yes or no value that indicates the format of the text sent to the remote destination. The yes or no value is referred to as the *raw send* flag.

### ***Example Entry***

---

ATALLA1 , \$BOX1 , Y

In this example entry, ATALLA1 is the assign name and \$BOX1 is the PPD name of the Atalla Boxcar process. The Y value indicates that raw text is sent to the destination. When the indicator is set to a value of N, text is sent with XPNET headers.

## Security Device

For each security device (e.g., Atalla, Bull, Eracom, or Thales) assigned on the Security Device Configuration window, there must be a corresponding entry in the MDSDEST. The physical location of the MDSDEST is specified by the SIMDS\_DEST\_FILENAME parameter set using the Metadata Manager tool.

In the case of an Atalla security device connected using the TCP/IP protocol, each MDSDEST entry points to a Boxcar PPD name. For example, if a customer has three Atalla security devices using Boxcar processes (e.g., \$BOX1, \$BOX2, and \$BOX3) and the ACI desktop user interface Security Device Configuration assigns are ATALLA1, ATALLA2, and ATALLA3, the MDSDEST edit file would contain the following lines:

```
ATALLA1 , $BOX1 , Y
ATALLA2 , $BOX2 , Y
ATALLA3 , $BOX3 , Y
```

**Note:** All Atalla implementations using the TCP/IP protocol require the Boxcar process.

For Bull BNTng hardware security modules, each entry points to the Remote Application Manager (RAM) process. In the following example MDSDEST entries, the names of the devices on the Security Device Configuration window are BNTNG1 and BNTNG2:

```
BNTNG1 , $RAM , Y
BNTNG2 , $RAM , Y
```

For Eracom security devices, each MDSDEST entry points to the Eracom device itself, to a Blocker process PPD name, or to a Remote Application Manager (RAM) process, depending on whether the device is connected using an asynchronous, Ethernet, or TCP/IP communications protocol, respectively.

In the following example MDSDEST entry for an asynchronous connection, the PPD name of the device is \$EMS2000.

```
ERACOM1 , $EMS2000 . #TERM , Y
```

In the following example MDSDEST entry for an Ethernet connection, the entry points to a Blocker process.

```
ERACOM1 , $BLKR1 , Y
```

In the next example MDSDEST entry for a TCP/IP connection, the entry points to a Remote Application Manager (RAM) process:

```
ERACOM1 , $RAM2 . #MARKII , Y
```

For Thales e-Security HSMs, each entry points to the Racal Access Module (RAM) process. In the following example MDSDEST entries, the names of the devices on the Security Device Configuration window are RACAL1 and RACAL2:

```
RACAL1 , $RAM , Y  
RACAL2 , $RAM , Y
```

## Timer Process

For each timer process, there must be a corresponding entry in the MDSDEST.  
For example:

```
#STP , $PPD , N
```

## IS Process

An entry must exist in the MDSDEST for at least one Integrated Server (IS) process to which you want to send STATUS and INFO process control commands. These entries are needed in the MDSDEST to support synchronous messaging between the XML Server process and an IS process. STATUS and INFO commands are handled through inter-process communications rather than through the XPNET process. For STATUS and INFO commands, the XML Server process must receive a response from an IS process before it can send the command response back to the ACI desktop user interface client. If an IS process does not have an entry in the MDSDEST, the XML Server process cannot send STATUS and INFO commands to that process when the destination for the command is IS.

You can either configure just one IS process in the MDSDEST to handle all STATUS and INFO commands, or you can configure all IS processes in the MDSDEST.

If you configure just one IS process, STATUS and INFO commands are sent to that process when you specify “IS” as the destination.

If you configure all IS processes in the MDSDEST, you can then enter the XPNET symbolic name of an individual IS process (e.g., P1A^ISL001) as the destination for a STATUS or INFO command to send the command to a specified IS process. An entry must exist in the MDSDEST for the IS process specified as the destination. The assign names for IS processes must be unique for each IS process and start with “IS.” Example process entries are shown below:

```
IS1 , $TSI1 , N  
IS2 , $TSI2 , N  
IS3 , $TSI3 , N  
IS4 , $AKI1 , N
```

**Note:** IS processes include all Transaction Security Services processes and AKDS processes in the TSS environment.

## Creating the MDSDEST

MDSDEST entries are made to an edit-type file (e.g., MDSDESTS). It then needs to be converted to an unstructured, uneditable file. This is done using the SIS Remote Dest File Loader program. The following is an example command for running this program:

```
TACL> run $vol.subvol.sirdfll/in MDSDESTS/MDSDEST
```

This command creates a MDSDEST if one does not exist, otherwise it inserts the new entries into the existing MDSDEST.

# Modifying CONFCSV Parameters

Two parameters in the Metadata Configuration File (CONFCSV) control processing in the TSS environment.

The `SIMDS_DEST_FILENAME` parameter specifies the location of the MDSDEST. This parameter must be modified during installation to contain the physical location of the MDSDEST on your system. A sample of the `SIMDS_DEST_FILENAME` parameter in the CONFCSV is shown below.

```
"SIMDS_DEST_FILENAME", "1.0", "$vol.subvol.mdsdest",  
"bt_stringf", 1, 32, False, ""
```

The `SIMDS_DEST_WARMBOOT_DELTA` parameter specifies the amount of time the System Interface Services (SIS) layer of processes running in the TSS environment waits to retry after a connection is lost to an external destination specified in the MDSDEST. This parameter also specifies the interval between warmbooting the MDSDEST when a send and wait is performed. The default value is 60 (seconds). If this parameter is set to a value of 0, you must stop and restart the User Interface (XML Server) process before a modified MDSDEST file takes effect in processing.

When a connection is lost to a destination, the SIS layer retries to open the connection when a request is received for that destination. You can specify a delta timestamp if you want the SIS layer to wait a specified amount of time before it retries the connection. When you set the `SIMDS_DEST_WARMBOOT_DELTA` parameter to a value of 0, the SIS layer attempts a connection retry for each message sent to the failed destination. When you set the `SIMDS_DEST_WARMBOOT_DELTA` parameter to a nonzero number, the SIS layer waits the specified number of seconds before attempting a connection retry for each message sent to the failed destination. Samples of the `SIMDS_DEST_WARMBOOT_DELTA` parameter in the CONFCSV are shown below. In the first example, the parameter is set to 0. In the second example, the parameter is set to wait 45 seconds before attempting to retry a connection to a failed destination.

```
"SIMDS_DEST_WARMBOOT_DELTA", "1.0", "0", "bt_stringv", 1, 5, True, ""  
"SIMDS_DEST_WARMBOOT_DELTA", "1.0", "45", "bt_stringv", 1, 5, True, ""
```

**Note:** Any time you change the CONFCSV, you must stop and restart the process that uses the changed data. To ensure that no transactions currently in progress are lost, you should stop the process in an orderly fashion. The change does not take effect until after the process is stopped and restarted.

## Adding a New Security Device

For performance and throughput purposes, you may need to add additional security devices after your TSS environment is initially installed. Perform the following steps to add a new security device.

1. Install and configure the security device according to the vendor documentation and the information provided in [section 4](#).
2. Edit the MDSDESTS file and add an assign and PPD entry for the new security device.  
  
`ATALLA4 , $BOX4 , Y`
3. Recreate the MDSDEST. Refer to the “Creating the MDSDEST” topic for more information.
4. Add the new security device on the Security Device Configuration window (accessed from **Configure > Transaction Security > Security Device**). Refer to [section 3](#) and [section 4](#) for detailed information on configuring security devices.
5. From the Process Control window, warmboot the Security Device Configuration File (SECD) data source (SEC\_DEV\_CONFIG) using the Alter command, a destination of IS, and the Data Source component. Refer to [section 2](#) for detailed information on warmbooting data sources accessed as internal memory tables.

# Adding Transaction Security Services Processes

For performance and throughput purposes, you may need to add Transaction Security Services processes after your TSS environment is initially installed. Perform the following steps to add a Transaction Security Services process.

**Note:** Regular Transaction Security Services processes use the TSSISLO executable object while AKDS processes use the AKDSISLO executable object.

1. Add the new Transaction Security Services process to the XPNET node. Use the following commands to create a Transaction Security Services process from a network control facility. Modify the commands as needed for site-specific information.

**Note:** The `-STRT` param for the `STARTOPTIONS` attribute must be set to the volume and subvolume where the Metadata files (`CONFCSV` and `MDBC`) are located.

```
RESET PROCESS
SET PROCESS LIBRARY $VOLUME.XNLIBN.XNLIBN
SET PROCESS PROGRAM $VOLUME.TSS6TOBJ.TSSISLO
SET PROCESS PPD $TIS4
SET PROCESS PRIORITY 147
SET PROCESS CPU 1
SET PROCESS QAT 30
SET PROCESS QMT 75
SET PROCESS HOMETERM $SUDO
SET PROCESS SERVICE IS
SET PROCESS STARTOPTIONS "-MDBV=CSV -AI=TSS -STRT=$VOLUME.
TSSTCNFG"
SET PROCESS REPLACEDST OFF
SET PROCESS NSKSTARTSEQ ON
ADD PROCESS P1A^TSSIS4, UNDER SYSNAME \TANDEM, UNDER NODE
P1A^NODE
```

2. Edit the `MDSDESTS` file and add an assign and PPD entry for the new process.  
  
IS3, \$TSI4, N
3. Recreate the `MDSDEST`. Refer to the “Creating the `MDSDEST`” topic for more information.
4. Add or modify the `LCONF` assigns and params for the `BASE24` processes that you want to start using the new Transaction Security Services process. Most notably, the `SECURE-DEV1` and/or `SECURE-DEV2` assigns, or a `SECURE-DEV-xx` assign, in the `LCONF` must be set to the PPD of the new Transaction Security Services process.

5. Warmboot the BASE24 processes for which the LCONF assigns and params were added or altered in the previous step.



# Adding Timer Processes

For performance and throughput purposes, or for redundancy, you may need to add Timer processes after your TSS environment is initially installed. Perform the following steps to add a Timer process. The TSS environment accesses Timer processes in a round-robin manner when starting timers. If a Timer process is unavailable, the Message Delivery Subsystem (MDS) of the System Interface Services (SIS) within the TSS environment uses the next Timer process in its table to start the timer.

1. Add the new Timer process to the XPNET node. Use the following commands to create a Timer process from a network control facility. Modify the commands as needed for site-specific information.

**Note:** The `-STRT` param for the `STARTOPTIONS` attribute must be set to the volume and subvolume where the Metadata files (`CONFCSV` and `MDBC`) are located.

```
RESET PROCESS
SET PROCESS LIBRARY $VOLUME.XNLIBN.XNLIBN
SET PROCESS PROGRAM $VOLUME.TSS6TOBJ.SISTPLO
SET PROCESS PPD $TTIM2
SET PROCESS PRIORITY 147
SET PROCESS CPU 1
SET PROCESS QAT 30
SET PROCESS QMT 75
SET PROCESS HOMETERM $SUDO
SET PROCESS STARTOPTIONS "-MDBV=CSV -AI=TSS -STRT=$VOLUME.
TSSTCNFG"
SET PROCESS REPLACEDST OFF
SET PROCESS NSKSTARTSEQ ON
SET PROCESS SAVEABEND ON
ADD PROCESS P1A^TIMR02, UNDER SYSNAME \HP, UNDER NODE P1A^NODE
```

2. Edit the `MDSDESTS` file and add an assign and `PPD` entry for the new process.

```
#STP, $TTIM2, N
```

3. Recreate the `MDSDEST`. Refer to the “Creating the `MDSDEST`” topic for more information.

*ACI Worldwide Inc.*

## 8: TSS Event Management

---

This section discusses the facilities and procedures used to monitor event messages generated by components in the TSS environment. It includes an overview of configuring events, recognizing events, determining the cause, effect, and remedy for an event, and detailed procedures for configuring and suppressing events.

## Event Configuration Overview

An event is a condition of the application (e.g., initialization, termination, missing file, disk full, etc.) that warrants notification to the user. Notification occurs in the form of an event message. Event messages provide information which give clear indication as to the state or status of a component or other system entity at the instant an event is triggered. There are generally five aspects involving events. These are described below.

## Event Generation and Suppression

Component-generated events are triggered according to conditions set by software developers. Component software developers decide if and when events will be triggered.

Event messages may be suppressed (i.e., not logged), however. An event may be suppressed always, never, or generated once for a specified time period and/or number of times the event has been triggered. When an event is suppressed, there is no indication the event occurred.

## Event Logging

When an event is triggered, a component supplies the subsystem ID, event number, and any required parameters that are used for text substitution. The Event component retrieves the event message information (i.e., text and emphasis) from the database and formats an event message using the supplied parameters in the message text. The event message is then sent to a logging destination. This may be a file, application, or output device.

## Event Monitoring

Event monitoring is the action of observing an event. This may be performed manually or automatically by external software and hardware systems.

## Event Response or Action

When events are observed, decisions must be made as to whether or not a response or action is required. Some events are information-only events and do not require action on the part of the observer while other events indicate symptoms of a problem (e.g., unable to write records to a file because the disk device is full).

Often times, human operators do not observe events until well after a failure has occurred. That's why many large payment system environments also employ an automated event monitoring system. Automated monitoring systems continuously monitor events and are capable of correcting many of the resulting situations. Of course, the automated monitoring system is only as good as the rules by which it operates.

## Log Destination

The destination to which events are logged must be specified via the EVT\_DEST parameter in the Environment Management window of the ACI desktop user interface. The parameter value must be the name of a valid terminal, printer, file, or application.

## Recognizing Events

Events appear as event messages and may be observed on any output device (i.e., terminal and printer) capable of receiving such messages. An example of an event message is shown below.

```
02-05-13;10:55:53.332 \ARGUS.$W3E10      ACI.1022.1000      20
Mgr PPD: \ARGUS.$WEDU  Generator: P3E^XML01
PCTL: Command = STATUS not supported by service = EVT
```

Ordinarily, event messages contain data such as date and time, system and subsystem, event number, and event text. The message sample above is the *brief* form of an event message. In the detailed form of an event message, additional data are available. The preceding event message can be dissected and explained as follows.

| Data                | Meaning                                                                 |
|---------------------|-------------------------------------------------------------------------|
| 02-05-13            | Date the event message was generated.                                   |
| 10:55:53.332        | Time the event message was generated.                                   |
| \ARGUS.\$W3E10      | System where the event message was generated and process name.          |
| P3E^XML01           | Symbolic process name of the process which generated the event message. |
| 1022                | Subsystem ID of the event message.                                      |
| 20                  | Event message number                                                    |
| \ARGUS.\$WEDU       | System where the event message was generated and network name.          |
| PCTL:               | Component that generated the event message.                             |
| Command = STATUS... | Event message text                                                      |

Date, time, system, subsystem, process name, network name, and event number are provided by the component at the time the event is generated. The text and emphasis of the event message is retrieved from the database. Additional event message information is contained in the database and is available for viewing via the ACI desktop user interface. This information includes event cause, effect, and remedy.

## Cause, Effect, and Remedy

Each event message is configured with a cause, effect, and remedy information. This information is intended to provide you adequate investigative information to determine the operational status of the environment and resolve any problems. Cause, effect, and remedy may be viewed using the Event Management window of the ACI desktop user interface.

## Subsystem ID and Event Number

To determine the cause, effect, and remedy, you will first need the following two pieces of data from the event message:

- Subsystem ID
- Event number

Given the following event message sample, the first line indicates the Subsystem ID is 1022 and the Event number is 20.

```
02-05-13;10:55:53.332 \ARGUS.$W3E10      ACI.1022.1000      20
Mgr PPD: \ARGUS.$WEDU  Generator: P3E^XML01
PCTL: Command = STATUS not supported by service = EVT
```

**Note:** Unlike other BASE24 software, subsystem IDs for components in the TSS environment are not defined in the EMCLDDLS file on the SCRIBE subvolume of your HP NonStop system. Instead, they are defined in the Event Document File (EVD OCD) and mapped to events in the Event File (EVENTD). You can view all the subsystem IDs in the EVD OCD for TSS components from the Event Subsystem ID Configuration window of the ACI desktop user interface.

## Integrated Server Event Configuration and Management Window

Using the subsystem ID and event number, you can observe additional information about the event on the Integrated Server Event Configuration and Management window.

### Event Message Tab

On the Event Message tab of the Integrated Server Event Configuration and Management window, you can view documentation of any tokens used in the event message. The following image shows descriptions of the tokens used in the preceding event message example. In the TSS environment, tokens are variables for which the generating component substitutes specific text or values.

**Integrated Server Event Configuration and Management - 1022 - 20**

**Subsystem ID** 1022 **PCTLI** Process Control Implementation class

**Event Number** 20

**Event Message** **Event Message Documentation** **Event Suppression Information**

**Message Text**

Command = <1> not supported by service = <2>

**Severity** Informational

| Token ID | Token Type             | Length | Description           |
|----------|------------------------|--------|-----------------------|
| <1>      | Variable Length String | 0      | The full command text |
| <2>      | Variable Length String | 0      | Service identifier    |

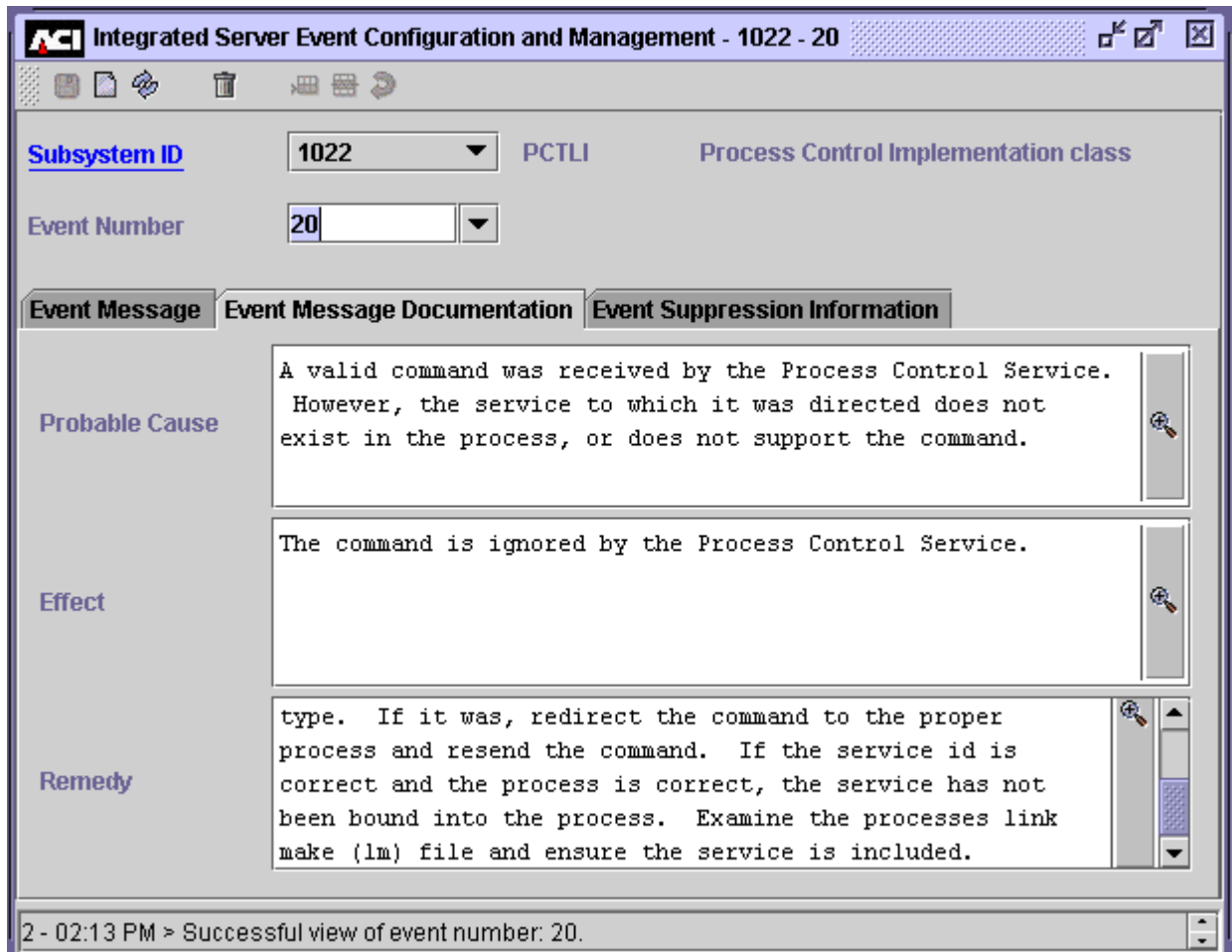
Token descriptions

2 - 02:13 PM > Successful view of event number: 20.



## Event Message Documentation Tab

On the Event Message Documentation tab of the Integrated Server Event Configuration and Management window, you can view the probable cause, effect, and remedy for an event. The following image shows the cause, effect, and remedy information that corresponds to the preceding event message example.



## Suppressing an Event Message

Event suppression provides a means of reducing the number of times an event message is generated by components in the TSS environment. Event suppression may be desirable when multiple components or processes report the same error condition or when a component or process reports the same error continually.

This subsection describes how event suppression works and how it is configured in the TSS environment and includes the following topics:

- An overview of event suppression
- A discussion of event suppression methods
- A discussion of event suppression configuration on the Event Suppression Information tab of the Integrated Server Event Configuration and Management window.
- A discussion of event suppression performance impacts

**Warning:** Indiscriminate use of event suppression may adversely affect your ability to monitor system processing by limiting your ability to detect significant processing events if and when they should occur. Therefore, any attempts at event suppression should not be undertaken until after you have had significant experience running your TSS software in a production environment and you are familiar with the types of event messages generated.

Event suppression control is available to indicate whether the event is logged and whether to skip events. Skipping allows a number of events to be skipped once an event suppression threshold has been reached. By default, all events are logged.

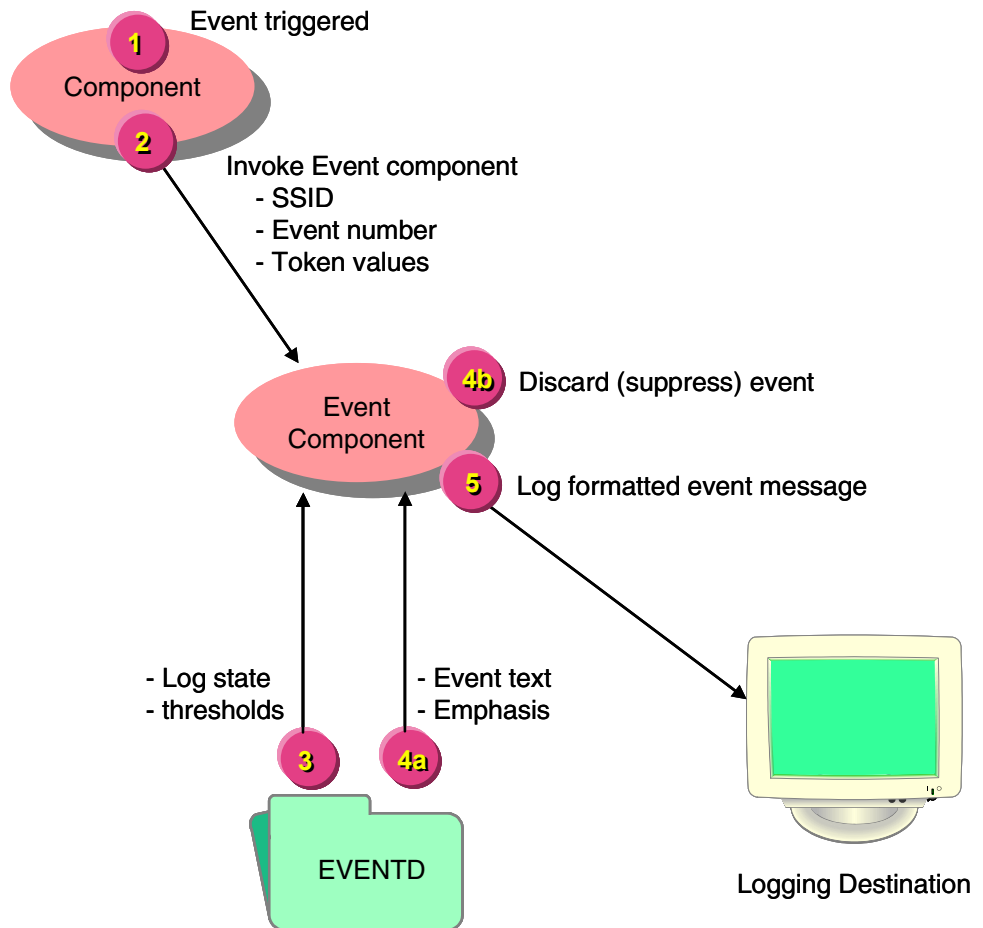
### Overview

This subsection provides an overview of event suppression processing and introduces each component involved in determining whether an event is suppressed.

**Note:** Events associated with file errors should never be suppressed, because the same event can be generated with different file codes.

Event suppression is activated by subsystem IDs (SSIDs) and event numbers. The Event component suppresses events for the TSS environment. When an event is generated, the component that generates the event invokes the Event component using the Log Event Message member function. The Event component uses the

SSID and event number provided in the request to read the Event File (EVENTD) to determine whether the event is logged or suppressed. Suppression data entered on the Event Suppression Information tab of the Integrated Server Event Configuration and Management window is stored in the EVENTD. The Event component then either logs or suppresses (discards) the event accordingly. The processing involved in suppressing an event is depicted in the following illustration.



1. An event is triggered by a component.
2. The event-generating component invokes the Event component to log an event message. The originating component provides the subsystem ID, event number, and token values in the request.
3. Using the SSID and event number, the Event component retrieves the event suppression information from the Event File and determines whether the event is logged or discarded (suppressed).

4. The Event component logs or discards the event.
  - a. If the event is to be logged, the component retrieves the event text and emphasis from the Event File and formats the event.
  - b. If the event is to be suppressed, the component discards the event.
5. If the event is not suppressed, the Event component sends the formatted event to a designated logging destination.

## Event Suppression Information Tab

Event suppression is configured on the Event Suppression Information tab of the Integrated Server Event Configuration and Management window. A sample of the Event Suppression Information tab is shown below.

The screenshot displays the 'Integrated Server Event Configuration and Management - 1022 - 20' window. The 'Event Suppression Information' tab is active. The 'Subsystem ID' is set to '1022' and the 'Event Number' is '20'. The 'PCTLI' is 'Process Control Implementation class'. The 'Time Suppression Unit' is 'Minutes' and the 'Suppression Option' is 'Always Log the ...'. The 'Time Suppression Interval' is '0' and the 'Event Suppression Number' is '0'. The status bar at the bottom indicates '2 - 02:13 PM > Successful view of event number: 20.'

|                                      |         |                          |                                      |
|--------------------------------------|---------|--------------------------|--------------------------------------|
| Subsystem ID                         | 1022    | PCTLI                    | Process Control Implementation class |
| Event Number                         | 20      |                          |                                      |
| <b>Event Suppression Information</b> |         |                          |                                      |
| Time Suppression Unit                | Minutes | Suppression Option       | Always Log the ...                   |
| Time Suppression Interval            | 0       | Event Suppression Number | 0                                    |

## Log State

The Log State field indicates the event suppression action, if any, to be taken when logging an event. Four states of event suppression are supported:

- Always Log the Event
- Never Log the Event
- Log at Network Threshold
- Log at Process Threshold

If you select either the Log at Network Threshold or Log at Process Threshold value in the Log State field, the event is suppressed according to configured thresholds. A threshold can be a count as defined in the Number of events to skip field or a time duration as defined in the Time Unit and Time Interval fields. Either or both types of thresholds can be configured for these log states.

When an event is configured to be suppressed at the network threshold, the event thresholds apply to the entire system. When an event is configured to be suppressed at the process threshold, the event thresholds apply only to a single running process so multiple events can occur across processes.

If you select the Always Log the Event value, the event is always logged. If you select the Never Log the Event value, the event is never logged.

The default log state value for an event is Always Log the Event.

Each of these log state values are described in more detail below.

### Always Log the Event

When you select the “Always Log the Event” log state for an event, the event is never suppressed. Thus, an event will **always** be logged when this level of event suppression is selected. Since this is the default log state for handling an event, it is not necessary to configure all SSIDs and event numbers for this state. You might manually set this value for an event if you were changing from logging at the network or process threshold and you want to always allow logging of the event while retaining the threshold information used with these logging states.

## Never Log the Event

When you select the “Never Log the Event” log state, the event is always suppressed. Thus, an event will never be logged if this log state is selected. This log state effectively cancels out an event completely.

**Caution:** Extreme caution and careful consideration should be made before any event is specified as *not logged*, since this level effectively prohibits the event from ever being logged.

## Log at Process Threshold

When you select the “Log at Process Threshold” log state, the event is suppressed by each individual process running in the TSS environment. There is no global coordination of events with this log state because process-controlled suppression is implemented within the Event component in the local process. Each process monitors its own event stream. When an event is logged for which the “Log at Process Threshold” log state is used, subsequent occurrences of the event will be suppressed, based on the time interval or the number of events to skip. Thus with this level, any event will appear at least once per process before it is suppressed.

## Log at Network Threshold

When you select the “Log at Network Threshold” log state, the event is suppressed across multiple processes running in the TSS environment. Thus, logging at a network threshold is analogous to logical network control. With this log state, the global coordination of events is performed by the Event component, which monitors events being logged within a logical network. For this log state, the event is logged when the count or time threshold is reached across all running processes. A single event is logged once during a time threshold and each running process accumulates into one count threshold. Fewer events are logged when suppressing at the network level than when suppressing at the process level, which reduces overall system resource usage.

## Event Suppression Methods

Event suppression methods determine how an event is suppressed—either by time interval or by the number of events to skip. These event suppression methods are only used for the following log states:

- Log at Process Threshold
- Log at Network Threshold

These event suppression methods are not used for the following log states:

- Always Log the Event
- Never Log the Event

One or both of the methods can be configured for an event. If both methods are configured, whichever condition occurs first causes the event to be logged again.

### Suppression by Number of Events to Skip

Suppression by the number of events to skip causes some attempts to log an event to be suppressed. Each attempt to log the event increases a counter by one. When the counter reaches the defined threshold, the event is logged and the counter is reset. The threshold for this event suppression method is set in the Number of events to skip field, which can be set from 0–9999 and defaults to 0. When this field is set to 0, suppression by the number of events to skip is not used.

### Suppression by Time Interval

Suppression by time interval causes all subsequent occurrences of an event to be suppressed for a specified time interval, after the event is initially logged once. This threshold is set in the Time Interval and Time Unit fields:

- The Time Unit field indicates whether time interval value is measured in seconds or minutes.
- The Time Interval field defaults to a value of 0, which indicates that suppression by time interval is not used.

When the Time Unit field is set to Minutes, you can set the Time Interval field to a value from 0–166 (minutes). When the Time Unit field is set to Seconds, you can set the Time Interval field to a value from 0–9999 (seconds).

**Note:** If you select Seconds in the Time Unit field and enter a number in the Time Interval field that is evenly divisible by 60, the system will convert that number to minutes. When you redisplay the event suppression configuration information, the Time Unit field will display Minutes and the Time Interval field will contain a value in minutes. If the number of seconds that you originally enter in the Time Interval field is not evenly divisible by 60, the system will not convert the value to minutes.



# Configuring Events

There are two areas of configuration related to event management. These are:

- Environment Management

The Environment Management window is used to configure the event message destination device name parameter.

- Event

The Integrated Server Event Configuration and Management window is used to configure the event message text, emphasis, cause, effect, and remedy. Additionally, control is available that enables suppression of event messages.

The Event Subsystem ID Configuration window is used to configure subsystem ID, name, and description.

Using the ACI desktop:

1. Access **System Operations > Environment**

Configure the environment parameter called EVT\_DEST so that the parameter value is a valid event message destination.

The following illustration shows an example Environment Configuration window.

**Note:** Many of the attributes displayed in this example are specific to BASE24-es and are not used in a TSS environment.

The screenshot shows the 'ACI Environment Configuration' window. It contains a table with three columns: 'Short Name', 'Long Name', and 'Value'. Below the table is a 'Comment' text area. At the bottom, a status bar displays '1 - 05:00 PM > Successful view of Environment Configuration .'

| Short Name             | Long Name                | Value          |
|------------------------|--------------------------|----------------|
| ALLOW_PVV_DISPLAY      | ALLOW THE PIN VERIFIC... | 0              |
| BASE_CRNCY             | NETWORK CURRENCY C...    | 840            |
| BASE_CRNCY_INST        | BASE CURRENCY INSTIT...  | BNK1           |
| CARD_REFRESH_IMPACT    | CARD REFRESH IMPACT ...  | Y              |
| DEFAULT_REFRESH_RE...  | DEFAULT REFRESH REP...   | REFRESH_REPORT |
| EVT_DEST               | EVENT DESTINATION        | \$0            |
| HASH_TBL_MEM_SIZE      | HASH TABLE MEMORY SI...  | 2000000        |
| HOURS_OFFSET           | HOURS_OFFSET             | 0              |
| JRNL_CUTOVER_DELAY...  | JOURNAL CUTOVER DEL...   | 10             |
| JRNL_POSN_CLEANUP      | JOURNAL POSITION CLE...  | Y              |
| JRNL_POSN_INFO_MAX...  | JOURNAL POSITION INFO... | 10             |
| JRNL_POSN_INFO_RET...  | JOURNAL POSITION INFO... | 5              |
| JRNL_POSN_INFO_TIMR... | JOURNAL POSITION INFO... | 08:00          |
| LGNT_COUNTRY_ABBR      | LOGICAL NET COUNTRY ...  | US             |
| LGNT_CUTOVER           | LOGICAL NET CUTOVER ...  | 23:55          |
| LOGICAL_NET            | LOGICAL NET              | LGNT           |

**Comment**

1 - 05:00 PM > Successful view of Environment Configuration .

## 2. Access **System Operations> Event > Event Management**

Though no configuration is required, text, emphasis, cause, effect, and remedy information may be reconfigured (i.e., information added, changed, or deleted) to suit local operational requirements. However, it is suggested that you do this with caution. Changing and deleting this information may impact your ability to resolve problems efficiently.

The following illustration shows an example of event message text and emphasis. The text with  $\langle n \rangle$  is a token, where  $n$  is a number where text substitution occurs with items that vary between event messages.

**ACI Integrated Server Event Configuration and Management - 1022 - 20**

Subsystem ID: 1022 PCTLI: Process Control Implementation class

Event Number: 20

Event Message | Event Message Documentation | Event Suppression Information

Message Text: Command = <1> not supported by service = <2>

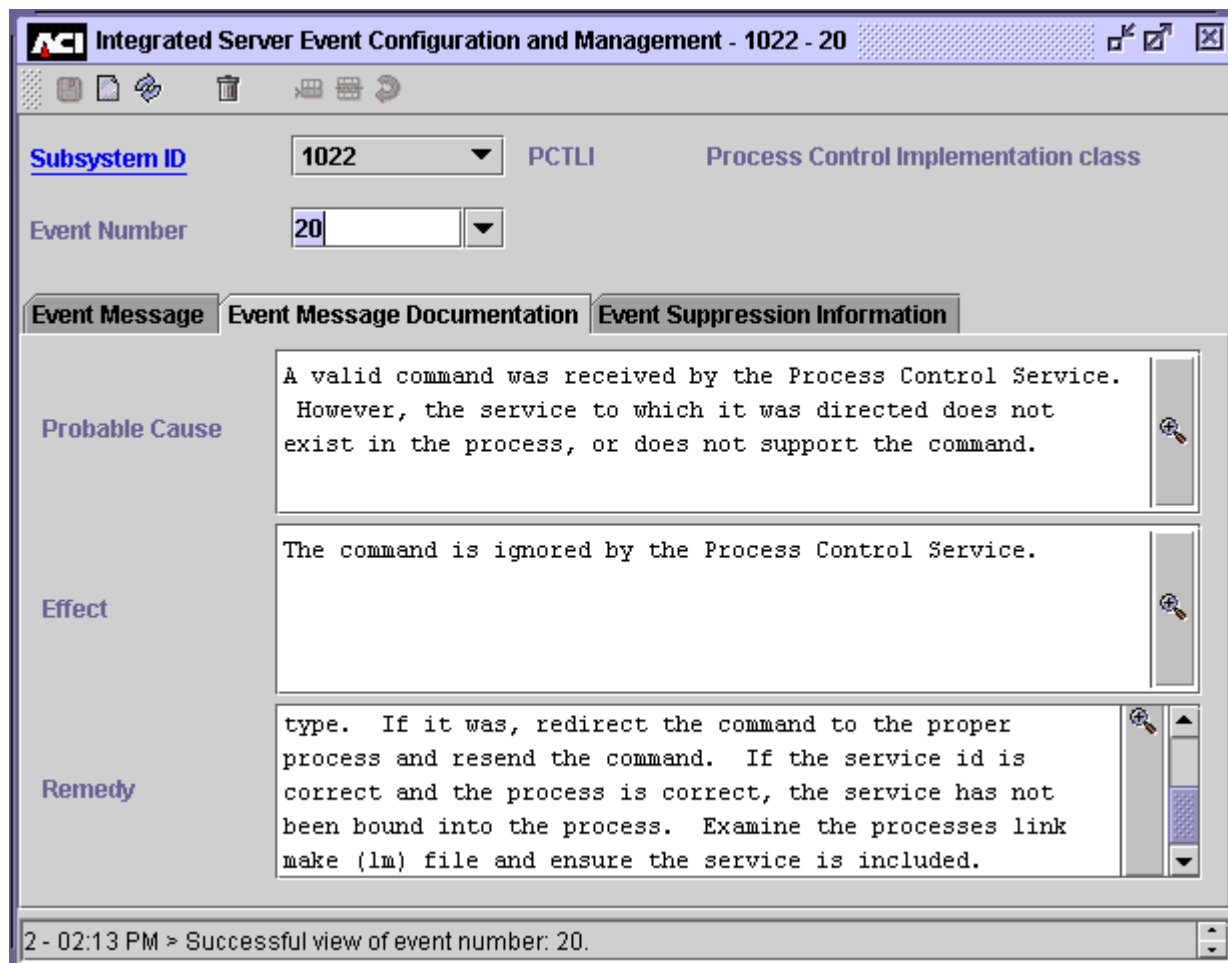
Severity: Informational

| Token ID | Token Type             | Length | Description           |
|----------|------------------------|--------|-----------------------|
| <1>      | Variable Length String | 0      | The full command text |
| <2>      | Variable Length String | 0      | Service identifier    |

2 - 02:13 PM > Successful view of event number: 20.

Event Message Documentation (i.e., cause, effect, and remedy information) is available on the Event Message Documentation tab.

The following illustration shows an example of event management documentation.



Event suppression control is available to indicate whether or not the event is logged and whether or not to skip events. If logged, further suppression control is available according the process and network threshold values. Skipping allows a number of events to be skipped once an event suppression threshold has been reached. By default, all events are logged. ACI Online Help, available from the ACI desktop user interface, provides an explanation of the event suppression fields.

The following illustration shows an example of event suppression information.

The screenshot shows a window titled "ACI Integrated Server Event Configuration and Management - 1022 - 20". The window has a menu bar with icons for file operations. Below the menu bar, there are two rows of configuration fields:

- Row 1: "Subsystem ID" (dropdown menu showing "1022"), "PCTLI" (text field), and "Process Control Implementation class" (text field).
- Row 2: "Event Number" (dropdown menu showing "20").

Below these fields are three tabs: "Event Message", "Event Message Documentation", and "Event Suppression Information". The "Event Suppression Information" tab is selected. Inside this tab, there are two rows of configuration fields:

- Row 1: "Time Suppression Unit" (dropdown menu showing "Minutes") and "Suppression Option" (dropdown menu showing "Always Log the ...").
- Row 2: "Time Suppression Interval" (text field showing "0") and "Event Suppression Number" (text field showing "0").

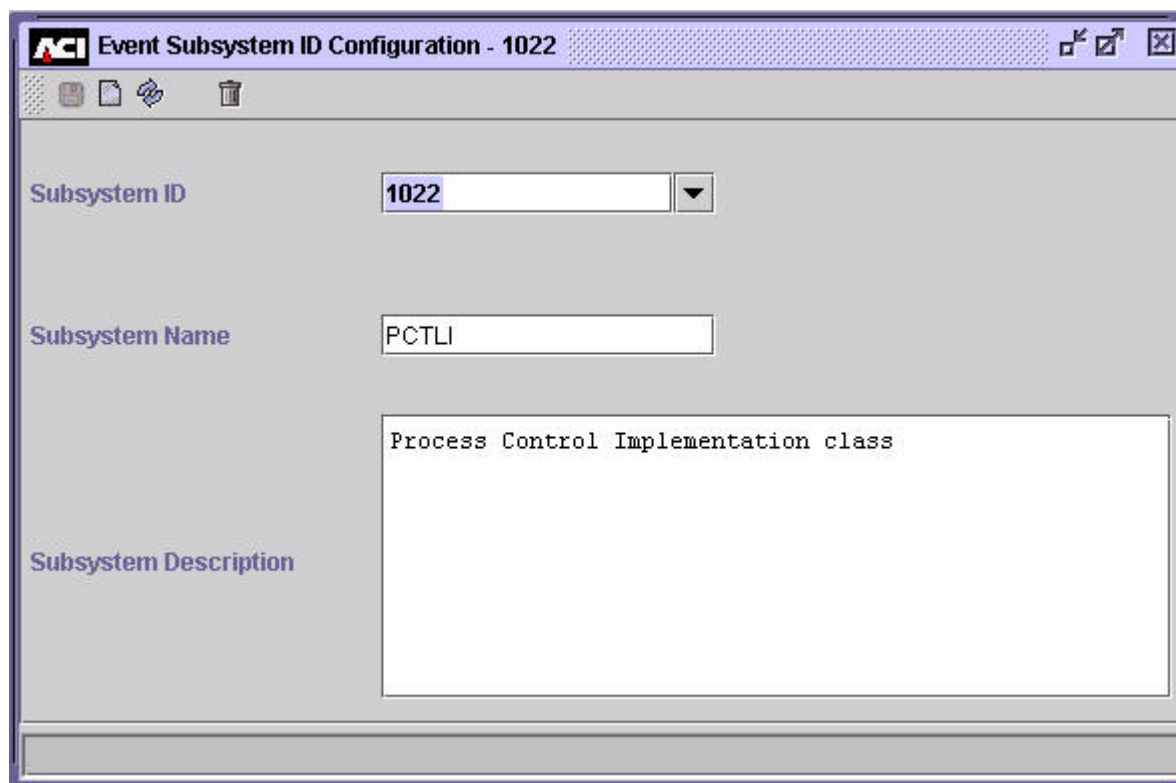
At the bottom of the window, there is a status bar that reads: "2 - 02:13 PM > Successful view of event number: 20."

In this example events are not suppressed. All events are allowed to the event logging destination.

3. Access **System Operations > Event > Subsystem ID**

No configuration is required here. Subsystems are preconfigured and should not be altered.

The following illustration shows an example Event Subsystem ID Configuration window.



The screenshot shows a window titled "Event Subsystem ID Configuration - 1022". The window has a standard Windows-style title bar with a logo on the left and maximize, minimize, and close buttons on the right. Below the title bar is a toolbar with icons for saving, opening, refreshing, and deleting. The main area of the window is divided into three sections:

- Subsystem ID:** A text box containing the value "1022" and a dropdown arrow button.
- Subsystem Name:** A text box containing the value "PCTLI".
- Subsystem Description:** A large text area containing the text "Process Control Implementation class".

At the bottom of the window is a horizontal bar, likely for status or navigation.

## 9: Process Control

---

This section describes the process control facilities provided in the TSS environment. It includes an overview of the functions you can perform from the Process Control window, describes the Process Control window and its fields, and lists the process control commands available in the TSS environment.

## Process Control Overview

Process Control is an interactive facility that provides users operational control of the TSS environment. Operational requirements vary from environment to environment.

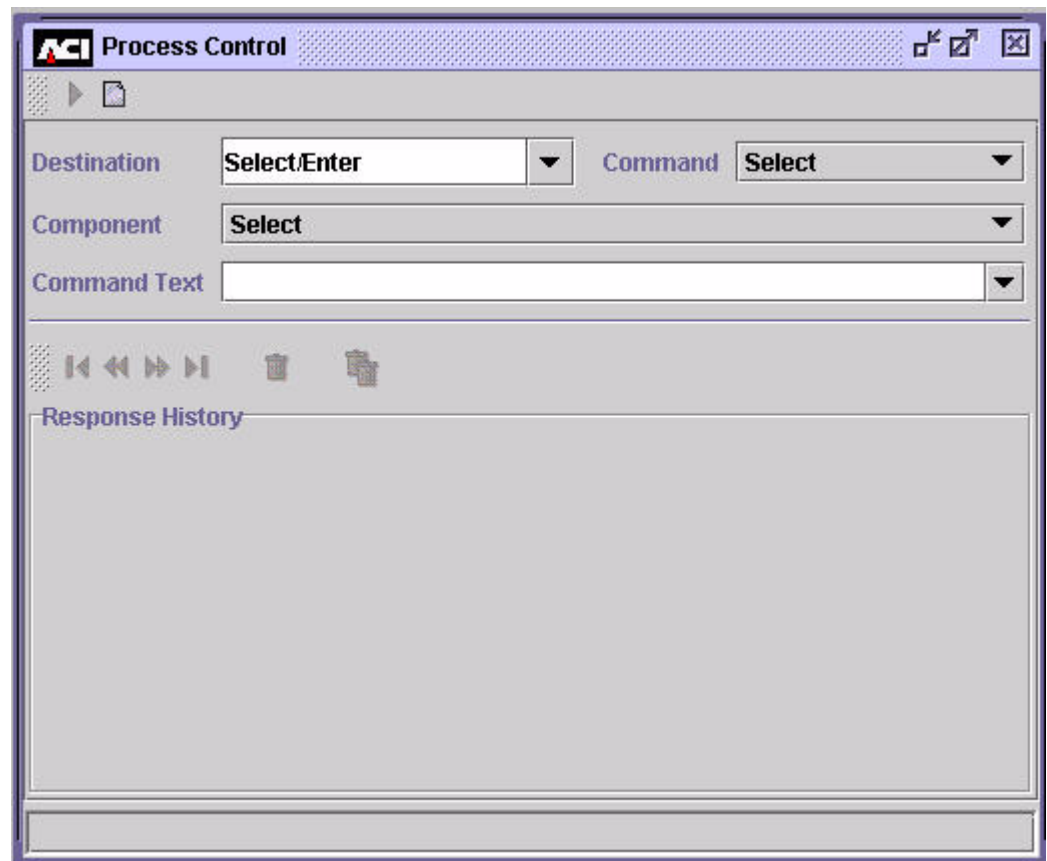
The following are some of the common operational control tasks you can perform.

- Obtaining and altering environment configuration information
- Obtaining status information
- Reloading an external or internal memory table
- Obtaining and resetting statistics
- Initializing, enabling, and disabling a security device
- Enabling or disabling audit store-and-forward processing



# Process Control Window

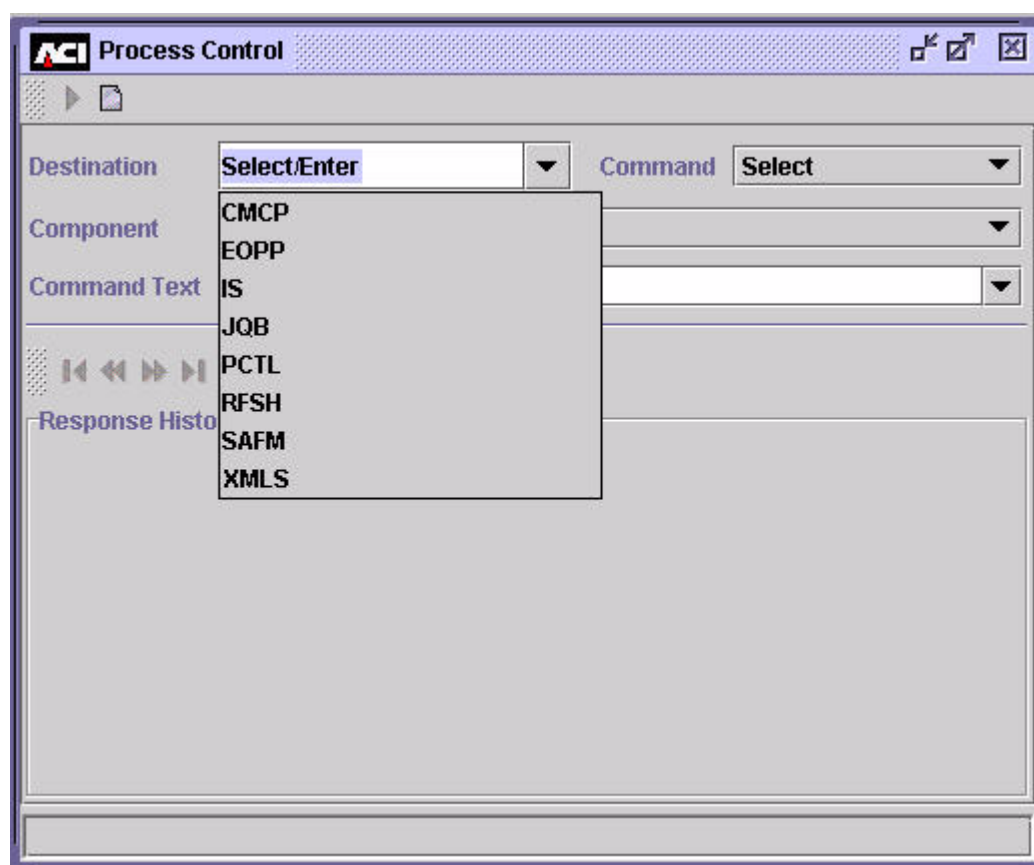
The Process Control window is available under the System Operations menu option of the ACI desktop user interface. Control is facilitated using selectable lists, destination entry, and keyword entry. A sample of the Process Control window is shown below.



## Destinations

Generic destinations provided in a selectable list include all the application servers available in a BASE24-es system. Only the Integrated Server (IS) process, Process Control (PCTL), and the C++ User Interface Server (XMLS) process destinations are used with the Transaction Security Services process in a BASE24 system. The destination of PCTL sends commands to all components registered to use the command, regardless of the type of process in which they are found.

**Note:** The CMCP, EOPP, JQB, RFSH, and SAFM selections are not applicable and are not used in the TSS environment.

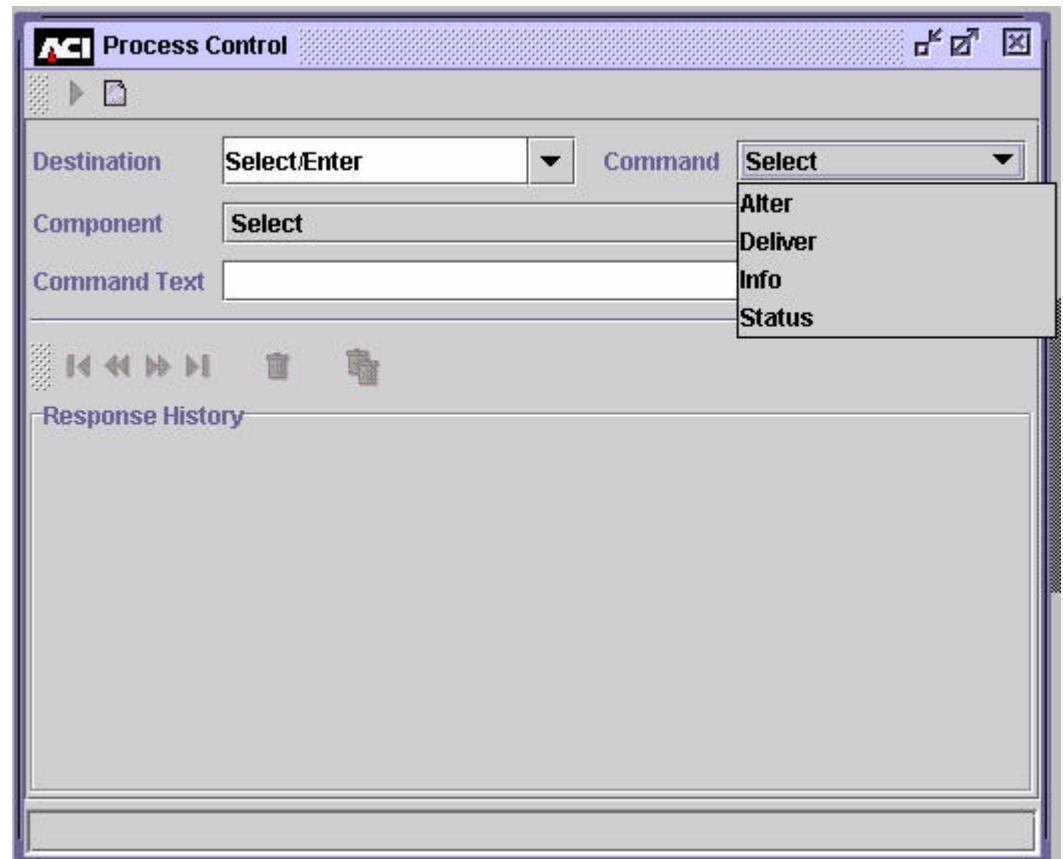


Besides the entries in the selectable list, you can also enter the symbolic name of a single process when entering an ALTER or DELIVER command. When you send an ALTER command to an entry in the selectable list (other than the PCTL destination), it is broadcast to all processes configured with a matching SERVICE attribute configured in the XPNET node. When you send an ALTER command to the symbolic name of an individual process, it is not broadcast to any other processes.

When entering an INFO or STATUS command, you must enter the assign specified for a single Transaction Security Services process in the System Interface Message Delivery Service Destination File (MDSDEST). For example, assume you have 12 Transaction Security Services processes with assign names of “IS $n$ ” in the MDSDEST, where  $n$  is a number from 1 to 12. To send an INFO or STATUS command to the first Transaction Security Services process, you would enter “IS1” in the Destination field. To send an INFO or STATUS command to the twelfth Transaction Security Services process, you would enter “IS12” in the Destination field.

## Commands

Allowed commands include Alter, Deliver, Info, and Status. Refer to ACI Online Help for an explanation of the commands.

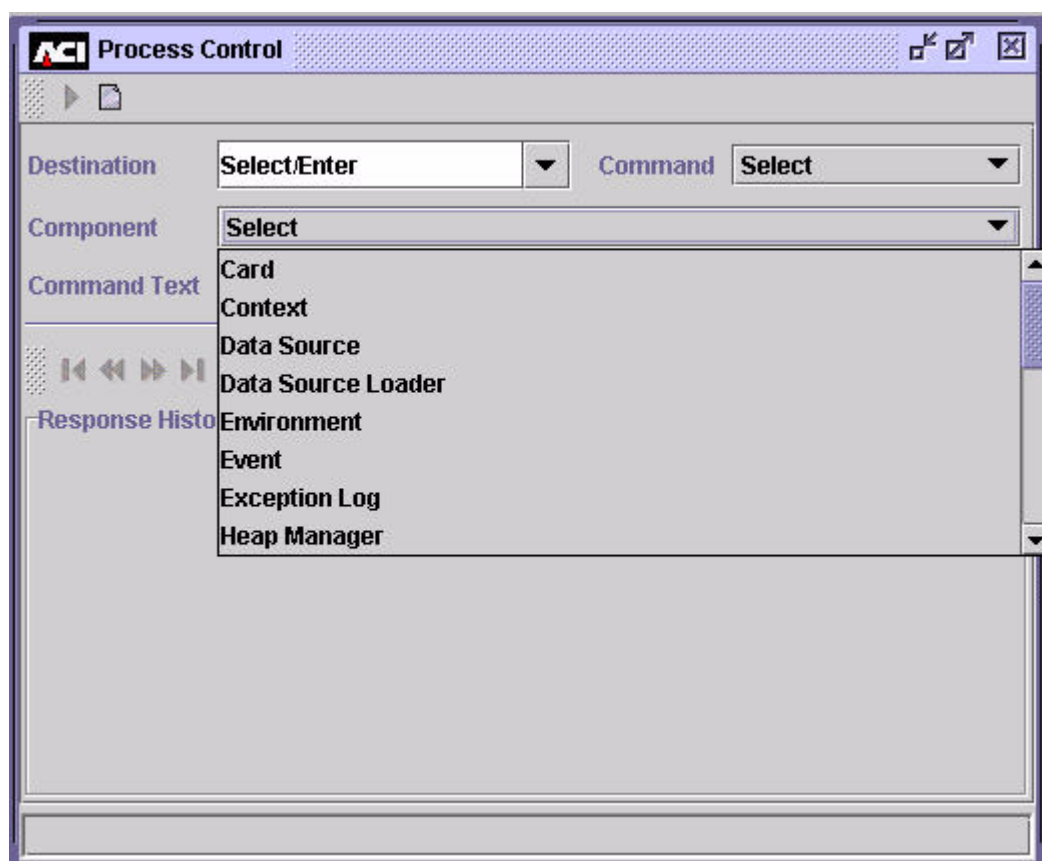


ALTER commands are broadcast to all processes that have been configured to the XPNET node with a service value equal to the value entered in the Destination field. For example, ALTER commands are sent to all Transaction Security

Services processes configured to the XPNET node with a SERVICE attribute value of “IS” when you enter “IS” in the Destination field. DELIVER commands are sent to a specified process. If you send an ALTER command to symbolic process name, it is only sent to the specified process. For DELIVER commands, you can enter a generic destination (i.e., IS or XMLS) or a symbolic process name (e.g., P1A^TSSIS1, P1A^TSSIS3, P1A^XMLS01) in the Destination field. INFO and STATUS commands are sent to a single Transaction Security Services process. For these commands, the value entered in the Destination field must match the assign name defined for the Transaction Security Services process in the MDSDEST.

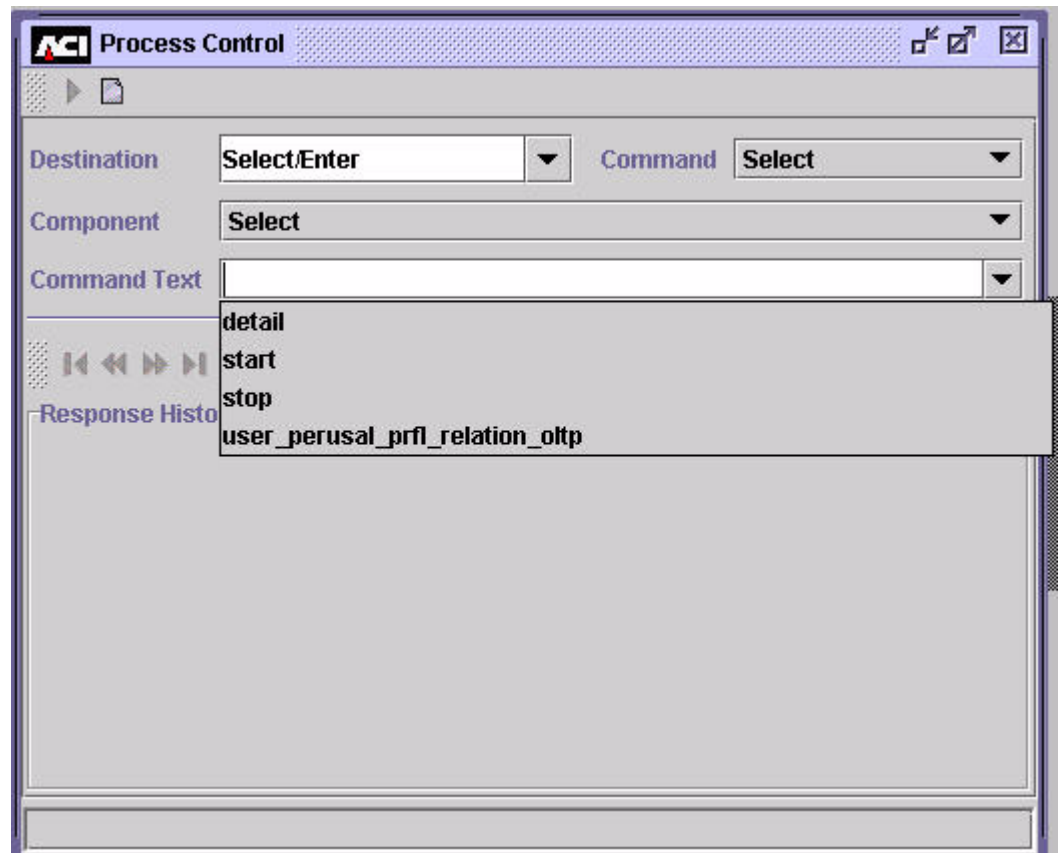
## Components

Available components vary from system to system and may include some or all of those shown below.



## Command Text

The default command text keywords are shown below. The supported command text keywords depend on the components available in the respective application servers. Refer to the “Process Control Commands” topic or ACI Online Help for the ACI desktop user interface for a list of components and the related command text keywords.



When the text for a command includes multiple items or variables, you can separate the items or variables using a comma or a space in the Command Text field.

## Process Control Commands

The following table describes the Process Control commands available for the TSS environment. These commands are also documented in ACI Online Help for the ACI desktop user interface.

You can enter all of these commands from the Process Control window on the ACI desktop. The syntax for entering commands on the Process Control window is provided for each command. You can also enter most of these commands from an XPNET network control facility (i.e., NCS screen, NCPCOM). If a command can be entered from a network control facility, the NCPCOM command syntax is also provided.

## Data Source Component

The following process control commands are available to the Data Source component.

### Alter Data Source

The Alter Data Source command alters the specified data source (e.g., memory table). It causes the component to reopen and close the data source. Use this command to start using an external memory table after it has been reloaded or to start rebuilding an internal memory table after its associated disk file has been modified. The memory tables used in the TSS environment are described in [section 2](#).

You can also use this command to open a new data source for any data source in the CONFCSV if the physical file has changed, the CONFCSV entry was changed, or the old file was renamed and a new file was put in its place. This applies to any data source in the CONFCSV rather than just those data sources with associated memory tables.

**Note:** When you send this command to the PCTL destination, processes of all types in the system attempt to close and reopen (re-read) the memory table, thereby accessing the new data in processing. When using the PCTL destination, you may see informational event messages logged for any processes that attempt to close and reopen the memory table if they do not actually use the table.

***Process Control Syntax***

---

Destination: IS or PCTL  
Command: Alter  
Component: Data Source  
Command Text: *data-source-name*

where: *data-source-name* is the assign name specified for the external memory table or disk file in the CONFCSV on the data subvolume. For example, the assign name for the IDESOD external memory table is IBM\_DES\_OLTP. The assign name for the Channel Security Configuration File (CSECD) is CHANNEL\_SECURITY.

***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, "ALTER DASRC *data-source-name*"

*object-name* — The symbolic name assigned to a Transaction Security Services process in the XPNET node.

*data-source-name* — See process control command text above.

## Data Source Loader Component

The following process control commands are available to the Data Source Loader component.

### Deliver Data Source

The Deliver Data Source command loads the specified data source (i.e., external memory table) from another data source (i.e., data file). Use this command to reload an external memory table after you have added, updated, or deleted data in the underlying data file. After the external memory table is reloaded, use the Alter Data Source command to begin using the external memory table in processing. The relationships between ACI desktop windows and the underlying data files they maintain are described in [section 2](#).

**Process Control Syntax**

---

Destination: XMLS

Commands to rebuild external memory tables should be sent to the User Interface (XML Server) process instead of a Transaction Security Services process to avoid any impact to online transaction processing performance.

Command: Deliver

Component: Data Source Loader

Command Text: *oltp-assign-name*

where: *data-source-name* is the assign name specified for the external memory table in the CONFCSV on the data subvolume. For example, the assign name for the IDESOD external memory table is IBM\_DES\_OLTP.

**NCPCOM Syntax**

---

DELIVER PROCESS *object-name*, "DELIVER DSLDR *oltp-assign-name*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*oltp-assign-name* — See process control command text above.

## Environment Component

The following process control commands are available to the Environment component.

### Alter Attribute

The following attribute change command rereads the specified attribute or all attributes (denoted by \*) and notifies registered clients to begin using the new values.

Use this command after you have added, updated, or deleted attributes on the Environment Configuration window. Modified attribute values are not used in processing until they are reread using this command.



---

**Process Control Syntax**


---

Destination: Any selection from the Destination field selectable list or a symbolic process name

Command: Alter

Component: Environment

Command Text: *attribute-name* | \*

where: *attribute-name* is the short name of a specified environment attribute. An asterisk (\*) specifies all environment attributes. The environment attributes available in the TSS environment are described in [section 3](#) and [appendix C](#).

---

**NCPCOM Syntax**


---

DELIVER PROCESS *object-name*, "ALTER ENVMT *attribute-name* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*attribute-name* — See process control command text above.

## Info Attribute

The Info Attribute command reports on the specified attribute or on all attributes (denoted by \*) used by a process. The command displays the current value and default value for the specified attribute or all attributes. The reported information for the attributes is displayed in the Response History area of the Process Control window. The report includes default attribute values for the process that are not overridden on the Environment Configuration window. The following example shows you the information reported for the EVT\_DEST attribute:

```
Optional configuration parameter: EVT_DEST
                                Value: $0
                                Default Value: NONE
```

If you specify XMLS as the destination, the XML Server process that receives the initial request from the ACI desktop client displays its own environment attributes and does not send the request out to any other process. If you want to view the environment attributes of a Transaction Security Services process, you must use the assign name defined for the process in the MDSDEST as the destination.

***Process Control Syntax***

---

Destination: XMLS or the assign name defined for a Transaction Security Services process in the MDSDEST

Command: Info

Component: Environment

Command Text: *attribute-name* | \*

where: *attribute-name* is the short name of a specified environment attribute. An asterisk (\*) specifies all environment attributes.

## Exception Log Component

The following process control commands are available to the Exception Log component.

### Purge EXLOGD Records

The Purge EXLOGD Records command purges all records from the Exception Log File (EXLOGD) that have a date prior to a specified date. The EXLOGD contains messages that various Integrated Server (IS) process types could not interpret or process.

***Process Control Syntax***

---

Destination: XMLS or a symbolic process name

Command: Deliver

Component: Exception Log

Command Text: PURGERECS *yyyymmdd*

where: *yyyymmdd* is the date prior to which records will be purged.

***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, "DELIVER ELOG PURGERECS  
*yyyymmdd*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*yyyymmdd* — The date prior to which records will be purged.

## Event Component

The following process control commands are available to the Event component.

### Reset Suppression

The Reset Suppression command clears all events in the current memory collection that contain event counts and timers for suppressed events.

---

#### ***Process Control Syntax***

|               |                                                                                     |
|---------------|-------------------------------------------------------------------------------------|
| Destination:  | Any selection from the Destination field selectable list or a symbolic process name |
| Command:      | Alter                                                                               |
| Component:    | Event                                                                               |
| Command Text: | RESETSUPPRESSION                                                                    |

---

#### ***NCPCOM Syntax***

DELIVER PROCESS *object-name*, "ALTER EVT RESETSUPPRESSION"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

## Heap Manager Component

The following process control commands are available to the Heap Manager component.

### Compare Heap Statistics

The Compare Heap Statistics command compares current heap statistics to the statistics that were generated when the previous compare command was sent. You can use the compare command to identify memory leaks.

***Process Control Syntax***

---

Destination: Any assign name of a Transaction Security Services process defined in the MDSDEST

Command: Status

Component: Heap Manager

Command Text: COMPARE

**Reset Heap Statistics**

The Reset Heap Statistics command resets the heap (dynamic) memory statistics maintained by the Heap Manager component.

***Process Control Syntax***

---

Destination: Any selection from the Destination field selectable list or a symbolic process name

Command: Deliver

Component: Heap Manager

Command Text: RESET

***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, "DELIVER HEAP RESET"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process, Timer process, or a Transaction Security Services process in the XPNET node.

**Start Monitoring Heap Memory**

The Start Monitoring Heap Memory command starts monitoring the amount of heap (dynamic) memory in use and resets collections.

***Process Control Syntax***

---

Destination: Any selection from the Destination field selectable list or a symbolic process name

Command: Deliver

Component: Heap Manager  
Command Text: MONITOR [ON]

### ***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, “DELIVER HEAP MONITOR [ON]”

*object-name* — The symbolic name assigned to the User Interface (XML Server) process, Timer process, or a Transaction Security Services process in the XPNET node.

## **Status Heap Statistics**

The Status Heap Statistics command provides heap statistics. When you enter the status command without the DETAIL key word, it reports the amount of heap memory that is allocated. When you enter the status command with the DETAIL key word, it reports detailed heap statistics that include the number of parcels allocated and the number of parcels available, according to the parcel size.

### ***Process Control Syntax***

---

Destination: Any assign name of a Transaction Security Services process defined in the MDSDEST  
Command: Status  
Component: Heap Manager  
Command Text: [DETAIL]

## **Status Monitoring Heap Memory**

The Status Monitoring Heap Memory command reports the amount of heap memory not released since the command to turn monitoring on was sent.

### ***Process Control Syntax***

---

Destination: Any assign name of a Transaction Security Services process defined in the MDSDEST  
Command: Status  
Component: Heap Manager  
Command Text: MONITOR

## Stop Monitoring Heap Memory

The Stop Monitoring Heap Memory command stops monitoring the amount of heap (dynamic) memory in use and resets collections.

---

### ***Process Control Syntax***

---

|               |                                                                                     |
|---------------|-------------------------------------------------------------------------------------|
| Destination:  | Any selection from the Destination field selectable list or a symbolic process name |
| Command:      | Deliver                                                                             |
| Component:    | Heap Manager                                                                        |
| Command Text: | MONITOR OFF                                                                         |

---

### ***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, "DELIVER HEAP MONITOR OFF"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process, Timer process, or a Transaction Security Services process in the XPNET node.

## Key Conversion Component

The following process control commands are available to the Key Conversion component.

### Convert AKB

The Convert AKB command converts all non-AKB formatted keys in the specified files into AKB format. This command requires the use of an NSP with AKB version software.

You can set optional Environment attributes that specify the AKB header values used when this command is entered. These attributes are described in appendix C.

---

### ***Process Control Syntax***

---

|              |                                                     |
|--------------|-----------------------------------------------------|
| Destination: | XMLS or the symbolic name of the XML Server process |
| Command:     | Deliver                                             |
| Component:   | Key Conversion                                      |

where:

*File-name* can be one or more of the following file acronyms, with each acronym separated by a comma:

If you specify `*` as one of the entries in the *file-name* list, all other entries are ignored.

***object-name*** — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

## Erase LMK or MFK

Dec-2005 R6.0v6, TSS 05.4 BA-AE000-14  
ACI Worldwide Inc.

Command: Deliver  
Component: Key Conversion  
Command Text: ERASE *security-device-name* | \*  
  
where: *security-device-name* is the name of an individual NSP. If you enter an asterisk (\*), the command is executed on all NSPs.

---

***NPCOM Syntax***

---

DELIVER PROCESS *object-name*, "DELIVER KEYCONV ERASE  
*security-device-name* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*security-device-name* — See process control command text above.

## **Validate Check Digits**

The Validate Check Digits command validates the check digits stored in the TSS database for keys encrypted by Atalla NSPs (using variant or AKB software). You can use this command to validate check digits in a single file or all TSS files. Valid check digits are required for non-AKB formatted keys before they can be converted by the AKB Conversion utility.

---

***Process Control Syntax***

---

Destination: XMLS or the symbolic name of the XML Server process  
Command: Deliver  
Component: Key Conversion



Command Text: VVK VALIDATE [HSM *security-device-name* | ALL | \*]  
*file-name, file-name,...* | \*

where:

*security-device-name* is the name of an individual NSP. If you enter ALL or an asterisk (\*), the command is executed on all NSPs.

*file-name* can be one or more of the following file acronyms, with each acronym separated by a comma:

\* (validates keys in all TSS files)

CRDVD (validates keys in the Card Verification File)

CSECD (validates keys in the Channel Security File)

DNTD (validates keys in the Diebold Number Table File)

DRVKD (validates keys in the DUKPT Derivation Key File)

EMVSD (validates keys in the EMV Security File)

IDESD (validates keys in the IBM DES Verification File)

ISECD (validates keys in the Interface Security File)

VPVVD (validates keys in the Visa PVV Verification File)

If you specify \* as one of the entries in the *file-name* list, all other entries are ignored.

---

### ***NCPCOM Syntax***

DELIVER PROCESS *object-name*, "DELIVER KEYCONV VVK VALIDATE  
[HSM *security-device-name* | ALL | \*] *file-name, file-name,...* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*security-device-name* — See process control command text above.

*file-name* — See process control command text above.

## **Generate Check Digits**

The Generate Check Digits command generates check digits for keys stored in the TSS database that are encrypted by Atalla NSPs (using variant or AKB software). You can use this command to generate check digits in a single file or all TSS files. If check digits already exist for a key in a TSS file, they are overwritten with the check digits generated by this command. Valid check digits are required for non-AKB formatted keys before they can be converted by the AKB Conversion utility.

**Process Control Syntax**

---

Destination: XMLS or the symbolic name of the XML Server process  
Command: Deliver  
Component: Key Conversion  
Command Text: VVK GENERATE [HSM *security-device-name* | ALL | \*]  
*file-name, file-name,...* | \*

where:

*security-device-name* is the name of an individual NSP. If you enter ALL or an asterisk (\*), the command is executed on all NSPs.

*file-name* can be one or more of the following file acronyms, with each acronym separated by a comma:

\* (generates check digits for keys in all TSS files)  
CRDVD (generates check digits for keys in the Card Verification File)  
CSECD (generates check digits for keys in the Channel Security File)  
DNTD (generates check digits for keys in the Diebold Number Table File)  
DRVKD (generates check digits for keys in the DUKPT Derivation Key File)  
EMVSD (generates check digits for keys in the EMV Security File)  
IDESD (generates check digits for keys in the IBM DES Verification File)  
ISECD (generates check digits for keys in the Interface Security File)  
VPVVD (generates check digits for keys in the Visa PVV Verification File)

If you specify \* as one of the entries in the *file-name* list, all other entries are ignored.

**NCPCOM Syntax**

---

DELIVER PROCESS *object-name*, "DELIVER KEYCONV VVK  
GENERATE [HSM *security-device-name* | ALL | \*] *file-name, file-name,...* |  
\*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*security-device-name* — See process control command text above.

*file-name* — See process control command text above.

## Timer Component

The following process control commands are available to the Timer component.

### Status Timer Statistics

The Status Timer Statistics command returns timer statistics information for named timers.

---

#### ***Process Control Syntax***

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| Destination: | The assign name of a Transaction Security Services process defined in the MDSDEST |
| Command:     | Status                                                                            |
| Component:   | Timer                                                                             |

## Trace Component

The Trace component is the interface to the process tracing capabilities of the system.

The following command can be sent to the Trace component from the Process Control window.

### Status Information on Trace

The following information command returns status information on a trace. When you enter the command without detail, it returns information that the trace has or has not been started. When you enter the command with detail, it returns information about all the trace levels with the corresponding components which have been started.

***Command Syntax***

---

Destination: Any selection from the Destination field.  
Command: Status  
Component: Trace  
Command Text: [DETAIL]

## Transaction Security Component

The following process control commands are available to the Transaction Security component.

### Disable Security Device

The Disable Security Device command disables the specified security device or all security devices (denoted by \*) by setting the status of the specified devices to inactive, thereby removing the devices from service. The Transaction Security component generates an event message, indicating that the device has been deactivated. This command also cancels the automatic reset timer if the Transaction Security component is currently attempting to enable the device at the intervals specified in the Reset Timer Limit field on the Security Configuration window. This prevents the Transaction Security component from continuing to attempt to re-establish communications with a device when it is to be taken out of service for an extended period of time.

---

#### **Process Control Syntax**

|               |                                         |
|---------------|-----------------------------------------|
| Destination:  | IS or a symbolic process name           |
| Command:      | Alter                                   |
| Component:    | Transaction Security                    |
| Command Text: | DISABLE <i>security-device-name</i>   * |

where: *security-device-name* is the name of an individual security device. If you enter an asterisk (\*), all security devices are disabled.

---

#### **NCPCOM Syntax**

DELIVER PROCESS *object-name*, "ALTER TSEC DISABLE *security-device-name* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*security-device-name* — See process control command text above.

## Enable Security Device

The Enable Security Device command causes the Transaction Security component to send a status or echo test message to the specified security device or all security devices (denoted by \*). If this message is successfully processed for a device, the component changes the status of the device from inactive to active and sets the consecutive error count to zero, thereby returning the device to service. If this message is not successfully processed for a device, the Transaction Security component generates an event message that indicates the error and specifies whether the status or echo test message will be automatically retried. The Reset Timer Limit field on the Security Device Configuration window specifies the number of seconds the component waits before retrying the security device call. If the Reset Timer Limit field is set to 0, the component does not automatically attempt to retry the call.

---

### Process Control Syntax

Destination: IS or a symbolic process name  
Command: Alter  
Component: Transaction Security  
Command Text: ENABLE *security-device-name* | \*

where: *security-device-name* is the name of an individual security device. If you enter an asterisk (\*), all security devices are enabled.

---

### NPCOM Syntax

DELIVER PROCESS *object-name*, "ALTER TSEC ENABLE *security-device-name* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*security-device-name* — See process control command text above.

## Load DNT

The Load DNT command loads the Diebold number table stored in the Diebold Number Table File (DNTD) into the memory of the specified security device or all security devices (denoted by \*).

---

**Process Control Syntax**


---

Destination: IS or XMLS or a symbolic process name  
 Command: Deliver  
 Component: Transaction Security  
 Command Text: DNTLOAD *security-device-name* | \*

where: *security-device-name* is the name of an individual security device. If you enter an asterisk (\*), the DNT is loaded in all security devices.

---

**NCPCOM Syntax**


---

DELIVER PROCESS *object-name*, "DELIVER TSEC DNTLOAD *security-device-name* | \*\*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*security-device-name* — See process control command text above.

## Key Generate

The Key Generate command generates an RSA key pair for the BASE24 system. This key pair is required for implementing automated key distribution using public key cryptography. Each variable in the command text must be separated by a comma in the command text line, regardless of whether data for the variable is provided.

**Note:** The security device used for the KEYGEN command is disabled during key generation, which may take up to 90 seconds. For the *host-name*, *common-name*, and *org-name* variables in the KEYGEN command syntax, the following characters are allowed: A–Z, a–z, 0–9, *b*, (, ), +, -, ., /, :, ?, and #, where *b* represents a blank space. You cannot use a comma “,” or exclamation point “!” within these variables because they are reserved to denote the end of a variable value and to override the current status in the Host Public Key Security Files, respectively. For shared Wincor Nixdorf signature devices, the *host-name* variable in this command must be set to the value of the KEY SIGNER field on screen 22 of the BASE24-atm Terminal Data files (ATD). If this field is not configured, it defaults to a value of HOST. For NCR and Diebold devices, the *host-name* variable must be set to a value of HOST for the Automated Key Distribution System to work properly.

***Process Control Syntax***

---

Destination: XMLS or symbolic name of the XML Server process  
Command: Deliver  
Component: Transaction Security



Command Text:    *KEYGEN host-name, dev-typ, [common-name, organization-name][, !]*

where:

*host-name* — For shared Wincor Nixdorf signature devices, this is the name of the host BASE24 system as configured in the KEY SIGNER field on screen 22 of the BASE24-atm Terminal Data files (ATD). If this field is not configured, it defaults to a value of HOST. For NCR and Diebold devices, this variable must be set to a value of HOST. The key pair is stored under this name in the Host Public Key File. 1–16 alphanumeric characters.

*dev-typ* — The type of ATM device to which keys are to be distributed. Valid values are:

|                         |                                                                                  |
|-------------------------|----------------------------------------------------------------------------------|
| NCR or 0                | = NCR 5XXX devices                                                               |
| DIEBOLD, DBLD, or 1     | = Diebold 10XX/478X devices                                                      |
| WINCORSIG, WINSIG, or 2 | = Shared Wincor Signature devices running in NDC+ or Diebold 912 emulation mode. |

*common-name* — The specific entity for which the BASE24 system public key certificate is being requested. This value is used in generating the BASE24 system public key certificate for use with Diebold devices. This variable is not used for NCR devices or shared Wincor Nixdorf signature devices running in Diebold 912 emulation or NCR NDC+ emulation modes; however, the intervening comma used to separate this variable is always required. 1–64 alphanumeric characters.

*organization-name* — The name of the organization or company for which the BASE24 system public key certificate is being requested. This value is used in generating the BASE24 system public key certificate for use with Diebold devices. This variable is not used for NCR devices or shared Wincor Nixdorf signature devices running in Diebold 912 emulation or NCR NDC+ emulation modes; however, the intervening comma used to separate this variable is always required, if the command line includes the override flag. 1–64 alphanumeric characters.

Command Text:     ! — An optional override flag. Forces the KEYGEN  
*continued*           command to be performed regardless of the current status in  
                      the Host Public Key Security File. Normally, you would  
                      only use this flag if the database has been corrupted. See the  
                      note below for further usage information.

Command Text Examples:

```
KEYGEN HOST,NCR,,!  
KEYGEN HOST,DBLD,BASE24,BANK1
```

**Note:** The “!” optional override flag for the Key Generate command was designed to prevent multiple Key Generate commands from being performed at the same time. This can occur if the command is issued (typically using NCPCOM) to two different processes (either XMLS or IS) or if you issue the command to the XMLS destination using the Process Control window and more than one XMLS process is configured.

If Key Generate commands are issued to two different processes at the same time, the following event message is generated (assuming that you have not included the “!” flag in either command):

```
YY-MM-DD;HH:MM:SS.sss \ARGUS.$W4X12      ACI.1502.1000      70  
Mgr PPD: \ARGUS.$W4XN  Generator: P4X^TSS07  
TSEC: RSA Key Generation already in progress. Host Name = DUMMY1  
. Implementation = 0.
```

You should only use the “!” override flag when the above event message is generated and you know that a key generation is not in progress (this can occur if the process that was executing the Key Generate command ended abnormally before completing the operation).

---

***NCPCOM Syntax***

---

DELIVER PROCESS *object-name*, “DELIVER TSEC KEYGEN *host-name*, *dev-typ*, [*common-name*, *organization-name*][, !]”

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

*host-name*, *dev-typ*, *common-name*, *organization-name* — See process control command text above.

## Initialize Security Device

The Initialize Security Device command initializes the specified security device or all security devices (denoted by \*). Initializing a security device causes the TSS application to send the text strings configured in the Initialization String and Configuration String fields for the device on the Security Device Configuration window to the device. In addition, if the status of the device is Active, this command causes the Transaction Security component to send a status or echo test message to the security device. If this message is successfully processed, the component sets the consecutive error count to zero. If this message is not successfully processed for a device, the Transaction Security component generates an event message that indicates the error and specifies whether the status or echo test message will be automatically retried. The Reset Timer Limit field on the Security Device Configuration window specifies the number of seconds the component waits before retrying the security device call. If the Reset Timer Limit field is set to 0, the component does not automatically attempt to retry the call.

This command also causes the firmware version and maximum buffer length for the device to be displayed on the Security Device Configuration window. This information is not displayed until you initialize the device.

---

### Process Control Syntax

|               |                                      |
|---------------|--------------------------------------|
| Destination:  | IS or XMLS or symbolic process name  |
| Command:      | Deliver                              |
| Component:    | Transaction Security                 |
| Command Text: | INIT <i>security-device-name</i>   * |

where: *security-device-name* is the name of an individual security device. If you enter an asterisk (\*), all security devices are initialized.

---

### NCPCOM Syntax

DELIVER PROCESS *object-name*, "DELIVER TSEC INIT *security-device-name* | \*"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process or a Transaction Security Services process in the XPNET node.

*security-device-name* — See process control command text above.

## Refresh Individual CSEC Records

If you maintain Channel Security data source (CSECD) records in memory (i.e., the CSEC\_MEMORY\_USE Environment attribute is set to Y), you can use the CSEC Refresh command to reread an individual channel security record from disk into memory, without deleting any other CSECD records from memory.

If you use the Alter Data Source command on the CSECD (CHANNEL\_SECURITY assign) when CSECD records are maintained in memory, all records are deleted from internal memory and are read from disk back into memory as they are needed in processing. Using the CSEC Refresh command, you can avoid flushing all CSECD records from memory when only one channel security record has been modified. You do not need to use this command for a new CSECD record because it read from disk and added to memory on the first transaction received that requires the record. You can optionally use this command after deleting a CSECD record to remove it from memory.

---

### Command Syntax

Destination: IS or a symbolic process name  
Command: Alter  
Component: Transaction Security  
Command Text: CSEC REFRESH *channel-id*

where:

*channel-id* is the channel ID of the CSECD record to be warmbooted from disk.

## Start Statistics

The Start Statistics command resets the transaction counter to zero and starts monitoring the number of transactions processed by the specified Transaction Security Services process. The counter is increased each time the Transaction Security Services process receives a message. You can use this command to monitor transaction load balancing across the Transaction Security Services processes configured in your system.

---

### Process Control Syntax

Destination: IS or a symbolic process name

Command: Alter or Deliver

If you set this field to Alter and the Destination field is set to IS, the command is broadcast to all processes configured in the XPNET node with a service value of IS.

If you set this field to Deliver, the command is sent to a single process.

Component: Transaction Security

Command Text: STATISTICS ON

---

***NPCOM Syntax***

---

DELIVER PROCESS *object-name*, "[ALTER | DELIVER] TSEC  
STATISTICS ON"

*object-name* — The symbolic name assigned to a Transaction Security Services process in the XPNET node.

## Status Security Device

The Status Security Device command indicates whether the specified security device or all security devices (denoted by \*) are in service and reports the consecutive error count and the maximum error count for the specified device.

---

***Process Control Syntax***

---

Destination: The assign name of a Transaction Security Services process defined in the MDSDEST

Command: Status

Component: Transaction Security

Command Text: *security-device-name* | \*

where: *security-device-name* is the name of an individual security device. If you enter an asterisk (\*), status information for all security devices is returned.

## Status Statistics

The Status Statistics command returns the current number of transactions processed by the specified Transaction Security Services process. The transaction counter is increased each time the Transaction Security Services process receives a message. You can use this command to monitor transaction load balancing across the Transaction Security Services processes configured in your system.

---

### Process Control Syntax

|               |                                                                                    |
|---------------|------------------------------------------------------------------------------------|
| Destination:  | The assign name of a Transaction Security Services process defined in the MDSDEST. |
| Command:      | Status                                                                             |
| Component:    | Transaction Security                                                               |
| Command Text: | STATISTICS                                                                         |

## Stop Statistics

The Stop Statistics command stops monitoring the number of transactions processed by the specified Transaction Security Services process.

---

### Process Control Syntax

|               |                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Destination:  | IS or a symbolic process name                                                                                                                                             |
| Command:      | Alter or Deliver                                                                                                                                                          |
|               | If you set this field to Alter and the Destination field is set to IS, the command is broadcast to all processes configured in the XPNET node with a service value of IS. |
|               | If you set this field to Deliver, the command is sent to a single process.                                                                                                |
| Component:    | Transaction Security                                                                                                                                                      |
| Command Text: | STATISTICS OFF                                                                                                                                                            |

---

### NCPCOM Syntax

DELIVER PROCESS *object-name*, "[ALTER | DELIVER] TSEC  
STATISTICS OFF"

*object-name* — The symbolic name assigned to a Transaction Security Services process in the XPNET node.

## User Interface Auditing Component

The following process control commands are available to the User Interface Auditing component.

**Note:** With the advent of TSS release 05.4, the User Interface Auditing component is no longer used. The configuration required to write auditing messages to the Audit Store and Forward File (ASAFD) for transmission to the UAUD application and subsequent transmission to the ESWEB process is not installed in this release. Therefore, you should always configure the USE\_LOCAL\_USER\_AUDIT Environment attribute to a value of Y (The C++ User Interface (XML Server) process writes auditing records directly to the UALOGD).

### Disable Audit Store and Forwarding

The Disable Audit Store and Forwarding command disables the sending of audit store-and-forward messages to the UAUD application running on the server machine.

---

#### ***Process Control Syntax***

|               |                                                 |
|---------------|-------------------------------------------------|
| Destination:  | XMLS or symbolic name of the XML Server process |
| Command:      | Deliver                                         |
| Component:    | User Interface Auditing                         |
| Command Text: | SAFSTOP                                         |

### Enable Audit Store and Forwarding

The Enable Audit Store and Forwarding command enables the sending of audit store-and-forward messages to the UAUD application running on the server machine.

---

#### ***Process Control Syntax***

|               |                                                 |
|---------------|-------------------------------------------------|
| Destination:  | XMLS or symbolic name of the XML Server process |
| Command:      | Deliver                                         |
| Component:    | User Interface Auditing                         |
| Command Text: | SAFSEND                                         |

## Remove Audit Store and Forward Record

The Remove Audit Store and Forward Record command removes a specified record from the Audit Store and Forward File (ASAFD). You can use this command to remove an invalid record.

### ***Process Control Syntax***

---

Destination: XMLS or symbolic name of the XML Server process

Command: Deliver

Component: User Interface Auditing

Command Text: SAFPOP *timestamp session-number*

where:

*timestamp* is the timestamp of the record to be removed.

*session-number* is the session number of the record to be removed.



# 10: Process Tracing

---

BASE24-es provides an extensive means of tracing transaction processing for analysis and problem solving. Using the BASE24-es trace facilities, you can trace messages through your production system and provide information to ACI to assist in the remote diagnosis of problems when required.

Through the ACI desktop user interface, you can configure the types of information to be traced, execute traces, and filter the transactions to be included in the trace output. Information recorded during a trace is written to the Trace data source (TRCD).

This section covers the following topics related to tracing:

- What is a trace?
- Configuring traces
- Configuring trace filter profiles
- Trace operations
- Running traces
- Examining trace output

**Note:** The Trace facility was designed as a diagnostic tool for development and test environments. ACI strongly recommends that it be used in a controlled and limited manner in a production environment due to the overhead and impacts on performance. In addition, there are potential security risks for fraud when using the Trace facility in a production environment. Using the Trace facility, sensitive customer data (e.g., PANs, clear PIN offsets, encrypted PIN blocks, etc.) can be written to the trace output without any masking of data.

## What is a Trace?

A trace provides various *snapshots* of a transaction as it is processed through different components of the BASE24-es system. These snapshots can highlight different aspects of a transaction (e.g., TDEs, external messages, decision points, etc.) as configured for the trace. These *snapshots* are recorded in a trace data source that can be examined later for analysis and diagnosis of transaction processing problems. You can configure the locations at which the *snapshots* are taken (i.e., the processes and components to be traced) and the type of information to be included in the *snapshots* (i.e., the trace levels to be included in the trace). Through the configuration of filters, you can control which transactions are included in the trace output (e.g., you could trace only transactions with a specified PAN, processing code, or institution ID, or other filter criteria).

## Configuring Traces

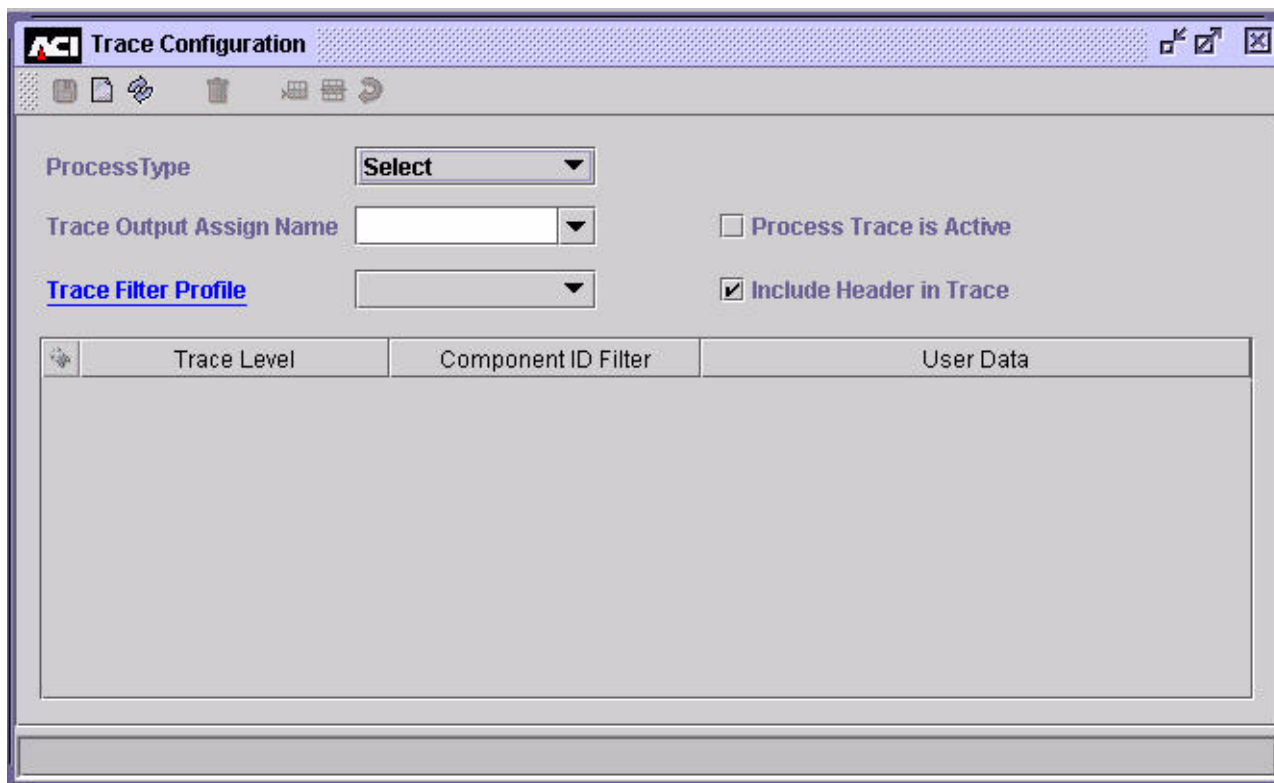
Use the Trace Configuration window on the ACI desktop user interface (accessed from **System Operations > Trace > Trace**), to configure a trace to be executed. On this window, you can configure the following:

- The type of process on which the trace is to be performed (e.g., IS or XMLS)
- The assign associated with the output data source for the trace
- The trace filter profile to be used for the trace
- Whether header data is included in the trace output
- The levels of tracing to be performed
- The component IDs of components to be traced
- Any optional user data to be used in the trace

You can specify multiple combinations of trace levels, components, and user data to be included in a single trace.

The Trace Configuration window maintains records stored in the Trace Configuration data source (TRCCFD). Because BASE24-es processes only read the TRCCFD during startup, you must warmboot them to reread (alter) the TRCCFD after making any changes on the window before the changes take effect.

A sample of the Trace Configuration window is shown below, followed by descriptions of its fields.



## Process Type

Trace commands are sent to a single process of a specified process type. The process type identifies the type of executable to be traced. You select the process type in the Process Type field. The following are all valid process types for this field. The descriptive name of the destination is shown in parentheses.

- CMCP (Channel Merchant Cutover process)
- CTXM (Context Manager process)
- EOPP (End-of-Period process)
- IS (Transaction Security Services process, which is an Integrated Server process)
- JQB (Journal Query Batch process)
- RFSH (Partial Refresh process)

- SAFM (SAF Manager process)
- XMLS (C++ User Interface process)

**Note:** Although all the above values are displayed for entry, only the IS and XMLS values apply to the TSS environment.

## Trace Output Assign

You specify the assign name of the Trace data source (TRCD) for the trace in the Trace Output Assign Name field. All assign names currently configured in any Trace Configuration data source (TRCCFD) records are displayed in the selectable list for this field. You can select one of these assign names or enter a new assign name in this field. In either case, the assign name you select or enter must match the assign name configured for the TRCD in the Metadata Configuration file (CONFCSV). You can configure multiple TRCD assigns in the CONFCSV. The following is an example assign definition for a TRACE assign in the CONFCSV.

```
"TRACE", "Trace", "1.0", "$vol.subvol.trcd", "Enscribe", "", ""
```

**Note:** Any time you change the CONFCSV, you must stop and restart the process that uses the changed data. To ensure that no transactions currently in progress are lost, you should stop the process in an orderly fashion. The change does not take effect until after the process is stopped and restarted.

## Process Trace is Active

The Process Trace is Active checkbox lets you enable and disable the trace. A checkmark indicates that the trace is currently enabled. You must warmboot the target processes to reread the TRCCFD before a change to this checkbox takes effect. When you enable a trace and warmboot the TRCCFD, the trace is started. When you disable a trace and warmboot the TRCCFD, the trace is stopped.

When you read a trace configuration record on the Trace Configuration window, a trace that is marked as enabled may or may not be running and a trace marked as disabled may or may not be stopped, depending on when the TRCCFD was last warmbooted and a process was started.

## Trace Filter Profile

A trace filter profile is an optional feature that enables you to filter the transactions to be included in the trace output. In this field, you can assign a trace filter profile to the trace from the selectable list of all configured profiles. You can also use the [hypertext link](#) to navigate directly to the Trace Filter Profile Configuration window if you need to configure a new trace filter profile or you need to view or edit an existing trace filter profile.

If a transaction matches any of the criteria configured in the trace filter profile, it is included in the trace output. If a transaction does not match any of the criteria configured in the trace filter profile, it is not included in the trace output.

## Include Header in Trace

The Include Header in Trace checkbox indicates whether headers are added to each trace record written to the TRCD. A checkmark indicates that headers are enabled.

When trace headers are enabled, the following header information is added to each trace record written to the TRCD.

```
DATE AND TIME: MMM DD, YYYYHH:MM:SS  
LEVEL: trace-level COMPONENT: component-id
```

where:

*trace-level* is one of the following:

|       |                      |
|-------|----------------------|
| DCSNP | = Decision point     |
| EM    | = External message   |
| ENVMT | = Environment        |
| HEAP  | = Heap               |
| JTRC  | = Journal trace      |
| OPRND | = Script operand     |
| SCRLN | = Script line number |
| SCRPT | = Script             |
| TDE   | = TDE                |

*component-ID* is an identifier of the component that generated the trace data. Some examples of component IDs are:

ENV = Environment  
HEAP = Heap Manager  
MD = Message Delivery  
TSEC = Transaction Security

## Trace Levels

You can configure one or more trace levels in a single trace. Each trace level traces a different aspect of transaction processing as described below.

**Note:** Because the Transaction Security Services environment uses only a portion of BASE24-es functionality, the following trace levels do not trace any data in the TSS environment: Decision Point, Journal Trace, Script, Script Line Number, and Script Operand.

## External Message

External message tracing enables you to trace all external request and response messages exchanged between a BASE24-es process and all external entities or just specified external entities. External entities include security devices, processes, stations, services (HP NonStop platform only), or other BASE24-es process types.

Each message included in the trace output is separated by a blank line and specifies the destination, source, and text of the message. The first nine columns of the trace output is binary data in hexadecimal format. The first column contains a two-byte word counter for the message. Every new message traced starts with the counter set to 000. Counter values are in hexadecimal format (e.g., 010 = 016) and each row in the trace output contains 8-byte words (16 bytes). The last column of the trace output contains the character format of the message. The Message Delivery component always traces in both binary and character format. Trace data for all other components is provided in character format only.

## Environment

Environment tracing lists the current value and default value for all the environment attributes in the destination process that have been registered. Attributes are written to the TRCD in the following format:

```
ATTRIBUTE NAME = name
                VALUE = value
                DEFAULT VALUE = default-value
```

## Heap

Heap tracing causes each Heap Supervisor to keep track of memory allocations and to write trace records when you enter a Status Monitor process control command. These trace records report the memory allocated and their contents to assist in memory leak detection.

**Note:** Heap tracing is only available in executables that were not compiled with the NDEBUG option due to performance.

## TDE

Transaction Data Element (TDE) tracing writes a record to the TRCD with the TDE's contents each time a TDE is added, removed, or modified for a transaction. These records are formatted with one of the following lines preceding the TDE's contents:

```
TDE Added. tde-name
TDE Removed. tde-name
TDE Changed. tde-name
```

## Trace Level, Component ID Filter, and User Data

The Trace Level column of the trace configuration table on the Trace Configuration window defines the type of information to be included in the trace by the component specified in the Component ID Filter column of the same table row. The Component ID Filter column specifies the components that will generate trace data for the trace level specified in the Trace Level column of the same table row. If you specify {none} in the Component ID Filter column, all appropriate components in the transaction path will generate the specified trace data. You can configure up to 100 rows for any allowed combination of trace level, component, and user data to be included in the trace.

You must configure at least one trace level row before enabling a trace.



Only certain combinations of trace level, component ID, and user data are supported, as illustrated in the following table.

| Trace Level      | Component ID Filter                                                                                                                                                                                                                                                                                                                                                                                                                                                | User Data                                                                                 |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Decision Point*  | {none}<br>Prefix<br>Route<br>Application Parameter Management Interface<br>BASE24 HISO 8583 Host Interface<br>BankNet ISO Interface<br>ISO 93 Host Interface<br>MDS Interface<br>VisaNet ISO Interface<br>ISO93 Interface<br>Contingency Interface<br>IMT Interface<br>ATM Device Handler Interface<br>Real Time Feed Interface<br>Star ISO Interface<br>Visa DPS Interface                                                                                        | Not allowed.                                                                              |
| External Message | {none}<br>Message Delivery<br>Application Parameter Management Interface<br>BASE24 HISO 8583 Host Interface<br>BankNet ISO Interface<br>ISO 93 Host Interface<br>MDS Interface<br>VisaNet ISO Interface<br>ISO93 Interface<br>Contingency Interface<br>ATM Device Handler Interface<br>Real Time Feed Interface<br>Star ISO Interface<br>Visa DPS Interface<br><br><b>Note:</b> The only values that apply to the TSS environment are {none} and Message Delivery. | Optional for the Message Delivery component.<br><br>Not allowed for all other components. |
| Environment      | {none}<br>Environment                                                                                                                                                                                                                                                                                                                                                                                                                                              | Not allowed.                                                                              |

| Trace Level                        | Component ID Filter       | User Data    |
|------------------------------------|---------------------------|--------------|
| Heap                               | { none }<br>Heap          | Not allowed. |
| Journal Trace <sup>*</sup>         | { none }<br>Journal Trace | Required.    |
| TDE                                | { none }<br>TDE           | Not allowed. |
| Script <sup>*</sup>                | { none }<br>Script        | Not allowed. |
| Script Line<br>Number <sup>*</sup> | { none }<br>Script        | Not allowed. |
| Script Operand <sup>*</sup>        | { none }<br>Script        | Not allowed. |

<sup>\*</sup> Because the TSS environment does not use the Prefix or Router components, the Journal data source (JRNLD), or scripts, these trace levels do not trace any data.

## External Message User Data

For external message tracing, you can optionally specify entities external to the process type for which external messages are traced. The external entity can be the assign name of a security device, the symbolic name of a process, station, or service (HP NonStop platform only), or another process type.

- If you specify a security device, the assign name must match the assign name configured for a security device in the message delivery subsystem.
- If you specify a process, station, or service, the symbolic name must match the symbolic name configured to the middleware for your system.

If you do not specify an external entity to be traced, external messages for all external entities (e.g., all security devices, processes, and stations) are traced.

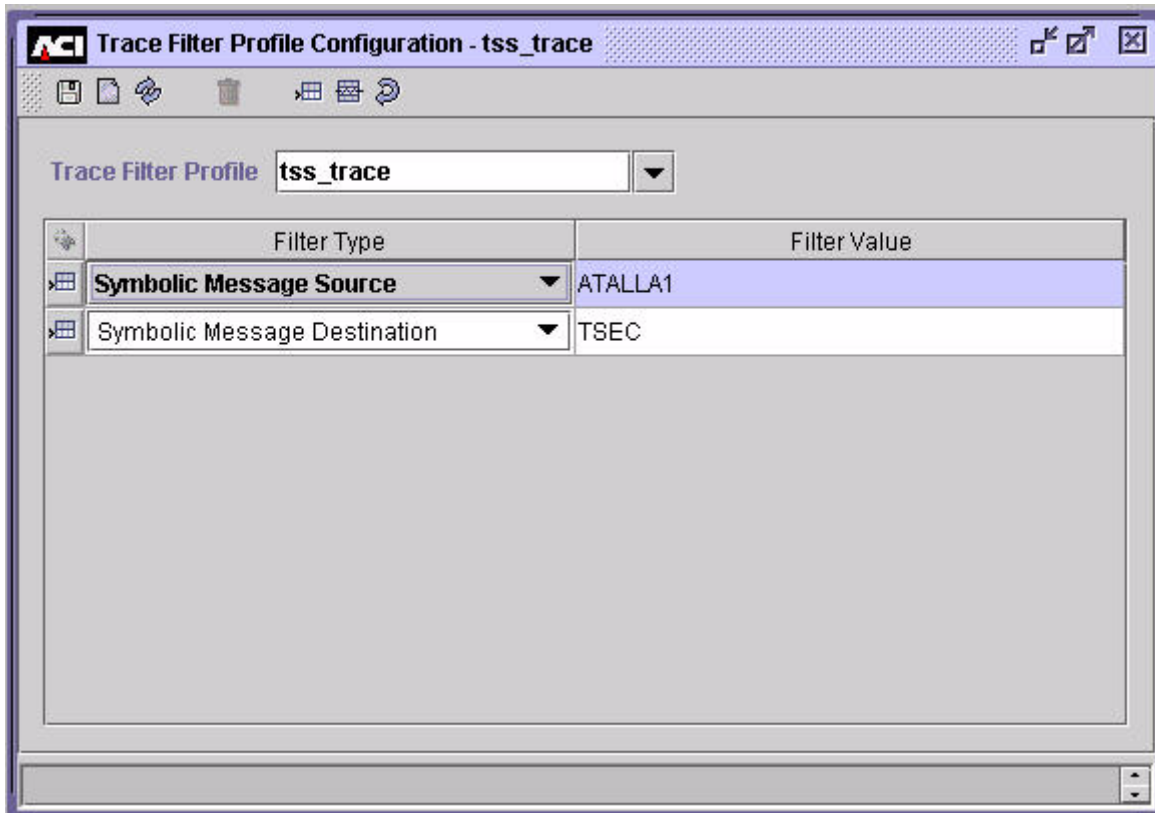
# Configuring Trace Filter Profiles

Use the Trace Filter Profile Configuration window on the ACI desktop user interface (accessed from **System Operations > Trace > Trace Filter Profile**), to configure a trace filter profile. The information entered on this window is stored in the Trace Filter Configuration data source (TRCFLD). A trace filter profile is an optional feature that enables you to filter the transactions to be included in the trace output based on one or more of the following filter types:

- Action code
- Acquiring endpoint name
- Acquiring institution
- Issuing endpoint name
- Issuing institution
- Primary account number
- Partial primary account number
- Processing code
- Symbolic message destination\*
- Symbolic message source\*
- Terminal ID

\* Because the TSS environment only uses a subset of the transaction data elements (TDEs) in BASE24-es, only these filter types apply for tracing in the Transaction Security Services process.

You can assign one or more values to one or more of these filters in the table displayed on this window. An example of the Trace Filter Profile Configuration window is shown below, followed by a discussion of its fields



## Trace Filter Profile

You enter or select the name of the trace filter profile in the Trace Filter Profile field. You can select the name from the selectable list if the profile is already configured.

## Filter Type

The Filter Type column of the table specifies the type of information on which transaction data is to be filtered for the trace. You can select one of the following values from the selectable list for this column in the table:

|                                           |                                                                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Action Code                               | = A one- to three-digit code indicating the outcome of authorization.                                                                    |
| Acquiring End Point Name                  | = The name of the acquirer endpoint.                                                                                                     |
| Acquiring Institution                     | = The ID of the acquirer institution.                                                                                                    |
| Issuing End Point Name                    | = The name of the issuer endpoint.                                                                                                       |
| Issuing Institution                       | = The ID of the issuer institution.                                                                                                      |
| Symbolic Message Destination <sup>*</sup> | = The component ID (e.g., TSEC) or symbolic name (e.g., ATALLA1, PIA^AUTH1) of the TSS message destination.                              |
| Symbolic Message Source <sup>*</sup>      | = The component ID (e.g., TSEC) or symbolic name (e.g., ATALLA1, PIA^AUTH1) of the TSS message source.                                   |
| Primary Account Number                    | = The primary account number (PAN) used in the transaction.                                                                              |
| Partial Primary Account Number            | = The first <i>n</i> digits of the PAN used in the transaction, where <i>n</i> is a variable number of configured digits for the filter. |
| Processing Code                           | = The processing code for the transaction.                                                                                               |
| Terminal ID                               | = The terminal ID associated with the transaction.                                                                                       |

<sup>\*</sup> Because the TSS environment only uses a subset of the transaction data elements (TDEs) in BASE24-es, only these filter types apply for tracing in the Transaction Security Services process.

If you want to filter on multiple values for one of these filter types, you must enter a separate table row for each value assigned to the filter type. If a transaction matches any row in the filter table, it is included in the trace output.

## Filter Value

Enter values associated with the filter type on the same row of the table in the Filter Value column. A transaction must match the value in one of the rows to be included in the trace output.

## Trace Operations

Whenever you make a change to the TRCCFD from the Trace Configuration window or the TRCFLD from the Trace Filter Configuration window, the change does not take effect in processing until after you have saved the change and either warmbooted (altered) Transaction Security Services processes to reread the TRCCFD and TRCFLD or stopped and restarted the Transaction Security Services processes. Transaction Security Services processes read the TRCCFD and associated TRCFLD records during initialization processing.

You alter a process to reread the TRCCFD (and associated TRCFLD records) by sending an alter data source process control command to the process. The syntax for this command is as follows:

---

***Command Syntax***

---

|               |              |
|---------------|--------------|
| Destination:  | PCTL         |
| Command:      | Alter        |
| Component:    | Data Source  |
| Command Text: | TRACE_CONFIG |

For convenience, you can right-click anywhere on the Trace Configuration window or Trace Filter Configuration window and click the “Alter Trace Configuration Data Source” action. When you do so, the Process Control window is displayed with all the fields preset to the values you need to warmboot the appropriate processes to reread TRCCFD (and associated TRCFLD records). At that point, you need only click Submit (▶) to warmboot the trace configuration.

# Running Traces

Using the ACI desktop, you can start, modify, and stop traces.

## Starting a Trace

After you have configured a trace, you can execute it by placing a check mark in the Process Trace is Active checkbox on the Trace Configuration window, clicking Save (📁), and then warmbooting the appropriate processes to reread the Trace Configuration data source (TRCCFD). For convenience, you can right-click anywhere on the Trace Configuration window and click the “Alter Trace Configuration Data Source” action. When you do so, the Process Control window is displayed with all the fields preset to the values you need to warmboot the appropriate processes to reread the TRCCFD. At that point, you need only click Submit (▶) to start the trace or wait for the process that is to be traced to be started.

## Modifying a Running Trace

While a trace is active, you can modify the trace levels, component ID filters, user data for the trace by editing the trace level, component ID filter, and user data table on the Trace Configuration window, or the trace filter profile in the Trace Filter Profile field by editing the trace filter table on the Trace Filter Configuration window. After you have made the desired changes, click Save (📁) and warmboot the appropriate processes to reread the Trace Configuration data source (TRCCFD). For convenience, you can right-click anywhere on the Trace Configuration window or the Trace Filter Configuration window and click the “Alter Trace Configuration Data Source” action. When you do so, the Process Control window is displayed with all the fields preset to the values you need to warmboot the appropriate processes to reread TRCCFD. At that point, you need only click Submit (▶) for the modified trace levels, component ID filters, user data, and trace filter profile to start being used for the trace or wait for the process that is to be traced to be started.

## Stopping a Trace

When you are done running the trace, remove the check mark from the Process Trace is Active checkbox on the Trace Configuration window, click Save (📁), and then warmboot the appropriate processes to reread the Trace Configuration data source (TRCCFD). For convenience, you can right-click anywhere on the Trace

Configuration window and click the “Alter Trace Configuration Data Source” action. When you do so, the Process Control window is displayed with all the fields preset to the values you need to warmboot the appropriate processes to reread the TRCCFD. At that point, you need only click Submit (▶) to stop the trace or the trace will be disabled the next time the process that is to be traced is started.



# Examining Trace Output

All trace output is written to the Trace data source (TRCD). The TRCD is an entry-sequenced file with the following primary keys:

- Date and time
- Symbolic process ID
- Symbolic process name
- Symbolic thread ID
- Record Number

The date and time of the transaction is provided in YYYYMMDDhhmmsslllccc format.

The symbolic process ID is the symbolic process ID as defined by SIS, which may represent a PPD name, task, transaction type, or whatever uniquely identifies a specific process on a given platform. The symbolic process name is the symbolic name of the process.

The symbolic thread ID identifies the particular thread of execution traced in the process. For the HP NonStop platform, this field is always set to a value of 1.

The record number uniquely identifies multiple rows in the TRCD when the trace data generated by the process exceeds the trace buffer limit of 3988 bytes and is written to multiple TRCD rows.

You can view trace data written to the TRCD using an ACI-provided trace viewer utility or the DAL Conversational Interface (DALCI). Instructions for using either method are provided below as well as a sample of external message trace output.

## Viewing Trace Data Using the Trace Viewer Utility

The ACI-supplied Trace Viewer utility program (TRCVLO executable) displays TRCD data in a user-friendly format and is the preferred method for viewing trace data. You can execute this utility using the appropriate run command for your platform. Because the Trace Viewer utility uses some of the same configuration design as the DAL Conversational Interface (DALCI), you must include the -mdbv, -ai, and -strt DALCI startup options when running this utility. The utility displays all the data in the data source defined for the TRACE assign in the CONFCSV. Note that this is a hardcoded value in the utility so you must enter a

value of TRACE in the Trace Output Assign Name field on the Trace Configuration window if you want to view the trace output with this utility. An example of the run command for the Trace Viewer utility is shown below.

```
run $vol.subvol.trcvlo/name, &  
lib $vol.xnlib.xnlibn/ &  
-MDBV=CSV &  
-AI=TSS &  
-STRT=$vol.subvol (location of the CONFCSV file)
```

The output from the Trace Viewer utility is written to the file location specified in the MTRCRC assign in the CONFCSV.

An example of the Trace Viewer utility output for an external message trace is provided under the [“Example External Message Trace Output”](#) topic later in this section. For detailed information on DALCI startup options, refer to the ***DAL Conversational Interface (DALCI) User Guide***.

## Viewing Trace Data Using DALCI

Another method for viewing TRCD data is to use the DALCI SELECT command with or without the INSERTFORM option. Use the INSERTFORM option if you want to first select the trace data to be viewed from the TRCD and then write the selected data to a stream data source. You can then use the Trace Viewer utility to view the data. In this case, you would insert the selected data using DALCI into the stream data source and enter the assign for the stream data source in the Trace Output Assign Name field on the Trace Configuration window. The TRACE assign name for the stream data source must be configured in the Metadata Configuration file (CONFCSV). You could also use the DALCI SELECT command without the INSERTFORM option to view the trace data directly from the TRCD, however, the data is not easily viewed in this format.

An example of the SELECT command with the INSERTFORM option is shown below.

```
select * from trcd insertform;
```

This command selects all data from the TRCD. You can also qualify the data selected using the WHERE constraint. An example of the selected output in INSERT command format for an external message trace is shown below. For detailed information on DALCI commands and options, refer to the ***DAL Conversational Interface (DALCI) User Guide***.

**External Message Trace Example:**

```

INSERT into TRACE
yyyyymmddhhmmsslllccc = "20050318212103287387"
sym_pro_id = "\\ARGUS.$W4A2"
sym_thrd_id = "1"
rec_num_bin = 0
trc_buf = "20050318212103287387.000: 4465 7374 696E 6174 696F 6E" \
"20 3D20 2020 Destination = .008: 494E 5446 4939 3320 2020 " \
" 2020 2020 2020 INTFI93 .010: 536F 7572 6365 203D 2" \
"020 2020 2020 2020 Source = .018: 5034 415E 5349 4D5E" \
" 415E 4939 3320 2020 P4A^SIM^A^I93 .020: 5465 7874 203D " \
"2020 2020 2020 2020 2020 Text = .028: 3132 3030 373" \
"0 3330 3035 3431 3030 3230 1200703005410020.030: 3830 3030 " \
" 3138 3832 3131 3131 3131 3131 8000188211111111.038: 3131 31" \
"32 3334 3536 3030 3230 3030 3030 1112345600200000.040: 3030 " \
" 3030 3030 3130 3030 3030 3030 3630 00000010000000060.048: 3" \
"035 3033 3138 3135 3231 3033 3030 3131 0503181521030011.050" \
": 3031 3030 3031 3230 3230 3035 3436 3231 0100012020054621" \
".058: 3031 3030 3035 3536 3637 3634 344D 7920 010005566764" \
"4My .060: 4261 6B65 7279 5C31 3233 3420 4F61 6B20 Bakery\\" \
"1234 Oak .068: 5374 5C4F 6D61 6861 5C36 3831 3036 3535 St\\" \
"\\Omaha\\6810655.070: 3535 3533 3333 5553 4138 3430 " \
" 555333USA840 ..Parsing External ISO Message for component = IF93_DEF" \
"AULT ...MTI = 1200..Primary Bit Map = 7030054100208000..18 Field 002 = " \
"8211111111111123456..Field 003 = 002000..Field 004 = 000000001000..Field 01" \
"1 = 000060..Field 012 = 050318152103..Field 022 = 001101000120..Field 024 " \
"= 200..Field 026 = 5462..10 Field 032 = 1000556676..44 Field 043 = My Bake" \
"ry\\1234 Oak St\\Omaha\\681065555333USA..Field 049 = 840..Building Extern" \
"al ISO Message for component = IF93_DEFAULT ...MTI = 1210..Field 002 = " \
"18 8211111111111123456..Field 003 = 002000..Field 004 = 000000001000..Field" \
" 011 = 000060..Field 012 = 050318152103..Field 032 = 10 1000556676..Field " \
"039 = 000..Field 049 = 840..Primary Bit Map = 7030000102008000..000: 4465" \
" 7374 696E 6174 696F 6E20 3D20 2020 Destination = .008: " \
"5034 415E 5349 4D5E 415E 4939 3320 2020 P4A^SIM^A^I93 .01" \
"0: 536F 7572 6365 203D 2020 2020 2020 2020 Source = " \
" .018: 494E 5446 4939 3320 2020 2020 2020 2020 INTFI93 " \
" .020: 5465 7874 203D 2020 2020 2020 2020 2020 Text = " \
" .028: 3132 3130 3730 3330 3030 3031 3032 3030 121" \
"0703000010200.030: 3830 3030 3138 3832 3131 3131 3131 3131 " \
" 8000188211111111.038: 3131 3132 3334 3536 3030 3230 3030 3" \
"030 1112345600200000.040: 3030 3030 3030 3130 3030 3030 3030" \
" 3630 00000010000000060.048: 3035 3033 3138 3135 3231 3033 " \
"3130 3130 0503181521031010.050: 3030 3535 3636 3736 3030 303" \
"8 3430 00556676000840 ."
;

```

## Example External Message Trace Output

A sample of the trace output for tracing external messages for a specified simulator process and a security device is shown below. For this trace, the trace level is set to EM (external message), the component ID is MD for the Message Delivery component, and the *user-data* specifies two external entities (p3c^sim06 and atalla1). The external messages sent to and received from a security device are highlighted in [blue text](#).

Each message included in the trace output is separated by a blank line and specifies the destination, source, and text of the message. The first nine columns of the trace output is binary data in hexadecimal format. The first column contains a two-byte word counter for the message. Every new message traced starts with the counter set to 000. Counter values are in hexadecimal format (e.g., 010 = 016) and each row in the trace output contains 8-byte words (16 bytes). The last column of the trace output contains the character format of the message.

This sample trace includes seven messages. The first message is a key generate request for a channel device sent from a simulator process to the Transaction Security component in the Transaction Security Services process. The second message is a key generate command for a PIN encryption key sent from the Transaction Security component in the Transaction Security Services process to a security device. The third message is the response returned from the security device to the Transaction Security Services process. The fourth message is change transaction key command sent from the Transaction Security component in the Transaction Security Services process to a security device. This command encrypts the PIN encryption key generated in the previous message for downloading to a channel device. The fifth message is the response returned from the security device to the Transaction Security Services process. The sixth message is the response message sent from the Transaction Security component in the Transaction Security Services process back to the simulator process. The last message is a command sent from a network control facility to the Transaction Security Services process to stop the trace.

---

|      |      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------|------------------|
| 000: | 4465 | 7374 | 696E | 6174 | 696F | 6E20 | 3D20 | 2020 | Destination =    |
| 008: | 5453 | 4543 | 2020 | 2020 | 2020 | 2020 | 2020 | 2020 | TSEC             |
| 010: | 536F | 7572 | 6365 | 203D | 2020 | 2020 | 2020 | 2020 | Source =         |
| 018: | 5033 | 435E | 5349 | 4D30 | 3620 | 2020 | 2020 | 2020 | P3C^SIM06        |
| 020: | 5465 | 7874 | 203D | 2020 | 2020 | 2020 | 2020 | 2020 | Text =           |
| 028: | 3034 | 3031 | 2043 | 4841 | 4E4E | 454C | 5F52 | 4143 | 0401 CHANNEL_RAC |
| 030: | 414C | 2020 | 2032 | 3132 | 3334 | 3536 | 3738 | 3930 | AL 21234567890   |
| 038: | 3132 | 3334 | 3536 | 3020 |      |      |      |      | 1234560          |
|      |      |      |      |      |      |      |      |      |                  |
| 000: | 4465 | 7374 | 696E | 6174 | 696F | 6E20 | 3D20 | 2020 | Destination =    |

---

```

008: 4154 414C 4C41 3120 2020 2020 2020 2020 ATALLA1
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 5453 4543 2020 2020 2020 2020 2020 2020 TSEC
020: 5465 7874 203D 2020 2020 2020 2020 2020 Text =
028: 3C31 3023 3123 3030 3030 3030 3030 3030 <10#1#0000000000
030: 3030 3030 3030 2353 233E 000000#S#>

000: 4465 7374 696E 6174 696F 6E20 3D20 2020 Destination =
008: 5453 4543 2020 2020 2020 2020 2020 2020 TSEC
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 4154 414C 4C41 3120 2020 2020 2020 2020 ATALLA1
020: 5465 7874 203D 2020 2020 2020 2020 2020 Text =
028: 3C32 3023 3738 4246 3633 4245 3934 3236 <20#78BF63BE9426
030: 4242 3836 2330 4634 4542 4437 3445 3346 BB86#0F4EBD74E3F
038: 3235 4441 3123 4442 4634 233E 25DA1#DBF4#>

000: 4465 7374 696E 6174 696F 6E20 3D20 2020 Destination =
008: 4154 414C 4C41 3120 2020 2020 2020 2020 ATALLA1
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 5453 4543 2020 2020 2020 2020 2020 2020 TSEC
020: 5465 7874 203D 2020 2020 2020 2020 2020 Text =
028: 3C31 3523 3223 3738 4246 3633 4245 3934 <15#2#78BF63BE94
030: 3236 4242 3836 2335 3342 4531 3342 4341 26BB86#53BE13BCA
038: 3736 3044 3936 3223 3E 760D962#>

000: 4465 7374 696E 6174 696F 6E20 3D20 2020 Destination =
008: 5453 4543 2020 2020 2020 2020 2020 2020 TSEC
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 4154 414C 4C41 3120 2020 2020 2020 2020 ATALLA1
020: 5465 7874 203D 2020 2020 2020 2020 2020 Text =
028: 3C32 3523 3223 3139 3644 3532 3336 4444 <25#2#196D5236DD
030: 3245 3539 3631 2332 3445 3632 3638 3332 2E5961#24E626832
038: 3636 3341 3537 3723 3E 663A577#>

000: 4465 7374 696E 6174 696F 6E20 3D20 2020 Destination =
008: 5033 435E 5349 4D30 3620 2020 2020 2020 P3C^SIM06
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 5453 4543 2020 2020 2020 2020 2020 2020 TSEC
020: 5465 7874 203D 2020 2020 2020 2020 2020 Text =
028: 3034 3030 3031 2044 4246 3420 2031 2020 040001 DBF4 1
030: 2020 2020 2020 3139 3644 3532 3336 4444 196D5236DD
038: 3245 3539 3631 2020 2020 2020 2020 2020 2E5961
040: 2020 2020 2020 2020 2020 2020 2020 2020
048: 2020 2020 2020 2020 2020 2020 2020 2020
050: 2020 2020 2020 4348 414E 4E45 4C5F 5241 CHANNEL_RA
058: 4341 4C20 2020 3234 4536 3236 3833 3236 CAL 24E6268326
060: 3633 4135 3737 63A577

000: 4465 7374 696E 6174 696F 6E20 3D20 2020 Destination =
008: 5033 435E 4941 5330 3120 2020 2020 2020 P3C^IAS01
010: 536F 7572 6365 203D 2020 2020 2020 2020 Source =
018: 234E 4554 434E 544C 2020 2020 2020 2020 #NETCNTL

```

## Process Tracing

---

|      |      |      |      |      |      |      |      |      |                  |
|------|------|------|------|------|------|------|------|------|------------------|
| 020: | 5465 | 7874 | 203D | 2020 | 2020 | 2020 | 2020 | 2020 | Text =           |
| 028: | 6465 | 6C69 | 7665 | 7220 | 7472 | 6320 | 7374 | 6F70 | deliver trc stop |

# A: Database Conversion

---

To allow existing BASE24 customers to use the Transaction Security Services database files, the existing BASE24 files are converted and the hardware encrypted key data is moved to the new database. The hardware encrypted key fields in existing BASE24 database files are either set to the key value for the record (e.g., the terminal ID for hardware key fields in the BASE24-atm Terminal Data files and BASE24-pos Terminal Data files), an index value created by the conversion program, or are set to binary zeros (e.g., for the Key Authorization File).

TSS database conversion programs also use a runtime parameter called LN-IND to create a key locator value when converting existing BASE24 files. This parameter contains a two-character hexadecimal identifier for a logical network, which allows the CSECD and ISECD to be physically partitioned by logical network.

When a database conversion program is run on a BASE24-atm Terminal Data file or BASE24-pos Terminal Data file, the two-character LN-IND value is added to the beginning of the 1–16 character terminal ID. If the terminal ID for the terminal record being converted has more than 14 characters, the last one or two characters are truncated when the key locator value is created.

When converting Key File (KEYF) or Key 6 File (KEY6) records to the TSS database, the two-character LN-IND runtime param value is added to the beginning of a 14-character key locator index value generated by the conversion program.

If you need to partition the TSS database by logical network, the value of the LN-IND runtime parameter must be set to the same value specified in the LN-IND LCONF param. For more information on the LN-IND LCONF param, refer to [section 3](#).

If you do not need to partition the TSS database by logical network, the LN-IND runtime parameter must be set to a value of 00.

When set to the key value for the record or an index value, this value is referred to as the *key locator* value. The key locator is then used as the primary key value in the new database files, which contain the actual hardware key field values. For fields set to binary zeros, the primary key for the existing BASE24 file record is

copied to the new database and is used as the key locator value. The following table shows the relationship of existing BASE24 database files to the new database files. The first column lists the existing BASE24 files and the second column lists the new files accessed by the Transaction Security Services process.

| <b>BASE24 File</b>                                         | <b>TSS File</b>                     |
|------------------------------------------------------------|-------------------------------------|
| BASE24-atm Terminal Data Dynamic File—general data (ATDD1) | Channel Security File (CSECD)       |
| Terminal Data File (TDF)                                   |                                     |
| BASE24-pos Terminal Data Dynamic File—general data (PTDD1) |                                     |
| POS Terminal Data File (PTDF)                              |                                     |
| Derivation Key File (KEYD)                                 | Derivation Key File (DRVKD)         |
| Key Authorization File (KEYA)                              | Card Verification File (CRDVD)      |
|                                                            | Diebold Number Table File (DNTD)    |
|                                                            | IBM DES Verification File (IDESD)   |
|                                                            | Identikey Verification File (IDNTD) |
|                                                            | Visa PVV Verification File (VPVVD)  |
| ICC Key File (KEYI)                                        | EMV Authorization File (EMVSD)      |
| Key File (KEYF)                                            | Interface Security File (ISECD)     |
| Key 6 File (KEY6)                                          |                                     |

ACI provides a complete set of utility programs to convert the hardware key fields in existing BASE24 records and add corresponding records to the new database files. Records in the KEYA are converted to one of four new database files, depending on the record type. If the KEY6 is converted, a new record is created in the KEYF for each record converted from the KEY6. After the KEY6 is converted, it is no longer used in processing.

ACI also provides an additional utility program to convert the keys in the new database to the new Atalla key block format, if necessary.



The conversion programs provided by ACI are listed on the following page. For more information on using these conversion programs, refer to the ***BASE24 Release 6.0 Implementation Guide***.

## Database Conversion Programs

ACI provides a conversion program for each BASE24 file that contains hardware encrypted key information. These programs are run using obey files. The following table lists the obey files and programs provided by ACI. These files are located on the AT60UC05, BA60UC05, or PS60UCO5 subvolumes.

| BASE24 File                                                | Obey File | Conversion Program |
|------------------------------------------------------------|-----------|--------------------|
| BASE24-atm Terminal Data Dynamic File—general data (ATDD1) | TSSATDR   | TSSATD             |
| Terminal Data File (TDF)                                   | TSSTDFR   | TSSTDF             |
| BASE24-pos Terminal Data Dynamic File—general data (PTDD1) | TSSPTDR   | TSSPTD             |
| Derivation Key File (KEYD)                                 | TSSKEYDR  | TSSKEYD            |
| POS Terminal Data File (PTDF)                              | TSSPTDFR  | TSSPTDF            |
| Key Authorization File (KEYA)                              | TSSKEYAR  | TSSKEYA            |
| Key File (KEYF)                                            | TSSKEYFR  | TSSKEYF            |
| Key 6 File (KEY6)                                          | TSSKEY6R  | TSSKEY6            |
| ICC Key File (KEYI)                                        | TSSKEYIR  | TSSKEYI            |

For detailed procedures on running these conversion programs, refer to the ***BASE24 Release 6.0 Implementation Guide*** and the ***BASE24 Classic 6.0 Installation Guide***.

**Note:** The TSSATDR obey file creates an entry-sequenced file named TRMF that is simply an output listing of ATMs that are configured for EMV support.

After the existing BASE24 files are converted, all security data maintenance is performed through the ACI desktop user interface, except for certain situations described in the note below. User access to the following BASE24 files

maintenance screens must be turned off in the Security File (SEC): KEYA, KEYI, KEYF, and KEY6. Changes to non-hardware encrypted key information in the KEYF are made using the BASE24 KEYF Configuration window accessed from the ACI desktop user interface.

**Note:** If an interchange encrypts PINs in hardware and the BASE24 system processes PINs internally in the clear, or if an interchange uses clear PINs while the BASE24 system uses PINs encrypted in software, an intermediate key must be configured in the KEYF using the Pathway-based Key File (KEYF) screens. The ACI desktop user interface does not grant access to intermediate keys. In this case, user access to the KEYF screens must be retained in the SEC.

*ACI Worldwide Inc.*

# **B: Automated Key Distribution System**

---

This appendix describes the security-related configuration and implementation requirements for the BASE24-atm Automated Key Distribution System (AKDS) add-on product.

It describes the public key cryptography keys and concepts used for the BASE24-atm Automated Key Distribution System add-on product, how it is configured in the BASE24 system, the out-of-band and online procedures for implementing this add-on product, the windows and files used for AKDS processing, and the AKDS Capacity Report.

# Public Key Cryptography

Public key cryptography, often referred to as PKI, differs from DES and triple DES in that key pairs instead of shared keys are used between the authenticating entities. A unique key pair, consisting of a public key and a private key, is created for each authenticating entity. The two parties then exchange their public keys in a secure manner. The sender encrypts the data using the receiver's public key. The receiver decrypts the data using its private key. The sender and receiver need not share the same key. Thus, public key cryptography is referred to as an *asymmetrical* security scheme as opposed to the *symmetrical* security scheme of DES, where the same keys must be known by both the sender and receiver of a message. PKI provides the following advantages over symmetrical security schemes:

- The receiver of a secured message uses a public key for verifying a message rather than the private (i.e., secret) key that the sender uses to secure the message. This lowers the need for expensive hardware devices at the receiver's site to store shared private keys.
- Each sender requires only one public and private key pair for secure communication with  $n$  number of receivers. In symmetric security schemes, the sender would require  $n$  number of different keys to secure the communication with  $n$  number of receivers. This lowers the complexity (costs) of the security architecture (key management).
- The sender and the receiver do not need to share a private key before actually exchanging secure messages when using public key certificates. This increases the flexibility for unrelated parties to securely exchange information.

For the BASE24-atm Automated Key Distribution System add-on product, the authenticating entities are individual ATM terminals and the BASE24 system. The ATM's private key, also known as the secret key, is stored within a tamper-resistant module in the ATM terminal. The BASE24 private key is stored in the TSS database. The public keys are signed by a certificate authority's private key. The entities then authenticate each other and exchange their public keys, signed by the certificate authority's private key. In this case, the entities are each ATM in your network and the BASE24 system.

Signatures provide a method of preventing data from being changed during transmission and of preventing the sender from being impersonated. A secure hashing algorithm is applied to the data and then the result is encrypted. With Rivest, Shamir, and Adleman (RSA) encryption, the signature is generated using the private key. Only the holder of the private key can generate the signature but anybody can verify it using the associated public key.

After both authenticating entities have successfully exchanged public keys, the receiver can use its private key to securely decrypt data received. In this case, a new ATM master key can be generated by the BASE24 system and securely downloaded to an ATM.

Some differences exist in the way Diebold, NCR, and Wincor Nixdorf have implemented public key cryptography. For example, Diebold uses certificates and a third-party certificate authority while NCR does not. Devices using the shared Wincor Nixdorf signature specification can emulate NCR and Diebold devices and use the same signature-based PKI remote key loading procedures as NCR, but with some minor differences. All of these differences are noted where appropriate.

## **NCR Devices**

NCR creates the public and private keys for the encrypting PIN pad (EPP) in each ATM during manufacture. NCR also assigns a unique serial number to each EPP and creates a signature for the EPP's public key.

The NCR trusted center serves as a certificate authority that signs the public keys for both the EPP and the BASE24 system. The trusted center sends the root public key of the trusted center (through an offline method) to both ACI and its customers. The NCR trusted center refers to their root public and private keys as the Global NCR public verification key and the Global NCR private signature key, respectively.

The EPP's public key, private key, public key signature, serial number, serial number signature, and the NCR trusted center's public key are stored within the EPP.

## **Diebold Devices**

Diebold EPPs create two RSA key pairs when they are first powered up. One key pair is used for signature verification and generation. The second key pair is used for encipherment and decipherment. The public keys for each of these two key pairs are signed by a certificate authority. Diebold has chosen Digital Signature Trust Co. (DST) as their certificate authority.

For each public key created, the certificate authority creates two certificates—a primary certificate and a secondary certificate. These certificates contain the EPP's public keys and signatures. Primary and secondary certificates are features of Diebold's disaster recovery procedure; however, BASE24 uses the primary certificate only. The EPP's public key certificates, private signature generation

key, private decipherment key, and two certificates (primary and secondary) containing the certificate authority's public signature verification key and signature, are stored in the EPP.

## Devices Supporting Wincor Nixdorf's Shared Signature Specification

Any device that is compliant with Wincor Nixdorf's shared signature specification, ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL***, and is equipped with an encrypting PIN pad (EPP), can support the BASE24-atm Automated Key Distribution System add-on product. The device vendor does not have to be Wincor Nixdorf as long as the vendor's ATMs can run the necessary firmware.

**Note:** For simplicity, Wincor Nixdorf is assumed to be the device vendor and devices compliant with the ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL*** specification are referred to as *shared Wincor Nixdorf signature devices* throughout this appendix.

Wincor Nixdorf creates the public and private keys for the encrypting PIN pad (EPP) in each ATM during manufacture. Wincor Nixdorf also assigns a unique serial number to each EPP and creates a signature for the EPP's public key.

The Wincor Nixdorf Customer Key Center – International (CKCI) serves as a certificate authority that signs the public keys for both the EPP and the BASE24 system. CKCI sends a customer-specific Wincor Nixdorf public signature key (through an offline method) to its customers. This key is referred to as the root Wincor Nixdorf public key throughout this appendix.

The EPP's public key, private key, public key signature, serial number, serial number signature, and the root Wincor Nixdorf public key are stored within the EPP.

Wincor Nixdorf devices running in NCR NDC+ or Diebold 912 emulation mode implement signature-based public key cryptography remote key loading in the same manner as NCR, but with the following differences.

- The BASE24 system public key is not self-signed when sent to the certificate authority.
- The Wincor Nixdorf Customer Key Center – International (CKCI) serves as the certificate authority that signs the public keys for both the EPP and the BASE24 system.



- CKCI creates customer-specific public and private signature keys.
- A device type of “Wincor Signature” is used with the KEYGEN command, and on the Certificate Management, Host Public Key Perusal, Channel Public Key Perusal, EPP Serial Number Configuration, and Disable KEYGEN windows for Wincor Nixdorf devices.
- You cannot migrate from single-length keys to double-length keys using the BASE24-atm Automated Key Distribution System due to limitations in Wincor Nixdorf firmware.

## BASE24 System

The BASE24 system serves as the host in public key exchanges and authentication with ATM terminals. Using the KEYGEN command entered from the Process Control window in the ACI desktop and a hardware security module, a unique RSA key pair is generated for the host. The public key is sent to a certificate authority to be verified and signed.

For NCR devices, the BASE24 system’s public key, private key, and public key signature, as well as the NCR trusted center’s public key are stored in the TSS database.

For Diebold devices, the BASE24 system’s primary and secondary public signature verification key certificates, private signature generation key, the root Diebold certificate authority’s public verification key, and the primary and secondary Diebold certificate authority’s public keys for signature verification and generation are stored in the TSS database.

For shared Wincor Nixdorf signature devices, the BASE24 system’s public key, private key, and public key signature, as well as the Wincor Nixdorf Customer Key Center – International (CKCI)’s public key are stored in the TSS database.

## Terminology

A number of terms are unique to public key cryptography. These terms are described below.

**A-Key** — *See* master key.

**Abstract Syntax Notation One (ASN.1)** — A formal language used to abstractly describe messages to be exchanged between distributed computer systems. Previously, ASN.1 was used to write application, national and international standards. However more recently with the advent of ASN.1 software tools, ASN.1 has been used to generate programming language code that forms the core of a wide variety of messaging systems applications. ASN.1 encoding rules are sets of rules used to transform data specified in the ASN.1 language into a standard format that can be decoded on any system that has a decoder based on the same set of rules. Diebold uses ASN.1. NCR and Wincor Nixdorf use ASN.1 only when transmitting messages to the certificate authority (i.e., the NCR trusted center or the Wincor Nixdorf Customer Key Center – International).

**Base64 Encoding** — A method of encoding used by Diebold and shared Wincor Nixdorf signature devices to convert a sequence of octets (bytes) into printable ASCII characters. Base64 encoding allows RSA cryptographic fields to be transmitted in messages without exceeding Diebold's 1920 byte limit. Other encoding methods double or triple the size of the original data, while the Base64 encoding method results in data that is only about 33 percent larger than the original un-encoded data.

Diebold and shared Wincor Nixdorf signature devices use Abstract Syntax Notation One (ASN.1) when transmitting cryptographic fields. This is a standard syntax for defining and encoding data for an open systems environment. The ASN.1 values and types are encoded using the Basic Encoding Rules (BER) and Distinguished Encoding Rules (DER).

ASN.1 produces binary BER and DER data that cannot be directly transmitted over 7-bit ASCII networks. The solution is to convert the binary data into 7-bit ASCII before transmission using Base64 encoding.

Diebold's implementation of the base64 encoding algorithm follows the standard set forth in RFC 1521, section 5.2 - Base64 Content-Transfer-Encoding (N. Borenstein, 1993).

**Base94 Encoding** — A method of encoding data used by NCR to convert binary data into the 94 displayable ASCII characters, ranging from hex 20 (space) to hex 7E (tilde). Using these characters, a base94 algorithm can produce a 5 to 4 ratio of encoded bytes to binary bytes. Given the block size (256 bytes for RSA-encrypted data) and message length constraints, NCR has determined that other encoding methods are unsuitable for transmitting RSA blocks.

**Basic Encoding Rules (BER)** — This set of encoding rules was created in the early 1980s and used in a wide range of applications, such as Simple Network Management Protocol (SNMP) for management of the Internet, Message Handling Services (MHS) for exchange of electronic mail and TSAPI for control of telephone/computer interactions. See also Abstract Syntax Notation One.

**Certificate** — An electronic document binding some pieces of information together, such as an ATM's or host's identity and public key. Diebold uses certificates; NCR and shared Wincor Nixdorf signature devices do not use certificates.

**Certificate Authority (CA)** — A person or organization that validates and signs public keys. The CA could be an ATM manufacturer, a financial institution, or a third party. Diebold uses a third party certificate authority—Digital Signature Trust Co. (DST). The NCR trusted center serves as the certificate authority for NCR, while the Customer Key Center – International serves as the certificate authority for Wincor Nixdorf.

**Certificate Signing Request (CSR)** — A request for a certificate to be signed by a certificate authority. This request is always in PKCS #10 Certificate Request Message format.

**Cipher** — An encryption-decryption algorithm.

**Ciphertext** — Encrypted data.

**Cryptosystem** — An encryption-decryption algorithm (cipher), together with all possible plaintexts, ciphertexts and keys.

**Distinguished Encoding Rules (DER)** — This set of encoding rules is a specialized form of BER that is used in security-conscious applications. These applications, such as electronic commerce, typically involve cryptography, and require that there be one and only one way to encode and decode a message. *See also* Abstract Syntax Notation One.

**Encrypting PIN Pad (EPP)** — A tamper-proof ATM component comprised of a PIN pad, a security module to perform cryptographic processing, data encryption algorithms (e.g., DES, Triple DES, MAC, RSA), and the cryptographic keys. All cryptographic keys are stored in the EPP. All cryptographic functions are performed by the EPP.

**Master Key** — A key used to encrypt and decrypt the working keys that are downloaded from the host to the ATM. Also known as the A-Key, Key Exchange Key (KEK) and Terminal Master Key (TMK).

**Message Digest** — The representation of text in the form of a single string of digits, created using a formula called a one-way hash function. Encrypting a message digest with a private key creates a digital signature, which is an electronic means of authentication. *See also* Secure Hash Algorithm (SHA-1).

**Nonce** — A randomly generated value used to defeat *playback* attacks in communication protocols. One party randomly generates a nonce and sends it to the other party. The receiver encrypts it using the agreed upon secret key and returns it to the sender. Since the sender randomly generated the nonce, this defeats playback attacks because the replier cannot know in advance the nonce the sender will generate. The receiver denies connections that do not have the correctly encrypted nonce. Diebold uses nonces; NCR and shared Wincor Nixdorf signature devices do not.

**Out-of-Band** — A term describing a function that is performed offline, rather than using online transaction processing. For example, manually loading keys at an ATM is a function that is performed out-of-band. Sending a key to a certificate authority to be validated and signed is an example of another out-of-band function.

**PKCS #7 Cryptographic Message** — A message standard for data that may have cryptography applied to it, such as digital signatures and digital envelopes. The Diebold certificate authority returns certificates that have been registered with the certificate authority using this format. The Diebold EPP transports certificates and other data to and from the BASE24 system using this message. Diebold uses the PKCS #7 cryptographic message; NCR and Wincor Nixdorf do not.

**PKCS #10 Certificate Request Message** — A message standard for a request for certification of a public key. Also referred to as a certificate signing request (CSR). The BASE24 system public key must be sent in this format when it is sent to the Diebold certificate authority. When it is returned, it is in PKCS #7 cryptographic message format.

**Plaintext** — The data to be encrypted.

**Private Key** — One-half of a cryptographic key pair used in a public key algorithm such as Rivest-Shamir-Adleman (RSA). The other half of the cryptographic key pair is the public key. The private key is uniquely associated

with an entity and is not available publicly. It is used to decrypt data that was encrypted using its associated public key. It is also used to create signatures that can only be verified using its associated public key. A mathematical relationship exists between the private key and the public key, but it is virtually impossible to derive one from the other. The private key is also known as the secret key.

**Public Key** — One-half of a cryptographic key pair used in a public key algorithm such as Rivest-Shamir-Adleman (RSA). The other half of the cryptographic key pair is the private key (also known as the secret key). The public key is uniquely associated with an entity and is available publicly. It is used to encrypt data that can only be decrypted using its associated private key. It is also used to verify signatures that were created with its associated private key. A mathematical relationship exists between the public key and the private key, but it is virtually impossible to derive one from the other.

**Public Key Cryptography Standards (PKCS)** — A series of cryptographic standards dealing with public key issues, published by RSA Laboratories.

**Public Key Infrastructure (PKI)** — A security infrastructure whose services are implemented and delivered using public key concepts and techniques.

**RSA** — A public key algorithm based on the factoring problem. RSA stands for Rivest-Shamir-Adleman, the developers of the RSA public key algorithm.

**Secret Key** — *See* private key.

**Secure Hash Algorithm (SHA-1)** — An algorithm for computing a *condensed representation* of a message or a data file. The *condensed representation* is of fixed length and is known as a *message digest* or *fingerprint*. Because it is conjectured to be computationally infeasible to produce two messages having the same message digest, a SHA-1 hash value can be used to verify the integrity of a file. NCR and Wincor Nixdorf use SHA-1 hash values to ensure the integrity of public key data transported between the certificate authority (i.e., NCR trusted center or Wincor Nixdorf Customer Key Center – International) and an NCR or Wincor Nixdorf customer.

**Signature** — A method of preventing data from being changed during transmission and of preventing the sender from being impersonated. A secure hashing algorithm is applied to the data and then the result is encrypted. With

RSA encryption, the signature is generated using the private key. Only the holder of the private key can generate a valid signature, but anybody can verify it using the associated public key.

For NCR and Wincor Nixdorf, signatures are generated using the RSAES-PKCS1-v1\_5 scheme. A 256-byte signature block is created. For transmission to the ATM, the block is base94 encoded for NCR devices and base64 encoded for shared Wincor Nixdorf signature devices.

**Working Key** — A category of keys used to perform specific cryptographic operations. This key is encrypted under the master key prior to being downloaded from the host to the ATM. PIN encryption keys, MAC keys and PIN verification keys are all examples of working keys.

# Implementation Requirements

The following paragraphs discuss various hardware and software requirements for implementing the BASE24-atm Automated Key Distribution System add-on product.

## NCR 5XXX Devices

The NCR 5XXX device must be running NCR NDC+ 7.00 (or greater) or APTRA Advance NDC 3.00 (or greater) and APTRA Edge environment NDC Business Services software and be equipped with an encrypting PIN pad (EPP) to support the BASE24-atm Automated Key Distribution System add-on product. NCR supports the BASE24-atm Automated Key Distribution System add-on product on most 56XX and PersonaS models that meet the above requirements

The EPP must contain a public and private key pair, a signature of the EPP's public key, the NCR trusted center's public key, and a unique serial number. All of these are created when the EPP is manufactured. See phases 1 and 2 of the implementation procedures for NCR devices.

ATM master keys generated for NCR 5XXX devices can be single length (16 hexadecimal characters) or double length (32 hexadecimal characters). The ATM uses a single or double length key depending on the key entry mode selected in supervisor mode or downloaded using the Load Key Entry Mode command entered from the DCT. Refer to the "Key Entry Mode Download" topic provided later in this appendix for more information. NCR recommends that all DES keys be double-length keys. A double-length master key is required to support triple DES encryption between the BASE24 system and an NCR 5XXX device.

**Note:** NCR's enhanced BAPE software that mimics triple DES encryption does not support the loading of initial keys using the RSA method.

## NDC+ 7.00 Environment

Previous releases of NDC+ using the BAPE usually preserved the encryption keys stored in the encryptor. No special action was required on the part of NDC+ for this. When NDC+ 7.00 is installed on an ATM with a BAPE encryptor, the initial keys entered using a previous release of NDC+ are preserved. The only exception is release 6.05 where the allocation of key slots differed from other releases (this is corrected in release 6.05.01).

With the EPP, special action is required to preserve the keys in the encryptor when migrating to release 7.00. The EPP is provided in two variants that can emulate either a BAPE (EPPB) or an EKC (EPPE). For NDC+, the EPP must be an EPPB, which is referred to as an EPP throughout this appendix. The EPP can operate in either its emulation mode or as an EPP. If a previous release of NDC+ has been installed on an ATM with an EPPB, it will operate in BAPE emulation mode. NDC+ release 7.00 will detect this condition, and rather than reprogramming the encryptor as an EPP, it will continue to run in BAPE emulation mode, preserving the encryption keys, until the encryption key entry mode is changed. You can change the key entry mode either locally through supervisor mode or remotely using the Load Key Entry Mode command entered from the DCT. In either case, changing the key entry mode causes all encryption keys to be deleted and the encryptor to be re-programmed as an EPP.

While the EPP is operating in BAPE emulation mode, all the triple DES operations are emulated in software. RSA initial key download is not supported in BAPE emulation mode. No attempt is made to preserve keys loaded onto an EPP; it is always reprogrammed as an EPP.

## APTRA Environment

APTRA Advance NDC 3.00 (or greater) and APTRA Edge environment NDC Business Services software only supports triple DES and RSA cryptography in hardware and the EPP must operate in EPP mode.

## Protocol Support

The message size required for AKDS messages dictates the protocols supported for NCR 5XXX devices. The following table lists the protocols supported for NCR 5XXX devices as identified by NCR and the corresponding protocol used by the XPNET process.

| NCR               | XPNET                                                    |
|-------------------|----------------------------------------------------------|
| SNA (LU0)/SDLC*   | SNA Device Support - Primary LU Type 0, 2                |
| X.25 LAPB/X.21bis | X.25 Network Interface with the HOST or NONE end to ends |



| NCR                | XPNET                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------|
| SNA/X.25 (QLLC)    | X.25 Network Interface with the HOST or NONE end to ends where the HP NonStop system handles the SNA layer. |
| TCP/IP             | TCP/IP                                                                                                      |
| TC500 Asynchronous | 6100 Burroughs/NCR Multipoint Supervisor                                                                    |
| TC500 Synchronous  |                                                                                                             |

\* The XPNET process no longer supports straight SDLC.

## Diebold 10XX/478X Devices

The Diebold 10XX/478X device must be running Terminal Control Software level 1.3 (or greater) and be equipped with the EPP4 encrypting PIN pad. Agilis 91X, TCS Plus, and TCS/OS2 systems that meet the above requirements will support the BASE24-atm Automated Key Distribution System add-on product.

The EPP must contain two sets of public and secret key pairs. One key pair is used for signature verification and generation. The second key pair is used for encipherment and decipherment. Both key pairs are signed by a certificate authority and are stored in the EPP as certificates. For each public key, the certificate authority creates two certificates—a primary certificate and a secondary certificate—for disaster recovery purposes. BASE24 accepts and stores both primary and secondary certificates during registration and certification, but the secondary certificate is never used. In addition to the four certificates that contain the EPP's public keys and signatures, two certificates (primary and secondary) containing the certificate authority's public key and signature are stored in the EPP. See phase 1 of the implementation procedures for Diebold devices.

ATM master keys generated for Diebold 10XX/478X devices can be single length (16 hexadecimal characters) or double length (32 hexadecimal characters). If the terminal is set to use single-length keys and the new master key downloaded to the terminal is a valid double-length key, the terminal automatically switches to double-length keys. If the terminal is set to use double-length keys and the new master key downloaded to the terminal is a valid single-length key, the terminal automatically switches to single-length keys. Thus, there is no need to configure the key length at the terminal prior to remotely downloading the master key.

## Protocol Support

The message size required for AKDS messages dictates the protocols supported for Diebold 10XX/478X devices. The following table lists the protocols supported for Diebold 10XX/478X devices as identified by Diebold and the corresponding protocol used by the XPNET process.

| Diebold                                          | XPNET                                                    |
|--------------------------------------------------|----------------------------------------------------------|
| Burroughs Poll Select asynchronous               | 6100 Burroughs/NCR Multipoint Supervisor                 |
| Burroughs Poll Select synchronous                |                                                          |
| IBM 3271 Bisync (ASCII)                          | 6100 Bisync Multipoint Supervisor                        |
| IBM 3271 Bisync (EBCDIC)                         |                                                          |
| IBM SNA/SDLC LU types 0 and 2                    | SNA Device Support - Primary LU Type 0, 2                |
| TCP/IP Stream Sockets (client/server)            | TCP/IP                                                   |
| TCP/IP (Diebold TBAPP)                           |                                                          |
| TCP/IP Dial                                      |                                                          |
| IBM SNA over X.25 (QLLC) LU types 0 and 2 (CCS)* | X.25 Network Interface with the HOST or NONE end to ends |

\* SNA over X.25(QLLC) is standard X.25 to XPNET (X.25 Network Interface), using either the NONE or HOST end to ends. The SNA part over X.25 is configured and handled by the HP NonStop system. The XPNET process has no knowledge of the SNA layer. The X.25 Network Interface with either the HOST or NONE end to ends are used for standard SVC and PVC circuits.

## Shared Wincor Nixdorf Signature Devices

The Wincor Nixdorf device must be running firmware compliant with release 1.5 of Wincor Nixdorf's shared signature specification, ***Wincor Nixdorf NDC/Diebold D91x Message Format Extension for RKL***, and be equipped with an encrypting PIN pad (EPP) to support the BASE24-atm Automated Key Distribution System add-on product.

**Note:** The device vendor need not be Wincor Nixdorf as long as the device is compliant with Wincor Nixdorf's shared signature specification and is equipped with an encrypting PIN pad (EPP) to support the BASE24-atm Automated Key Distribution System add-on product. Any other device vendor whose ATMs can run the necessary firmware can also be used. For simplicity, Wincor Nixdorf is assumed to be the device vendor throughout this appendix.

The EPP must contain a public and private key pair, a signature of the EPP's public key, the customer-specific Wincor Nixdorf public key, a unique serial number, and a signature of the serial number. The signatures are created by signing the EPP's public key and serial number using the customer-specific Wincor Nixdorf secret key. All of this data is created when the EPP is manufactured. See phases 1 and 2 of the implementation procedures for Wincor Nixdorf devices.

The message size required for AKDS messages dictates the protocols supported for Wincor Nixdorf devices running in NDC+ or Diebold 912 emulation mode. Refer to the "Protocol Support" topics for NCR 5XXX and Diebold 10XX/478X devices above for lists of the protocols supported.

## BASE24 System

The BASE24-atm Automated Key Distribution System add-on product requires BASE24 release 6.0, version 3 or greater for NCR 5XXX devices, BASE24 release 6.0, version 4 or greater for Diebold 10XX/478X devices, BASE24 release 6.0, version 6 or greater for Wincor Nixdorf signature devices, a special Transaction Security Services process called the AKDS process, and a hardware security module with the appropriate firmware or software. An RSA public/private key pair must be created for the BASE24 system. This key pair is created in a hardware security module and stored in the Host Public Key Security Primary File (HSPKPD). Use the "Key Generate (KEYGEN) Command" to generate and store this RSA key pair. For further information on how this command is used for NCR, Diebold, and Wincor Nixdorf devices, see [phase 3](#) of the implementation procedures for NCR devices, [phase 2](#) of the implementation procedures for Diebold devices, or phase 3 of the implementation procedures for Wincor Nixdorf devices.

**Note:** The AKDS process is a special Transaction Security Services process that contains all of the components of the Transaction Security Services process plus the components required to support the BASE24-atm Automated Key Distribution System add-on product. The AKDS process uses the AKDSISLO executable object while the regular Transaction Security Services process uses the TSSISLO executable object.

## Security Device Configuration

For each security device to be used for public key cryptography (PKI) functions, you must check the Enable PKI field on the Security Device Configuration window. If this field is not checked for a security device, it is not used for PKI functions. This enables you to consolidate all PKI functions in just one or a few devices.

## Atalla Public Key Cryptography Commands

An Atalla NSP must be running a minimum of AKB version 2.40 or 1.4 software to support public key cryptography commands using the Rivest-Shamir-Adleman (RSA) scheme. The 2.40 version software is used for A10100 AKB models while the 1.4 version software is used for A10000 AKB models. All of these commands are defaulted to off in the security policy delivered from the factory due to security exposure. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of the BASE24 application), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands required for the BASE24-atm Automated Key Distribution System add-on product must be enabled in your security policy using the 108 and 109 commands:

- Generate RSA key pair (Command 120)
- Generate AKB for root public key (Command 122)
- Verify public key and generate AKB of public key (Command 123)
- Generate digital signature (Command 124)
- Verify digital signature (Command 125)
- Generate message digest (Command 126)
- Generate ATM master key and encrypt with public key (Command 12F)

## AKB Headers For RSA Keys

The following table lists the AKB header assignments to use for RSA public and private keys.

| Header Assignment                  | Description                                                          |
|------------------------------------|----------------------------------------------------------------------|
| 1 K R E E 0 0 0<br>1 K R E N 0 0 0 | The RSA public key used to encrypt key information                   |
| 1 K R D E 0 0 0<br>1 K R D N 0 0 0 | The RSA private key used to decrypt key information                  |
| 1 S R G E 0 0 0<br>1 S R G N 0 0 0 | The RSA private key pair used to sign (generate a digital signature) |
| 1 S R V E 0 0 0<br>1 S R V N 0 0 0 | The RSA public key used to verify a digital signature                |

## Thales e-Security Public Key Cryptography Commands

Thales e-Security version 6.02 firmware and the optional DSP plug-in (for RSA functions) provides public key cryptography commands using the RSA scheme. BASE24 requires a number of these commands to support the BASE24-atm Automated Key Distribution System add-on product. Some of these commands are used out-of-band (outside of BASE24), nonetheless, they are all required to support the BASE24-atm Automated Key Distribution System add-on product. The following commands are required for the BASE24-atm Automated Key Distribution System add-on product:

- Generate a key (Command A0)
- Generate RSA key set (Command EI)
- Generate a MAC on a public key (Command EO)
- Validate a certificate and generate MAC on its public key (Command ES)
- Generate a signature (Command EW)
- Validate a signature (Command EY)
- Export a DES key (Command GK)
- Hash a Block of Data (GM)

**Note:** Public keys sent to the Thales HSM must be in the Abstract Syntax Notation One (ASN.1) format using the Distinguished Encoding Rules (DER). The format is as follows:

|                     |             |                    |                |         |                    |                 |          |
|---------------------|-------------|--------------------|----------------|---------|--------------------|-----------------|----------|
| Sequence Identifier | Byte Length | Integer Identifier | Modulus Length | Modulus | Integer Identifier | Exponent Length | Exponent |
|---------------------|-------------|--------------------|----------------|---------|--------------------|-----------------|----------|

A default value of 65537 is assumed for all public key exponent values.

The AKDS process automatically handles any necessary HSM message format conversions.

# LCONF Params

The BASE24-atm Automated Key Distribution System add-on product uses the following LCONF params in addition to the LCONF assigns and params discussed in section 3.

**ATM-DH-AKDS-TSS-TIMER param** — The number of seconds the NCR 5XXX Device Handler process or Diebold 10XX/478X Device Handler process waits for a reply from the AKDS process before it considers the Automated Key Distribution System (AKDS) request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.

This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window.

**ATM-DH-1000-AKDS-SERIAL-NUM-VRFY** — An optional param indicating whether the Diebold 10XX/478X Device Handler process verifies the EPP serial numbers received from the ATM when authenticating and exchanging public keys. If the values match valid EPP serial numbers stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:

Y = Yes, validate the EPP serial numbers  
N = No, do not validate the EPP serial numbers

If this param is not configured or contains an invalid value, it defaults to a value of N.

This param applies when using Diebold 10XX/478X devices or Wincor Nixdorf devices running in Diebold 912 emulation mode.

**ATM-DH-N50-AKDS-SERIAL-NUM-VRFY** — An optional param indicating whether the NCR 5XXX Device Handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:

Y = Yes, validate the EPP serial number

N = No, do not validate the EPP serial number

If this param is not configured or contains an invalid value, it defaults to a value of N.

This param applies when using NCR 5XXX devices or Wincor Nixdorf devices running in NCR NDC+ emulation mode.



# NCR 5XXX Terminal Implementation Procedures

Implementation of remote initial key processing for NCR 5XXX terminals is accomplished in five phases, each of which are described below. The first two phases are performed by NCR alone; ACI and its customers are not involved. These phases are performed out-of-band of the BASE24 system.

## Phase 1—Generate an NCR RSA Key Pair

The NCR trusted center generates a Rivest-Shamir-Adleman (RSA) key pair composed of a private (secret) key and a public key. These keys are used for signature authentication between an NCR ATM and the BASE24 system. The private key is used to generate signatures, while the public key is used to verify signatures. This key pair is referred to as the *root* NCR key pair. The root public key of the NCR trusted center is sent (through an offline method) to you during phase 3.

## Phase 2—Manufacture Encrypting PIN Pad (EPP)

When the EPP for a 5XXX terminal is manufactured, NCR generates an RSA public and private key pair for each EPP. These keys are used for the encryption and decryption of keys exchanged between the EPP and the BASE24 system. The EPP's public and private keys are stored in the EPP. The public key for the EPP is signed with the root NCR private key. This signature allows the BASE24 system to validate the EPP using the root NCR public key created in phase 1. The signature is stored in the EPP along with the root NCR public key. The root NCR public key is used to validate a signature received from the BASE24 system. Finally, the serial number stored in the EPP is signed by the root NCR private key. This signature is used for additional authentication between the EPP and the BASE24 system. At this point, the following key information is stored in the EPP:

- The EPP's public and private keys
- The EPP's public key signature generated by the root NCR private key
- The root NCR public key
- The EPP's serial number
- The EPP's serial number signature generated by the root NCR private key

## Phase 3—Generate an RSA Key Pair for the BASE24 System

To support automated key distribution for ATMs, you must also generate an RSA key pair for the BASE24 system using a hardware security module. This is accomplished by entering the KEYGEN command on the Process Control window of the ACI desktop. After the key pair is generated, the NCR trusted center must sign the public key for the BASE24 system and return the resulting signature to you along with NCR trusted center's root public key. You must use this key to verify the signature returned from NCR. Finally, you must generate a MAC of the self-signed root NCR public key and store all these keys in the TSS database.

**Note:** The procedures are essentially the same whether you are generating an RSA key pair for a test or production environment on the BASE24 system. For testing, a test EPP provided by NCR must be used instead of a production EPP. When you initially contact NCR to start the process of submitting a public key and getting it signed by NCR, you must specify whether the key to be signed is for testing or production. You should use the same forms and procedures for getting both test and production keys signed, except that for a test key, the key data exchanged with the NCR trusted center need not be delivered on diskette or CD-ROM in tamper-resistant envelopes or using a courier service. Data can be exchanged using an e-mail message instead. The NCR trusted center signs test keys using a root public key that is different from the one it uses to sign production keys.

The above tasks are accomplished in the following steps:

1. To start the process of submitting a public key and getting it signed by NCR, contact your NCR/Reseller Account Manager. The account manager will then establish your name, mailing address, e-mail address, telephone and fax numbers and advise NCR Customer and Sales Support to send you the following forms:
  - Signature Request Process (NCR\_001)
  - NCR Approved Contacts (NCR\_002)

This form will be completed by NCR before it is sent to you.

  - Signature Request (NCR\_003)
2. Generate an RSA public and private key pair for the BASE24 system. Enter the KEYGEN command on the Process Control window to generate the RSA key pair for the BASE24 system. Refer to the “Key Generate (KEYGEN) Command” topic provided later in this appendix for detailed information on using this command.

3. Access the Certificate Management window on the ACI desktop (access **Configure > Transaction Security > Automated Key Distribution > Certificate Management**) to retrieve the BASE24 system public key and format it in the manner that the NCR trusted center expects it. When the Certificate Management window is displayed, select the NCR device type in the Device Type field. The BASE24 system public key is retrieved, signed, formatted, and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The BASE24 system public key is self-signed and displayed in ASN.1 format. This is the message format required when sending the public key to the NCR trusted center.
4. Copy the BASE24 system public key cryptogram in the Host Public Key Data field, format it as required, and send it on the appropriate media (e.g, on a diskette or CD-ROM in a tamper-resistant envelope) to your primary NCR contact specified on form NCR\_002. Follow the format and procedures prescribed on form NCR\_001.

At the same time, complete form NCR\_003 and send it by e-mail message or by facsimile (fax) to all Key Signing Individuals (i.e., the primary and all secondary NCR contacts specified on form NCR\_002). Make sure you enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Host Public Key Data tab in the SHA-1 (Fields 1–6) field on form NCR\_003.

5. The NCR trusted center validates the BASE24 system public key signature and SHA-1 hash value. The Key Signing Individual will send an e-mail message to you to acknowledge receipt of the diskette or CD-ROM and to indicate whether the signature and hash value are correct. If either are incorrect, you must resend both form NCR\_003 and the BASE24 system public key. If both are correct, the trusted center will sign the BASE24 system public key, generate SHA-1 hash values for both the BASE24 system public key signature and the self-signed root NCR public key, and return them back to you on a diskette or CD-ROM delivered by a courier service. NCR will also complete and issue the following two forms by e-mail message or by facsimile (fax) to the approved customer contact. You must use these forms to verify that the SHA-1 hash values over the self-signed root NCR public key and BASE24 system public key signature are valid.
  - NCR Public Key (NCR\_004)
  - NCR Signature of Customer Public Key (NCR\_005)

The diskette or CD-ROM returned to you will contain several files. Each file contains a key or hash value. You must copy the contents of some of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table.

| Value (NCR Filename)                                                  | Certificate Management Field                                                                                                                                         |
|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Self-signed root NCR public key (PK_NCR.txt)                          | Generate Message Digest > Certificate Authority Public Key (step 6a)<br><br>and<br><br>Validate Certificates/Signatures > Certificate Authority Public Key (step 9a) |
| SHA-1 hash of self-signed NCR public key (PK_NCR_digest.txt)          | Generate Message Digest > SHA-1 Hash Value (step 6b)                                                                                                                 |
| BASE24 system public key signature (PK_(cust)_ncr_sig)                | Validate Certificates/Signatures > Host Signature (step 9a)                                                                                                          |
| SHA-1 hash of BASE24 system public key (PK_(cust)_NCR_sig_digest.txt) | Validate Certificates/Signatures > SHA-1 Hash Value (step 9b)                                                                                                        |

6. On the Generate Message Digest tab of the Certificate Management window, perform the following:
  - a. Copy the self-signed root NCR public key cryptogram returned on diskette or CD-ROM from the NCR trusted center into the Certificate Authority Public Key field. You can click Expand View () to expand this field if needed.
  - b. Enter the SHA-1 hash value of the self-signed NCR public key into the SHA-1 Hash Value field.
  - c. Click Save ()

If you are using an Atalla NSP, the NSP validates the signature of the self-signed root NCR public key, validates the SHA-1 hash value for the key, and generates a message digest of the self-signed root NCR public key, which is displayed in the Message Digest field.

If you are using a Thales e-Security HSM, the HSM validates the signature and SHA-1 hash value of the self-signed root NCR public key. Thales HSMs do not create a message digest.

Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the self-signed root NCR public key, SHA-1 hash value, and form NCR\_004. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab.

7. If you are using a Thales e-Security HSM, you can skip this step.

For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. You must enter this variant when using the SCT. The SCA automatically uses variant 30. On the SCT, use the Utilities Menu--> Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the 32-byte message digest as two 16-byte values (Left Challenge and Right Challenge on the SCT or Challenge on the SCA). Because the SCA only allows single-length keys for a challenge/response, you must input the 32-byte message digest value as two separate key components (left 16 bytes and right 16 bytes) and encrypt each separately in the Challenge field. You must then recombine the two encrypted 16 byte values to form the root certificate hash value. Refer to the *Atalla Secure Configuration Assistant User Guide* for detailed SCA procedures.

8. If you are using a Thales e-Security HSM, you can skip this step.

For an Atalla NSP only, copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab.

9. On the Validate Certificates/Signatures tab of the Certificate Management window, perform the following:
  - a. Copy the self-signed root NCR public key and BASE24 system public key signature returned from the NCR trusted center into the Certificate Authority Public Key and Host Signature fields, respectively, on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View () to expand either of these fields if needed.
  - b. Enter the SHA-1 hash value for the BASE24 system public key signature into the SHA-1 Hash Value field.
  - c. Click Save ()

The Atalla NSP or Thales HSM validates the signature and SHA-1 hash value of the BASE24 system public key received from NCR. If validated successfully, the BASE24 system public key signature is stored in the Host Public Key Security Primary File (HSPKPD).

For an Atalla NSP only, the self-signed root NCR public key is validated using the hash value input from step 8, imported, and stored in the HSPKPD along with the message digest.

Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the BASE24 system public key signature, SHA-1 hash value, and form NCR\_005.

After the above steps are completed, the following key data is now entered into the database for the AKDS process.

- The BASE24 system's public and private keys
- The signature of the BASE24 system's public key signed by the root NCR private key
- The self-signed root NCR public key
- A MAC of the self-signed root NCR public key (for Thales e-Security HSMs only)

## Phase 4—Authenticate and Exchange Public Keys

Before you can automatically distribute (download) an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. This is accomplished online by sending the Load Public Key command (entered as LOADPUBLIC or LPUBLIC) to the terminal's NCR 5XXX Device Handler process using the Device Control Terminal (DCT) or a network control facility. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for detailed Load Public Key command syntax and processing steps performed during this phase.

The Load Public Key command initiates an online exchange of public keys between the BASE24 system and the EPP of a specified terminal.

When you enter the Load Public Key command, the NCR 5XXX Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM's configuration response message, it determines whether the ATM terminal has an encryptor type of 7 or 8. It also

checks the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the BASE24-atm Terminal Data file (ATD) to determine whether the ATM is using the B (Bisync), C (Bisync – ASCII), or S (SDLC) protocols. An encryptor type of 7 or 8 indicates that the ATM has an AKDS-capable EPP installed. A protocol value other than B, C, or S indicates that the ATM is using a protocol that supports the BASE24-atm Automated Key Distribution System add-on product. If the ATM has an AKDS-capable EPP, the requested load command is allowed to proceed. If the ATM is not capable of processing AKDS commands and is using a protocol that supports AKDS, the Load Public Key command is not performed. In addition, one or more event messages are generated and the process reopens the ATM.

During this phase, the AKDS process interacts with the hardware security module to perform the following:

- Verify the signature of the ATM's EPP serial number using the self-signed root NCR public key retrieved from the Host Public Key Security Primary File (HSPKPD).
- Optionally verify the EPP serial number itself against valid EPP serial numbers entered in the EPP Serial Number File (EPNUMD). The value of the ATM-DH-N50-AKDS-SERIAL-NUM-VRFY LCONF param determines whether this additional validation is performed. Valid EPP serial numbers must be manually entered on the EPP Serial Number Configuration window before this validation is performed. Information entered on the EPP Serial Number Configuration window is stored in the EPNUMD. Refer to the "Data Access" topic provided later in this appendix for more information on the EPP Serial Number Configuration window and the EPNUMD.
- Retrieve the BASE24 system public key and signature from the HSPKPD. These are sent to the ATM after the EPP serial number signature has been successfully validated.
- Verify the signature of the EPP's public key using the self-signed root NCR public key retrieved from the HSPKPD. If the signature is valid, the process generates a MAC of the EPP public key and stores the EPP public key, and the EPP public key MAC in the CHNPKD.

The NCR 5XXX Device Handler process sends the BASE24 system public key and signature to the ATM's EPP, where they are verified and stored.

At this point, the following key information is stored in the EPP:

- The EPP's public and private keys
- The EPP's public key signature generated by the root NCR private key
- The self-signed root NCR public key

- The EPP's serial number
- The EPP's serial number signed by the root NCR private key
- The BASE24 system's public key and signature

At this point, the following key data is entered into the database for the AKDS process.

- The BASE24 system's public and private keys
- The signature of the BASE24 system's public key signed by the root NCR private key
- The self-signed root NCR public key
- A MAC of the root NCR public key
- The EPP's public key
- The EPP's serial number
- A MAC of the EPP's public key

## Key Entry Mode Download

After the BASE24 system and the EPP have authenticated each other and exchanged their public keys, the key entry mode can be downloaded to the ATM. This is accomplished by entering the Load Key Entry Mode command (entered as LOADKEYMODE or LKEYMODE) from the Device Control Terminal (DCT) or a network control facility.

**Note:** This procedure is not necessary if the key entry mode is set through supervisor mode at the terminal itself, prior to authenticating and exchanging public keys.

Changing the key entry mode of an EPP deletes all the DES keys in the encryptor. To reduce the chances of accidentally changing the mode and deleting the keys, the Load Key Entry Mode command is accepted only when the "Authenticate and Exchange Public Keys" phase has been successfully performed since the last power up. This ensure that the DES keys can be restored after they are deleted by changing the key mode.



**Note:** NCR allows one exception to the above restriction. To assist in the migration from a previous release of NDC+ to release 7.00, the Load Key Entry Mode command is accepted one time only without having to perform the “Authenticate and Exchange Public Keys” phase if the EPP is operating in Basic Alpha PIN pad and Encryptor (BAPE) emulation mode.

If the EPP is installed with NDC+ 7.00 software and is running in BAPE emulation mode when the key entry mode is changed, either through supervisor mode or by entering the Load Key Entry Mode command, it is also reprogrammed as an EPP and moves out of BAPE emulation mode.

Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for detailed Load Key Entry Mode command syntax and processing steps performed during this phase.

## Phase 5—Download the Master Key

After an EPP has been successfully authenticated and exchanged public keys with the BASE24 system, and the key entry mode has been set, you can securely download the ATM master key to the terminal using the Load Master Key command (entered as LOADMASTER or LMASTER). You can enter this command from the DCT or a network control facility. This command specifies the ID of the terminal to which the ATM master key is to be downloaded. After the master key is successfully loaded in both the EPP and the TSS database, you can securely download working keys to the EPP. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for detailed Load Master Key command syntax and processing steps performed during this phase.

When you enter the Load Master Key command, the NCR 5XXX Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM’s configuration response message, it determines whether the ATM terminal has an encryptor type of 7 or 8. It also checks the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the BASE24-atm Terminal Data file (ATD) to determine whether the ATM is using the B (Bisync), C (Bisync – ASCII), or S (SDLC) protocols. An encryptor type of 7 or 8 indicates that the ATM has an AKDS-capable EPP installed. A protocol value other than B, C, or S indicates that the ATM is using a protocol that supports the BASE24-atm Automated Key Distribution System add-on product. If the ATM has an AKDS-capable EPP, the requested load command is allowed to proceed. If the ATM is not capable of processing AKDS commands and is using a protocol that supports AKDS, the Load Master Key command is not performed. In addition, one or more event messages are generated and the process reopens the ATM.

During this phase, the AKDS process interacts with the hardware security module to generate a new master key encrypted under the EPP's public key and sign it using the BASE24 system private key. The new master key and associated check digits are stored in the Channel Security File (CSECD). The new master key and signature are also sent to the ATM.

At this point, both the BASE24 system and the EPP have the terminal's master key stored in their respective databases. Working keys (i.e., the PIN encryption key and MAC key) can now be securely downloaded to the EPP.

## Financial Institution Table (FIT) Data

The PIN encryption key in the FIT is encrypted under the master key. If a new master key is generated and a PIN encryption key exists in the FIT for local PIN verification, you must manually translate the PIN encryption key from encryption under the old master key to encryption under the new master key, replace the PIN encryption key cryptogram in the FIT, and download the updated FIT data to the ATM.

The PIN encryption key is entered in the PEKEY (PIN ENCRYPTION KEY) field on the Financial Institution Table screen, which you can access from the NCR 5XXX Configuration File Menu in the Device Control Terminal (DCT). For additional information on this field and downloading FIT data, refer to the *BASE24-atm NCR 5XXX Device Support Manual*.

## Migrating from Single-Length to Double-Length Keys

After you have successfully completed all phases of AKDS implementation for an NCR 5XXX terminal, you can use the following procedure to migrate the terminal to double-length keys.

1. Send the Load Key Entry Mode command to the terminal with a value of 3 (Double length with XOR). You can enter this command from the Device Control Terminal (DCT) NCR 5XXX Load Commands screen or from a network control facility. Refer to the *BASE24-atm NCR 5XXX Device Support Manual* for detailed command syntax.

Changing the key entry mode of an EPP deletes all the DES keys in the encryptor.

An event message is generated when the command is successfully completed.

2. Send the Load Public Key command to the terminal to re-establish the BASE24 system's public key that was erased in step 1 in the terminal's EPP. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for detailed Load Public Key command syntax. Another event message is generated when the processing for this command is successfully completed.
3. On the ACI desktop, read the terminal's current record on the Channel Security Configuration window. Change the Key Length field for the exchange key and inbound key from a value of Single to a value of Double.

If you are using Thales e-Security HSMS, you may also need to change the External Key Block Format field from a value of Variant method to ANSI X9.17 method. If you are using Atalla NSPs, the External Key Block Format field must be set to Standard Atalla.

Click Save () to save the changes in the CSECD.

4. Warmboot the Transaction Security Services and AKDS processes to reread the Channel Security data source (CSECD) record into memory. You can warmboot these processes to either reread all CSECD records from disk into memory using the [Alter Data Source](#) process control command or just reread the specified CSECD terminal record into memory using the [CSEC Refresh](#) process control command.
5. Send the Load Master Key command to the terminal to generate and load a new ATM master key. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** for detailed Load Master Key command syntax. Another event message is generated when the processing for this command is successfully completed.
6. Generate and download new double-length communications keys (i.e., PIN and/or MAC keys) using the appropriate commands entered from the DCT or a network control facility. As before, check the event message log to make sure the command completed successfully.
7. Bring the ATM back into service. Transactions entered at the ATM should now be processed using the new double-length communications key.

# Diebold 10XX/478X Terminal Implementation Procedures

Implementation of remote initial key processing for Diebold 10XX/478X terminals is accomplished in four phases, each of which are described below. This implementation uses digital certificates initially signed by a third-party certificate authority—Digital Signature Trust Co. (DST). The first phase is performed by Diebold and the initial certificate authority; ACI is not involved.

All certificates for the Diebold 10XX/478X implementation follow the International Telecommunications Union (ITU) X.509 standard, *Information Technology - Open Systems Interconnection - The Directory: Public-Key and Attribute Certificate Frameworks*.

## Phase 1—Initial Registration and Certification for the EPP

Diebold EPPs create two RSA key pairs when they are first powered up. One key pair is used for signature verification and generation. The second key pair is used for encipherment and decipherment. The public keys for each of these two key pairs are signed by a certificate authority. Diebold has chosen Digital Signature Trust Co. (DST) as their certificate authority.

For each public key created, the certificate authority creates two certificates—a primary certificate and a secondary certificate. These certificates contain the EPP's public keys and signatures. Primary and secondary certificates are features of Diebold's disaster recovery procedure; however, BASE24 uses the primary certificate only. The EPP's public key certificates, private signature generation key, private decipherment key, and two additional certificates (primary and secondary) containing the certificate authority's public signature verification key and signature, are stored in the EPP.

## Phase 2—Initial Registration and Certification for the BASE24 System

To support automated key distribution for ATMs, you must also generate an RSA key pair for signature verification and generation for the BASE24 system using a hardware security module. This is accomplished by entering the KEYGEN command on the Process Control window of the ACI desktop. After the key pair is generated, the Diebold certificate authority—Digital Signature Trust Co. (DST)—must verify the BASE24 system's public key and create certificates of the BASE24

system's public key, signed by the certificate authority's private keys. The BASE24 system's certificate are referred to as the Host Bank server certificates by the certificate authority. The certificate authority also creates certificates of its own public signature verification keys, and root certificate authority public key, signed by the certificate authority's private keys. All of these certificates are returned to you. You must then validate these certificates and store them in the Host Public Key Security Files. These tasks are accomplished in the following steps:

1. Generate an RSA key pair for the BASE24 system for signature verification or generation. Enter the KEYGEN command on the Process Control window to generate the RSA key pair certificates for the BASE24 system. Refer to the "Key Generate (KEYGEN) Command" topic provided later in this appendix for detailed information on using this command.
2. Access the Certificate Management window on the ACI desktop (access **Configure > Transaction Security > Automated Key Distribution > Certificate Management**) to retrieve the BASE24 system public key certificate and format it the manner that the target certificate authority expects it. When the Certificate Management window is displayed, select Diebold as the device type in the Device Type field. The BASE24 system public key certificate is retrieved and displayed in the Host Public Key Data field. For Diebold, the key is displayed in PKCS#10 Certificate Request format, Base64 encoded. This is the message format required when sending the public key to Diebold's certificate authority.
3. Copy the BASE24 system public key certificate in the Host Public Key Data field and paste it in the appropriate field in the DST-hosted Diebold Host Bank registration pages. Host bank registration involves three phases: authorization, application, and retrieval.

For the authorization phase, you must enter a unique authorization code distributed by Diebold to ensure that you are permitted to receive a Host Bank server certificate. Ask your Diebold System Engineer to contact the Diebold Product Manager for their Remote Key product using an e-mail message. The Diebold Product Manager will send you an e-mail message containing the address of the DST-hosted web site, your authorization code (password), and an HTML demonstration of the registration website. An example of this e-mail message is shown below. Each time you need new certificates (e.g., for test, production, and certification) you must contact your Diebold System Engineer to request a new authorization code.

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This e-mail contains the voucher (Authorization Code) that allows \*Financial Institution\* to receive host certificates.  
The information in this message should be regarded with the appropriate

level of security, as this voucher retrieves certificates that allow the holder to be identified as a legitimate ATM host by the EPP.

Website: <https://secure.digsigtrust.com/tsapp/dbold-ssl-start.jsp>  
<<https://secure.digsigtrust.com/tsapp/dbold-ssl-start.jsp>>  
Authorization Code: 1234567-XXXX-XXXX

Attached is a demonstration of the host registration site. Please use these test html pages to become familiar with the registration site. Unzip the attached file into a directory before beginning at the file named "Start\_Here".

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For the application phase, you must enter some required and optional contact information (e.g., e-mail address, phone number, and mailing address) for the headquarters of your organization. You must also enter some required and optional contact information about yourself (e.g., your name, job title, and office phone number) and choose a DST passphrase. A DST passphrase is required to renew or revoke your digital certificate, as well as to update your personal information. Finally, you must copy the BASE24 system public key certificate from the Host Public Key Data field on the Certificate Management window and paste it into the CSR field on the DST registration pages.

At this point, the application phase is complete. DST verifies the BASE24 system public key certificate and creates two certificates of the BASE24 system public key, signed by the certificate authority's primary and secondary private keys, respectively. The certificate authority also creates two additional certificates. One certificate contains the primary certificate authority's public signature verification key, signed by the certificate authority's primary private key. The second certificate contains the secondary certificate authority's public signature verification key, signed by the certificate authority's secondary private key.

For the retrieval phase, you must copy the following certificates generated or displayed by DST in your browser and paste them in the appropriate fields on the Certificate Management window in the following steps. All certificates are in PKCS #7 Cryptographic Message format. The following table shows

you which field on the Certificate Management window to paste each certificate from DST and in which of the following steps you paste the certificate.

| <b>DST Certificate</b>                 | <b>Certificate Management Field</b>                                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DST Root CA X3 Certificate             | Generate Message Digest > Root Certificate Authority Certificate (step 4)<br><br>and/or<br>Validate Certificates/Signatures > Root Certificate Authority Certificate (step 7) |
| DST Primary Sub-CA Certificate         | Validate Certificates/Signatures > CA Primary Certificate (step 7)                                                                                                            |
| DST Secondary Sub-CA Certificate       | Validate Certificates/Signatures > CA Secondary Certificate (step 7)                                                                                                          |
| Primary Host Bank Server Certificate   | Validate Certificates/Signatures > Host Primary Certificate (step 7)                                                                                                          |
| Secondary Host Bank Server Certificate | Validate Certificates/Signatures > Host Secondary Certificate (step 7)                                                                                                        |

4. If you are using a Thales e-Security HSM, you can skip to step 7. If you are using an Atalla NSP, create a message digest of the root certificate authority certificate (DST Root CA X3 Certificate). Copy the root certificate authority certificate returned from DST into the Root Certificate Authority Certificate field on the Generate Message Digest tab of the Certificate Management window and click Save (📁). You can click Expand View (🔍) to expand this field if needed. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab.
5. Using the SCT, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK. Use the Utilities Menu--> Encrypt Challenge function on the SCT to encrypt the message digest value. You must input the message digest as two 16 byte values (Left Challenge and Right Challenge). The results are two 16 byte values that comprise the root certificate hash value.
6. Copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab.

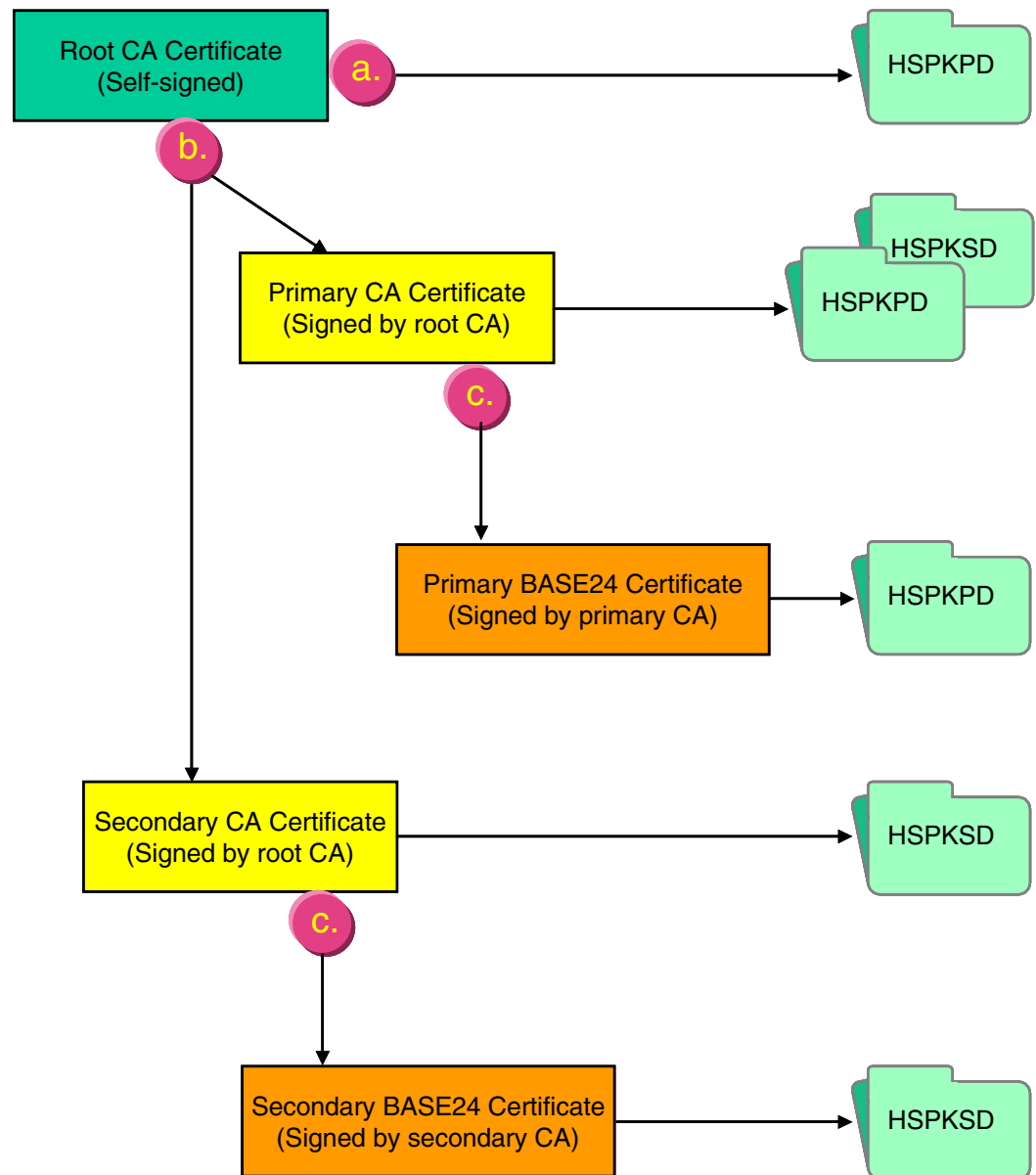
7. Copy the root certificate authority certificate, primary and secondary certificate authority's public signature verification key certificates, and the primary and secondary BASE24 system public signature verification key certificates returned from DST in step 3 into the appropriate fields on the Validate Certificates/Signatures tab of the Certificate Management window and click Save (  ).

When you click Save (  ), the following processing is performed. This processing is illustrated in the graphic following the steps.

- a. The root certificate authority certificate is validated using the hash value input from step 6 (for Atalla NSPs only), the self-signed signature of the certificate is validated, and the root certificate authority public key is stored in the Host Public Key Security Primary File (HSPKPD).
- b. The signatures of the primary and secondary certificate authority's public signature verification keys are validated using the root certificate authority public key stored in step a. The primary certificate authority public key is stored in the HSPKPD. The secondary certificate authority public key is stored in the Host Public Key Security Secondary File (HSPKSD).
- c. The signatures of the BASE24 system (host) primary and secondary certificates are validated using the primary and secondary certificate authority's public signature verification keys stored in step b,



respectively. The primary BASE24 system public key signature is stored in the HSPKPD. The secondary BASE24 system public key signature is stored in the HSPKSD.



Note that the certificates for the certificate authority are parsed and only the public key values are stored in the HSPKPD. For the BASE24 system (host) public keys, the full certificate signatures are stored in the HSPKPD or HSPKSD.

After the above steps are completed, the following key data is entered into the database for the AKDS process or in the tamper-resistant memory of the hardware security module.

- Root certificate authority public key
- Primary and secondary BASE24 system public key signatures (signed by the certificate authority's primary and secondary private keys, respectively)
- Primary and secondary certificate authority's public signature verification keys
- The BASE24 system's private signature generation key

## Phase 3—Exchange Certificates

After the keys and certificates required for automated key distribution have been established at the EPP and the BASE24 system, you can initiate an online exchange of public signature verification key certificates and the EPP public encipherment key certificate. This is accomplished by sending the Load Public Key command (entered as LOADPUBLIC) to the terminal's Diebold 10XX/478XX Device Handler process using the Device Control Terminal (DCT) or a network control facility, or by sending an Unsolicited Status message from the terminal with a status indicating that a new network certificate is required. Refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for detailed Load Public Key command syntax and processing steps performed during this phase.

The Load Public Key command initiates an online exchange of public signature verification key certificates and the EPP public encipherment key certificate.

When you enter the Load Public Key command, the Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM's configuration response message, it searches the Additional Hardware Configuration Status area for the value "EP0104RK". The presence of this value indicates the ATM has an AKDS-capable EPP installed. If this value is found, the requested load command is allowed to proceed. If this value is not found, the Load Public Key command is not performed. In addition, one or more event messages is generated and the process reopens the ATM.

During this phase, the AKDS process interacts with the hardware security module to perform the following:

- When the EPP sends its signature to the BASE24 system, optionally verify the EPP serial number itself against valid EPP serial numbers entered in the EPP Serial Number File (EPNUMD). The value of the ATM-DH-1000-AKDS-SERIAL-NUM-VRFY LCONF param determines whether this additional validation is performed. This optional verification is performed during verification of both the EPP public signature verification key certificate and the EPP public encipherment verification key certificate. The EPP serial numbers in these certificates are appended with the letters V and E, respectively, to indicate which certificate they are used in. Valid EPP serial numbers must be manually entered on the EPP Serial Number Configuration window before this validation is performed. These serial numbers must exactly match the values contained in the certificates. Therefore, you must enter two EPP serial numbers on the entries EPP Serial Number Configuration window for each Diebold 10XX/478X device. For example, if the EPP serial number for a Diebold 10XX/478X device is 05010070, you must enter serial number values of 05010070 V and 05010070 E, where *b* is a blank space. Information entered on the EPP Serial Number Configuration window is stored in the EPNUMD. Refer to the “Data Access” topic provided later in this appendix for more information on the EPP Serial Number Configuration window and the EPNUMD.
- Retrieve the primary BASE24 system’s primary public signature verification key from the Host Public Key Security Primary File (HSPKPD). This key is sent to the ATM along with a request for the terminal’s EPP public signature verification key certificate.
- Verify the signature of the EPP’s public signature verification key certificate using the primary certificate authority public key retrieved from the HSPKSD. If the signature is valid, the process stores the EPP public signature verification key in the Channel Public Key Security File (CHNPKD).
- Verify the signature of the EPP’s public encipherment key certificate using the primary certificate authority public key retrieved from the HSPKSD. If the signature is valid, the process stores the EPP public encipherment key in the CHNPKD.

At this point, the following key information is stored in the EPP:

- The EPP’s primary and secondary public signature verification key certificates (signed by the certificate authority’s primary and secondary private keys, respectively)
- The EPP’s private signature generation key

- The EPP's primary and secondary public encipherment key certificates (signed by the certificate authority's primary and secondary private keys, respectively)
- The EPP's private decipherment key
- The certificate authority's primary and secondary public signature verification key certificates (signed by the certificate authority's primary and secondary private keys, respectively)
- The BASE24 system's public signature verification key certificate (signed by the certificate authority's primary private key)

At this point, the following key data is entered into the database for the AKDS process or in the tamper-resistant memory of the hardware security module.

- Root certificate authority's public key
- Primary and secondary BASE24 system public signature verification keys (signed by the certificate authority's primary and secondary private keys, respectively)
- Primary and secondary certificate authority's public signature verification keys
- The BASE24 system's private signature generation key
- The EPP's public signature verification key (signed by the certificate authority's primary private key)
- The EPP's public encipherment key (signed by the certificate authority's primary private key)

## Phase 4—Download the Master Key

After an EPP has been successfully authenticated and exchanged public key certificates with the BASE24 system, you can securely download the ATM master key to the terminal. You can initiate the download by either entering the Load Master Key command (entered as LOADMASTER) from the DCT or a network control facility, or by sending an Unsolicited Status message from the ATM terminal to the BASE24 system with a status indicating that remote key transport is required. Refer to the *BASE24-atm Diebold 10XX/478X Device Support Manual* for detailed Load Master Key command syntax and processing steps performed during this phase.

When you enter the Load Master Key command, the Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM's configuration response message, it searches the Additional Hardware Configuration Status area for the value "EP0104RK". The presence of this value indicates the ATM has an AKDS-capable EPP installed. If this value is found, the Load Master Key command is allowed to proceed. If this value is not found, the Load Master Key command is not performed. In addition, an event message is generated and the process reopens the ATM.

If the ATM is set to use single-length keys and BASE24 downloads a double-length master key, the ATM will automatically switch to double-length keys. If the ATM is set to use double-length keys and BASE24 downloads a single-length master key, the ATM will automatically switch to single-length keys. There is no need to configure the key length at the ATM prior to remotely transporting the master key.

During this phase, the AKDS process interacts with the hardware security module to generate a new master key, encrypt it under the EPP's public encipherment key retrieved from the Channel Public Key Security File (CHNPKD), and sign it using the BASE24 system's private signature generation key retrieved from the Host Public Key Security Primary File (HSPKPD).

At this point, both the BASE24 system and the EPP have the terminal's master key stored in their respective databases. Working keys (i.e., the PIN encryption key and MAC key) can now be securely downloaded to the EPP.

After the AKDS process has generated a new master key and stored it in the TSS database, any errors in subsequent processing (e.g., a failure of the message sent to the ATM to load the new master key or the ATM rejecting the new master key), causes the AKDS process to back out the new master key (i.e., restore it to its previous value). No operator intervention is required to destroy the new master key.

## **Financial Institution Table (FIT) Data**

The PIN encryption key in the FIT is encrypted under the master key. If a new master key is generated and a PIN encryption key exists in the FIT for local PIN verification, you must manually translate the PIN encryption key from encryption under the old master key to encryption under the new master key, replace the PIN encryption key cryptogram in the FIT, and download the updated FIT data to the ATM terminal.

The PIN encryption key is entered in the PEKEY field on the Financial Institution Table screen, which you can access from the Diebold Entry Program Menu. For additional information on this field and downloading FIT data, refer to the *BASE24-atm Diebold 10XX/478X Device Support Manual*.

## Migrating from Single-Length to Double-Length Keys

After you have successfully completed all phases of AKDS implementation for a Diebold 10XX/478X terminal, you can use the following procedure to migrate the terminal to double-length keys.

1. On the ACI desktop, read the terminal's current record on the Channel Security Configuration window. Change the Key Length field for the exchange key and inbound key from a value of Single to a value of Double.

If you are using Thales e-Security HSMS, you may also need to change the External Key Block Format field from a value of Variant method to ANSI X9.17 method. If you are using Atalla NSPs, the External Key Block Format field must be set to Standard Atalla.

Click Save () to save the changes in the CSECD.

2. Warmboot the Transaction Security Services and AKDS processes to reread the Channel Security data source (CSECD) record into memory. You can warmboot these processes to either reread all CSECD records from disk into memory using the [Alter Data Source](#) process control command or just reread the specified CSECD terminal record into memory using the [CSEC Refresh](#) process control command.
3. Send the Load Master Key command to the terminal to generate and load a new ATM master key. Refer to the *BASE24-atm Diebold 10XX/478X Device Support Manual* for detailed Load Master Key command syntax. Another event message is generated when the processing for this command is successfully completed.
4. Generate and download new double-length communications keys (i.e., PIN and/or MAC keys) using the appropriate commands entered from the DCT or a network control facility. As before, check the event message log to make sure the command completed successfully.
5. Bring the ATM back into service. Transactions entered at the ATM should now be processed using the new double-length communications key.

# Shared Wincor Nixdorf Signature Terminal Implementation Procedures

Implementation of remote initial key processing for Wincor Nixdorf terminals running in NCR NDC+ or Diebold 912 emulation mode is accomplished in five phases, each of which are described below. The first two phases are performed by Wincor Nixdorf alone; ACI and its customers are not involved. These phases are performed out-of-band of the BASE24 system.

**Note:** The device vendor does not have to be Wincor Nixdorf as long as the device is compliant with Wincor Nixdorf's shared signature specification and is equipped with an encrypting PIN pad (EPP) to support the BASE24-atm Automated Key Distribution System add-on product. The following procedures assume Wincor Nixdorf as the device vendor. If you are using another vendor's device, the implementation procedures may differ from those presented here.

## Phase 1—Generate a Wincor Nixdorf RSA Key Pair

The Wincor Nixdorf Customer Key Center – International (CKCI) generates a customer-specific RSA key pair composed of a private (secret) key and a public key. These keys are used for signature authentication between a Wincor Nixdorf ATM and the BASE24 system. The private key is used to generate signatures, while the public key is used to verify signatures. This key pair is referred to as the root Wincor Nixdorf key pair throughout this appendix. The root Wincor Nixdorf public key is sent (through an offline method) to you during phase 3.

## Phase 2—Manufacture Encrypting PIN Pad (EPP)

When the EPP for a Wincor Nixdorf terminal is manufactured, Wincor Nixdorf generates an RSA public and private key pair for each EPP. These keys are used for the encryption and decryption of keys exchanged between the EPP and the BASE24 system. The EPP's public and private keys are stored in the EPP. The public key for the EPP is signed with the root Wincor Nixdorf private key. This signature allows the BASE24 system to validate the EPP using the root Wincor Nixdorf public key created in phase 1. The signature is stored in the EPP along with the root Wincor Nixdorf public key. The root Wincor Nixdorf public key is used to validate a signature received from the BASE24 system. Finally, the serial number stored in the EPP is signed by the root Wincor Nixdorf private key. This signature is used for additional authentication between the EPP and the BASE24 system. At this point, the following key information is stored in the EPP:

- The EPP's public and private keys
- The EPP's public key signature generated by the root Wincor Nixdorf private key
- The root Wincor Nixdorf public key
- The EPP's serial number
- The EPP's serial number signature generated by the root Wincor Nixdorf private key

## Phase 3— Generate an RSA Key Pair for the BASE24 System

To support automated key distribution for ATMs, you must also generate an RSA key pair for the BASE24 system using a hardware security module. This is accomplished by entering the KEYGEN command on the Process Control window of the ACI Desktop. Once the key pair is generated, the Wincor Nixdorf Customer Key Center – International (CKCI) must sign the public key for the BASE24 system and return the resulting signature to you along with Wincor Nixdorf's root public key. You must use this key to verify the signature returned from Wincor Nixdorf. Finally, you must store all these keys in the TSS database.

**Note:** The procedures are essentially the same whether you are generating an RSA key pair for a test or production environment on the BASE24 system. For testing, a test EPP provided by Wincor Nixdorf must be used instead of a production EPP. When you initially contact Wincor Nixdorf to start the process of submitting a public key and getting it signed by Wincor Nixdorf, you must specify whether the key to be signed is for testing or production. You should use the same forms and procedures for getting both test and production keys. Wincor Nixdorf Customer Key Center – International (CKCI) signs test keys using a root public key that is different from the one it uses to sign production keys.

The above tasks are accomplished in the following steps:

1. To start the process of submitting a public key and getting it signed by Wincor Nixdorf, contact your Wincor Nixdorf representative and request a Key Form.
2. Generate an RSA public and private key pair for the BASE24 system. Enter the KEYGEN command on the Process Control window of the ACI Desktop to generate the RSA key pair for the BASE24 system. Refer to the Key Generate (KEYGEN) Command topic provided later in this document for detailed information on using this command.



3. Access the Certificate Management window on the ACI Desktop (access **Configure > Transaction Security > Automated Key Distribution > Certificate Management**) to retrieve the BASE24 system public key and format it in the manner expected by the Wincor Nixdorf Customer Key Center – International (CKCI). When the Certificate Management window is displayed, select the Wincor Signature device type in the Device Type field. The BASE24 system public key is retrieved, formatted, and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The BASE24 system public key is displayed in ASN.1 format. This is the message format required when sending the public key to the Wincor Nixdorf Customer Key Center – International (CKCI).
4. Enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Host Public Key Data tab in the Digest (SHA-1) of Customer's Public Key field on the Wincor Nixdorf Key Form. Choose a Key Length of 2048 bits, and an Exchange Format of ASCII Hex. Choosing the ASCII Hex option ensures that the signature of your BASE24 system public key returned by Wincor Nixdorf will be in the correct format to enter into the ACI Desktop. After completing the remainder of the Key Form, fax it to the Wincor Nixdorf Customer Key Center – International (CKCI).
5. When requested by Wincor Nixdorf, copy the BASE24 system public key cryptogram from the Host Public Key Data field on the Certificate Management window into a text file. Send the text file on the appropriate media to Wincor Nixdorf following the format and procedures requested by the Wincor Nixdorf Customer Key Center – International (CKCI).
6. The Wincor Nixdorf Customer Key Center – International (CKCI) validates the SHA-1 hash value for your BASE24 system. Wincor Nixdorf will contact you to indicate whether the hash value is correct. If it is incorrect, you must resend the BASE24 system public key and hash value. If it is correct, the Wincor Nixdorf Customer Key Center – International (CKCI) creates a signature of the BASE24 system public key, generates a SHA-1 hash value for the root Wincor Nixdorf public key, and returns them back to you in the media format you requested. You will receive two files. Each file contains a

key or hash value in ASCII Hex format. You must copy the contents of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table.

| <b>Value (Wincor Nixdorf Filename)</b>                                                                                                                                                           | <b>Certificate Management Field</b>                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Root Wincor Nixdorf public key<br>pkSignWN<br>(pkSignWN<cust>T.mpk if this is a test key, or<br>pkSignWN<cust>P.mpk if this is a production key)                                                 | Generate Message Digest ><br>Certificate Authority Public Key<br>(step 7a)<br>and<br>Validate Certificates/Signatures ><br>Certificate Authority Public Key<br>(step 11a) |
| SHA-1 hash of Wincor Nixdorf public key<br>Digest public key, calculated with SHA1<br>(pkSignWN<cust>T.mpk if this is for a test key, or<br>pkSignWN<cust>P.mpk if this is for a production key) | Generate Message Digest > SHA-1 Hash Value (step 7b)                                                                                                                      |
| BASE24 system public key signature<br>(siHSM<cust>T.sig if this is for a test key, or<br>siHSM<cust>P.sig if this is for a production key)                                                       | Validate Certificates/Signatures > Host Signature (step 11b)                                                                                                              |

7. On the Generate Message Digest tab of the Certificate Management window, perform the following:
  - a. Copy the root Wincor Nixdorf public key cryptogram from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field. Remove all spaces between the digits of the key. You can click Expand View to expand this field if needed.
  - b. Enter the SHA-1 hash value of the root Wincor Nixdorf public key into the SHA-1 Hash Value field.
  - c. Click Save.

8. If you are using an Atalla NSP, the NSP validates the SHA-1 hash value of the root Wincor Nixdorf public key and generates a message digest of the root Wincor Nixdorf public key, which is displayed in the Message Digest field. Any validation errors are reported in screen messages on the Certificate Management window. If the SHA-1 hash value is incorrect, you must request Wincor Nixdorf to resend the root Wincor Nixdorf public key and SHA-1 hash value. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab.
9. For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. On the SCT, use the Utilities Menu -> Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the message digest as two 16-byte values (Left Challenge and Right Challenge on the SCT or Component block 1 and Component block 2 on the SCA). The results are two 16-byte values that comprise the root certificate hash value.
10. For an Atalla NSP only, copy or enter the 32-byte (16-byte left and 16-byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab.
11. On the Validate Certificates/Signatures tab of the Certificate Management window, perform the following:
  - a. Copy the root Wincor Nixdorf public key returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.
  - b. Copy the BASE24 system public key signature returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Host Signature field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.
  - c. Click Save.

The Atalla NSP validates the signature of the BASE24 system public key received from Wincor Nixdorf. If validated successfully, the BASE24 system public key signature is stored in the Host Public Key Security Primary File (HSPKPD).

For an Atalla NSP only, the root Wincor Nixdorf public key is validated using the hash value input from step 10, imported, and stored in the HSPKPD along with the message digest. Any validation errors are reported in screen messages on the Certificate Management window. If the key signature is incorrect, you must request that Wincor Nixdorf resend the BASE24 system public key signature.

Once the above steps are completed, the following key data is now entered into the database for the AKDS process:

- The BASE24 system's public and private keys
- The signature of the BASE24 system's public key signed by the root Wincor Nixdorf private key
- The root Wincor Nixdorf public key

## Phase 4—Authenticate and Exchange Public Keys

Before you can automatically distribute (download) an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. You authenticate and exchange public keys with a Wincor Nixdorf terminal (running in NDC+ or Diebold 912 emulation mode) online by sending the Load Public Key command (entered as LOADPUBLIC or LPUBLIC) to the terminal's NCR 5XXX or Diebold 10XX/478X Device Handler process using the Device Control Terminal (DCT) or a network control facility. Refer to the *BASE24-atm NCR 5XXX Device Support Manual* or the *BASE24-atm Diebold 10XX/478X Device Support Manual* for detailed Load Public Key command syntax and processing steps performed during this phase.

The Load Public Key command initiates an online exchange of public keys between the BASE24 system and the EPP of a specified terminal.

When you enter the Load Public Key command, the NCR 5XXX or Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. This message requests the ATM to send remote key loading (RKL) information and EPP capabilities to the BASE24 system. If the ATM does not have an AKDS-capable EPP installed, it rejects this request. If the ATM responds, the NCR 5XXX or Diebold 10XX/478X Device Handler process checks the Signature/Certificate status in the response. If the status is not 1 (Signature) or 3 (Signature and certificate), the process aborts the load and re-opens the ATM. Otherwise, it continues the load and requests the EPP serial number and signature from the ATM by sending a Terminal Command message with a command code indicating "Send RKL Information, EPP Serial Number and Signature." The ATM sends the EPP serial number and signature to the BASE24

system. The Device Handler requests the Transaction Security Services process to verify the signature using the root Wincor Nixdorf public key and store the EPP serial number and signature. The Transaction Security Services process returns the BASE24 system public key and signature, which the Device Handler process encodes and downloads to the terminal. The EPP in the terminal verifies the signature of the BASE24 system public key using the root Wincor Nixdorf public key and stores the BASE24 system public key in the EPP. The Device Handler process then requests the terminal to send its EPP public key to the BASE24 system. The EPP sends the EPP public key and signature to the BASE24 system. The Device Handler requests the Transaction Security Services process to verify the signature and store the EPP public key in the TSS database. In addition, one or more event messages are generated and the process reopens the ATM.

During this phase, the AKDS process interacts with the hardware security module to perform the following:

- Verify the signature of the ATM's EPP serial number using the root Wincor Nixdorf public key retrieved from the Host Public Key Security Primary File (HSPKPD).
- Optionally verify the EPP serial number itself against valid EPP serial numbers entered in the EPP Serial Number File (EPNUMD). The value of the ATM-DH-N50-AKDS-SERIAL-NUM-VRFY or ATM-DH-1000-AKDS-SERIAL-NUM-VRFY LCONF param determines whether this additional validation is performed. Valid EPP serial numbers must be manually entered on the EPP Serial Number Configuration window before this validation is performed. Information entered on the EPP Serial Number Configuration window is stored in the EPNUMD. Refer to the "Data Access" topic provided later in this appendix for more information on the EPP Serial Number Configuration window and the EPNUMD.
- Retrieve the BASE24 system public key and signature from the HSPKPD. These are sent to the ATM after the EPP serial number signature has been successfully validated.
- Verify the signature of the EPP's public key using the root Wincor Nixdorf public key retrieved from the HSPKPD. If the signature is valid, the process generates a MAC of the EPP public key and stores the EPP public key, and the EPP public key MAC in the CHNPKD.

The NCR 5XXX or Diebold 10XX/478X Device Handler process sends the BASE24 system public key and signature to the ATM's EPP, where they are verified and stored.

At this point, the following key information is stored in the EPP:

- The EPP's public and private keys
- The EPP's public key signature generated by the root Wincor Nixdorf private key
- The root Wincor Nixdorf public key
- The EPP's serial number
- The EPP's serial number signed by the root Wincor Nixdorf private key
- The BASE24 system's public key and signature

At this point, the following key data is entered into the database for the AKDS process:

- The BASE24 system's public and private keys
- The signature of the BASE24 system's public key signed by the root Wincor Nixdorf private key
- The root Wincor Nixdorf public key
- A MAC of the root Wincor Nixdorf public key
- The EPP's public key
- The EPP's serial number
- A MAC of the EPP's public key

## Phase 5—Download the Master Key

After an EPP has been successfully authenticated and exchanged public keys with the BASE24 system and the key entry mode has been set, you can securely download the ATM master key to the terminal using the Load Master Key command (entered as LOADMASTER or LMASTER). You can enter this command from the DCT or a network control facility. This command specifies the ID of the terminal to which the ATM master key is to be downloaded. After the master key is successfully loaded in both the EPP and the TSS database, you can securely download working keys to the EPP. Refer to the ***BASE24-atm NCR 5XXX Device Support Manual*** or ***BASE24-atm Diebold 10XX/478X Device Support Manual*** for detailed Load Master Key command syntax and processing steps performed during this phase.

When you enter the Load Master Key command, the NCR 5XXX or Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. This message requests the ATM to send remote

key loading (RKL) information and EPP capabilities to the BASE24 system. If the ATM does not have an AKDS-capable EPP installed, it rejects this request. If the ATM responds, the NCR 5XXX or Diebold 10XX/478X Device Handler process checks the Signature/Certificate status in the response. If the status is not 1 (Signature) or 3 (Signature and certificate), the process aborts the load and re-opens the ATM. Otherwise, it continues the load by requesting the Transaction Security Services process to generate a new master key and signature and downloading the new key and signature to the terminal. The EPP in the terminal verifies the signature using the BASE24 system public key, decrypts the master key using the EPP's private key, and stores the master key in the EPP. The terminal then sends a six-byte key verification value (KVV) for the master key just loaded back to the BASE24 system. Finally, the Device Handler process requests the Transaction Security Services process to validate the KVV.

During this phase, the AKDS process interacts with the hardware security module to generate a new master key encrypted under the EPP's public key and sign it using the BASE24 system private key. The new master key and associated check digits are stored in the Channel Security File (CSECD). The new master key and signature are also sent to the ATM.

At this point, both the BASE24 system and the EPP have the terminal's master key stored in their respective databases. Working keys (i.e., the PIN encryption key and MAC key) can now be securely downloaded to the EPP.

## **Financial Institution Table (FIT) Data**

The PIN encryption key in the FIT is encrypted under the master key. If a new master key is generated and a PIN encryption key exists in the FIT for local PIN verification, you must manually translate the PIN encryption key from encryption under the old master key to encryption under the new master key, replace the PIN encryption key cryptogram in the FIT, and download the updated FIT data to the ATM.

If your Wincor Nixdorf terminal is running in NDC+ emulation mode, the PIN encryption key is entered in the PEKEY (PIN ENCRYPTION KEY) field on the Financial Institution Table screen, which you can access from the NCR 5XXX Configuration File Menu in the Device Control Terminal (DCT). For additional information on this field and downloading FIT data, refer to the ***BASE24-atm NCR 5XXX Device Support Manual***.

If your Wincor Nixdorf terminal is running in Diebold 912 emulation mode, the PIN encryption key is entered in the PEKEY field on the Financial Institution Table screen, which you can access from the Diebold Entry Program Menu. For additional information on this field and downloading FIT data, refer to the ***BASE24-atm Diebold 10XX/478X Device Support Manual***.



# Load All Keys Command

The Load All Keys command performs the same processing as the Load Public Key and Load Master Key commands as well as downloading the PIN and MAC encryption keys to the ATM. The command syntax and processing performed for the Load All Keys command is slightly different between NCR 5XXX terminals, Diebold 10XX/478X terminals, and Wincor Nixdorf terminals running in NDC+ or Diebold 912 emulation mode. Therefore, they are described separately below.

## NCR 5XXXLoad All Keys Command

When you enter the Load All Keys command (entered as LOADALLKEYS or LALLKEYS), the NCR 5XXX Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM's configuration response message, it determines whether the ATM terminal has an encryptor type of 7 or 8. It also checks the TERMINAL LINE PROTOCOL field on NCR 5XXX screen 7 of the BASE24-atm Terminal Data file (ATD) to determine whether the ATM is using the B (Bisync), C (Bisync – ASCII), or S (SDLC) protocols. An encryptor type of 7 or 8 indicates that the ATM has an AKDS-capable EPP installed. A protocol value other than B, C, or S indicates that the ATM is using a protocol that supports the BASE24-atm Automated Key Distribution System add-on product. If the ATM has an AKDS-capable EPP, the requested load command is allowed to proceed. If the ATM is not capable of processing AKDS commands and is using a protocol that supports AKDS, the Load All Keys command is not performed. In addition, one or more event messages are generated and the process reopens the ATM.

If the ATM supports AKDS, the NCR 5XXX Device Handler process sequentially performs the following for the Load All Keys command:

1. Performs [phase 4](#)—authenticate and exchange public keys.
2. Performs [phase 5](#)—download the master key
3. Downloads the MAC encryption key if the MAC SUPPORT TYPE field on NCR 5XXX screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 3 for the ATM.
4. Downloads the PIN encryption key if the PIN ENCRYPT TYPE field on NCR 5XXX screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 01, 03, 04, or 05 for the ATM.
5. Reopens the ATM.

If the ATM does not support AKDS, the NCR 5XXX Device Handler process sequentially performs only steps 3 and 4 above for the Load All Keys command.

Refer to the *BASE24-atm NCR 5XXX Device Support Manual* for the detailed syntax of the Load All Keys command.

## Diebold 10XX/478X Load All Keys Command

When you enter the Load All Keys command (entered as LOADALLKEYS), the Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. When the process receives the ATM's configuration response message, it searches the Additional Hardware Configuration Status area for the value "EP0104RK". The presence of this value indicates the ATM has an AKDS-capable EPP installed. If this value is found, the requested load command is allowed to proceed. If this value is not found, the Load All Keys command only loads the PIN and MAC encryption keys. In addition, one or more event messages is generated and the process reopens the ATM.

If the ATM supports AKDS, the Diebold 10XX/478X Device Handler process sequentially performs the following for the Load All Keys command:

1. Performs [phase 3](#)—exchange certificates
2. Performs [phase 4](#)—download the master key
3. Downloads the MAC encryption key if the MAC SUPPORT TYPE field on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 3 for the ATM.
4. Downloads the PIN encryption key if the PIN ENCRYPT TYPE field on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 01, 03, 04, or 05 for the ATM.
5. Reopens the ATM.

If the ATM does not support AKDS, the Diebold 10XX/478X Device Handler process sequentially performs only steps 3 and 4 above for the Load All Keys command.

Refer to the *BASE24-atm Diebold 10XX/478X Device Support Manual* for the detailed syntax of the Load All Keys command

## Wincor Nixdorf Load All Keys Command

When you enter the Load All Keys command (entered as LOADALLKEYS or LALLKEYS), the NCR 5XXX or Diebold 10XX/478X Device Handler process closes the specified ATM and sends a configuration request message. This message requests the ATM to send remote key loading (RKL) information and EPP capabilities to the BASE24 system. If the ATM does not have an AKDS-capable EPP installed, it rejects this request. If the ATM responds, the NCR 5XXX or Diebold 10XX/478X Device Handler process checks the Signature/Certificate status in the response. If the status is not 1 (Signature) or 3 (Signature and certificate), the process aborts the load, generates one or more event messages, and reopens the ATM.

If the ATM supports AKDS, the NCR 5XXX or Diebold 10XX/478X Device Handler process sequentially performs the following for the Load All Keys command:

1. Performs [phase 4](#)—authenticate and exchange public keys.
2. Performs [phase 5](#)—download the master key
3. Downloads the MAC encryption key if the MAC SUPPORT TYPE field on NCR 5XXX screen 9 or on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 3 for the ATM.
4. Downloads the PIN encryption key if the PIN ENCRYPT TYPE field on NCR 5XXX screen 9 or on Diebold 10XX/478X screen 9 of the BASE24-atm Terminal Data files (ATD) is set to a value of 01, 03, 04, or 05 for the ATM.
5. Reopens the ATM.

If the ATM does not support AKDS, the NCR 5XXX or Diebold 10XX/478X Device Handler process sequentially performs only steps 3 and 4 above for the Load All Keys command.

Refer to the *BASE24-atm NCR 5XXX Device Support Manual* or *BASE24-atm Diebold 10XX/478X Device Support Manual* for the detailed syntax of the Load All Keys command.

## Key Generate (KEYGEN) Command

The KEYGEN command enables you to generate an RSA key pair for the BASE24 system. This command is only entered one time, unless the existing BASE24 system RSA key pair is corrupted in the TSS database. After you have successfully generated an RSA key pair for the BASE24 system and had the public key signed by a certificate authority, you can disable use of the KEYGEN command from the Disable KEYGEN window. You can also re-enable the command from this window if it is needed at a later time. See the “Disabling the KEYGEN Command” topic for detailed information on the Disable KEYGEN window.

The security device used for the KEYGEN command is disabled during key generation, which may take up to 90 seconds. Therefore, you should check to make sure you always have at least one other hardware security module in an enabled state in your production system to authorize transactions before entering this command. Otherwise, a situation could arise where no hardware security modules are available for online transaction processing. This is especially important when testing and production systems share the same hardware security modules.

For the *host-name*, *common-name*, and *org-name* variables in the KEYGEN command syntax, the following characters are allowed: A–Z, a–z, 0–9, *b*, (, ), +, -, ., /, :, ?, and #, where *b* represents a blank space. You cannot use a comma “,” or exclamation point “!” within these variables because they are reserved to denote the end of a variable value and to override the current status in the Host Public Key Security Files, respectively. For shared Wincor Nixdorf signature devices, the *host-name* variable in this command must be set to the value of the KEY SIGNER field on screen 22 of the BASE24-atm Terminal Data files (ATD). If this field is not configured, it defaults to a value of HOST. For NCR and Diebold devices, the *host-name* variable must always be set to a value of HOST for the Automated Key Distribution System to work properly.

---

### Syntax

---

KEYGEN *host-name*, *dev-typ*, [*common-name*, *org-name*] [,!]

*host-name* — For shared Wincor Nixdorf signature devices, this is the name of the host BASE24 system as configured in the KEY SIGNER field on screen 22 of the BASE24-atm Terminal Data files (ATD). If this field is not configured, it defaults to a value of HOST. For NCR and Diebold devices, this variable must always be set to a value of HOST. The key pair is stored under this name in the Host Public Key Security Files. 1–16 alphanumeric characters. Required.

*dev-typ* — The type of ATM device to which keys are to be distributed. Required.  
Valid values are:

|                            |                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------|
| NCR or 0                   | = NCR 5XXX devices                                                                             |
| DIEBOLD, DBLD, or 1        | = Diebold 10XX/478X devices                                                                    |
| WINCORSIG,<br>WINSIG, or 2 | = Shared Wincor Nixdorf signature devices<br>running in NDC+ or Diebold 912 emulation<br>mode. |

*common-name* — The specific entity for which the BASE24 system public key certificate is being requested. This value is used in generating the BASE24 system public key certificate for use with Diebold devices. This variable is not used for NCR devices or shared Wincor Nixdorf signature devices running in Diebold 912 emulation or NCR NDC+ emulation modes; however the intervening comma used to separate this variable is always required. 1–64 alphanumeric characters.

*org-name* — The name of the organization or company for which the BASE24 system public key certificate is being requested. This value is used in generating the BASE24 system public key certificate for use with Diebold devices. This variable is not used for NCR devices or shared Wincor Nixdorf signature devices running in Diebold 912 emulation or NCR NDC+ emulation modes; however the intervening comma used to separate this variable is always required if the command line includes the override flag. 1–64 alphanumeric characters.

! — An optional override flag. This flag forces the KEYGEN command to be performed regardless of current status in the Host Public Key Security Files. Normally, you would only use this flag if the database has been corrupted.

**Note:** You can enter the KEYGEN command from the Process Control window on the ACI desktop or from an XPNET network control facility. The following procedure assumes the use of the Process Control window. For more information on entering this command from a network control facility, refer to [section 9](#).

To use the KEYGEN command, perform the following steps:

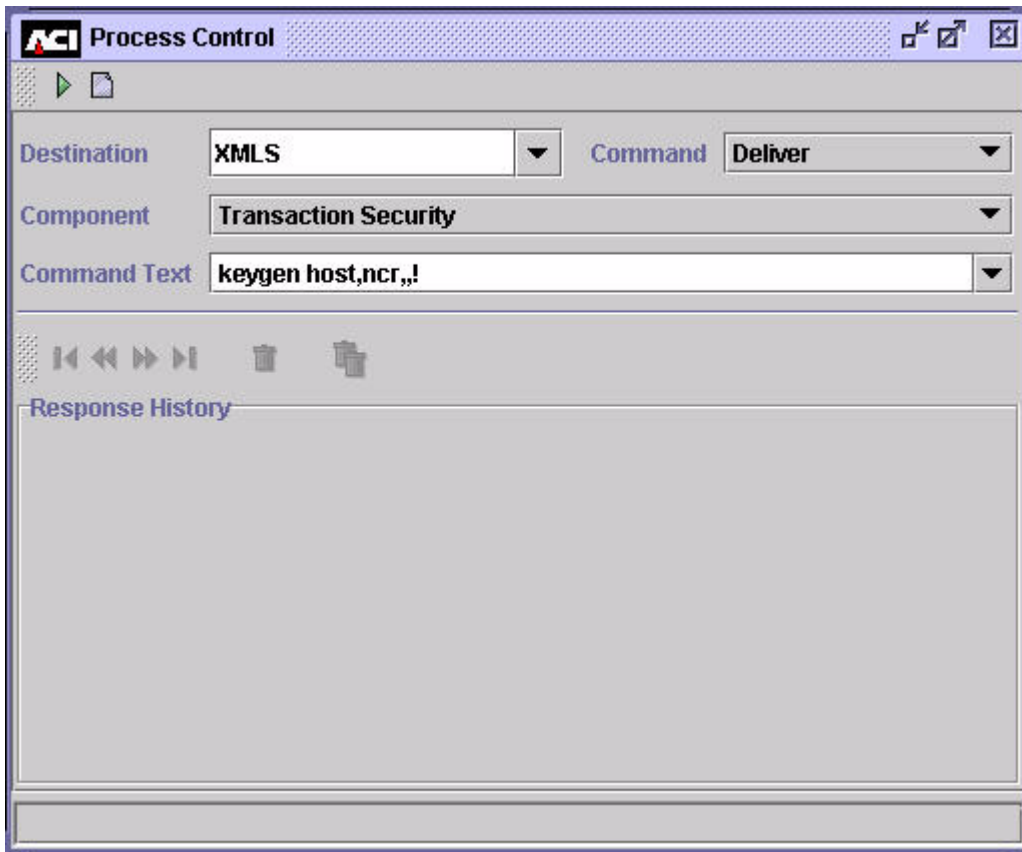
1. Log on to the ACI desktop user interface and navigate to the Process Control window (access **System Operations > Process Control**). If necessary, make sure the appropriate hardware security module (i.e., one with PKI-enabled software or firmware) is enabled.

2. On this window, set the Destination field to XMLS or the symbolic name of the XML Server process, the Command field to Deliver, the Component field to Transaction Security, and the Command Text field according to the command syntax described above. The following are examples of the command text:

```
KEYGEN HOST,NCR,,!  
KEYGEN HOST,DBLD,BASE24,BANK1  
KEYGEN B24PRO1,WINSIG,,
```

3. Click Submit (▶) to send the command to the hardware security device. The device generates the key pair and returns the keys to the Transaction Security component, which stores them in the Host Public Key Security Primary File (HSPKPD). Check the command response to make sure the command was successfully processed.

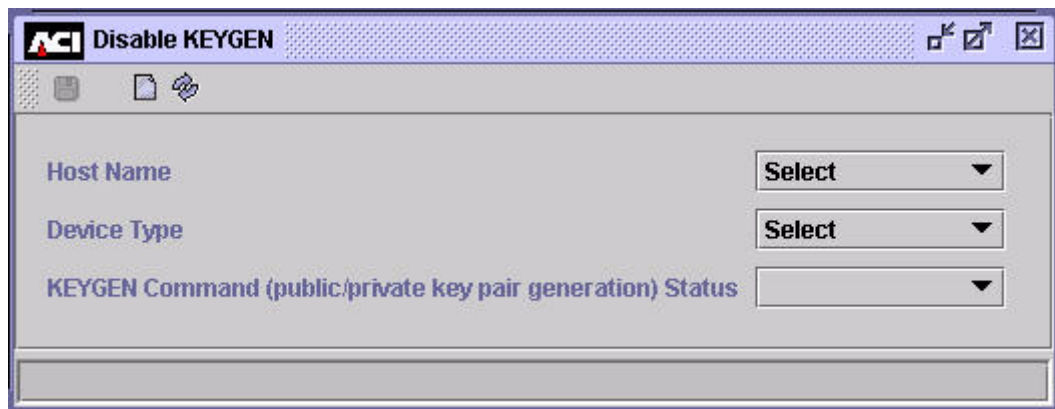
An example of the Process Control window with the KEYGEN command is shown below.



## Disabling the KEYGEN Command

The Disable KEYGEN window allows you to enable or disable the KEYGEN process control command from the Process Control window or from an XPNET network control facility. You access the Disable KEYGEN window from **Configure > Transaction Security > Automated Key Distribution > Disable KEYGEN**.

If you enter the KEYGEN command after you have already used it to successfully generate an RSA key pair for the BASE24 system (host) and had the public key signed by a certificate authority, the currently signed keys in the Host Public Key Security Primary File (HSPKPD) are deleted and replaced with the new keys. In this case, the new keys must be signed by the certificate authority, which can take a number of days to complete. To prevent this from happening inadvertently, you can use the Disable KEYGEN window to disable the KEYGEN process control command. You would do this after you have used the command once to successfully generate the RSA key pair for the BASE24 system, had the key pair signed by a certificate authority, and stored the keys in the HSPKPD. A sample of this window is shown below, followed by descriptions of its fields.



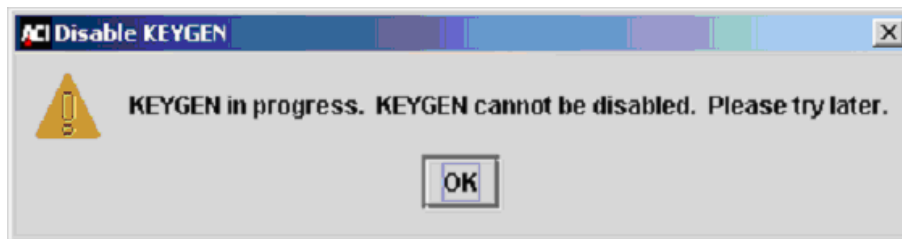
The Host Name field specifies the name of the BASE24 system (host) public key owner in the HSPKPD.

The Device Type field allows you to select the type of device (NCR, Diebold, or Wincor Signature) for which the KEYGEN command is to be disabled or enabled.

The Enable/Disable KEYGEN command (public/private key pair generation) field allows you to change the status of the public/private key generation for the BASE24 system (host) public key pair to a value of Enabled or Disabled. When the KEYGEN command is entered, the Transaction Security Services process

checks the status value in the HSPKPD to determine whether the command can be performed. The command is executed only if an RSA key generation is not currently in progress and the KEYGEN command is enabled.

The Disabled value effectively disables the use of the KEYGEN command from the Process Control window or from an XPNET network control facility, while the Enabled value enables the command. To disable the KEYGEN command, select the desired device type in the Device Type field, select the Disabled value in the Enable/Disable KEYGEN command field, and click Save (📁). The value is stored in the HSPKPD. To re-enable the KEYGEN command, select the desired device type in the Device Type field, select the Enabled value in the Enable/Disable KEYGEN command field, and click Save (📁). The value is stored in the HSPKPD. If you attempt to disable the KEYGEN command while an RSA key generation is in progress, the following dialog box is displayed.



Because of the sensitive nature of the KEYGEN command, user access to the Disable KEYGEN window should be highly restricted. The task ID and default permission ID associated with this window are KEYGENLockdown and FullKEYGENLockdown, respectively. In addition, users granted security access to the Disable KEYGEN window must also be granted at least view access to the Host Public Key Perusal window to retrieve the drop-down list for the Host Name field.



## Data Access

When the BASE24-atm Automated Key Distribution System add-on product is installed, two additional TSS database files are required to support NCR, Diebold, or shared Wincor Nixdorf signature terminals— the Host Public Key Security Primary File (HSPKPD) and Channel Public Key Security File (CHNPKD). An additional TSS database file is also required for Diebold terminals to store secondary keys— the Host Public Key Security Secondary File (HSPKSD). Another TSS database file—the EPP Serial Number File (EPNUMD)—is optional for all three types of terminals.

Data is entered into the HSPKPD and HSPKSD using the KEYGEN command entered on the Process Control window and data entered on the Certificate Management window of the ACI desktop when generating an RSA key pair for the BASE24 system. Data is automatically entered into the CHNPKD when authenticating and exchanging public keys with an ATM terminal. After data is entered in these files, you can only peruse the data using the Host Public Key Perusal and Channel Public Key Perusal windows on the ACI desktop (access **View > Automated Key Distribution > Host Public Key Perusal** or **Channel Public Key Perusal**).

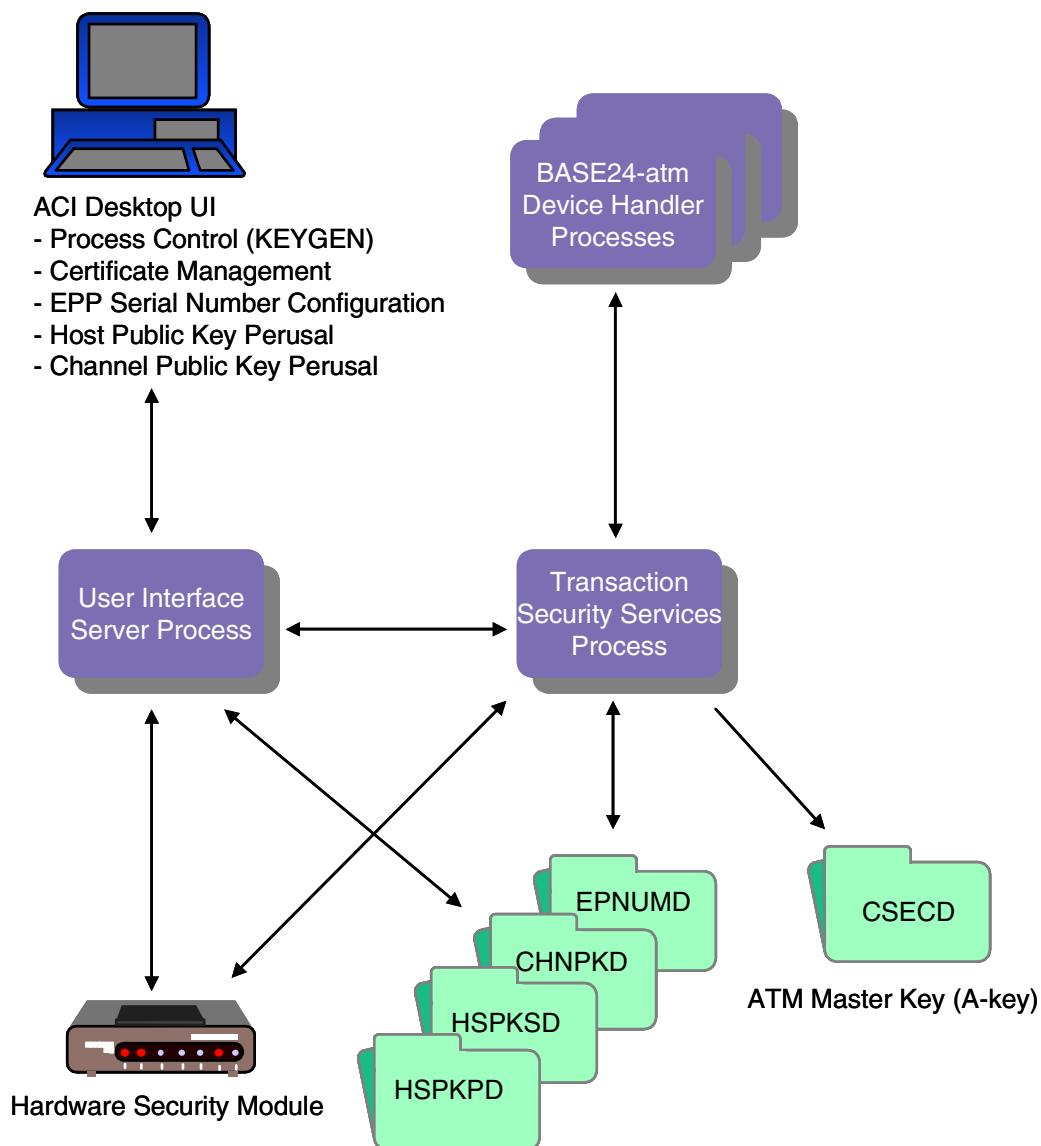
Data is entered into the EPNUMD using the EPP Serial Number Configuration window on the ACI desktop (access **Configure > Transaction Security > EPP Serial Number**).

**Note:** The HSPKPD, HSPKSD, CHNPKD, and EPNUMD are not accessed as memory tables.

When the AKDS process generates a new master key, it stores the new key in the Channel Security File (CSECD).

The diagram on the following page shows you the relationships between the HSPKPD, HSPKSD, CHNPKD, and EPNUMD and the processing components that access them.

Descriptions of the Certificate Management window, EPP Serial Number Configuration window, Host Public Key Perusal window, and Channel Public Key Perusal window are provided following the diagram.



## Certificate Management Window

The Certificate Management window enables you to perform several functions when generating an RSA key pair for the BASE24 system. For NCR devices, these functions are performed during [phase 3](#). For Diebold devices, these functions are performed during [phase 2](#). For shared Wincor Nixdorf signature devices, these functions are performed during [phase 3](#). The Certificate Management window has three tabs, each of which are described below.

### Host Public Key Data

The Host Public Key Data tab on the Certificate Management window enables you to retrieve the BASE24 system public key after it is generated using the KEYGEN command and format it the manner that the target certificate authority expects it.

When the Certificate Management window is displayed, select the device type in the Device Type field. The BASE24 system public key is retrieved and displayed in the Host Public Key Data field. For NCR and shared Wincor Nixdorf signature devices, the key is displayed in ASN.1 format. For NCR devices, the BASE24 system public key is self-signed; for shared Wincor Nixdorf signature devices, the BASE24 system public key is not self-signed. In addition, a 40-byte Secure Hash Algorithm 1 (SHA-1) hash value of the BASE24 system public key is displayed in the SHA-1 Hash Value field for NCR and shared Wincor Nixdorf signature devices. Note that the signature of the key is included in the SHA-1 hash calculation for NCR devices. For Diebold devices, the key is displayed in PKCS#10 Certificate Request format, Base64 encoded. This is the message format required when sending the public key to the Diebold certificate authority. For Diebold devices, the SHA-1 Hash Value field is not used.

For NCR devices, the certificate authority is the NCR trusted center. To start the process of submitting a public key and getting it signed by NCR, contact your NCR/Reseller Account Manager. The account manager will then establish your name, mailing address, e-mail address, telephone and fax numbers and advise NCR Customer and Sales Support to send you the proper forms, format, and procedures.

For shared Wincor Nixdorf signature devices, the certificate authority is the Wincor Nixdorf Customer Key Center – International (CKCI). To start the process of submitting a public key and getting it signed by the Wincor Nixdorf Customer Key Center – International (CKCI), contact your Wincor Nixdorf representative and request a Key Form.

For Diebold, the certificate authority is the Digital Signature Trust Co. (DST). To begin the process of submitting a public key certificate and getting it signed by DST, ask your Diebold System Engineer to contact the Diebold Product Manager for their Remote Key product using an e-mail message. The Diebold Product Manager will then send you an e-mail message containing the address of the DST-hosted web site, your authorization code (password), and an HTML demonstration of the registration website. Each time you need new certificates (e.g., for test, production, and certification) you must contact your Diebold System Engineer to request a new authorization code.

**NCR Example.** A sample of the Certificate Management window after selecting the host name and NCR device type is shown below. In this example, the BASE24 system public key is a self-signed key displayed in ASN.1 format. You can copy the BASE24 system public key cryptogram in the Host Public Key Data field, format it as required, and send it on the appropriate media (e.g, diskette or CD-ROM) to the NCR trusted center to be validated and signed.

For NCR devices, the SHA-1 Hash Value field displays the SHA-1 of the ANSI.1 formatted BASE24 system public key. You must enter the value displayed in this field into the SHA-1 (Fields 1–6) field on form NCR\_003 - Signature Request sent to the NCR trusted center.

**ACI Certificate Management - HOST-NCR**

Host Name: **HOST** Device Type: **NCR**

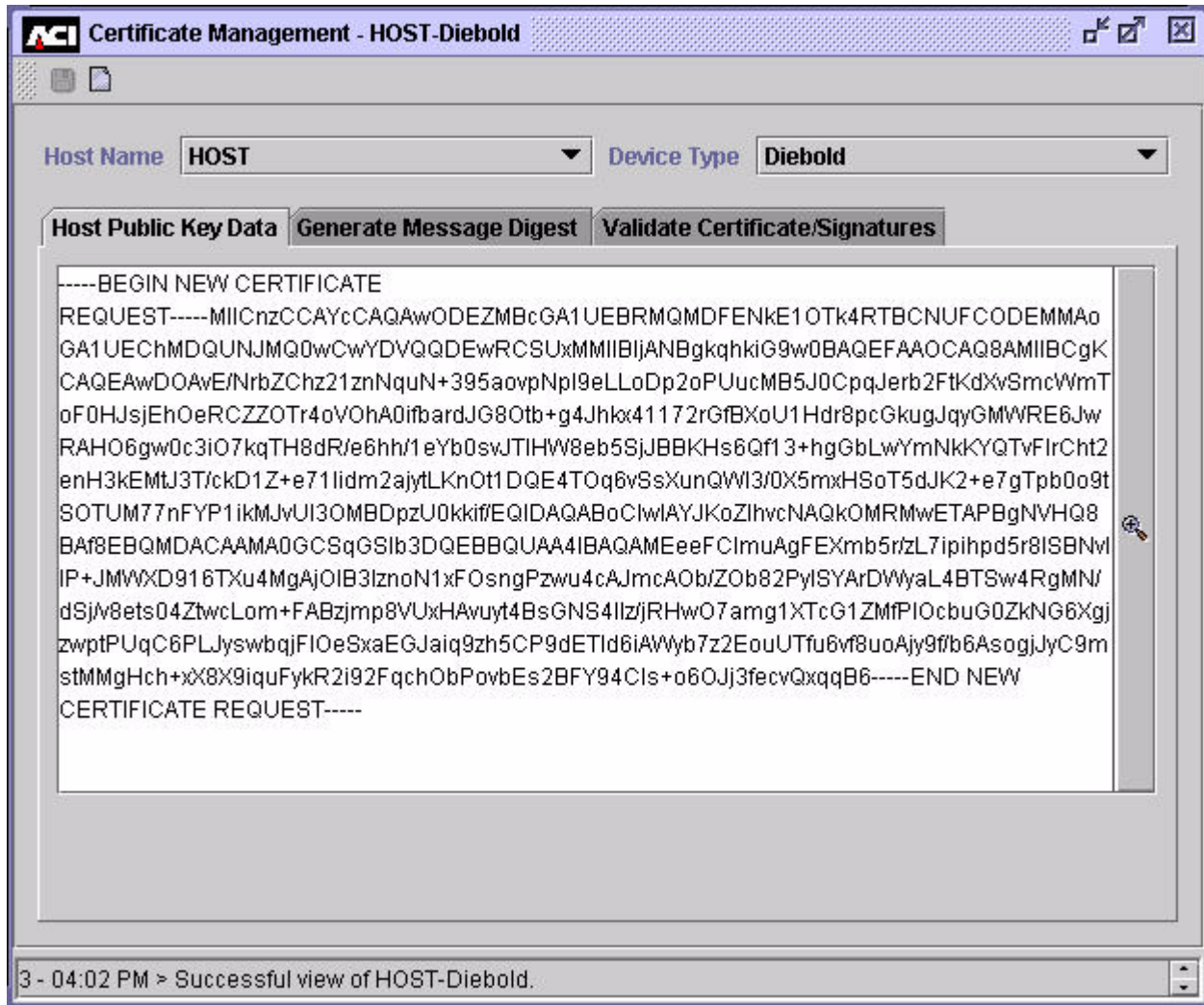
Host Public Key Data | Generate Message Digest | Validate Certificate/Signatures

3082010A0282010100C65A10B240B9026BFDDB1A3E368F5010D5748A8A88BC6586876986B5B  
 2E27D60398660ACC85AA89D91592B7275F71A450B19149DE10A64F2569F0672B11729FAA5EB  
 971680E197E1D31F4886CFEEFCBBA7DE7894A31A8A74C76237BBBE09D6614220DB4D7B4367  
 803CEECF2518C9C24FD9B0BFC76AFE6E99A0BA59010B829DDE68965A614685F606783CCF6F  
 227B38DD1BDBDFE86B3E1BA132198275F188C2A126E6B97D19B7FA1AB4FB05D9DB91D80CA  
 917EFD08E9837E0D955BF3076278577C12EF927B3D2C57B63FEE1CA9AE035B924489FA4B8B  
 3AC00DBE81D08D165D25B2F044894089EE01F3670993999FFFA5538C1ED3E419416859D936  
 204B4A726250203010001A309BD3B9FE7228C4837C0C95A5AB0348A91B6CA0F4307F1E71538  
 5996E9873F1899FE03A0FC39C0B729C9AF424754730C3460D6A80C576E6880536EBE644E88F  
 0C26F40378808E52E9C296108288B9468047E8B34945F772E85CFC46E3D7E0345C7BDCB4D1  
 51E454702D1DA1A35AC9123EED164A64AF11D37B0E8B425D31870A27B75A35A5901D62F6830  
 DE5AF7ADC2FE9F2957C81F9E82BF0DDA37828D0F47B7F3202D0E837A4317BF25ECD1E42205  
 78FFCB843E4028198B3B563F5DE84EB851499952EA90F84B78538E0A05447107A517DEFFFC8  
 F7D05E28FC28C76C7ED0D5BA47FB0D452FD1568DD4A515EBD77F616ADF6E4372FC0041A79  
 AB2407DFEAD1

SHA-1 Hash Value: 96897993EB3C5917A9B27C11965DB24727FAF9AD

1 - 03:46 PM > Successful view of HOST-NCR.

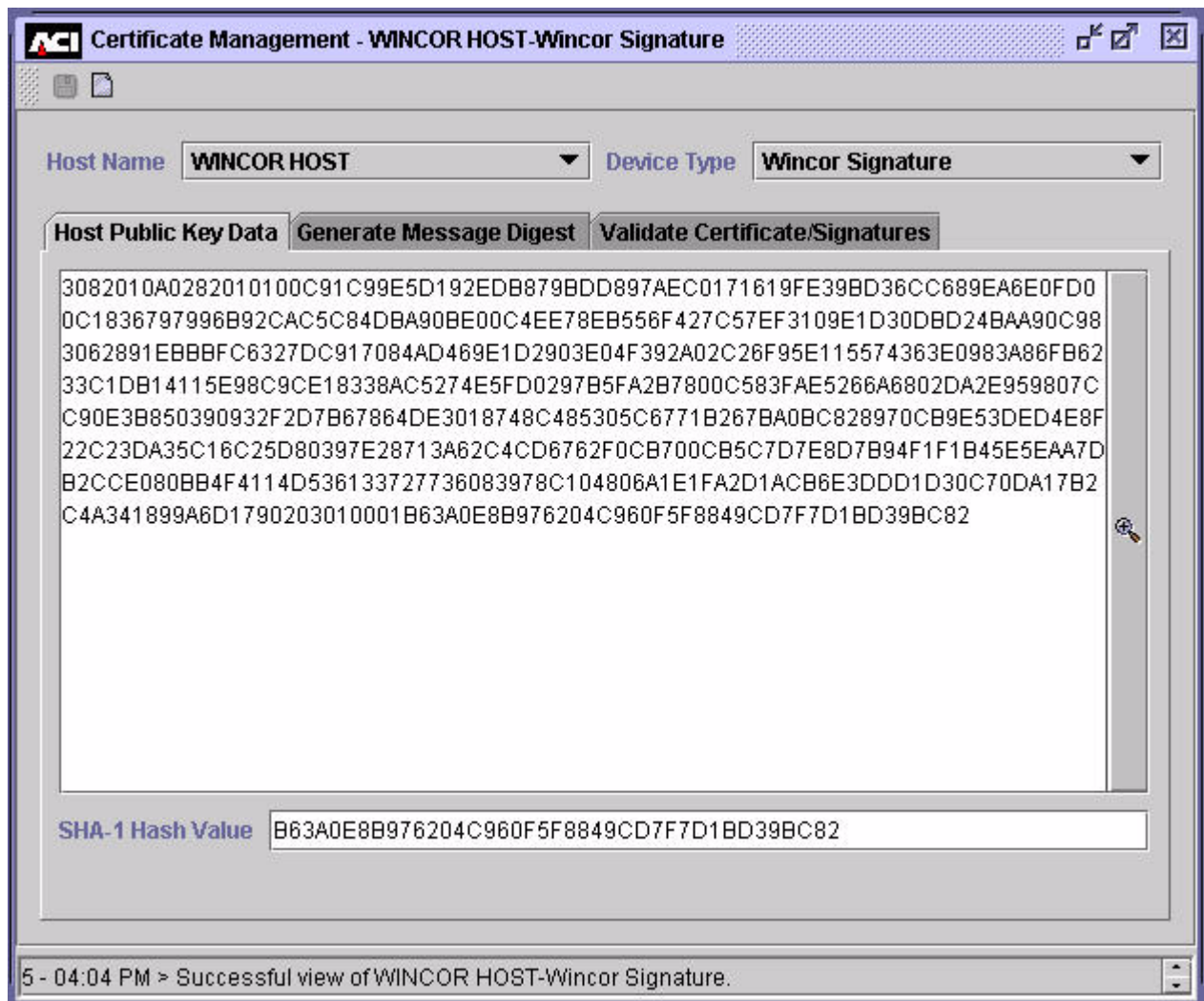
**Diebold Example.** A sample of the Certificate Management window after selecting the Diebold device type is shown below. The BASE24 system public key is displayed in PKCS#10 Certificate Request format, Base64 encoded. Note that the SHA-1 Hash Value field is not displayed for Diebold devices.



**Shared Wincor Nixdorf Signature Example.** A sample of the Certificate Management window after selecting the host name and Wincor Signature device type is shown below. In this example, the BASE24 system public key is displayed in ASN.1 format, but is not self-signed. You can copy the BASE24 system public key cryptogram in the Host Public Key Data field, format it as required, and send it on the appropriate media (e.g, e-mail or diskette) to the Wincor Nixdorf Customer Key Center – International (CKCI) to be validated and signed.



For shared Wincor Nixdorf signature devices, the SHA-1 Hash Value field displays the SHA-1 of the ANS.1 formatted BASE24 system public key. You must enter the value displayed in this field into the Digest (SHA-1) of Customers Public Key field on the Key Form sent to the Wincor Nixdorf Customer Key Center – International (CKCI).



## Generate Message Digest

The Generate Message Digest tab on the Certificate Management window enables you to create a message digest of the public key returned from the certificate authority. For NCR devices, a message digest is created for the self-signed root NCR public key returned from the NCR trusted center. For Diebold devices, a

message digest is created for the root certificate authority certificate returned from DST for Diebold devices. For shared Wincor Nixdorf signature devices, a message digest is created for the root Wincor Nixdorf public key returned from Wincor Nixdorf Customer Key Center – International (CKCI). A message digest is only required if you are using Atalla NSPs in your BASE24 system.

For NCR and shared Wincor Nixdorf signature devices, this tab also enables you to validate the SHA-1 hash value of the root certificate authority public key (i.e., the root NCR public key returned to you on form NCR\_004 - NCR Public Key or the root Wincor Nixdorf public key returned to you from the Wincor Nixdorf Customer Key Center – International (CKCI). The SHA-1 hash value only needs to be validated for NCR and shared Wincor Nixdorf signature devices.

If you are using Diebold devices and Thales e-Security HSMs in your BASE24 system, you do not need to use this tab.

On this tab, you can click Expand View () to expand the Certificate Authority Public Key or Root Certificate Authority Certificate field if needed.

**NCR or Shared Wincor Nixdorf Signature Devices.** To use this tab for NCR or Shared Wincor Nixdorf Signature devices, perform the following:

1. Copy the root certificate authority public key cryptogram into the Certificate Authority Public Key field. For NCR devices, this is the self-signed root NCR public key cryptogram returned from the NCR trusted center. For shared Wincor Nixdorf signature devices, this is the root Wincor Nixdorf public key cryptogram returned from the Wincor Nixdorf Customer Key Center – International (CKCI).
2. Enter the SHA-1 hash value of the root certificate authority public key cryptogram into the SHA-1 Hash Value field. For NCR devices, this is provided in the SHA-1 (Fields 1–6) field on form NCR\_004 - NCR Public Key. For shared Wincor Nixdorf signature devices, this is the “digest public key, calculated with SHA1” returned in the (pkSignWN<cust>T.mpk or pkSignWN<cust>P.mpk file, or the “Digest (SHA-1 Calculation of pkSignWNWINT Field 1-5)” if returned by fax.
3. Click Save ()

If you are using Atalla NSPs, the NSP creates a value that is displayed in the Message Digest field. You must copy this value and encrypt it using the SCT or SCA as described in [step 7 of phase 3](#) for NCR devices or in [step 9 of phase 3](#) for shared Wincor Nixdorf signature devices. The Message Digest field is not displayed or used for Thales e-Security HSMs.



The Atalla NSP or Thales e-Security HSM verifies the signature to ensure that the root public key cryptogram is authentic.

The Atalla NSP or Thales e-Security HSM generates a SHA-1 hash value for the NCR public key cryptogram and compares it to the hash value you entered in the SHA-1 Hash Value field. If the values match, the public key is valid. If the values do not match, either the public key was corrupted or changed during transport or the SHA-1 hash value you entered was incorrectly entered.

A message is displayed at the bottom of the window to indicate the success or failure of the signature validation and SHA-1 hash value comparison.

Examples of the Generate Message Digest tab for NCR devices and shared Wincor Nixdorf signature devices with Atalla NSPs and Thales e-Security HSMs are shown on the following pages.

The following is a sample of the Generate Message Digest tab displayed for NCR devices and Atalla NSPs.

The screenshot shows a software window titled "Certificate Management - HOST-NCR". At the top, there are two dropdown menus: "Host Name" set to "HOST" and "Device Type" set to "NCR". Below these are three tabs: "Host Public Key Data", "Generate Message Digest" (which is selected), and "Validate Certificate/Signatures". The "Generate Message Digest" tab contains three input fields: "Certificate Authority Public Key" (a large text area), "SHA-1 Hash Value" (a single-line text box), and "Message Digest" (a single-line text box). A status bar at the bottom of the window displays the text "7 - 04:07 PM > Successful view of HOST-NCR."

The following is a sample of the Generate Message Digest tab displayed for NCR devices and Thales e-Security HSMs.

The screenshot shows a software window titled "ACI Certificate Management - HOST-NCR". At the top, there are two dropdown menus: "Host Name" with the value "HOST" and "Device Type" with the value "NCR". Below these are three tabs: "Host Public Key Data", "Generate Message Digest" (which is the active tab), and "Validate Certificate/Signatures". The "Generate Message Digest" tab contains two input fields. The first is labeled "Certificate Authority Public Key" and is a large, empty text area. The second is labeled "SHA-1 Hash Value" and is a single-line text input field. At the bottom of the window, a status bar displays the message "9 - 02:13 PM > Successful view of HOST-NCR."

The following is a sample of the Generate Message Digest tab displayed for shared Wincor Nixdorf signature devices and Atalla NSPs.

The screenshot shows a software window titled "ACI Certificate Management - WINCOR HOST-Wincor Signature". At the top, there are two dropdown menus: "Host Name" set to "WINCOR HOST" and "Device Type" set to "Wincor Signature". Below these are three tabs: "Host Public Key Data", "Generate Message Digest" (which is selected), and "Validate Certificate/Signatures". The "Generate Message Digest" tab contains three input fields: "Certificate Authority Public Key" (a large empty box with a key icon on the right), "SHA-1 Hash Value" (a single-line text box), and "Message Digest" (a single-line text box). The status bar at the bottom of the window displays the text "9 - 04:11 PM > Successful view of WINCOR HOST-Wincor Signature."

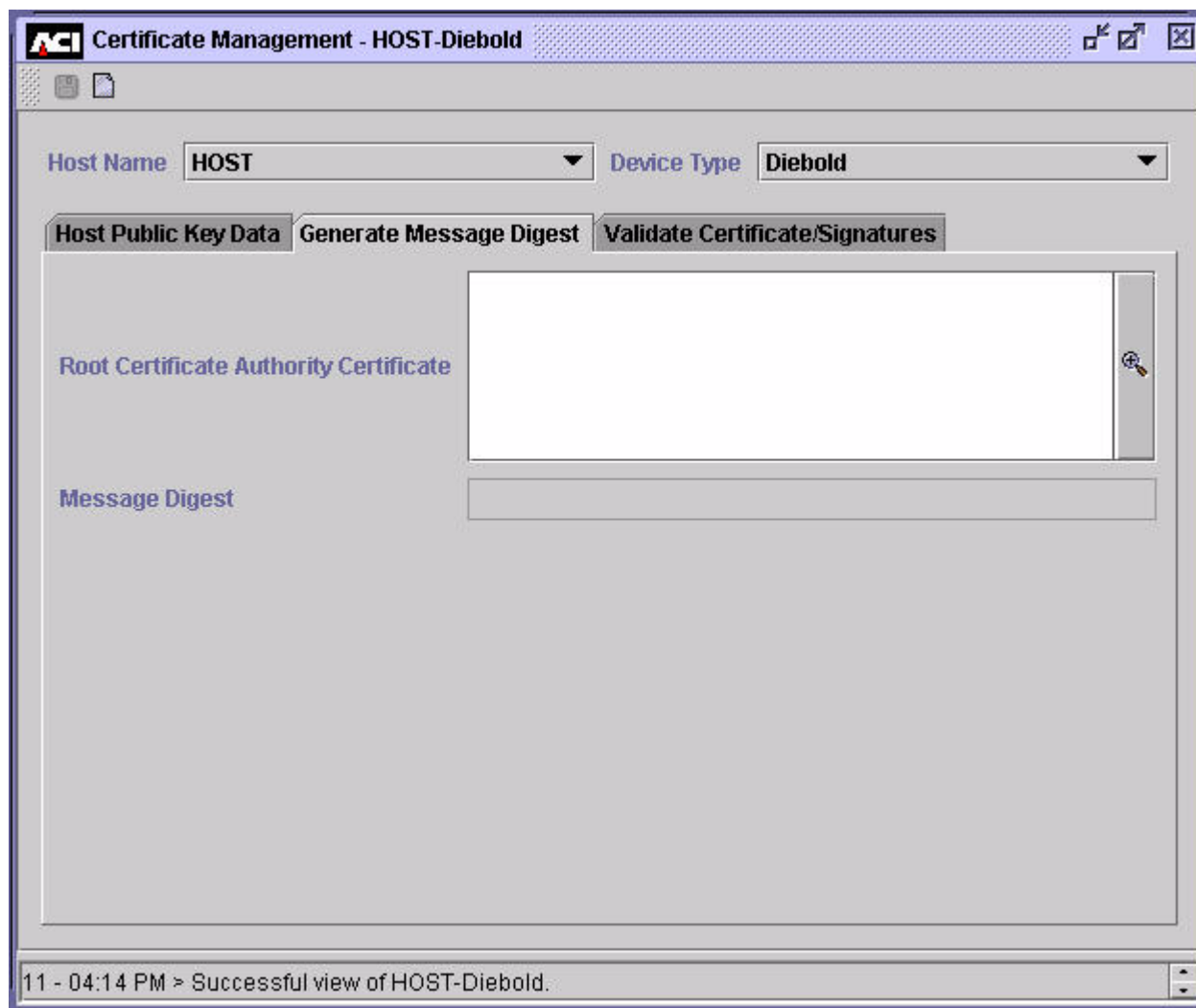
The following is a sample of the Generate Message Digest tab displayed for shared Wincor Nixdorf signature devices and Thales e-Security HSMs.

The screenshot shows a web application window titled "ACI Certificate Management - WINCORHOST-Wincor Signature". The window has a header bar with the ACI logo and title. Below the header, there are two dropdown menus: "Host Name" set to "WINCORHOST" and "Device Type" set to "Wincor Signature". Below these are three tabs: "Host Public Key Data", "Generate Message Digest" (which is selected), and "Validate Certificate/Signatures". The "Generate Message Digest" tab contains two input fields: "Certificate Authority Public Key" and "SHA-1 Hash Value". The "Certificate Authority Public Key" field is a large text area with a small icon on the right. The "SHA-1 Hash Value" field is a single-line text input. At the bottom of the window, a status bar displays the message: "7 - 02:11 PM > Successful view of WINCORHOST-Wincor Signature."

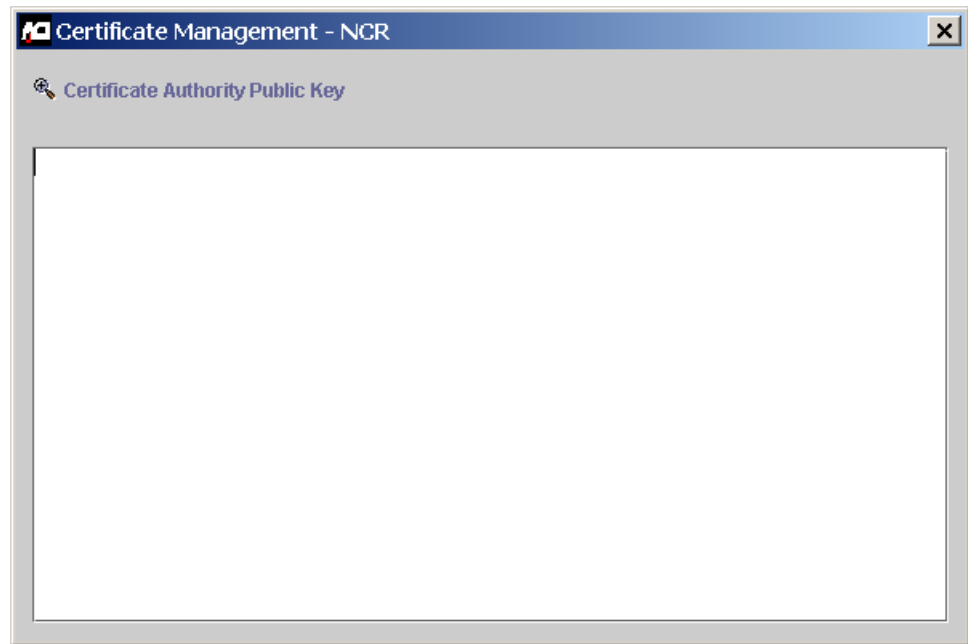
**Diebold Devices.** To use this tab for Diebold devices and Atalla NSPs, perform the following:

1. Copy the root certificate authority certificate returned from DST into the Root Certificate Authority Certificate field.
2. Click Save (📁). The NSP creates a value that is displayed in the Message Digest field. You must copy this value and encrypt it using the SCT as described in [step 5 of phase 2](#) for Diebold devices.

The following is a sample of the Generate Message Digest tab displayed for Diebold devices and Atalla NSPs.



**Expanded Field.** The following image shows you the Certificate Authority Public Key field after it has been expanded.



## Validate Certificates/Signatures

The Validate Certificates/Signatures tab on the Certificate Management window enables you to validate and enter certificate authority and BASE24 system (host) public key information into the TSS database. The fields displayed on this tab differ, depending on the ATM device type selected and type of security devices used.

When you select the NCR device type, the following fields are displayed:

- Root Certificate Hash Value (for Atalla NSPs only)
- Certificate Authority Public Key
- Host Signature
- SHA-1 Hash Value

When you select the Diebold device type, the following fields are displayed:

- Root Certificate Hash Value (for Atalla NSPs only)
- Root Certificate Authority Certificate
- Certificate Authority Primary Certificate
- Certificate Authority Secondary Certificate
- Host Primary Certificate
- Host Secondary Certificate

When you select the Wincor Signature device type, the following fields are displayed:

- Root Certificate Hash Value (for Atalla NSPs only)
- Certificate Authority Public Key
- Host Signature

You can expand the fields for which Expand View () is displayed.

If you are using Atalla NSPs, you must enter the root certificate hash value encrypted from the SCA or SCT into the Root Certificate Hash Value field as described in [step 8 of phase 3](#) for NCR devices, in [step 6 of phase 2](#) for Diebold devices, or in [step 10 of phase 3](#) for shared Wincor Nixdorf signature devices.

For NCR devices, you must also enter the following:

- Self-signed root NCR public key in the Certificate Authority Public Key.
- BASE24 system public key signature returned from the NCR trusted center in the Host Signature field.
- SHA-1 hash value for the BASE24 system public key returned from the NCR trusted center in the SHA-1 (Field 1) field on form NCR\_005 - NCR Signature of Customer Public Key.

For Diebold devices, you must also enter the following:

- The root certificate authority certificate in the Root Certificate Authority Certificate field.
- The primary and secondary certificate authority's public signature verification key certificates in the Certificate Authority Primary Certificate and Certificate Authority Secondary Certificate fields, respectively.



- The host (BASE24 system) primary and secondary public signature verification key certificates in the Host Primary Certificate and Host Secondary Certificate fields, respectively.

For shared Wincor Nixdorf signature devices, you must also enter the following:

- Root Wincor Nixdorf public key in the Certificate Authority Public Key.
- BASE24 system public key signature returned from the Wincor Nixdorf Customer Key Center – International (CKCI) in the Host Signature field.

After all the required data is entered, click Save () to validate signatures and hash values and store data in the Host Public Key Security Primary File (HSPKPD) and Host Public Key Security Secondary File (HSPKSD).

The Atalla NSP or Thales e-Security HSM verifies the signature or certificates returned from the certificate authority to ensure their authenticity.

For NCR devices, the Atalla NSP or Thales e-Security HSM generates a SHA-1 hash value for the BASE24 system public key cryptogram and compares it to the hash value you entered in the SHA-1 Hash Value field. If the values match, the public key is valid. If the values do not match, either the public key was corrupted or changed during transport or the SHA-1 hash value you entered was incorrectly entered.

A message is displayed at the bottom of the window to indicate the success or failure of the signature or certificate validation and SHA-1 hash value comparison.

A sample of the Validate Certificates/Signatures tab is shown below for NCR devices and Atalla NSPs.

The screenshot displays a software window titled "ACI Certificate Management - HOST-NCR". At the top, there are two dropdown menus: "Host Name" set to "HOST" and "Device Type" set to "NCR". Below these are three tabs: "Host Public Key Data", "Generate Message Digest", and "Validate Certificate/Signatures", with the third tab being the active one. The main area contains four input fields with labels to their left: "Root Certificate Hash Value", "Certificate Authority Public Key", "Host Signature", and "SHA-1 Hash Value". The "Certificate Authority Public Key" and "Host Signature" fields have a small key icon on their right side. At the bottom of the window, a status bar shows the text "13 - 04:17 PM > Successful view of HOST-NCR."

A sample of the Validate Certificates/Signatures tab is shown below for NCR devices and Thales e-Security HSM.

The screenshot shows a web application window titled "Certificate Management - HOST-NCR". At the top, there are two dropdown menus: "Host Name" set to "HOST" and "Device Type" set to "NCR". Below these are three tabs: "Host Public Key Data", "Generate Message Digest", and "Validate Certificate/Signatures". The "Validate Certificate/Signatures" tab is active, displaying three input fields on the right side of the window:

- Certificate Authority Public Key**: A large text area with a search icon on the right.
- Host Signature**: A large text area with a search icon on the right.
- SHA-1 Hash Value**: A single-line text input field.

At the bottom of the window, a status bar displays the message: "13 - 02:16 PM > Successful view of HOST-NCR."

A sample of the Validate Certificates/Signatures tab is shown below for Diebold devices and Atalla NSPs.

The screenshot displays the 'ACI Certificate Management - HOST-Diebold' application window. At the top, there are two dropdown menus: 'Host Name' set to 'HOST' and 'Device Type' set to 'Diebold'. Below these are three tabs: 'Host Public Key Data', 'Generate Message Digest', and 'Validate Certificate/Signatures', with the third tab being the active one. The main area contains six rows of input fields for certificates, each with a label on the left and a text box on the right. The labels are: 'Root Certificate Hash Value', 'Root Certificate Authority Certificate', 'Certificate Authority Primary Certificate', 'Certificate Authority Secondary Certificate', 'Host Primary Certificate', and 'Host Secondary Certificate'. Each text box has a small icon (a magnifying glass over a document) to its right, likely for file selection. At the bottom of the window, a status bar shows the text '15 - 04:24 PM > Successful view of HOST-Diebold.'

| Field Name                                  | Input Type                        |
|---------------------------------------------|-----------------------------------|
| Root Certificate Hash Value                 | Text box                          |
| Root Certificate Authority Certificate      | Text box with file selection icon |
| Certificate Authority Primary Certificate   | Text box with file selection icon |
| Certificate Authority Secondary Certificate | Text box with file selection icon |
| Host Primary Certificate                    | Text box with file selection icon |
| Host Secondary Certificate                  | Text box with file selection icon |

15 - 04:24 PM > Successful view of HOST-Diebold.

A sample of the Validate Certificates/Signatures tab is shown below for Diebold devices and Thales e-Security HSMs.

The screenshot shows a software window titled "ACI Certificate Management - HOST-Diebold". At the top, there are two dropdown menus: "Host Name" set to "HOST" and "Device Type" set to "Diebold". Below these are three tabs: "Host Public Key Data", "Generate Message Digest", and "Validate Certificate/Signatures". The "Validate Certificate/Signatures" tab is active, displaying a list of certificate types on the left and corresponding empty input fields on the right. Each input field has a small icon with a magnifying glass and a key, indicating a search or selection function. The certificate types listed are: "Root Certificate Authority Certificate", "Certificate Authority Primary Certificate", "Certificate Authority Secondary Certificate", "Host Primary Certificate", and "Host Secondary Certificate". At the bottom of the window, a status bar displays the message: "11 - 02:15 PM > Successful view of HOST-Diebold."

| Host Name | Device Type |
|-----------|-------------|
| HOST      | Diebold     |

| Host Public Key Data                        | Generate Message Digest | Validate Certificate/Signatures |
|---------------------------------------------|-------------------------|---------------------------------|
| Root Certificate Authority Certificate      |                         |                                 |
| Certificate Authority Primary Certificate   |                         |                                 |
| Certificate Authority Secondary Certificate |                         |                                 |
| Host Primary Certificate                    |                         |                                 |
| Host Secondary Certificate                  |                         |                                 |

11 - 02:15 PM > Successful view of HOST-Diebold.

A sample of the Validate Certificates/Signatures tab is shown below for shared Wincor Nixdorf signature devices and Atalla NSPs.

The screenshot displays a software window titled "Certificate Management - WINCOR HOST-Wincor Signature". The window has a menu bar with "File" and "Edit" options. Below the menu bar, there are two dropdown menus: "Host Name" set to "WINCOR HOST" and "Device Type" set to "Wincor Signature". Below these are three tabs: "Host Public Key Data", "Generate Message Digest", and "Validate Certificate/Signatures", with the third tab being the active one. The main area of the window contains three labels on the left: "Root Certificate Hash Value", "Certificate Authority Public Key", and "Host Signature". To the right of each label is a large, empty rectangular field for data entry. Each of these fields has a small icon (a magnifying glass over a document) on its right side. At the bottom of the window, a status bar displays the text "17 - 04:26 PM > Successful view of WINCOR HOST-Wincor Signature."

A sample of the Validate Certificates/Signatures tab is shown below for shared Wincor Nixdorf signature devices and Thales e-Security HSM.

The screenshot displays a software window titled "ACI Certificate Management - WINCORHOST-Wincor Signature". The window has a menu bar with "File" and "Edit" options. Below the menu bar, there are two dropdown menus: "Host Name" set to "WINCORHOST" and "Device Type" set to "Wincor Signature". A tabbed interface is present with three tabs: "Host Public Key Data", "Generate Message Digest", and "Validate Certificate/Signatures", with the latter being the active tab. The main content area is divided into two sections: "Certificate Authority Public Key" and "Host Signature", each with a large text input field and a small icon on the right. At the bottom, a status bar shows the message "15 - 02:17 PM > Successful view of WINCORHOST-Wincor Signature."

## Host Public Key Perusal Window

The Host Public Key Perusal window (access **View > Automated Key Distribution > Host Public Key Perusal**) enables you to view the BASE24 system public key or certificate and the root public key or certificate of the certificate authority. For Diebold devices, you can also view the primary and secondary certificate authority public key certificates for signature validation.

The BASE24 system public key or certificate is displayed on the Host Public Key tab. The root public key or certificate of the certificate authority is displayed on the Root CA Public Key tab. The primary and secondary certificate authority public key certificates for signature validation are displayed on the CA Primary Public Key and CA Secondary Public Key tabs, respectively. The CA Primary Public Key and CA Secondary Public Key tabs are not applicable when the device type is NCR or Wincor Signature.

When you click on a key element field in the upper display on any tab, the full value for that element is displayed in the Key Value Display portion of the window. For some key elements, you can copy the unformatted key element value onto the clipboard by clicking on the Copy Unformatted button. This can be helpful if you need to enter the value in another application. The following illustration shows



you a sample of the Host Public Key tab of the Host Public Key Perusal window for NCR devices and Atalla NSPs. The fields displayed for Wincor Signature devices are similar.

**Host Public Key Perusal - HOST NCR**

Host Name: **HOST** Device Type: **NCR**

Host Public Key | Root CA Public Key | CA Primary Public Key | CA Secondary Public Key

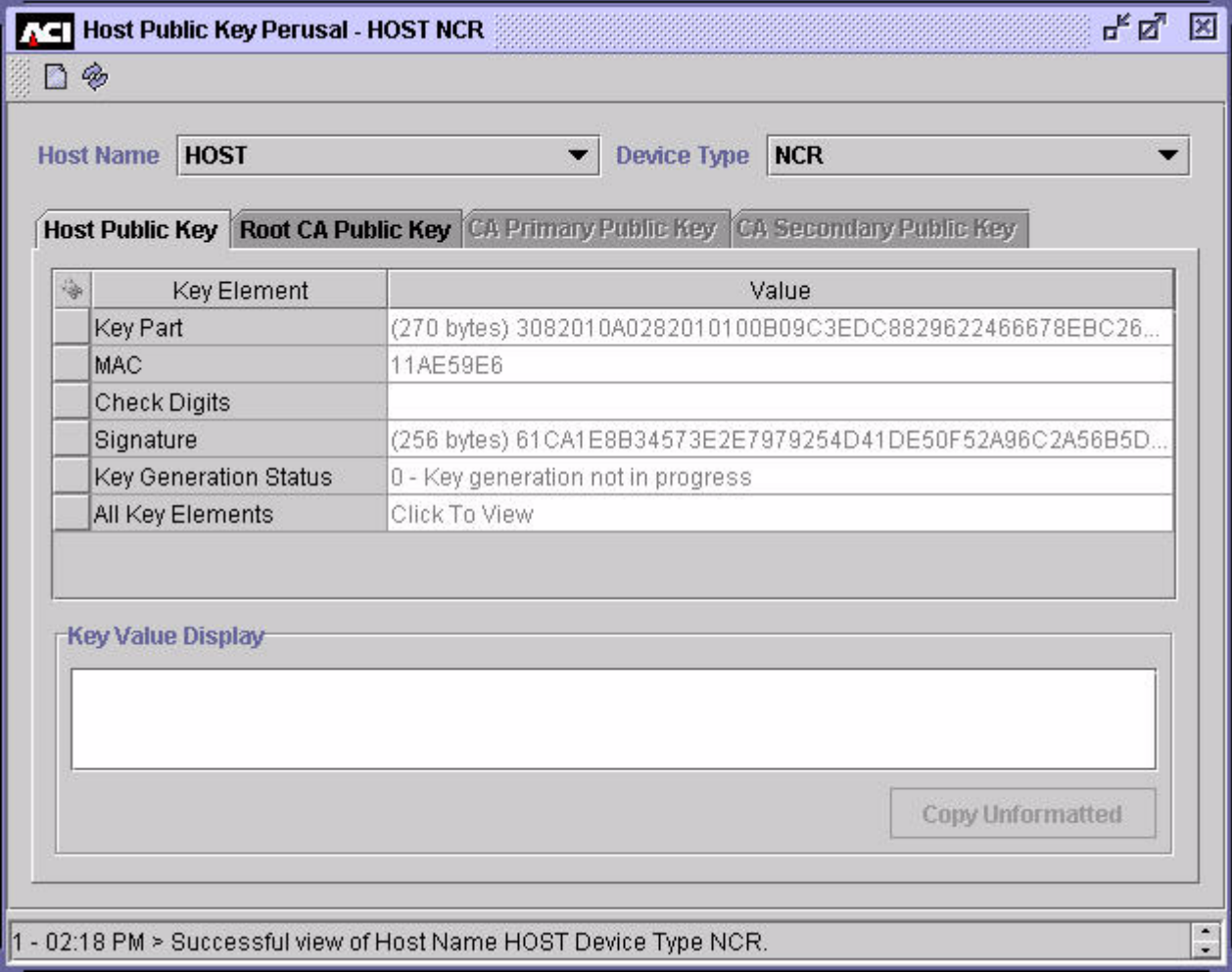
| Key Element           | Value                                                        |
|-----------------------|--------------------------------------------------------------|
| AKB Header            | 1SRVE000                                                     |
| Key Part              | (263 bytes) 00030100010100C65A10B240B9026BFDBB1A3E368F501... |
| AKB MAC               | 9F8E7ACCDFF78ACE                                             |
| Check Digits          | AABD8F                                                       |
| Signature             | (256 bytes) 1905A4CC5E03F2E38662D8358993686D1D15539253D0...  |
| Key Generation Status | 0 - Key generation not in progress                           |
| All Key Elements      | Click To View                                                |

**Key Value Display**

Copy Unformatted

1 - 04:30 PM > Successful view of Host Name HOST Device Type NCR.

The following illustration shows you a sample of the Host Public Key tab of the Host Public Key Perusal window for NCR devices and Thales e-Security HSMs. The fields displayed for Wincor Signature devices are similar.



**Host Public Key Perusal - HOST NCR**

Host Name: **HOST** Device Type: **NCR**

Host Public Key | Root CA Public Key | CA Primary Public Key | CA Secondary Public Key

| Key Element           | Value                                                       |
|-----------------------|-------------------------------------------------------------|
| Key Part              | (270 bytes) 3082010A0282010100B09C3EDC8829622466678EBC26... |
| MAC                   | 11AE59E6                                                    |
| Check Digits          |                                                             |
| Signature             | (256 bytes) 61CA1E8B34573E2E7979254D41DE50F52A96C2A56B5D... |
| Key Generation Status | 0 - Key generation not in progress                          |
| All Key Elements      | Click To View                                               |

**Key Value Display**

Copy Unformatted

1 - 02:18 PM > Successful view of Host Name HOST Device Type NCR.

The next illustration shows you a sample of the Host Public Key tab of the Host Public Key Perusal window for Diebold devices and Atalla NSPs.

**Host Public Key Perusal - HOST Diebold**

Host Name: **HOST** Device Type: **Diebold**

Host Public Key | Root CA Public Key | CA Primary Public Key | CA Secondary Public Key

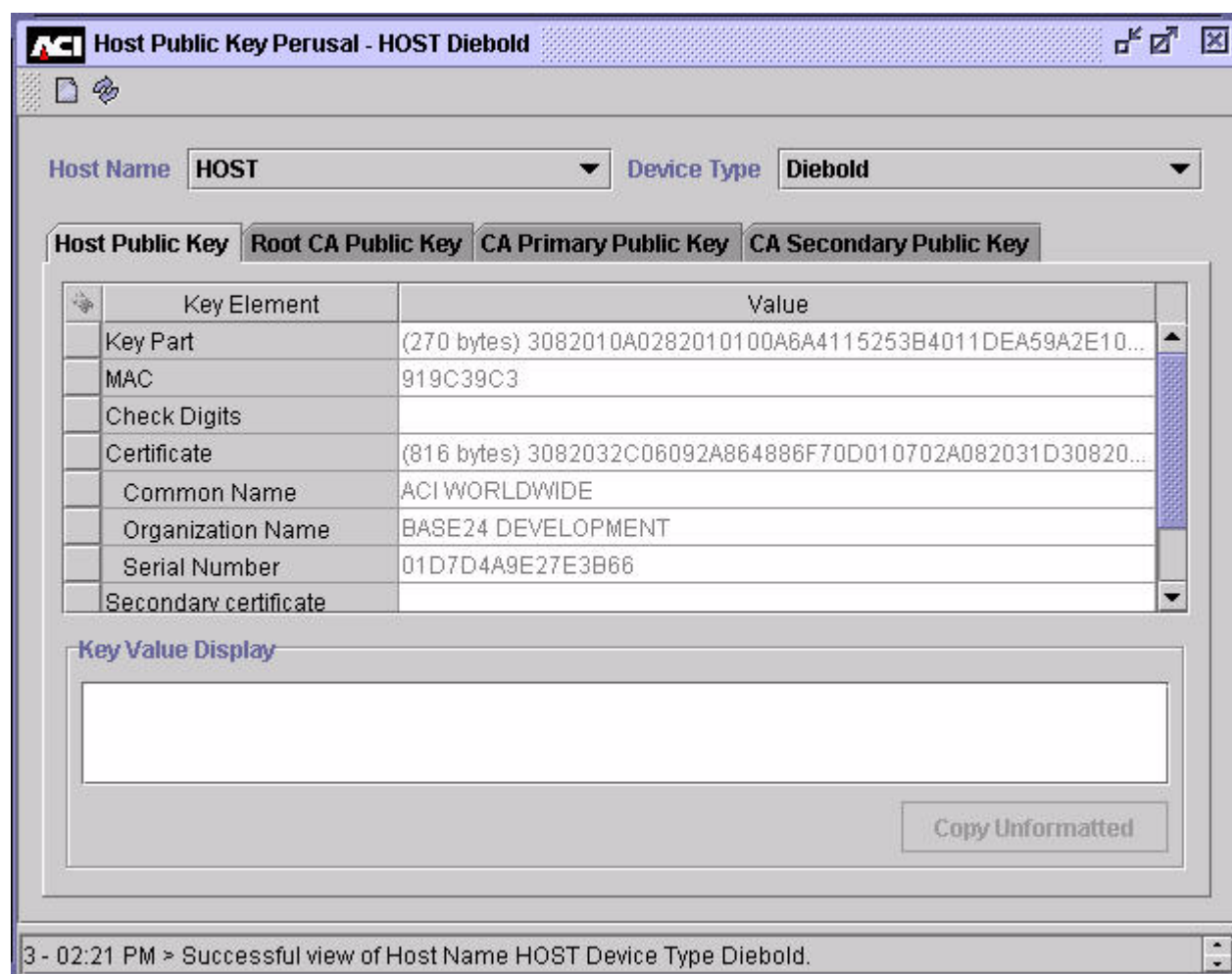
| Key Element       | Value                                                      |
|-------------------|------------------------------------------------------------|
| AKB Header        | 1SRVE000                                                   |
| Key Part          | (263 bytes) 00030100010100C03380BC4FCDADB642873DB5CE7...   |
| AKB MAC           | CA02CB326993913F                                           |
| Check Digits      | F256C5                                                     |
| Certificate       | (816 bytes) 3082032C06092A864886F70D010702A082031D30820... |
| Common Name       | BILL                                                       |
| Organization Name | ACI                                                        |
| Serial Number     | 01D6A5998E0B5AB8                                           |

**Key Value Display**

Copy Unformatted

3 - 04:34 PM > Successful view of Host Name HOST Device Type Diebold.

The next illustration shows you a sample of the Host Public Key tab of the Host Public Key Perusal window for Diebold devices and Thales e-Security HSMs.

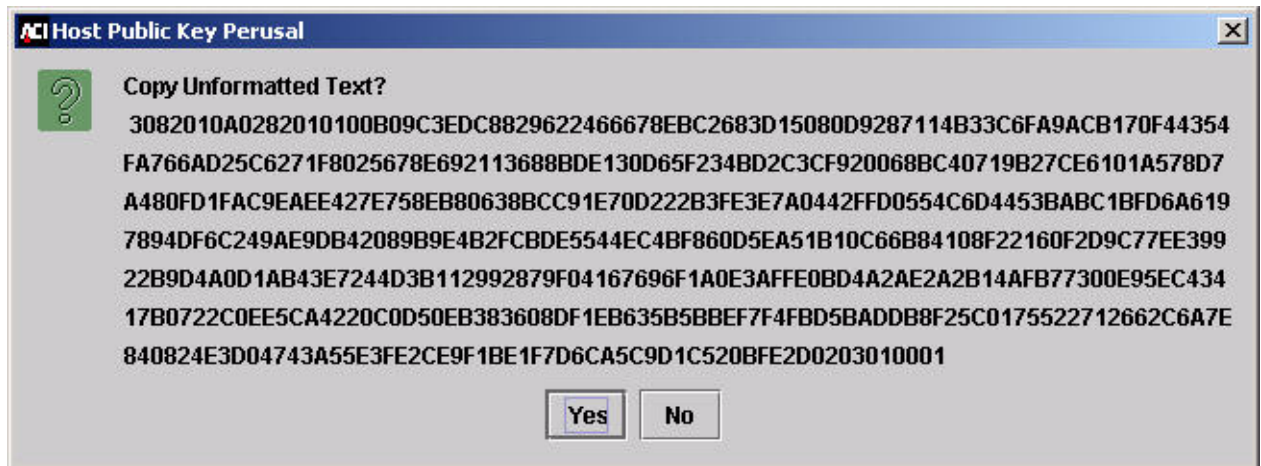


You can copy the key text as formatted or unformatted text from the Host Public Key Perusal window.

To copy the key text as formatted text, display the complete text of the key element in the Key Value Display area. Then copy it to the clipboard by selecting the text and pressing the **Control-C** keys.

To copy the key text as unformatted text, display the complete text of the key element in the Key Value Display area and click the Copy Unformatted button.

The full unformatted value of the selected key element is displayed in a pop-up window, from which you can copy the value to your clipboard by clicking Yes. You can then easily paste the value into another application. A sample of this pop-up window for the Key Part key element is shown below.



## Channel Public Key Perusal Window

The Channel Public Key Perusal window (access **View > Automated Key Distribution > Channel Public Key Perusal**) enables you to view the EPP encipherment and signing public keys for individual terminal IDs (channel IDs). The EPP's public encipherment key is displayed on the EPP Encipherment Key tab. The EPP's public signature verification key is displayed on the EPP Signing Key tab. The EPP Signing tab is only displayed for Diebold devices and is not used for NCR or Wincor Signature devices.

As with the Host Public Key Perusal window, you can display the full value for a key element in the Key Value Display portion of the window and copy key text as formatted or unformatted text using the **Control-C** keys or the Copy Unformatted button. If you do not know the channel ID for the record you want to view, you can search for CHNPKD records on the Channel Public Key Perusal window by clicking Find ()

The following illustration shows you a sample of the EPP Encipherment Key tab for an NCR device and Atalla NSPs on the Channel Public Key Perusal window. The fields displayed for Diebold and Wincor Signature devices are similar.

The screenshot shows a window titled "Channel Public Key Perusal - CHAN\_1K\_NCR". Inside, there is a "Channel ID" field with the value "CHAN\_1K\_NCR" and a "Device Type" field with the value "NCR". Below these fields are two tabs: "EPP Encipherment Key" (selected) and "EPP Signing Key". The "EPP Encipherment Key" tab contains a table with the following data:

| Key Element      | Value                                                              |
|------------------|--------------------------------------------------------------------|
| AKB Header       | 1KREE000                                                           |
| Key Part         | (264 bytes) 0003010001010100C65A10B240B9026BFDBB1A3E368F5010D57... |
| AKB MAC          | 952F62046779E6C1                                                   |
| Check Digits     |                                                                    |
| All Key Elements | Click To View                                                      |

Below the table is a "Key Value Display" section with a large empty text area and a "Copy Unformatted" button.

At the bottom of the window, a status bar displays: "1 - 04:40 PM > Successful view of Channel ID CHAN\_1K\_NCR."



The following illustration shows you a sample of the EPP Encipherment Key tab for an NCR device and Thales e-Security HSMs on the Channel Public Key Perusal window. The fields displayed for Diebold and Wincor Signature devices are similar.

The screenshot shows a software window titled "Channel Public Key Perusal - CHAN\_1K\_NCR". Inside, there's a "Channel ID" field with the value "CHAN\_1K\_NCR" and a "Device Type" field with the value "NCR". Below these are two tabs: "EPP Encipherment Key" (selected) and "EPP Signing Key". The "EPP Encipherment Key" tab contains a table with the following data:

| Key Element      | Value                                                              |
|------------------|--------------------------------------------------------------------|
| Key Part         | (270 bytes) 3082010A0282010100C65A10B240B9026BFDDB1A3E368F5010D... |
| MAC              | DE9334E5                                                           |
| Check Digits     |                                                                    |
| All Key Elements | Click To View                                                      |

Below the table is a section titled "Key Value Display" with a sub-label "Key Part". It contains a text area displaying the following hexadecimal values:

```
3082010A 02820101 00C65A10 B240B902 6BFDDB1A 3E368F50 10D5748A
8A88BC65 86876986 B5B2E27D 60398660 ACC85AA8 9D91592B 7275F71A
450B1914 9DE10A64 F2569F06 72B11729 FAA5EB97 1680E197 E1D31F48
86CFEEFC BBA7DE78 94A31A8A 74C76237 BBBE09D6 614220DB 4D7B4367
```

At the bottom right of this section is a button labeled "Copy Unformatted". The status bar at the bottom of the window displays the message: "3 - 02:55 PM > Successful view of Channel ID CHAN\_1K\_NCR."



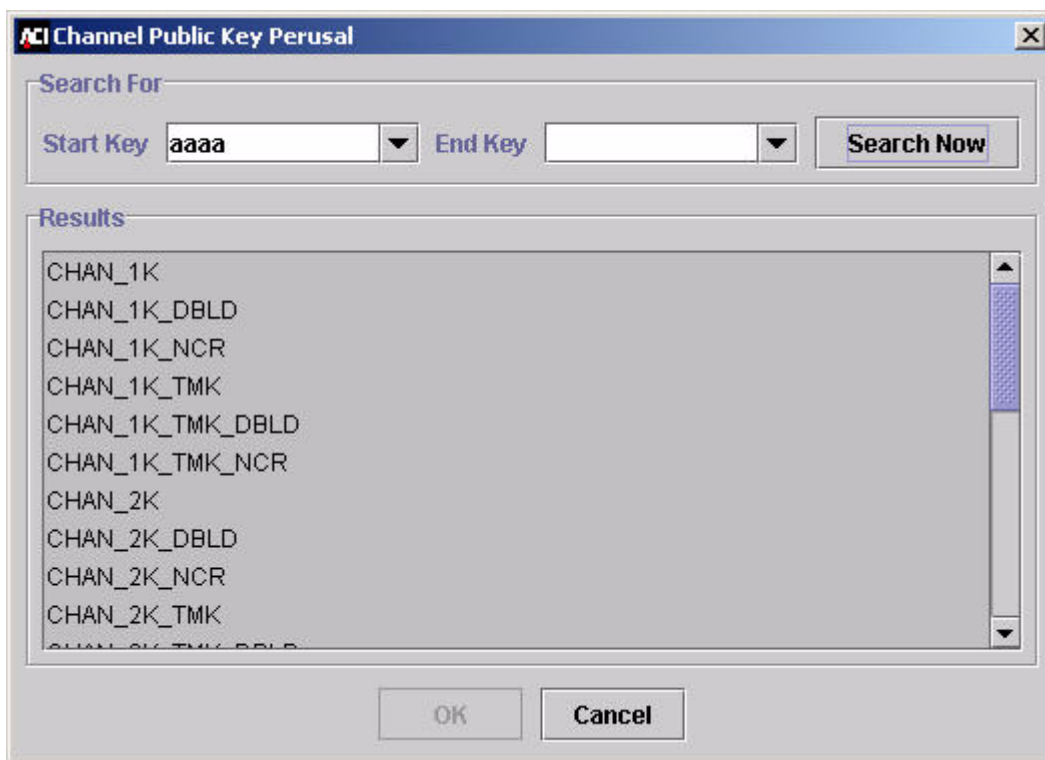
The following illustration shows you a sample of the EPP Signing Key tab for a Diebold device and Atalla NSPs on the Channel Public Key Perusal window. This tab is not used for NCR and Wincor Signature devices.

The screenshot shows a software window titled "Channel Public Key Perusal - CHAN\_2K\_DBLD". At the top, there are fields for "Channel ID" (CHAN\_2K\_DBLD) and "Device Type" (Diebold). Below these are two tabs: "EPP Encipherment Key" and "EPP Signing Key", with the latter being selected. The "EPP Signing Key" tab contains a table with two columns: "Key Element" and "Value".

| Key Element      | Value                                                              |
|------------------|--------------------------------------------------------------------|
| AKB Header       |                                                                    |
| Key Part         | (270 bytes) 3082010A0282010100BA11DA272657E7C15A36F5C8F347AA197... |
| AKB MAC          | F7FB297E                                                           |
| Check Digits     |                                                                    |
| All Key Elements | Click To View                                                      |

Below the table is a section titled "Key Value Display" which contains a large empty text area. To the right of this area is a button labeled "Copy Unformatted". At the bottom of the window, a status bar displays the message: "5 - 02:20 PM > Successful view of Channel ID CHAN\_2K\_DBLD."

When you click Find (  ) on the Channel Public Key Perusal window, the following search window is displayed.



On this window, you can search for CHNPKD records within a specified range of channel ID values. After you click the Search Now button, the channel IDs for CHNPKD records within the specified range are displayed in the Results portion of the window. You can then select individual records to be displayed on the Channel Public Key Perusal window by highlighting a row in the Results area and clicking OK.

## EPP Serial Number Configuration Window

The EPP Serial Number Configuration window (access **Configure > Transaction Security > Automated Key Distribution > EPP Serial Number**) is an optional window that enables you to view, add, and delete valid EPP serial numbers in the EPNUMD. On this window you can also search for existing records in the EPNUMD within a specified serial number range. After a serial number record is added to the EPNUMD from this window, you can clear out the nonkey field values for the record (i.e., everything but the serial number). New values are assigned to these fields when a Device Handler process sends the EPP serial number to the AKDS process in an online request. This enables you to reset information for an EPP, for example, when it needs to be repaired and temporarily taken out of the system.

The AKDS process optionally validates the EPP serial number returned from an ATM against valid numbers stored in the EPNUMD when authenticating and exchanging public keys with an ATM. If the EPP serial number sent from the terminal is found in the EPNUMD, it is considered to be authentic. Otherwise, it is not. The LCONF params ATM-DH-N50-AKDS-SERIAL-NUM-VERFY and ATM-DH-1000-AKDS-SERIAL-NUM-VERFY determine whether this additional validation processing is performed for NCR 5XXX and Diebold 10XX/478X terminals, respectively.

When entering records on the EPP Serial Number Configuration window, the value you enter in the Institution ID field should match the FIID value configured in the BASE24-atm Terminal Data file (ATD) for the terminal.

For Diebold 10XX/478X terminals, you must enter two serial numbers for each channel (terminal) ID. One serial number entry must end with “*b*V” and the other must end with “*b*E”, where *b* is a blank space. The serial number ending with “*b*V” must match the serial number received in the EPP public signature verification key certificate sent from the terminal, while the serial number ending in “*b*E” must match the serial number received in the EPP public encipherment verification key certificate sent from the terminal.

A sample of the EPP Serial Number Configuration window is shown below.

| Serial Number | Channel ID | Validation Date | Validation Time |
|---------------|------------|-----------------|-----------------|
| 05010070 E    | S1AC100003 | July 9, 2003    | 06:09 pm        |
| 05010070 V    | S1AC100003 | July 16, 2003   | 04:06 pm        |

1 - 09:53 AM > Successful view of Institution ID bk32, Device Type Diebold .

# AKDS Capacity Report

When using the BASE24-atm Automated Key Distribution System (AKDS) add-on product, you must generate the AKDS Capacity Report on a monthly basis and send it to ACI. The report indicates the total number of ATM terminals that have been authenticated and exchanged public keys with the BASE24 system. ACI uses this report to reconcile your AKDS usage (number of ATM terminals) against your AKDS capacity license. ACI recommends that you generate and send this report in conjunction with your monthly BASE24 Transaction Based Pricing reports.

The following paragraphs describe the AKDS Capacity Report, how to configure it's output location and the number of lines printed on each page of the report, and how to generate the report using a process control command.

When run, the AKDS Capacity Report reads through the Channel Public Key Security File (CHNPKD), determines the total number of ATM terminals defined in the CHNPKD, and writes the total to specified report stream location (e.g., a file). You can optionally specify the report to list the channel ID of each ATM terminal defined in the CHNPKD.

The following are samples of the AKDS Capacity Report, with and without the channel ID listing. The date, in YY/MM/DD format, is printed at the top of each report page. The page number is printed at the bottom of each report page. If you specify to include a listing of channel IDs, they are printed in alphabetical order from left to right in four columns across the report. The total number of ATM terminals defined in the CHNPKD is listed in the "Total Devices:" line at the end of the report.

The first sample, which consists of two report pages, shows you a report where the channel IDs have been listed.

The second sample, which consists of only one report page, shows you the report run against the same CHNPKD without the optional channel IDs listed.

| AKDS Capacity Report |             |         |
|----------------------|-------------|---------|
| YY/MM/DD             | Device Ids: |         |
| DIEBOLD001           | NCR001      | TERM01  |
|                      | TERM04      | TERM05  |
|                      | TERM07      | TERM09  |
|                      | TERM100     | TERM102 |
|                      | TERM104     | TERM106 |
|                      | TERM108     | TERM111 |
|                      | TERM111     | TERM112 |
|                      | TERM115     | TERM116 |
|                      | TERM119     | TERM12  |
|                      | TERM122     | TERM123 |
|                      | TERM126     | TERM127 |
|                      | TERM13      | TERM130 |
|                      | TERM133     | TERM134 |
|                      | TERM137     | TERM138 |
|                      | TERM140     | TERM141 |
|                      | TERM144     | TERM145 |
|                      | TERM148     | TERM149 |
|                      | TERM151     | TERM152 |
|                      | TERM155     | TERM156 |
|                      | TERM159     | TERM16  |
|                      | TERM162     | TERM163 |
|                      | TERM166     | TERM167 |
|                      | TERM17      | TERM170 |
|                      | TERM173     | TERM174 |
|                      | TERM177     | TERM178 |
|                      | TERM180     | TERM181 |
|                      | TERM184     | TERM185 |
|                      | TERM188     | TERM189 |
|                      | TERM20      | TERM21  |
|                      | TERM24      | TERM25  |
|                      | TERM28      | TERM29  |
|                      | TERM32      | TERM33  |
|                      |             | TERM02  |
|                      |             | TERM06  |
|                      |             | TERM10  |
|                      |             | TERM103 |
|                      |             | TERM107 |
|                      |             | TERM110 |
|                      |             | TERM114 |
|                      |             | TERM118 |
|                      |             | TERM121 |
|                      |             | TERM125 |
|                      |             | TERM129 |
|                      |             | TERM132 |
|                      |             | TERM136 |
|                      |             | TERM14  |
|                      |             | TERM143 |
|                      |             | TERM147 |
|                      |             | TERM150 |
|                      |             | TERM154 |
|                      |             | TERM158 |
|                      |             | TERM161 |
|                      |             | TERM165 |
|                      |             | TERM169 |
|                      |             | TERM172 |
|                      |             | TERM176 |
|                      |             | TERM175 |
|                      |             | TERM18  |
|                      |             | TERM183 |
|                      |             | TERM187 |
|                      |             | TERM190 |
|                      |             | TERM23  |
|                      |             | TERM27  |
|                      |             | TERM31  |
|                      |             | TERM35  |
| Page # 1             |             |         |

| AKDS Capacity Report |        |        |
|----------------------|--------|--------|
| YY/MM/DD             |        |        |
| TERM36               | TERM37 | TERM38 |
| TERM40               | TERM41 | TERM42 |
| TERM44               | TERM45 | TERM46 |
| TERM48               | TERM49 | TERM50 |
| TERM52               | TERM53 | TERM54 |
| TERM56               | TERM57 | TERM58 |
| TERM60               | TERM61 | TERM62 |
| TERM64               | TERM65 | TERM66 |
| TERM68               | TERM69 | TERM70 |
| TERM72               | TERM73 | TERM74 |
| TERM76               | TERM77 | TERM78 |
| TERM80               | TERM81 | TERM82 |
| TERM84               | TERM85 | TERM86 |
| TERM88               | TERM89 | TERM90 |
| TERM92               | TERM93 | TERM94 |
| TERM96               | TERM97 | TERM98 |
| Total Devices: 192   |        |        |
|                      | TERM39 | TERM43 |
|                      | TERM47 | TERM51 |
|                      | TERM55 | TERM59 |
|                      | TERM63 | TERM67 |
|                      | TERM71 | TERM75 |
|                      | TERM79 | TERM83 |
|                      | TERM87 | TERM91 |
|                      | TERM95 | TERM99 |

|                    |                      |
|--------------------|----------------------|
| YY/MM/DD           | AKDS Capacity Report |
| Total Devices: 192 |                      |
| Page # 1           |                      |



## Configuring the Report Output Location

You must configure the output location of the AKDS Capacity Report in the AKDS\_CAPACITY\_REPORT assign in the CONFCSV. The AKDS Capacity Report must be written to a valid Stream file. Stream files are write-only files to which single lines of variable length string data (up to 4062 bytes) are written as the data is generated. Stream files must be configured in the CONFCSV with the TableName attribute set to "Stream132". Samples of the AKDS\_CAPACITY\_REPORT assign in the CONFCSV are shown below. The first example directs the output to a spooler location. The second directs the output to a file.

```
"AKDS_CAPACITY_REPORT", "Stream132", "1.0", "$S.#AKDSCAP", "WOF", "", ""  
"AKDS_CAPACITY_REPORT", "Stream132", "1.0", "$VOL.SUBVOL.RPT", "WOF", "", ""
```

## Configuring the Report Page Length

You can configure the number of report lines printed on each page of the AKDS Capacity Report using the optional AKDS\_CAP\_RPT\_PG\_LGTH Environment attribute. If you do not configure this attribute on the Environment Configuration window, it defaults to 60 lines per page. The AKDS Capacity Report component inserts a page break whenever the number of lines specified by this attribute is reached in the report output.

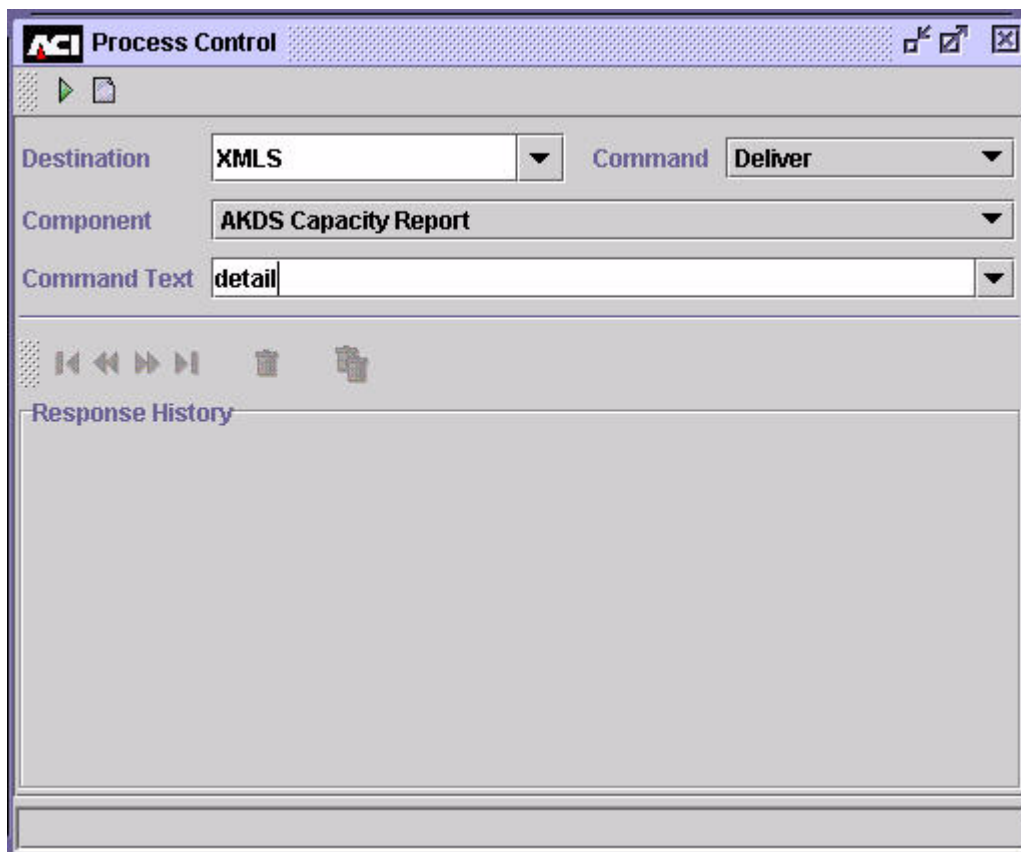
## Generating the AKDS Capacity Report

You generate the AKDS Capacity Report using a DELIVER process control command entered from the Process Control window on the ACI desktop user interface or from an XPNET network control facility.

### Process Control Window

When entering this command on the Process Control window, set the Destination field to "XMLS", the Command field to "Deliver", and the Component field to "AKDS Capacity Report". You can optionally enter "detail" or "DETAIL" in the Command Text field to include the channel ID of each ATM terminal in the report output. If you leave the Command Text field blank, only the total number of ATM

terminals configured in the CHNPKD is output to the report. An example of the Process Control window for generating the AKDS Capacity Report with a listing of channel IDs is shown below.



## XPNET Network Control Facility

To generate the AKDS Capacity Report from an XPNET network control facility, use the following text command.

### **Syntax**

---

DELIVER PROCESS *object-name*, "DELIVER AKDSRPT [DETAIL]"

*object-name* — The symbolic name assigned to the User Interface (XML Server) process in the XPNET node.

***Example***

---

The following is an example of a DELIVER PROCESS command sent to the XML Server process to generate the AKDS Capacity Report with a list of the channel IDs for each ATM terminal using AKDS.

```
deliver process p1a^xml,"deliver akdsrpt detail"
```

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# C: Security Device Utilities and Operations

---

This appendix describes the utilities and operations provided in the TSS environment for security device management.

The following utilities are described in this appendix:

- Check Digit Validation and Generation utility
- AKB Conversion utility
- Master File Key Conversion utility

The following operations procedures performed on hardware security devices are also provided in this appendix:

- Validating check digits for keys in the TSS database
- Generating check digits for keys in the TSS database
- Converting Atalla keys from non-AKB format to AKB format
- Changing the master file key in a security device
- Disabling a security device
- Enabling a security device
- Initializing a security device
- Obtaining the current status of a security device

## Check Digit Validation and Generation Utility

ACI provides a check digit validation and generation program to validate and generate check digits for hardware-encrypted keys stored in the TSS database. The program enables you to validate or generate the check digits for keys in one or more files by entering a command on the Process Control window of the ACI desktop.

**Note:** Currently, this utility only supports check digit validation and generation of keys for Atalla NSPs in either variant or AKB format.

### Prerequisites

The Check Digit Validation and Generation utility requires that the Generate Check Digits (7E) command be enabled in the Atalla NSP. This command is enabled in the default factory security policy. The `ATALLA_CHK_DGTS_LGTH` Environment attribute specifies whether the utility generates and validates four or six check digits. To use the leftmost six digits method in the 7E command, you must also enable option 88 in your Atalla NSP security policy.

For Atalla NSPs running AKB software, the 7E command accepts single-length keys only if the Atalla premium option 6C is enabled in your security policy using commands 108 and 109. This option does not apply and is not needed for Atalla NSPs running variant software.

### Procedures

The following paragraphs provide procedures for validating and generating the check digits of keys in one or more files.

#### Validating Check Digits

You can validate the check digits of all keys in one or more files using the `VVK VALIDATE` command entered from the Process Control window (accessed from **System Operations > Process Control**) or from an XPNET network control facility. For each key's check digits validated, the utility requests the Atalla NSP to generate check digits for the key and compares the check digits returned with those stored in the TSS database. If the check digits match, the check digits in the database are valid. Otherwise, they are invalid. You can optionally specify the name of an individual Atalla NSP in the `VVK VALIDATE` command to perform

all check digit validation processing. If you do not specify an individual Atalla NSP in the command, check digit validation requests are distributed in a round-robin manner among all Atalla NSPs in the system.

The syntax for the VVK VALIDATE command is as follows:

```
VVK VALIDATE [HSM security-device-name] filename, filename,...
```

where:

*security-device-name* is the name of an individual Atalla NSP configured on the Security Device Configuration window, ALL, or \*. If you specify ALL or \* for the HSM keyword in the command, check digit validation requests are distributed in a round-robin manner among all Atalla NSPs in the system. This has the same effect as excluding the optional HSM keyword syntax altogether from the command.

*filename* is one or more of the following acronyms for files:

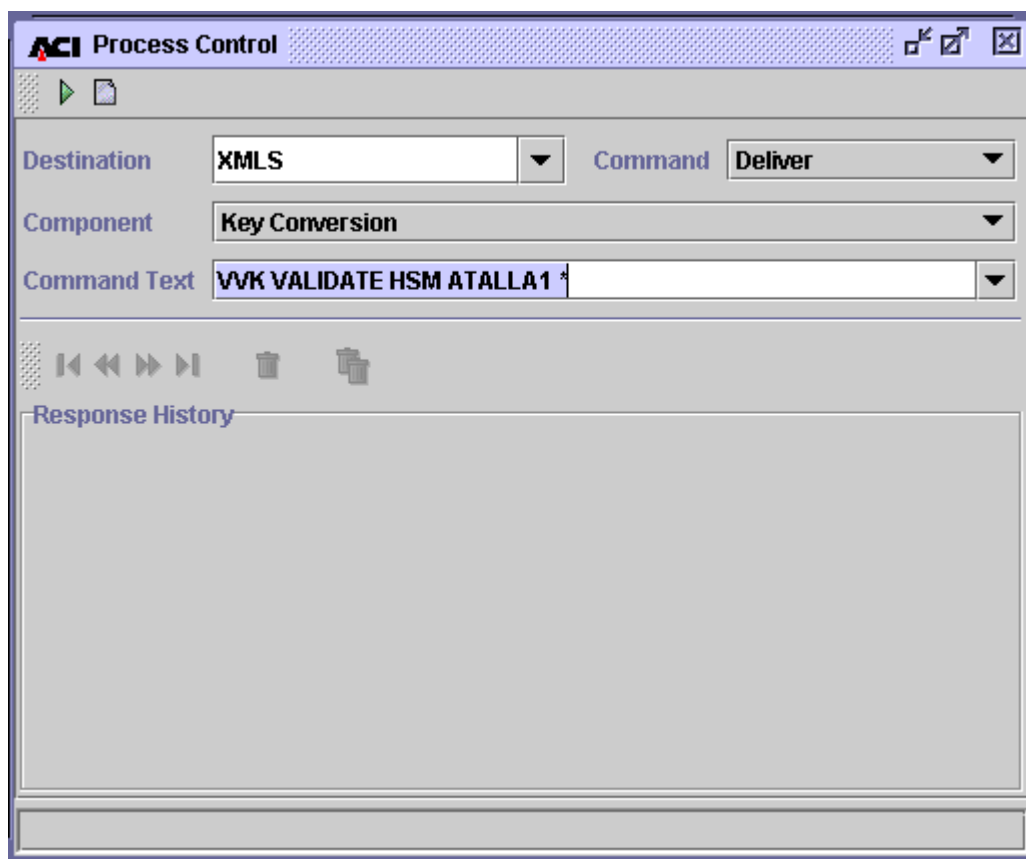
|       |                                                                     |
|-------|---------------------------------------------------------------------|
| *     | = Validates check digits for keys in all files                      |
| CRDVD | = Validates check digits for keys in the Card Verification File     |
| CSECD | = Validates check digits for keys in the Channel Security File      |
| DNTD  | = Validates check digits for keys in the Diebold Number Table File  |
| DRVKD | = Validates check digits for keys in the DUKPT Derivation Key File  |
| EMVSD | = Validates check digits for keys in the EMV Authorization File     |
| IDESD | = Validates check digits for keys in the IBM DES Verification File  |
| ISECD | = Validates check digits for keys in the Interchange Security File  |
| VPVVD | = Validates check digits for keys in the Visa PVV Verification File |

If you specify more than one file name, each file name must be separated by a comma. If you specify \* in the file name list, all other entries in the list are ignored.

When you enter this command from the Process Control window, you must set the Destination, Command, and Component fields as described below. You must enter the VVK VALIDATE command syntax described above in the Command Text field.

|               |                               |
|---------------|-------------------------------|
| Destination:  | XMLS or IS                    |
| Command:      | Deliver                       |
| Component:    | Key Conversion                |
| Command Text: | See VVK VALIDATE syntax above |

A sample Process Control window for sending the VVK VALIDATE command is shown below.



A sample NCPCOM text string for sending the VVK VALIDATE command is shown below.

```
DELIVER PROCESS P1A^TSSXML001,"DELIVER KEYCONV VVK VALIDATE HSM  
ATALLA2 *"
```

## Generating Check Digits

You can generate the check digits of all keys in one or more files using the VVK GENERATE command entered from the Process Control window (accessed from **System Operations > Process Control**) or from an XPNET network control facility. For each key's check digits generated, the utility requests the Atalla NSP to generate check digits for the key and stores the check digits returned in the TSS database. If a check digit value already exists for a key in a file, it is overwritten by the check digits generated by the NSP. You can specify the name of an individual Atalla NSP in the VVK GENERATE command to perform all check digit



generation processing. If you do not specify an individual Atalla NSP in the command, check digit generation requests are distributed in a round-robin manner among all Atalla NSPs in the system. The syntax for the VVK GENERATE command is as follows:

```
VVK GENERATE [HSM security-device-name] filename, filename,...
```

where:

*security-device-name* is the name of an individual Atalla NSP configured on the Security Device Configuration window, ALL, or \*. If you specify ALL or \* for the HSM keyword in the command, check digit generation requests are distributed in a round-robin manner among all Atalla NSPs in the system. This has the same effect as excluding the optional HSM keyword syntax altogether from the command.

*filename* is one or more of the following acronyms for files:

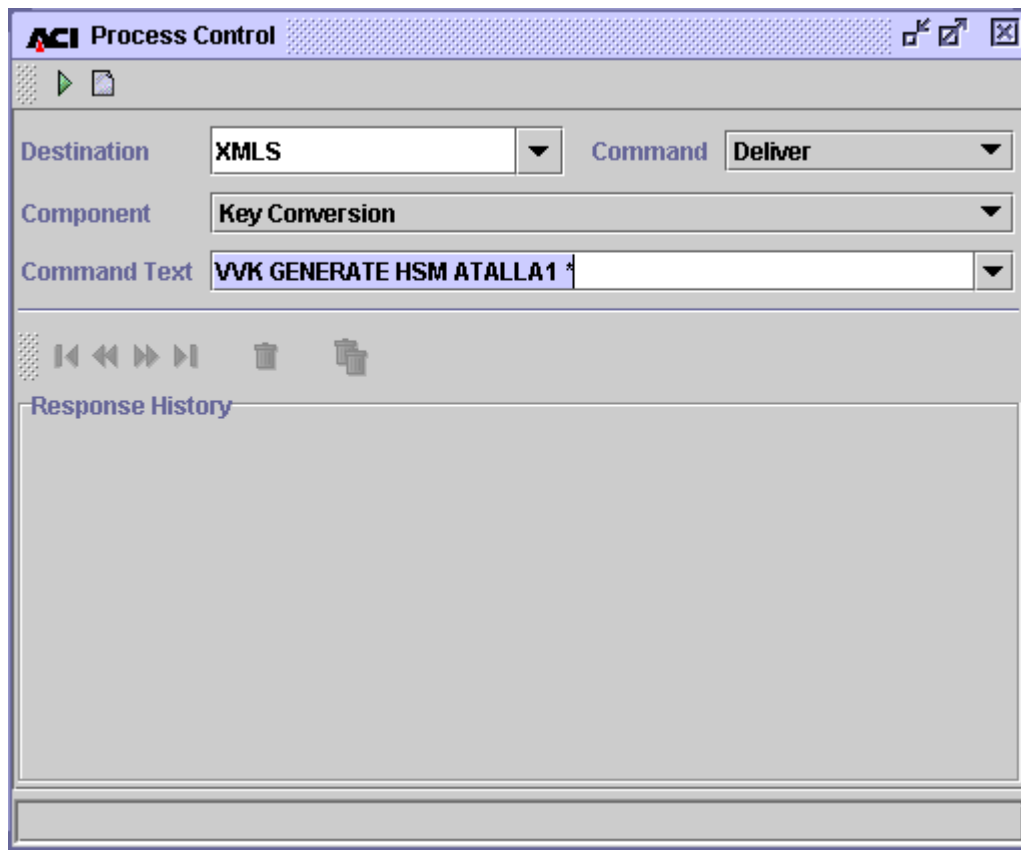
|       |                                                                     |
|-------|---------------------------------------------------------------------|
| *     | = Generates check digits for keys in all files                      |
| CRDVD | = Generates check digits for keys in the Card Verification File     |
| CSECD | = Generates check digits for keys in the Channel Security File      |
| DNTD  | = Generates check digits for keys in the Diebold Number Table File  |
| DRVKD | = Generates check digits for keys in the DUKPT Derivation Key File  |
| EMVSD | = Generates check digits for keys in the EMV Authorization File     |
| IDESD | = Generates check digits for keys in the IBM DES Verification File  |
| ISECD | = Generates check digits for keys in the Interchange Security File  |
| VPVVD | = Generates check digits for keys in the Visa PVV Verification File |

If you specify more than one file name, each file name must be separated by a comma. If you specify \*in the file name list, all other entries in the list are ignored.

When you enter this command from the Process Control window, you must set the Destination, Command, and Component fields as described below. You must enter the VVK VALIDATE command syntax described above in the Command Text field.

|               |                               |
|---------------|-------------------------------|
| Destination:  | XMLS or IS                    |
| Command:      | Deliver                       |
| Component:    | Key Conversion                |
| Command Text: | See VVK GENERATE syntax above |

A sample Process Control window for sending the VVK GENERATE command is shown below.



A sample NCPCOM text string for sending the VVK GENERATE command is shown below.

```
DELIVER PROCESS P1A^TSSXML001, "DELIVER KEYCONV VVK GENERATE HSM  
ATALLA2 * "
```

## Processing Report

A processing report for the Check Digit Validation and Generation utility is printed to the location specified by the KEY\_CONVERSION\_REPORT assign in the CONFCSV (e.g., \$\$.#KEYCONV).

## Description

For each file in which check digits were validated or generated, the following is reported:

- The name of the data source (file) being processed, the date and time the utility started processing the file, and text indicating whether check digits are being validated or generated for the file.
- The total number of records read from the file.
- For each type of key stored in the file, the number of keys for which check digits were successfully validated or generated.
- For each type of key stored in the file, the number of keys for which check digit validation or generation failed.
- The total number of keys processed in the records read for which check digits were successfully validated or generated.
- The total number of keys processed in the records read for which the check digits were invalid or processing failed.

If the utility encounters an error while performing check digit validation or generation, an error message is written to the processing report. For each error message, the utility also writes the primary key fields of the record on which the error occurred. The following are examples of error messages that can be reported.

Error opening cursor. Unable to process data source.

<key name> CHECK DIGIT MISMATCH. Generated: <check digit value returned from 7E command> Current: <current check digit value in file>

Error Generating <key name> Check Digits: <error text>

where:

<key name> is the type of key on which the error occurred.

<error text> is one of the following:

No HSM available  
Error formatting request to HSM  
Unable to send message to HSM  
HSM Response: <native HSM response message>

The types of keys that are reported on for each TSS file are shown in the following table.

| <b>TSS File</b> | <b>Keys</b>                     |
|-----------------|---------------------------------|
| CRDVD           | Key Encrypt 1                   |
|                 | Key Encrypt 2                   |
| CSECD           | KKT PIN                         |
|                 | KKT MAC                         |
|                 | Inbound PIN                     |
|                 | Inbound MAC                     |
|                 | KKT DT                          |
|                 | Outbound KDT Lvl1               |
| DNTD            | DNT Key                         |
| DRVKD           | Derivation Key                  |
| EMVSD           | Secure Message Key              |
|                 | Secure Message Confidential Key |
|                 | Crypto Key                      |
| IDESD           | Key Encrypt                     |

| <b>TSS File</b> | <b>Keys</b>    |
|-----------------|----------------|
| ISECD           | KKT PIN MASTER |
|                 | KKT MAC MASTER |
|                 | KKT IMPORT PIN |
|                 | KKT EXPORT PIN |
|                 | KKT IMPORT MAC |
|                 | KKT EXPORT MAC |
|                 | Inbound PIN    |
|                 | Outbound PIN   |
|                 | Inbound MAC    |
|                 | Outbound MAC   |
|                 | KKT DT IMPORT  |
|                 | KKT DT EXPORT  |
|                 | Outbound KDT   |
|                 | Inbound KDT    |
| VPVVD           | Key Encrypt    |

## Sample

The following pages show a sample processing report for the Check Digit Validation and Generation utility when validating check digits.

Check Digit Report for Data Source CARD\_VRFCTN  
 Validating Check Digits  
 Oct 09, 2003 08:30:19

Total Records: 3

| Key<br>-----         | Success<br>----- | Fail<br>----- |
|----------------------|------------------|---------------|
| Key Encrypt 1        | 3                | 0             |
| Key Encrypt 2        | 3                | 0             |
| Total Keys Processed | 6                | 0             |

Check Digit Report for Data Source IBM\_DES  
 Validating Check Digits  
 Oct 09, 2003 08:30:19

Total Records: 6

| Key<br>-----         | Success<br>----- | Fail<br>----- |
|----------------------|------------------|---------------|
| Key Encrypt          | 6                | 0             |
| Total Keys Processed | 6                | 0             |

Check Digit Report for Data Source VISA\_PVV  
 Validating Check Digits  
 Oct 09, 2003 08:30:19

Total Records: 3

| Key<br>-----         | Success<br>----- | Fail<br>----- |
|----------------------|------------------|---------------|
| Key Encrypt          | 36               | 0             |
| Total Keys Processed | 36               | 0             |

Check Digit Report for Data Source CHANNEL\_SECURITY  
 Validating Check Digits  
 Oct 09, 2003 08:30:19

KKT PIN Key CHECK DIGIT MISMATCH. Generated: 55F7 Current: '  
 '

KKT MAC Key CHECK DIGIT MISMATCH. Generated: 3F8B Current: '  
'  
Inbound PIN Key CHECK DIGIT MISMATCH. Generated: CB02 Current:  
' '  
KKT DT Key CHECK DIGIT MISMATCH. Generated: 12C1 Current: '  
'  
Outbound KDT Lvl1 Data Key CHECK DIGIT MISMATCH. Generated: CA50  
Current: ' '  
Channel ID BEADDECCBEADDECC

Total Records: 8

| Key                  | Success | Fail  |
|----------------------|---------|-------|
| -----                | -----   | ----- |
| KKT PIN              | 15      | 1     |
| KKT MAC              | 15      | 1     |
| Inbound PIN          | 15      | 1     |
| Inbound MAC          | 16      | 0     |
| KKT DT               | 15      | 1     |
| OUTBOUND KDT         | 31      | 1     |
| Total Keys Processed | 107     | 5     |

Check Digit Report for Data Source ICHG\_SECURITY  
Validating Check Digits  
Oct 09, 2003 08:30:19

Total Records: 5

| Key                  | Success | Fail  |
|----------------------|---------|-------|
| -----                | -----   | ----- |
| KKT IMPORT PIN       | 5       | 0     |
| KKT EXPORT PIN       | 5       | 0     |
| KKT IMPORT MAC       | 5       | 0     |
| KKT EXPORT MAC       | 5       | 0     |
| Inbound PIN          | 10      | 0     |
| Outbound PIN         | 10      | 0     |
| Inbound MAC          | 10      | 0     |
| Outbound MAC         | 10      | 0     |
| KKT DT IMPORT        | 5       | 0     |
| KKT DT EXPORT        | 5       | 0     |
| Outbound KDT         | 20      | 0     |
| Inbound KDT          | 20      | 0     |
| Total Keys Processed | 110     | 0     |

Check Digit Report for Data Source EMV\_SECURITY  
Validating Check Digits  
Oct 09, 2003 08:30:19

Total Records: 3

| Key                             | Success | Fail  |
|---------------------------------|---------|-------|
| -----                           | -----   | ----- |
| Secure Message Key              | 3       | 0     |
| Secure Message Confidential Key | 3       | 0     |
| Crypto Key                      | 72      | 0     |
| Total Keys Processed            | 78      | 0     |



# AKB Conversion Utility

ACI provides a conversion program to convert working keys stored in the database in the variant key format (non-AKB format) to the Atalla Key Block (AKB) format. The program enables you to convert the keys in one or more files by entering a command on the Process Control window of the ACI desktop.

## Prerequisites

An NSP configured with AKB version 2.30 or later or 1.3 or later software (referred to as AKB NSP throughout this document) is required for the AKB Conversion utility. The 2.30 version software is used for A10100 AKB models while the 1.3 version software is used for A10000 AKB models.

## 11C Command

AKB conversion requires that the Atalla DES Key to 3DES AKB Format (11C) command be enabled in the AKB NSP. This command is not enabled in the default factory security policy. Note that with AKB version 2.51 or later software, you must enable the 11C command with a specific count of the number of times the command can be executed. To use this command, you must add it to your security policy using commands 108 and 109.

## Command Options

The following options must be enabled for the AKB Conversion utility:

- Option 6C must be enabled for the 11C command if any of the keys (KPE, KEK, MAC, etc.) to be converted to AKB format are single length keys or if any keys to be converted are comprised of key pairs (CVV, PVV) and have the same value for key 1 and key 2. This option is not enabled in the default factory security policy. To use this option, you must add it to your security policy using commands 108 and 109.
- Option E3 must be enabled for commands 10 and 1A to support ATM master key headers. This option is not enabled in the default factory security policy. To use this option, you must add it to your security policy using commands 108 and 109.

- Premium option 8C must be enabled for the Verify PIN—Visa PVV Method (32#3) and PIN Change—Visa PVV (37#3) commands to allow single-length Visa PVV keys to be replicated in triple DES operations. Therefore, if you convert single-length Visa PVV keys to AKB format using the AKB Conversion utility, you must enable option 8C to use the 32#3 and 37#3 commands. You can obtain option 8C, free of charge, in a 105 command from Atalla.

## 11B Command

The AKB\_KEY\_IMPORT\_VARIANT\_SOURCE environment attribute allows PIN, MAC, and data key exchange keys in the Interface Security File (ISECD) to be converted to AKB format using the key variant or the key type. The value of this attribute indicates whether the Transaction Security Services process passes the key variant or key type in the variant field of the 11B command to the NSP. Valid values for this attribute are as follows:

- 0 = Pass the key type in the variant field of the 11B command. Default value.
- 1 = Pass the key variant in the variant field of the 11B command.

When this attribute is set to a value of 0, one of the following key type values is passed as the key variant in the 11B command:

- 1 = PIN key
- 2 = Data encryption key
- 3 = MAC key

When this attribute is set to a value of 1, the key variant value configured in the Key Variant field on the Interface Security Configuration window for the key is passed in the 11B command.

If your system uses key exchange keys with a key variant of 0, you must set this attribute to a value of 1 to correctly convert key exchange keys. Otherwise, you can set this attribute to 0 or allow it to default to 0 by not configuring it.

## Check Digits

**Note:** All keys to be converted in a file must have valid check digits or the conversion will fail. It is possible that the check digits fields in the TSS database may contain blanks or invalid values after BASE24 files are converted because BASE24 does not validate check digits when keys are entered into the BASE24

database. If so, use the “Check Digit Validation and Generation Utility” to generate and store check digits in the TSS database for all keys before running the AKB Conversion utility.

The AKB Conversion utility uses the 7E command to generate check digits using the leftmost four digits or the leftmost six digits method. The ATALLA\_CHK\_DGTS\_LGTH Environment attribute specifies whether the utility generates and validates four or six check digits. To use the leftmost six digits method in the 7E command, you must also enable option 88 in your Atalla NSP security policy.

## Procedures

The following procedures show you how to use the AKB Conversion utility when both a non-AKB NSP (i.e., an NSP configured with variant version software) and an AKB NSP are available, or when just an AKB NSP is available. The non-AKB NSP is referred to as the Variant NSP throughout this section.

### Variant NSP and AKB NSP Available

To convert working keys from non-AKB format to AKB format using both a Variant NSP and an AKB NSP, perform the following steps:

1. Back up the current database in case the keys need to be restored to non-AKB format.
2. Generate a triple-length or double-length SuperKey and MFK in AKB format and load them into the AKB NSP using the Secure Configuration Terminal (SCT) or Secure Configuration Assistant (SCA) under appropriate supervision (e.g., split knowledge and dual control). Refer to the appropriate Atalla documentation for detailed information on generating keys and use of the SCT and SCA.

**Note:** Although the AKB NSP supports both triple- and double-length MFKs, Atalla strongly encourages that you use a triple-length MFK. If you do use a double-length MFK, you cannot use any triple-length keys.

3. Load the first two key parts of the triple-length MFK or both parts of the double-length MFK generated in the above step as the Pending MFK (PMFK) on the Variant NSP using the SCT or SCA.

4. Using the Variant NSP, convert all current keys from encryption under the MFK to encryption under the PMFK using the Master File Key Conversion utility. (Note that this step assumes that the existing TSS database and system are still configured to use the Variant NSP). See the “Master File Key Conversion” topic for detailed procedures.
5. Set any optional Environment attributes on the Environment Configuration window for AKB header values that you want to be different from the default value. See the Environment Attributes topic below for a list of these optional Environment attributes and their associated default AKB header values.
6. Modify the Security Device Configuration File to set all Subtype fields to an AKB subtype.
7. Warmboot the Transaction Security Services process to start using the modified Security Device Configuration File (e.g., ALTER DASRC SEC\_DEV\_CONFIG).
8. Modify all HSM assigns in the MDSDEST file to specify AKB NSPs (or their associated Boxcar processes).

**Note:** All implementations using the TCP/IP protocol require the Boxcar process.

9. If the SIMDS\_DEST\_WARMBOOT\_DELTA parameter in the CONFCSV is set to a value of 0, then you must stop and restart the User Interface (XML Server) process before the modified MDSDEST file takes effect.
10. Convert all keys from variant format to AKB format using the CONVERT AKB command from the Process Control window or from an XPNET network control facility. You can specify the name of an individual AKB NSP in the CONVERT AKB command to perform all conversion processing. If you do not specify an individual AKB NSP in the command, conversion function requests are distributed in a round-robin manner among all AKB NSPs in the system. The syntax for the CONVERT AKB command is as follows:

```
CONVERT AKB [HSM security-device-name] filename, filename,...
```

where *security-device-name* is the name of an individual AKB NSP configured on the Security Device Configuration window and *filename* is one or more of the following acronyms for files:

|       |                             |
|-------|-----------------------------|
| *     | = Convert all files         |
| CRDVD | = Card Verification File    |
| CSECD | = Channel Security File     |
| DNTD  | = Diebold Number Table File |
| DRVKD | = DUKPT Derivation Key File |
| ISECD | = Interchange Security File |

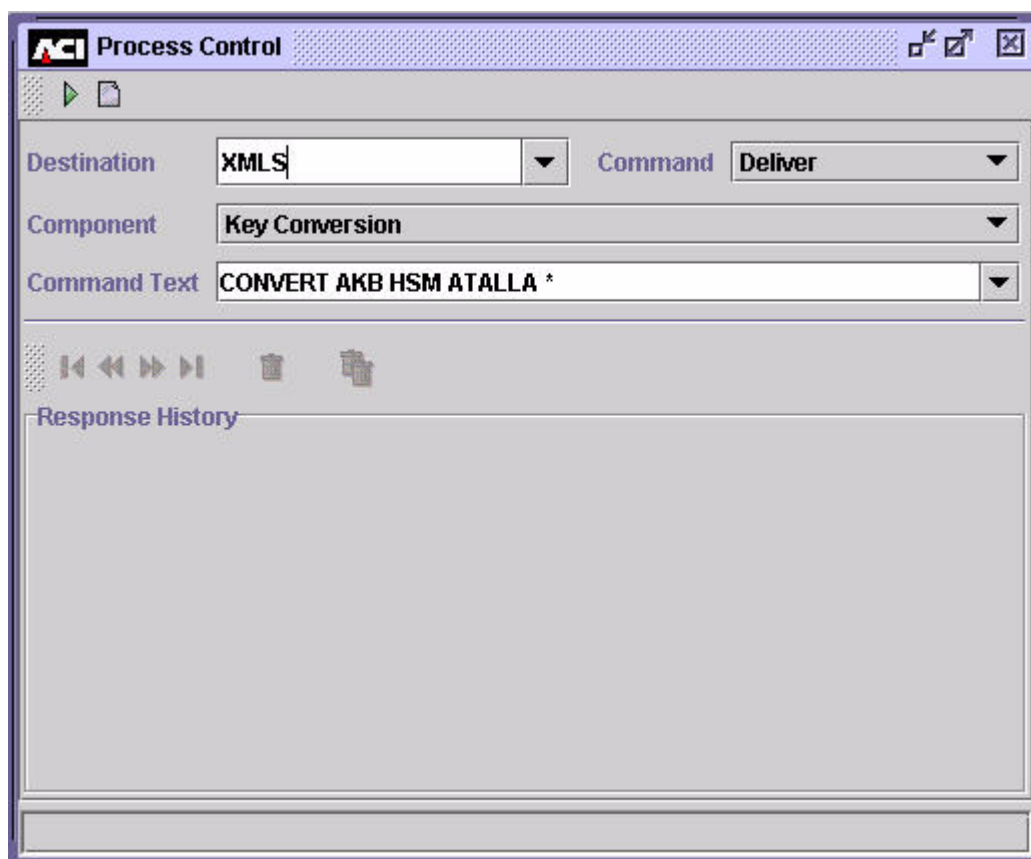
EMVSD = EMV Authorization File  
 IDESD = IBM DES Verification File  
 VPVVD = Visa PVV Verification File

If you specify more than one file name, each file name must be separated by a comma. If you specify \* in the file name list, all other entries in the list are ignored.

**Note:** Although DNTD is a valid file name in the CONVERT AKB command syntax, the utility does not currently convert keys in the DNTD.

Event messages are generated when the key conversion processing is started and completed.

A sample Process Control window for sending the CONVERT AKB command is shown below.



A sample NCPCOM text string for sending the CONVERT AKB command is shown below.

```
DELIVER PROCESS P1A^TSSXML001, "DELIVER KEYCONV CONVERT AKB HSM
ATALLA2 * "
```

## AKB NSP Only

To convert working keys from non-AKB format to AKB format using only an AKB NSP, perform the following steps:

1. Back up the current database in case the keys need to be restored to non-AKB format.
2. Generate a triple-length or double-length SuperKey and MFK in AKB format and load them into the AKB NSP using the Secure Configuration Terminal (SCT) or Secure Configuration Assistant (SCA) under appropriate supervision (e.g., split knowledge and dual control). Refer to the appropriate Atalla documentation for detailed information on generating keys and use of the SCT and SCA.

**Note:** Although the AKB NSP supports both triple- and double-length MFKs, Atalla strongly encourages that you use a triple-length MFK. If you do use a double-length MFK, you cannot use any triple-length keys.

3. Using the SCT or SCA, create the key portion of the variant MFK under which keys in the TSS database are currently encrypted and enter it as an AKB-formatted key using three Environment attributes.
  - a. Generate the key portion under appropriate supervision (e.g., split knowledge and dual control). For the SCT, you must use the Keyblock Utility provided by Atalla to generate this key portion. You must modify the Keyblock Utility configuration files in order to use a value of 1KDDN0M0 as the AKB header for this key portion. The following modifications are required:

- 1) Add an entry to the headers.fil file using a text editor such as Notepad. The entry to be added is underlined below.

```
#23456789.123456789.123456789.123456789.123456789.123456789
#Name      Descriptions                               Header
MFK      Master File Key                            1KDDN0M0
KEK        Key Encryption Key                          1KDNE000
```

- 2) Add an entry to the options.fil file using a text editor such as Notepad. The entry to be added is underlined below.

```
** BYTE 6-Special Handling
0*Regular key header
I Import-KEK header.
N KEK that can import or export other keys.
M MFK
```

- b. Optionally enter the AKB\_CNV\_DES\_3DES\_HDR attribute on the Environment Configuration window of the ACI desktop user interface. If you do not enter this attribute, it defaults to a value of 1KDDN0M0.

- c. Enter the resulting 48-byte key block (from step 3a) into the AKB\_CNV\_DES\_3DES\_KEY attribute on the Environment Configuration window of the ACI desktop user interface.
- d. Enter the 16-character MAC generated for the key portion (from step 3a) into the AKB\_CNV\_DES\_3DES\_MAC attribute on the Environment Configuration window.

If the above Environment attributes are configured when you enter the CONVERT AKB command, the Key Conversion component includes the MFK cryptogram defined by these attributes in the 11C command sent to the AKB NSP, along with the cryptogram of the key to be translated. The AKB NSP then decrypts the key with this MFK before translating the key to AKB format. Otherwise, only the cryptogram of the key to be translated is included in the 11C command.

4. Set any optional Environment attributes on the Environment Configuration window for AKB header values that you want to be different from the default value. See the Environment Attributes topic below for a list of these optional Environment attributes and their associated default AKB header values.
5. Modify the Security Device Configuration File to set all Subtype fields to an AKB subtype.
6. Warmboot the Transaction Security Services process to start using the modified Security Device Configuration File (e.g., ALTER DASRC SEC\_DEV\_CONFIG).
7. Modify all HSM assigns in the MDSDEST file to specify AKB NSPs (or their associated Boxcar processes).

**Note:** All implementations using the TCP/IP protocol require the Boxcar process.

8. If the SIMDS\_DEST\_WARMBOOT\_DELTA parameter in the CONFCSV is set to a value of 0, then you must stop and restart the User Interface (XML Server) process before the modified MDSDEST file takes effect.
9. Convert all keys from variant format to AKB format using the CONVERT AKB command from the Process Control window or from an XPNET network control facility. You can specify the name of an individual AKB NSP in the CONVERT AKB command to perform all conversion processing. If you do not specify an individual AKB NSP in the command, conversion function requests are distributed in a round-robin manner among all AKB NSPs in the system. The syntax for the CONVERT AKB command is as follows:

```
CONVERT AKB [HSM security-device-name] filename, filename,...
```

where *security-device-name* is the name of an individual AKB NSP configured on the Security Device Configuration window and *filename* is one or more of the following acronyms for files:

|       |                              |
|-------|------------------------------|
| *     | = Convert all files          |
| CRDVD | = Card Verification File     |
| CSECD | = Channel Security File      |
| DNTD  | = Diebold Number Table File  |
| DRVKD | = DUKPT Derivation Key File  |
| ISECD | = Interchange Security File  |
| EMVSD | = EMV Authorization File     |
| IDESD | = IBM DES Verification File  |
| VPVVD | = Visa PVV Verification File |

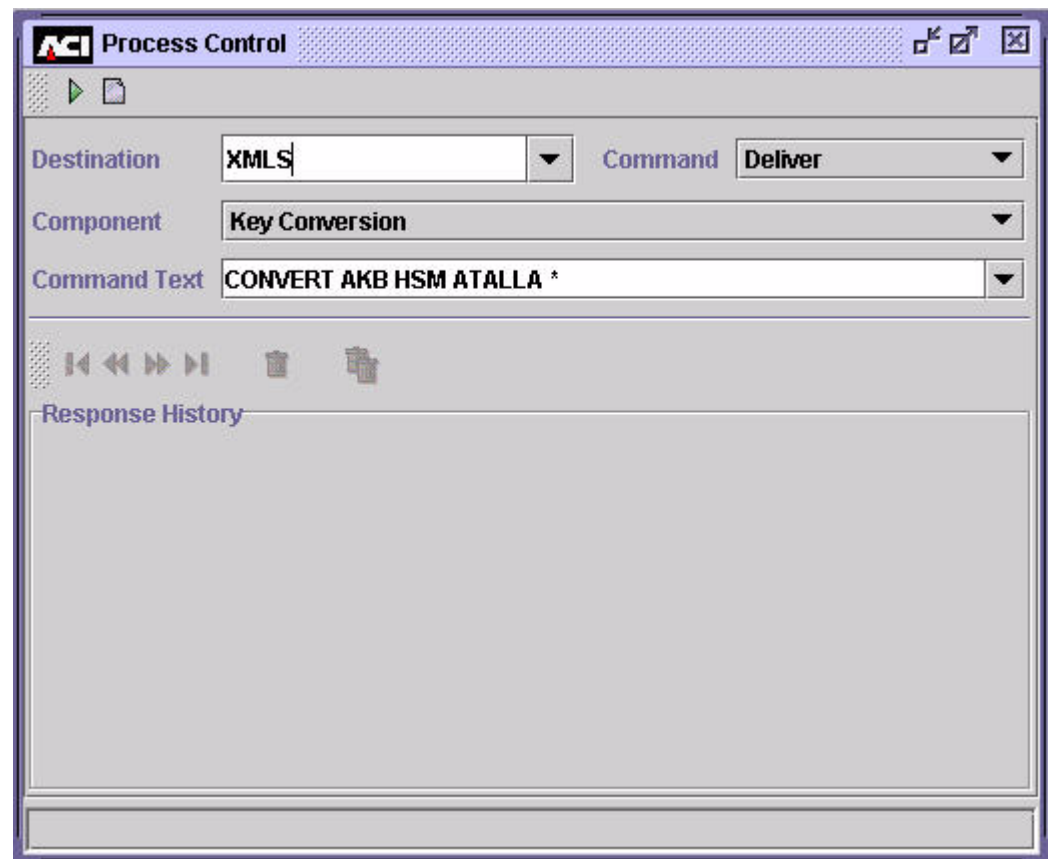
If you specify more than one file name, each file name must be separated by a comma. If you specify \* in the file name list, all other entries in the list are ignored.

**Note:** Although DNTD is a valid file name in the CONVERT AKB command syntax, the utility does not currently convert keys in the DNTD.

Event messages are generated when the key conversion processing is started and completed.



A sample Process Control window for sending the CONVERT AKB command is shown below.



A sample NCPCOM text string for sending the CONVERT AKB command is shown below.

```
DELIVER PROCESS P1A^TSSXML001, "DELIVER KEYCONV CONVERT AKB HSM
ATALLA2 * "
```

## Processing Report

A processing report for the AKB Conversion utility is printed to the location specified by the KEY\_CONVERSION\_REPORT assign in the CONFCSV (e.g., \$\$.#KEYCONV).

For each data file converted, the following counts are reported after a data file has been completely processed:

- The number of records read

- The number of records successfully converted
- The number of records not converted

For each record not converted, the primary key value for the record is listed.

**Note:** Although DNTD is a valid file name in the CONVERT AKB command syntax, the utility does not currently convert keys in the DNTD. If you include the DNTD file name in the command syntax, the text “Command not supported for DNTD” is written to the processing report.

## Optional AKB Header Environment Attributes

The following table lists the optional Environment attributes you can configure if you want to use an AKB header value that differs from the default value when translating keys. The first column of the table lists the name of the Environment attribute. The second column indicates the default AKB header value of the attribute. The third column indicates the file in which the key is stored, and the fourth column indicates the type of key.

| Attribute Name         | Default Value | TSS File | Key Type                                                                              |
|------------------------|---------------|----------|---------------------------------------------------------------------------------------|
| AKB_HDR_ATM_MSTR       | 1ADNE000      | CSECD    | PIN and MAC key exchange key when the Key Variant field is set to 5 (ATM Key Variant) |
| AKB_HDR_CARD_VRFY      | 1CDNE000      | CRDVD    | Card verification                                                                     |
| AKB_HDR_AMEX_CARD_VRFY | 1mXNE000      | CRDVD    | American Express Card Security Code                                                   |
| AKB_HDR_DNT            | 1ndNE000      | DNTD     | Diebold number table                                                                  |
| AKB_HDR_DERIVATION     | 1dDNE000      | DRVKD    | DUKPT derivation key                                                                  |
| AKB_HDR_EMV_CRYPTO     | 1mENE000      | EMVSD    | EMV ACC master keys (cryptographic keys)                                              |
| AKB_HDR_EMV_SECURE     | 1mENE00M      | EMVSD    | EMV MAC master key (secure message key for integrity)                                 |

| Attribute Name          | Default Value | TSS File       | Key Type                                                                                        |
|-------------------------|---------------|----------------|-------------------------------------------------------------------------------------------------|
| AKB_HDR_EMV_SECURE_CONF | 1mENE00E      | EMVSD          | PIN encryption key<br>(secure message key for confidentiality)                                  |
| AKB_HDR_IBM_DES         | 1V3NE000      | IDESD          | IBM DES PIN<br>verification key                                                                 |
| AKB_HDR_INBND_MAC       | 1MDNE000      | CSECD<br>ISECD | Inbound MAC<br>encryption key                                                                   |
| AKB_HDR_INBND_DT        | 1DDNE000      | ISECD          | Inbound data encryption<br>key                                                                  |
| AKB_HDR_INBND_IV        | 1IDNE000      | ISECD          | Inbound initialization<br>vector                                                                |
| AKB_HDR_INBND_PIN       | 1PUNE000      | CSECD<br>ISECD | Inbound PIN encryption<br>key                                                                   |
| AKB_HDR_OUTBND_DT       | 1DDNE000      | CSECD<br>ISECD | Outbound data<br>encryption key                                                                 |
| AKB_HDR_OUTBND_IV       | 1IDNE000      | CSECD<br>ISECD | Outbound initialization<br>vector                                                               |
| AKB_HDR_KKT_DT_EXPORT   | 1KDNE000      | ISECD          | Export data encryption<br>key exchange key                                                      |
| AKB_HDR_KKT_DT_IMPORT   | 1DDNN0I0      | CSECD<br>ISECD | Import data encryption<br>key exchange key                                                      |
| AKB_HDR_KKT_PIN         | 1KDNE000      | CSECD          | PIN key exchange key<br>when the Key Variant<br>field is set to 0 (Key<br>Exchange Key Variant) |
| AKB_HDR_KKT_PIN_IMPORT  | 1PUNN0I0      | ISECD          | Import PIN key<br>exchange key                                                                  |
| AKB_HDR_KKT_MAC         | 1KDNE000      | CSECD          | MAC key exchange key<br>when the Key Variant<br>field is set to 0 (Key<br>Exchange Key Variant) |

| <b>Attribute Name</b>  | <b>Default Value</b> | <b>TSS File</b> | <b>Key Type</b>               |
|------------------------|----------------------|-----------------|-------------------------------|
| AKB_HDR_KKT_MAC_IMPORT | 1MDNN0I0             | ISECD           | Import MAC key exchange key   |
| AKB_HDR_KKT_PIN_EXPORT | 1KDNE000             | ISECD           | Export PIN key exchange key   |
| AKB_HDR_KKT_MAC_EXPORT | 1KDNE000             | ISECD           | Export MAC key exchange key   |
| AKB_HDR_OUTBND_PIN     | 1PUNE000             | CSECD<br>ISECD  | Outbound PIN encryption key   |
| AKB_HDR_OUTBND_MAC     | 1MDNE000             | CSECD<br>ISECD  | Outbound MAC encryption key   |
| AKB_HDR_VISA_PVV       | 1VVNE000             | VPVVD           | Visa PVV PIN verification key |

# Master File Key Conversion

Good security practice as well as several network and card associations require the master file key in a hardware security module to be replaced periodically. Master file key usage varies according to the hardware security module vendor as described below:

- In an Atalla NSP, the master file key is referred to as the Master File Key (MFK). Different types of working keys stored in the database are encrypted and stored under different variants of the MFK when using an NSP.
- In a Thales e-Security HSM, the master file key is referred to as a Local Master Key (LMK), which actually consists of a pair of keys. When using a Thales e-Security HSM, different types of working keys stored in the database are encrypted and stored under different LMK key pairs.

When you change the master file key in a hardware security file, you also have to translate each working key from encryption under the current master file key to encryption under the new master file key. Because of this, manual translation of each working key in the database from one master key to another can be a time-consuming and error-prone process. To alleviate this manual work, ACI provides the Master File Key Conversion utility to automate these conversion tasks. Using the Master File Key Conversion utility, you can translate all the hardware-encrypted keys stored in the database or just the hardware-encrypted keys of a particular type (e.g., PIN verification keys, card verification keys, interface keys, etc.).

**Note:** When you convert a working key from encryption under the current master file key to encryption under the new master file key, the value of the working key itself is not changed. Therefore, there is no need to download working keys to devices after running this utility on the Channel Security File (CSECD).

The material in this document is not meant to be a tutorial on Atalla NSPs or Thales e-Security HSMs; it presumes a familiarity with the device capabilities and nomenclature as discussed in documentation published by Atalla or Thales.

**Note:** The key translation options of the Master File Key Conversion utility result in the modification of security-related information in the database. The use of this utility by anyone other than qualified, trusted network security staff is discouraged, and should not be attempted without a full understanding of the processing involved.

The keys listed in section 4 of this guide for each of the supported vendors (i.e., [Atalla](#) or [Thales](#)) are converted using this utility.

The public and private keys described in appendix B of this guide can also be converted using this utility when the BASE24-atm Automated Key Distribution System add-on product is installed. Note that for public keys, the conversion of public keys is actually only a re-calculation of the MAC value.

## Prerequisites

Before you can use the Master File Key Conversion utility, the following prerequisites must be met:

- The network must be shut down while the key translation function is performed. Thus, use of the utility requires careful planning and should only be performed by authorized personnel.
- Atalla NSPs must contain the old MFK as the current MFK and contain the new MFK as the pending MFK before this utility is used. The utility requires the 9E command (Translate Working Key From Under the Current MFK to the Pending MFK) to translate keys.
- For Atalla NSPs, the Master File Key Conversion utility uses the 7E command to generate check digits using the leftmost four digits or the leftmost six digits method. The ATALLA\_CHK\_DGTS\_LGTH Environment attribute specifies whether the utility generates and validates four or six check digits. To use the leftmost six digits method in the 7E command, you must also enable option 88 in your Atalla NSP security policy.
- For Atalla NSPs, you must enable the 12B (Translate Public or Private Key AKB from MFK to PMFK) premium command in your Atalla NSP security policy to support MFK conversion for keys stored in the Host Public Key Security Primary File (HSPKPD), Host Public Key Security Secondary File (HSPKSD), and Channel Public Key Security File (CHNPKD). The HSPKPD, HSPKSD, and CHNPKD are used with the BASE24-atm Automated Key Distribution System add-on product.
- For Thales e-Security HSMs, the old LMK pair must be moved into key change storage before this utility is used. The utility requires the BW command (Translate Keys From Old LMK to New LMK) to translate keys.
- For Thales e-Security HSMs, you must set the DECIMAL\_TBL\_ENCRYPT optional environment attribute to a value of Y if you are using encrypted decimalization tables in your system. This attribute allows the Master File Key Conversion utility to properly translate encrypted decimalization tables from the old LMK pair to the new LMK pair. If you are not using encrypted decimalization tables in your system, you do not need to configure this attribute or you can configure it with a value of N.

## Accessing the Utility

The Master File Key Conversion window on the ACI desktop enables you to run the conversion utility. You can access this window by selecting **Configure > Transaction Security > Master File Key Conversion**. When the window is displayed, enter the name of the security device to be used or ALL from the selectable list in the Security Device Name field. Select ALL if you want to translate keys for all security devices. The security device type is displayed in the Security Device Type field and “to be determined” is displayed in the Report Location field. The report location is controlled by the KEY\_CONVERSION\_REPORT assign value (e.g., \$S.#KEYCONV) in the Metadata Configuration File (CONFCSV). This is the location where the utility generates reports that list the keys converted in each file. The options displayed in the Select Key Type(s) To Translate area of the window vary according to the type of security device used. Refer to ACI Online Help for detailed descriptions of the Master File Key Conversion window fields and functions.

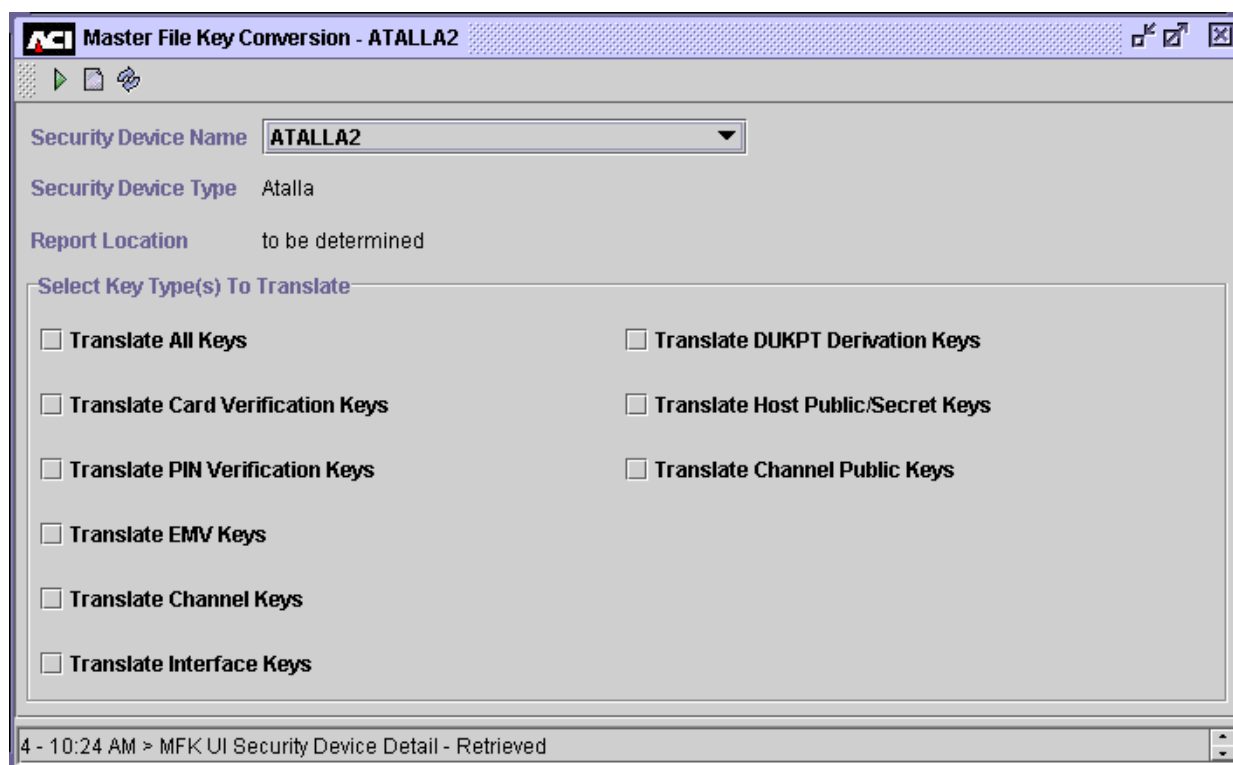
Event messages are generated when the key conversion processing is started and completed.

## Atalla Options

For an Atalla NSP, you can translate all keys or any of the following types of keys in the database by clicking on the types of keys you want to translate and then clicking Submit (▶). The acronyms of the files in which each type of key is stored is provided in parentheses:

- Card verification keys (CRDVD)
- PIN verification keys (IDESD, VPVVD)
- EMV keys (EMVSD)
- Channel keys (CSECD)
- Interface keys (ISECD)
- DUKPT derivation keys (DRVKD)
- Host (BASE24 system) public and secret keys (HSPKPD)
- Channel (ATM terminal) public keys (CHNPKD)

A sample of the Master File Key Conversion window for an Atalla NSP is shown below.



## Thales Options

For a Thales e-Security HSM, keys must be translated according to the LMK pair under which they are encrypted. You can translate all keys or select any of the following types of keys in the database to be translated by clicking on each type of key to be translated and clicking Submit (▶):

- Zone master keys (LMK pair 04–05)
- Zone working keys (LMK pair 06–07)
- Card verification, PIN verification, terminal master, and terminal PIN keys (LMK pair 14–15)
- Terminal MAC keys (LMK pair 16–17)
- IBM DES PIN verification decimalization table (LMK pair 18–19)
- NCR PIN Verification Keys (LMK pair 20–21)
- EMV and DUKPT derivation keys (LMK pair 28–29)



- Host (BASE24 system) public and secret keys (LMK pairs 34–35 and 36–37)  
Private keys are encrypted under LMK pair 34–35 while the MAC values for public keys are encrypted under LMK pair 36–37.
- Channel public keys (LMK pair 36–37)

The HSM can translate only one LMK pair at a time, regardless of the number of LMK pairs that need to be translated. As a result, it is important to know which files must be processed when a particular LMK pair changes. LMK pairs affect the following files.

| LMK Pair | CSECD | IDESD | VPVVD | NCRD | CRDVD | EMVSD | ISECD | DRVKD | HSPKPD/<br>HSPKSD | CHNPKD |
|----------|-------|-------|-------|------|-------|-------|-------|-------|-------------------|--------|
| 04-05    |       |       |       |      |       |       | ✓     |       |                   |        |
| 06-07    |       |       |       |      |       |       | ✓     |       |                   |        |
| 14-15    | ✓     | ✓     | ✓     |      | ✓     |       |       |       |                   |        |
| 16-17    | ✓     |       |       |      |       |       | ✓     |       |                   |        |
| 18-19    |       | ✓     |       |      |       |       |       |       |                   |        |
| 20-21    |       |       |       | ✓    |       |       |       |       |                   |        |
| 28-29    |       |       |       |      |       | ✓     |       | ✓     |                   |        |
| 34-35    |       |       |       |      |       |       |       |       | ✓                 |        |
| 36-37    |       |       |       |      |       |       |       |       | ✓                 | ✓      |

A sample of the Master File Key Conversion window for a Thales HSM is shown below.

**Master File Key Conversion - AATHALES1**

Security Device Name: **AATHALES1**

Security Device Type: **Thales**

Report Location: **to be determined**

**Select Key Type(s) To Translate**

- ☐ Translate All Keys
- ☐ Translate Zone Master Keys (LMK pair 04 / 05)
- ☐ Translate Zone Working Keys (LMK pair 06 / 07)
- ☐ Translate Card Verification, PIN Verification, Terminal Master and Terminal PIN keys (LMK pair 14 / 15)
- ☐ Translate Terminal MAC Keys (LMK pairs 16 / 17)
- ☐ Translate NCR PIN Verification Keys (LMK pair 20 / 21)
- ☐ Translate EMV, DUKPT Derivation Keys (LMK pairs 28 / 29)
- ☐ Translate Decimalization Table (LMK pairs 18/19)
- ☐ Translate Host Public/Secret Keys (LMK pairs 34/35 and 36/37)
- ☐ Translate Channel Public Keys

4 - 10:33 AM > MFK UI Security Device Detail - Retrieved

## Key Translation Report

The Master File Key Conversion utility generates a Key Translation Report for each file in which keys were converted. The report lists the total number of keys successfully converted for each type of key in the file, as well as the total number of keys on which errors were encountered. The report displays error messages for any errors encountered during processing. Each error message indicates the type of error that occurred, the response returned from the security device, and the primary key values of the record on which the error occurred.

The format of the Key Translation Report varies slightly between an Atalla device and a Thales e-Security device. Examples of both formats are shown below, followed by descriptions of the fields on the report.

## Atalla Format

The format of the Key Translation Report for an Atalla NSP is shown below.

```
Key Translation Report for Data Source <data source name>
Translating keys from MFK to Pending MFK
Month DD, YYYY HH:MM:SS
```

(any processing errors appear here)

Total Records:

| Key          | Success | Fail  |
|--------------|---------|-------|
| -----        | -----   | ----- |
| <key name 1> |         |       |
| <key name n> |         |       |

Total Keys Processed

## Thales e-Security Format

The format of the Key Translation Report for a Thales e-Security HSM is shown below.

```
Key Translation Report for Data Source <data source name>
LMK Key Pair <xx/xx>
Translating keys from Old LMK to Current LMK
Month DD, YYYY HH:MM:SS
```

(any processing errors appear here)

Total Records:

| Key          | Success | Fail  |
|--------------|---------|-------|
| -----        | -----   | ----- |
| <key name 1> |         |       |
| <key name n> |         |       |

Total Keys Processed

## Field Descriptions

**<data source name>** — The name of the file in which keys were translated.

**<xx/xx>** — The name of the LMK pair for which keys were translated. This applies to Thales e-Security HSMs only.

**Total Records** — The total number of records processed for the indicated file and LMK pair.

**Key** — A descriptive name of the type of key translated for the file.

**Success** — The total number of keys successfully translated of the indicated key type.

**Fail** — The total number of keys of the indicated key type for which translation failed.

**Total Keys Processed** — The total number of key records processed for the file.

## Post-Translation Procedures

After the selected keys have been successfully translated from encryption under the current master file key to encryption under the pending master file key, enter the ERASE command on the Process Control window or from an XPNET network control facility to erase the old LMK key pair in a Thales e-Security HSM or make the pending MFK the current MFK in an Atalla NSP. The ERASE command sends the 9A (Network Security Processor Status Key) and 9F (Replace the Current MFK with the Pending MFK) commands to an Atalla NSP and the BS (Erase the Key Change Storage) command to a Thales e-Security HSM.

### Process Control Window

To enter the ERASE command on the Process Control window, perform the following steps:

1. Log on to the ACI desktop user interface and access System Operations > Process Control. The Process Control window is displayed.
2. In the Destination field, select XMLS from the list of values.
3. In the Command field, select Deliver from the list of values.
4. In the Component field, select Key Conversion from the list of values.

5. In the Command Text field, enter the following command. If you want to execute the ERASE command on a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to execute the ERASE command on all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

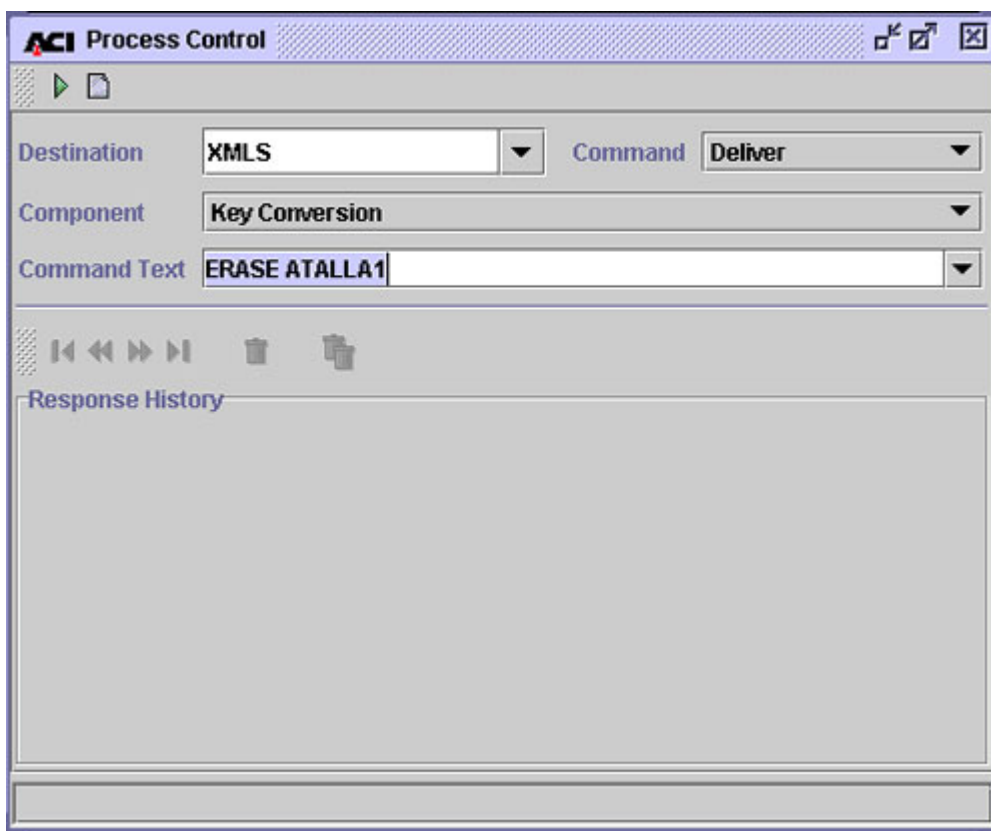
```
ERASE sec-dev-assign-name | *
```

**Examples:**

```
ERASE ATALLA1
```

```
ERASE *
```

The following illustration shows the Process Control window values for sending the ERASE command.



6. Click the Send button at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window.

## Network Control Facility

An example of the ERASE command entered from an XPNET network control facility (e.g., NCPCOM) is shown below.

```
DELIVER PROCESS P1A^TSSXML001, "DELIVER KEYCONV ERASE THALES1"
```

## Disabling a Security Device

Disabling a security device changes the status of the device from active to inactive, thereby removing the device from service. The Transaction Security component generates an event message, indicating that the device has been deactivated. This command also cancels the automatic reset timer if the Transaction Security component is currently attempting to enable the device at the intervals specified in the Reset Timer Limit field on the Security Configuration window. This prevents the Transaction Security component from continuing to attempt to re-establish communications with a device when it is to be taken out of service for an extended period of time.

The Transaction Security component does not send requests to inactive devices; it only alternates requests among active devices with a consecutive error count that is less than the maximum.

You can disable a security device by entering a command on the Process Control window or from an XPNET network control facility. Both methods are described below. You do not need to warmboot the Transactions Security Services process for the Security Device Configuration File (SECDCD) after disabling the security device.

**Note:** If you want to permanently disable a security device, set the Status field on the Security Device Configuration window for the device to Inactive and warmboot the SECDCD (Destination: PCTL, Command: Alter, Component: Data Source, Command Text: SEC\_DEV\_CONFIG).

## Process Control Window

To disable one or all security devices from the Process Control window, perform the following:

1. Log on to the ACI desktop user interface and access **System Operations > Process Control**. The Process Control window is displayed.
2. In the Destination field, select IS from the list of values or enter the symbolic name of the XML Server process or a Transaction Security Services process.
3. In the Command field, select Alter from the list of values.
4. In the Component field, select Transaction Security from the list of values.



5. In the Command Text field, enter the following command. If you want to disable a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to disable all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

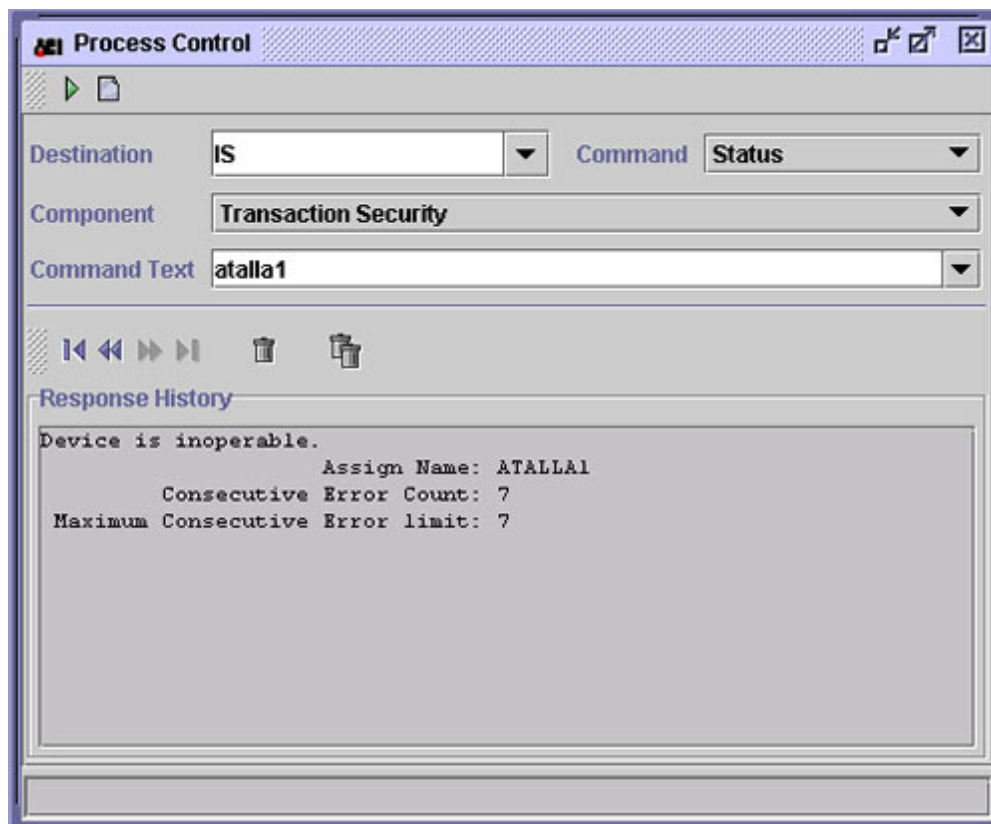
```
DISABLE sec-dev-assign-name | *
```

**Examples:**

```
DISABLE ATALLA1  
DISABLE THALES1  
DISABLE ERACOM1  
DISABLE BNTNG1  
DISABLE *
```

6. Click the Send button at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window. If the specified security device was successfully disabled, the status of the device on the Security Device Configuration window is changed from “Active” to “Inactive.”

The following illustration shows the response for the Status command after a security device has been disabled. Note that the device is now inoperable and that the consecutive error count has been set to the maximum consecutive error limit value.



## Network Control Facility

An example of the DISABLE command entered from an XPNET network control facility (e.g., NCPCOM) is shown below.

```
DELIVER PROCESS P1A^TSSISLO01, "ALTER TSEC DISABLE ATALLA1"
```

## Enabling a Security Device

Enabling a security device causes the Transaction Security component to send a status or echo test message to the security device. If this message is successfully processed, the component changes the status of the device from inactive to active and sets the consecutive error count to zero, thereby returning the device to service. If this message is not successfully processed for a device, the Transaction Security component generates an event message that indicates the error and specifies whether the status or echo test message will be automatically retried. The Reset Timer Limit field on the Security Device Configuration window specifies the number of seconds the component waits before retrying the security device call. If the Reset Timer Limit field is set to 0, the component does not automatically attempt to retry the call. The Transaction Security component does not send online transaction requests to inactive devices; it only alternates requests among active devices with a consecutive error count that is less than the maximum.

You can enable a security device by entering a command on the Process Control window or from an XPNET network control facility. Both methods are described below. You do not need to warmboot the Transactions Security Services process for the Security Device Configuration File (SECDCD) after enabling the security device.

**Note:** If you want to permanently enable a security device, set the Status field on the Security Device Configuration window for the device to Active and warmboot the SECDCD (Destination: PCTL, Command: Alter, Component: Data Source, Command Text: SEC\_DEV\_CONFIG).

## Process Control Window

To enable one or all security devices from the Process Control window, perform the following:

1. Log on to the ACI desktop user interface and access **System Operations > Process Control**. The Process Control window is displayed.
2. In the Destination field, select IS from the list of values or enter the symbolic name of the XML Server process or a Transaction Security Services process.
3. In the Command field, select Alter from the list of values.
4. In the Component field, select Transaction Security from the list of values.

5. In the Command Text field, enter the following command. If you want to enable a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to enable all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

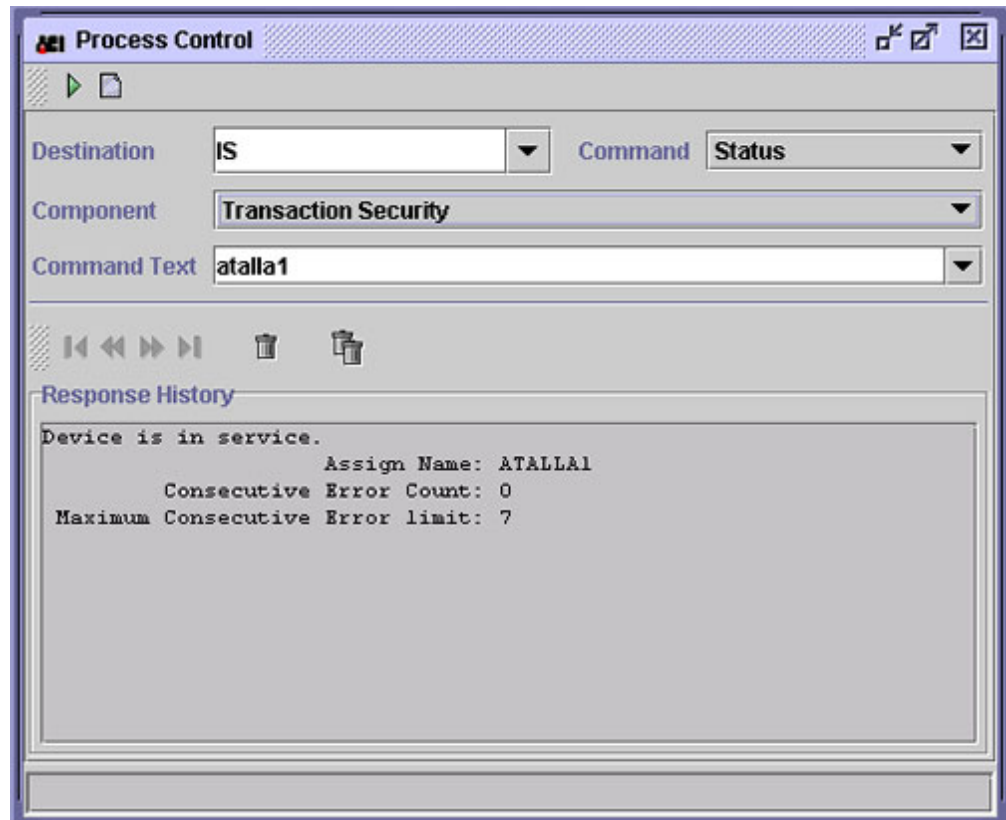
```
ENABLE sec-dev-assign-name | *
```

**Examples:**

```
ENABLE ATALLA1  
ENABLE THALES1  
ENABLE ERACOM1  
ENABLE BNTNG1  
ENABLE *
```

6. Click the Send button at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window. If the specified security device was successfully enabled, the status of the device on the Security Device Configuration window is changed from “Inactive” to “Active.”

The following illustration shows the response for the Status command after a security device has been enabled. Note that the device is now in service and that the consecutive error count has been reset to zero.



## Network Control Facility

An example of the ENABLE command entered from an XPNET network control facility (e.g., NCPCOM) is shown below.

```
DELIVER PROCESS P1A^TSSISLO01, "ALTER TSEC ENABLE  ATALLA1"
```

## Initializing a Security Device

Initializing a security device causes the TSS application to send the text strings configured in the Initialization String and Configuration String fields for the device on the Security Device Configuration window to the device. In addition, if the status of the device is Active, this command causes the Transaction Security component to send a status or echo test message to the security device. If this message is successfully processed, the component sets the consecutive error count to zero and uses the response to determine the firmware version and maximum buffer length displayed on the Security Device Configuration window for the device. If this message is not successfully processed for a device, the Transaction Security component generates an event message that indicates the error and specifies whether the status or echo test message will be automatically retried. The Reset Timer Limit field on the Security Device Configuration window specifies the number of seconds the component waits before retrying the security device call. If the Reset Timer Limit field is set to 0, the component does not automatically attempt to retry the call.

This command causes the firmware version and maximum buffer length for the device to be displayed on the Security Device Configuration window. This information is not displayed until you initialize the device.

For Bull BNTng HSMs, this command also loads data from the applicable record in the BNTng Initialization data source (BNTID) into the HSM.

You can initialize a security device by entering a command on the Process Control window or from an XPNET network control facility. Both methods are described below.

## Process Control Window

To initialize one or all security devices from the Process Control window, perform the following:

1. Log on to the ACI desktop user interface and access System Operations > Process Control. The Process Control window is displayed.
2. In the Destination field, select IS or XMLS from the list of values or enter the symbolic name of the XML Server process or a Transaction Security Services process.
3. In the Command field, select Deliver from the list of values.
4. In the Component field, select Transaction Security from the list of values.

5. In the Command Text field, enter the following command. If you want to initialize a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to initialize all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

```
INIT sec-dev-assign-name | *
```

**Examples:**

```
INIT ATALLA2
INIT THALES1
INIT ERACOM1
INIT BNTNG1
INIT *
```

6. Click the Send button at the top of the Process Control window. A “Command sent” message is displayed in the Response History area of the window.

## Network Control Facility

An example of the INIT command entered from an XPNET network control facility (e.g., NCPCOM) is shown below.

```
DELIVER PROCESS P1A^TSSISLO01, "DELIVER TSEC INIT ATALLA1"
```

## Obtaining the Current Status of a Security Device

The status of a security device indicates whether the device is in service or inoperable, enabled or disabled, and lists the consecutive error count, maximum consecutive error count, firmware version, and maximum buffer length for the device. The Transaction Security component does not send requests to inoperable devices; it only alternates requests among devices in service. To obtain the status of one or all security devices, perform the following:

1. Log on to the ACI desktop user interface and access **System Operations > Process Control**. The Process Control window is displayed.
2. In the Destination field, enter the assign name of a Transaction Security Services process defined in the System Interface Message Delivery Service Destination File (MDSDEST).
3. In the Command field, select Status from the list of values.
4. In the Component field, select Transaction Security from the list of values.
5. In the Command Text field, enter the assign name for the device or an asterisk (\*). If you want to obtain the status for a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to obtain the status for all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

*sec-dev-assign-name | \**

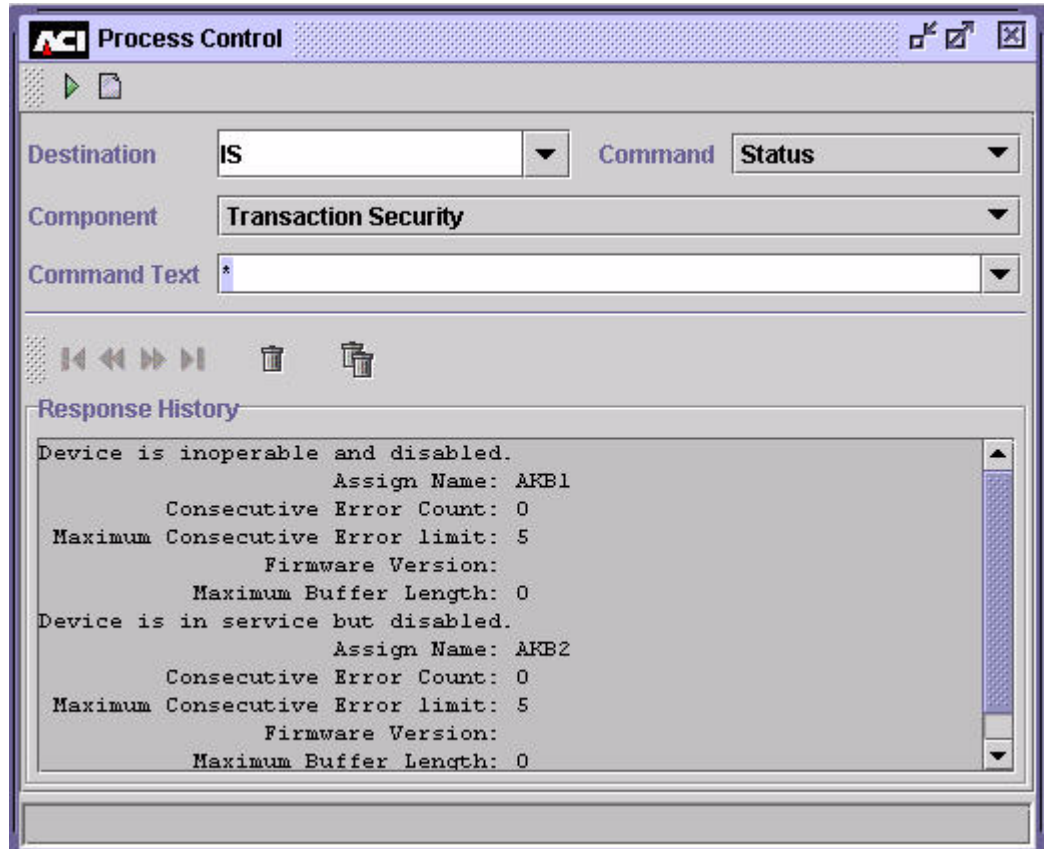
Examples:

ATALLA1  
THALES1  
ERACOM1  
BNTNG1  
\*

6. Click the Send button at the top of the Process Control window. The status information for the device is displayed in the Response History area of the window. If the specified security device is active, “Device is in service” is displayed. If the specified device is inactive, “Device is inoperable” is displayed. The status response also displays the assign name for the device,



the consecutive error count, the maximum consecutive error limit, firmware version, and maximum buffer length. A sample of the response text for the status of multiple security devices is shown below.



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## **D: Europay, MasterCard, and Visa (EMV)**

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This appendix contains miscellaneous topics related to EMV processing when using the Transaction Security Services process.

## Cryptographic Support

EMV cryptographic support is provided for the following security functions specified by MasterCard and Visa:

- For request and response cryptogram processing, Visa uses the derived key stored on the card, whereas MasterCard generates a session key from the derived key.
- For secure messaging, Visa has a different method of generating the session key from that used by MasterCard.
- Visa supports secure messaging with both integrity and confidentiality.

To support different security implementations, the Transaction Security Services process uses the EMV Scheme field on the EMV tab of the Authentication Profile Configuration window, and the cryptogram version number passed in the EMV request, to determine the key and transaction data fields to use in cryptogram functions. See the “EMV Scheme and Cryptogram Version Number” topic below.

Both Visa and MasterCard permit the use of issuer-proprietary cryptogram processing. This is identified by a specific cryptogram version number. The data fields required to generate the issuer-proprietary cryptogram are set up based on the cryptogram version number in the request.

## EMV Scheme and Cryptogram Version Number

To support different security implementations for individual payment schemes, the EMV Scheme field on the EMV tab of the Authentication Profile Configuration window identifies the specific EMV implementation associated with an EMV profile (i.e., authentication data). This field, in conjunction with the cryptogram version number passed in requests, identifies the specific implementation associated with an EMV Authorization File (EMVSD) record and is used by the Transaction Security Services process to determine the specific security module calls and security functions to be performed. Each of the valid EMV schemes are explained below.

- Visa—Issuer application data (IAD) format defined by Visa in the *Visa Integrated Circuit Card Specification*. (Note this format is used for UKIS cards).
- MasterCard/Europay—IAD format defined for MasterCard Chip Payment Application (MCPA) version 2.1 (M/Chip 2.1) cards.

- M/CHIP 4—IAD format defined for MasterCard Chip Payment Application (MCPA) version 4 (M/Chip 4) cards.

**Note:** On the Authentication Profile Configuration window, only values of Visa and MasterCard/Europay are available in the EMV Scheme field. For MasterCard/Europay, the cryptogram version number is sufficient to differentiate between M/Chip 2.1 and M/Chip 4 cards. For M/Chip 4 cards, the valid cryptogram version number values are %h10, %h11, %h12 and %h13. The first nibble set to “1”, effectively identifies the card as an M/Chip 4 card. For M/Chip 2.1 cards, the first nibble of the cryptogram version number is always set to “0”.

## Cryptogram Version Numbers in EMV

The Cryptogram Version Number is used (in conjunction with the EMV scheme) to identify the security requirements in EMV online transaction processing. In particular, the Cryptogram Version Number defines the required processing in the following areas:

- Card key derivation method
- Session key derivation method for ARQC calculation
- Data format in ARQC/TC calculation
- Session key derivation method for ARPC calculation
- Method used to calculate ARPC
- Session key derivation method for secure messaging
- Data format in secure messaging MAC calculation
- Padding of encrypted data for secure messaging
- Method used to encrypt data in secure messaging

The tables below indicate the security processing options, data formats, and methods used for each supported combination of EMV scheme and cryptogram version number. You can obtain further details on each of these security processing options, data formats, and methods in the EMV specifications.

Throughout these tables, *x* represents the characters 0–9 and A–F in cryptogram version number %h0*x*, and *y* represents the characters A–F in cryptogram version number %h*y*4.

## Card Key Derivation Method

| Security Processing Options                                                                                                                                                                                                                                                                                                                                                                                                                                    | EMV Application |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | EMV Scheme      | CVN  |
| <p>Option A – Format a 16-byte data block by concatenating the PAN and application PAN sequence number (APSN), using padding or truncation if required. The left half of the key is computed by 3DES encrypting the PAN/APSN with the master key. The right half of the key is computed by 3DES encrypting the inverted (bitwise) PAN/APSN with the master key.</p> <p>If the APSN is not present in the transaction, then it is replaced by a '00' byte).</p> | VIS             | %h0A |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | VIS             | %h0E |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h0x |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h10 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h11 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h12 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h13 |

## Session Key Derivation Method For ARQC Calculation

| Security Processing Options                                                                                                                                                                                                                                                                                                                                                                    | EMV Application |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                                                                                                                                                                                | EMV Scheme      | CVN  |
| None – Use card key.                                                                                                                                                                                                                                                                                                                                                                           | VIS             | %h0A |
| <p>EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.</p>                                                                                                                                                      | VIS             | %h0E |
|                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h12 |
|                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h13 |
| <p>EMV 1996 – Format an 8-byte data block, and use it to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key.</p> | M/Chip          | %h0x |
|                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h10 |
|                                                                                                                                                                                                                                                                                                                                                                                                | M/Chip          | %h11 |

## Data Format In ARQC/TC Calculation

| Data Format                                                                                                                                                                                                                                                                                                                               | EMV Application |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                                                                                                                           | EMV Scheme      | CVN  |
| Amount, Auth<br>Amount, Other<br>Terminal Country Code<br>Terminal Verification Results (TVR)<br>Transaction Currency Code<br>Transaction Date<br>Transaction Type<br>Unpredictable Number<br>Application Interchange Profile (AIP)<br>Application Transaction Counter (ATC)<br>Card Verification Results (4 bytes)                       | VIS             | %h0A |
|                                                                                                                                                                                                                                                                                                                                           | VIS             | %h0E |
|                                                                                                                                                                                                                                                                                                                                           | M/Chip          | %h0x |
| Amount, Auth<br>Amount, Other<br>Terminal Country Code<br>Terminal Verification Results (TVR)<br>Transaction Currency Code<br>Transaction Date<br>Transaction Type<br>Unpredictable Number<br>Application Interchange Profile (AIP)<br>Application Transaction Counter (ATC)<br>Card Verification Results (6 bytes)                       | M/Chip          | %h10 |
|                                                                                                                                                                                                                                                                                                                                           | M/Chip          | %h12 |
| Amount, Auth<br>Amount, Other<br>Terminal Country Code<br>Terminal Verification Results (TVR)<br>Transaction Currency Code<br>Transaction Date<br>Transaction Type<br>Unpredictable Number<br>Application Interchange Profile (AIP)<br>Application Transaction Counter (ATC)<br>Card Verification Results (6 bytes)<br>Counters (8 bytes) | M/Chip          | %h11 |
|                                                                                                                                                                                                                                                                                                                                           | M/Chip          | %h13 |

## Session Key Derivation Method For ARPC Calculation

| Derivation Method                                                                                                                                                                                                                  | EMV Application |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                    | EMV Scheme      | CVN  |
| None – Use card key.                                                                                                                                                                                                               | VIS             | %h0A |
|                                                                                                                                                                                                                                    | M/Chip          | %h0x |
|                                                                                                                                                                                                                                    | M/Chip          | %h10 |
|                                                                                                                                                                                                                                    | M/Chip          | %h11 |
| EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC. | VIS             | %h0E |
|                                                                                                                                                                                                                                    | M/Chip          | %h12 |
|                                                                                                                                                                                                                                    | M/Chip          | %h13 |

## Method Used To Calculate ARPC

| Calculation Method                                                                                                                                 | EMV Application |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                    | EMV Scheme      | CVN  |
| Format an 8-byte data block. Calculate an 8-byte ARPC by XORing the data block with the ARQC, and 3DES encrypting the result with the session key. | VIS             | %h0A |
|                                                                                                                                                    | VIS             | %h0E |
|                                                                                                                                                    | M/Chip          | %h0x |
|                                                                                                                                                    | M/Chip          | %h10 |
|                                                                                                                                                    | M/Chip          | %h11 |
|                                                                                                                                                    | M/Chip          | %h12 |
|                                                                                                                                                    | M/Chip          | %h13 |



## Session Key Derivation Method For Secure Messaging

| Derivation Method                                                                                                                                                                                                                                                                                                                                                                                  | EMV Application |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                                                                                                                                                                                    | EMV Scheme      | CVN  |
| EMV 1996 – Format an 8-byte data block (the ARQC), and use it to generate the left and right halves of the session key. The left half is computed by replacing the third byte of the data block with %hF0, and 3DES encrypting the result with the card key. The right half is computed by replacing the third byte of the data block with %h0F, and 3DES encrypting the result with the card key. | M/Chip          | %h0x |
|                                                                                                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h10 |
|                                                                                                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h11 |
| EMV 2000 – Use a specified function that contains a branch factor, a <i>parent</i> key, and a <i>grandparent</i> key. Use the function to take bits from the ATC to derive a session key that is unique for each value of the ATC.                                                                                                                                                                 | VIS             | %h0E |
|                                                                                                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h12 |
|                                                                                                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h13 |
| Format an 8-byte data block (including the ATC), and use it to generate the left and right halves of the session key. The left half is computed by XORing the data block with the left half of the card key. The right half is computed by performing a bitwise inversion and XORing the result with the right half of the card key.                                                               | VIS             | %h0A |

## Data Format In Secure Messaging MAC Calculation

| Data Format                                                                                                                                                                                                                                                                                                        | EMV Application |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                                                                                                                                    | EMV Scheme      | CVN  |
| The data elements (including the MAC) contained in the data field of the secured command are not composed of TLV data objects. Transaction-specific data, (e.g., the ATC), is inserted into the data field (immediately before any command data) to ensure that the MAC value generated is unique per transaction. | VIS             | %h0A |
|                                                                                                                                                                                                                                                                                                                    | VIS             | %h0E |
|                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h0x |
|                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h10 |
|                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h11 |
|                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h12 |
|                                                                                                                                                                                                                                                                                                                    | M/Chip          | %h13 |

## Padding Of Encrypted Data For Secure Messaging

| Padding Method                                                                                                                                                                                              | EMV Application |      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
|                                                                                                                                                                                                             | EMV Scheme      | CVN  |
| Prefix encrypted data with 1 byte containing the length of the encrypted data. Append encrypted data with 1 byte containing %h80. Append bytes of %h00 until data block is a multiple of 8 bytes in length. | VIS             | %h0A |
|                                                                                                                                                                                                             | VIS             | %h0E |
| Pad only if the encrypted data is not a multiple of 8 bytes in length.                                                                                                                                      | M/Chip          | %h0x |
|                                                                                                                                                                                                             | M/Chip          | %h10 |
|                                                                                                                                                                                                             | M/Chip          | %h11 |
|                                                                                                                                                                                                             | M/Chip          | %h12 |
|                                                                                                                                                                                                             | M/Chip          | %h13 |

**Method Used To Encrypt Data In Secure Messaging**

| Encryption Method            | EMV Application |      |
|------------------------------|-----------------|------|
|                              | EMV Scheme      | CVN  |
| Electronic Code Book (ECB).  | VIS             | %h0A |
|                              | VIS             | %h0E |
| Cipher Block Chaining (CBC). | M/Chip          | %h0x |
|                              | M/Chip          | %h10 |
|                              | M/Chip          | %h11 |
|                              | M/Chip          | %h12 |
|                              | M/Chip          | %h13 |

## Issuer Script Commands

Issuer script commands allow the EMV issuer to alter the data or functionality of cards already in circulation. A script is a command or a string of commands transmitted by the issuer to the terminal for the purpose of being sent serially to the integrated circuit card (ICC) as commands. The terminal forwards a script command to the card, if provided from the issuer. The card uses the script data to reconfigure itself. For example, issuer script commands can be used to support offline PIN unblocking. PIN unblocking is needed when the PIN try counter on the card has reached zero and needs to be reset to the normal value (i.e., the PIN try limit value).

Script data passed between the EMV issuer and the card is secured with a message authentication code (MAC) generated by a hardware security module. The Secure Message Key for Integrity field on the EMV tab of the Authentication Profile Configuration window contains a key used to derive the key used to generate the MAC added to secure the script command.

EMV issuer script commands are supported in pass-through mode by the BASE24 system.

# Offline PIN Change

Offline PIN validation allows a cardholder PIN to be verified by the EMV card at an ATM. The management of offline PIN validation requires two administrative functions for card issuers:

- Support for offline PIN changes (i.e., the cardholder changes his or her PIN to a new value at an ATM). The new PIN must be transmitted to the card so that the PIN value on the card can be synchronized with the new PIN offset value in the issuer's database.
- Support for offline PIN unblocking. In this case, the PIN try counter on the EMV card has reached zero and needs to be reset to the normal value (i.e., the maximum PIN try limit).

The Secure Message Key for Confidentiality field on the EMV tab of the Authentication Profile Configuration window contains a key used to derive the key used to encrypt the PIN block for secure transmission to the card. The Destination PIN Block Type field on this tab specifies the PIN block format in which the new PIN is transmitted to the card. Offline PIN unblocking is supported using an issuer script command. A sample of the EMV tab on the Authentication Profile Configuration window is provided in [section 2](#).

**Note:** An EMV PIN change function is not currently supported by Bull BNTng HSMs.

## Notification to Host

If you need to notify a card management system of offline PIN changes, set the OFFLINE PIN TRANSPORT KEY field on Card Prefix File (CPF) screen 11 to a PIN block encryption key locator value and the OFFLINE PIN MANAGEMENT ACTION field to a value of 2. The PIN block encryption key locator value must correspond to a primary key value (interface name) for a record in the ISECD. This record must specify a PIN encryption key in the Outbound Keys field on the PIN information tab of the Interface Security Configuration window. The Transaction Security Services process translates the new PIN block to encryption under this key. The translated PIN block is then written to the PIN Change token in the Transaction Log File (TLF). When the card management system receives the new PIN block (e.g. in a TLF extract), it uses the same key as stored in the ISECD to decrypt the new PIN block.

## Destination PIN Block

The Transaction Security Services process uses different destination PIN blocks, based on the EMV scheme selected for EMV processing. If you select the Visa scheme, the Transaction Security Services process uses the Visa Format Without Using Current PIN for the destination PIN block. If you select the MasterCard/Europay scheme, the Transaction Security Services process uses the Standard EMV PIN Block for the destination PIN block. The EMV scheme is set using the EMV Scheme option button on the EMV tab of the Authentication Profile Configuration window. Refer to the appropriate vendor documentation for detailed information on these PIN block formats.

The Destination PIN Block Type field on the EMV tab of the Authentication Profile Configuration window is currently a display-only field. It displays the PIN block format value appropriate to the EMV scheme selected.

## EMV 2000

The Transaction Security Services process supports EMV2000 processing for Atalla NSPs, Bull BNTng HSMs, and Thales e-Security hardware security modules (HSMs). The key derivation method value passed in messages to the Transaction Security Services process indicates whether normal EMV processing (1 - EMV 1996) or EMV 2000 processing (2 - EMV 2000) is performed.

### Atalla NSPs

EMV 2000 processing is supported on the following Atalla NSPs: Axx100-V or Axx100 (AKB) running version 2.62 or later software.

For Atalla NSPs, the Transaction Security Services process sends the branch/height parameter in 350, 351, and 352 commands requests for EMV 2000 processing. In addition, for the 350 and 351 command requests, the Transaction Security Services process passes the ATC instead of a random number for EMV 2000 processing.

### Bull BNTng HSMs

EMV 2000 processing is supported on Bull BNTng HSMs running version 7 or later software. For Bull BNTng HSMs, the Transaction Security component sends the branch/height parameter in E2, E3, and E4 command requests for EMV 2000 processing.

### Thales e-Security HSMs

For Thales e-Security hardware security modules (HSMs), the Transaction Security Services process supports EMV 2000 processing for version 2.0a (firmware version 0352.E001) or later software on the HSM 8000 and version 7.0a (firmware version 1053.0801) or later software on the RG7x00.

The following three commands are used for EMV 2000 processing instead of the KQ and KU EMV authorization commands:

- Generate Secure Message with Integrity and optional Confidentiality and PIN Change (KY)
- ARQC (or TC/ACC) Verification or ARPC Generation (KW)

- Verify Encrypted Counters - M/Chip 4 (K0)

The KY command performs a similar function to the KU command and the KW command performs a similar function to the KQ command. The KY and KW commands, however, use the EMV 2000 method (session key derivation function) for generating the session key.

## Session Key Derivation Function

This method uses the session key derivation function, which generates a unique session key for each ICC application transaction by generating a *tree* of keys. The ICC master key entered in the Cryptographic Keys field on the EMV tab of the Authentication Profile Configuration window serves as the base of this tree. Several levels of intermediate keys reside above it, with each intermediary key derived from the keys beneath it. At the very top of the tree are the session keys, with one session key per value of the application transaction counter (ATC). For detailed information on session key derivation for EMV 2000, refer to your hardware security module vendor documentation and the appropriate EMV 2000 specifications.

The session key derivation function uses two parameters—the height of the tree (i.e., the number of levels of intermediary keys in the tree excluding the base level) and the *branch* factor (i.e., the number of *child* keys that can be derived from a *parent* key). The parent key must be one level lower in the tree to the child key it derives. You configure the tree height and branch factor in the Branch/Height Parameters field on the EMV tab of the Authentication Profile Configuration window. Possible values for this field are:

- Branch factor 2; Tree Height 16
- Branch factor 4; Tree Height 8

The Transaction Security Services process passes the branch factor and tree height parameters you configure in extended 350, 351, and 352 command requests sent to an Atalla NSP, in E2, E3, and E4 command requests sent to a Bull BNTng HSM, and in KY, KW, and K0 command requests to a Thales e-Security HSM.

Atalla NSPs also require a 16-byte initializing value (IV) and index parameters in the extended 350, 351, and 352 commands. The Transaction Security Services process does not send these parameters in command requests. Therefore, the Atalla NSP uses default values of zeros for the IV and 7 for the index.



## Verify Counters

For the M/Chip 4 scheme, the Transaction Security Services process also provides the ability to verify an encrypted counter that the issuer can optionally include in Issuer Application Data in Thales e-Security HSMs. The Transaction Security Services process sends the K0 command request to verify encrypted counters.

## Data Block Padding

In the Atalla 350 command, Bull BNTng E2 command, and Thales KW command, the Transaction Security Services process appends the data block with the x80 pad character and additional padding of binary zeros to ensure it is a multiple of 8 bytes.

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# **E: Diebold PIN Verification**

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This appendix contains miscellaneous topics related to Diebold PIN verification processing when using the Transaction Security Services process.

## Diebold Number Table

A Diebold number table (DNT) is a random set of 256 nonrepeating values used in Diebold PIN verification.

Diebold number table values must be entered in the Diebold Number Table fields under the Diebold tab on the Authentication Profile Configuration window of the ACI desktop user interface. The Diebold number table is downloaded into HSM memory when the Transaction Security component is initialized or when an operator enters the DNTLOAD command from the Process Control window or an XPNET network control facility.

Each value in the Diebold number table consists of a pair of hexadecimal characters (e.g., AA, A1, 9F, etc.). There are 256 possible combinations of hexadecimal pairs, and these combinations can be scrambled to derive a unique set of table entries, subject to the following rules:

- There must be 256 values in the table. Eight values are defined in each row in the table. Thus, 32 rows must be defined in the table.
- No hexadecimal pair can be repeated.
- The first character of the first hexadecimal pair of the table must be a C. Thus, the first hexadecimal pair can be any one of the 16 combinations from C0 to CF

**Note:** This table functions in much the same way as a DES key; therefore, it must be kept confidential.

## Atalla NSPs

For an Atalla NSP, each of the 32 rows in the Diebold number table must be encrypted under the MFK before they are loaded into the NSP's memory. You can enter the Diebold number table values on the Authentication Profile Configuration window either in the clear or in encrypted format. In either case, the values must be entered in AKB format. You cannot enter Diebold number table values in non-AKB (i.e., variant) format.

## Encrypted Data Entry

To enter or update the table in encrypted format, you must first encrypt the Diebold number table values using the Atalla Secure Configuration Terminal (SCT) or Secure Configuration Assistant (SCA) and copy the resulting cryptograms into the Diebold Number Table fields under the Diebold tab on the Authentication Profile Configuration window. Connect the SCT or SCA to the serial port of your PC. Use the Atalla Helper application with the SCT or the Calculate AKB Cryptogram menu on the SCA to set up the AKB header and display the resulting encrypted key. Set the Encrypt Type field on the Authentication Profile Configuration window to a value of Encrypted and copy the encrypted data for each DNT row into the DNT fields. When using the SCA, you can copy the encrypted data from the SCA Remote Management application on your PC. You can copy and paste the entire key into a row of the Diebold number table. The ACI desktop automatically parses the key parts into the correct fields (i.e., Header, Key Part 1, Key Part 2, Key Part 3, MAC, and Check Digits). After you have copied all encrypted data for all rows, click on the Save button to store the table.

## Clear Data Entry

To enter the Diebold number table in the clear, the premium command 75 (Enter Key Part) must be enabled in the security policy of the NSP. If you enter or update the Diebold number table in the Diebold Number Table fields in clear format, set the Encrypt Type field to a value of Clear and click on the Save button after the clear values are entered. The Diebold number table values are automatically encrypted under the MFK using command 75 for storage in the TSS database. After the Diebold number table is entered or updated, you can only view it in encrypted format. When viewing the Diebold number table after it is entered or updated, the Encrypt Type field value indicates whether the table was originally entered in clear or encrypted format.

**Warning:** The entry of clear Diebold number table values is not secure and is not in compliance with major card association PIN security rules. Entering the Diebold number table in encrypted format is the most secure method for entering the table. In addition, Atalla command 75, which must be enabled to support the entry of clear values, could be used by an adversary to seriously breach the security of an NSP. Therefore, if you do enter Diebold number table values in the clear, command 75 should be immediately disabled after the clear table values are entered and saved.

## Thales e-Security HSMs

For a Thales e-Security HSM, the Diebold number table must be entered and updated in encrypted format in the Diebold Number Table fields under the Diebold tab on the Authentication Profile Configuration window. Thus, the Encrypted Type field is not used when entering or updating Diebold number table data. When viewing Diebold number table data, the Encrypted Type field will always display the Encrypted value. To enter the Diebold number table on a Thales e-Security HSM, enter the table data and the location of the data in user storage using the R command on the HSM console. The data is encrypted under the appropriate LMK key pair and the encrypted data is displayed on the screen. Copy the encrypted data from the HSM console and enter it on the Authentication Profile Configuration window. After all the data is entered, click the Save button to store the encrypted Diebold number table in the TSS database. An Environment attribute, RSM\_KEY\_STORE\_LGTH, specifies the length of the Diebold number table key values stored in HSM memory. The valid values for this attribute are as follows:

- 1 = Single length keys (Default)
- 2 = Double length keys
- 3 = Triple length keys

## Relative Location

The Relative Location field under the Diebold tab of the Authentication Profile Configuration window specifies the relative location of the Diebold Number Table (DNT) in the security device. When the security device contains one DNT, the value entered in this field is 1. When the security device contains multiple DNTs, the relative location of the DNT used with this authentication profile is entered in this field. The Transaction Security component calculates the actual table index based on the value specified in this field.

# Downloading the Diebold number table (DNT)

The DNT entered on the Authentication Profile Configuration window must be downloaded into the memory of your security modules before you can perform PIN verification. This must be done any time you change the DNT values on the Authentication Profile Configuration window. The DNT is not automatically downloaded when you stop and restart a Transaction Security Services process due to the fact that several Transaction Security Services processes can be running in the TSS environment. Therefore, you must manually download the DNT using this procedure.

You can download the DNT by entering a command on the Process Control window or from an XPNET network control facility. Both methods are described below.

## Process Control Window

To download the DNT into the memory of a hardware security device from the Process Control window, perform the following steps using the ACI desktop:

1. Log on to the ACI desktop user interface and access **System Operations > Process Control**. The Process Control window is displayed.
2. In the Destination field, select IS from the list of values.
3. In the Command field, select Deliver from the list of values.
4. In the Component field, select Transaction Security from the list of values.
5. In the Command Text field, enter the following command. If you want to download the DNT to a single device, enter the name of the device as it is configured on the Security Configuration window in the command text. If you want to download the DNT to all security devices in the BASE24 system, enter an asterisk (\*) in the command text.

```
DNTLOAD sec-dev-assign-name | *
```

### Examples:

```
DNTLOAD ATALLA1
DNTLOAD THALES1
```

6. Click Submit (▶) at the top of the Process Control window. A “command sent” message is displayed in the Response History area of the window.

## Network Control Facility

An example of the DNTLOAD command entered from an XPNET network control facility (e.g., NCPCOM) is shown below.

```
DELIVER PROCESS P1A^TSSISLO01, "DELIVER TSEC DNTLOAD ATALLA1"
```



# F: NCR PIN Verification Support

---

This appendix describes BASE24 support of the NCR PIN verification method.

NCR PIN verification can be performed only within Thales e-Security RG7x00 HSMs equipped with version 2.01/6.01 or later firmware.

## NCR PIN Verification

NCR PIN verification is based on the use of a PIN verification key, a PIN data block, and a PIN offset. NCR PIN verification can be performed using a hardware security module linked to the processor on which the BASE24 system is running.

At card-issuing time, the card issuer generates a unique PIN and PIN offset combination for each card, using validation data taken from the cardholder's account number. If the PIN is issuer-assigned or chosen by the cardholder, the PIN offset is generated to correspond to the PIN; if the PIN offset is a set value, the PIN is generated to correspond to the PIN offset. The PIN offset is then placed in the magnetic stripe data of the card.

The HSM performs the following steps to verify a PIN using the NCR PIN verification method.

1. Decrypts the single- or double-length PIN encryption key from under the appropriate LMK pair as designated by the PIN source flag.  
If the key does not have odd parity on each byte, the HSM returns error code 10 (PIN key parity error).
2. Decrypts the \*PVK from under LMK pair 20-21(32H) or (1A+32H).  
If the key does not have odd parity on each byte, the HSM returns error code 11 (\*PVK parity error).
3. Decrypts the PIN block from under the PIN key.
4. Disassembles the PIN block using the format defined by the PIN block format code and, if necessary, the 12 right-most digits of the PAN after the check digit has been removed.  
If the PIN block does not contain valid values, the HSM returns error code 20 (PIN block does not contain valid values).  
If the PIN length is invalid, the HSM returns error code 24 (PIN < 4 or > 12 digits).
5. Calculates the PIN from the PIN data block and PIN offset according to the NCR PIN verification algorithm.
6. Compares the derived PIN with the customer PIN. If the PIN is valid, the HSM returns error code 00 (No errors). If the PIN is not valid, the HSM returns error code 01 (Verification failure).

## Input Data

The NCR PIN verification algorithm uses the following input data to verify a PIN:

- PIN encryption key (the key under which the PIN block is encrypted)
- PIN verification key
- PIN source flag
- PIN block
- PIN block format
- PIN data block
- PAN (12 right-most digits, excluding the check digit)
- PIN offset

## PIN Encryption Key and PIN Source Flag

BASE24 can receive an NCR PIN block from a terminal or from Interpay. When the transaction originates from an ATM terminal, the PIN block is encrypted under the terminal PIN key. When the transaction originates from Interpay, the PIN block is encrypted under the interface PIN key.

The PIN source flag indicates whether the PIN is encrypted under the Interpay PIN encryption key or the terminal PIN encryption key. Valid values are:

- 0 = PIN block from Interpay. The PIN key is encrypted under LMK pair 06–07.
- 1 = PIN block from terminal. The PIN key is encrypted under LMK pair 14–15.

Before verifying the PIN, the HSM decrypts the PIN block accordingly.

## PIN Verification Key

NCR PIN verification requires that the validation data be encrypted for processing. The PIN verification key is the key used to encrypt that validation data. In this case, encryption is used to generate a different sequence of digits and to tie that string of digits to a secret value (the key).

For confidentiality, an encrypted key (encrypted by or for the hardware module) is required that can be passed to the HSM for encrypting the validation data. The encrypted PIN verification key must be entered in the Encryption Keys table on the NCR tab of the Authentication Profile Configuration window. The Key Scheme field in the Encryption Keys table specifies whether the Thales e-Security HSM is configured to use the old (ANSI X9.17 method) non-variant (32H) key encryption scheme or the variant (1A+32H) key encryption scheme.

## PIN

In NCR PIN verification, the PIN entered by the cardholder is compared to the PIN computed by the hardware security module.

PINs are always verified by a hardware module, and must be passed to the module in encrypted form, along with the key used to encrypt them. The module then decrypts the PIN for comparing to the PIN it computes.

BASE24-atm allows cardholders to select their own PINs. The requirements for this are that PIN offsets be maintained in the CAF and that they initially be blank.

Upon receiving the first transaction from a cardholder, BASE24-atm computes an appropriate PIN offset based on the PIN entered by the cardholder and places that offset in the CAF. From that point on, that PIN entered by the cardholder is the only one allowed.

PINs can be from four to 12 digits in length. However, the length of the PIN offset is the minimum required length of the PIN. For example, if the PIN offset is four digits, the PIN must be at least four digits as well. The PIN can be longer than the PIN offset; however, if the PIN is longer than the PIN offset, only the right-most digits of the PIN are used in the verification.

## PIN Block and PIN Block Format

The PIN block is 16 hexadecimal characters that includes the encrypted PIN. NCR PIN verification supports PIN blocks in PIN/PAN (ANSI or ISO Format 0) or PIN PAD format, depending on the transaction source. PINs received from ATMs are in PIN PAD format (Thales PIN block format 03), encrypted under the \*TPK. PINs received from Interpay are in PIN/PAN (ANSI or ISO Format 0) (Thales PIN block format 01), encrypted under the \*KSO. The PIN source flag indicates the source of the transaction.

## **PIN Data Block**

The PIN data block contains the PIN identifier derived from track 3 of the card. The PIN data block is 16 hexadecimal characters in length. The NCR PIN verification algorithm uses the PIN identifier along with the PIN offset from track 3 to calculate the PIN compared to the PIN entered by the cardholder.

## **PAN**

A primary account number (PAN) is the card number used to identify each card. It must be carried in the magnetic stripe data of the card and is used to disassemble the PIN block if necessary. The NCR PIN verification algorithm uses the 12 right-most digits of the PAN, excluding the check digit.

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